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Sukanta Das
Biplab K. Sikdar

Non-Uniform Cellular Automata

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Preface

Forty years ago, non-uniform cellular automata were first considered as the tools to find solutions in VLSI design and test. To cater to the need, since then only, the theories of non-uniform cellular automata have been nurtured. The cellular automata theories, developed during the last 30 years, are to be presented under two different heads. Such a requirement necessitates the emergence of this book. It is to cover the theories developed till date under the umbrella of non-uniform cellular automata. The Additive Cellular Automata: Theory and Applications Volume 1, by P. Pal Chaudhuri et al., published in 1997, can be considered to be the first one that accumulates non-uniform cellular automata theories. Against this backdrop, this book targets to come up with the pointers towards the investigable non-uniform cellular automata theories and their application domains.

The works of many researchers are culminated in this book. However, all these are not from our research group. We are grateful to our students, co-researchers and other scholars who have contributed to develop theory of non-uniform cellular automata. Some of the topics discussed in this book have not been published before. These are only available in the Ph.D. theses.

This book can be used as a text book on non-uniform cellular automata. From that perspective, this is the first book on non-uniform cellular automata. The chapter-notes provide the pointers to the literature and future research problems. We would be happy if the young researchers consider those problems as their research topic.

Cellular automata, introduced by John von Neumann in 1950's, are the simple models of biological self-reproduction. These consist of a large number of simple, homogeneous, locally interacting components and these can model complex systems. Our main goal is to present the fundamental theoretical concepts of non-uniform cellular automata necessary for a researcher to address advanced applications. We, therefore, include characterization tools of non-uniform cellular automata, details on reversibility, analysis and synthesis of cellular automata with specific cycle structure as well as maximal length cellular automata. Further details on number conservation, convergence of cellular automata and identification of chaotic cellular automata can also be found in this book. We hope that this will benefit researchers in diverse

disciplines who employ different modeling and simulation methodologies to study complex systems.

The department of Information Technology, and Computer Science and Technology of Indian Institute of Engineering Science and Technology have provided us with the scope to carry on our research. Special acknowledgement is due to these two departments. The other organizations such as SERB, AICTE and SPARC, under the Ministry of Education, Govt. of India have provided grant-in-aid to carry on the research. Finally, we sincerely acknowledge and appreciate the support and cooperation received from the Springer Nature executives for publication of this book.

Kolkata, India
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Sukanta Das
Biplab K. Sikdar

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Chapter 1
Introduction

A cellular automaton (CA) is a collective system. It is made with a large number of anonymous cells, organized as a regular network. The cells are ordinary and have the limited computational abilities. They interact with their neighbors only and act locally. In a CA, no cell is superior, none is inferior, and there is no master-slave relationship among the cells. The effect of co-operation of its cells is the strength of a CA.

Cellular automata (CAs) are discrete in time and space. At a particular time step t, each cell of a CA assumes a state s from a set of states S, defined for the CA. Depending on the states of its neighbors, a cell updates its state in the next time step $t + 1$. The collection of states of all the cells of a CA at time t is the configuration of the CA at t. Whenever a CA is set to an initial configuration, its cells move from one state to other in discrete steps—that is, the CA hops from one configuration to other.

One of the key properties of CAs is considered to be the uniformity, where all the cells uniformly follow the same *rule* while updating their states. This uniformity constraint of a CA simplifies the characterization of its behavior. Although this model exhibits superlative skills, a group of researchers have started questioning this uniformity constraint since 1980's, and as a result the idea of *Non-uniform Cellular Automata* has been emerged.

1.1 A Brief History of Cellular Automata

Although cellular automata have been reinvented several times (often in different names), John von Neumann is respected as the pioneer of today's CAs [1]. In the early 1950's, von Neumann was involved in developing a machine model of biological self-reproduction, and at the end, the machine was found to be a cellular automaton. This CA is an infinite 2-dimensional (2-D) square grid, where each square box, called

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cell, can be in any of the predefined 29 states. A cell changes its state depending on the states of itself and its four orthogonal neighbors. It is self-reproducing in the sense that a (finite) *pattern*, observed in an initial configuration, is reproduced as a result of the evolution of the CA. This CA cannot only model the biological self-reproduction, but also is computationally universal [2].

After its invention, the CA research has been flourished mainly in three directions: (1) development of new CAs with lesser number of states per cell, keeping the self-reproduction and universality properties intact (Direction 1), (2) discovery of various set theoretic properties of CAs such as injectivity, surjectivity, etc. (Direction 2), and (3) employing CAs to model physical systems [3] (Direction 3).

In Direction 1, von Neumann's self-reproducing CA has been simplified in terms of the number of states. In [4], Lee has reduced the state count of the CA to 16, having the von Neumann neighborhood—that is, 5. Subsequently in [5], Codd has drastically reduced the complexity of the automaton. A cell of such a CA can have 8 states. Arbib, in his work [6], has provided a simple description of self-reproducing CA with only 2 states. Banks in [7], has proved the universal computability of such 2-state CA. However, each of these constructions has been for 2-dimensional infinite CA, and considered 5-neighborhood, where each cell moves to its next state depending on the present states of itself and its four orthogonal neighbors. Apart from the 5-neighborhood, Moore neighborhood is also in place. It is 9-cell (unit cube) neighborhood interconnection [8]. The Game of Life, a famous cellular automaton designed by John Conway, follows Moore neighborhood [9].

In 2-dimensional CAs, there are other variations, such as Margolus neighborhood [10]. Generally, cells of a CA are assumed to be square boxes. However, in [11, 12], hexagonal CAs are defined over 2-D grid, where each cell is of the form of a hexagon. Further, in [13, 14], the CA is defined in hyperbolic plane.

In Direction 2, the set theoretic properties of CAs have been investigated. A CA can be viewed as a function (G) which maps one configuration to other. A CA is *injective* if G is injective. Similarly, it is *surjective* if G is surjective. The questions of injectivity, surjectivity and bijectivity of CAs have remained one of the central issues in CA research since 1960s. One of the earliest discovered results on CAs is the *Garden-of-Eden* theorems contributed by [8, 15]. These are correlated with injectivity and surjectivity. A configuration of a CA is a Garden-of-Eden configuration if it does not have a predecessor. In 1962, Moore has shown that the existence of a configuration with more than one predecessor ensures existence of a configuration without the predecessor. Myhill has proved the reverse in 1963 [15].

Bijective CAs are popularly known as reversible CAs. The research on reversibility of CA has been started with Hedlund in 1969 [16]. In their seminal work, Amoroso and Patt have reported an effective way to decide reversibility of 1-dimensional infinite CA [17]. An elegant scheme, based on the de Bruijn graph, for deciding reversibility of a 1-dimensional CA has been presented in [18]. However, Kari has shown (in 1990) that no algorithm can decide whether an arbitrary 2-D CA is reversible [19]. This result can also be extended to higher dimensional CAs. But, it is shown in [20] that a $(D+1)$-dimensional reversible CA can simulate any D-dimensional CA.

In Direction 3, CAs have been utilized as to model physical systems. For example, CAs are popularly used to model road traffic systems. A good account of CA models for different physical systems can be found in [3].

During the early phase of development, the CAs have been mainly defined over 2-D grid. Nowadays, however, the major research works are on 1-D CAs. Exploring 1-D CAs has gained popularity after Wolfram [21], who has systematically studied 1-D binary CAs in the early 1980s. He has shown that these simple CAs have beautiful features [22, 23]. However, the 1-D CAs can behave differently than the two or higher dimensional CAs. For example, reversibility, a well addressed problem in 1-D CA framework, is undecidable for two or higher dimensional CAs [19].

Although the CAs are classically defined over infinite grid, a group of researchers have started to work with finite CAs since 1980s. These researchers are mostly the technologists. They have considered the CAs for solving different real-life problems. It is observed that the finite CAs exhibit different behavior than the classical CAs in many occasions. However, many theoretical issues of finite CAs are yet to be explored. This book primarily focuses on the finite CAs.

1.2 Non-uniformity in Cellular Automata

Conventionally, all the variants of CAs possess three basic properties—*uniformity, synchronicity* and *locality*. The uniformity refers to that all the cells of a CA update their states following the same local rule. The synchronicity implies that all the cells update their states simultaneously whereas, locality refers to the fact that the rule acts locally. The *locality* is the core property of a CA.

It can be described that synchronicity is a special form of uniformity, where all the cells are updated simultaneously and uniformly. In fact, uniformity is everywhere in a CA—in local rule, in cell update and in the lattice structure. These uniformities can be summarised as:

- Uniformity in update: All the cells of a CA are updated simultaneously in each discrete time step.
- Uniformity in lattice structure and neighborhood dependency: The lattice structure is uniform and all the cells of the CA follow similar neighborhood structure.
- Uniformity in local rule: The cells follow same rule to update their states.

Over the years, the CAs are being used as successful modelling tools. Indeed, collective effort of individual cells of a CA to achieve a goal seems very natural and consistent with the history of human civilization. Not by few intellectuals, the civilization is structured due to the collective efforts of unnumbered common people. However, considering common people as uniform entities is not rational. Many natural phenomena, such as growth of trees in a forest, cannot be uniform. The modelling of such non-uniform physical/natural systems has necessitated a new variant of the CA and as a result, non-uniformity in CAs has been introduced. Following are the

three major variants of non-uniformity in CAs that can be obtained after diminishing the constraints of uniformity.

1. Asynchronous cellular automata: All cells of the CA do not update their states at the same time step—the uniform update constraint is denied.
2. Network automata: The CA is defined on a network and the cells (nodes) evolve with neighborhood defined for the network—it denies the uniform lattice and neighborhood constraint.
3. Hybrid or non-uniform cellular automata: The cells of a CA can assume different rules to update their states—it denies the uniform local rule constraint.

The asynchronous CAs and their computational ability have been first studied by Nakamura [24]. Further studies are also reported in [25–29]. The Network automata and their potential have been investigated in [30–33]. However, among the three variants, the most popular one is the *Hybrid* or *Non-uniform* CA. The non-uniform CA has been first investigated by Pries et al. [34].

In early 1980s, Stephen Wolfram has extensively studied the properties of 1-dimensional 3-neighborhood binary CAs [21, 35]. These are of simple structure and each cell of such a CA depends only on its left, self and right neighbors while updating its state. Wolfram has shown in a number of works that these CAs can exhibit chaos, generate random sequence, perform universal computation, develop traffic model etc. [22, 23]. One of the most attractive features of these CAs is that these are straight forwardly implementable in hardware. A cell of such a CA can be implemented by a flip-flop (which stores the state of the cell) and a combinational logic (referred to as the next state logic, input to which is the present states of left neighbor, self and right neighbor). The output of the combinational logic is the next state of the cell.

The simple and modular hardware design of CA acts as the catalyst to evolve the CA technology in the era of VLSI design. The uniformity constraint is slackened as there is no significant additional advantage of uniform CA over the non-uniform one so far as the hardware design is concerned. On the other hand, if such constraint is diminished, a wide variety of CAs can be at hand to solve the real-life problems. This is the first point which motivates the researchers to explore the non-uniform CAs.

It has been indicated in the very first work of non-uniform CAs that they can have many engrossing properties that the classical CAs do not have [34]. This point has been well-taken by a group of researchers, and they have started exploring those properties, such as maximality of a non-uniform CA. These properties of the CA make it suitable for pseudo-random number generation, encryption, VLSI design and test, etc. Prof. Pal Chaudhuri and his team have popularised the non-uniform CAs as technology, which is reported in [36]. The research, in this direction, has been continued and many more CA based applications have been reported.

The CAs that have been explored in [36], as well as in the subsequent works, are *linear/additive*. From hardware design or application point of view, there is no definite reason of using the linear/additive CAs only. The advantage of choosing of such a class of CAs can be the availability of standard algebraic tools for characterization.

Some tools are now available to characterize non-linear CAs [37, 38]. As a result, non-linear CAs are also used in many applications.

The dichotomy between uniform (classical) and non-uniform CAs, however, has been partly resolved in [39]. A non-uniform CA can be simulated with the uniform CAs having more states. A programmable CA structure has been developed in [36] that can simulate any non-uniform CA with uniform CAs. The similar results can be found in [38, 39]. It is shown that a non-uniform CA with d states, configured with k rules, can be simulated with a uniform CA of $k \cdot d$ states. However, this is a costly alternative if used to replace a non-uniform CA.

1.3 Cellular Automata Technology

Technology refers to the collection of techniques, methods or processes used to provide some services or solutions to the problems, or in the accomplishment of an objective, such as scientific invigilation.

The dynamical behavior of finite (non-uniform) CAs with simple cellular structure has an appeal to use them as the solutions to many real-life problems. One such dynamical behavior is the reversibility. It ensures that an initial configuration of a (finite) CA repeats after a finite number of time steps. That is, the configurations form cycles. In *maximal length CA*, all but a single configuration form a cycle. Such a CA is chosen to design Pseudo-Random Number Generator (PRNG).

Hortensius [40–42] has first proposed the hybrid/non-uniform CA based PRNG for built-in self-test (BIST) in VLSI circuits. The other notable contributions in the same line can be found in [43–46]. The maximal length CAs and CAs with large cycles have been used for the design of PRNGs [47, 48]. Apart from for the applications in VLSI domain, CAs have been utilized for general purpose PRNGs [49–51]. A comprehensive survey on PRNGs can be found in [52].

Cellular automata as technology have gained popularity due to their effectiveness in VLSI design and test. The CAs are proposed as a framework for BIST structures [38, 43, 44, 53, 54] and as the deterministic test pattern generator [38, 44, 55, 56]. Utilizing the scalability of reversible CA, a test solution for multi-core chips has been proposed in [57]. These reversible CAs are not the maximal length CAs, but having large length cycles. The reversible CAs with large length cycles contain some smaller cycles which are not used for test pattern generation. These are utilized in random pattern generation without the patterns *prohibited* for an application [58, 59]. A universal test pattern generator, referred to as the Universal BIST (*UBIST*), has been proposed in [60]. The *UBIST* can effectively generate all of the four kinds of test patterns, namely pseudo-random, pseudo-exhaustive [61], pseudo-random without prohibited patterns and deterministic. The CA based testable FSMs (Finite State Machines) have also been developed in [62–65]. The other notable contributions, related to the electronic circuit design, are the CA based error correcting codes [66, 67], signature analysis [68, 69], etc.

A reversible CA partitions its configurations space into a set of cycles which do not overlap. That is, a configuration in a cycle of a reversible CA is not reachable from a configuration of other cycle. This property is utilized in clustering. The data clustering algorithms are designed around the reversible CAs in [70–73]. The reversible CAs with cycles of equal length have been used in [74] for cryptographic solutions. A number of works are also reported that deal with the cellular automata based encryption, cryptography and authentication [75–79].

The convergence to a point attractor (a cycle of length one) is an important dynamical property of a class of CAs. Such a CA settles to a point attractor from any initial configuration. The configurations of such a CA form tree(s) rooted at the point attractor(s). If the number of such trees is more than one, then the CA is a multiple attractor CA (MACA). The MACA can act as a natural classifier. This establishes that the CA can be effective tool for pattern recognition and classification [36, 80–83]. The correlation between MACA and *Hamming Hash Family* (*HHF*) is established in [84]. Although most of these works are around linear CA, the non-linear CAs based classifiers can be found in [85–87].

The CA based text image compression technology has also been developed [88–92]. A technology, called *Encompression*, where encryption and compression are married, has been developed around CAs [93, 94]. A group of researchers have also explored the CAs for image processing. Specially, the 2-dimensional CAs are employed for image processing solutions. These address the issue of rotation, zooming, translation, segmentation, thinning, edge detection, compression and noise reduction [95–99]. In [100], a CA based transform coding scheme for still images has been proposed.

The CAs have always been an effective tool for modelling biological phenomena, such as evolution of DNA sequences [101], modelling of amino acid [102], etc. For CGI (CpG Islands) detection in DNA sequence [103], prediction and classification of enzymes [104], prediction of protein structure [105, 106], the cellular automata have also been effectively used. A detail report and prospect of CA based biological modelling can be found in the book "*A New Kind of Computational Biology*" [107].

1.4 Organization of the Book

This book deals with the theories of 1-dimensional hybrid or non-uniform CAs with different rules for their cells. Most of the contents revolve around the finite binary CAs. Many of the theories, reported in this book, keep an eye on the CA technology.

The cellular automata and non-uniform cellular automata are formally introduced in Chap. 2. It covers different classes of non-uniform CAs with a special emphasis on finite cellular automata. Priority is given on the elementary CAs (ECAs) as most of the non-uniform CAs follow ECA rules. Finally, the hardware implementation of CAs is briefed.

To characterize the non-uniform CAs, tools have been developed over the years. The de Bruijn graph, matrix algebra and reachability tree are well known tools for such characterization. The tools are introduced in Chap. 3.

One of the most considered properties of the non-uniform CAs is the reversibility, which is covered in Chap. 4. Reversibility of non-uniform CAs over infinite lattice, of linear and complemented finite CAs, as well as of non-linear (finite) CAs have been elaborated separately. An n-cell non-uniform reversible CA can be extended to get an $(n + m)$-cell reversible CA, which makes the reversible CAs scalable. The scalability issue is also included in this chapter.

Another well-known property of CAs is the number conservation. Chapter 5 explores the number conserving CAs in non-uniform domain. The analysis and synthesis of such number conserving CAs are presented. The framework provided in this chapter can also be used to explore the number increasing and decreasing CAs.

The maximal length cellular automata have been primary focus of research during the early phase of development of the theories of non-uniform CAs. Chapter 6 deals with the maximal length cellular automata. It elaborates how a maximal length CA can be used as the generator of primitive polynomials. The matrix algebra and polynomial theory are used for characterization of this class of CAs.

Chapter 7 investigates the cycle structure of linear and complemented CAs employing theories reported in Chap. 6 and vector-space theoretic analysis. For obtaining the cycle structure of non-linear CAs, reachability tree is applied. The cyclic behavior of non-linear CAs are detailed out in Chap. 8.

Chapter 9 deals with the convergence of CAs. The CAs that converge to point attractor from any configuration are studied here. These CAs can act as the hash function. This chapter provides theoretical framework for CA based hash function, designed around linear CAs. The theory developed in Chap. 8 is used in this chapter to study the convergence of finite CAs.

Chapter 10 is the final chapter of this book. It explores the chaos in non-uniform CAs. Unlike other chapters, this chapter considers the CAs defined only on infinite lattice, as the chaos can only be observed in CAs with infinite lattice. A theoretical framework is reported for the chaos in CAs, based on which a parameter is presented to predict on the chaotic behavior of CAs.

Each of the succeeding chapters is concluded with chapter notes, which provide a brief list of pointers to the available literature on the topic of the chapter. Some research problems, related to the chapter, are appended at the end.

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Chapter 2
Cellular Automata

This chapter introduces the notion of cellular automata (CAs). Although this book deals with the one-dimensional CAs, a generalized definition of cellular automaton (CA) is set in the next section. The non-uniform CAs are then defined after formal presentation of the elementary CAs (ECAs) in Sect. 2.2. The foundation of finite CAs is provided in Sect. 2.4. Finally, the hardware implementation of finite CAs are briefed in Sect. 2.5.

2.1 Basic Definitions

Let $\mathscr{L} \subseteq \mathbb{Z}^D$ denote the D-dimensional cellular space, where $\mathbb{Z}$ is the set of integers. Each element of $\mathscr{L}$ is called a cell. A cell assumes a state from a set S. To change its state, a cell considers the present states of its neighbors and based on that, it updates its state.

Definition 1 A cellular automaton (CA) is a quadruple $(\mathscr{L}, S, \mathcal{N}, R)$, where

- $\mathscr{L} \subseteq \mathbb{Z}^D$ is the D-dimensional cellular space
- S is a finite set of states
- $\mathcal{N} = (\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_m})$ is the neighborhood vector of m distinct elements of $\mathscr{L}$ which associates one cell to its *neighbors*. The neighbors of a cell at location $\overrightarrow{x} \in \mathscr{L}$ are the m cells at locations $(\overrightarrow{x} + \overrightarrow{x_i}) \in \mathscr{L}$, $1 \leq i \leq m$.
- $R : S^m \rightarrow S$ is a local rule or transition function of the automaton. The next state of a cell is given by $R(a_1, a_2, \ldots, a_m)$, where a_i is the state of ith neighbor and $1 \leq i \leq m$.

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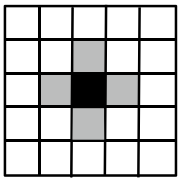
In cellular automata, both space and time are discrete. All the cells of a CA update their states simultaneously at each discrete time. An assignment of states to the cells is called a *configuration* of the CA. That is, at any given time, a configuration of the automaton is the mapping $c : \mathscr{L} \rightarrow S$ that specifies the states of all cells. The set of all possible configurations is denoted as $S^{\mathscr{L}}$. If s is the state of cell $\overrightarrow{x} \in \mathscr{L}$ at time t and c is the configuration, then $c_{\overrightarrow{x}} = s$.

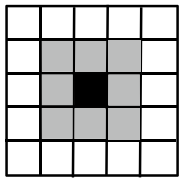
Instead of specifying neighborhood vector ($\mathcal{N}$), neighbors of cells of a CA can be identified by *radius* (**r**). The **r** is the number of consecutive cells in each direction on which a cell depends on. If **r** is considered for defining a CA, then the CA is a quadruple: ($\mathscr{L}$, S, **r**, R). In case of one-dimensional CA with radius **r**, each cell depends on its consecutive **r** left neighbors and consecutive **r** right neighbors. Considering the cell depends on itself, the neighborhood vector $\mathcal{N}$ of the CA can be obtained from the radius: $\mathcal{N} = (-\mathbf{r}, -\mathbf{r}+1, \ldots, -1, 0, 1, \ldots, \mathbf{r}-1, \mathbf{r})$. Hence, the size of neighborhood is $m = 2\mathbf{r}+1$. Figure 2.1a shows 3-neighborhood dependency with $\mathbf{r} = 1$ for a 1-dimensional CA. Here, the black cell is the cell under consideration and the shaded cells are its neighbors. In two-dimension, however, two types of neighborhood dependencies are popular—von Neumann's 5-neighborhood dependency, where a cell depends on its four orthogonal neighbors as shown in Fig. 2.1b, and Moore's 9-neighborhood dependency, where a cell depends on its eight neighbors (Fig. 2.1c). Hence, for von Neumann neighborhood $\mathcal{N} = ((-1, 0), (0, -1), (0, 0), (0, 1), (1, 0))$, and for Moore neighborhood $\mathcal{N} = ((-1, -1), (-1, 0), (0, -1), (1, -1), (0, 0), (-1, 1), \quad (0, 1), (1, 0), (1, 1))$. Here, $(0, 0)$ is the relative position of the cell under consideration. Although neighborhood vector $\mathcal{N}$ can be obtained from the given radius, it is convenient to include $\mathcal{N}$ in the definition. For example, in two dimension, radius **r** is not sufficient to specify the neighborhood. For defining neighborhood dependency, we have to mention either von Neumann neighborhood or Moore neighborhood.

When a CA evolves, it hops from one configuration to another in discrete time. Therefore, a CA can be seen as a mapping $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$. If c is the *initial* configuration of the CA, then successive configurations are $G(c)$, $G^2(c)$, $G^3(c)$, ... The G is referred to as *global transition function* of the CA. A CA can be identified by its

(a) 3-neighborhood CA



(b) von Neumann neighborhood



(c) Moore neighborhood

Fig. 2.1 Black cell is under consideration and shaded cells are its neighbors. Here $\mathbf{r} = 1$

global transition function G only. The local transition function (R) of a CA induces the G. The next configuration $G(c)$ of c is computed as

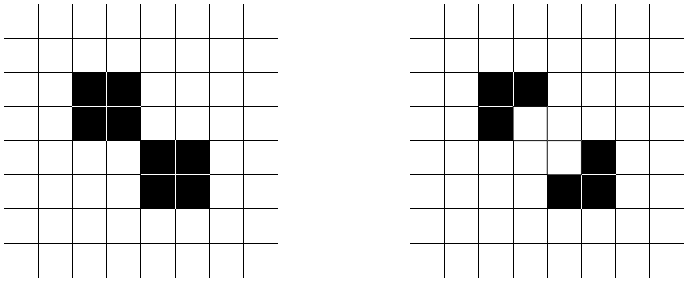
$$G(c)(\overrightarrow{x}) = R(c(\overrightarrow{x} + \overrightarrow{x_1}), c(\overrightarrow{x} + \overrightarrow{x_2}), \ldots, c(\overrightarrow{x} + \overrightarrow{x_m}))$$

for all $\overrightarrow{x} \in \mathscr{L}$.

Example 2.1 Let us consider Conway's *Game of Life* which is a popular CA, developed in 1970. The game is designed on a two-dimensional square grid. The square cells can use two colors, *white* and *black*, representing *dead* cell and *alive* cell respectively. When the game runs, a dead cell may remain dead or become alive in the next time step. Similarly, an alive cell may remain alive or may die. However, whether a cell will be dead or alive in the next step that depends on the neighboring condition of the cell. The game is actually a two-dimensional cellular automaton with $\mathscr{L} = \mathbb{Z}^2$, $S = \{0, 1\}$ where 0 represents dead/white and 1 represents alive/black. Here, $\mathbf{r} = 1$ with Moore neighborhood. The local rule $R : \{0, 1\}^9 \rightarrow \{0, 1\}$ is the following.

- Birth: A dead cell can alive in the next time step if exactly three of its eight neighbors are alive. Otherwise, a dead cell remains dead.
- Death: A live cell can die in the next time step due to
  - Overcrowding: if four or more of its neighbors (excluding itself) are also alive, or
  - Exposure: if only one of its neighbors (excluding itself) is alive.
- Survival: A cell survives in the next time step iff two or three of its neighbors (excluding itself) are alive.

Figure 2.2a shows a configuration of the CA, which consists of only eight live cells. Now, if the rule R is applied to all the cells simultaneously, it settles to the next configuration as shown in Fig. 2.2b . Here, out of eight live cells, six survive in the next time step and two die due to overcrowding. Further, no dead cell becomes alive in the next configuration. If G is the global transition function of the CA, and c is



(a) Present configuration **(b)** Next configuration

Fig. 2.2 Evolution of game of life

the present configuration as shown in Fig. 2.2a , then the configuration of Fig. 2.2b is $G(c)$.

Classically, the cellular space is infinite, therefore, the configurations of CA are also infinite. There are two special types of configurations—*finite configurations* and *periodic configurations*. To define finite configurations, we need to define the *quiescent state* $q \in S$: q is a quiescent state if $R(q, q, \ldots, q) = q$, where R is the local rule of the CA.

Definition 2 A configuration of a CA is called a *finite configuration* if all but a finite number of cells are in q, where q is a quiescent state of the CA, that is, $R(q, q, \ldots, q) = q$, and the cellular space is infinite.

In Game of Life (Example 2.1), 0 (white) is the quiescent state, and the configuration of Fig. 2.2a is an example of finite configuration. The configuration of Fig. 2.2b is also a finite configuration. In fact, if present configuration of a CA is a finite configuration, then the next configuration is also a finite configuration. Because, only a finite number of cells can assume a non-quiescent state in next time step. So, all but some cells remain in quiescent state in the next time state. It implies that the next configuration is a finite configuration. Applying this argument, it can be concluded that if the initial configuration of a CA is a finite configuration, then any new configuration, no matter how many time steps the CA runs, is always a finite configuration.

Definition 3 A configuration x of a D-dimensional infinite CA is *periodic* if the configuration is invariant under D linearly independent translations. That is, there exist D natural numbers $n_1, n_2, \ldots, n_D$ so that x is same as n_i times shift of x in dimension i, for each $i \in \{1, 2, \ldots, D\}$.

In case of one-dimension, if x is a periodic configuration, then there exists a *period length* $p \geq 1$ such that $x_i = x_{p+i}$ for any $i \in \mathbb{Z}$. That is, a finite string of states with length p is infinitely repeated in a periodic configuration. If x' is the next configuration of x, then $x'_i = x'_{p+i}$, for all $i \in \mathbb{Z}$. Hence, the next configuration is also a periodic configuration with a period length p.

Although CAs are defined over any dimension D, the applications of CA are mostly confined in the one and two dimensional CAs. Specifically, in CA technology, the one dimensional CA has the major share. Among the one-dimensional CAs, the *Elementary Cellular Automata* (ECAs) are mostly explored till now.

2.2 Elementary Cellular Automata

The Elementary Cellular Automata (ECAs) are one-dimensional binary CAs (two states per cell) with 3-neighborhood dependency. That is, in case of ECAs

$$\mathscr{L} = \mathbb{Z}, \quad S = \{0, 1\}, \quad \mathcal{N} = (-1, 0, 1)\ [\mathbf{r} = 1], \quad R : S^3 \to S$$

Table 2.1 Example rules of Elementary Cellular Automata

Rule min term (RMT)	111	110	101	100	011	010	001	000	Rule
	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
R	0	1	0	1	1	0	1	0	90
	1	0	0	1	0	1	1	0	150
	0	0	0	1	1	1	1	0	30
	0	0	0	0	0	1	0	1	5
	0	1	0	0	1	0	0	1	73
	1	1	0	0	1	0	0	0	200
	0	1	0	1	0	0	0	0	80

The ECAs are simple in many aspects. In a series of works during 1980s, Stephen Wolfram has shown that these simple CAs can exhibit many complex behaviors.

There are 256 ECAs, as the number of possible local rule is $2^{2^3} = 256$. An ECA is generally denoted by its local rule. Let us represent a rule in tabular form, where the elements of domain (that is, S^3) are written in one row and the next row shows the function values against the elements of domain. Therefore, the table contains eight entries against the eight possible combinations 111, 110, ..., 000. The outputs of the functions, noted as

$$R(111)\ R(110)\ R(101)\ R(100)\ R(011)\ R(010)\ R(001)\ R(000)$$

form an 8-bit binary sequence. Wolfram, in [1], has introduced the ECA naming scheme which is the decimal equivalent of this binary sequence. Table 2.1 shows seven such ECA rules. The last column of the table notes the decimal equivalents of the rules (the meaning of *Rule Min Term* will be discussed in Sect. 2.3.3). If an ECA considers the rule 90 as its next state function, then it is called ECA 90.

When a CA evolves, the new configurations are generated in the subsequent time steps. The behavior of a CA can be studied by plotting the configurations against time. Such a plot is called the *space-time diagram* of the CA. For a D dimensional CA, the space-time diagram is of dimension $D + 1$. In case of ECAs, it is two dimensional. Space-time diagrams are massively explored by Wolfram to study the properties of ECAs.

Example 2.2 Let us consider that the ECA 90 starts evolving from the initial configuration $\cdots 000010000 \cdots$. That is, at time $t = 0$, the configuration is $\cdots 000010000 \cdots$. As $90(010) = 0$ ('90' represents the local function of the ECA), the cell with state 1 becomes 0 in the next time step. However, the left and right neighbors of the cell assume state 1 as $90(001) = 90(100) = 1$. The rest of the cells remain at state 0 ($90(000) = 0$). Hence, at $t = 1$, configuration of the CA is $\cdots 000101000 \cdots$. Sim-

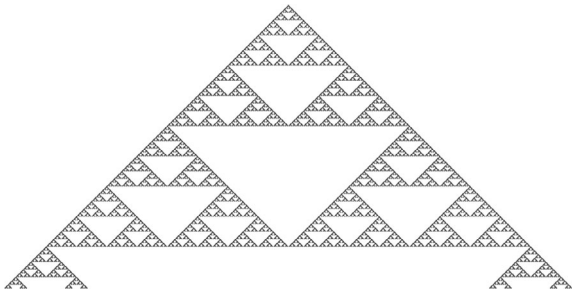


Fig. 2.3 Space-time diagram of ECA 90

ilarly, the configuration at $t = 2$ is $\cdots 001000100 \cdots$, and so on. Now let us plot the configurations as

$$\cdots 0001000 \cdots$$

$$\cdots 000101000 \cdots$$

$$\cdots 001000100 \cdots$$

$$\vdots$$

where t is in downward direction. This is an example of space-time diagram of ECA 90. Figure 2.3 shows the same diagram replacing 0 and 1 by white box and black box respectively for a number of time steps. This points to the fact that the automaton can generate *complex* patterns from a simple initial configuration.

In spite of the simplicity, many interesting patterns can be generated by the ECAs. Depending upon the capability of forming patterns in space-time diagrams, the ECAs are classified as *simple*, *complex* and *chaotic*. Many of the simple ECAs are also capable of showing interesting behavior. For example, the ECA 184 is simple, but it can simulate basic traffic flow on road. On the other hand, the ECA 110 is complex, and capable of universal computation. Whereas, ECA 30, ECA 90 etc. are the chaotic CAs. In general, the ECAs are very effective tools to model various physical systems.

Before concluding this section, let observe the behavior of the ECA 30. It is one of the most explored ECAs that has been exploited to generate pseudo-random sequences. Till date, the ECA 30 remains as one of the best sources of pseudo-random numbers . The following example shows one of the ways of extracting bit sequence from the ECA 30.

Example 2.3 Let us fix ECA 30 with initial configuration $\cdots 00100 \cdots$. When the ECA evolves, new configurations are generated, and from each configuration only one bit is extracted. Consider that initially cell 0 is at state 1 while rest of the cells are at state 0. Further, consider that the state of cell 0 is the bit contributed to the proposed bit-sequence. Now if the ECA evolves for n time steps, then the bit sequence at cell 0 (of length n) is:

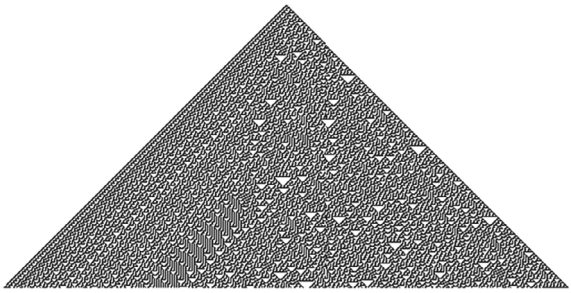


Fig. 2.4 Evolution of ECA 30

1100111111000011110000001100111100001100001111110011001111110000111111001100110011111000011100111100111111111111000000001111111100000000111001100111111000000000011000011001111000000111111100000011110011110011110011001111111000000000011000000111111111100111111001100001111 · · ·

It is interesting to observe that the above sequence is aperiodic. If the CA is evolved for one billion steps, then also the sequence remains aperiodic. However, whether the sequence is aperiodic for ever is yet to be proved. Figure 2.4 shows the space time diagram of ECA 30.

2.3 Non-uniform Cellular Automata

The CAs, defined in the earlier section, are characterized by the three fundamental properties:

- *Uniformity*: All the cells of a CA are updated by the same local rule R.
- *Synchronicity*: The cells of a CA are updated simultaneously.
- *Locality*: The rule R of the CA acts locally, depending only on the neighbors of a cell.

When the uniformity property is relaxed, a new class of CAs, referred to as the *Non-uniform Cellular Automata* is obtained. These CAs are also called *hybrid* CAs. A non-uniform CA allows its cells to use different rules to update their states. Since the neighborhood dependency and rules are correlated, in a non-uniform CA, the neighborhood dependency may also vary. The non-uniform CAs can be of any dimension. However, only the one-dimensional CAs are considered in this book.

Definition 4 A (one-dimensional) *Non-uniform Cellular Automaton* is a quadruple $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$, where $\mathscr{L} \subseteq \mathbb{Z}$, $i \in \mathscr{L}$, $\mathbf{r}_i$ and $\mathcal{R}_i : S^{2\mathbf{r}_i+1} \rightarrow S$ are the radius and the local transition function (rule) of cell i respectively.

The configuration of a non-uniform CA is also an assignment of states to the cells, and $S^{\mathscr{L}}$ is the set of all configurations of the non-uniform CA. The set of local rules $\{\mathcal{R}_i\}_{i\in\mathscr{L}}$ induces the global transition function $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ of the non-uniform CA. The G is defined as:

$$G(c)_i = \mathcal{R}_i(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}), \forall c \in S^{\mathscr{L}}, \forall i \in \mathscr{L}$$

Therefore, a non-uniform CA adopts a collection of rules $\{\mathcal{R}_i\}_{i\in\mathscr{L}}$. It signifies that the cell $i \in \mathscr{L}$ of a CA adopts the rule $\mathcal{R}_i$. It points to that fact that the *uniform* CAs, are also the special class of non-uniform CAs, where $\{\mathcal{R}_i\}_{i\in\mathscr{L}}$ is a singleton set.

2.3.1 Subclasses of Non-uniform Cellular Automata

When $\mathscr{L} = \mathbb{Z}$, Definition 4 becomes too general to deal with. So, three subclasses of the non-uniform CAs have been specified. These are d-ν CA, p-ν CA and $\mathbf{r}$-ν CA. In a d-ν CA, all but finitely many cells adopt a single rule.

Definition 5 d-ν CA: A non-uniform CA $(\mathbb{Z}, S, \mathbf{r}_i, \mathcal{R}_i)$ is a d-ν CA if there exist $k, \mathbf{r} \in \mathbb{N}$ and a rule $h : S^{2r+1} \rightarrow S$ such that $\mathcal{R}_i = h$ and $\mathbf{r}_i = \mathbf{r}$ for any $i \in \mathbb{Z}$ with $|i| > k$. Here, h is the default rule for the CA.

The p-ν CAs, defined below, are the generalization of d-ν CAs.

Definition 6 p-ν CA: A non-uniform CA $(\mathbb{Z}, S, \mathbf{r}_i, \mathcal{R}_i)$ is a p-ν CA if there exist $k \geq 0$, $p > 0$ such that, for any $i > k$, $\mathcal{R}_i = \mathcal{R}_{i+p}$ and $\mathcal{R}_{-i} = \mathcal{R}_{-i-p}$.

Therefore, a p-ν CA adopts two sets of local rules—$\{\mathcal{R}_i\}_{i>k}$ and $\{\mathcal{R}_i\}_{i<-k}$ that are periodically applied to the right and left cells respectively but are not in between k and $-k$. If $p = 1$, then $\mathcal{R}_{k+1}$ is the default rule for the right side and $\mathcal{R}_{-k-1}$ for the left side. When $\mathcal{R}_{k+1} = \mathcal{R}_{-k-1}$, then the CA is d-ν CA.

Definition 7 $\mathbf{r}$-ν CA: A non-uniform CA $(\mathbb{Z}, S, \mathbf{r}_i, \mathcal{R}_i)$ is an $\mathbf{r}$-ν CA if there exists a radius $\mathbf{r} \in \mathbb{N}$ such that, $\mathbf{r}_i = \mathbf{r}$ for any $i \in \mathbb{Z}$. In this case, the radius of the CA is $\mathbf{r}$, and the CA can be defined as $(\mathbb{Z}, S, \mathbf{r}, \mathcal{R}_i)$.

As classically the cellular space is infinite, the cellular space of the CAs of these classes are also infinite. However, most of the research works on non-uniform CAs assume finite cellular space. The non-uniform CAs have been historically developed to solve the technological problems. Possibly, this puts constraints on the researchers to consider finite cellular space for easy realization in VLSI design.

When the radius (**r**) of all rules of a non-uniform CA is same and the cellular space $\mathscr{L}$ is finite, then the automaton ($\mathscr{L}$, S, **r**, $\mathcal{R}_i$) is defined by a sequence of rules, called *rule vector* of the CA.

Definition 8 A *rule vector* of an n-cell non-uniform CA ($\mathscr{L}$, S, **r**, $\mathcal{R}_i$) is $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$, where $\mathscr{L} = \{0, 1, \ldots, n-1\}$ and $\mathcal{R}_i$ is a rule used by cell $i \in \mathscr{L}$.

When $\mathscr{L}$, S and **r** are understood from the context, then a CA is simply defined by its rule vector. Such a definition is widely used for *Non-uniform ECAs*.

Definition 9 Non-uniform ECA: A non-uniform CA ($\mathscr{L}$, S, **r**, $\mathcal{R}_i$) is the Non-uniform ECA if $S = \{0, 1\}$, $\mathbf{r} = 1$ and $\mathcal{R}_i$ is an ECA rule for each $i \in \mathscr{L}$. Usually $\mathscr{L}$ is considered as finite for non-uniform ECAs.

Let us consider a non-uniform ECA with $\mathscr{L} = \{0, 1, 2, 3\}$ and rule vector (165, 73, 200, 90). It denotes that the cell 0 adopts rule 165, cell 1 adopts 73, cell 2 adopts rule 200 and for the last cell, it is rule 90.

Although several types of non-uniform CAs have been proposed in the literature, the major share of non-uniform CA research goes to non-uniform ECAs, where $\mathscr{L}$ is finite. This book also primarily deals with the non-uniform ECAs.

2.3.2 Linear Cellular Automata

Cellular automata, uniform or non-uniform, can also be classified as *linear* and *non-linear*. A CA is linear iff its global transition function G is a linear map. Otherwise, the CA is non-linear. Although the linear CAs can be of any dimension, only one-dimensional linear CAs are considered here.

Let W and V be the two vector spaces over a field $\mathbb{K}$. A function $f : W \to V$ is a linear map if for any two vectors $x, y \in W$ and any scalar $\alpha \in \mathbb{K}$, the following two conditions are satisfied:

- Additivity: $f(x + y) = f(x) + f(y)$
- Homogeneity with degree 1: $f(\alpha.x) = \alpha f(x)$.

Let S be the vector space over field $\mathbb{K}$. Then, for a CA ($\mathscr{L}$, S, $\mathbf{r}_i$, $\mathcal{R}_i$), a rule $\mathcal{R}_i : S^{2\mathbf{r}_i+1} \to S$ is linear if it satisfies the above two conditions. When the CA ($\mathscr{L}$, S, $\mathbf{r}_i$, $\mathcal{R}_i$) is linear, then the set of configurations $S^{\mathscr{L}}$ forms a vector space over $\mathbb{K}$ and the global transition function $G : S^{\mathscr{L}} \to S^{\mathscr{L}}$ satisfies the above two conditions. The following is an important result.

Theorem 2.1 *A CA ($\mathscr{L}$, S, $\mathbf{r}_i$, $\mathcal{R}_i$) is linear iff, for each $i \in \mathscr{L}$, $\mathcal{R}_i$ is linear.*

Proof Let assume that ($\mathscr{L}$, S, $\mathbf{r}_i$, R_i) is linear—that is, its global transition function $G : S^{\mathscr{L}} \to S^{\mathscr{L}}$ is a linear map. It is to be proved that for each $i \in \mathscr{L}$, $\mathcal{R}_i : S^{2\mathbf{r}_i+1} \to S$ is linear.

For an arbitrary $i \in \mathscr{L}$, let $(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}), (s'_{i-\mathbf{r}_i}, \ldots, s'_i, \ldots, s'_{i+\mathbf{r}_i}) \in S^{2\mathbf{r}_i+1}$. Then,

$$\begin{aligned}
&\mathcal{R}_i((s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}) + (s'_{i-\mathbf{r}_i}, \ldots, s'_i, \ldots, s'_{i+\mathbf{r}_i})) \\
&= G(c_1 + c_2)_i \qquad [\text{where } c_1, c_2 \in S^{\mathscr{L}}] \qquad [\text{definition of } G] \\
&= G(c_1)_i + G(c_2)_i \qquad [\text{following from the assumption}] \\
&= \mathcal{R}_i(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}) + \mathcal{R}_i(s'_{i-\mathbf{r}_i}, \ldots, s'_i, \ldots, s'_{i+\mathbf{r}_i})
\end{aligned}$$

That is, $\mathcal{R}_i$ is additive.

Now, for a scalar $\alpha_1 \in S$, $\alpha_1.\mathcal{R}_i(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i})$

$$\begin{aligned}
&= \alpha_1.G(c)_i \\
&= G(\alpha_1.c)_i \qquad [\because G \text{ is linear}] \\
&= \mathcal{R}_i(\alpha_1.(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}))
\end{aligned}$$

$\therefore \mathcal{R}_i$ is homogeneous. Hence, $\mathcal{R}_i$ is linear. Since $i \in \mathscr{L}$ is arbitrary, each $\mathcal{R}_i$ of the CA is linear.

Conversely, let assume that for each $i \in \mathscr{L}$, $\mathcal{R}_i$ is linear and $x, y \in S^{\mathscr{L}}$. Now for an arbitrary $i \in \mathscr{L}$, $G(x + y)_i$

$$\begin{aligned}
&= \mathcal{R}_i((s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}) + (s'_{i-\mathbf{r}_i}, \ldots, s'_i, \ldots, s'_{i+\mathbf{r}_i})) \\
&= \mathcal{R}_i(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}) + \mathcal{R}_i(s'_{i-\mathbf{r}_i}, \ldots, s'_i, \ldots, s'_{i+\mathbf{r}_i}) \\
&\qquad\qquad [\because R_i \text{ is linear}] \\
&= G(x)_i + G(y)_i
\end{aligned}$$

That is, G is additive.

Let assume $\alpha_2 \in S$. Hence, $\alpha_2.G(x)_i$

$$\begin{aligned}
&= \alpha_2.\mathcal{R}_i(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i}) \\
&= \mathcal{R}_i(\alpha_2.(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i})) \qquad [\because \mathcal{R}_i \text{ is linear}] \\
&= G(\alpha_2 \dot{.} x)_i
\end{aligned}$$

That is, G is homogeneous. Hence, G is linear. Consequently the theorem is proved. □

The following corollary is directly derived from Theorem 2.1.

Corollary 2.1 *A CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ is additive iff for each $i \in \mathscr{L}$, the $\mathcal{R}_i$ is additive.*

If a local rule $\mathcal{R}_i : S^{2\mathbf{r}_i+1} \to S$ is linear, then it takes the following form:

$$\mathcal{R}_i(x_0, x_1, \ldots, x_{2\mathbf{r}_i}) = a_0 x_0 + a_1 x_1 + \cdots + a_{2\mathbf{r}_i} x_{2\mathbf{r}_i}$$

where $a_0, a_1, \ldots, a_{2\mathbf{r}_i} \in S$. For ECAs, the linear rules are of the following form:

$$\mathcal{R}_i(x, y, z) = ax + by + cz$$

where $a, b, c \in \{0, 1\}$. Depending on the values of a, b and c, there are $2^3 = 8$ linear rules. These rules are 0, 60, 90, 102, 150, 170, 204 and 240 (Table 2.2). Here, addition is the modulo 2 addition which is equivalent to logical exclusive OR (XOR), denoted as $\oplus$.

Table 2.2 List of linear and complemented ECA rules ($\mathcal{R}_i(x, y, z)$)

Rule	Linear	Rule	Complemented
60:	$x \oplus y$	195:	$\overline{x \oplus y}$
90:	$x \oplus z$	165:	$\overline{x \oplus z}$
102:	$y \oplus z$	153:	$\overline{y \oplus z}$
150:	$x \oplus y \oplus z$	105:	$\overline{x \oplus y \oplus z}$
170:	z	85:	$\overline{z}$
204:	y	51:	$\overline{y}$
240:	x	15:	$\overline{x}$
0:	0	255:	1

When $S = \{0, 1\}$, then additivity implies linearity. So, for binary cases, linear CAs and additive CAs are synonymous. These CAs are referred to as the *linear/additive CAs*. The rest are *non-linear CAs*.

Definition 10 A CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ is *non-linear* if there exists an $i \in \mathscr{L}$ such that $\mathcal{R}_i$ is non-linear.

In case $S = \{0, 1\}$, a class of CAs are identified which are named as *Complemented CAs*. A CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ is complemented if its rule vector includes linear/additive and *complemented rules*. A rule is *complemented* if it is logical complement of a linear/additive rule. Table 2.2 shows the complemented ECA rules, along with their linear counterpart. The complemented CAs are not linear.

2.3.3 Rule Min Terms

Let re-look at Table 2.1, which represents some ECA rules. The first row of the table shows all possible combinations of the present states of the three neighbors of a cell whereas, the next states against the combinations form the rules. Borrowing vocabulary from *Switching Theory*, each combination of the present states can be called as the Rule Min Term (RMT)—that is, *minterm* of a 3-variable switching function.

Definition 11 Let $R : S^m \rightarrow S$ be a local rule of a CA. A tuple $(s_1, s_2, \ldots, s_m) \in S^m$ is called the *Rule Min Term* (RMT) of the rule R, and is usually represented by its decimal equivalent. If $S = \{0, 1, \ldots, d-1\}$, then the RMT is $r = s_1.d^{m-1} + s_2.d^{m-2} + \cdots + s_{m-1}.d + s_m$. $R(s_1, s_2, \ldots, s_m)$ is also represented as $R[r]$.

The column 011 of Table 2.1 is the RMT 3. The next states corresponding to this RMT are 1 for rule 90 and 0 for rule 150. Hence, 90[3] = 1 and 150[3] = 0. These can also be referred to as the "RMT 3 of rule 90 is 1" and "RMT 3 of rule 150 is 0".

The RMTs are atomic elements of a rule. An RMT along with its next state is also referred to as the *transitions*. In case of an ECA rule, a transition is a quadruplet $(x, y, z, R(x, y, z))$.[1] The transition (and the RMT) is *active* if $R(xyz) \neq y$. Otherwise, it is *passive*. A passive RMT is also known as *self-replicating* RMT. For example, RMT 3 of rule 90 is passive (self-replicating) whereas, RMT 3 of rule 150 is active.

A configuration can alternatively be seen as collection of RMTs. If c is the configuration of a D-dimensional CA and $\mathcal{N} = (\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_m})$ is its neighborhood vector, then the cell $\overrightarrow{x} \in \mathscr{L}$ is acting on the RMT $r_{\overrightarrow{x}} = (c(\overrightarrow{x} + \overrightarrow{x_1}), c(\overrightarrow{x} + \overrightarrow{x_2}), \ldots, c(\overrightarrow{x} + \overrightarrow{x_m}))$. So, the configuration can be presented by the RMTs, on which the cells are acting. That is, the configuration can be seen as $(r_{\overrightarrow{x}})_{\overrightarrow{x} \in \mathscr{L}}$. In one-dimension, if a configuration is presented in terms of RMTs, then an *RMT sequence* can be found.

Definition 12 Let $c = (s_i)_{i \in \mathcal{L}}$ be a configuration of a one-dimensional CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$. The *RMT sequence* of c, denoted as $\tilde{c}$, is $(r_i)_{i \in \mathcal{L}}$, where r_i is the RMT $(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i})$.

To get an RMT sequence of a configuration of a (non-uniform) ECA, an imaginary 3-bit window can be considered that slides over the configuration. The content of the window is an RMT. For example, RMT sequence of the configuration $\cdots 0001000 \cdots$ is $(\cdots 0, 1, 2, 4, 0, \ldots)$, which is also simply represented as $\cdots 01240 \cdots$.

The non-uniform CAs possess a few properties that the classical CAs, with the same $\mathscr{L}$, S and $\mathbf{r}$, do not have. In the subsequent discussions of the book, these properties are explored. Whenever the non-uniformity is allowed in cellular structure, a large number of possible automata can be found. For example, if $\mathscr{L}$ is finite and only the ECA rules are considered, then about 256^n non-uniform CAs can be found (instead of only 256 for classical case), where $|\mathscr{L}| = n$.

2.4 Finite Cellular Automata

A CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ is finite if its cellular space $\mathscr{L}$ is finite. The finite CA has the boundary cells which are to be treated specially. For a (one-dimensional) finite CA with $n \geq 1$ cells, $\mathscr{L} = \{0, 1, \ldots, n-1\}$. Generally, two boundary conditions are assumed—*open* boundary and *periodic* boundary.

In open boundary condition, some neighbors of boundary cells, as well as close to a boundary cell, are missing. A missing neighbor is assumed to be always in a fixed state. Among the open boundary conditions, the most popular one is the *null* boundary, where a missing neighbor is in state 0. In case of one-dimensional CA with radius 1, the missing neighbors of cell 0 and cell $n-1$ are cell -1 and cell n respectively, which are assumed to be fixed in state 0. Figure 2.5a shows the block diagram of a 1-dimensional null boundary CA.

[1] Throughout the text $R(x, y, z)$ and $R(xyz)$ are synonymous.

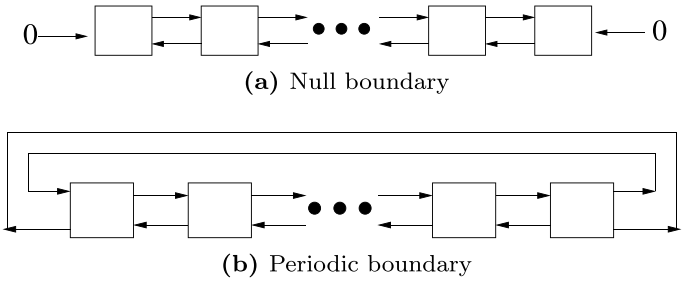


Fig. 2.5 Boundary conditions of one-dimensional finite CA in 3-neighborhood

In periodic boundary CA, the boundary cells are neighbors of some other boundary cells. For one-dimensional CA, the cells form a circular array of size n. That is, the rightmost and leftmost cells are neighbors of each other (Fig. 2.5b). Mathematically, the elements of $\mathscr{L}$ form a *ring of integers modulo* n, and is denoted as $\mathbb{Z}/n\mathbb{Z}$. Hence, under periodic boundary, $\mathscr{L} = \mathbb{Z}/n\mathbb{Z}$.

In case of one-dimension, periodic configuration (Definition 3) of a (uniform) CA $(\mathbb{Z}, S, \mathbf{r}, R)$ over infinite lattice is equivalent to periodic boundary condition of finite (uniform) CA $(\mathbb{Z}/n\mathbb{Z}, S, \mathbf{r}, R)$. The behavior of the first CA can be understood by observing the second one. This is one instance, where infinite case handshakes with the finite case. However, the RMTs on which the boundary cells act at a given time depend on the boundary condition. Therefore, the RMT sequence of a configuration differs from one boundary condition to other. To illustrate this difference, let us consider a configuration 1110 of a 4-cell binary CA in 3-neighborhood. Under null boundary condition, the RMT sequence of the configuration is 3764 whereas, under periodic boundary condition, it is 3765. Figure 2.6 explores the RMT sequences under both the conditions.

Example 2.4 Let us consider a 4-cell non-uniform ECA $(165, 73, 200, 90)$ with periodic boundary. The rules are shown in Table 2.3. Now, consider that the initial configuration of the CA is 1110. To get the next configuration, the RMT sequence of the configuration can be found out. Since the left neighbor of the first cell and the right neighbor of last cell are the last and the first cell respectively,

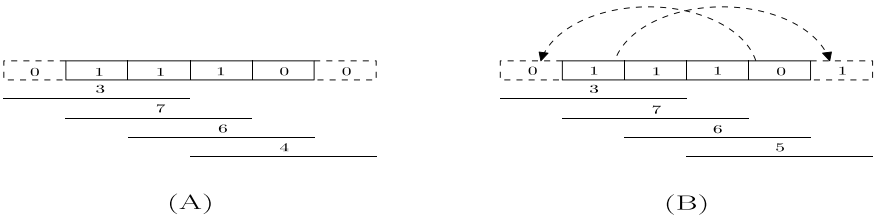


Fig. 2.6 RMT sequence of 1110 under **a** Null boundary condition and **b** Periodic boundary condition

Table 2.3 RMTs of the rules of CA (165, 73, 200, 90)

Present state:	111	110	101	100	011	010	001	000	Rule
(RMT)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
(i) Next state:	1	0	1	0	0	1	0	1	165
(ii) Next state:	0	1	0	0	1	0	0	1	73
(iii) Next state:	1	1	0	0	1	0	0	0	200
(iv) Next state:	0	1	0	1	1	0	1	0	90

the RMT sequence is 3765. Here, $165[3] = 0, 73[7] = 0, 200[6] = 1, 90[5] = 0$. That is, the next configuration is 0010. The RMT sequence of 0010 is 0124, and $165[0] = 1, 73[1] = 0, 200[2] = 0, 90[4] = 1$. Hence, the next configuration of 0010 is 1001. Similarly, the next of 1001 is 0001, the next of 0001 is 0100, and so on.

In case of open boundary, some of the RMTs of rules, applied on the cells close to the boundary cannot appear during evolution of the CA. For example, in case of non-uniform ECA in null boundary, the RMTs 4, 5, 6 and 7 cannot act on first cell, because left neighbor of the first cell is assumed to be always in state 0. These RMTs are *don't care* RMTs for the first rule. Similarly, the RMTs 1, 3, 5 and 7 of the last rule are the *don't care* RMTs.

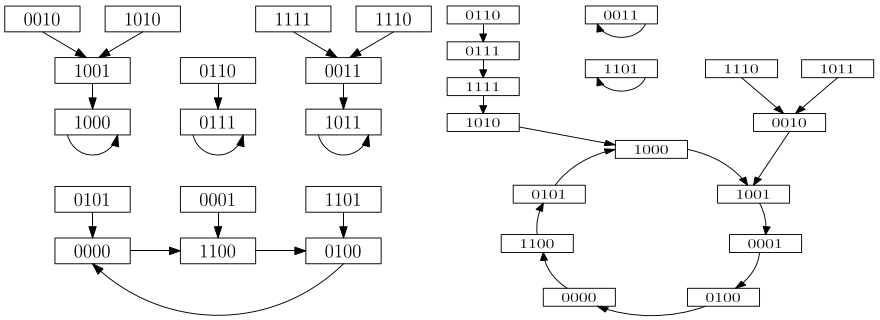
Example 2.5 Let us consider the CA (165, 73, 200, 90) with null boundary. If 1110 is the initial configuration of the CA, then the RMT sequence of the configuration is 3764. Since, $165[3] = 0, 73[7] = 0, 200[6] = 1, 90[4] = 1$, the next configuration is 0011. The RMT sequence of 0011 is 0136, and $165[0] = 1, 73[1] = 0, 200[3] = 1, 90[6] = 1$. So, the next configuration of 0011 is 1011.

In Example 2.5, if the first rule 165 is replaced by rule 5, then the behavior of the CA remains same as the RMTs 4, 5, 6 and 7 of the two rules 165 and 5 have no role on the configurations. Similarly, rule 90 of the last cell can be replaced by rule 80 without affecting the behavior of the CA. Therefore, for null boundary condition (or, for any fixed boundary condition), a set of rules are equivalent when used for the first cell. Similar argument is applicable to the rule of the last cell.

The *transition diagram* is frequently used to explore the behavior of a finite CA. It is an effective tool to study the behavior of a CA when the number of cells of the CA is less.

Definition 13 Let $\mathcal{C}$ be a set of all configurations of a finite CA G. Then, transition diagram of the CA is a graph (V, E), where $V = \mathcal{C}$ and $E = \{(c, G(c)) | c \in \mathcal{C}\}$.

The transition diagram is a directed graph, and the direction shows predecessor-successor relationship of configurations. Let consider the CA (165, 73, 200, 90) of Example 2.4. Figure 2.7 shows the transition diagrams of the CA under the null boundary and periodic boundary.



(a) Under null boundary (b) Under periodic boundary

Fig. 2.7 Transition diagrams of CA (165, 73, 200, 90)

Cellular automata are as powerful as Turing machine and, therefore, capable of universal computation. But, these CAs are infinite CAs. On the other hand, the finite CAs have similarity with Finite State Machine (FSM). They have finite number of configurations. So, the finite CAs can be considered as the FSMs. As a consequence, a configuration of CA is seen as a state. To distinguish them from cells' states, first one is called *global states* and the latter is called *local states*. However, in this book, the term *configuration* is used for both the finite and infinite CAs.

2.5 Hardware Realization

This section describes hardware implementation of one-dimensional finite CA. It is assumed that the CAs are binary and the cells use ECA rules. That is, each cell of a CA depends on its left and right neighbors, and on itself for the next configuration. A cell of such a CA should have a 1-bit memory to store its state, and a logic circuit to realize the rule. Hence, a cell of the CA can be the collection of a flip-flop (*FF*) for 1-bit memory, and a combinational logic which realizes the rule. The states of the cell and its two neighbors are the input to the combinational logic. The output of the combination logic is fed to the FF of the cell (Fig. 2.8). A global clock is connected to all the cells of a CA. In an n-cell CA, the outputs of all the FFs constitute a configuration of the CA. When a clock is received by the cells, the next configuration is generated. Figure 2.9 shows the schematic diagram of an n-cell CA under null boundary.

The combinational logic for an ECA rule can be found with the help of rule table (Table 2.3) and K-map (*Karnaugh map*). Let us consider that S_i^t is the state of cell i at time t and S_i^{t+1} is the next state (output of the combinational logic) of the cell while applying the ECA rule. Then, from the K-map S_i^{t+1} can be expressed in terms of S_{i-1}^t (state of left neighbor of cell i), S_i^t and S_{i+1}^t (state of right neighbor of cell i). For example, for

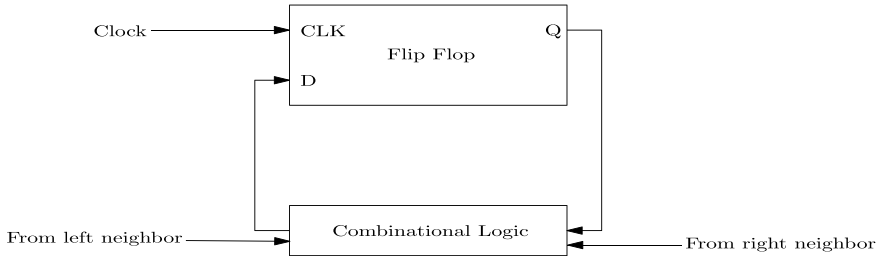


Fig. 2.8 A typical cell of a CA

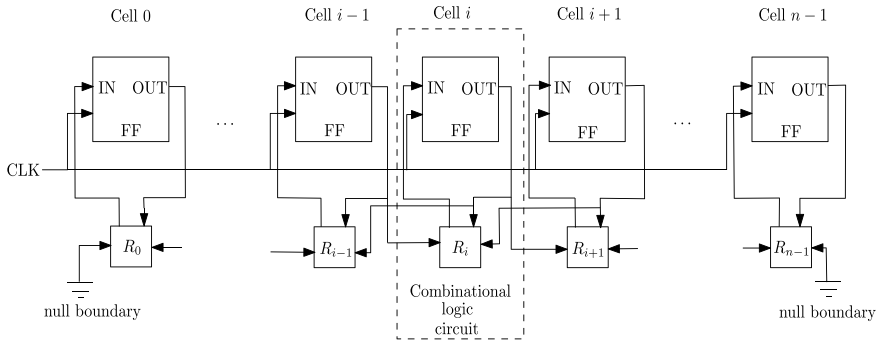


Fig. 2.9 An n-cell null boundary CA

Rule 90: $S_i^{t+1} = S_{i-1}^t \oplus S_{i+1}^t$
Rule 150: $S_i^{t+1} = S_{i-1}^t \oplus S_i^t \oplus S_{i+1}^t$
Rule 75: $S_i^{t+1} = S_i^t.(S_{i-1}^t \oplus S_{i+1}^t) + (\overline{S_{i-1}^t}).(\overline{S_i^t})$.

Example 2.6 Let us realize the CA (165, 73, 200, 90) in null boundary. The (switching) expressions for the four rules are found out. Consider that S_i^t is the present state of cell $i \in \{0, 1, 2, 3\}$, and S_i^{t+1} is its next state. The expression for S_0^{t+1} (following rule 165), S_1^{t+1} (following rule 73), S_2^{t+1} (following rule 200) and S_3^{t+1} (following rule 90) are—

$$\begin{aligned} S_0^{t+1} &= \overline{S_{-1}^t}.\overline{S_0^t}.\overline{S_1^t} + \overline{S_{-1}^t}.S_0^t.\overline{S_1^t} + S_{-1}^t.\overline{S_0^t}.S_1^t + S_{-1}^t.S_0^t.S_1^t \\ &= \overline{S_{-1}^t}.\overline{S_1^t} + S_{-1}^t.S_1^t \\ &= S_{-1}^t \odot S_1^t \end{aligned}$$

$$\begin{aligned} S_1^{t+1} &= \overline{S_0^t}.\overline{S_1^t}.\overline{S_2^t} + \overline{S_0^t}.S_1^t.S_2^t + S_0^t.S_1^t.\overline{S_2^t} \\ &= \overline{S_0^t}.(\overline{S_1^t}.\overline{S_2^t} + S_1^t.S_2^t) + S_0^t.S_1^t.\overline{S_2^t} \\ &= \overline{S_0^t}.(S_1^t \odot S_2^t) + S_0^t.S_1^t.\overline{S_2^t} \end{aligned}$$

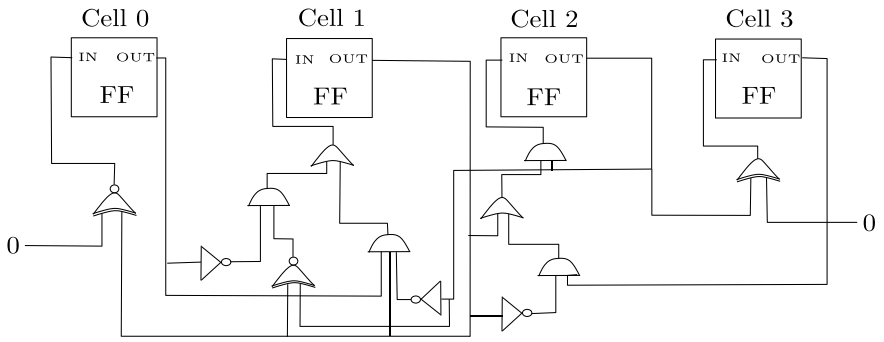


Fig. 2.10 Realization of the (165, 73, 200, 90) in null boundary

$$\begin{aligned} S_2^{t+1} &= \overline{S_1^t}.S_2^t.S_3^t + S_1^t.S_2^t.\overline{S_3^t} + S_1^t.S_2^t.S_3^t \\ &= \overline{S_1^t}.S_2^t.S_3^t + S_1^t.S_2^t \\ &= S_2^t.(\overline{S_1^t}.S_3^t + S_1^t) \end{aligned}$$

$$\begin{aligned} S_3^{t+1} &= \overline{S_2^t}.\overline{S_3^t}.S_4^t + \overline{S_2^t}.S_3^t.S_4^t + S_2^t.\overline{S_3^t}.\overline{S_4^t} + S_2^t.S_3^t.\overline{S_4^t} \\ &= S_2^t.\overline{S_4^t} + \overline{S_2^t}.S_4^t \\ &= S_2^t \oplus S_4^t \end{aligned}$$

From the above expressions, the combinational logic diagrams for all the cells can be drawn. If the CA is under periodic boundary, then $S_{-1}^t = S_3^t$ and $S_4^t = S_0^t$. But if it is in null boundary, then $S_{-1}^t = S_4^t = 0$. Figure 2.10 shows the realization of the CA under null boundary.

The above design uses a dedicated hardware for implementing a single CA. A specialized hardware can be developed that can implement *any* CA of the same number of cells. Such a structure is called *Programmable CA* (Fig. 2.11). Each cell of a programmable CA is having an 8×1 MUX and an 8-bit shift register. The binary value of the rule is stored in the shift register. It is the input to the *MUX*. The states of the left neighbor of the cell, the cell itself and the right neighbor are the select lines of the *MUX*. These three select the RMT of the rule stored in the shift register, on which the cell changes its state in the next time steps. The value stored against this RMT, in the shift register, is then fed to IN of the cell. The boundary condition and clock signal are omitted from the figure.

An n-cell programmable CA can be programmed with any rule vector of length n. Whenever a CA is to be run, each shift register is to be loaded with the binary value of corresponding rule of the rule vector. Although shift registers demand hardware overhead, this structure is more flexible than the realization shown in Fig. 2.10. It is to be observed that the cells of programmable CA are of uniform structure.

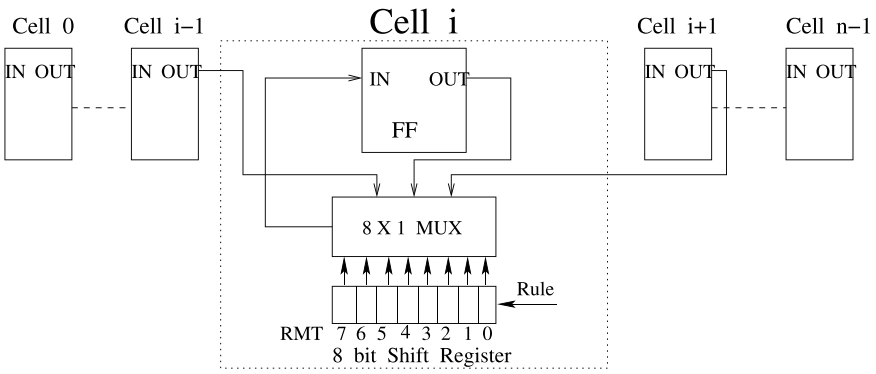


Fig. 2.11 Structure of programmable CA

2.6 Chapter Notes

Historical notes on the development of cellular automata are briefly given in Chap. 1. It is noted that researches on non-uniform ECAs have been started in the second half of 1980s [2–4]. It has taken a pace from 1990s (see Refs. [5, 6] and the relevant references therein). These CAs are finite. Non-uniformity on infinite lattice has later been dealt in [7, 8] and in other works reported by the same group. The term RMT has been first coined in [9], which has subsequently been used in the later works on non-uniform CAs. Details of hardware realization and applications of non-uniform CAs have been reported in [6].

However, future research may be carried out in the following directions.

1. Identify other subclasses of non-uniform cellular automata, which are not defined in this chapter.
2. Develop an alternative way of implementing a CA in hardware.

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Chapter 3
Characterization Tools

In '*A New Kind of Science*', Stephen Wolfram has argued that, to know how a particular cellular automaton (CA) is behaving, one has to observe what is happening just by running the CA. Predicting behavior of a system by means of (mathematical) analysis can only be possible for the special system with simple behavior (page 6 of [1]). In spite of this observation of some researchers, a few characterization tools have been proposed over the years to analyze and predict the behavior of some special classes of cellular automata (CAs). Needless to say, all kinds of dynamic behaviors of a CA may not be analyzed by a tool; but the tools are used to discover some specific properties. In this chapter, three characterization tools—*de Bruijn graph*, *Matrix algebra* and *Reachability tree* are presented for one-dimensional CAs. The de Bruijn graph has been first used to study one-dimensional uniform CA. Later it has been extended to study the non-uniform CA. Whereas, the other two have been primarily developed to study the non-uniform CA. As uniform CAs are the special case of non-uniform CAs, unless otherwise specified, the "CA" refers to the non-uniform CA throughout this chapter.

3.1 The de Bruijn Graph

A de Bruijn graph is a directed graph that represents the overlaps between sequences of symbols. A node of the graph is a sequence of symbols and an edge shows the overlap between two sequences. If Seq_1 and Seq_2 are the sequences representing two adjacent nodes, then the suffix of Seq_1 and prefix of Seq_2 are the same.

Definition 14 Let Σ be a set of symbols, and $s \geq 1$ be a number. Then the *de Bruijn graph* is $B(s, \Sigma) = (V, E)$, where $V = \Sigma^s$ is the set of vertices, and $E = \{(ax, xb) | a, b \in \Sigma, x \in \Sigma^{s-1}\}$ is the set of edges.

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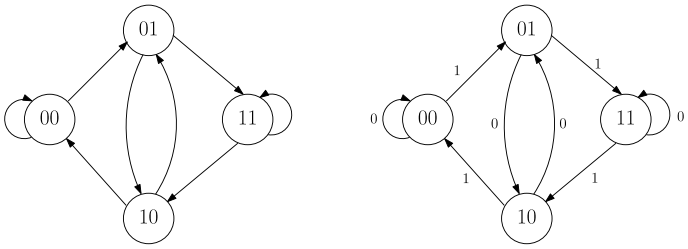
Complexity and Computation 49, https://doi.org/10.1007/978-3-032-00257-0_3



(a) A de Bruijn graph $B(2, \{0, 1\})$ **(b)** de Bruijn graph for rule 90

Fig. 3.1 The de Bruijn graph of ECA with rule 90

Figure 3.1a shows a de Bruijn graph $B(2, \{0, 1\})$. In the graph, one symbol of two adjacent nodes overlaps. For example, nodes 01 and 10 contain 1 as the overlapping symbol.

A de Bruijn graph has k^s vertices, consisting all possible length-s sequences of Σ, where k is the number of symbols in Σ. The graph is balanced in the sense that each vertex has both the in-degree and out-degree as s. A one-dimensional uniform CA can be represented by a de Bruijn graph. Let consider, $s = m - 1$ and $\Sigma = S$, where m is the size of neighborhood and S is the state-set of the one-dimensional CA. The edges (ax, xb) of $B(m - 1, S)$ show the overlap of nodes and $axb \in S^m$. Now, if each edge (ax, xb) of $B(m - 1, S)$ is labelled as $R(axb) \in S$, where $R : S^m \to S$ is the local rule of a CA, then this labelled graph represents the m-neighborhood one-dimensional CA with rule R and state set S.

Figure 3.1b is the de Bruijn graph for ECA with rule 90. This graph shows that if the left, self and right neighbors of a cell are all 0s, then the next state of the cell (that is, $R(000)$) is 0. If the neighbors are 0, 0 and 1 respectively, then the next state is 1, and so on.

An edge (ax, xb) of a de Bruijn graph represents an element (axb) of the domain of $R : S^m \to S$ which is called *Rule Min Term* or RMT (see Sect. 2.3.3). For example, the self loop at node 00 of Fig. 3.1b represents 000 (that is, RMT 0) whereas, the edge from 00 to 01 represents 001 (RMT 1). Against each node of the graph, there are d incoming RMTs and d outgoing RMTs, where d is the number of elements of S. If the de Bruijn graph is traversed, then a node can be reached through any one of the d incoming edges, hence all edges are *equivalent* with respect to the reachability to the node. Similarly, there are d outgoing edges which come out from a single mother node. The RMTs corresponding to the the incoming edges are called *equivalent* RMTs, and those of the outgoing edges are called *sibling* RMTs.

Definition 15 Let $R : S^m \to S$ be a rule and $|S| = d$. A set of d RMTs $r_1, r_2, \ldots, r_d$ are *equivalent* to each other if the RMTs differ only in first symbol. Another set of d RMTs $s_1, s_2, \ldots, s_d$ are the *sibling* to each other if they differ only in last symbol.

If axb represents an RMT of rule $R : S^m \rightarrow S$, where $S = \{z_0, z_1, \ldots, z_{d-1}\}$, then for any $a, b \in S$ and $x \in S^{m-2}$, $\{z_0xb, z_1xb, \ldots, z_{d-1}xb\}$ is a set of equivalent RMTs and $\{axz_0, axz_1, \ldots, axz_{d-1}\}$ is a set of sibling RMTs.

Let us consider $S = \{0, 1, \ldots, d-1\}$. If the RMTs are presented by their decimal equivalents, then the equivalent RMTs and sibling RMTs obey the following relations.

For equivalent RMTs, $r_1 d \equiv r_2 d \equiv \cdots \equiv r_d d \pmod{d^m}$.
For sibling RMTs, $\lfloor \frac{s_1}{d} \rfloor = \lfloor \frac{s_2}{d} \rfloor = \cdots = \lfloor \frac{s_d}{d} \rfloor$.

The $Equi_i = \{i, d^{m-1} + i, 2d^{m-1} + i, \ldots, (d-1)d^{m-1} + i\}$ is denoted as a set of equivalent RMTs that includes RMT i and all of its equivalents ($0 \leq i \leq d^{m-1} - 1$). Similarly, $Sibl_j = \{d.j, d.j + 1, \ldots, d.j + d - 1\}$ ($0 \leq j \leq d^{m-1} - 1$) is a set of sibling RMTs. For ECA, the RMTs 0 (000) and 4 (100) are equivalent to each other whereas, the RMTs 0 (000) and 1 (001) are sibling to each other. Here, $Equi_0 = \{0, 4\}$ and $Sibl_0 = \{0, 1\}$.

A configuration $c \in S^{\mathscr{L}}$ can be traced in the de Bruijn graph. Let consider a window of size m slides over c. The content of the window is an RMT which refers to an edge of the graph. If the window slides towards right (one symbol position), then the next RMT, referring to an edge of the graph, can be found. The first and second edges are incoming and outgoing edges of the node. This can point to a path in the graph, corresponding to c, which is a sequence of edges. If $\mathscr{L}$ is an infinite set, then there is a bi-infinite path for a c. Since the de Bruijn graph, representing a CA, is a labelled graph, the next configuration of c can be obtained if each edge is replaced by its label.

As pointed out in Sect. 2.3.3, a configuration c of a one-dimensional CA can be seen as an RMT sequence $\tilde{c} = (r_i)_{i \in \mathcal{L}}$, where r_i is the RMT $(s_{i-\mathbf{r}_i}, \ldots, s_i, \ldots, s_{i+\mathbf{r}_i})$. This RMT sequence is equivalent to a path—that is, sequence of edges in the de Bruijn graph. The relation between two consecutive RMTs in an RMT sequence can be identified by observing the path in the de Bruijn graph—that is, if $r_j \in Equi_i$, then $r_{j+1} \in Sibl_i$. This also points to the fact that an arbitrary placement of RMTs does not form a valid RMT sequence. Rather, to form an RMT sequence if r_j is chosen from $Equi_i$, then r_{j+1} is to be taken from $Sibl_i$.

The de Bruijn graphs that represent uniform CAs can be generalized to represent non-uniform CAs. It is considered that all the rules of non-uniform CA are of same radius—that is, the non-uniform CA is $(\mathscr{L}, S, \mathbf{r}, \mathcal{R}_i)$ and the set of rules $\{\mathcal{R}_i\}_{i \in \mathscr{L}}$ is finite.

Definition 16 Let $\{\mathcal{R}_i\}_{i \in \mathscr{L}}$ be a finite set of rules of radius $\mathbf{r}$. The *de Bruijn graph* of $\{\mathcal{R}_i\}_{i \in \mathscr{L}}$ is a labelled multi-edge graph (V, E), where $V = S^{2\mathbf{r}}$ is the set of vertices and edges in E are all the pairs (ax, xb) with label $(\mathcal{R}_i, \mathcal{R}_i(axb))$, obtained while varying $a, b \in S, x \in S^{2\mathbf{r}-1}$.

Let consider a non-uniform CA is formed with two ECA rules, 90 and 204. Rule 204 is an identity rule, where $204(x, y, z) = y$ for any $x, y, z \in \{0, 1\}$ whereas,

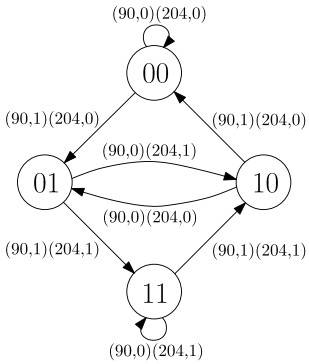


Fig. 3.2 The de Bruijn graph for the rule set {90, 204}

$90(x, y, z) = x + z \pmod 2$. Figure 3.2 shows the de Bruijn graph for these two rules. The labels on the edges are having two parts, each of which corresponds to a rule.

The de Bruijn graph can show evolution of the CA. Since the graph does not relate lattice size, it can be used to study both the finite and infinite one-dimensional CAs.

Example 3.1 Let consider a 4-cell CA (90, 204, 90, 90) in periodic boundary and 1011 is its initial configuration. In Fig. 3.2, a path can be identified against this configuration and it is $(11, 10) \rightarrow (10, 01) \rightarrow (01, 11) \rightarrow (11, 11)$. From the labels of edges, the next configuration of 1011 can be identified. There are two parts in the labels and the appropriate part is to be considered depending on the rule vector. Since the first rule in the rule vector is 90, the first part of the label of edge (11,10) is to be accepted. It points to 1 as the next state of first cell. The second rule is 204. So, the last part of the label of edge (10,01) is selected and this gives 0 as the next state of second cell. At the end, the next configuration is identified as 1010.

3.2 Matrix Algebra

Matrix algebra is used to characterize linear cellular automata. Unlike de Bruijn graph, it has been primarily developed for characterization of non-uniform CA. Let $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ be the global transition function of a CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$. The CA is *linear* if G is a linear map, and the G is linear iff $\mathcal{R}_i$ is linear for each $i \in \mathscr{L}$ (Theorem 2.1). In case of linear CA, the set of all configurations $S^{\mathscr{L}}$ forms a vector space over a field $\mathbb{K}$, and the set S has a natural structure of vector space over $\mathbb{K}$.

Let consider the finite CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$. The vector space $S^{\mathscr{L}}$ of it is of finite dimension. Now, the computations with linear function G can be translated into computations with the *matrix* having scalar elements.

Theorem 3.1 *If $G : S^{\mathscr{L}} \to S^{\mathscr{L}}$ is the global transition function of a finite linear CA with $|\mathscr{L}| = n$, then there exists an $n \times n$ matrix T such that for any configuration $x \in S^{\mathscr{L}}$*

$$y = G(x) \quad \textit{iff} \quad y = T.x$$

where $y \in S^{\mathscr{L}}$[1].

Proof Let $\{\overrightarrow{e_1}, \overrightarrow{e_2}, \ldots \overrightarrow{e_n}\}$ be a basis of $S^{\mathscr{L}}$. So, $G(\overrightarrow{e_1}), G(\overrightarrow{e_2}), \ldots, G(\overrightarrow{e_n})$ are n vectors of $S^{\mathscr{L}}$. Let $G(\overrightarrow{e_i}) = \overrightarrow{v_i}$ for some $\overrightarrow{v_i} \in S^{\mathscr{L}}$, where $i \in \{1, 2, \ldots, n\}$.

Now, for any $\overrightarrow{x} \in S^{\mathscr{L}}$, $\overrightarrow{x} = \sum_{i=1}^{n} x_i \overrightarrow{e_i}$, where $x_i \in S$ and $i \in \{1, 2, \ldots, n\}$. That is,

$$\begin{aligned}
G(\overrightarrow{x}) &= G(\sum_{i=1}^{n} x_i \overrightarrow{e_i}) \\
&= \sum_{i=1}^{n} x_i G(\overrightarrow{e_i}) [\because \ G \text{ is linear}] \\
&= \sum_{i=1}^{n} x_i \overrightarrow{v_i} \\
&= x_1 \overrightarrow{v_1} + x_2 \overrightarrow{v_2} + \cdots x_n \overrightarrow{v_n} \\
&= x_1 \begin{bmatrix} v_{11} \\ v_{21} \\ \vdots \\ v_{n1} \end{bmatrix} + x_2 \begin{bmatrix} v_{12} \\ v_{22} \\ \vdots \\ v_{n2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} v_{1n} \\ v_{2n} \\ \vdots \\ v_{nn} \end{bmatrix},
\end{aligned}$$

Assuming

$$\begin{aligned}
\overrightarrow{v_i} &= \begin{bmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{in} \end{bmatrix} \\
&= \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\
&= T.\overrightarrow{x}
\end{aligned}$$

So, for some $\overrightarrow{y} \in S^{\mathscr{L}}$, $G(\overrightarrow{x}) = \overrightarrow{y}$ iff $T.\overrightarrow{x} = \overrightarrow{y}$ when G is a linear map. Hence the proof. □

[1] When multiplied with T, the configurations x and y are considered as column vectors.

The matrix $T = [a_{ij}]$ is called the *characteristic matrix* of linear CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$, where $i, j \in \mathscr{L}$ and $a_{ij} \in S$. Since $\mathcal{R}_i : S^{2\mathbf{r}_i+1} \to S$ is linear,

$$\mathcal{R}_i(x_{-\mathbf{r}_i}, \ldots, x_0, \ldots, x_{\mathbf{r}_i}) = a_{i(-\mathbf{r}_i)}x_{-\mathbf{r}_i} + \cdots + a_{i0}x_0 + \cdots + a_{i\mathbf{r}_i}x_{\mathbf{r}_i}$$

for some $a_{i(-\mathbf{r}_i)}, \ldots, a_{i0}, \ldots, a_{i\mathbf{r}_i} \in S$. Here, the assumption is that $S = \{0, 1, \ldots, d-1\}$ and $\mathscr{L} = \{0, 1, \ldots, n-1\}$. If the CA is of *null boundary*, the characteristic matrix can be expressed as:

$$T = \begin{bmatrix} a_{00} & \cdots & a_{0\mathbf{r}_0} & 0 & & & & \\ & & & \vdots & & & & \\ & 0 & a_{i(-\mathbf{r}_i)} & \cdots & a_{i0} & \cdots & a_{i\mathbf{r}_i} & 0 \\ & & & \vdots & & & & \\ & & & & 0 & a_{(n-1)(-\mathbf{r}_{n-1})} & \cdots & a_{(n-1)0} \end{bmatrix}$$

The blank spaces of the matrix are filled with 0s. The rows of the matrix show the dependence of cells on their neighbors. For example, the ith row denotes that cell i depends only on its $\mathbf{r}_i$ consecutive left and right neighbors. According to this construction of T, $a_{ij} = 0$ if cell i does not depend on cell j. Otherwise, a_{ij} assumes a non-zero value. As a cell depends only on its nearby neighbors, the characteristic matrices are generally sparse.

Let consider that $S = \{0, 1\}$. That is, the CA is binary. Underlying field of this CA is $GF(2)$ and the characteristic matrix is

$$T_{i,j} = \begin{cases} 1 & \text{if cell } i \text{ depends on cell } j \\ 0 & \text{otherwise} \end{cases} \tag{3.1}$$

For example, the rules 90 and 150 of the CA (150, 150, 90, 150) are linear (Table 2.2). These are expressed as $150(x, y, z) = x + y + z$, and $90(x, y, z) = x + z$ (additions are modulo 2 additions). The characteristic matrix of the CA in null boundary is

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

The successor of any configuration of the CA can be found out by multiplying T with the configuration. For example, the successor of the configuration 0011 is 0110, because

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

Hence, the evolution of a finite linear CA can be obtained by multiplying T with an initial configuration. If x is an initial configuration, then its successive configurations are: $T.x$, $T^2.x$, $T^3.x$, ...

In case of periodic boundary, the representation of characteristic matrix marginally differs. For the CA (150, 90, 150, 90) in null boundary, the first cell with rule 150 depends on the left, self and right cells. In periodic boundary, the left neighbor of the first (left most) cell is the last (right most) cell. The same argument is applicable to the right most cell. Hence, the characteristic matrix of the CA (150, 150, 90, 150) in periodic boundary is

$$T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

There is a class of binary CAs, called *complemented CAs*, which are not linear but close to linear CAs. Rule vector of a complemented CA consists of linear rules and the complements of linear rules. Such a CA is represented by a combination of characteristic matrix and an *Inversion Vector*. The inversion vector F for an n-cell CA is defined as

$$F_i = \begin{cases} 1 & \text{if complemented rule is employed on cell } i \\ 0 & \text{otherwise} \end{cases} \tag{3.2}$$

If there are k linear rules, then there are k complemented rules. For ECAs, there are 8 complemented rules (see Table 2.2). The CA (150, 150, 165, 150) is a complemented CA. Rule 165 is the complement of rule 90. The T and F for this CA (in null boundary) are

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad F = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Let denote the matrix operator of a complemented CA as $\overline{T}$ which is the combination of T matrix and F vector. If y is successor of a configuration x of the CA, then $y = \overline{T}.x = T.x + F$. For the CA (150, 150, 165, 150), if $x = 0011$, then $y = 0100$, as

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

The linear CAs as well as the complemented CAs are efficiently characterized by the matrix algebra.

3.3 Reachability Tree

Reachability tree is a tool, developed for characterizing the one-dimensional finite CAs. It represents all possible *reachable configurations* of a CA. Unlike matrix algebra, this tool can deal with any linear and non-linear CA.

Definition 17 Let $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ be global transition function of a CA. A configuration $c \in S^{\mathscr{L}}$ is *reachable* if there exists a configuration $x \in S^{\mathscr{L}}$ such that $c = G(x)$. If there exists no such $x \in S^{\mathscr{L}}$, for which $c = G(x)$, then the c is a *non-reachable* configuration.

This section first defines the reachability tree for binary CAs that use ECA rules only. Reachability tree is sensitive to the boundary condition. Section 3.3.1 introduces reachability tree for null boundary condition. It is then defined for the CAs in periodic boundary in Sect. 3.3.2. The reachability tree for non-binary CA is introduced in Sect. 3.3.3. In all cases, the radius of a CA is considered as 1.

3.3.1 Null Boundary Condition

The reachability tree is developed by scanning a rule vector $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$ from left to right. For an n-cell CA, there are $n + 1$ levels in the tree—root is at level 0, and the leaves are at level n. A node of the tree is represented by $N_{i.j}$, where i $(0 \leq i \leq n)$ is the level index and j $(0 \leq j \leq 2^i - 1)$ is the node number at ith level. The nodes at each level are numbered from left to right. In the reachability tree, a node is a set of RMTs of a rule. The root is formed with the RMTs of $\mathcal{R}_0$. The nodes of level $n - 1$ are formed with the RMTs of $\mathcal{R}_{n-1}$. The leaves are empty. The edges of the tree are denoted as $E_{i.2j} = (N_{i.j}, N_{i+1.2j}, l_{i.2j})$ and $E_{i.2j+1} = (N_{i.j}, N_{i+1.2j+1}, l_{i.2j+1})$ $(0 \leq i \leq n - 1,\ 0 \leq j \leq 2^{i+1} - 1)$, where $l_{i.2j} \subseteq N_{i.j}$ and $l_{i.2j+1} \subseteq N_{i.j}$ are the labels of the edges, and $l_{i.2j} \cup l_{i.2j+1} = N_{i.j}$. If $\mathcal{R}_i[r] = 0$ (resp. $\mathcal{R}_i[r] = 1$), then the RMT r appears at label $l_{i.2j}$ (resp. $l_{i.2j+1}$).

Definition 18 *Reachability tree* of a finite CA $(\mathscr{L}, \{0, 1\}, 1, \mathcal{R}_i)$ with rule vector $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$, in null boundary, is a rooted and edge-labelled binary tree with $n + 1$ levels, where $E_{i.2j} = (N_{i.j}, N_{i+1.2j}, l_{i.2j})$ and $E_{i.2j+1} = (N_{i.j}, N_{i+1.2j+1}, l_{i.2j+1})$ are the edges between the nodes $N_{i.j}$ and $N_{i+1.2j}$ with label $l_{i.2j} \subseteq N_{i.j}$, and between nodes $N_{i.j}$ and $N_{i+1.2j+1}$ with label $l_{i.2j+1} \subseteq N_{i.j}$ respectively $(0 \leq i \leq n - 1, 0 \leq j \leq 2^i - 1)$. A node $N_{i.j}$ is a subset of the set of possible RMTs of $\mathcal{R}_i$. The following relations exist in the tree:

1. [For root] $N_{0.0} = \{0, 1, 2, 3\}$.
2. $\forall r \in N_{i.j}$, RMT r of $\mathcal{R}_i$ is in $l_{i.2j}$ (resp. $l_{i.2j+1}$), if $\mathcal{R}_i[r] = 0$ (resp. 1). That means, $l_{i.2j} \cup l_{i.2j+1} = N_{i.j}$ $(0 \leq i \leq n - 1, 0 \leq j \leq 2^i - 1)$.
3. $\forall r \in l_{i.j}$, RMTs $2r \pmod 8$ and $2r + 1 \pmod 8$ of $\mathcal{R}_{i+1}$ are in $N_{i+1.j}$ $(0 \leq i \leq n - 3, 0 \leq j \leq 2^{i+1} - 1)$.

4. [For level $n-1$] $\forall r \in l_{n-2.j}$, RMT $2r \pmod 8$ of $\mathcal{R}_{n-1}$ is in $N_{n+1.j}$ $(0 \leq j \leq 2^{n-1}-1)$.
5. [For level n] $N_{n,j} = \emptyset$, for any j, $0 \leq j \leq 2^n - 1$.

The procedure for construction of the reachability tree of a CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$ (in null boundary) is laid out in the following steps. The root is formed with the RMTs 0, 1, 2 and 3 of $\mathcal{R}_0$. Out of the 4 RMTs, the RMTs having next state 0 are in the label of left edge of the root, and the rest RMTs are on the label of the right edge. Now, the nodes at level 1 are formed. If an RMT r is in a label of an edge, then the node connected through the edge is formed with the RMTs $2r \pmod 8$ and $2r+1 \pmod 8$ of $\mathcal{R}_1$. Similarly, the next edges and nodes at level 2, level 3, ... are constructed. However, for the $\mathcal{R}_{n-1}$, all the RMTs are not *valid* in null boundary. Only the even RMTs of $\mathcal{R}_{n-1}$ can be in the nodes at level $n-1$. The edges from level $n-1$ to the leaves are, therefore, labelled by even RMTs.

Definition 19 The edge $E_{i.2j}$ (resp. $E_{i.2j+1}$) with label $l_{i.2j}$ (resp. $l_{i.2j+1}$) is called 0-edge (resp. 1-edge).

For an RMT $r \in l_{i.2j}$ (resp. $l_{i.2j+1}$), $\mathcal{R}_i[r] = 0$ (resp. $\mathcal{R}_i[r] = 1$), where $\mathcal{R}_i$ is the corresponding rule.

Definition 20 An edge $E_{i.j}$ is *non-reachable* if $l_{i.j} = \emptyset$.

Let consider the 4-cell binary CA (165, 73, 200, 90) in null boundary. For ease of reference, the binary values of the rules are given in Table 3.1. Figure 3.3 is the reachability tree of the CA. For null boundary, RMTs 0, 1, 2 and 3 (resp. RMTs 0, 2, 4 and 6) are valid for the left most (resp. right most) cell. So, the root $N_{0.0} = \{0, 1, 2, 3\}$. The label of edge $E_{0.0}$ is $l_{0.0} = \{1, 3\}$, as the RMTs 1 and 3 of rule 165 are 0. The RMTs on an edge are shown within a bracket. The $l_{0.1}$ of the tree is $\{0, 2\}$. The label of edge $E_{2.7}$ is empty,—that is, $l_{2.7} = \emptyset$. This edge, represented by dotted lines, is non-reachable. It cannot connect any node at the next level. Figure 3.3 represents the non-reachable edges. The number of leaves (excluding shaded leaves as they do not exit) in Fig. 3.3 is 8. A sequence of edges from the root to a leaf node represents a reachable configuration, when 0-edge and 1-edge are replaced by 0 and 1 respectively. Hence, the number of reachable configurations of the CA is 8.

Table 3.1 Binary values of the rules 165, 73, 200 and 90

Present state:	111	110	101	100	011	010	001	000	Rule
(RMT)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
(i) Next state:	1	0	1	0	0	1	0	1	165
(ii) Next state:	0	1	0	0	1	0	0	1	73
(iii) Next state:	1	1	0	0	1	0	0	0	200
(iv) Next sate:	0	1	0	1	1	0	1	0	90

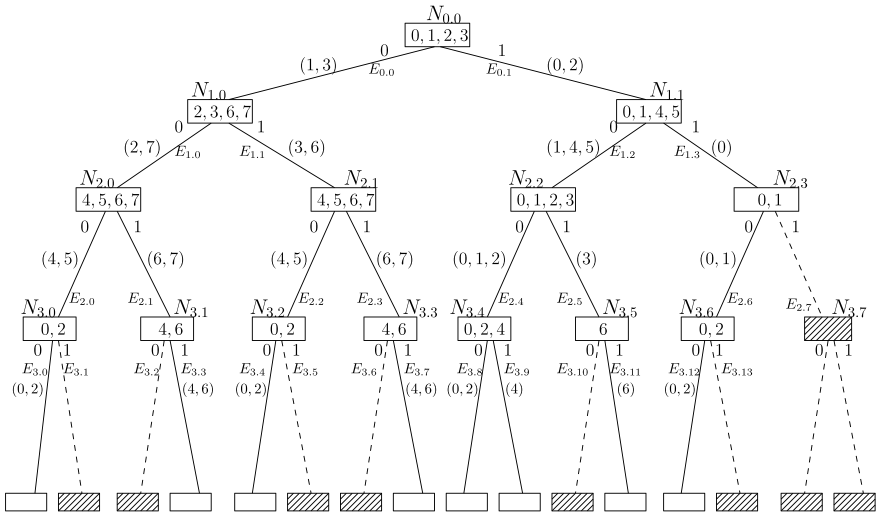


Fig. 3.3 Reachability Tree of CA (165, 73, 200, 90) in null boundary

Let consider the configuration 0011 of the 4-cell CA (165, 73, 200, 90). In Fig. 3.3, this configuration can be traced by the edge sequence ($E_{0.0}$, $E_{1.0}$, $E_{2.1}$, $E_{3.3}$). Hence the configuration is reachable. On the other hand, no such sequence can be found for the configuration 1101. For the first three bits of 1101, the edges $E_{0.1}$, $E_{1.3}$ and $E_{2.6}$ can be traced from the root. But there is no 1-edge to represent the last (right most) 1 of the configuration 1101. The corresponding 1-edge ($E_{3.13}$) is a non-reachable edge (shown by dotted line in the figure). Hence, the configuration 1101 is non-reachable.

Definition 21 The edge $E_{i.j}$, which connects two nodes at levels i and $i+1$, is said to be an *edge of level* i.

The edges $E_{0.0}$ and $E_{0.1}$ are the edges of level 0. Similarly, $E_{n-1.0}$, $E_{n-1.1}$, etc. are the edges of level $n-1$. The term "level" of an edge is frequently used in the next discussions.

As a sequence of edges $\langle E_{0.j_0} E_{1.j_1} \cdots E_{i.j_i} E_{i+1.j_{i+1}} \cdots E_{n-1.j_{n-1}} \rangle$, from root to a leaf, represents a reachable configuration $d_0 d_1 \cdots d_{n-1}$, there is at least one RMT sequence $\langle r_0 r_1 \cdots r_i r_{i+1} \cdots r_{n-1} \rangle$, where $r_i \in l_{i.j_i}$ and $r_{i+1} \in l_{i+1.j_{i+1}}$ ($0 \leq i < n-1$, $0 \leq j_i \leq 2^i - 1$, and $j_{i+1} = 2j_i$ or $2j_i + 1$), and the next configuration of which is the $d_0 d_1 \cdots d_{n-1}$. That is, the sequence of edges represents at least two configurations—the reachable configuration $d_0 d_1 \cdots d_{n-1}$ and its predecessor, represented by the RMT sequence $\langle r_0 r_1 \cdots r_i r_{i+1} \cdots r_{n-1} \rangle$. There are 2^n RMT sequences in the tree, but the number of reachable configurations may be less than 2^n. As a sequence of edges may associate m number of RMT sequences ($m \geq 1$), therefore, such a configuration can be reachable from m different configurations.

Example 3.2 Let consider the sequence of edges representing the reachable configuration 0011 of the CA of Fig. 3.3. There are 2 RMT sequences, 3764 and 3776,

representing the configurations 1110 and 1111, are the predecessor of 0011. This can be verified from the transition diagram (Fig. 2.7a).

Lemma 1 *At any level i of a reachability tree of an n-cell CA in null boundary, $\sum_{j=0}^{2^{i+1}-1} |l_{i.j}| \leq 2^{i+2}$, where $0 \leq i < n-1$ and $\sum_{j=0}^{2^n-1} |l_{n-1.j}| \leq 2^n$.*

Proof The upper limit can be shown from Definition 18. At level 0—that is, for $i = 0$, there are 4 RMTs which are distributed in two labels—$l_{0.0}$ and $l_{0.1}$. Hence, $|l_{0.0}| + |l_{0.1}| = 4 = 2^{0+2}$. Now, the number of RMTs are doubled in each next level (except level $n-1$). Hence, upper limit of $\sum_{j=0}^{2^{i+1}-1} |l_{i.j}|$ is 2^{i+2} when $0 \leq i < n-1$. Since half of the RMTs are absent at level $n-1$, $\sum_{j=0}^{2^n-1} |l_{n-1.j}| = 2^n$.

Now, if two equivalent RMTs are in a label $l_{i.j}$, then $|N_{i+1.j}| < 2 \times |l_{i.j}|$. In that case, the total number of RMTs in the labels is less than the above upper limit. Hence the proof. □

3.3.2 Periodic Boundary Condition

The structure of reachability tree is altered when boundary condition changes from null to periodic. In particular, structure of nodes changes, though the basic properties of reachability tree remain unchanged.

Definition 22 *Reachability tree* of a CA $(\mathscr{L}, \{0,1\}, 1, \mathcal{R}_i)$ with rule vector $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$ in periodic boundary is a rooted and edge-labelled binary tree with $n+1$ levels, where each node $N_{i.j}$ $(0 \leq i \leq n,\ 0 \leq j \leq 2^i - 1)$ is an ordered list of 4 sets of RMTs, and the root $N_{0.0}$ is the ordered list of all sets of sibling RMTs. The edges between $N_{i.j}$ $(0 \leq i \leq n-1,\ 0 \leq j \leq 2^i - 1)$ and its children are denoted as $E_{i.2j} = (N_{i.j}, N_{i+1.2j}, l_{i.2j})$ and $E_{i.2j+1} = (N_{i.j}, N_{i+1.2j+1}, l_{i.2j+1})$, where $l_{i.2j}$ and $l_{i.2j+1}$ are the labels of the edges. Like nodes, the labels are also ordered list of 4 sets of RMTs. Let ${\Gamma_p}^{N_{i.j}}$ be the p^{th} set of the node $N_{i.j}$, and ${\Gamma_q}^{E_{i.2j+m}}$ be the q^{th} set of the label on edge $E_{i.2j+m}$ $(0 \leq p, q \leq 3)$. Then, $N_{i.j} = ({\Gamma_p}^{N_{i.j}})_{0 \leq p \leq 3}$ and $l_{i.2j+m} = ({\Gamma_q}^{E_{i.2j+m}})_{0 \leq q \leq 3}$ $(m \in \{0,1\})$. The following relations hold in the tree.

1. [For root] $N_{0.0} = ({\Gamma_k}^{N_{0.0}})_{0 \leq k \leq 3}$, where $\Gamma_0^{N_{0.0}} = \{0,1\}$, $\Gamma_1^{N_{0.0}} = \{2,3\}$, $\Gamma_2^{N_{0.0}} = \{4,5\}$ and $\Gamma_3^{N_{0.0}} = \{6,7\}$.
2. $\forall r \in {\Gamma_k}^{N_{i.j}}$, RMT r of $\mathcal{R}_i$ is included in ${\Gamma_k}^{E_{i.2j}}$ (resp. ${\Gamma_k}^{E_{i.2j+1}}$), if $\mathcal{R}_i[r] = 0$ (resp. 1), where $\mathcal{R}_i$ is the rule of the ith cell of the CA. That means, ${\Gamma_k}^{N_{i.j}} = {\Gamma_k}^{E_{i.2j}} \bigcup {\Gamma_k}^{E_{i.2j+1}}$ $(0 \leq k \leq 3)$.
3. $\forall r \in {\Gamma_k}^{E_{i.2j}}$ (resp. ${\Gamma_k}^{E_{i.2j+1}}$), RMTs $2r \pmod 8$ and $2r+1 \pmod 8$ of $\mathcal{R}_{i+1}$ are in ${\Gamma_k}^{N_{i+1.2j}}$ (resp. ${\Gamma_k}^{N_{i+1.2j+1}}$).
4. [For level $n-2$] ${\Gamma_k}^{N_{n-2.j}} = \{s \mid$ if $r \in {\Gamma_k}^{E_{n-3.j}}$, then $s \in \{2r \pmod 8, 2r+1 \pmod 8\}\} \cap \{i, i+2, i+4, i+6\}\}$ $(i = \lfloor \frac{k}{2} \rfloor, 0 \leq k \leq 3, 0 \leq j \leq 2^{n-2} - 1)$.

5. [For level $n-1$] ${\Gamma_k}^{N_{n.j}} = \{s \mid$ if $r \in \Gamma_k^{E_{n-2.j}}$, then $s \in \{2r \pmod 8, 2r+1 \pmod 8\} \cap \{k, k+4\}\}, 0 \le k \le 3$.
6. [For level n] ${\Gamma_k}^{N_{n.j}} = \{2r \pmod 8, 2r+1 \pmod 8 | r \in \Gamma_k^{E_{n-1.j}}\}$. ${\Gamma_k}^{N_{n-1.j}}$ is a set of sibling RMTs of $\mathcal{R}_0$, and $\bigcup_j \Gamma_k^{N_{n.j}} = {\Gamma_k}^{N_{0.0}}$

The nodes of levels $n-2$ and $n-1$ are different from the other intermediate nodes (Points 4 and 5 of Definition 22). Only $\frac{1}{2}$ of the selective RMTs can be in ${\Gamma_k}^{N_{i.j}}$ of a node $N_{i.j}$ $(0 \le k \le 3, 0 \le j \le 2^i - 1)$ when $i = n-2$ or $n-1$. Further, the leaves are resulted with ${\Gamma_k}^{N_{n.j}}$, where ${\Gamma_k}^{N_{n.j}}$ is either empty or a set of sibling RMTs. The ${\Gamma_k}^{N_{0.0}}$ is a set of sibling RMTs (Point 1 of Definition 22) and ${\Gamma_k}^{N_{0.0}} = \bigcup_j \Gamma_k^{N_{n.j}}$. That is, the RMTs of leaves are the RMTs of $\mathcal{R}_0$. As in null boundary, the $E_{i.2j}$ is the 0-edge (the next states of all the RMTs at the label of $E_{i.2j}$ are 0, Point 2 of Definition 22). Similarly, $E_{i.2j+1}$ is the 1-edge.

Let consider the CA (165, 73, 200, 90) in periodic boundary (see Table 3.1). The reachability tree of the CA is shown in Fig. 3.4. The root of the tree is $N_{0.0} = (\Gamma_0^{N_{0.0}}, \Gamma_1^{N_{0.0}}, \Gamma_2^{N_{0.0}}, \Gamma_3^{N_{0.0}})$. The root is independent of the CA rule whereas, other nodes are rule dependent. Here $l_{0.0}$, the label of $E_{0.0}$, is $(\{1\}, \{3\}, \{4\}, \{6\})$ and the corresponding child $N_{1.0}$ is $(\{2, 3\}, \{6, 7\}, \{0, 1\}, \{4, 5\})$ (Point 3 of Definition 22). If no RMT is the member of $\Gamma_k^{N_{i.j}}$ or $\Gamma_k^{E_{i.j}}$ $(0 \le k \le 3, 0 \le i \le n-1, 0 \le j \le 2^i - 1)$ (i.e., the set is empty), then $\Gamma_k^{N_{i.j}}$ or $\Gamma_k^{E_{i.j}}$ is noted as '_' in the tree. For example $l_{1.1}$, the label of $E_{1.1}$, is $(\{3\}, \{6\}, \{0\}, \emptyset)$ and the corresponding node $N_{2.1}$ is $(\{6\}, \{4\}, \{1\}, \emptyset)$. However, an arbitrary RMT cannot be a part of the nodes of level $n-2$ and $n-1$. For example, $l_{1.1} = (\{3\}, \{6\}, \{0\}, \emptyset)$ but the corresponding node $N_{2.1}$ (that is, $N_{n-2.1}$, since $n = 4$) is $(\{6\}, \{4\}, \{1\}, \emptyset)$. It can be observed that $\Gamma_0^{N_{2.1}}$ of the node $N_{2.1}$ of Fig. 3.4 is $\{6\}$. If Point 3 of Definition 22 is followed, then RMT 7 should also be the part of $\Gamma_0^{N_{2.1}}$. But it cannot happen as the node is at level $n-2$ (Point 4 of Definition 22). Here, ${\Gamma_k}^{N_{0.0}} = \cup_j {\Gamma_k}^{N_{4.j}}$ for each $k \in \{0, 1, 2, 3\}$.

Definition 23 An edge $E_{i.j}$ in the reachability tree of a CA, in periodic boundary, is *non-reachable* if $\Gamma_0^{E_{i.j}} = \Gamma_1^{E_{i.j}} = \Gamma_2^{E_{i.j}} = \Gamma_3^{E_{i.j}} = \emptyset$.

For example, $E_{3.6}$ of Fig. 3.4 is a non-reachable edge. As in null boundary, a sequence of edges $\langle E_{0.j_0} E_{1.j_1} \cdots E_{i.j_i} E_{i+1.j_{i+1}} \cdots E_{n-1.j_{n-1}} \rangle$ associates a reachable configuration and at least one RMT sequence $\langle r_0 r_1 \cdots r_i r_{i+1} \cdots r_{n-1} \rangle$, where $r_i \in \Gamma_p^{E_{i.j_i}}$ and $r_{i+1} \in \Gamma_p^{E_{i+1.j_{i+1}}}$ for some $p \in \{0, 1, 2, 3\}$ $(0 \le i < n-1, 0 \le j_i \le 2^i - 1$, and $j_{i+1} = 2j_i$ or $2j_i + 1)$.

3.3.3 Non-binary Cellular Automata

For a d-state CA, $d \ge 2$, the reachability tree is a d-ary tree. The local rules of d-state CAs can also be expressed in tabular form. Table 3.2 shows a few rules of 3-state CAs with radius 1. A rule is represented by the sequences of next states $f[r]$ for RMT r

with $f[0]$ as the right most digit. Although these CAs can be non-uniform, here only uniform CAs are considered. Throughout this section, the CAs are assumed to be in periodic boundary and radius as 1.

Definition 24 *Reachability tree* of an n-cell d-state CA is a rooted and edge-labelled d-ary tree with $(n+1)$ levels where each node $N_{i,j}$ $(0 \leq i \leq n,\ 0 \leq j \leq d^i - 1)$ is an ordered list of d^2 sets of RMTs, and the root $N_{0,0}$ is the ordered list of all sets of sibling RMTs. The edges between $N_{i,j}$ $(0 \leq i \leq n-1,\ 0 \leq j \leq d^i - 1)$ and its possible d children are denoted as $E_{i.dj+m} = (N_{i.j}, N_{i+1.dj+m}, l_{i.dj+m})$, where $l_{i.dj+m}$ is the label of the edge and $0 \leq m \leq d-1$. Like nodes, the labels are also ordered list of d^2 sets of RMTs. Let consider that ${\Gamma_p}^{N_{i.j}}$ is the pth set of the node $N_{i.j}$, and ${\Gamma_q}^{E_{i.dj+m}}$ is the qth set of the label on edge $E_{i.dj+m}$ $(0 \leq p, q \leq d^2 - 1)$—that is, $N_{i.j} = ({\Gamma_p}^{N_{i.j}})_{0 \leq p \leq d^2-1}$ and $l_{i.dj+m} = ({\Gamma_q}^{E_{i.dj+m}})_{0 \leq q \leq d^2-1}$. The following relations exist in the tree:

1. [For root] $N_{0.0} = ({\Gamma_k}^{N_{0.0}})_{0 \leq k \leq d^2-1}$, where ${\Gamma_k}^{N_{0.0}} = Sibl_k$.
2. $\forall r \in {\Gamma_k}^{N_{i.j}}$, r is included in ${\Gamma_k}^{E_{i.dj+m}}$, if $f[r] = m, m \in \{0, 1, \ldots, d-1\}$, where f is the rule of the CA. That is, ${\Gamma_k}^{N_{i.j}} = \bigcup_m {\Gamma_k}^{E_{i.dj+m}}$ $(0 \leq k \leq d^2 - 1)$.
3. $\forall r$, if $r \in {\Gamma_k}^{E_{i.dj+m}}$, then RMTs $d.r \pmod{d^3}, d.r+1 \pmod{d^3}, \ldots, d.r+(d-1) \pmod{d^3}$ are in ${\Gamma_k}^{N_{i+1.dj+m}}$ $(0 \leq m \leq d-1)$.

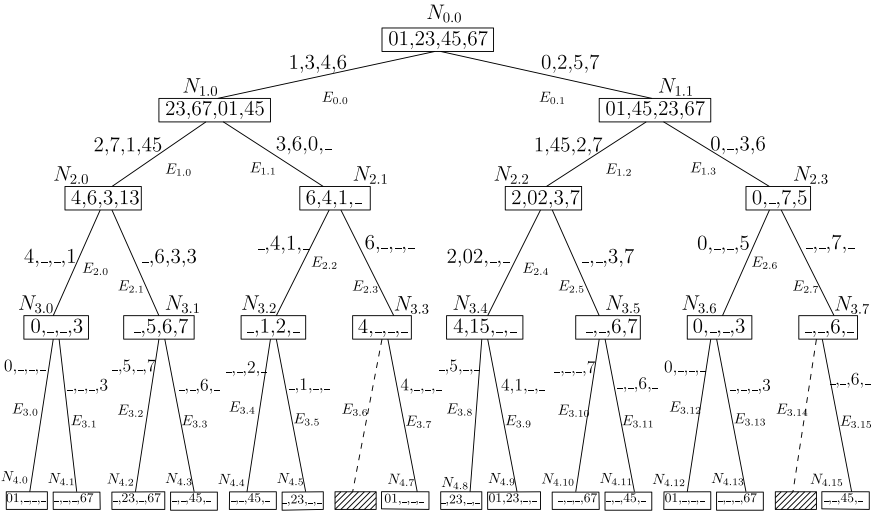


Fig. 3.4 Reachability tree of CA (165, 73, 200, 90) in periodic boundary

Table 3.2 Some rules of 3-state CAs

222	221	220	212	211	210	202	201	200	122	121	120	112	111	110	102	101	100	022	021	020	012	011	010	002	001	000
(26)	(25)	(24)	(23)	(22)	(21)	(20)	(19)	(18)	(17)	(16)	(15)	(14)	(13)	(12)	(11)	(10)	(9)	(8)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)
2	0	1	2	1	0	2	1	0	2	0	1	2	1	0	2	1	0	2	0	1	2	1	0	2	1	0
2	0	1	0	1	2	2	1	0	2	0	1	0	1	2	2	1	0	2	0	1	0	1	2	2	1	0
0	2	0	1	2	0	1	1	2	1	2	2	0	1	0	1	2	0	1	1	0	1	2	2	0	2	0
1	0	2	2	2	1	0	1	0	1	0	2	2	2	1	0	1	0	1	0	2	2	2	1	0	1	0
1	1	2	2	2	1	0	1	0	1	1	2	2	2	1	0	0	0	1	1	2	2	2	1	0	0	0

4. [For level $n-2$] ${\Gamma_k}^{N_{n-2.j}} = \{s \mid \text{if } r \in {\Gamma_k}^{E_{n-3.j}}, \text{then } s \in \{d.r \pmod{d^3}, d.r+1 \pmod{d^3}, \ldots, d.r+(d-1) \pmod{d^3}\} \cap \{i, i+d, \ldots, i+(d^2-1)d\}\}$ $(i = \lfloor \frac{k}{d} \rfloor, 0 \leq k \leq d^2-1, 0 \leq j \leq d^{n-2}-1)$.
5. [For level $n-1$] ${\Gamma_k}^{N_{n-1.j}} = \{s \mid \text{if } r \in {\Gamma^{E_{n-2.j}}}_k, \text{then } s \in \{d.r \pmod{d^3}, d.r+1 \pmod{d^3}, \ldots, d.r+(d-1) \pmod{d^3}\} \cap \{k + i \times d^2 | 0 \leq i \leq d-1\}\}, 0 \leq k \leq d^2-1$.

Due to periodic boundary, the nodes of level $n-2$ and $n-1$ are different from the other intermediate nodes (Points 4 and 5 of Definition 24). Only a subset of selective RMTs can be in ${\Gamma_k}^{N_{i.j}}$ in a node $N_{i.j}$, $(0 \leq k \leq d^2-1, 0 \leq j \leq d^i-1)$, when $i = n-2$ or $n-1$. If A number of RMTs are the possible at $N_{i.j}$, following Point 3 of Definition 24, then only $\frac{A}{d}$ RMTs can exist in $N_{i.j}$ for $i = n-2$ or $n-1$ as per the points 4 and 5 of Definition 24. The leaves are formed with ${\Gamma_k}^{N_{n.j}}$, where ${\Gamma_k}^{N_{n.j}}$ is either empty or a set of sibling RMTs. It is to be noted that the ${\Gamma_k}^{N_{0.0}}$ is a set of sibling RMTs (Point 1 of Definition 24) and ${\Gamma_k}^{N_{0.0}} = \bigcup_j {\Gamma_k}^{N_{n.j}}$.

Example 3.3 Reachability tree of a 3-state 4-cell CA is shown in Fig. 3.5. As it is of 3 states, a node $N_{i,j}$ can have at most 3 children—$N_{i+1.3j}$, $N_{i+1.3j+1}$ and $N_{i+1.3j+2}$. Hence, the maximum number of possible nodes in the tree, for a 4-cell CA, is $\frac{3^{4+1}-1}{3-1} = 121$. Figure 3.5, however, contains 105 nodes. The root of the tree is $N_{0.0} = ({\Gamma_k}^{N_{0.0}})_{0 \leq k \leq 8}$, where ${\Gamma_k}^{N_{0.0}}$ is a set of sibling RMTs $(Sibl_k)$. That means, $N_{0.0} = (\{0,1,2\}, \{3,4,5\}, \{6,7,8\}, \{9,10,11\}, \{12,13,14\}, \{15,16,17\}, \{18,19,20\}, \{21,22,23\}, \{24,25,26\})$. It is to be noted that the root is independent of the CA rule (it depends only on d, the number of states per cell) whereas, other nodes are rule dependent. Here, $l_{0.0}$—that is, the label of $E_{0.0}$ is $(\{0\}, \{5\}, \{7\}, \{9\}, \{14\}, \{16\}, \{18\}, \{23\}, \{25\})$ and

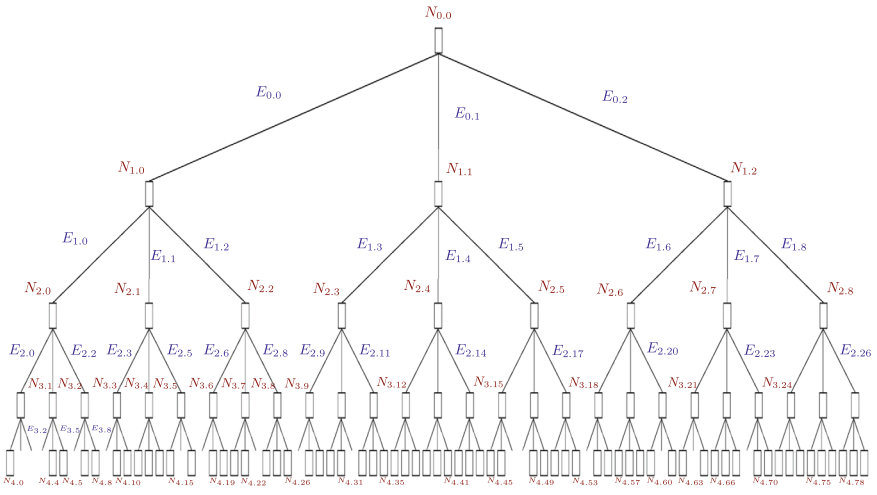


Fig. 3.5 Reachability tree of a 3-state 4-cell CA with rule 201012210201012210201012210

the corresponding child $N_{1.0}$ is $(\{0, 1, 2\}, \{15, 16, 17\}, \{21, 22, 23\}, \{0, 1, 2\}, \{15, 16, 17\}, \{21, 22, 23\}, \{0, 1, 2\}, \{15, 16, 17\}, \{21, 22, 23\})$. However, an arbitrary RMT cannot be the part of a node of level $n-2$ or $n-1$. For example, $l_{1.1} = (\{1\}, \{15\}, \{22\}, \{1\}, \{15\}, \{22\}, \{1\}, \{15\}, \{22\})$, but the corresponding node $N_{2.1}$ (that is, $N_{n-2.1}$, since $n = 4$) is $(\{3\}, \{18\}, \{12\}, \{4\}, \{19\}, \{13\}, \{5\}, \{20\}, \{14\})$. It can be observed that, $\Gamma_1^{N_{2.1}}$ of the node $N_{2.1}$ of Fig. 3.5 is $\{18\}$. As per point 3 of Definition 24, the RMTs 19 and 20 are to be the part of $\Gamma_1^{N_{2.1}}$. But, as the node is at level $n-2$, these are not included (Point 4 of Definition 24). Similarly at level 3, $l_{2.3} = (\emptyset, \{18\}, \emptyset, \emptyset, \emptyset, \emptyset, \{5\}, \emptyset, \{14\})$ and $N_{3.3} = (\emptyset, \{1\}, \emptyset, \emptyset, \emptyset, \emptyset, \{15\}, \emptyset, \{17\})$ (Point 5). That is, for node $N_{3.3}$, in the set ${\Gamma_1}^{N_{3.3}}$, only RMT 1 is present. So, only $\frac{1}{3}$ of the possible RMTs can be the part of any set at level $n-2$ or $n-1$.

Here, an edge $E_{i.j}$ is non-reachable if $\bigcup_{0 \le k \le d^2-1} \Gamma_k^{E_{i.j}} = \emptyset$. The edges of the reachability tree associate the states of CA cells, and a sequence of edges from the root to a leaf represents a reachable configuration. Since d number of edges can come out from a parent node, the left most edge is marked as 0-edge which represents state 0, second left most edge as 1-edge which represents state 1, and so on. The right most edge ($(d-1)$-edge) represents the state $d-1$.

Definition 25 An edge $E_{i.j}$ is called m-edge if $f[r] = m$, $r \in \bigcup_{0 \le k \le d^2-1} \Gamma_k^{E_{i.j}}$, where $0 \le m \le d-1$ and f is the CA rule.

Example 3.4 In Fig. 3.5, there are some non-reachable edges—$E_{3.1}$, $E_{3.2}$, $E_{3.3}$, $E_{3.6}$ etc. on which the labels are empty. The corresponding nodes $N_{3.1}$, $N_{3.2}$, $N_{3.3}$, $N_{3.6}$ etc. are non-reachable nodes. However, the sequence $\langle E_{0.0}, E_{1.1}, E_{2.3}, E_{3.11} \rangle$ represents the reachable configuration 0102.

3.4 Non-reachable Configurations

Reachability tree depicts the reachable configurations of a CA. In this section, an algorithm to count the non-reachable configurations of a binary CA with null-boundary is introduced.

If non-reachable configurations exist in a finite CA, then the CA is *irreversible*. Otherwise, the CA is *reversible* (see Chap. 4 for detail discussion). The next two theorems set the lower bounds of non-reachable configurations in finite irreversible CAs.

Theorem 3.2 *An n-cell irreversible CA $(\mathscr{L}, \{0, 1\}, 1, \mathcal{R}_i)$ contains at least 2^{n-3} non-reachable configurations.*

Proof For ECA rules, $\frac{1}{8}$ of the total RMT sequences, representing configurations of a binary CA, use each out of 8 RMTs of $\mathcal{R}_i$. Since the CA is irreversible, there is at least one RMT r of $\mathcal{R}_i$ that causes the existence of a non-reachable edge at level i

in the reachability tree. The RMT r is used in $\frac{2^n}{8} = 2^{n-3}$ RMT sequences, which are non-reachable configurations. Hence, the number of non-reachable configurations is at least 2^{n-3}. □

Definition 26 A rule is *balanced* if it contains equal number RMTs for each of the possible states as their next state; otherwise it is an *unbalanced* rule.

For example, ECA rule 165 is a balanced rule, whereas rule 73 is unbalanced (see Table 3.1). Next theorem considers only balanced rules in its rule vector.

Theorem 3.3 *An n-cell irreversible CA $(\mathscr{L}, \{0, 1\}, 1, \mathcal{R}_i)$, where each rule $\mathcal{R}_i$ is balanced, contains at least 2^{n-2} non-reachable configurations.*

Proof Let consider that the reachability tree of an n-cell CA, configured only with balanced rules, has no non-reachable edge up to the ith level, but some of the edges of level i are non-reachable ($1 \leq i \neq n-1$). The rule $\mathcal{R}_i$ is responsible for that. Since $\mathcal{R}_i$ is balanced, therefore, there exist at least 2 RMTs that cause the existence of a non-reachable edge at level i in the reachability tree. As $\frac{1}{8}$ of the total configurations are related to an RMT, the total number of non-reachable configurations for such CA is $\frac{1}{4}$ $(\frac{1}{8} + \frac{1}{8})$ of the total configurations. That is, the irreversible CA contains at least 2^{n-2} non-reachable configurations. Hence the proof. □

Corollary 3.1 *An n-cell linear/additive irreversible CA contains at least 2^{n-2} non-reachable configurations.*

Proof Since a linear/additive rule is balanced, the proof is directly followed from Theorem 3.3. □

An algorithm can be developed to count the number of non-reachable configurations in an n-cell CA by exploring the reachability tree. However, the exponential growth of reachability tree can be the major hindrance. This is avoided by exploring only the *unique* nodes while eliminating the *duplicate* nodes at a level.

Let consider the two nodes $N_{i.j}$ and $N_{i.k}$ $(j \neq k)$ at level i $(1 \leq i < n-1)$ of a reachability tree. Further, assume that $N_{i.j} = N_{i.k}$. So, two subtrees rooted at $N_{i.j}$ and $N_{i.k}$ are similar. That is, the number of non-reachable configurations contributed by one subtree is equal to that of the other. Hence, the tree can be explored either for $N_{i.j}$ or $N_{i.k}$. However, the number of non-reachable configurations in the subtree, considered for exploring, is to be doubled. If $N_{i.j}$ is explored, then $N_{i.k}$ is marked as the *duplicate* node and $N_{i.j}$ is the *unique* node. Any two *unique* nodes are dissimilar.

Lemma 2 *Number of unique nodes in the reachability tree of a CA with null boundary, at a level, is less than 15.*

Proof Since the sibling RMTs are always paired in a node (Point 3 of Definition 18), there are four sets of RMTs (that is, {0, 1}, {2, 3}, {4, 5} and {6, 7}) which form the nodes. Using these four sets, $2^4 - 1 = 15$ unique nodes (excluding $\emptyset$ as a unique node) can only be formed.

To show that these 15 unique nodes cannot be generated at a level simultaneously, the following six unique nodes—$N_1 = \{0, 1, 2, 3\}$, $N_2 = \{4, 5, 6, 7\}$, $N_3 = \{0, 1, 4, 5\}$, $N_4 = \{2, 3, 6, 7\}$, $N_5 = \{0, 1, 6, 7\}$ and $N_6 = \{2, 3, 4, 5\}$ can be considered. These six nodes cannot be generated simultaneously at any level i for any rule as $\mathcal{R}_{i-1}$. Hence the proof. □

The input to Algorithm 1 that counts the number of non-reachable configurations in a CA, is a rule vector of an n-cell CA in null boundary. The $l_0, l_1, \ldots$ and $l'_0, l'_1, \ldots$ are the labels of edges of two consecutive levels. These labels are the sets of RMTs for null boundary condition (for periodic boundary, the labels are not sets, rather a collection of 4 sets; see Sect. 3.3.2). The variable *NoUL* is used to store the number of unique labels at a level that is related to the number of unique nodes. The algorithm does not construct the whole tree at a time, but finds the labels of edges at level $i + 1$ from that of level i. The NS is a variable that stores the non-reachable configurations. The algorithm further uses two arrays—W and W'. The W_j (resp. W'_k) is the number of configurations that can be represented as the bit sequence in the tree by an edge labelled by l_j (resp. l'_k). Example 3.5 illustrates the execution of Algorithm 1.

Algorithm 1: Count non-reachable configurations

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: Number of non-reachable configurations.

Step 1: Find the labels l_0 and l_1 of the edges of level 0 using $\mathcal{R}_0$.
(i) If $l_i = \emptyset$, $i \in \{0, 1\}$, then consider the nonempty label as l_0, and set $NS \leftarrow 2^{n-1}$, $W_0 \leftarrow 2^{n-1}$, $NoUL \leftarrow 1$.
(ii) Else if $l_0 = l_1$, then set $NS \leftarrow 0$, $W_0 \leftarrow 2^n$ and $NoUL \leftarrow 1$.
(iii) Else, set $NS \leftarrow 0$, $W_0 \leftarrow W_1 \leftarrow 2^{n-1}$ and $NoUL \leftarrow 2$.
Step 2: For $i = 1$ to $n - 1$, repeat Step 3 to Step 6
Step 3: For each l_j, $j \in \{0, 1, \cdots, NoUL - 1\}$
(i) Find RMTs of the corresponding node using Point 3 (or, Point 4 if $i = n - 1$) of Definition 18.
(ii) Distribute the RMTs in the labels l'_{2j} and l'_{2j+1} of edges of level i using $\mathcal{R}_i$ (Point 2 of Definition 18).
(iii) Set $W'_{2j} \leftarrow W'_{2j+1} \leftarrow \frac{W_j}{2}$.
(iv) If $l'_k = \emptyset$, $k \in \{2j, 2j + 1\}$, set $NS \leftarrow NS + W'_k$.
(v) Replace RMTs 4, 5, 6, and 7 of l'_{2j} and l'_{2j+1}, if any, by RMTs 0, 1, 2 and 3 respectively.
Step 4: Set $k \leftarrow 0$. For $j = 0$ to $2NoUL - 1$, repeat Step 5.
Step 5: If $l'_j \neq \emptyset$, then do the following
(i) Set $l_k \leftarrow l'_j$, $W_k \leftarrow W'_j$.
(ii) If there is any set l'_m in the list $l'_{j+1}, l'_{j+2}, \cdots l'_{2NoUL-1}$ so that $l_k = l'_m$, then set $W_k \leftarrow W_k + W'_m$ and $l'_m \leftarrow \emptyset$.
(iii) Set $k \leftarrow k + 1$.
Step 6: Set $NoUL \leftarrow k$.
Step 7: Report NS as the number of non-reachable configurations.

Example 3.5 Let consider that the 4-cell CA (165, 73, 200, 90) (Fig. 3.3) is input to Algorithm 1. Step 1 of the algorithm results in $l_0 = \{1, 3\}$ and $l_1 = \{0, 2\}$; $W_0 = W_1 = 2^{4-1} = 8$ and $NoUL \leftarrow 2$. From l_0, the node with RMTs 2, 3, 6 and 7 (Step 3(i)), and then $l'_0 = \{2, 7\}$ and $l'_1 = \{3, 6\}$ following rule 73 (Step 3(ii)) are generated. Now, $W'_0 = W'_1 = 4$. In Step 3(v), the labels are modified as $l'_0 = \{2, 3\}$ and $l'_1 = \{3, 2\}$. Hence, $l'_0 = l'_1$. So, the algorithm proceeds with only one label.

Following the same procedure, the $l'_2 = \{1, 4, 5\}$ and $l'_3 = \{0\}$, where $W'_2 = W'_3 = 4$, can be identified. Now, Step 3(v) generates $l'_2 = \{0, 1\}$ and $l'_3 = \{0\}$. The steps 4 and 5 compute $l_0 = \{2, 3\}$, $l_1 = \{0, 1\}$ and $l_2 = \{0\}$; and $W_0 = 8$, $W_1 = 4$ and $W_2 = 4$. Hence, the new $NoUL$ is 3. The execution now returns to Step 3 with the newly generated labels.

Likewise, the $l'_0 = \{4, 5\}, l'_1 = \{6, 7\}, l'_2 = \{0, 1, 2\}, l'_3 = \{3\}, l'_4 = \{0, 1\}$ and $l'_5 = \emptyset$ can be found. Further, $W'_0 = W'_1 = 4$, $W'_2 = W'_3 = 2$ and $W'_4 = W'_5 = 2$. Since $l'_5 = \emptyset$, the NS is 2 (Step 3(iv)). However, the new (unique) labels are $l_0 = \{0, 1\}$, $l_1 = \{2, 3\}$, $l_2 = \{0, 1, 2\}$ and $l_3 = \{3\}$ with $W_0 = 6$, $W_1 = 4$, $W_2 = 2$ and $W_3 = 2$.

The next i is 3 which is $n - 1$. Hence, only the even numbered RMTs are generated in the corresponding nodes (Point 4 of Definition 18). As a result, $l'_0 = \{0, 2\}$ ($W'_0 = 3$), $l'_1 = \emptyset$ ($W'_1 = 3$), $l'_2 = \emptyset$ ($W'_2 = 2$), $l'_3 = \{4, 6\}$ ($W'_3 = 2$), $l'_4 = \{0, 2\}$ ($W'_4 = 1$), $l'_5 = \{5\}$ ($W'_5 = 1$), $l'_6 = \emptyset$ ($W'_6 = 1$) and $l'_7 = \{6\}$ ($W'_7 = 1$) are obtained. Hence, the NS is $2 + W'_1 + W'_2 + W'_6$—that is 8. It is the total number of non-reachable configurations of the CA. This fact can be verified from Fig. 3.3.

Complexity: The maximum possible value of $NoUL$ is related to the maximum possible unique nodes at any level, and it is <15 (Lemma 2). In Algorithm 1, there is a main loop in Step 2 that depends on n, and two nested loops in Step 3 and Step 4 that depend on $NoUL$. Therefore, the computational complexity of Algorithm 1 is $O(n)$.

3.5 Chapter Notes

The de Bruijn graph has been first used by Sutner [2] to study the reversibility of one-dimensional CAs. Later, it has attracted the researchers to study the surjectivity, number conservation, equicontinuity etc. of one-dimensional CAs (see for example [2–4]). The authors of [5] have explored cyclic properties and inverse of a CA with pair diagram of de Bruijn graph. To study parity problem [6], the periods of pre-images of spatially periodic configurations in surjective CA [7], the de Bruijn graphs have also been used. It has been extended to represent non-uniform CAs in [8].

The matrix algebra has been developed by Das et al. [9] to characterize non-uniform CAs which are finite and linear. In their book, Pal Chaudhuri et al. [10] have reported several results of linear/additive and complemented CAs employing the matrix algebra. The authors of [11–13] have also used matrix algebra to explore various issues of linear/additive and complemented CAs. For 2-dimensional linear

CAs also, a number of works have been reported that apply matrix algebra as the characterization tool (see [14–17]).

Reachability tree has been first introduced in [18] to characterize the non-linear binary CAs in null boundary. It has been utilized specially to discover the reversibility, convergence, etc., of non-uniform ECAs [19, 20]. The reachability tree for binary non-uniform ECAs with periodic boundary has been introduced in [21]. Its extension is reported in [22] to study the properties of non-binary finite CAs. Some variants of reachability tree are also developed, such as Rule Vector Graph [23], Next State RMT Transition Diagram (NSRTD) [24].

Discovery of new characterisation tools that can ease to understand the global behaviors of CAs is an active research area. Following are few topics on which future research may be carried out.

1. Extend reachability tree to represent CAs over infinite lattice.
2. Develop reachability tree for other state machines, such as LFSR, Counter, etc., as well as for one-dimensional CAs with boundary condition other than null and periodic.
3. Is it possible to develop reachability tree for two-dimensional finite CAs? Explore.

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Chapter 4
Reversibility

This chapter deals with the reversibility of non-uniform cellular automata (CAs). It starts with the basics of reversibility and then discusses on the decidability issues of reversible CAs, defined over infinite lattice (Sect. 4.1). The finite non-uniform ECAs—that is, the finite CAs with only ECA rules are the major consideration of this chapter. The linear and complemented CAs are dealt in Sect. 4.2 whereas, the reversibility of an arbitrary non-uniform ECA (linear, complemented and non-linear) is studied in Sect. 4.3. The reachability tree, introduced in Sect. 3.3, is the instrumental in this study. Section 4.4 reports the analysis and synthesis of reversible CAs utilizing the reachability tree. Based on the potential of forming reversible CAs, the ECA rules are classified in Sect. 4.5. This classification enables the design of *scalable* reversible CAs (Sect. 4.6).

4.1 Reversibility and Decidability

Let $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ be the global transition function of a cellular automaton $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$. The cellular automaton (CA) is reversible iff G is a bijective map. When G is bijective, it is possible to completely identify not only the future configuration of the system but also the previous configurations from its current configuration. If $x \in S^{\mathscr{L}}$ is the current configuration of the CA, then the previous configurations are $G^{-1}(x), G^{-2}(x), G^{-3}(x), \cdots$, where G^{-1} is the inverse of G.

To decide the bijectivity of G, the injectivity as well as the surjectivity of G are to be decided. In general, different algorithms are needed to test the injectivity and surjectivity. For some special cases, however, one test algorithm may suffice to decide the reversibility. In case of finite CAs, surjectivity implies injectivity, and hence bijectivity. In case G is not surjective, there exist some configurations which

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Complexity and Computation 49, https://doi.org/10.1007/978-3-032-00257-0_4

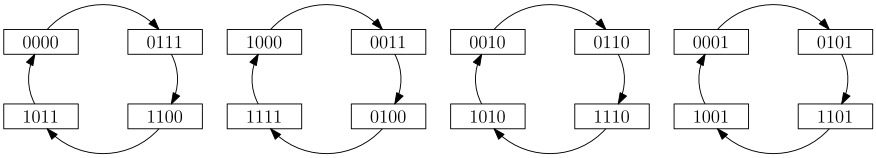


Fig. 4.1 Transition diagram of CA (90, 15, 85, 15) with null boundary

have no predecessor. These configurations are the *non-reachable* configurations,[1] and existence of a non-reachable configuration implies that the CA is not reversible.

When a CA is finite, the reversibility implies that a configuration x returns back after some time steps. That is, $x = G^k(x)$ for some $k \geq 1$. Let consider the 4-cell CA (90, 15, 85, 15) with null boundary. Figure 4.1 is the transition diagram of the CA. This CA is reversible. Each configuration of the CA is reachable from itself. On the other hand, the CA of Fig. 2.7 is not reversible. There are some configurations in Fig. 2.7 that are not reachable from themselves. Based on this property of finite CA, configurations are classified as *cyclic* and *acyclic*.

Definition 27 Let $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ be the global transition function of a finite CA. A configuration $x \in S^{\mathscr{L}}$ is called *cyclic* if there exists a $k \geq 1$ such that $x = G^k(x)$. Otherwise, the configuration is *acyclic*. A finite CA is *reversible* if all of its configurations are cyclic. Otherwise, it is *irreversible*.

For example, all the configurations of Fig. 4.1 are cyclic whereas, some configurations of Fig. 2.7 are cyclic and others are acyclic. A few of acyclic configurations, however, are *non-reachable*.

It is, therefore, possible to develop an algorithm to decide the reversibility of a finite CA. On the other hand, if $\mathscr{L}$ is infinite and distribution of rules on $\mathscr{L}$ of a CA is unknown, then it is not possible in general to decide on the reversibility of the CA. However, if distribution of rules of a CA is arbitrary over a finite places of $\mathscr{L}$ and maintains an order on the rest part of $\mathscr{L}$, then the reversibility may be decidable. In p-ν CAs, which is a subclass of non-uniform CAs (see Sect. 2.3.1), two sequences of rules are periodically applied to the cells in left and right side of a finitely many places of $\mathscr{L}$. A subclass of p-ν CAs is the d-ν CAs, where all but finitely many cells use a single rule. Reversibility of p-ν and d-ν CAs is decidable.

To show the decidability of d-ν or p-ν CAs, the de Bruijn graphs are used. A de Bruijn graph helps to represent a configuration as a sequence of vertices, and its successor is obtained by following the rules of cells of the CA (Sect. 3.1). Two schemes, based on the de Bruijn graph, are presented next to test injectivity and surjectivity of the d-ν and p-ν CAs.

[1] They are also called *surjective* configurations.

4.1.1 Injectivity Test

To decide on the injectivity of a CA, the corresponding de Bruijn graph is searched to check whether there are two configurations having the same successor. If such configurations exist, then the CA is not injective, hence not bijective. Otherwise, the CA is injective. To facilitate the search of such many-to-one mapping, a *product graph* of the de Bruijn graph is developed.

Definition 28 Let $\mathcal{R}$ be a finite set of rules of radius $\mathbf{r}$ and (V, E) be the de Bruijn graph of $\mathcal{R}$. The *Product Graph* of $\mathcal{R}$ is the labelled graph $(V \times V, W)$, where $((u, u'), (v, v')) \in W$ with $f \in \mathcal{R}$, iff there exists $a \in S$ such that (u, v) and (u', v') belong to E, both with same label (f, a).

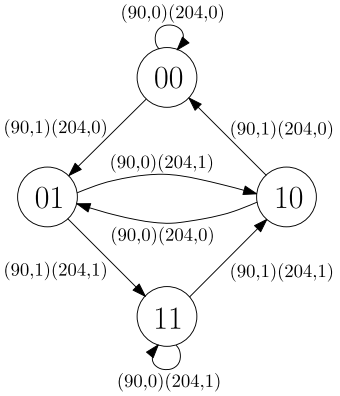
In the product graph, two configurations can be traced by a single sequence of nodes. For example, consider the construction of product graph for a set of two ECA rules 90 and 204. The de Bruijn graph of $\{90, 204\}$ is shown in Fig. 3.2 which is reproduced in Fig. 4.2a. Figure 4.2b shows the product graph of the de Bruijn graph, removing all the second components of edge labels. The graph has sixteen nodes and each node consists of two components from the two graphs. Any sequence of edges, therefore, represents parts of two configurations—that is, two *sub-configurations*. For example, the sequence $(00/01) \to (01/11) \to (11/11) \to (11/10) \to (11/01)$ is for two sub-configurations—001111 and 011101. A bi-infinite path represents two configurations of an (infinite) CA. Now, if a cycle in the product graph is found, then that cycle corresponds to two configurations. In the above sequence, if two nodes—(10/10) and (00/01) are appended, then a cycle can be found that represents two configurations $\cdots 001111001111 \cdots$ and $\cdots 011101011101 \cdots$.

In the product graph, any bi-infinite path with label $(\mathcal{R}_i, z_i)_{i \in \mathscr{L}}$ corresponds to two configurations x and y such that $G(x) = G(y) = z = (z_i)_{i \in \mathscr{L}}$. If such a path exists in the product graph and $x \neq y$, the CA cannot be injective. Conversely, if the CA is non-injective—that is, if there exist two distinct configurations x and y such that $G(x) = G(y)$, then by its construction, the product graph contains a unique bi-infinite path. Hence, if no bi-infinite path with label $(\mathcal{R}_i, z_i)_{i \in \mathscr{L}}$ exists in the product graph, then the CA $(\mathscr{L}, S, \mathbf{r}, \mathcal{R}_i)$ is injective.

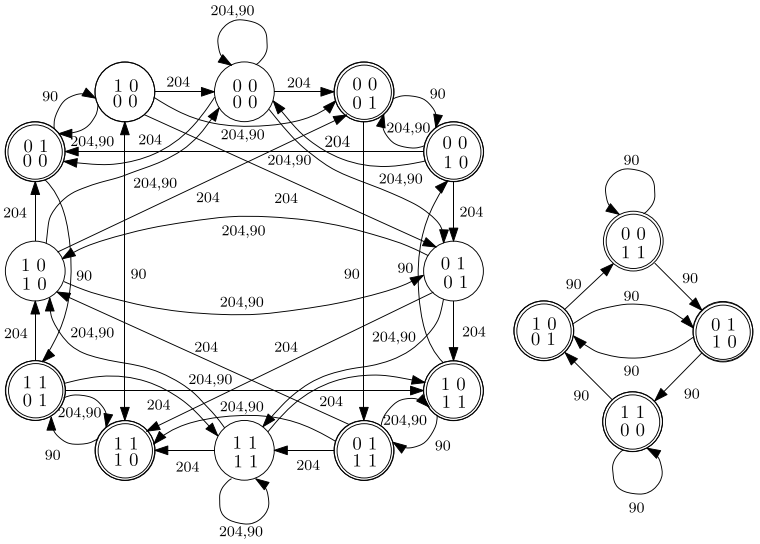
Therefore, the product graph can be exploited to decide the injectivity of a CA. Let consider a p-ν CA which applies h_l, the left default rule, to cell i when $i < -k$ and h_r, the right default rule, to cell j when $j > k$ for some positive k. For the rest of the cells, the CA uses a sequence of rules, say $\mathcal{R}_{-k}, \mathcal{R}_{-k+1}, \ldots, \mathcal{R}_k$. Assume that the CA uses a single radius. To decide the injectivity of the CA, the search for a bi-infinite path that consists of the following components is needed.

1. A left-infinite sequence of edges with label $(h_l, z_i)_{i<-k}$, then
2. A sequence of $2k + 1$ edges with label $(\mathcal{R}_i, z_i)_{-k \le i \le k}$, and then
3. A right-infinite sequence of edges with label $(h_r, z_i)_{i>k}$.

If no such path that represents two different configurations, having the same next configuration, exists in the product graph, the CA is injective; otherwise, it is non-injective.



(a) The de Bruijn graph for the rule set $\{90, 204\}$



(b) Product graph of {90, 204}

Fig. 4.2 Product graph of {90, 204} removing all second components of edge labels

Now, existence of an infinite sequence of edges implies the existence of a cycle in the product graph, and vice-versa. Therefore, to get a left-infinite sequence of edges with label $(h_l, z_i)_{i<-k}$, search for a cycle having the first component of its edge labels as h_l is needed. If such a cycle exists, then a sequence of $2k + 1$ edges with label $(\mathcal{R}_i, z_i)_{-k \le i \le k}$ from any node of the cycle or from a node that can be reached from the cycle by a path with h_l as the first component of the label is to be identified. Finally, the cycle with h_r as the first component of edge labels that is reached from the end of the second path by a sequence of edges having h_r as the first component

of their labels is computed. Therefore, to search for a bi-infinite path for the search of two distinct configurations, having same next configuration, the following steps are followed.

1. Construct product graph of the de Bruijn graph of the set of rules of the non-uniform CA.
2. Find cycle(s) in the product graph, if any, with the first component of its edge labelled as h_l.
3. Find a sequence of $2k+1$ edges with $(\mathcal{R}_i)_{-k\le i\le k}$ as the first component of labels from a node of the cycle, or from a node that can be reached from the cycle by a path with h_l as the first component of the label.
4. From the terminating node of the above sequence, find a path that includes a cycle with h_r as the first component of the labels.
5. If a bi-infinite path is found following the above steps and the path represents two distinct configurations, then the CA is non-injective. Otherwise, it is injective.

If there exists a cycle representing two distinct configurations in the product graph and both the configurations have the same next configuration, then the CA is not injective. In case of a uniform CA, existence of such a cycle is sufficient to show the non-injectivity of the CA. For non-uniform CAs, the rule sequence is considered to find the next configurations. To simplify the process of testing the injectivity, the product graph is *reduced* with the following steps.

- If a node is the reflection of another node, then merge these two to a single node. That is, if (ab/cd) and (cd/ab) are the two nodes, then consider only one, say, (ab/cd).
- Merge the corresponding edges of the merged nodes. While merging the edges, if an outgoing merged edge represents two edges with dissimilar labels, then concatenate them to label the merged edge.

In reduced product graph, there can be always a component which is isomorphic to the original de Bruijn graph. Figure 4.3 is the reduced graph of Fig. 4.2b. The isomorphic component of the de Bruijn graph (Fig. 4.2a) is marked by dotted circle in Fig. 4.3. Any bi-infinite path containing only the nodes of isomorphic component points to only one unique configuration instead of two. Hence, to find a many-to-one mapping, it is needed to find a bi-infinite path in the reduced product graph that has at least one node outside of this isomorphic component. Further, it is always possible to form cycles using the nodes of the isomorphic component, where the first component is any of the rules used in the de Bruijn graph. Therefore, instead of exclusively searching for the bi-infinite path with the first component of its edge labels as h_l (resp. h_r), it can be started (resp. ended) with any node of the isomorphic component and look for a path from (resp. to) this node containing at least one node outside the isomorphic component for which the first components of the edge labels follow the rule sequence $(\mathcal{R}_i)_{-k\le i\le k}$ prefixed (resp. suffixed) with rule h_l (resp. h_r), if necessary. So, for this purpose, the nodes inside the isomorphic component are called as the *start nodes* and *final nodes*.

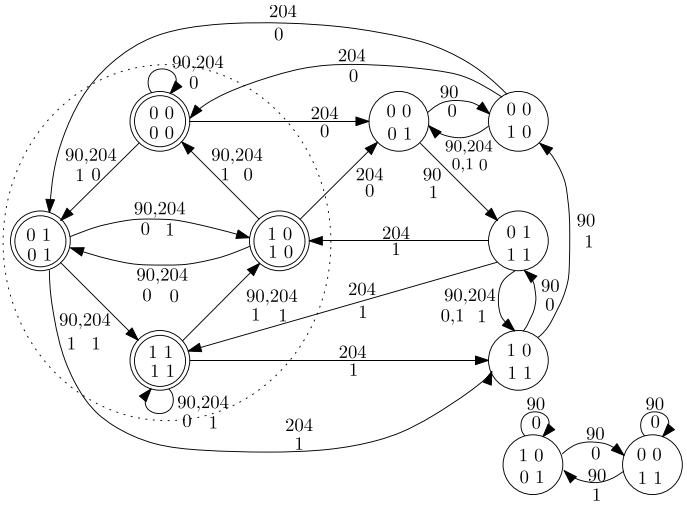


Fig. 4.3 Reduced product graph of {90, 204}

Example 4.1 This example illustrates the process to decide the injectivity of a CA from reduced product graph. Let consider that the cells -1, 0 and 1 of a CA $(\mathbb{Z}, \{0, 1\}, 1, \mathcal{R}_i)$ follow the rules 90, 204 and 90 respectively. The rest of the cells follow the rule 204. By definition, this is a d-v CA. Since the CA uses only two rules, the reduced product graph for the rule set {90, 204} is first constructed. Figure 4.3 shows the graph. The nodes in the isomorphic component are marked with double circle. These nodes are *start* and *final nodes*. Any cycle in this component represents a single configuration (not two). To decide the injectivity, a cycle is to be identified. If exists, then that starts and ends with a start/final node. However, while searching for such cycle, the rules of the cells are to be considered. In this example, cycles exist.

The following result follows from the above discussion.

Lemma 3 *Injectivity of p-v CA with period 1 is decidable.*

4.1.2 Surjectivity Test

To decide the surjectivity of a CA, a non-reachable configuration is to be identified. If a configuration is not non-reachable, it can be associated with its predecessor. If such association fails for a configuration, then the CA is not surjective. The *powerset automaton*, derived from the de Bruijn graph of a set of rules, is used for this purpose.

Definition 29 Let $\mathcal{R}$ be a finite set of rules of radius **r** and (V, E) be the de Bruijn graph of $\mathcal{R}$. The *powerset automaton* of $\mathcal{R}$ is a deterministic finite automaton represented by the 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

- Q, the set of states, is the powerset of V—that is, all subsets of V are states.
- $\Sigma = \{(f, a) |\ a \in S, f \in \mathcal{R}\}$
- $q_0 = V$
- $F = \emptyset$
- δ is the transition function. For any state $X, Y \in Q$, δ defines the transition from X to Y with label (f, a), iff in the de Bruijn graph of $\mathcal{R}$, for all $v \in Y$, there exists $u \in X$ such that $(u, v) \in E$ with label (f, a).

Figure 4.4 shows the powerset automaton of the de Bruijn graph of {90, 204}, shown in Fig. 4.2a . The start node is {00, 01, 10, 11} and the final node is $\emptyset$. Although the powerset automaton can have $2^4 = 16$ nodes, the nodes that are reachable from the start node are shown only in the figure. So, in Fig. 4.4, the graph has only six nodes. Each node has four outgoing edges corresponding to each of the two possible state transitions from each rule used in the de Bruijn graph.

This automaton recognizes the language of all the non-reachable configurations for the particular set of rules. Any sequence of edges from the start node to the final node represents a string that is accepted by this powerset automaton. That is, all such configurations, containing that string, are non-reachable. However, likewise the de Bruijn graph and product graph, this powerset automaton also depicts all possible non-uniform CAs formed with any combination of the rules of the rule-vector. Hence, to test for surjectivity of a particular CA, the edge-sequence with the particular rule sequence is considered. That is, the acceptable strings against the rule sequence are checked. If such a string exists, then the CA is not surjective.

For example, in Fig. 4.4, the sequence of edges $(204, 0) \rightarrow (90, 0) \rightarrow (204, 1)$ represents that the string 001 is accepted by this automaton. So, for a d-v CA $(\mathbb{Z}, \{0, 1\}, 1, \mathcal{R}_i)$ with default rule as 90 and only the cell 0 with rule 204, all

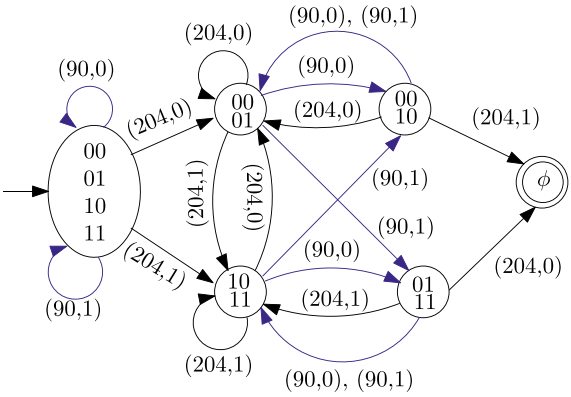


Fig. 4.4 Powerset automaton of {90, 204}

the configurations containing the string 001 are non-reachable. That is, the CA is non-surjective. Some other accepting edge-sequences are $(90, 0)^* \to (204, 0) \to (204, 0)^* \to (90, 0) \to ((204, 0) \to (90, 0))^* \to (204, 1), (90, 0)^* \to (204, 0) \to (204, 0)^* \to ((90, 0) \to (204, 0))^* \to (90, 1) \to (204, 0), (90, 1)^* \to (204, 1) \to (204, 1)^* \to ((90, 0) \to (204, 1))^* \to (90, 1) \to (204, 1), (90, 1)* \to (204, 1) \to (204, 1)* \to (90, 0) \to ((90, 0) \to (90, 0))^* \to (204, 0)$, etc. representing the corresponding non-reachable sub-configurations 0*00*0(00)*1, 0*00*(00)*10, 1*11*(01)*11, 1*11*0(00)*0, etc. respectively.

Following are the steps to decide the surjectivity of a CA from its de Bruijn graph.

1. Build powerset automaton for the de Bruijn graph from the start/initial state (the set of all possible nodes of the de Bruijn graph) to the accepting/final state (the empty set) with all the rules used in the CA.
2. Start from the start node with the left-default rule h_l as the first component of the label.
3. Find a sequence of $2k + 1$ edges with $(\mathcal{R}_i)_{-k \le i \le k}$ as the first component of the edge labels that can be reached from the start node through a path with h_l as the first component of the label.
4. From the terminating node of the above sequence, find a path with h_r as the first component of its labels to the final node $\emptyset$.
5. If such a path exists, the CA is not surjective for the given rule-sequence.

Example 4.2 Consider that the cells -1, 0 and 1 of a CA $(\mathbb{Z}, \{0, 1\}, 1, \mathcal{R}_i)$ are following ECA rules 90, 204 and 90 respectively. The rest of the cells apply rule 204. For this d-v CA, the powerset automaton (shown in Fig. 4.4) contains an accepting edge-sequence $(204, 0) \to (90, 0) \to (204, 0) \to (90, 1) \to (204, 0)$. It shows that all the configurations containing the string 00010 are non-reachable. So, the CA is non-surjective.

Example 4.3 Let consider the non-uniform CA with ECA rules 30 and 204. For this CA, the powerset automaton is shown in Fig. 4.5. This graph has eight reachable nodes

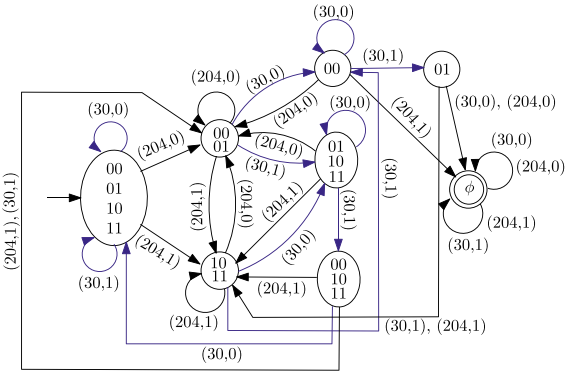


Fig. 4.5 Powerset automaton of {30, 204}

from the start node. It also has accepting edge-sequences $(30, 0)^* \rightarrow (204, 0) \rightarrow (204, 0)^* \rightarrow (30, 0) \rightarrow (30, 0)^* \rightarrow (30, 1) \rightarrow (204, 0)$, $(30, 0)^* \rightarrow (204, 0) \rightarrow (204, 0)^* \rightarrow (30, 0) \rightarrow (30, 0)^* \rightarrow (204, 1)$, etc. So, for a rule sequence (30, 204, 30) with default rule 204, it is non-surjective. The smallest accepting non-reachable sub-configuration for this CA is 00001. It is for the edge-sequence $(204, 0) \rightarrow (30, 0) \rightarrow (204, 0) \rightarrow (30, 0) \rightarrow (204, 1)$.

The above discussion leads to the following lemma.

Lemma 4 *Surjectivity of p-ν CA with period 1 is decidable.*

Theorem 4.1 *Reversibility is decidable for p-ν CA.*

Proof A CA is reversible iff its global transition function G is bijective. By Lemmas 3 and 4 respectively, injectivity and surjectivity of p-ν CA with period 1 is decidable. Hence, reversibility of p-ν CA with period 1 is decidable. Let us now consider a p-ν CA $\mathcal{A}$ with period $p \geq 2$. Then it always possible to find a p-ν CA $\mathcal{A}'$ with period 1, which is dynamically equivalent to $\mathcal{A}$. Injectivity and surjectivity of $\mathcal{A}'$ is decidable by Lemmas 3 and 4. Hence, reversibility of p-ν CA with any period p is decidable. □

4.2 Linear and Complemented Cellular Automata

Let consider that the CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ is finite and linear. This CA can be represented by a *characteristic matrix* (Sect. 3.2). The *Invertible Matrix Theorem* can result in some basic properties of the characteristic matrix. Following theorem, presented without proof, is derived from the invertible matrix theorem.

Theorem 4.2 *Let $G : S^{\mathscr{L}} \rightarrow S^{\mathscr{L}}$ be the global transition function of a finite linear CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$, and T be its characteristic matrix over a field $\mathbb{K}$. The following statements are equivalent.*

1. *T is invertible—that is, T has an inverse.*
2. *Determinant of T is non-zero—that is, T is non-singular.*
3. *The linear transformation G is one-to-one (injective).*
4. *The linear transformation G is surjective.*
5. *$T.x = 0$ has only the trivial solution $x = 0$.*

Therefore, reversibility (bijectivity) of the linear CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ can be decided by testing determinant of T. The computational complexity of finding determinant of an $n \times n$ matrix is $O(n^3)$ in worst case. So, the (time) complexity of deciding reversibility of an n-cell CA is also $O(n^3)$.

Example 4.4 Let consider a 4-cell CA (150, 150, 90, 150) in null boundary. This CA is reversible, because the

$$det[T] = det \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} = 1$$

On the other hand, the 4-cell CA (90, 150, 90, 90) in periodic boundary is irreversible, as the

$$det[T] = det \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = 0$$

Here, all the operations are GF(2) operations.

If a finite linear CA is reversible, then for any configuration x of the CA, there exists a $k \geq 1$ such that $x = T^k.x$. That is, $T^k = I$ (the identity matrix). Here, x is a cyclic configuration and the other elements of the cycle can be found as $T.x, T^2.x, \ldots, T^{k-1}.x$. The set of operators $\{I, T, T^2, \ldots, T^{k-1}\}$ forms a cyclic group. If for any $k' < k$, $x \neq T^{k'}.x$, then k is the order of the group. This order refers to the length of cycle of the CA. For this cyclic group, the following relation is maintained.

$$I + T + T^2 + \cdots + T^{k-1} = 0 \tag{4.1}$$

There are *complemented* CAs that are represented by characteristic matrix T and the *inversion vector* F (Sect. 3.2). The operator for a complemented CA is denoted as $\overline{T}$. That is, for any configuration x of the CA

$$\overline{T}.x = T.x \oplus F$$

$$\overline{T}^2 = T[T.x \oplus F] \oplus F = T^2.x \oplus [I \oplus T].F$$

Therefore, if $\overline{T}^p$ denotes p times application of the operator $\overline{T}$, then

$$\overline{T}^p.x = T^p.x \oplus [I \oplus T \oplus T^2 \oplus \cdots \oplus T^{p-1}].F \tag{4.2}$$

Here $\oplus$ represents modulo 2 addition. For binary CAs, the underlying field is GF(2) and so, all operations are GF(2) operations.

Lemma 5 *If a linear binary CA is reversible, then all of its complements are also reversible.*

Proof Let us consider that T is the characteristic matrix of a linear CA, and $\overline{T}$ is the operator of an arbitrary complement of the linear CA, having inversion vector $F \neq 0$. From Eq. 4.2, for any configuration x of the complemented CA, the following is true

$$\overline{T}^p.x = T^p.x \oplus [I \oplus T \oplus T^2 \oplus \cdots \oplus T^{p-1}].F$$

for some $p \geq 1$. Since the linear CA is reversible, there exists a $k \geq 1$ such that $T^k = I$. To prove reversibility of the complemented CA, it is sufficient to show that there exists an integer q such that

$$I \oplus \overline{T} \oplus \overline{T}^2 \oplus \cdots \oplus \overline{T}^{q-1} = 0 \quad \text{and} \quad \overline{T}^q = I$$

Here,

$$\begin{aligned}
&(I \oplus \overline{T} \oplus \overline{T}^2 \oplus \cdots \oplus \overline{T}^{q-1}).x \\
&= x \oplus \overline{T}.x \oplus \overline{T}^2.x \oplus \cdots \oplus \overline{T}^{q-1}.x \\
&= x \oplus [T.x \oplus F] \oplus [T^2.x \oplus (I \oplus T).F] \oplus \cdots \oplus [T^{q-1}.x \\
&\quad \oplus (I \oplus T \oplus \cdots \oplus T^{q-2}).F] \\
&= [x \oplus T.x \oplus T^2.x \oplus \cdots \oplus T^{q-1}.x] \oplus [I \oplus (I \oplus T) \\
&\quad \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{q-2}].F
\end{aligned}$$

Considering $q = 2k$ and putting $I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1} = 0$,

$$\begin{aligned}
&I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{2k-2}) \\
&= I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-1}) \oplus \\
&\qquad (I \oplus T \oplus \cdots \oplus T^k) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{2k-2}) \\
&= I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-2}) \oplus 0 \oplus \\
&\qquad T^k \oplus (T^k \oplus T^{k+1}) \oplus \cdots \oplus (T^k \oplus T^{k+1} \oplus \cdots \oplus T^{2k-2}) \\
&= I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-2}) \oplus \\
&\qquad [I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-2})].T^k \\
&= I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-2}) \oplus \\
&\qquad [I \oplus (I \oplus T) \oplus \cdots \oplus (I \oplus T \oplus \cdots \oplus T^{k-2})].I \\
&= 0
\end{aligned}$$

Hence,

$$\begin{aligned}
&(I \oplus \overline{T} \oplus \overline{T}^2 \oplus \cdots \oplus \overline{T}^{2k-1}).x \\
&= x \oplus T.x \oplus T^2.x \oplus \cdots \oplus T^{2k-1}.x \\
&= [I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1} \oplus T^k \oplus T^{k+1} \oplus T^{k+2} \oplus \cdots \oplus T^{2k-1}].x \\
&= [(I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1}) \oplus (I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1}).T^k].x \\
&= [0 \oplus 0.I].x \\
&= 0
\end{aligned}$$

This proves that $I \oplus \overline{T} \oplus \overline{T}^2 \oplus \cdots \oplus \overline{T}^{2k-1} = 0$. Following is the next part of the proof.

$$\begin{aligned}
\overline{T}^{2k}.x &= T^{2k}.x \oplus [I \oplus T \oplus T^2 \oplus \cdots \oplus T^{2k-1}].F \\
&= T^{2k}.x \oplus [(I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1}) \\
&\quad \oplus (I \oplus T \oplus T^2 \oplus \cdots \oplus T^{k-1}).T^k].F \\
&= T^{2k}.x \\
&= (T^k)^2.x \\
&= I^2.x \\
&= x
\end{aligned}$$

That is, $\overline{T}^{2k} = I$. Hence the proof. □

Theorem 4.3 *A complemented (binary) CA is reversible iff its linear counterpart is reversible.*

Proof Sufficiency of the theorem—that is, "a complemented CA is reversible if its linear counterpart is reversible" is directly followed from Lemma 5. To prove the necessity, let us consider that a complemented (binary) CA with operator $\overline{T}$ is reversible. Assume that the inversion vector of the CA is F and the characteristic matrix of the corresponding linear CA is T. Hence, for a configuration x of the CA, $\overline{T}.x = T.x \oplus F$. This implies,

$$\overline{T}.x \oplus F = T.x \oplus F \oplus F = T.x$$

Hence,

$$T^p.x = \overline{T}^p.x \oplus [I \oplus \overline{T} \oplus \overline{T}^2 \oplus \cdots \oplus \overline{T}^{p-1}].F$$

for some $p \geq 1$, by interchanging $\overline{T}$ and T in Eq. 4.2. Now, from Lemma 5, if $\overline{T}$ as generator forms a cyclic group, then T as generator also forms a cyclic group. Hence the proof. □

Example 4.5 The 4-cell CA (150, 150, 90, 150) of Example 4.4 is reversible. A complemented CA (150, 150, 165, 150) in null boundary is also reversible. The rule 165 of the complemented CA is the complement of rule 90.

Corollary 4.1 *Let an n-cell uniform CA with linear ECA rule R be reversible and* $\overline{R}$ *be the complement rule of R. Then, an n-cell CA with rule vector as a permutation of R and* $\overline{R}$ *is reversible.*

Proof A CA with any permutation of R and $\overline{R}$ as rule vector is a complemented CA of the n-cell linear CA with ECA rule R. If the linear CA is reversible, then by Theorem 4.3, the complemented CA is also reversible. □

4.3 Reversibility and Reachability Tree

This section explores reachability tree to study the reversibility of non-uniform finite ECAs—that is, reversibility of an n-cell CA $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ where each $\mathcal{R}_i$, $i \in \{0, 1, \ldots, n-1\}$, is an ECA rule. Unlike matrix algebra, reachability tree can deal with any finite CA—linear, complemented or non-linear. The reachability tree represents reachable configurations of a CA. In case of reversible (finite) CA, all the configurations are cyclic (Definition 27). This implies that each configuration of the CA has exactly one predecessor.

Theorem 4.4 *The reachability tree of a reversible CA is complete.*

Proof Since all the configurations of a reversible CA are reachable, the number of leaves in the reachability tree of an n-cell reversible (binary) CA is 2^n (total number of configurations). Therefore, the tree is complete as it is a binary tree of $(n+1)$ levels. □

Example 4.6 Let consider a 4-cell CA (90, 15, 85, 15) with null boundary (Table 4.1). The reachability tree of the CA, shown in Fig. 4.6, is a complete binary tree, and the CA is reversible. Transition diagram of the CA (Fig. 4.1) verifies the fact. On the other hand, the CA (202, 195, 105, 165) with periodic boundary is also reversible. Figure 4.7 is the reachability tree of the CA.

Table 4.1 RMT values of the 4-cell CA (90, 15, 85, 15)

RMT	111	110	101	100	011	010	001	000	Rule
	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
Cell 0	0	1	0	1	1	0	1	0	90
Cell 1	0	0	0	0	1	1	1	1	15
Cell 2	0	1	0	1	0	1	0	1	85
Cell 3	0	0	0	0	1	1	1	1	15

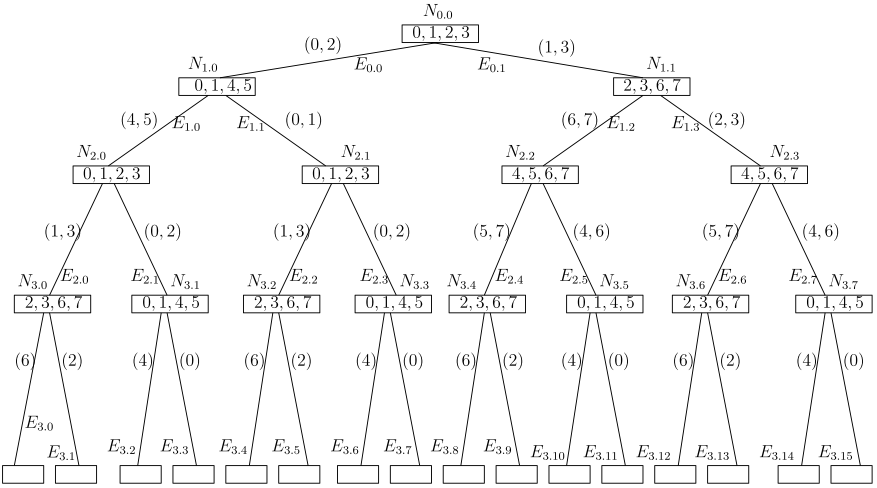


Fig. 4.6 Reachability tree of CA (90, 15, 85, 15) in null boundary

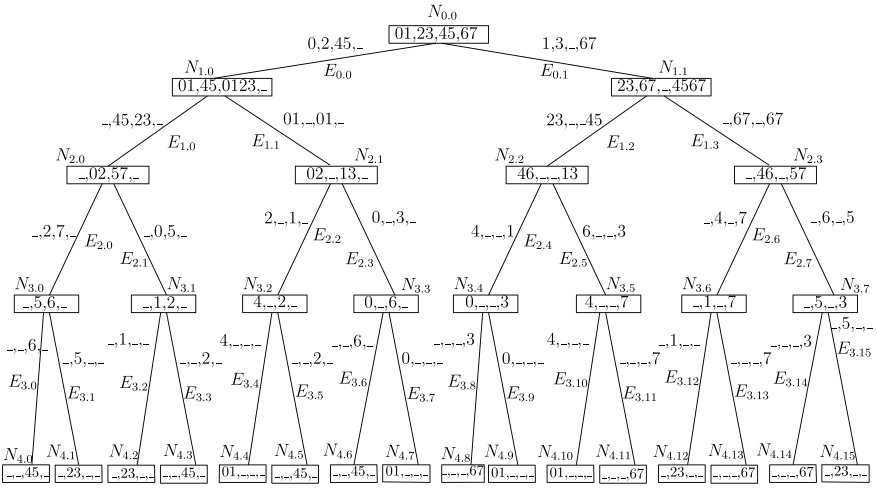


Fig. 4.7 Reachability tree of the CA (202, 195, 105, 165) in periodic boundary

To be a complete tree, the internal nodes and edge labels of a reachability tree need to satisfy some conditions. These conditions vary with boundary conditions. The following theorem is applicable for the CAs in null boundary.

Theorem 4.5 *If the reachability tree of an n-cell CA* $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ *in null boundary is complete, then for each edge* $E_{i.j}$ *of level* i, *where* $j \in \{0, 1, \cdots, 2^{i+1} - 1\}$, $|l_{i.j}|$ *is*

(i) 1 *if* $i = n - 1$, *and*
(ii) 2 *if* $0 \leq i \leq n - 2$.

Proof To prove this theorem by contradiction, let us consider that the reachability tree of the CA is complete, but in the label of an edge $E_{i.j}$, the number of RMTs is less than that is mentioned in (i) and (ii). This means:

(i) When $i = n - 1$, the edge $E_{n-1.j}$ is non-reachable as there is no RMT in $l_{n-1.j}$. Hence, the tree is incomplete; a contradiction.
(ii) When $i < n - 1$, the node $N_{i+1.j}$ consists of two RMTs only (point 3 of Definition 18). Now, two cases may arise: (a) Based on the rule $\mathcal{R}_{i+1}$, both the RMTs are on a single label, either on $l_{i+1.2j}$ or on $l_{i+1,2j+1}$. This leads to the contradiction that the tree is incomplete. (b) In the other case, the RMTs are equally distributed to $l_{i+1.2j}$ and $l_{i+1,2j+1}$. That is, both the labels contain single RMT. Hence, each of the nodes $N_{i+2.4j}$ and $N_{i+2,4j+3}$ contains two RMTs. In the best case, these two nodes may have two children each, and this scenario may continue upto level $n - 1$. But in this level, the corresponding nodes consist of single RMT, which implies that the nodes have a single child. This makes the tree incomplete—this is also a contradiction. Hence, the number of RMTs cannot be less than the number mentioned in (i) and (ii).

Let us now consider that the number of RMTs in $l_{i.j}$ is greater than that is mentioned in (i) and (ii). If $i = 0$ and any one of $l_{0.0}$ and $l_{0.1}$ consists of more than two RMTs, then the other has less than two RMTs, as the root consists of four RMTs. Since one label has more than two RMTs, the tree is incomplete by the above case. In general, if an $l_{i.j}$ for any $i \in \{0, 1, \ldots, n - 1\}$ consists of RMTs more than that is mentioned in (i) and (ii), then there exist some edges with label $l_{i.j1}$ that has less in number of RMTs than that is mentioned in (i) and (ii). Hence, the tree is to be incomplete and contradicts the assumption. Therefore, when the reachability tree is complete, the cardinality of $l_{i.j}$ is 1 if $i = n - 1$, and 2 if $i < n - 1$. □

Example 4.7 Consider the 4-cell CA (90, 15, 85, 15) of Example 4.6. The CA is reversible. The labels $l_{3.j}$ consist of single RMT, and rest of the labels consist of two RMTs (Fig. 4.6). The RMTs of labels are noted within the bracket.

Corollary 4.2 *In the reachability of an n-cell CA* $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ *in null boundary, the cardinality of a node* $N_{i.j}$ *is*

(i) 2 *if* $i = n - 1$, *and*
(ii) 4 *if* $i < n - 1$.

Proof If an n-cell CA $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ in null boundary is reversible, the reachability tree is complete (Theorem 4.4). In this tree, two edges with labels $l_{i.2j}$ and $l_{i.2j+1}$ are there against a node $N_{i.j}$. If $i = n-1$, then for null boundary, both the labels are singleton sets (Theorem 4.5). Hence, cardinality of $N_{n-1.j}$ is 2. On the other hand, if $i < n-1$, then $|l_{i.2j}| = |l_{i.2j+1}| = 2$—that is, the cardinality of $N_{i.j}$ is 4. □

Theorem 4.6 is the counter part of Theorem 4.5 for reachability tree of reversible CA with periodic boundary.

Theorem 4.6 *If the reachability tree of an n-cell CA* $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ *in periodic boundary is complete, then for each edge* $E_{i.j}$ *of level* i *where* $j \in \{0, 1, \cdots, 2^{i+1} - 1\}$, $|\bigcup_{k=0}^{3} \Gamma_k^{E_{i.j}}|$ *is*

(i) *1 if* $i = n-1$,
(ii) *2 if* $i = n-2$, *and*
(iii) *4 if* $0 \leq i \leq n-3$.

Proof The proof is similar to that of Theorem 4.5. Let us consider that the reachability tree of an n-cell CA $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ in periodic boundary, is complete but the number of RMTs in the label of an edge $E_{i.j}$ is less than that is mentioned in (i) to (iii). That is:

(1) If $i = n-1$, then $\bigcup_{k=0}^{3} \Gamma_k^{E_{n-1.j}} = \emptyset$. Hence, the edge $E_{n-1.j}$ is non-reachable. This implies that the tree is incomplete—a contradiction.
(2) If $i = n-2$, then $\bigcup_{k=0}^{3} \Gamma_k^{E_{n-2.j}}$ is a singleton set. Then, within the node $N_{n-1.j}$, there is a single RMT (point 4 of Definition 22), which implies that the node can have a single child. Hence, the tree is incomplete—this is also a contradiction.
(3) If $i \leq n-3$, then $|\bigcup_{k=0}^{3} \Gamma_k^{E_{n-1.j}}| = 3$—that is, the edge is resulted from 3 RMTs. Therefore, there can be at most 6 RMTs within the node $N_{i+1,j}$ (if a pair of equivalent RMTs stay together in a label, then only a single pair of sibling RMTs is contributed). However, the node may have two children, connected by edges $E_{i+1.2j}$ and $E_{i+1.2j+1}$ which may have any one of the following shares of these six RMTs:
1 and 5, 2 and 4, or 3 and 3 RMTs.

In best case, the tree may remain complete upto level $n-2$. In level $n-2$, there can be at least one node that comes from the edge which is resulted from 1 or 2 or 3 RMTs. Therefore, the node is constructed with 1, 2 or 3 RMTs (point 4 of Definition 22). If the number of RMT is one, then the tree is incomplete. Otherwise, at least one edge of that node is resulted from the single RMT, which results in an incomplete tree [by (2)]. Hence a contradiction.

Let us now consider that the number of RMTs in the label of an edge $E_{i.j}$ is greater than that is mentioned in (i) to (iii). Since the tree is complete, there can be an edge $E_{i.j1}$ with number of RMTs in the label as less than that is mentioned in (i) to (iii). This leads to the contradiction by above argument. Hence, when the tree is complete, then $|\bigcup_{k=0}^{3} \Gamma_k^{E_{i.j}}|$ is

(i) 1 if $i = n - 1$,
(ii) 2 if $i = n - 2$, and
(iii) 4 if $0 \leq i \leq n - 3$. □

Corollary 4.3 *In the reachability of an n-cell CA* $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ *in periodic boundary, for a node* $N_{i.j}$, $|\bigcup_{k=0}^{3} \Gamma_k^{N_{i.j}}|$ *is*

(i) 2 if $i = n$,
(ii) 4 if $i = n - 1$, *and*
(iii) 8 otherwise.

The proof is omitted as it is followed from Theorems 4.4 and 4.6, and the argument is similar to Corollary 4.2.

Definition 30 A node N_{ij} is called *balanced* if it has equal number of RMTs with next states 0 and 1 for the rule $\mathcal{R}_i$.

For example, the root of Fig. 4.6 is balanced. It follows rule 90. Here, the RMTs 0 and 2 are 0 and the RMTs 1 and 3 are 1.

Corollary 4.4 *The nodes of reachability tree of a reversible CA are balanced.*

This corollary directly follows from Corollaries 4.2 and 4.3. All the nodes of Figs. 4.6 and 4.7 are balanced.

4.4 Analysis and Synthesis of Reversible CA

In this section, two fundamental algorithms for reversible CAs are presented—one is the decision algorithm and the other is a synthesis algorithm. Both the algorithms are developed around reachability tree. The decision algorithm tests whether a given CA is reversible. On the other hand, the synthesis algorithm synthesizes an n-cell reversible CA for a given n (the number of cells).

4.4.1 Testing Reversibility of a CA

A finite CA is reversible if all of its configurations are cyclic. When all the configurations are cyclic, then there is no non-reachable configuration. Conversely, if there exists no non-reachable configuration, then the finite CA is reversible. An efficient algorithm to count non-reachable configurations of a finite CA can easily be developed by exploiting the reachability tree (Algorithm 1).

An improved decision algorithm is further reported in this section. The algorithm virtually constructs a reachability tree to make a decision. Although the structure of

reachability tree varies with the boundary condition, the theories related to reversibility of CAs under both the boundary conditions are very similar. So, a single algorithm is presented for both the boundary conditions.

Reachability tree, in general, grows exponentially. There may be 2^i nodes at level i of the reachability tree for an n-cell CA, $i \leq n$. However, all the nodes are not unique. If $N_{i.j} = N_{i.k}$ for $j \neq k$, then two subtrees rooted at $N_{i.j}$ and $N_{i.k}$ are identical. In Fig. 4.6, $N_{2.0} = N_{2.1}$ (and $N_{2.2} = N_{2.3}$). So, two subtrees are identical. If the unique nodes are considered only, then the tree of Fig. 4.6 is transformed to Fig. 4.8. To decide the reversibility, it is sufficient to deal only with the unique nodes. Interestingly, the number of unique nodes at a level of reachability tree of a reversible CA is very limited. Following lemma is related to the null boundary condition.

Lemma 6 *At each level, except root, the number of unique nodes of the reachability tree of a reversible CA is either 2 or 4 when the CA is with null boundary.*

Proof Each node of the reachability tree for a reversible CA with null boundary consists of 4 RMTs (Corollary 4.2) and the sibling RMTs are associated with the same node. Since there are 4 sets of sibling RMTs (0 & 1, 2 & 3, 4 & 5, and 6 & 7), 3 different organizations of RMTs for the nodes are possible—{0, 1, 2, 3} & {4, 5, 6, 7}, {0, 1, 4, 5} & {2, 3, 6, 7} and {0, 1, 6, 7} & {2, 3, 4, 5}. The root of the tree is {0, 1, 2, 3}. Depending on the rule ($\mathcal{R}_0$) of first cell, any one of the three organisations can be the possible nodes of level 1. So, the number of nodes is 2. For level 2, at most four unique nodes can appear, which implies that two of the above organisations can be the nodes at that level. Due to the reversibility condition, given in Theorem 4.5, at most two organisation can appear at level 3. So, at most 4 unique nodes are possible. This scenario is true for all other levels. □

Example 4.8 Consider the reversible CA of Example 4.6. Figure 4.8 shows the unique nodes of the reachability tree for the CA. Each level, except the root, contains 2 unique nodes.

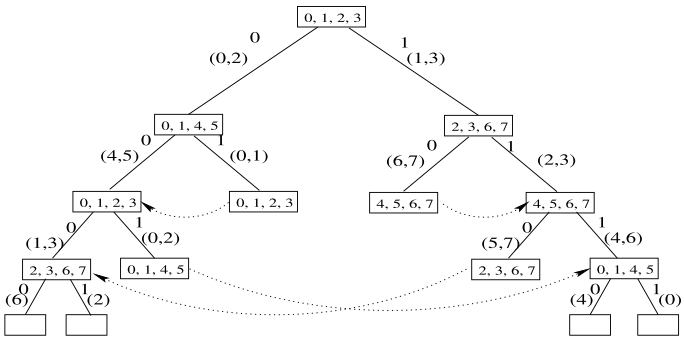


Fig. 4.8 Reachability tree of CA (90, 15, 85, 15) in null boundary without identical subtrees

In case of periodic boundary condition, the maximum number of unique nodes at an arbitrary level is marginally higher, but it is still bounded by a small integer.

Algorithm 2 describes the steps of the decision algorithm. To check the reversibility of a CA, it does not develop the whole reachability tree at-a-time; rather it finds only the possible edges against a level and then tests whether the conditions of Theorem 4.5/Theorem 4.6 are satisfied. Therefore, it needs to store the information about the labels of two consecutive levels. The $l_0, l_1, \cdots$ and $l'_0, l'_1, \cdots$ are the labels of edges of the two consecutive levels. For null boundary, l_j (and l'_k) is a set of RMTs whereas, for periodic boundary, it is a collection of four sets of RMTs. A variable, $NoUL$ is used to store the number of unique labels at a level, which is obviously related to the number of unique nodes. The labels of edges of level $i+1$ are found from the level i. The $l_0, l_1, \cdots$ and $l'_0, l'_1, \cdots$ correspond to the labels of edges of level i and level $i+1$ respectively. If the algorithm is tuned for null boundary, the condition of Theorem 4.5 is applied. Theorem 4.6 is for the periodic boundary. If any label violets the condition, the CA is irreversible.

Algorithm 2: Testing reversibility

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$.
Output: "Yes" if the CA is reversible; "No" otherwise.

Step 1: Find the labels l_0 and l_1 of the edges of level 0 using $\mathcal{R}_0$.
(i) If the labels do not satisfy the condition of Theorem 4.5/Theorem 4.6, report "No" and exit.
(ii) Set $NoUL \leftarrow 2$.
Step 2: For $i = 1$ to $n-1$, repeat Step 3 to Step 5
Step 3: For each l_j, $j \in \{0, 1, \cdots, NoUL - 1\}$
(i) Find the labels l'_{2j} and l'_{2j+1} of edges of level i using $\mathcal{R}_i$.
(ii) If l'_{2j} and l'_{2j+1} do not satisfy the conditions of Theorem 4.5/Theorem 4.6, report "No" and exit.
(iii) Replace RMTs 4, 5, 6, and 7 of l'_{2j} and l'_{2j+1}, if any, by RMTs 0, 1, 2 and 3 respectively.
Step 4: Remove the duplicates from the labels $l'_0, l'_1, \cdots, l'_{2NoUL-1}$, and consider the unique labels as $l_0, l_1, \cdots$.
Step 5: Set NoUL as the number of unique labels.
Step 6: Report "Yes" and exit.

Example 4.9 This example illustrates the execution steps of Algorithm 2. Let consider the CA (90, 15, 85, 15) in null boundary (Fig. 4.6) as input to Algorithm 2. From Step 1, we obtain $l_0 = \{0, 2\}, l_1 = \{1, 3\}$ and $NoUL = 2$. Here, $|l_0| = |l_1| = 2$—that is, the condition of Theorem 4.5 is satisfied.

Now from l_0, a node with RMTs 0, 1, 4 and 5 is formed. These RMTs are distributed in $l'_0 = \{4, 5\}$ (for edge $E_{1.0}$) and $l'_1 = \{0, 1\}$ (for edge $E_{1.1}$) following the rule 15 (Step 3(i)). Similarly, $l'_2 = \{6, 7\}$ and $l'_3 = \{2, 3\}$ (Fig. 4.8). All the new labels follow the condition of Theorem 4.5. Now, according to Step 3(iii), the new labels are

modified as $l'_0 = \{0, 1\}$, $l'_1 = \{0, 1\}$, $l'_2 = \{2, 3\}$ and $l'_3 = \{2, 3\}$. Here, $l'_0 = l'_1$ and $l'_2 = l'_3$. So, the algorithm proceeds with two unique labels and considers $l_0 = \{0, 1\}$ and $l_1 = \{2, 3\}$ (Step 4). Further, $NoUL = 2$ (Step 5).

Similarly, the labels of edges of the next level ($i = 2$) $l'_0 = \{1, 3\}$, $l'_1 = \{0, 2\}$, $l'_2 = \{5, 7\}$ and $l'_3 = \{4, 6\}$ are found. The unique labels after Step 4 are $l_0 = \{1, 3\}$ and $l_1 = \{0, 2\}$.

Finally, the labels of edges of level $n - 1$, where $n = 4$ can be generated. So, the labels are $l'_0 = \{2\}, l'_1 = \{6\}, l'_2 = \{0\}$ and $l'_3 = \{4\}$ (odd RMTs are excluded; see Definition 18). Each label contains single RMT—that is, the condition of Theorem 4.5 is satisfied. In the next step, the algorithm outputs "Yes". That is, the CA is reversible.

Example 4.10 Consider the 4-cell CA (202, 195, 105, 165) in periodic boundary (Fig. 4.7) for testing the reversibility. For periodic boundary, a label is a quadruple. Let assume that a label l_i is $(s_i^0, s_i^1, s_i^2, s_i^3)$. For rule 202 ($\mathcal{R}_0$), $s_0^0 = \{0\}, s_0^1 = \{2\}, s_0^2 = \{4, 5\}$ and $s_0^3 = \{\}$, and $s_1^0 = \{1\}, s_1^1 = \{3\}, s_1^2 = \{\}$ and $s_1^3 = \{6, 7\}$ (Table 4.2). Hence, $|s_0^0| + |s_0^1| + |s_0^2| + |s_0^3| = 4 = |s_1^0| + |s_1^1| + |s_1^2| + |s_1^3|$. The labels of edges at level 1 are generated from l_0 and l_1 in Step 3(i). The nodes of level 1 can be found from l_0 and l_1, which are $(\{0, 1\}, \{4, 5\}, \{0, 1, 2, 3\}, \{\})$ and $(\{2, 3\}, \{6, 7\}, \{\}, \{4, 5, 6, 7\})$ (level 1 of Fig. 4.7). Algorithm 2 does not explicitly form the nodes. It directly generates the edge labels by distributing the RMTs of sets of l_i. From l_0, the $l'_0 = (\{\}, \{4, 5\}, \{2, 3\}, \{\})$ (0-edge) and $l'_1 = (\{0, 1\}, \{\}, \{0, 1\}, \{\})$ (1-edge) can be derived following the rule 195. Similarly, from l_1, $l'_2 = (\{2, 3\}, \{\}, \{\}, \{4, 5\})$ and $l'_3 = (\{\}, \{6, 7\}, \{\}, \{6, 7\})$ are found. These four labels satisfy the conditions of Theorem 4.6 (Step 3(*ii*)). Now, following Step 3(*iii*), the $l'_0 = (\{\}, \{0, 1\}, \{2, 3\}, \{\})$, $l'_1 = (\{0, 1\}, \{\}, \{0, 1\}, \{\})$, $l'_2 = (\{2, 3\}, \{\}, \{\}, \{0, 1\})$ and $l'_3 = (\{\}, \{2, 3\}, \{\}, \{2, 3\})$ can be identified. All these labels are unique (Step 4) and so $NoUL$ is 4.

For this example, level 2 is the $(n - 2)$th level. So, nodes of this level are formed with four RMTs (Point 4 of Definition 22). By Step 3(i), we get $l'_0 = (\{\}, \{2\}, \{7\}, \{\})$ and $l'_1 = (\{\}, \{0\}, \{5\}, \{\})$ using $\mathcal{R}_2 = 105$. Similarly, the labels $l'_2, l'_3, \ldots, l'_7$ can be found. There is also no violation of the condition of Theorem 4.6.

Similarly, the labels of $i = 3$ are generated following the rule $\mathcal{R}_2 = 165$. Since they do not violate the condition of Theorem 4.6, the CA is reversible.

Complexity: *Step 2* is the main loop in Algorithm 2. It contains inner loops (*Steps 3* and *4*) with expected *NoUE* number of iterations. The maximum value of *NoUE*

Table 4.2 RMT values of the 4-cell CA (202, 195, 105, 165)

RMT	111 (7)	110 (6)	101 (5)	100 (4)	011 (3)	010 (2)	001 (1)	000 (0)	Rule
Cell 0	1	1	0	0	1	0	1	0	202
Cell 1	1	1	0	0	0	0	1	1	195
Cell 2	0	1	1	0	1	0	0	1	105
Cell 3	1	0	1	0	0	1	0	1	165

is limited by the possible number of unique nodes at a level, hence it is a constant. Therefore, the time complexity of the algorithm depends only on n, and hence it is $O(n)$.

Algorithm 2 deals with the edges of reachability tree. An alternative algorithm of deciding reversibility of a CA can be designed, where nodes instead of edges of the tree are considered and Corollary 4.4 is applied.

4.4.2 Synthesis of Reversible CA

A synthesis algorithm takes n as input, and the output is an arbitrarily n-cell reversible CA. To synthesize the CA in null boundary, the rules of cells are selected in such a way that the conditions of Theorem 4.5 are not violated. For that, the RMTs of the rule $\mathcal{R}_i$ are appropriately set. The steps are noted in Algorithm 3. Like Algorithms 2 and 3 also considers two sets of labels of edges for two consecutive levels—$l_0, l_1, \cdots$ and $l'_0, l'_1, \cdots$. The array N is used to store the unique nodes of a level.

Algorithm 3: Synthesize reversible CA in null boundary

Input : n.
Output: An n-cell reversible CA in null boundary.

Step 1: Distribute two 0s and two 1s arbitrarily as the next states of RMTs 0, 1, 2 and 3 of $\mathcal{R}_0$.
(i) Consider, $l_0 = \{r \mid 0 \leq r \leq 3, \mathcal{R}_0[r] = 0\}$ and $l_1 = \{s \mid 0 \leq s \leq 3, \mathcal{R}_0[s] = 1\}$.
(ii) Set $NoUL \leftarrow 2$
Step 2: For $i = 1$ to $n - 2$, repeat Step 3 to Step 6
Step 3: For $j = 0$ to $NoUL - 1$
Get $N_j = \{2r, 2r + 1 \mid r \in l_j\}$.
Step 4: Distribute four 0s and four 1s randomly as the next state of the RMTs of $\mathcal{R}_i$, such that
(i) Each N_j is balanced (that is, the number of RMTs with next states 0 and 1 are equal).
(ii) If r and s are equivalent RMTs, and $r, s \in N_j$, then $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$.
Step 5: For each N_j, $j \in \{0, 1, \cdots, NoUL - 1\}$,
(i) Find the labels $l'_{2j} = \{r \mid \mathcal{R}_i[r] = 0\}$ and $l'_{2j+1} = \{s \mid \mathcal{R}_i[s] = 1\}$ of edges of next level.
(ii) Replace RMTs 4, 5, 6 and 7, if exist in l'_{2j} and l'_{2j+1}, by their equivalent RMTs – that is, RMTs 0, 1, 2 and 3 respectively.
Step 6: Remove duplicates from the labels $l'_0, l'_1, \cdots, l'_{2NoUL-1}$ and consider the unique labels as $l_0, l_1, \cdots$;
Set $NoUL$ as the number of unique labels.
Step 7: For $j = 0$ to $NoUL - 1$
Get the node $N_j = \{2r \mid r \in l_j\}$ of level $n - 1$.
Step 8: Distribute two 0s and two 1s randomly as the next state of the even RMTs of $\mathcal{R}_{n-1}$, such that the RMTs of each N_j have different next states.
Step 9: Assign 0 to the unfilled positions of $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$.
Report $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ as an n-cell reversible CA.

Example 4.11 This example illustrates the synthesis of a 4-cell reversible CA following Algorithm 3. Let consider, two 0s and two 1s are distributed arbitrarily in the least significant 4 RMTs of the first rule (Step 1 of Algorithm 3). Say, the RMTs 0 and 2 are set to 0—that is, $l_0 = \{0, 2\}$, and the RMTs 1 and 3 are 1—that is, $l_1 = \{1, 3\}$. Therefore, the number of nodes at level 1 is 2; one is generated from the edge with label l_0 and the other from the edge labelled l_1. Here, the first rule ($\mathcal{R}_0$) is 10, considering the next states of the RMTs 4, 5, 6 and 7 as 0. And, the nodes (of level 1) are $N_0 = \{0, 1, 4, 5\}$ and $N_1 = \{2, 3, 6, 7\}$ (Step 3). Four 0s and four 1s are randomly distributed in the RMTs of $\mathcal{R}_1$ such that the equivalent RMTs (RMT 0 & 4, 1 & 5, etc.) cannot be the same (Step 4). Suppose, the RMTs 0, 1, 2 and 3 are 1, and the RMTs 4, 5, 6 and 7 are 0. This sets $\mathcal{R}_1$ as 15. Hence, $l'_0 = \{4, 5\}$ & $l'_1 = \{0, 1\}$ and $l'_2 = \{6, 7\}$ & $l'_3 = \{2, 3\}$ (Step 5(i)). Therefore, the number of labels is 4 and each label contributes a node for the next level. Since the RMTs 0 & 4, 1 & 5, 2 & 6, and 3 & 7 are equivalent, the RMT 4 is replaced by 0, 5 by 1, 6 by 2 and 7 by 3 (Step 5(ii)). This makes l'_0 & l'_3 and l'_1 & l'_2 identical and the number of unique labels is reduced to $NoUL = 2$ ($\{0, 1\}$ and $\{2, 3\}$) (Step 6).

Similarly, the nodes for the level 2 are $N_0 = \{0, 1, 2, 3\}$ and $N_1 = \{4, 5, 6, 7\}$. Consider that the RMTs 1, 3, 5 and 7 of the rule are 0, and the RMTs 0, 2, 4 and 6 are 1. That is, $l'_0 = \{1, 3\}$ and $l'_1 = \{0, 2\}$ and $l'_2 = \{5, 7\}$ and $l'_3 = \{4, 6\}$. Hence, the next rule ($\mathcal{R}_2$) is 85. Here, the number of unique labels is $NoUL = 2$ ($\{1, 3\}$ and $\{0, 2\}$).

The unique nodes for the next level (level 3)—that is, the predecessors of the leaves, are $\{2, 3, 6, 7\}$ and $\{0, 1, 4, 5\}$. However, the RMTs 1, 3, 5 and 7 of the last rule are *don't cares*, as it is the right most cell of a null boundary CA. Therefore, the *effective* RMTs of the nodes are $\{2, 6\}$ and $\{0, 4\}$. Now, the RMTs of each set are randomly taken as 1 and 0 (Step 8). Say, the RMT 4 and 6 are 0, and the RMT 0 and 2 are 1 (Step 6). If 0 is chosen for the *don't care* RMTs, then the last rule ($\mathcal{R}_3$) is 5 (Step 9).

Therefore, the synthesized 4-cell CA is (9, 15, 85, 5). However, the CA is equivalent to (90, 15, 85, 15) in null boundary (as that of Example 4.9). The reachability tree of the CA and the reachability tree with unique nodes only are shown in Figs. 4.6 and 4.8 respectively.

Correctness of the algorithm can be understood from the previous discussions. The algorithm proceeds with only unique labels (and nodes) of the tree. The number of unique nodes at each level (except root) of such a tree is either 2 or 4. The rule is chosen in such a way that these nodes always remain balanced. In addition to that, the equivalent RMTs, if exist in a node, are not set to the same next state. These conditions ensure that each node has two children and the nodes of the next level are balanced. Finally, a complete reachability tree is constructed inherently during the run of the algorithm. Hence, the synthesized CA is reversible. It can also be observed that the synthesized CA is not a trivial CA. Any reversible CA with null boundary can be synthesized with Algorithm 3.

Complexity: Step 2 of Algorithm 3 contains a loop that depends on n (the number of CA cells). However, the inner loops (in Steps 3, 4, 5 and 6) and the loops in Steps 7 and 8 iterate based on $NoUL$ (the number of unique labels of edges). Since the maximum value of $NoUL$ is constant (4), the algorithm depends only on n. Hence, the complexity of Algorithm 3 is $O(n)$.

A similar algorithm can be developed for the synthesis of a reversible CA in periodic boundary. In that case, the label of an edge is to be considered as a quadruple of sets of the RMTs. Individual rules are to be synthesized in such a fashion that the conditions of Theorem 4.6 are satisfied in each step.

4.4.3 Rules for Reversible CAs

While designing the rule vector of a reversible CA, an arbitrary rule (from the set of 256 rules) cannot be chosen. Only a few rules can form the rule vectors of reversible CAs. Based on the role in forming a reversible CA, the rules are classified as *reversible rules* and *irreversible rules*.

Definition 31 A rule is *reversible rule* in null (resp. periodic) boundary, if there exists a reversible CA in null (resp. periodic) boundary which uses the rule in its rule vector. Otherwise, it is an *irreversible rule*.

Example 4.12 The 4-cell CA $(202, 195, 105, 165)$ with periodic boundary is reversible. Therefore, all the four rules are reversible rules in periodic boundary. On the other hand, the CA $(202, 196, 105, 165)$ is an irreversible CA. It is due to the rule 196. No reversible CA can be designed by incorporating rule 196 into the rule vector. So, 196 is an irreversible rule.

Reversible rules are boundary dependent. A reversible rule in periodic boundary may not be reversible in null boundary. For example, rule 226 is a reversible rule in periodic boundary but is an irreversible rule in null boundary. Indeed, there are 8 reversible rules in periodic boundary which are irreversible rules in null boundary. However, following theorem holds for both the boundary conditions.

Theorem 4.7 *An unbalanced rule is irreversible.*

Proof Let consider that $\mathcal{R}_i$, the rule of cell i of an n-cell CA, is unbalanced. That is, the domain of the rule consists of $k > 4$ RMTs that are having the same next state $d_i \in \{0, 1\}$. Now, there are $k * 2^{n-3}$ current configurations for which the next configurations are in $S = \{\cdots d_i \cdots\}$. The maximum possible such next configurations is clearly 2^{n-1}. Since $k * 2^{n-3} > 2^{n-1}$ $(k > 4)$, the number of next configurations is lesser than that of the current configurations. It implies that there is at least a configuration in S which contains more than one predecessor. Therefore, the CA with unbalanced rule is an irreversible CA. □

If the reversible rules are identified (in any boundary condition), then it is to be examined that whether any sequence of reversible rules form a reversible CA or not.

Theorem 4.8 *Only specific sequences of reversible rules form reversible CA.*

Proof The theorem is proved by showing that a sequence of reversible rules form a reversible CA but the other sequence with the same set of rules does not form reversible CA.

Let first consider an n-cell ($n > 3$) CA, in null boundary, with rule vector $(90, 15, 85, \mathcal{R}_3, \mathcal{R}_4, \cdots, \mathcal{R}_{n-1})$. The CA can be a reversible CA if $\mathcal{R}_3, \mathcal{R}_4, \cdots, \mathcal{R}_{n-1}$ are chosen properly. However, the CA $(90, 85, 15, \mathcal{R}_3, \mathcal{R}_4, \cdots, \mathcal{R}_{n-1})$ can never be reversible. Algorithm 2 can show that the condition of reversibility is violated by the edges of level 1.

Similar argument can be put for the CAs in periodic boundary. □

Example 4.13 The CA (90, 15, 85, 15) in null boundary is reversible (Example 4.9). However, the CA (90, 85, 15, 15) is irreversible under the same boundary condition.

4.5 Reversible Rules in Null Boundary

The reversible rules are the basic building blocks of a reversible CA. Characterization of reversible rules simplifies the analysis and synthesis of reversible CA. This section reports such characterization of the rules.

There are ${}^8C_4 = 70$ balanced ECA rules. However, all of them are not the reversible rules in null boundary. Only 62 are the reversible rules and the rest 8 are balanced irreversible rules. The following theorem characterizes the balanced irreversible rules.

Theorem 4.9 *A balanced rule with the same next state for the RMTs 0, 2, 3, 4, or RMTs 0, 4, 6, 7, or RMTs 0, 1, 2, 6, or RMTs 0, 1, 3, 7 is an irreversible rule in null boundary.*

Proof Let consider that the RMTs of a balanced rule R are clustered as $g_1 = \{0, 2, 3, 4\}$ and $g_2 = \{1, 5, 6, 7\}$ depending on their next states. That is, the next state of each RMT of g_1 is $d \in \{0, 1\}$ and for g_2 it is d'. Now the following four cases may arise—

Case I: R is the first rule of a rule vector—Since the RMTs 4, 5, 6 and 7 are *don't cares* for the first rule in a null boundary CA, the clustering of RMTs effectively becomes $g_1 = \{0, 2, 3\}$ and $g_2 = \{1\}$. Hence, the 0-edge (1-edge) of first level of the reachability tree is resulted either from 3 or 1 (1 or 3) RMTs of R, which violates the condition of Theorem 4.5. Hence, the CA with rule R is irreversible—that is, such R is an irreversible rule.

Case II: R is the second rule—Consider that the first rule is balanced over its least significant 4 RMTs 0, 1, 2 and 3. Therefore, the possible clustering of the RMTs to form the 0-edge and 1-edge from the root can be:

$f_1 = [\{0, 1\}\&\{2, 3\}]$,
$f_2 = [\{0, 2\}\&\{1, 3\}]$, and
$f_3 = [\{0, 3\}\&\{1, 2\}]$.

That is, for f_1, if the RMTs 0 and 1 are 1, then the RMTs 2 and 3 are 0. So, for f_1, the RMTs of level 1 nodes of the reachability tree are {0, 1, 2, 3} & {4, 5, 6, 7}. Similarly, for f_2, the nodes are {0, 1, 4, 5} & {2, 3, 6, 7}, and {0, 1, 6, 7} & {2, 3, 4, 5} are the two possible nodes for f_3. However, if the RMTs of first rule are clustered like f_1 or f_3, labels of edges, connecting the nodes of level 1 to their children are resulted from one or three RMTs. This implies, the reachability tree is not complete and the CA is irreversible.

If the RMTs of first rule are clustered as f_2, the RMTs constituting the nodes at level 1 are {0, 4} & {1, 5} and {2, 3} & {6, 7}. Therefore, the edges of reachability tree are resulted from the RMTs 0 & 4, 1 & 5, 2 & 3, and 6 & 7. Since the RMTs 0 & 4 (similarly 1 & 5) are equivalent, two nodes at level 2 are formed with 2 RMTs. It violets Corollary 4.2. Hence the CA is irreversible.

Case III: R is the ith rule—Let the reachability tree of the CA is complete up to level $(i - 1)$. Since the RMTs 0 & 4, 1 & 5, 2 & 6, and 3 & 7 are equivalent, without loss of generality, it can be considered that the level i nodes are generated from the RMT 0, 1, 2, and 3. But any combination of these RMTs leads to an incomplete tree (Case *II*).

Case IV: R is the last rule—Since the RMTs 1, 3, 5 and 7 are $don't\ care$s for the last rule, the clustering of RMTs of R effectively becomes $g_1 = \{0, 2, 4\}$ and $g_2 = \{6\}$. If the reachability tree is complete up to the level $(n - 1)$, a number of nodes at level $(n - 1)$ contain two RMTs out of 4 effective RMTs from g_1. These nodes will have only a single child. This leads to an incomplete reachability tree and the CA becomes irreversible.

The cases I to IV point to the fact that if the RMTs 0, 2, 3, 4 of a balanced rule have the same next state, then the CA formed with R is irreversible. Hence R is an irreversible rule. Similarly, it can also be shown that a balanced rule with the same next state for the RMTs {0, 4, 6, 7} or {0, 1, 2, 6} or {0, 1, 3, 7} is an irreversible rule. □

Corollary 4.5 *For null boundary condition, the number of balanced irreversible rules is 8.*

Proof As there are 4 clusterings of RMTs that lead to a balanced irreversible rule (Theorem 4.9) and each clustering corresponds to 2 rules, the total number of such balanced irreversible rules is $4 \times 2 = 8$. □

Table 4.3 List of reversible rules (null boundary)

15	23	27	30	39	43	45	51	53	54	57
58	60	75	77	78	83	85	86	89	90	92
99	101	102	105	106	108	113	114	120	135	141
142	147	149	150	153	154	156	163	165	166	169
170	172	177	178	180	195	197	198	201	202	204
210	212	216	225	228	232	240				

From the earlier discussion, it can be identified that the balanced irreversible rules are 29, 46, 71, 116, 139, 184, 209 and 226. For example, the next states of the RMTs 0, 2, 3 and 4 of rule 29 (00011101) are the same—that is, 1. The list of rest 62 balanced rules are shown in Table 4.3.

Conjecture 1 The 62 rules of Table 4.3 are the reversible rules in null boundary.

Theorem 4.8 further points out that the reversible rules are interrelated. The sequence of reversible rules that form a reversible CA follows a specific relation. The next subsections explore such relation and report classification of the rules to find the sequence of rules for a reversible CA.

4.5.1 Formation of Class

Let consider the rules $\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i$ are selected for the cell 0, cell 1, $\cdots$, cell i respectively to form an n-cell reversible CA satisfying Theorems 4.4 and 4.5. Now, the CA cell $(i+1)$ can support a set of rules S_j so that any one of S_j can be selected as the $\mathcal{R}_{i+1}$, satisfying the Theorems 4.4 and 4.5. Let refer the class of $(i+1)$th cell as C which supports the rules of S_j. The term *class* of cell $(i+1)$ as well as the class of S_j are used interchangeably. That is, the class of S_j is C.

Lemma 7 *There are 6 classes of reversible rules in null boundary.*

Proof Each node of the reachability tree of a reversible CA contains 4 RMTs (Corollary 4.2). Since the sibling RMTs are associated with the same node in the reachability tree and there are 4 sets of sibling RMTs (0 & 1, 2 & 3, 4 & 5, and 6 & 7), 3 different organizations of RMTs for the nodes are possible. The organizations are—{0, 1, 2, 3} & {4, 5, 6, 7}, {0, 1, 4, 5} & {2, 3, 6, 7}, and {0, 1, 6, 7} & {2, 3, 4, 5}. Therefore, if the reachability tree contains a node with RMTs {0, 1, 2, 3} at level i, it also contains a node with RMTs {4, 5, 6, 7}.

Each level of the reachability tree of a reversible CA can have either 2 or 4 unique nodes (Lemma 6). If at a level there are 2 unique nodes, then the nodes are any one of the 3 possible combinations of RMTs. Let denote that the class of the rule $\mathcal{R}_{i+1}$ as *I–III* based on the above possibilities. On the other hand, if a level consists of 4

unique nodes, then the nodes can be any two of the 3 possible organisations of RMTs. Whenever the nodes include the combinations of the RMTs of first and second type, first and third type, or second and third type, the class of that cell can be denoted as *IV–VI* respectively. Therefore, there are 6 classes of reversible rules. □

Rules under each class: Since each node of the reachability tree for a reversible CA is formed with the 4 RMTs (Corollary 4.2) and both the edges (0-edge and 1-edge) of the node are resulted from 2 RMTs (Theorem 4.5), as the 2 out of 4 RMTs are 0 and others are 1. Therefore, the RMTs of a node may be grouped as ${}^4C_2 = 6$ different ways. However, the RMTs of the nodes of class *II* cannot be grouped as any of the 6 ways.

For Class *II* (RMT partition is {0, 1, 4, 5} & {2, 3, 6, 7}), 0 & 4 (similarly 1 & 5, 2 & 6, and 5 & 7) are the equivalent RMTs (Definition 15) and both of these contribute the same set of RMTs for the next level. Hence, if the next state of equivalent RMTs are same, the number of RMTs of the corresponding node of the next level becomes 2, which makes the CA irreversible by Corollary 4.2. Therefore, the equivalent RMTs under the same node cannot have the same next state. Hence 4 groupings of RMTs, out of 6, are possible in each node for Class *II*. Therefore, the number of reversible rules of Class *II* is $4 \times 4 = 16$. Since the equivalent RMTs are not associated with the same node for Class *I* and Class *III*, all of the 6 groupings are possible for each node. Hence, the number of rules for those classes are $6 \times 6 = 36$. The classes and corresponding rules are shown in Table 4.4. The rules under the Class *IV–VI* are the common rules of *I* & *II*, *I* & *III*, and *II* & *III* respectively.

Table 4.4 Class table (null boundary)

Class	RMTs of nodes	Rules
I	{0, 1, 2, 3} {4, 5, 6, 7}	51, 53, 54, 57, 58, 60, 83, 85, 86, 89, 90, 92, 99, 101, 102, 105, 106, 108, 147, 149, 150, 153, 154, 156, 163, 165, 166, 169, 170, 172, 195, 197, 198, 201, 202, 204
II	{0, 1, 4, 5} {2, 3, 6, 7}	15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225, 240
III	{0, 1, 6, 7} {2, 3, 4, 5}	15, 23, 27, 39, 43, 51, 77, 78, 85, 86, 89, 90, 101, 102, 105, 106, 113, 114, 141, 142, 149, 150, 153, 154, 165, 166, 169, 170, 177, 178, 204, 212, 216, 228, 232, 240
IV	{0, 1, 2, 3} {4, 5, 6, 7} {0, 1, 4, 5} {2, 3, 6, 7}	60, 90, 105, 150, 165, 195
V	{0, 1, 2, 3} {4, 5, 6, 7} {0, 1, 6, 7} {2, 3, 4, 5}	51, 85, 86, 89, 90, 101, 102, 105, 106, 149, 150, 153, 154, 165, 166, 169, 170, 204
VI	{0, 1, 4, 5} {2, 3, 6, 7} {0, 1, 6, 7} {2, 3, 4, 5}	15, 90, 105, 150, 165, 240

4.5.2 Class Relationship Between $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$

This section determines the relationship between the class of $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$. From the known $\mathcal{R}_i$ and its class, the class of $\mathcal{R}_{i+1}$ can be found.

Let consider the class of $\mathcal{R}_i$ be I (Fig. 4.9). Therefore, two unique nodes having the RMTs {0, 1, 2, 3} and {4, 5, 6, 7} are available at level $(i-1)$ of the reachability tree. Now, consider that the RMTs of $\mathcal{R}_i$ are clustered as {0, 1, 4, 5} and {2, 3, 6, 7}, where the RMTs of a set have the same next state—that is, either 0 or 1. In Fig. 4.9a, the RMTs {0, 1, 4, 5} have the next state 0, and the RMTs {2, 3, 6, 7} have 1. Therefore, the RMTs are grouped as (0, 1), (2, 3), (4, 5) and (6, 7). Each edge of the nodes is resulted from any one of these groups. Hence, the two edges between the node $N_1 = \{0, 1, 2, 3\}$ with its children are resulted from (0, 1) and (2, 3). Therefore, the two children (N_1C_1 and N_1C_2) of N_1 are having the RMTs {0, 1, 2, 3} and {4, 5, 6, 7} (Fig. 4.9a). Similarly, the children N_2C_1 and N_2C_2 of the node $N_2 = \{4, 5, 6, 7\}$ are {0, 1, 2, 3} and {4, 5, 6, 7}—that is, the nodes N_2C_1 and N_2C_2 are the same as N_1C_1 and N_1C_2 of N_1. Therefore, the next level of the reachability tree contains two unique nodes having the RMTs {0, 1, 2, 3} and {4, 5, 6, 7} (Fig. 4.9a). Hence the class of $\mathcal{R}_{i+1}$ is I.

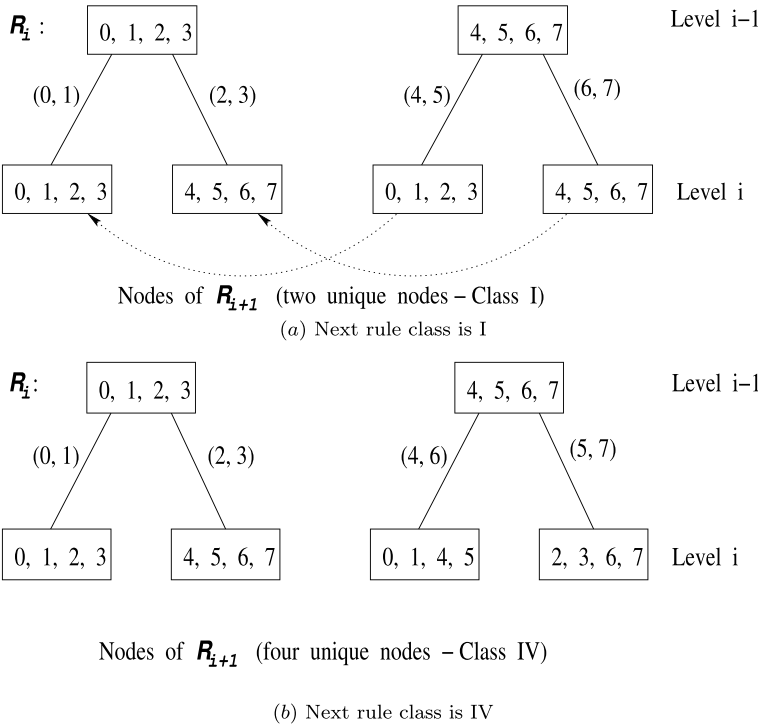


Fig. 4.9 Determination of class relationship between $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$

Further, if the RMTs of $\mathcal{R}_i$ are grouped as (0, 1), (2, 3), (4, 6), and (5, 7) (Fig. 4.9b), the nodes of N_1C_1 and N_1C_2 at level i, generated from the node N_1 of level $(i-1)$ with the RMTs {0, 1, 2, 3}, are having the RMTs {0, 1, 2, 3} and {4, 5, 6, 7}. The other two nodes (N_2C_1 and N_2C_2) at level i, generated from the node N_2 with RMTs {4, 5, 6, 7}, are having the RMTs {0, 1, 4, 5} and {2, 3, 6, 7}. In this case, the next level of reachability tree contains four unique nodes—{0, 1, 2, 3}, {4, 5, 6, 7}, {0, 1, 4, 5}, and {2, 3, 6, 7} (Fig. 4.9b)). Therefore, the organizations of RMTs support the property of both the Class *I* and Class *II*—that is, Class *IV* (Table 4.4, Column 2). So, the class of $\mathcal{R}_{i+1}$ is *IV*.

Table 4.5 displays the relationship (partly) between the reversible rules. Only 3 classes, *I*, *II*, and *IV* are selected to illustrate the relationship. First column shows the class of $\mathcal{R}_i$. Column 2 notes the RMTs of unique nodes at level $(i-1)$ whereas, Column 3 shows the grouping of RMTs for $\mathcal{R}_i$. The RMTs of unique nodes at level i are shown in Column 4. Based on the unique nodes at level i, the class of $\mathcal{R}_{i+1}$ is decided and is reported in Column 5.

The details of relationship among the classes are reported in Table 4.6. The first and second columns of the table represent the class of cell i and the rule $\mathcal{R}_i$ respectively whereas, the class of cell $(i+1)$ corresponding to this pair (the class of cell i and $\mathcal{R}_i$) is noted in the last column. It can be observed that a rule can be the member of

Table 4.5 Formation of class relationship between $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$ (null boundary)

(1)	(2)	(3)	(4)	(5)
Class of $\mathcal{R}_i$	RMTs of unique nodes at level $(i-1)$	Groupings of RMTs at level $(i-1)$	RMTs of unique nodes at level i	Class of $\mathcal{R}_{i+1}$
I	{0, 1, 2, 3} {4, 5, 6, 7}	(0, 1), (2, 3) (4, 5), (6, 7)	{0, 1, 2, 3} {4, 5, 6, 7}	I
		(0, 2), (1, 3) (4, 6), (5, 7)	{0, 1, 4, 5} {2, 3, 6, 7}	II
		(0, 3), (1, 2) (4, 7), (5, 6)	{0, 1, 6, 7} {2, 3, 4, 5}	1III
		{(0, 1), (2, 3) (4, 6), (5, 7)} or {(0, 2), (1, 3) (4, 5), (6, 7)}	{0, 1, 2, 3} {4, 5, 6, 7} {0, 1, 4, 5} {2, 3, 6, 7}	IV
II	{0, 1, 4, 5} {2, 3, 6, 7}	(0, 1), (4, 5) (2, 3), (6, 7)	{0, 1, 2, 3} {4, 5, 6, 7}	1I
IV	{0, 1, 2, 3} {4, 5, 6, 7} {0, 1, 4, 5} {2, 3, 6, 7}	(0, 1), (2, 3) (4, 5), (6, 7)	{0, 1, 2, 3} {4, 5, 6, 7}	I
		{(0, 1), (2, 3), (4, 6), (5, 7)} or {(0, 2), (1, 3), (4, 5), (6, 7)}	{0, 1, 2, 3} {4, 5, 6, 7} {0, 1, 4, 5} {2, 3, 6, 7}	IV

Table 4.6 Class relationship of $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$ (null boundary)

Class of $\mathcal{R}_i$	$\mathcal{R}_i$	Class of $\mathcal{R}_{i+1}$
I	51, 60, 195, 204	I
	85, 90, 165, 170	II
	102, 105, 150, 153	III
	53, 58, 83, 92, 163, 172, 197, 202	IV
	54, 57, 99, 108, 147, 156, 198, 201	V
	86, 89, 101, 106, 149, 154, 166, 169	VI
II	15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225, 240	I
III	15, 51, 204, 240	I
	85, 105, 150, 170	II
	90, 102, 153, 165	III
	23, 43, 77, 113, 142, 178, 212, 232	IV
	27, 39, 78, 114, 141, 177, 216, 228	V
	86, 89, 101, 106, 149, 154, 166, 169	VI
IV	60, 195	I
	90, 165	IV
	105, 150	V
V	51, 204	I
	85, 170	II
	102, 153	III
	86, 89, 90, 101, 105, 106, 149, 150, 154, 165, 166, 169	VI
VI	15, 240	I
	105, 150	IV
	90, 165	V

more than one class. For example, the rule 90, 105, 150 and 165 are the members of all the 6 classes. Such rules are referred to as the complete rules.

Definition 32 A rule is *complete* if it is the member of all the six classes.

First and Last rule: For null boundary CA, the RMTs 4, 5, 6 and 7 are the *don't cares* for $\mathcal{R}_0$ (left most rule) as the present state of left neighbor of cell 0 is always 0. So, there are only 4 *effective* RMTs (0, 1, 2, 3) for $\mathcal{R}_0$. Similarly, the RMTs 1, 3, 5 and 7 are the don't care RMTs for $\mathcal{R}_{n-1}$. The *effective* RMTs for $\mathcal{R}_{n-1}$ are, therefore, 0, 2, 4 and 6. That is, rule 105 and 9 are equivalent if selected as $\mathcal{R}_0$. Similarly, the rules 75 and 65 are effectively the same while chosen for the $(n-1)$th cell. There are $2^{2^2} = 16$ effective rules for $\mathcal{R}_0$ as well as for $\mathcal{R}_{n-1}$.

Corollary 4.6 *If $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ is a reversible CA, then $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$ are balanced over their effective 4 RMTs.*

Proof Let consider that the first rule is unbalanced over its 4 effective RMTs. That is, the next state of 3 RMTs out of 4 effective RMTs of $\mathcal{R}_0$ be d ($d = 0/1$). Therefore, there are $3 * 2^{n-2}$ present configurations for which the next configurations have the form $S = \{d \cdots\}$. The maximum possible number of such next configurations is clearly 2^{n-1}. Since the number of next configurations is lesser than that of the current configurations, there is at least one configuration in S which contains more than one predecessor. Hence the CA is irreversible. This is because of that the $\mathcal{R}_0$ is unbalanced over its 4 effective RMTs. Therefore, to form a reversible CA, $\mathcal{R}_0$ must be balanced over its 4 effective RMTs. With similar logic, it can be proved that the $\mathcal{R}_{n-1}$ has to be balanced over its 4 effective RMTs. □

Corollary 4.6 signifies that the unbalanced rule 3 is a reversible rule when it is selected as $\mathcal{R}_0$. The rule 3 (as $\mathcal{R}_0$) is balanced over its 4 effective RMTs. There are ${}^4C_2 = 6$ rules (out of total 16 effective rules for $\mathcal{R}_0$) which are balanced over their 4 effective RMTs. Table 4.7, referred to as the first rule table, identifies such 6 rules and the corresponding class of rule $\mathcal{R}_1$. The similar consideration is also true for the $\mathcal{R}_{n-1}$. Table 4.8, referred to as the last rule table, lists all such 6 reversible rules for the $\mathcal{R}_{n-1}$. Note that the next states of the *don't care* RMTs of $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$ are assumed to be 0 in Tables 4.7 and 4.8.

Table 4.7 First rule table (null boundary)

Rules for $\mathcal{R}_0$	Group of RMTs	RMTs of nodes for level 2	Class of $\mathcal{R}_1$
3, 12	(0, 1) (2, 3)	{0, 1, 2, 3} {4, 5, 6, 7}	I
5, 10	(0, 2) (1, 3)	{0, 1, 4, 5} {2, 3, 6, 7}	II
6, 9	(0, 3) (1, 2)	{0, 1, 6, 7} {2, 3, 4, 5}	III

Table 4.8 Last rule table (null boundary)

Rule class for $\mathcal{R}_{n-1}$	Rule set for $\mathcal{R}_{n-1}$
I	17, 20, 65, 68
II	5, 20, 65, 80
III	5, 17, 68, 80
IV	20, 65
V	17, 68
VI	5, 80

4.5.3 Reversible CA Synthesis

A reversible CA synthesis scheme is given in Algorithm 3 (Sect. 4.4.2). A simpler method to synthesize a reversible CA can be developed by exploiting the class relationships of reversible CA rules presented in the Tables 4.6, 4.7 and 4.8. Let consider the synthesis of a 4-cell reversible CA and say, the rule 9 is selected randomly as $\mathcal{R}_0$ from Table 4.7. Therefore, the class (obtained from Table 4.7) of the second rule ($\mathcal{R}_1$) is III. From Class III of Table 4.6, say rule 150 is selected randomly as $\mathcal{R}_1$. Therefore, the class of $\mathcal{R}_2$ is found to be II, since the class of $\mathcal{R}_1$ is III and $\mathcal{R}_1 = 150$ (Table 4.6). Now, rule 45 is selected as $\mathcal{R}_2$ from Class II of Table 4.6. The class of the last rule ($\mathcal{R}_3$) is, therefore, I. Rule 65 is selected randomly for $\mathcal{R}_3$ from Table 4.8. Therefore, the 4-cell reversible CA in null boundary is $\mathcal{R} = (9, 150, 45, 65)$.

4.6 Scalability of Reversible CA

The classification of reversible rules enables the design of *scalable* structure of reversible CA. That is, from the given rule vector of an n_1-cell reversible CA, the rule vector of an n_2-cell reversible CA, where $n_2 > n_1$, can be constructed. Similarly, from two reversible CAs of size n and m, an $(n + m)$-cell reversible CA can be designed. This section explores such designs for null boundary. The same can be applicable for the CAs in periodic boundary.

4.6.1 Design of n_2-Cell CA from n_1-Cell CA

The major advantage of the synthesis scheme of Sect. 4.5.3 is that an $(n + 1)$-cell reversible CA can be synthesized in two steps if an n-cell reversible CA along with class information of the rules is known. For example, consider the 4-cell reversible CA $\mathcal{R} = (9, 150, 45, 65)$ of Sect. 4.5.3. To synthesize a 5-cell reversible CA from $\mathcal{R}$, the rules $\mathcal{R}_3$ and $\mathcal{R}_4$ for the fourth and fifth cell respectively are to be selected. The first three rules are taken from $\mathcal{R}$. As the class of $\mathcal{R}_2$ is II, the class of $\mathcal{R}_3$ is to be I (Table 4.6). Now, if $\mathcal{R}_3$ is chosen randomly as 149 from Class I of Table 4.6, the class of the next rule ($\mathcal{R}_4$) is then VI. Let consider that the rule 80 is chosen as $\mathcal{R}_4$ following Table 4.8. It results in the 5-cell reversible CA (9, 150, 45, 149, 80) in null boundary.

The above scheme can be extended to synthesize an n_2-cell reversible CA from a given n_1-cell reversible CA, where $n_2 > n_1$. In the n_2-cell CA, the n_1-cell CA can be traced. So, the final rule vector can act as the n_2-cell CA as well as n_1-cell CA. A simple hardware can realize such CAs.

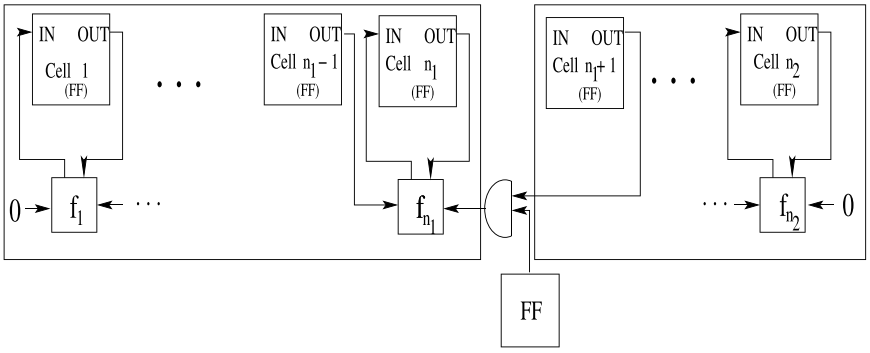


Fig. 4.10 Architecture of dual n_1-cell and n_2-cell CA

Figure 4.10 gives the general architecture for the scalable CA. It can act as n_1-cell as well as n_2-cell CA ($n_2 > n_1$). At the cell $(n_1 - 1)$ of the CA structure, a switch, implemented with an *AND* gate and a flip-flop (*FF*) is added. While considering the structure as n_1-cell CA, the *FF* is reset to 0. Since null boundary condition is considered, the right neighbor of the cell $n_1 - 1$ is to be 0. For implementing the n_2-cell CA, utilizing the same structure, the switch *FF* is set to 1 so that the next state logic of the cell $(n_1 - 1)$ can depend on the present state of the cell n_1.

To get n_1-cell and n_2-cell reversible CAs, first the rules for the n_1-cell CA are selected following the synthesis scheme of Sect. 4.5.3. The $(n_1 - 1)$th rule is chosen in such a way that it can act as the last cell of the n_1-cell CA as well as the $(n_1 - 1)$th cell of the n_2-cell CA. Then, the rule vector is extended to get n_2-cell CA following the same synthesis scheme.

Example 4.14 This example illustrates the design of a scalable CA structure that realizes the n_1-cell and n_2-cell CAs, where $n_1 = 4$ and $n_2 = 7$. To design the 4-cell and 7-cell reversible CAs in null boundary, the rules for the 4-cell CA are selected first following the synthesis scheme of Sect. 4.5.3. Let the rule vector of the CA be (9, 149, 105, 20). The rules 20 and 150 are equivalent while considered as the last rule of a CA with null boundary. So, rule 150 can be selected in place of 20 as the 4th rule of the 7-cell CA and then rest of the rules can be selected following Tables 4.6 and 4.8. Say, the resulted rule vector for the 7-cell reversible CA is (9, 149, 105, 150, 106, 150, 65). To get the 4-cell reversible CA, right neighbor of the 4th cell is set to 0 by resetting the switching *FF* of Fig. 4.10.

The following subsection illustrates the design of $(n + m)$-cell reversible CA from an n-cell and an m-cell reversible CAs.

4.6.2 Design of n-Cell, m-Cell and (n + m)-Cell Reversible CAs

This design realizes a structure that can be utilised as the n-cell reversible CA, m-cell reversible CA as well as the $(n + m)$-cell reversible CA. The synthesis of such a structure almost similar to that of the structure referred in the earlier subsection. This structure, however, can be utilized to realize two reversible CAs (n-cell and m-cell) simultaneously.

The design first synthesizes an n-cell CA following the scheme described in Sect. 4.5.3. As the right most cell of the n-cell CA should also act as the $(n - 1)$th cell of $(n + m)$-cell CA, the rule for the cell is selected carefully so that it can satisfy the both. Similarly, the first cell of the m-cell CA is selected in such a way that it can also act as the nth cell of $(n + m)$-cell CA. The other cells of the m-cell CA are then selected following the same synthesis scheme.

The hardware implementation of such CA structure is shown in Fig. 4.11. Two switches, implemented with flip-flops (FF_1 and FF_2) and *AND* gates, are employed to facilitate the design. Whenever both the FF_1 and FF_2 are set to 0, the CA can act as an n-cell and an m-cell reversible CAs concurrently. On the other hand, the structure acts as the $(n + m)$-cell reversible CA when both the flip-flops, FF_1 and FF_2 are set to 1.

Example 4.15 Let consider that the 4-cell reversible CA (9, 150, 45, 65) is synthesized following the synthesis scheme of Sect. 4.5.3. The rule 65 and 105 are equivalent while considered as the last cell rule of a CA in null boundary. So, rule 105 can be considered as the fourth rule for the desired 7-cell reversible CA. However, it is found that the class of the fourth cell of the 4-cell CA is I. To design 7-cell reversible CA, the fifth cell is to be selected in such a way that the cell can act also as the first cell of the desired 3-cell CA. Since the class of the fourth cell is I and the rule is 105, the rule 165 can be selected for the fifth cell rule (Table 4.6). Rule 165 and rule 5 are equivalent while they are treated as the first rule of any CA in null boundary. The class

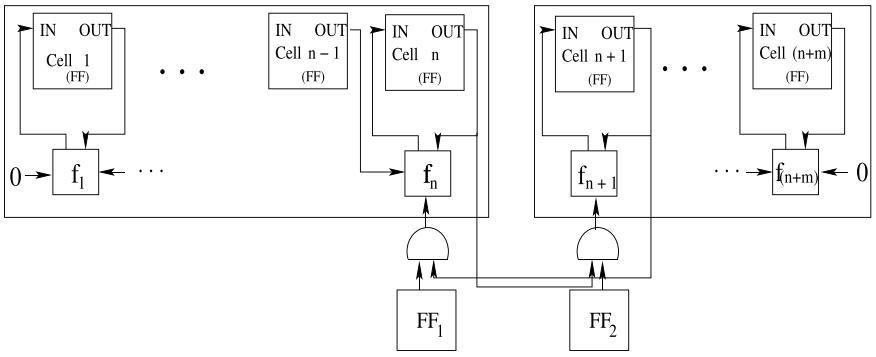


Fig. 4.11 Architecture of n-cell, m-cell and $(n + m)$-cell CA

of the next cell (that is, sixth cell of the 7-cell CA and second cell of the 3-cell CA) is found to be *II* (Tables 4.6 and 4.7). The remaining rules are selected from Tables 4.6, 4.7 and 4.8 to get the 7-cell reversible CA (9, 150, 45, 105, 165, 105, 20).

4.7 Chapter Notes

The group properties of finite cellular automata have been first explored in [1]. Matrix algebra as a tool of understanding reversibility of linear and complemented binary CAs has been introduced in [2]. The finite and linear reversible CAs are also known as *group* CA. A detail treatment to the group CAs (in null boundary) is given in [3]. Reachability tree has been primarily developed to check the reversibility of non-uniform ECAs in null boundary [4]. Reversibility of non-uniform ECAs with periodic boundary has been studied in [5]. Whether reversibility of an arbitrary non-uniform CA, defined over infinite lattice, is decidable or not can be found in [6, 7]. Various types of applications have been developed around reversible CAs, which are discussed in brief in [3, 8].

Reversibility in CAs is an active research area. The following are few topics, related to this chapter, on which future research may be carried out.

1. Develop an efficient algorithm to test the reversibility of non-uniform CAs over infinite lattice that are decidable.
2. Can reachability tree be tuned to decide the reversibility of non-uniform CAs over infinite lattice? If so, develop the scheme.
3. Study the reachability tree and reversibility of non-uniform binary CAs with boundary conditions other than null and periodic.
4. Develop the classes of reversible rules and their relationship (like Table 4.6) for d-state non-uniform CAs.
5. Let us consider a variant of open boundary condition, where the missing neighbors can stochastically assume a state at a time. That is, the missing neighbors can assume different states at different time. Study the reversibility of non-uniform ECAs under such boundary condition.

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Chapter 5
Number Conservation

Reversibility refers to a conservation property of a system, where the information in an initial configuration of the system is conserved during its evolution. Apart from such information conservation, the Number Conservation in cellular automata (CAs) is also studied. In a Number Conserving Cellular Automaton (NCCA), the sum of the states of the cells in an initial configuration is conserved during its evolution. Although the state set of a CA can be of different forms, the number conservation generally assumes the numeric states of a cellular automaton (CA). This chapter deals with the number conservation behavior of non-uniform CAs that primarily use ECA rules. The study relies on the reachability tree, defined in Sect. 3.3. Section 5.2 introduces the relation between the reachability tree and the NCCA, based on which a decision algorithm is reported in Sect. 5.3. The reachability tree-based theoretical framework enables us an efficient synthesis of a finite NCCA, which is reported in Sect. 5.4. This framework can also be extended to the study of the Number Non-Increasing and Number Non-Decreasing CAs (Sect. 5.5). Boundary condition has an impact on the finite NCCAs. The theoretical framework can be utilized for different boundary conditions with marginal modification (Sect. 5.6). Although most of the discussions of this chapter are around finite CAs, some lights on NCCAs over infinite lattice are thrown in Sect. 5.7.2. Finally, the logic design with NCCAs is briefed in Sect. 5.8.

5.1 Basics of Number Conservation

Let $S = \{0, 1, \cdots, d-1\}$ be the set of states of a (non-uniform) CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$. The CA is number conserving if, for every initial configuration, the sum of the states of its cells in a configuration remains invariant during its evolution. A configuration x can be represented as $(x_i)_{i\in\mathscr{L}}$, where $x_i \in S$ is the state of cell $i \in \mathscr{L}$. Let the global transition function of the CA be G and the CA be finite ($\mathscr{L}$ is finite). Then,

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the CA is number conserving if $\mu(x) = \mu(G(x))$ for each configuration $x \in S^{\mathscr{L}}$, where μ is the additive function and defined as $\mu(x) = \sum_{i \in \mathscr{L}} x_i$.

If $\mathscr{L}$ is infinite, $\mu(x)$ can be ∞ for some configuration x and the above definition of number conservation cannot work. However, if x is a finite configuration (on infinite $\mathscr{L}$), then $\mu(x)$ is finite and the above definition stands. That is, a CA (defined on infinite $\mathscr{L}$) is number conserving on finite configurations if for all the finite configurations x, $\mu(x) = \mu(G(x))$. For such a CA, $G(\mathbf{0}) = \mathbf{0}$, where $\mathbf{0}$ is the all 0s configuration. Now, the number conservation of a CA, defined on infinite $\mathscr{L}$, can be stated as: a CA is number conserving if it is number conserving on finite configurations. In rest of the chapter, if not stated otherwise, the $\mathscr{L}$ is considered as finite.

Let represent a homogeneous configuration[1] with state $s \in S$ in all cells as $\mathbf{s}$. Clearly, in any NCCA, the $G(\mathbf{s}) = \mathbf{s}$ for each $s \in S$.

Lemma 8 *Let $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$ be a finite number conserving CA (NCCA). Then, for each rule $\mathcal{R}_i$, $i \in \mathscr{L}$, $\mathcal{R}_i[ss \cdots s] = s$, where $(ss \cdots s) \in S^{2\mathbf{r}_i+1}$ is an RMT of $\mathcal{R}_i$ and $s \in S$.*

Proof For the configuration $\mathbf{s}$ of the CA $(\mathscr{L}, S, \mathbf{r}_i, \mathcal{R}_i)$, $\mu(\mathbf{s}) \neq \mu(G(\mathbf{s}))$ if $\mathcal{R}_i[ss \cdots s] \neq s$ for any $i \in \mathscr{L}$ and any $s \in S$. Since the CA is an NCCA, $\mathcal{R}_i[ss \cdots s] = s$ for each rule $\mathcal{R}_i$, $i \in \mathscr{L}$. □

If the CA is binary and all the rules are ECA rules, then obviously the RMT 0 and RMT 7 of each rule are 0 and 1 respectively. That is, $\mathcal{R}_i[0] = 0$ and $\mathcal{R}_i[7] = 1$ for each $i \in \mathscr{L}$.

Let consider the binary (finite) CAs only. If such a CA is number conserving, the number of 1s in an initial configuration remains invariant during its evolution. For example, the 4-cell CA (136, 252, 238, 192) in periodic boundary is an NCCA. Figure 5.1 shows the transition diagram of the CA for verification.

A CA is called *number non-increasing CA* (*number non-decreasing CA*) if, during its evolution, the number of 1s in a configuration does not increase (decrease). The 4-cell CA (136, 190, 64, 192) in periodic boundary is number non-increasing CA. Whereas, the CA (238, 136, 252, 238) is a non-decreasing CA. The transition diagrams, shown in Figs. 5.2 and 5.3, are to verify the condition of number non-increasing and non-decreasing CAs.

It is obvious that the set of number conserving CAs are the subset of the set of number non-increasing CAs, as well as the set of number non-decreasing CAs. That is, the intersection between the sets of number non-increasing CAs and number non-decreasing CAs is the set of number conserving CAs. An obvious extension of Lemma 8 is the following: For each rule $\mathcal{R}_i$ of a finite

[1] A configuration of a CA is homogeneous if all the cells of the CA are in the same state.

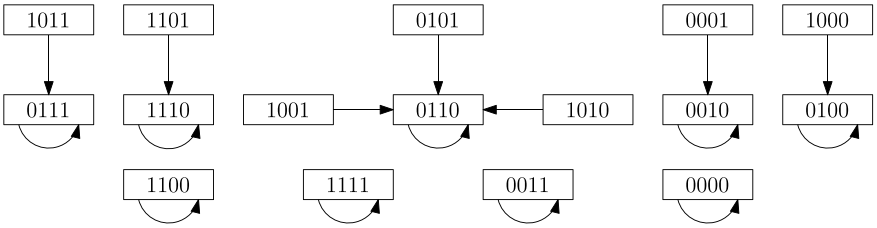


Fig. 5.1 Transition diagram of number conserving CA (136, 252, 238, 192) in periodic boundary

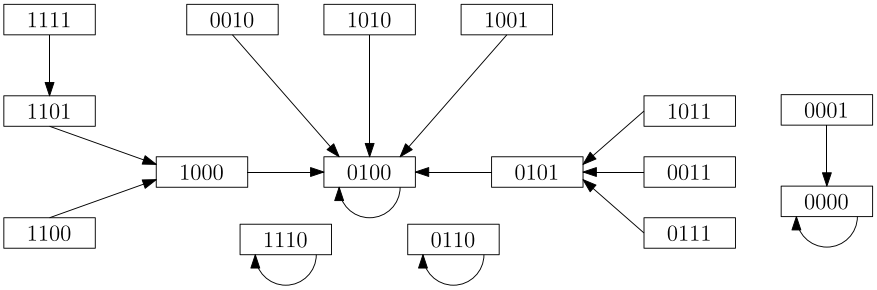


Fig. 5.2 Transition diagram of number non-increasing CA (136, 190, 64, 192) in periodic boundary

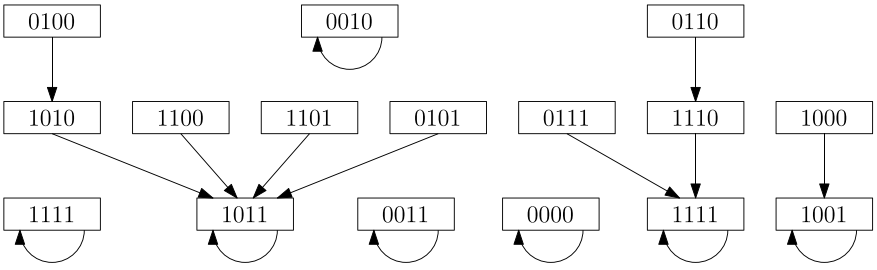


Fig. 5.3 Transition diagram of number non-decreasing CA (238, 136, 252, 238) in periodic boundary

- Number non-increasing CA, $\mathcal{R}_i[0] = 0$,
- Number non-decreasing CA, $\mathcal{R}_i[7] = 1$, and
- Number conserving CA, $\mathcal{R}_i[0] = 0$ and $\mathcal{R}_i[7] = 1$.

However, these three types of CAs always relate number of 1s in two consecutive configurations. If the number of 1s for each pair of two consecutive configurations is tested, then it can be decided that whether the CA is number conserving, number non-increasing or number non-decreasing. Reachability tree (see Sect. 3.3) pairs up all the consecutive configurations of a CA. So, it can be used for characterization of these classes of CAs.

Table 5.1 RMTs of the ECA rules 136, 252, 238 and 226

Present state:	111	110	101	100	011	010	001	000	Rule
(RMT)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
(i) Next state:	1	0	0	0	1	0	0	0	136
(ii) Next state:	1	1	1	1	1	1	0	0	252
(iii) Next state:	1	1	1	0	1	1	1	0	238
(iv) Next state:	1	1	0	0	0	0	0	0	192

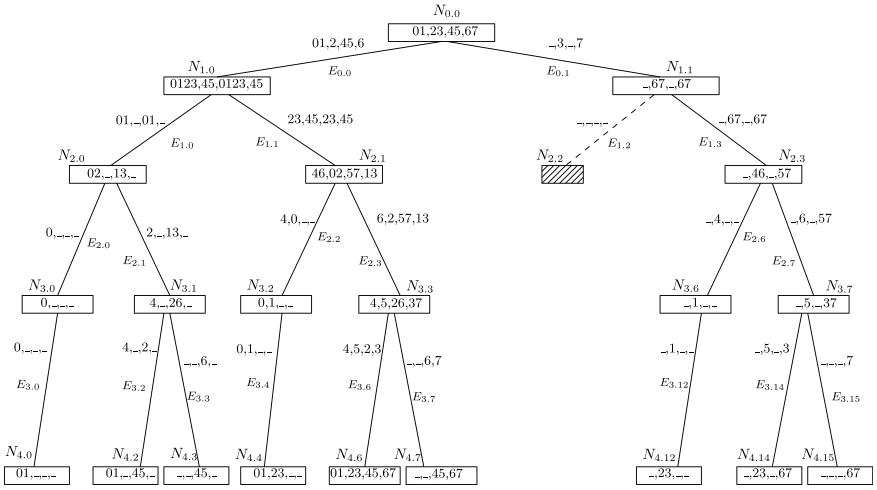


Fig. 5.4 Reachability tree for the CA (136, 252, 238, 192) in periodic boundary

5.2 Reachability Tree and Number Conservation

Reachability tree, introduced in Sect. 3.3, is an edge-labelled tree that represents the reachable configurations of a finite CA. It is a binary tree with $n+1$ levels for an n-cell binary CA. In this section, the CAs with periodic boundary are considered for characterization and to explore the number conservation. The following illustration is to revisit the structure of reachability tree of CA in periodic boundary.

Consider the CA (136, 252, 238, 192) in periodic boundary. The RMTs of the rules are noted in Table 5.1. Figure 5.4 shows the reachability tree of the CA, which is formed following Definition 22 of Chap. 3. It is a binary tree of 5 levels. A node at level i, denoted as $N_{i,j} = (\Gamma_0^{N_{i,j}}, \Gamma_1^{N_{i,j}}, \Gamma_2^{N_{i,j}}, \Gamma_3^{N_{i,j}})$ is an ordered list of four sets of RMTs. Similarly, the label of an edge $E_{i,j}$ is an ordered list – $l_{i,j} = (\Gamma_0^{E_{i,j}}, \Gamma_1^{E_{i,j}}, \Gamma_2^{E_{i,j}}, \Gamma_3^{E_{i,j}})$. The reachability tree provides the information about the reachable configurations of the CA. Figure 5.4 shows that the CA has 9 reachable configurations.

A sequence of edges $\langle E_{0.j_0}, E_{1.j_1}, \cdots, E_{i.j_i}, E_{i+1.j_{i+1}}, \cdots, E_{n-1.j_{n-1}} \rangle$ from the root to a leaf associates a reachable configuration and at least one RMT sequence $r_0 r_1 \cdots r_i r_{i+1} \cdots r_{n-1}$, where $r_i \in \Gamma_p^{E_{i.j_i}}$ and $r_{i+1} \in \Gamma_p^{E_{i+1.j_{i+1}}}$ for some $p \in \{0, 1, 2, 3\}$ ($0 \leq i <$n-1, $0 \leq j_i \leq 2^i - 1$, and $j_{i+1} = 2j_i$ or $2j_i + 1$). Since the RMT sequence is an alternative way of representing a configuration (see Definition 12), a sequence of edges represents at least two configurations. Further, if $\mathcal{R}_i[r_i] = 0$ (resp. 1), then $E_{i.j_i}$ is 0-edge (resp. 1-edge). Therefore, the reachable configuration is the next (resp. present) configuration of the current (resp. predecessor) configuration, represented as RMT sequence.

To decide on the number conservation of an n-cell CA, the following approach can be adopted: Develop the reachability tree of the given CA to get the RMT sequences. If any configuration, represented as the RMT sequence, does not have the same number of 1s in its next configuration, the CA is not an NCCA. The *weights* of an RMT, defined in the following subsection, provides better insights while examining NCCAs.

5.2.1 The Weight

The weight of an RMT notes the surplus or deficiency of 1s in the RMT sequence with respect to its next configuration. Let consider an RMT sequence $r_0 r_1 \cdots r_i r_{i+1} \cdots r_{n-1}$ and its next configuration $b_0 b_1 \cdots b_i \cdots b_{n-1}$ for a CA $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i, \cdots, \mathcal{R}_{n-1})$. That is, $\mathcal{R}_i[r_i] = b_i$, $i \in \{0, 1, \cdots, n-1\}$. Initially, there is no surplus or deficiency of 1s. So, weight of r_0 is 0. However, the surplus and deficiency of 1s in the first bit of these two configurations can be understood by comparing r_0 and b_0. For example, if $r_0 \in \{2, 3, 6, 7\}$ (that is, the first bit of the configuration, represented as $r_0 r_1 \cdots r_i \cdots r_{n-1}$ is 1) and $b_0 = 0$, then the first bit of the configuration $r_0 r_1 \cdots r_i \cdots r_{n-1}$ contributes one additional 1, which is carried by the RMT r_1. Then, weight 1 is assigned to RMT r_1. That is, if surplus (resp. deficiency) of 1 is observed in the first bit of the predecessor, r_1 is assigned weight 1 (resp. –1). Next, r_1 and b_1 are compared and weight to r_2 is assigned after taking the weight of the r_1 into consideration. For example, if $r_1 \in \{2, 3, 6, 7\}$ and $b_1 = 0$, and the weight of r_1 is 1, then weight of r_2 is 2. In this case, the first two bits of the configuration $r_0 r_1 \cdots r_i \cdots r_{n-1}$ are 11, but the next states of the RMTs r_0 and r_1 are 0. In this way, the weights are assigned recursively, and finally the weight of the RMT r_{n-1} is obtained. In case of NCCA, this weight is to be dismissed after comparing the r_{n-1} and b_{n-1}, to conclude that both the configurations have the equal number of 1s. For example, consider the two consecutive configurations, RMT sequence 4012 (0001) and its next configuration 0010 (of the CA of Fig. 5.1). Weights of RMTs of the sequence are 0, 0, 0, –1 respectively. The weight –1 is dismissed when the RMTs 2 and 0 are compared and so, both the configurations have the same number of 1s.

The weights to the RMTs of the nodes of reachability tree can be assigned. Initially, weights of the RMTs at root are 0. An RMT r_i, present in $\Gamma_k^{N_{i.j}}$ is also present at a

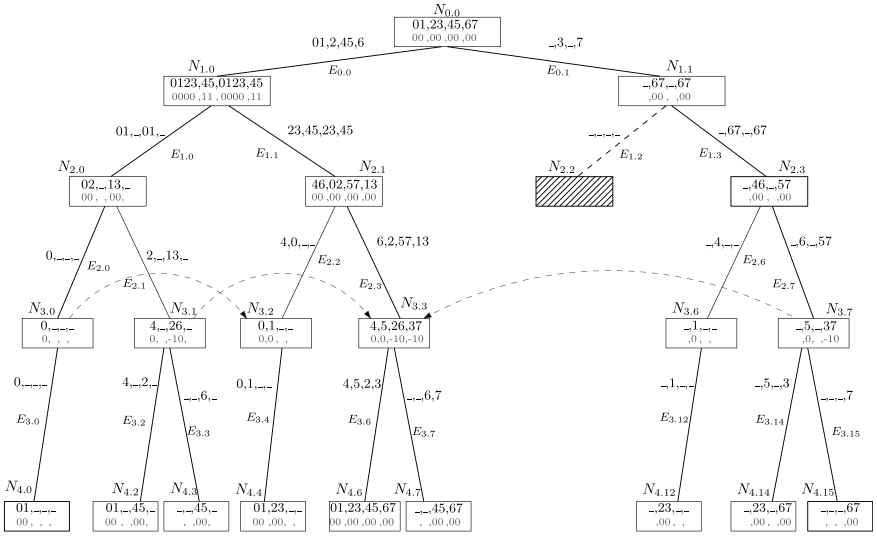


Fig. 5.5 Reachability tree of Fig. 5.4 with weights of RMTs at different nodes

label of an edge – either in $\Gamma_k^{E_{i.2j}}$ (of the 0-edge) or in $\Gamma_k^{E_{i.2j+1}}$ (of the 1-edge) (Point 2 of Definition 22). If the r_i is in $\Gamma_k^{E_{i.2j}}$ (resp. $\Gamma_k^{E_{i.2j+1}}$), $\mathcal{R}_i[r_i] = 0$ (resp. 1). So, the weight of RMT r_{i+1} can be determined considering the weight of r_i and observing the edge ($E_{i.2j}$ or $E_{i.2j+1}$) on which the RMT exists. However, the RMTs r_i and r_{i+1} are in an RMT sequence, and they are on some nodes as well. The weight $W_k(i, r)$ of RMT $r \in \Gamma_k^{N_{i.j}}$, $k \in \{0, 1, 2, 3\}$, can be defined as:

1. $W_k(0, r) = 0 \qquad : r \in \Gamma_k^{N_{0.0}}$
2. $W_k(i, 2r \pmod 8) = W_k(i, 2r + 1 \pmod 8)$

$$= \begin{cases} W_k(i-1, r) + 1 : r \in \{2, 3, 6, 7\} \text{ and } \mathcal{R}_{i-1}[r] = 0 \\ W_k(i-1, r) - 1 : r \in \{0, 1, 4, 5\} \text{ and } \mathcal{R}_{i-1}[r] = 1 \\ W_k(i-1, r) \qquad\qquad\qquad\qquad\quad : \text{otherwise} \end{cases}$$

Since the RMTs at leaves are the RMTs of $\mathcal{R}_0$–that is, of root, the weight of such an RMT r is to be 0. Therefore, if all the weights of leaves are 0, the CA is an NCCA.

Example 5.1 Consider the CA (136, 252, 238, 192) of Fig. 5.4. The weights of the RMTs at nodes of the reachability tree are assigned in Fig. 5.5 (noted at bottom of the RMTs). The weights of the RMTs of the sequence 4012 are 0, 0, 0, and –1 respectively. The weight –1 is nullified at the leaf $N_{4.2}$, as 192[2] = 0. Many of the RMTs of intermediate nodes have the non-zero weights. However, the weights at all leaves of the tree are zero. Hence, it is an NCCA.

5.2.2 Uniformity of Weights and the Rules

An RMT $r \in \Gamma_k^{E_{i.j}}$, in the label of an edge, contributes two RMTs $2r \pmod 8$ and $2r+1 \pmod 8$ to $\Gamma_k^{N_{i+1.j}}$ of a node (Point 3 of Definition 22). If another RMT s, which is equivalent to r, is present in $\Gamma_k^{E_{i.j}}$, then two more RMTs $2s \pmod 8$ and $2s+1 \pmod 8$ are added to $\Gamma_k^{N_{i+1.j}}$. It is mentioned earlier that the two RMTs r and s are equivalent, where $2r \equiv 2s \pmod 8$. Hence, the RMTs r and s contribute the same pair of RMTs. But, the weights of the RMTs $2r \pmod 8$ and $2s \pmod 8$ may not be the same. If this is happened, then the CA cannot be an NCCA.

Theorem 5.1 *The weight of an RMT in $\Gamma_k^{N_{i.j}}$, where $k \in \{0, 1, 2, 3\}$ and $N_{i.j}$ is any node in the reachability tree of an NCCA, is unique.*

Proof Suppose an RMT $r \in \Gamma_k^{N_{i.j}}$ has more than one weight. Then, the RMTs in $\Gamma_k^{N_{i+1.2j}}$ or in $\Gamma_k^{N_{i+1.2j+1}}$, contributed by r, can have more than one weight. Again these RMTs contribute the RMTs with multiple weights to the next level, and so on. Finally, in at least one leaf of the reachability tree, an RMT is associated with multiple weights. But to be an NCCA, the weights of all the RMTs on the leaves are to be 0. So, this violates the condition of NCCA. Hence, the weight of an RMT in $\Gamma_k^{N_{i.j}}$ is to be unique. □

Corollary 5.1 *The weights of RMT r of $\mathcal{R}_i$ that belongs to $\Gamma_k^{N_{i.a}}$ and $\Gamma_k^{N_{i.b}}$ are equal.*

Proof This is proved by contradiction. Let consider that r has two different weights in two nodes. Hence, the RMTs contributed by r (following Point 3 of Definition 22) to the nodes at the next level have different weights. As a result, at least one set of leaves, followed from $N_{i.a}$ or $N_{i.b}$, can have the RMTs with weight other than 0. The RMTs with non-zero weight in any leaf indicate that the CA is not an NCCA. Hence to be NCCA, the RMT r has to have the same weight in two nodes. □

Corollary 5.2 *For two equivalent RMTs $r \in \Gamma_k^{N_{i.a}}$ and $s \in \Gamma_k^{N_{i.b}}$ ($k \in \{0, 1, 2, 3\}$), $\mathcal{R}_i[r] = \mathcal{R}_i[s]$ iff $W_k(i, r) = W_k(i, s)$. Consequently, $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$ iff $W_k(i, r) \neq W_k(i, s)$.*

Proof Consider that $\mathcal{R}_i[r] = \mathcal{R}_i[s]$. Then, both the RMTs are either on 0-edge or on 1-edge. Since the RMTs are equivalent (i.e, $2r \equiv 2s \pmod 8$), they contribute the same set of RMTs (sibling) to $\Gamma_k^{N_{i+1.j_1}}$ ($j_1 = 2a$ or $2a+1$) and $\Gamma_k^{N_{i+1.j_2}}$ ($j_2 = 2b$ or $2b+1$) (Point 3 of Definition 22). So, the weights of RMTs r and s are to be the same. Otherwise, Theorem 5.1 and Corollary 5.1 are violated. Hence, $W_k(i, r) = W_k(i, s)$.

To prove the "if" part, assume that $W_k(i, r) \neq W_k(i, s)$ but $\mathcal{R}_i[r] = \mathcal{R}_i[s]$. Then, the weights of RMT $2r \in \Gamma_k^{N_{i.a}}$ and $2s \in \Gamma_k^{N_{i.b}}$, where $2r = 2s$ are not unique. It violates Theorem 5.1. Hence, $W_k(i, r) = W_k(i, s)$ when $\mathcal{R}_i[r] = \mathcal{R}_i[s]$.

The above argument points to the fact that $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$ iff $W_k(i, r) \neq W_k(i, s)$. □

Corollary 5.3 *$|W_k(i,r) - W_k(i,s)| = 0$ or 1, where $r \in \Gamma_k^{N_{i,a}}$ and $s \in \Gamma_k^{N_{i,b}}$ ($k \in \{0,1,2,3\}$) are the two equivalent RMTs.*

Proof According to Corollary 5.2, $W_k(i,r) = W_k(i,s)$ when $\mathcal{R}_i[r] = \mathcal{R}_i[s]$. But when $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$, then in case of NCCA, difference between the weights of RMTs r and s is 1. Otherwise, $W_k(i+1, 2r \pmod 8) \neq W_k(i+1, 2s \pmod 8)$. This violates Theorem 5.1 as $2r \equiv 2s \pmod 8$. Hence, $|W_k(i,r) - W_k(i,s)| = 0$ or 1. □

Corollary 5.4 *For two equivalent RMTs r and s that belong to both of the $\Gamma_{k_1}^{N_{i,a}}$ and $\Gamma_{k_2}^{N_{i,b}}$, where $k_1, k_2 \in \{0,1,2,3\}$ and $k_1 \neq k_2$, $W_{k_1}(i,r) - W_{k_1}(i,s) = W_{k_2}(i,r) - W_{k_2}(i,s)$.*

Proof Consider that for $k_1 \neq k_2$, $W_{k_1}(i,r) - W_{k_1}(i,s) \neq W_{k_2}(i,r) - W_{k_2}(i,s)$. According to Corollary 5.3, $|W_{k_1}(i,r) - W_{k_1}(i,s)| = 0$ or 1. So, if $|W_{k_1}(i,r) - W_{k_1}(i,s)| = 0$, then $|W_{k_2}(i,r) - W_{k_2}(i,s)| = 1$. Now, according to Corollary 5.2, $\mathcal{R}_i[r] = \mathcal{R}_i[s]$ when $|W_{k_1}(i,r) - W_{k_1}(i,s)| = 0$. But $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$ when $|W_{k_2}(i,r) - W_{k_2}(i,s)| = 1$. Hence, a contradiction. So, $|W_{k_1}(i,r) - W_{k_1}(i,s)|$ and $|W_{k_2}(i,r) - W_{k_2}(i,s)|$ are same. □

Corollary 5.5 *For any level i, and any $k \in \{0,1,2,3\}$:*

1. *$W_k(i,4) \geq W_k(i,0)$. If $W_k(i,4) = W_k(i,0)$, then $\mathcal{R}_i[4] = 0$. If $W_k(i,4) > W_k(i,0)$, then $\mathcal{R}_i[4] = 1$.*
2. *$W_k(i,5) \geq W_k(i,1)$. If $W_k(i,5) > W_k(i,1)$, then $\mathcal{R}_i[5] = 1$ and $\mathcal{R}_i[1] = 0$.*

Proof *Case 1:* From the definition of weight, $W_k(i+1,0)$ is either $W_k(i,4)$ –1 (if $\mathcal{R}_i[4] = 1$) or $W_k(i,4)$ (if $\mathcal{R}_i[4] = 0$). Further, $W_k(i+1,0) = W_k(i,0)$ as $\mathcal{R}_i[0] = 0$ for an NCCA. Hence, $W_k(i,4) \geq W_k(i,0)$. And, when $W_k(i,4) = W_k(i,0)$, then $\mathcal{R}_i[4] = 0$, and when $W_k(i,4) > W_k(i,0)$, then $\mathcal{R}_i[4] = 1$.

Case 2: As RMT 1 and RMT 5 are the sibling of RMT 0 and RMT 4 respectively, then at any level i, $W_k(i,5) \geq W_k(i,1)$ for each k. Further, when $W_k(i,5) > W_k(i,1)$, then $\mathcal{R}_i[5] = 1$ and $\mathcal{R}_i[1] = 0$. □

Corollary 5.6 *For any level i, and any $k \in \{0,1,2,3\}$:*

1. *$W_k(i,3) \leq W_k(i,7)$. If $W_k(i,3) = W_k(i,7)$, then $\mathcal{R}_i[3] = 1$. If $W_k(i,3) < W_k(i,7)$, then $\mathcal{R}_i[3] = 0$.*
2. *$W_k(i,2) \leq W_k(i,6)$. If $W_k(i,2) < W_k(i,6)$, then $\mathcal{R}_i[2] = 0$ and $\mathcal{R}_i[6] = 1$.*

Proof *Case 1:* According to the definition of weight, $W_k(i+1,7) = W_k(i,3)$ (if $\mathcal{R}_i[3] = 1$); $W_k(i+1,7) = W_k(i,3) + 1$ (if $\mathcal{R}_i[3] = 0$). Also, $W_k(i+1,7) = W_k(i,7)$ as $\mathcal{R}_i[7] = 1$ for an NCCA. Hence, $W_k(i,3) \leq W_k(i,7)$. And, when $W_k(i,3) < W_k(i,7)$, $\mathcal{R}_i[3] = 0$. Otherwise, $\mathcal{R}_i[3] = 1$.

Case 2: As RMT 2 and RMT 6 are the sibling of RMT 3 and RMT 7 respectively, then at any level i, $W_k(i,2) \leq W_k(i,6)$ for each k. Further, $\mathcal{R}_i[2] = 0$ and $\mathcal{R}_i[6] = 1$ when $W_k(i,2) < W_k(i,6)$. □

The above results specify the possible weights of the RMTs as well as the possible rules of NCCAs. The following theorem lays down some conditions that the RMTs of rule R have to satisfy while R participates in the rule vector of an NCCA.

Theorem 5.2 *Each rule (R) of an n-cell NCCA, with periodic boundary ($n \geq 5$), satisfies the following conditions–*

1. $R[0] = 0$ *and* $R[7] = 1$.
2. $R[0] = R[4]$ *and* $R[1] = R[5]$ ***or*** $R[0] = R[1]$ *and* $R[4] = R[5]$.
3. $R[2] = R[6]$ *and* $R[3] = R[7]$, ***or*** $R[2] = R[3]$ *and* $R[6] = R[7]$.

Proof *Case 1*: The condition 1 is obtained directly from Lemma 8.

Let consider that $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i, \cdots, \mathcal{R}_{n-1})$ ($n \geq 5$) is an NCCA but the $\mathcal{R}_i$ does not follow conditions 2 and 3 for an $i \geq 2$. Therefore, all the 8 RMTs are present in $\cup_j \Gamma_k^{N_{i,j}}$ for any $k \in \{0, 1, 2, 3\}$. Without loss of generality let further consider that all the properties of NCCA are maintained by the rules $\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{i-1}$. Here, Cases 2 and 3 are proved by method of contradiction.

Case 2: Let first consider that $\mathcal{R}_i[0] = \mathcal{R}_i[4]$ and $\mathcal{R}_i[1] \neq \mathcal{R}_i[5]$. Now, as per Corollary 5.2, $W_k(i, 1) \neq W_k(i, 5)$. Further, RMT 5 is the sibling of RMT 4, and RMT 1 is the sibling of RMT 0. That is, $W_k(i, 4) = W_k(i, 5)$ and $W_k(i, 0) = W_k(i, 1)$ (by definition of weight). So, $W_k(i, 0) \neq W_k(i, 4)$, which is not possible when $\mathcal{R}_i[0] = \mathcal{R}_i[4]$ (Corollary 5.2). Hence, the $\mathcal{R}_i$ cannot be in $\mathcal{R}$ when $\mathcal{R}_i[0] = \mathcal{R}_i[4]$ and $\mathcal{R}_i[1] \neq \mathcal{R}_i[5]$. Similarly, it can be proved that $\mathcal{R}_i[1] = \mathcal{R}_i[5]$ but $\mathcal{R}_i[0] \neq \mathcal{R}_i[4]$ is not possible.

Next consider that $\mathcal{R}_i[0] = \mathcal{R}_i[1]$ and $\mathcal{R}_i[4] \neq \mathcal{R}_i[5]$. By Lemma 8, $\mathcal{R}_i[0] = 0$, so $\mathcal{R}_i[1] = 0$. Now, $\mathcal{R}_i[0] = 0$ and $\mathcal{R}_i[4] = 0$ imply $W_k(i, 0) = W_k(i, 4)$ (Corollary 5.2). Further, the RMTs 1 and 5 are the sibling of RMTs 0 and 4 respectively, so $W_k(i, 1) = W_k(i, 5)$ as $W_k(i, 4) = W_k(i, 5)$ and $W_k(i, 0) = W_k(i, 1)$. But it violates Corollary 5.2 as $\mathcal{R}_i[1] = 0$ and $\mathcal{R}_i[5] = 1$. So, $\mathcal{R}_i[0] = \mathcal{R}_i[1]$ and $\mathcal{R}_i[4] \neq \mathcal{R}_i[5]$ are not possible. Similarly, $\mathcal{R}_i[4] = 1$ and $\mathcal{R}_i[5] = 0$ are also not possible. Hence, the $\mathcal{R}_i$ cannot be in $\mathcal{R}$ when $\mathcal{R}_i[0] = \mathcal{R}_i[1]$ and $\mathcal{R}_i[4] \neq \mathcal{R}_i[5]$. Similarly, it can be proved that $\mathcal{R}_i[4] = \mathcal{R}_i[5]$ but $\mathcal{R}_i[0] \neq \mathcal{R}_i[1]$ is not possible.

Finally consider that $\mathcal{R}_i[0] \neq \mathcal{R}_i[4], \mathcal{R}_i[1] \neq \mathcal{R}_i[5], \mathcal{R}_i[0] \neq \mathcal{R}_i[1]$ and $\mathcal{R}_i[4] \neq \mathcal{R}_i[5]$. That means $\mathcal{R}_i[0] = \mathcal{R}_i[5]$ and $\mathcal{R}_i[1] = \mathcal{R}_i[4]$. Since $\mathcal{R}_i[0] \neq \mathcal{R}_i[4]$ and $\mathcal{R}_i[1] \neq \mathcal{R}_i[5]$, by Corollary 5.2, $W_k(i, 0) \neq W_k(i, 4)$ and $W_k(i, 1) \neq W_k(i, 5)$. Now, as per Corollary 5.5, $W_k(i, 5) \geq W_k(i, 1)$. In this case, $W_k(i, 5) > W_k(i, 1)$. Now, $\mathcal{R}_i[0] = \mathcal{R}_i[5]$ and $\mathcal{R}_i[0] \neq \mathcal{R}_i[1]$ means $\mathcal{R}_i[5] = 0$ and $\mathcal{R}_i[1] = 1$. The RMTs 1 and 5 are equivalent RMTs, so both of them contribute the same set of RMTs. But, the weight of the RMTs, in both the cases, are different because $W_k(i, 5) > W_k(i, 1)$, $\mathcal{R}_i[5] = 0$ and $\mathcal{R}_i[1] = 1$. According to Theorem 5.1, an RMT in $\Gamma_k^{N_{i,j}}$ with different weight is not possible. So, the consideration is false.

Combining all these, it can be concluded that the $\mathcal{R}_i$ has to obey the condition of Case 2.

Case 3: The proof for the RMTs 2, 3, 6 and 7 is identical upto the 0/1 exchange of RMTs 0, 1, 4 and 5.

For the conditions of Cases 2 and 3, all the 8 RMTs are required. Since $\cup_j \Gamma_k^{N_{n-2,j}}$ and $\cup_j \Gamma_k^{N_{n-1,j}}$ do not contain all the 8 RMTs (Points 4 and 5 of Definition 22), we can get all the RMTs at least in one level if $n - 3 \geq 2$. This implies $n \geq 5$.

However, the conditions of Cases 2 and 3 are also applicable to the first rule ($\mathcal{R}_0$) and second rule ($\mathcal{R}_1$). In periodic boundary, any rule can be considered as the first rule and this consideration does not alter the behavior of CA if the sequence of the rules remains the same. That is, the dynamical behavior of the two CAs—$(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i, \cdots, \mathcal{R}_{n-1})$ and $(\mathcal{R}_i, \mathcal{R}_{i+1}, \cdots, \mathcal{R}_{n-1}, \mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{i-1})$ are the same. Hence the proof. □

There are nine rules that follow the conditions of Theorem 5.2. These are the 136, 170, 184, 192, 204, 226, 238, 240 and 252, and referred to as

$$\mathcal{S}_{NCCA} = \{136, 170, 184, 192, 204, 226, 238, 240, 252\}.$$

For $n < 5$, some additional rules can be found that can form NCCAs. For example, if $n = 4$, then there are six additional rules for NCCAs—160, 172, 202, 216, 228 and 250. The CAs (160, 170, 250, 170) and (202, 216, 172, 228) are NCCAs. However, an arbitrary sequence of these rules cannot form an NCCA.

5.3 Analysis of Number Conserving CAs

To decide on the NCCA, it is first checked whether each rule of the CA is from $\mathcal{S}_{NCCA}$. If no, the CA is not an NCCA. Otherwise, reachability tree of the CA is developed and the theories presented in the earlier section are followed for decision. To facilitate the process of decision, some nodes are identified as *sub-nodes* of some other nodes of the same level.

Definition 33 A node $N_{i.p}$ is a *sub-node* of a node $N_{i.q}$ if $\Gamma_k^{N_{i.p}} \subseteq \Gamma_k^{N_{i.q}}$, for each $k \in \{0, 1, 2, 3\}$. It is denoted as $N_{i.p} \subseteq N_{i.q}$.

In Fig. 5.5, $N_{3.0}$ is the sub-node of $N_{3.2}$ ($N_{3.0} \subseteq N_{3.2}$). Similarly, $N_{3.1} \subseteq N_{3.3}$ and $N_{3.7} \subseteq N_{3.3}$.

It is obvious from Corollary 5.1 that an RMT $r \in \Gamma_k^{N_{i.p}}$ has the same weight as it is in $\Gamma_k^{N_{i.q}}$, for any $k \in \{0, 1, 2, 3\}$, when $N_{i.p} \subseteq N_{i.q}$. This implies that if the weights of RMTs at leaves of the sub-tree rooted at $N_{i.q}$ are 0, then the weights of RMTs at leaves of the sub-tree rooted at $N_{i.p}$ are also 0. Hence, sub-nodes have no additional role in deciding the NCCA, and the nodes, excluding their sub-nodes, can successfully decide on whether the CA is NCCA. During the construction of the tree, therefore, if a sub-tree of a node is found, it can be removed. This saves time and space in constructing the reachability tree while deciding on the NCCA. Following are the steps to decide on NCCA.

1. If any rule is not from the $\mathcal{S}_{NCCA}$, output 'No' and stop.
2. Form the root of reachability tree. Assign weights to RMTs at root as 0.
3. Get the nodes of the next level and find weights of RMTs at each node.
4. If Theorem 5.1 or Corollary 5.1 is disobeyed, output 'No' and stop.
5. Remove the sub-nodes. Proceed to the next level with remaining nodes.
6. Repeat steps 3–5 until the leaves are encountered.
7. If there is an RMT at a leaf node with non-zero weight, output 'No'; otherwise, output 'Yes'.

The above procedure can successfully decide whether a finite CA is NCCA. The number of nodes in the reachability tree does not grow exponentially due to the presence of sub-nodes at a level. Practically, the number of nodes excluding the sub-nodes at any level is limited. The upper limit is $4^4 = 256$. Further, only the nodes of two consecutive levels are to be stored, which signifies that the upper limit of memory requirement is constant.

The decision procedure can further be improved if possible weights of the RMTs are found out. If possible weights of an RMT $r \in \Gamma_k^{N_{i,j}}$ are known but $W_k(i, r)$ is not the one of the possible weights, then it can be concluded that the CA is not an NCCA. In this case, it is not required to wait for the last level to get a decision.

5.3.1 Possible Weights of RMTs

To find the possible weights of the RMTs of rules that participate in NCCA, Lemma 8 is instrumental. For each rule $\mathcal{R}_i$ of an NCCA, $\mathcal{R}_i[0] = 0$ and $\mathcal{R}_i[7] = 1$. Obviously, $W_0(i, 0) = W_3(i, 7) = 0$ for each $i \in \{0, 1, \cdots, n\}$. Since sibling RMTs have the same weights in any node, $W_0(i, 1) = W_3(i, 6) = 0$. Using these weights, the possible weights of the other RMTs can be found. It is to be observed that all of the 8 RMTs can be present in $\Gamma_k^{N_{i,j}}$ when $i \geq 2$. From the definition of weight, it is understood that the weights of RMTs of different levels are related. For example, $W_k(i + 1, 0) = W_k(i, 4) - 1$ if $\mathcal{R}_i[4] = 1$, and $W_k(i + 1, 0) = W_k(i, 4)$ if $\mathcal{R}_i[4] = 0$. Now, since $W_0(i + 1, 0) = 0$, the weight of RMT $4 \in \Gamma_0^{N_{i,j}}$ has to be either 0 or 1. Similarly, the weights of RMTs $2, 3 \in \Gamma_0^{N_{i,j}}$ can either be -1 or 0 because $W_0(i - 1, 1) = 0$ and $W_0(i, 2) = W_0(i - 1, 1) - 1$ if $\mathcal{R}_{i-1}[1] = 1$ and $W_0(i, 2) = W_0(i - 1, 1)$ if $\mathcal{R}_{i-1}[1] = 0$. Now, since the possible weights of RMT 3 is –1 or 0, the possible weights of RMTs 6 and 7 are –1, 0 and 1. Hence, the possible weights of all the RMTs that belong to $\Gamma_0^{N_{i,j}}$, are $W_0(i, 0) = W_0(i, 1) = 0$. Similarly, the possible weights of all the RMTs that belong to $\Gamma_3^{N_{i,j}}$ can be obtained considering $W_3(i, 6) = W_3(i, 7) = 0$. These weights are noted in Table 5.2.

The weights of RMTs that are in $\Gamma_1^{N_{i,j}}$ and $\Gamma_2^{N_{i,j}}$ can also be found. In case of $\Gamma_1^{N_{i,j}}$, only the RMTs 2 and 3 exist when $i = n$. So, for NCCA, $W_1(n, 2) = W_1(n, 3) = 0$. Further, $W_1(0, 2) = W_1(0, 3) = 0$. Hence, $W_1(1, 4)$ can be either 0 or 1 depending on $\mathcal{R}_0$. Now, if $W_1(1, 4) = 0$ and $\mathcal{R}_1[4] = 0$, then $W_1(2, 0) = W_1(2, 1) = 0$; and if

Table 5.2 Possible weights of the RMTs

RMTs of $\mathcal{R}_i$	Possible weights of RMTs in			
	$\Gamma_0^{N_{i,j}}$	$\Gamma_1^{N_{i,j}}$	$\Gamma_2^{N_{i,j}}$	$\Gamma_3^{N_{i,j}}$
0	0	0, 1	–1, 0	–1, 0, 1
1	0	0, 1	–1, 0	–1, 0, 1
2	–1, 0	–1, 0, 1	–2, –1, 0	–1, 0
3	–1, 0	–1, 0, 1	–2, –1, 0	–1, 0
4	0, 1	0, 1, 2	–1, 0, 1	0, 1
5	0, 1	0, 1, 2	–1, 0, 1	0, 1
6	–1, 0, 1	0, 1	–1, 0	0
7	–1, 0, 1	0, 1	–1, 0	0

$W_1(1, 4) = 0$ but $\mathcal{R}_1[4] = 1$, then $W_1(2, 0) = W_1(2, 1) = -1$. On the other hand, if $W_1(1, 4) = 1$, then the weights of RMTs 0, $1 \in \Gamma_1^{N_{2,j}}$ can be 0 or 1. From the above logic, the possible weights of RMTs 0, 1 can be –1, 0 and 1. However, once the RMT 0 gets a weight, the weight is carried forward to the lower levels, as in NCCA $\mathcal{R}_i[0] = 0$ for any rule $\mathcal{R}_i$. Hence, $W_1(n-1, 0) = W_1(n-1, 1) = -1$, when $W_1(2, 0) = -1$. If this is happened, then the $W_1(n, 2)$ and $W_1(n, 3)$ can never be 0 for any rule $\mathcal{R}_{n-1}$. This implies that –1 cannot be the possible weight for the RMTs 0, $1 \in \Gamma_1^{N_{i,j}}$ for any i. Hence, the possible weights of the RMTs 0, $1 \in \Gamma_1^{N_{i,j}}$ are 0 and 1. With the similar logic, the possible weights of the RMTs 6, $7 \in \Gamma_1^{N_{i,j}}$ are 0 and 1.

Now, $W_1(i, 2)$ (or $W_1(i, 3)$) and $W_1(i-1, 1)$ are related. Depending on $\mathcal{R}_{i-1}$, the weights of the RMTs 2, $3 \in \Gamma_1^{N_{i,j}}$ can be -1, 0 and 1. On the other hand, the $W_1(i, 4)$ and $W_1(i, 5)$ are dependent on $W_1(i-1, 6)$ and $\mathcal{R}_{i-1}$. Hence, the possible weights of the RMTs 4, $5 \in \Gamma_1^{N_{i,j}}$ are 0, 1 and 2. Hence, the possible weights of all the RMTs that exist in $\Gamma_1^{N_{i,j}}$ can be obtained. Following the above argument, the possible weights of the RMTs that exist in $\Gamma_2^{N_{i,j}}$ can be found.

Table 5.2 notes the possible weights of all the possible RMTs of a node. The first column is the RMTs of a rule. The next four columns state the possible weights of the RMTs that are in four consecutive sets of a node. The possible weights of the RMTs can improve the decision algorithm. As soon as an RMT assumes a weight which is not consistent with Table 5.2, the CA is declared as non-NCCA. The following conditions, derived from Table 5.2, are used while deciding the NCCA.

For $\Gamma_0^{N_{i,j}}$,

1. If $W_0(i, 2) = -1$, then $\mathcal{R}_i[2] = 0$.
2. If $W_0(i, 4) = 0$, then $\mathcal{R}_i[4] = 0$. If $W_0(i, 4) = 1$, then $\mathcal{R}_i[4] = 1$.
3. If $W_0(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$.
4. If $W_0(i, 6) = -1$, then $\mathcal{R}_i[6] = 0$. If $W_0(i, 6) = 1$, then $\mathcal{R}_i[6] = 1$.

For $\Gamma_1^{N_{i.j}}$,

5. If $W_1(i, 2) = -1$, then $\mathcal{R}_i[2] = 0$.
6. If $W_1(i, 3) = -1$, then $\mathcal{R}_i[3] = 0$. If $W_1(i, 3) = 1$, then $\mathcal{R}_i[3] = 1$.
7. If $W_1(i, 4) = 0$, then $\mathcal{R}_i[4] = 0$. If $W_1(i, 4) = 2$, then $\mathcal{R}_i[4] = 1$.
8. If $W_1(i, 5) = 2$, then $\mathcal{R}_i[5] = 1$.

For $\Gamma_2^{N_{i.j}}$,

9. If $W_2(i, 2) = -2$, then $\mathcal{R}_i[2] = 0$.
10. If $W_2(i, 3) = -2$, then $\mathcal{R}_i[3] = 0$. If $W_2(i, 3) = 0$, then $\mathcal{R}_i[3] = 1$.
11. If $W_2(i, 4) = -1$, then $\mathcal{R}_i[4] = 0$. If $W_2(i, 4) = 1$, then $\mathcal{R}_i[4] = 1$.
12. If $W_2(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$.

For $\Gamma_3^{N_{i.j}}$,

13. If $W_3(i, 1) = -1$, then $\mathcal{R}_i[1] = 0$. If $W_3(i, 1) = 1$, then $\mathcal{R}_i[1] = 1$.
14. If $W_3(i, 2) = -1$, then $\mathcal{R}_i[2] = 0$.
15. If $W_3(i, 3) = -1$, then $\mathcal{R}_i[3] = 0$. If $W_3(i, 3) = 0$, then $\mathcal{R}_i[3] = 1$.
16. If $W_3(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$.

5.3.2 *The Decision Algorithm*

The theories presented in the earlier sections lead to the following observations.

i. Weights of an RMT r of $\mathcal{R}_i$, that belongs to $\Gamma_k^{N_{i.j_1}}, \Gamma_k^{N_{i.j_2}}, \cdots$, are the same,
ii. Weight of an RMT does not depend on the node number at a level, but depends on the fact whether it is in $\Gamma_0^{N_{i.j}}$, $\Gamma_1^{N_{i.j}}$, $\Gamma_2^{N_{i.j}}$ or $\Gamma_3^{N_{i.j}}$, and
iii. Final decision is taken based only on the weights of RMTs at leaves.

Now, consider $\Gamma_k^i = \cup_j \Gamma_k^{N_{i.j}}$. By definition, the weight of an RMT $r \in \Gamma_k^0$ is 0. Then, the weights of all the RMTs of Γ_k^1 can be set by following $\mathcal{R}_0$. Similarly, the weights of all the RMTs of Γ_k^i, for any i $(0 \leq i \leq n)$, can also be set after knowing $\mathcal{R}_{i-1}$. During the computation of weight, if it is observed that the weight of an RMT is not consistent with Table 5.2, then the CA cannot be an NCCA. It can be noted that Γ_k^i contains all the 8 RMTs for all $i \geq 2$ (but $\leq n - 3$).

The above discussion points to the fact that, to decide a CA as NCCA, it is not essential to get all the nodes of a reachability tree. Rather it is sufficient to work only with Γ_0^i, Γ_1^i, Γ_2^i and Γ_3^i at any level i. Here, a *super node* $\mathcal{N}_i = (\Gamma_0^i, \Gamma_1^i, \Gamma_2^i, \Gamma_3^i)$ is considered. Any node at a level is a sub-node of the super node. That is, $N_{i.j} \subseteq \mathcal{N}_i$ for any $j \in \{0, 1, \cdots, 2^i - 1\}$.

The decision algorithm develops the *super node* of level i and finds the weights of the RMTs of this node. The Γ_k in Algorithm 4 stores the RMTs present in Γ_k^i of level i, and W_k stores the weight of RMTs in Γ_k for each $k \in \{0, 1, 2, 3\}$. The procedure FindNextWeight is used to find the super node of the next level and the

weights of the RMTs of super node. Its arguments are the rule $\mathcal{R}_i$, a set of RMTs Γ_k^i and their weights (that is, $W_k(i, r)$ for each $r \in \Gamma_k^i$).

Procedure FindNextWeight(R, Γ, W)

```
Temp Γ ← ∅ ;
foreach RMT r ∈ Γ do
    if r ∈ {0, 1, 4, 5} then
        if R[r]=1 then
            TempW[2r (mod 8)] ← W[r] − 1 ;
            TempW[(2r + 1) (mod 8)] ← W[r] − 1 ;
        end
        else
            TempW[2r (mod 8)] ← W[r] ;
            TempW[(2r + 1) (mod 8)] ← W[r] ;
        end
    end
    else
        if R[r]=0 then
            TempW[2r (mod 8)] ← W[r] + 1 ;
            TempW[(2r + 1) (mod 8)] ← W[r] + 1 ;
        end
        else
            TempW[2r (mod 8)] ← W[r] ;
            TempW[(2r + 1) (mod 8)] ← W[r] ;
        end
    end
    Temp Γ ← Temp Γ ∪ {2r (mod 8), (2r + 1) (mod 8)} ;
end
Assign Γ ← Temp Γ ;
       W ← TempW ;
```

Step 1 of Algorithm 4 forms the root (which is also a super node of level 0) of the reachability tree, and initializes the weights of RMTs. Next, it checks whether the rule $\mathcal{R}_i$ is in $\mathcal{S}_{NCCA}$ or not (Step 2 and Step 4). If $\mathcal{R}_i$ is not in $\mathcal{S}_{NCCA}$, the algorithm stops with negative answer. Otherwise, for $i = 0$, the weights of RMTs of $\mathcal{N}_1$, the super node of level 1 are found out. For $i > 0$, the conditions relating the weights of RMTs of $\mathcal{N}_i$ and $\mathcal{R}_i$ are checked (Step 5), and then the conditions of Corollaries 5.5 and 5.6 are verified (Step 6). Next, the weights of RMTs of $\mathcal{N}_{i+1}$ are found out. Step 5 implements the 16 conditions presented in Sect. 5.3.1. The conditions of Step 5 and Step 6 can be read as "$W_0[2] = -1 \implies \mathcal{R}_i[2] = 0$" : If $W_0[2] = -1$, and $\mathcal{R}_i[2] = 0$, then the condition is true. But if $W_0[2] = -1$ but $\mathcal{R}_i[2] \neq 0$, the condition is false.

If any of the conditions of Steps 5 and 6 is violated, then the algorithm outputs 'No'. This procedure repeats when $i \leq n - 3$. For $i = n - 2$ and $i = n - 1$, it first executes Step 9 and then Step 6. Step 7 is followed after that. If any one condition of

Algorithm 4: Decide whether a CA in periodic boundary is NCCA

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in periodic boundary.
Output: 'Yes' if the CA is NCCA; 'No' otherwise.

Step 1 : Set $\Gamma_0 \leftarrow \{0, 1\}$, $\Gamma_1 \leftarrow \{2, 3\}$, $\Gamma_2 \leftarrow \{4, 5\}$, $\Gamma_3 \leftarrow \{6, 7\}$,
$W_k[r] \leftarrow 0$ if $r \in \Gamma_k$, for each k, $0 \leq k \leq 3$.
Step 2 : If $\mathcal{R}_0 \notin \{136, 170, 184, 192, 204, 226, 238, 240, 252\}$ then report 'No' and return.
Otherwise call *FindNextWeight*($\mathcal{R}_0$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$.
Step 3 : Set $i \leftarrow 1$.
Step 4 : If $\mathcal{R}_i \notin \{136, 170, 184, 192, 204, 226, 238, 240, 252\}$ then report 'No' and Return.
Step 5 : If any of the following conditions is not satisfied then report 'No' and Return.
(i) $W_0[2] = -1 \implies \mathcal{R}_i[2] = 0$;
(ii) $(W_0[4] = 0 \implies \mathcal{R}_i[4] = 0) \wedge (W_0[4] = 1 \implies \mathcal{R}_i[4] = 1)$;
(iii) $W_0[5] = 1 \implies \mathcal{R}_i[5] = 1$;
(iv) $(W_0[6] = -1 \implies \mathcal{R}_i[6] = 0) \wedge (W_0[6] = 1 \implies \mathcal{R}_i[6] = 1)$;
(v) $W_1[2] = -1 \implies \mathcal{R}_i[2] = 0$;
(vi) $(W_1[3] = -1 \implies \mathcal{R}_i[3] = 0) \wedge (W_1[3] = 1 \implies \mathcal{R}_i[3] = 1)$;
(vii) $(W_1[4] = 0 \implies \mathcal{R}_i[4] = 0) \wedge (W_1[4] = 2 \implies \mathcal{R}_i[4] = 1)$;
(viii) $W_1[5] = 2 \implies \mathcal{R}_i[5] = 1$;
(ix) $W_2[2] = -2 \implies \mathcal{R}_i[2] = 0$;
(x) $(W_2[3] = -2 \implies \mathcal{R}_i[3] = 0) \wedge (W_2[3] = 0 \implies \mathcal{R}_i[3] = 1)$;
(xi) $(W_2[4] = -1 \implies \mathcal{R}_i[4] = 0) \wedge (W_2[4] = 1 \implies \mathcal{R}_i[4] = 1)$;
(xii) $W_2[5] = 1 \implies \mathcal{R}_i[5] = 1$;
(xiii) $(W_3[1] = -1 \implies \mathcal{R}_i[1] = 0) \wedge (W_3[1] = 1 \implies \mathcal{R}_i[1] = 1)$;
(xiv) $W_3[2] = -1 \implies \mathcal{R}_i[2] = 0$;
(xv) $(W_3[3] = -1 \implies \mathcal{R}_i[3] = 0) \wedge (W_3[3] = 0 \implies \mathcal{R}_i[3] = 1)$;
(xvi) $(W_3[5] = 1 \implies \mathcal{R}_i[5] = 1)$;
Step 6 : If any of the following conditions is not satisfied for any $k \in \{0, 1, 2, 3\}$ then report 'No' and return;
(i) $W_k[4] = W_k[0] \implies \mathcal{R}_i[4] = 0$;
(ii) $W_k[4] > W_k[0] \implies \mathcal{R}_i[4] = 1$;
(iii) $W_k[5] > W_k[1] \implies (\mathcal{R}_i[5] = 1 \wedge \mathcal{R}_i[1] = 0)$;
(iv) $W_k[3] = W_k[7] \implies \mathcal{R}_i[3] = 1$;
(v) $W_k[3] < W_k[7] \implies \mathcal{R}_i[3] = 0$;
(vi) $W_k[2] < W_k[6] \implies (\mathcal{R}_i[2] = 0 \wedge \mathcal{R}_i[6] = 1)$;
Step 7 : Call *FindNextWeight*($\mathcal{R}_i$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;
Step 8 : $i \leftarrow i + 1$;
Step 9 : If $(i < n - 2)$ then goto Step 4 ;
If $(i = n - 2)$ then
Set $\Gamma_0 \leftarrow \Gamma_0 \cap \{0, 2, 4, 6\}$; $\Gamma_1 \leftarrow \Gamma_1 \cap \{0, 2, 4, 6\}$;
$\Gamma_2 \leftarrow \Gamma_2 \cap \{1, 3, 5, 7\}$; $\Gamma_3 \leftarrow \Gamma_3 \cap \{1, 3, 5, 7\}$;
goto Step 6;
If $(i = n - 1)$ then
Set $\Gamma_0 \leftarrow \Gamma_0 \cap \{0, 4\}$; $\Gamma_1 \leftarrow \Gamma_1 \cap \{1, 5\}$;
$\Gamma_2 \leftarrow \Gamma_2 \cap \{2, 6\}$; $\Gamma_3 \leftarrow \Gamma_3 \cap \{3, 7\}$;
goto Step 6;
Step 10 : If $W_k[r] \neq 0$ for any $r \in \Gamma_k$ and any $k \in \{0, 1, 2, 3\}$ then report 'No';
Otherwise, 'Yes';

Step 6 is violated, then it reports 'No'. At the end (for $i = n$), for each Γ_k, the weight of each RMT is checked. Finally, if any non-zero weight is found, then it reports 'No'; otherwise, 'Yes'.

Complexity: The time requirement of Algorithm 4 depends on n, the number of cells of the CA. Hence, the worst case time complexity of Algorithm 4 is $O(n)$. Whereas, the space complexity is $O(1)$.

Example 5.2 Let consider that the CA (192, 136, 184, 252, 204, 238) is input to Algorithm 4. The algorithm first forms root node $\mathcal{N}_0$ with four sibling pairs ({0,1}, {2,3}, {4,5}, {6,7}), and assigns the weight to each RMT as 0 (Step 1). Then the weights of RMTs are found out that are in $\mathcal{N}_1$ (level 1) for $\mathcal{R}_0 = 192$. The RMTs of level 1 and the weights of those RMTs are shown in node $\mathcal{N}_1$ of Fig. 5.6. The next rule $\mathcal{R}_1 = 136$ is in $\mathcal{S}_{NCCA}$ (Step 4). While checking conditions in Step 5, it is found that the only condition (x) of it is applicable. The condition (x) of Step 5 –that is, $W_2[3] = 0 \implies \mathcal{R}_i[3] = 1$, means that in Γ_2^1, if the weight of RMT 3 is 0, then $\mathcal{R}_1[3] = 1$. For rule 136, the next state of RMT 3 is 1, so the condition (x) of Step 5 is true. In this way, the conditions of Step 6 are also verified. Note that in

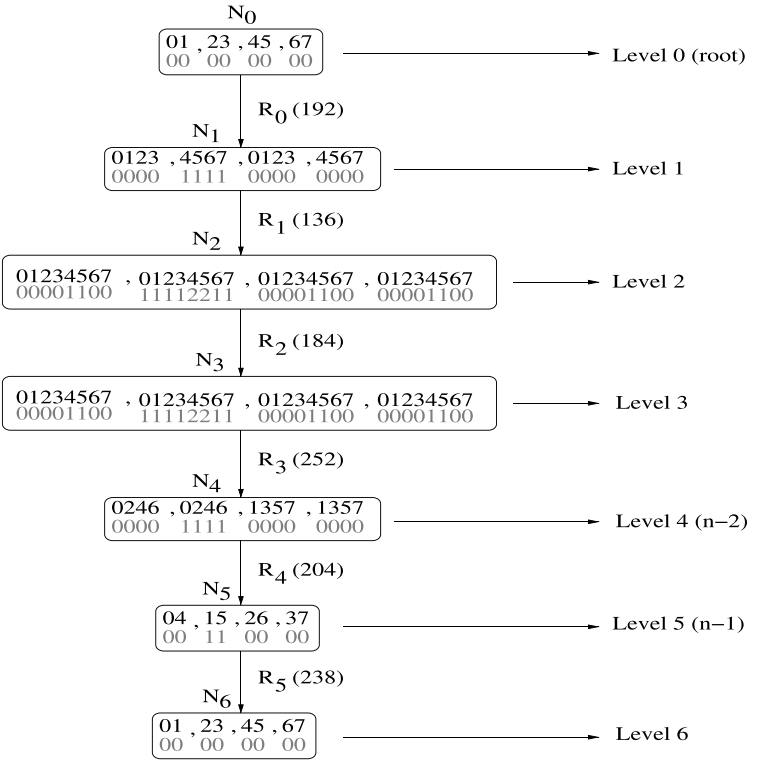


Fig. 5.6 Super node of each level for CA (192, 136, 184, 252, 204, 238)

case of rule 136, no condition of Step 5 and Step 6 is disobeyed. Now, in Step 7, the weights of RMTs that are in $\mathcal{N}_2$ are found out (level 2 of Fig. 5.6). For rule 184 ($\mathcal{R}_2$) and 252 ($\mathcal{R}_3$), the similar processing are done, and it is observed that no condition is disobeyed. For rule 204, which is $\mathcal{R}_{n-2}$, some RMTs are removed (Step 9) and then the conditions of Step 6 are verified. Here, only the conditions (i) and (iv) of Step 6 are applicable. From the node $\mathcal{N}_4$ (level $n-2$) of Fig. 5.6, it is observed that in Γ_0^{n-2} and Γ_1^{n-2}, the weights of RMTs 4 and 0 are equal, and $\mathcal{R}_{n-2}[4] = 0$. Similarly, in Γ_2^{n-2} and Γ_3^{n-2}, the weights of RMTs 3 and 7 are equal, and $\mathcal{R}_{n-2}[3] = 1$. Since $204[4] = 0$ and $204[3] = 1$, the conditions (i) and (iv) of Step 6 are true. Next, the weights of RMTs of $\mathcal{N}_5$ (level $n-1$ of Fig. 5.6) are found out. Lastly, for rule 238 ($= \mathcal{R}_{n-1}$), the conditions of Step 6 are tested. From the node $\mathcal{N}_5$ of Fig. 5.6, it is observed that in Γ_0^{n-1}, the weights of RMTs 4 and 0 are equal, and $\mathcal{R}_{n-1}[4] = 0$. Similarly, in Γ_3^{n-1}, the weights of RMTs 3 and 7 are equal, and $\mathcal{R}_{n-1}[3] = 1$. Since $238[4] = 0$ and $238[3] = 1$, the conditions (i) and (iv) of Step 6 are true. Again the weight of each RMT of $\mathcal{N}_6$ (level 6 of Fig. 5.6) is computed. Here, there is no RMT with non-zero weight (Step 10). So, Algorithm 4 returns 'Yes'. The reachability tree of the rule vector $\mathcal{R}$ is shown in Fig. 5.7.

Introduction of super node simplifies the decision algorithm by eliminating the need of explicit construction of reachability tree. Further, characterization has been done to get the sequence of rules of NCCAs and alternative scheme is developed to decide the NCCA.

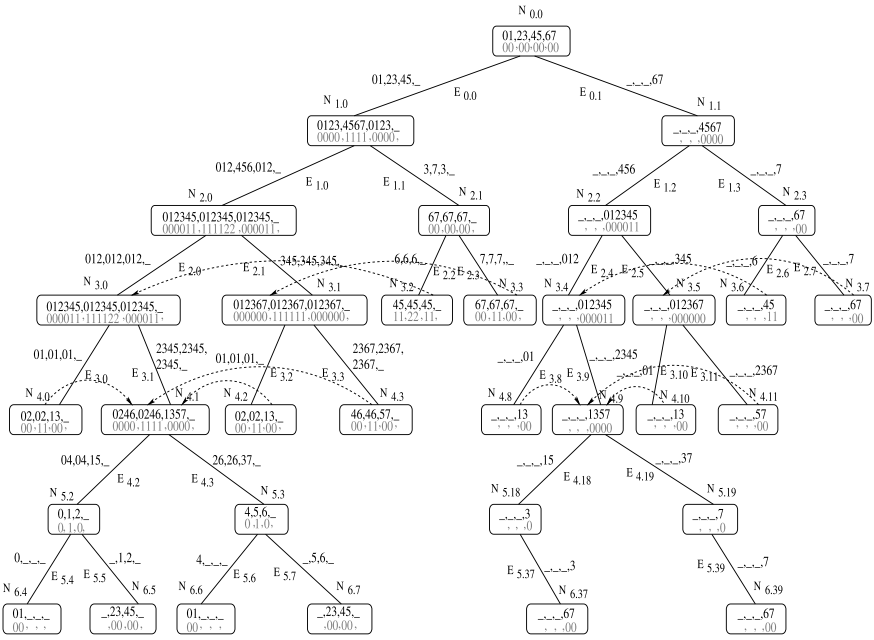


Fig. 5.7 The reachability tree of CA (192, 136, 184, 252, 204, 238) in periodic boundary

5.3.3 Rule Sequences for NCCA

It is clear from Table 5.2 that the RMTs can assume at most three possible weights. For example, a set of possible weights of the RMT 2 is $\{-2, -1, 0\}$. Hence, to test if an RMT is really assuming one of the possible weights, only three consecutive rules of a given rule vector are to be considered. It is shown that only nine rules—136, 170, 184, 192, 204, 226, 238, 240 and 252 can form NCCAs. Now, the target can be to find the conditions of getting a sequence of three rules (triplet) that obey the weights of Table 5.2. In fact, a computer program can be developed to exhaustively test which of the possible triplets can be part of an NCCA. Finding precise condition of getting triplets by applying the above theories of weights is abstained here. The only possible triplets are shown in Table 5.3. Any rule from the first column of a row, the rule of second column, and any rule from the third column of the row form a triplet. The number of possible triplets is, therefore, 66.

For a given CA, if all the sets of three consecutive rules are triplet of Table 5.3, the CA is an NCCA; otherwise, it is not an NCCA. For the example CA (192, 136, 184, 252, 204, 238), $\mathcal{R}_0 = 192$. Then, $\mathcal{R}_{-1} = \mathcal{R}_5 = 238$ and $\mathcal{R}_1 = 136$. The sequence of rules (238, 192, 136) is a triplet as per Table 5.3. The other sequences of three rules are also the triplets (Table 5.3). Therefore, the CA is an NCCA.

Table 5.3 can also be used to synthesize an NCCA of any $n \geq 5$. But the next section provides the microscopic details of the rules and synthesizes NCCA. Such a generic approach can guide to develop algorithms for other kind of CA synthesis.

Table 5.3 Triplets $\langle \mathcal{R}_{i-1}, \mathcal{R}_i, \mathcal{R}_{i+1} \rangle$ for NCCAs

$\mathcal{R}_{i-1}$	$\mathcal{R}_i$	$\mathcal{R}_{i+1}$
192, 204, 240, 252	136	184, 252
192, 204, 240, 252	170	240
136, 184	184	184, 252
226, 238	192	136, 170, 204, 238
192, 204, 240, 252	204	136, 170, 204, 238
226, 238	226	192, 226
192, 204, 240, 252	238	192, 226
170	240	136, 170, 204, 238
136, 184	252	136, 170, 204, 238
170	170	170
240	240	240

5.4 Synthesis of Number Conserving CAs

For synthesis of an n-cell NCCA, the rules $\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1}$ are to be selected following the theories reported in the earlier sections. As a first step, an arbitrary rule from $\mathcal{S}_{NCCA}$ is chosen as $\mathcal{R}_0$. To get $\mathcal{R}_1$ (and other rules), the next states of individual RMTs of the rule are appropriately set. For example, the next states of the RMTs 0 and 7 of a rule $\mathcal{R}_i$ are to be set to 0 and 1 respectively (Lemma 8). Similarly, if weight of the RMT $2 \in \Gamma_0^{N_{i,j}}$ is -1, $\mathcal{R}_i[2]$ is set as 0 to keep the weight of the RMT consistent with Table 5.2.

The synthesis procedure, like the decision algorithm, develops super node $\mathcal{N}_i$ for each level i, finds weights of the RMTs in Γ_k^i ($k \in \{0, 1, 2, 3\}$), and based on the weights, it *synthesizes* $\mathcal{R}_i$. However, special care is needed while choosing $\mathcal{R}_1, \mathcal{R}_{n-2}$ and $\mathcal{R}_{n-1}$.

5.4.1 Selection of $\mathcal{R}_i$

As a first step to select $\mathcal{R}_i$, the following are to be set: $\mathcal{R}_i[0] \leftarrow 0$ and $\mathcal{R}_i[7] \leftarrow 1$. Now, according to Corollary 5.5, for any k, $W_k(i, 4) \geq W_k(i, 0)$. However, if $W_k(i, 4) = W_k(i, 0)$, then in an NCCA, $\mathcal{R}_i[4] = \mathcal{R}_i[0]$ (Corollary 5.2). That is, to synthesize NCCA, we need to set $\mathcal{R}_i[4] \leftarrow 0$. On the other hand, if $W_k(i, 4) > W_k(i, 0)$, then $\mathcal{R}_i[4] \leftarrow 1$ (Corollary 5.2). If $W_k(i, 4) > W_k(i, 0)$, for a k, say $k = 0$, then $W_k(i, 4) > W_k(i, 0)$ for all other $k \in \{1, 2, 3\}$.

As the RMTs 1 and 5 are sibling of the RMTs 0 and 4 respectively, then if $\mathcal{R}_i[4] \neq \mathcal{R}_i[0]$, we set $\mathcal{R}_i[5] \leftarrow 1$ and $\mathcal{R}_i[1] \leftarrow 0$ following Theorem 5.2. However, if $\mathcal{R}_i[4] = \mathcal{R}_i[0]$, the next states of the RMTs 1 and 5 cannot be set (Theorem 5.2). Hence, when $\mathcal{R}_i[4] = \mathcal{R}_i[0]$, to set the next states of the RMTs 1 and 5, some additional conditions relating weights of these RMTs are to be checked. The following conditions of Sect. 5.3.1, followed by any NCCA, are to be checked.

(i) If $W_0(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$	(ii) If $W_1(i, 5) = 2$, then $\mathcal{R}_i[5] = 1$
(iii) If $W_2(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$	(iv) If $W_3(i, 5) = 1$, then $\mathcal{R}_i[5] = 1$
(v) If $W_3(i, 1) = 1$, then $\mathcal{R}_i[1] = 1$	(vi) If $W_3(i, 1) = -1$, then $\mathcal{R}_i[1] = 0$

Therefore, when any of the conditions (i)–(iv) is true, the $\mathcal{R}_i[5] \leftarrow 1$. As per Theorem 5.2, we can set $\mathcal{R}_i[1] \leftarrow 1$. If condition (v) is true, we set $\mathcal{R}_i[1] \leftarrow 1$ and to follow Theorem 5.2, we set $\mathcal{R}_i[5] \leftarrow 1$. Similarly, when condition (vi) is true, we set $\mathcal{R}_i[1] \leftarrow 0$ and $\mathcal{R}_i[5] \leftarrow 0$. It is to be noted that when condition (vi) is true, none of the conditions (i) to (v) can be true. Otherwise, weights of the RMTs become inconsistent.

Following the same rationale, next states of the rest of the RMTs of $\mathcal{R}_i$ are set. According to Corollary 5.6, for any k, $W_k(i, 3) \leq W_k(i, 7)$. So, if $W_k(i, 3) = W_k(i, 7)$,

then $\mathcal{R}_i[3] \leftarrow 1$ otherwise, $\mathcal{R}_i[3] \leftarrow 0$ (Corollary 5.2). As the RMTs 2 and 6 are the sibling of RMTs 3 and 7 respectively, so if $\mathcal{R}_i[3] \neq \mathcal{R}_i[7]$, then we set $\mathcal{R}_i[2] \leftarrow 0$ and $\mathcal{R}_i[6] \leftarrow 1$ (Theorem 5.2). When $\mathcal{R}_i[3] = \mathcal{R}_i[7]$, the next states of the RMTs 2 and 6 cannot be set following Theorem 5.2. The following conditions given in Sect. 5.3.1 also guide to set the $\mathcal{R}_i[2]$ and $\mathcal{R}_i[6]$.

(a) If $W_0(i, 2) = -1$ then $\mathcal{R}_i[2] = 0$	(b) If $W_0(i, 6) = -1$ then $\mathcal{R}_i[6] = 0$
(c) If $W_1(i, 2) = -1$ then $\mathcal{R}_i[2] = 0$	(d) If $W_2(i, 2) = -2$ then $\mathcal{R}_i[2] = 0$
(e) If $W_3(i, 2) = -1$ then $\mathcal{R}_i[2] = 0$	(f) If $W_0(i, 6) = 1$ then $\mathcal{R}_i[6] = 1$

If any of the conditions (a)–(e) is true, then the synthesis algorithm sets $\mathcal{R}_i[2] \leftarrow 0$ and $\mathcal{R}_i[6] \leftarrow 0$. Similarly, if condition (f) is true, then it sets $\mathcal{R}_i[2] \leftarrow 1$ and $\mathcal{R}_i[6] \leftarrow 1$ (Steps 6 and 7 of Algorithm 5).

While setting the RMTs of $\mathcal{R}_i$, it may so happen that some of the RMTs of $\mathcal{R}_i$ are not set. For such RMTs, the next states are set randomly satisfying the condition of Theorem 5.2.

5.4.2 *Selection of $\mathcal{R}_1$*

Though it is not mentioned explicitly, but selection process of a rule $\mathcal{R}_i$ expects the presence of all the RMTs in a set Γ_k^i of a super node $\mathcal{N}_i$. For example, if we want to check whether $W_k(i, 4)$ is greater than $W_k(i, 0)$, then RMTs 0 and 4 should be in Γ_k^i. However, in the super node $\mathcal{N}_1$, RMT 0 is in Γ_0^1 but not the RMT 4. In fact, all of the 8 RMTs are not present in any Γ_k^1 (Fig. 5.7). Therefore, if the selection process of $\mathcal{R}_i$ is applied to select $\mathcal{R}_1$, the next state of many RMTs may remain unfilled. If such RMTs are filled up arbitrarily, inconsistencies in weights may arise. Hence, extra measures are to be taken while choosing $\mathcal{R}_1$. As an option, let relate the weights of the RMTs, present in different sets of $\mathcal{N}_1$. As always, the $\mathcal{R}_1[0]$ is set to 0 and $\mathcal{R}_1[7]$ as 1. However, the following four conditions, out of sixteen, reported in Sect. 5.3.1, can arise in case of $\mathcal{N}_1$, when $\mathcal{R}_0$ is a number conserving rule.

(i) If $W_0(1, 2) = -1$, then $\mathcal{R}_1[2] = 0$	(ii) If $W_1(1, 4) = 0$, then $\mathcal{R}_1[4] = 0$
(iii) If $W_2(1, 3) = 0$, then $\mathcal{R}_1[3] = 1$	(iv) If $W_3(1, 5) = 1$, then $\mathcal{R}_1[5] = 1$

If any of the above conditions is true, then the next state of the corresponding RMT can be set directly. For example, if $W_1(1, 4) = 0$, then to get an NCCA, $\mathcal{R}_1[4]$ is to be set to 0. The important point is that the condition (i) and (iii) (similarly, condition (ii) and (iv)) are mutually exclusive—that is, both of these cannot be satisfied simultaneously. However, for synthesis, the next state of RMT 2 is set first. Here, Corollary 5.4 and Theorem 5.2 are instrumental. However, in this discussion,

the RMTs and their weights of two sets – Γ_0^1 and Γ_1^1 are mainly used. The similar outcome can be found if the other sets, like Γ_2^1 and Γ_3^1, are used.

Now, if $W_0(1, 2) = -1$, then $\mathcal{R}_1[2]$ is set as 0. Hence, $W_0(2, 4) = W_0(2, 5) = 0$. That is, $W_0(2, 0) - W_0(2, 4) = 0$. As per Corollary 5.4, $W_1(2, 0) - W_1(2, 4)$ is also to be 0. To get this, $\mathcal{R}_1[4]$ is set to 0 and $\mathcal{R}_1[6]$ as 1 when $W_1(1, 4) = W_1(1, 6)$. When $W_1(1, 4) = W_1(1, 6) = 0$, the condition (ii) is true. However, this condition needs to be checked, as $\mathcal{R}_1[4]$ is already set to 0 as per Corollary 5.4. It can be noticed that if $W_1(1, 4) = W_1(1, 6)$, then $W_3(1, 4) = W_3(1, 6)$ when $\mathcal{R}_0$ is in $\mathcal{S}_{NCCA}$. When $\mathcal{R}_1[6]$ is set to 1, then following Theorem 5.2, $\mathcal{R}_1[3]$ is set to 0. The rest of the RMTs—that is, RMTs 1 and 5 are arbitrarily set either as $\mathcal{R}_1[1] \leftarrow \mathcal{R}_1[5] \leftarrow 0$ or as $\mathcal{R}_1[1] \leftarrow \mathcal{R}_1[5] \leftarrow 1$.

However, when $W_1(1, 4) \neq W_1(1, 6)$, then the only possibility is – $W_1(1, 4) = 1$ and $W_1(1, 6) = 0$ (and, $W_3(1, 4) = 1$ and $W_3(1, 6) = 0$). Further, $W_3(1, 5) = W_3(1, 4) = 1$. So, the condition (iv) is followed, and $\mathcal{R}_1[5]$ is set to 1. To meet the condition of Corollary 5.4 at level 2, RMTs 4 and 6 are arbitrarily set: either $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 0$ or $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 1$, because when the condition (iv) is true, then the condition (ii) cannot be true. As the next step, the $\mathcal{R}_1[1]$ can be set 1 and $\mathcal{R}_1[3]$ as 1 if $\mathcal{R}_1[4] = 0$, set $\mathcal{R}_1[1] \leftarrow 0$ and $\mathcal{R}_1[3] \leftarrow 0$ otherwise (Theorem 5.2).

Now, if $W_0(1, 2) \neq -1$—that is, if $W_0(1, 2) = 0$, $\mathcal{R}_1[2]$ can be 0 or 1. Let us set $\mathcal{R}_1[2] \leftarrow 0$. Then, $W_0(2, 4) = W_0(2, 5) = 1$, and $W_0(2, 0) - W_0(2, 4) = -1$. As in the previous case, we need to set the same next state to $\mathcal{R}_1[4]$ and $\mathcal{R}_1[6]$ when $W_1(1, 4) = W_1(1, 6)$, in order to get $W_1(2, 0) - W_1(2, 4) = -1$ (otherwise, Corollary 5.4 is violated). But if $W_1(1, 4) = 0$, then to satisfy the condition (ii), we need to set $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 0$. Now as per Theorem 5.2, $\mathcal{R}_1[3] \leftarrow 1$. On the other hand, if $W_2(1, 3) = 0$, then following the condition (iii), $\mathcal{R}_1[3]$ is set to 1, $\mathcal{R}_1[6] \leftarrow 0$ (Theorem 5.2). Hence, $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 0$. However, if $W_1(1, 4) = W_1(1, 6) = 1$, then arbitrarily set either $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 0$ or $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 1$, and RMT 3 is set following Theorem 5.2. However, if $W_1(1, 4) = 1$ and $W_1(1, 6) = 0$ (that is, when $W_1(1, 4) \neq W_1(1, 6)$), then to meet the condition of Corollary 5.4 at level 2, $\mathcal{R}_1[4] \leftarrow 1$ and $\mathcal{R}_1[6] \leftarrow 0$. When $W_1(1, 4) \neq W_1(1, 6)$, then $W_3(1, 5) = 1$, so we set $\mathcal{R}_1[5] \leftarrow 1$. Now following Theorem 5.2, $\mathcal{R}_1[1] \leftarrow 0$ and $\mathcal{R}_1[3] \leftarrow 1$.

Now, let set $\mathcal{R}_1[2] \leftarrow 1$ when $W_0(1, 2) = 0$. Following Theorem 5.2, $\mathcal{R}_1[3] \leftarrow 1$ and $\mathcal{R}_1[6] \leftarrow 1$. When $W_0(1, 2) = 0$, then $W_0(2, 4) = W_0(2, 5) = 0$ and $W_0(2, 0) - W_0(2, 4) = 0$. Following that rationale, the rest of the RMTs an be as following: If $W_1(1, 4) = W_1(1, 6)$, then set $\mathcal{R}_1[4] \leftarrow 0$; otherwise, $\mathcal{R}_1[4] \leftarrow 1$. Now, if $\mathcal{R}_1[4] = 0$, then arbitrarily set either $\mathcal{R}_1[1] \leftarrow \mathcal{R}_1[5] \leftarrow 0$ or $\mathcal{R}_1[1] \leftarrow \mathcal{R}_1[5] \leftarrow 1$; otherwise, $\mathcal{R}_1[1] \leftarrow 0$ and $\mathcal{R}_1[5] \leftarrow 1$.

5.4.3 Selection of $\mathcal{R}_{n-2}$ and $\mathcal{R}_{n-1}$

For selection of $\mathcal{R}_{n-2}$ and $\mathcal{R}_{n-1}$, the general technique of choosing a rule $\mathcal{R}_i$ cannot be used. This is to avoid non-zero weights of RMTs at leaves of the reachability tree. Therefore, additional restrictions are to be imposed while selecting $\mathcal{R}_{n-2}$ and $\mathcal{R}_{n-1}$.

Selection of $\mathcal{R}_{n-1}$: All the RMTs cannot be present in $\Gamma_k^{N_{n-1.j}}$ of a node of level $n-1$ for any $k \in \{0, 1, 2, 3\}$ (Point 5 of Definition 22), and the RMTs of Γ_k^{n-1} cannot attain all the possible weights of Table 5.2. For example, no RMT of Γ_k^{n-1} can have weight 2 or –2. The possible weights of RMTs at the super node $\mathcal{N}_{n-1} = (\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\})$ are the following:

(i) $W_0(n-1, 4) = 0$ or 1	(ii) $W_1(n-1, 1) = 0$ or 1
(iii) $W_1(n-1, 5) = 0$ or 1	(iv) $W_2(n-1, 2) = 0$ or –1
(v) $W_2(n-1, 6) = 0$ or –1	(vi) $W_3(n-1, 3) = 0$ or –1

Hence, $\mathcal{R}_{n-2}$ has to be selected in such a way that the RMTs can attain the above possible weights. That is, if $W_0(n-1, 4) = 0$, then set $\mathcal{R}_{n-1}[4] \leftarrow 0$; otherwise, set $\mathcal{R}_{n-1}[4] \leftarrow 1$. Similarly, set $\mathcal{R}_{n-1}[3] \leftarrow 1$ if $W_3(n-1, 3) = 0$, otherwise set $\mathcal{R}_{n-1}[3] \leftarrow 0$. However, the following restrictions are imposed while selecting $\mathcal{R}_{n-1}$.

1. If $W_1(n-1, r) = 0$, where $r \in \{1, 5\}$, then set $\mathcal{R}_{n-1}[r] \leftarrow 0$. Otherwise, set $\mathcal{R}_{n-1}[r] \leftarrow 1$.
2. If $W_2(n-1, r) = 0$, where $r \in \{2, 6\}$, then set $\mathcal{R}_{n-1}[r] \leftarrow 1$. Otherwise, set $\mathcal{R}_{n-1}[r] \leftarrow 0$.

Selection of $\mathcal{R}_{n-2}$: As per Point 5 of Definition 22, all the RMTs cannot be present in $\Gamma_k^{n-2.j}$ of an arbitrary node of level $n-2$ for any $k \in \{0, 1, 2, 3\}$. In general, the super node $\mathcal{N}_{n-1} = (\{0, 2, 4, 6\}, \{0, 2, 4, 6\}, \{1, 3, 5, 7\}, \{1, 3, 5, 7\})$. Though the RMTs of $\mathcal{N}_{n-2}$ can attain all possible weights as described in Table 5.2, the next states of the RMTs of $\mathcal{R}_{n-2}$ are to be selected in such a fashion that the possible weights of RMTs of $\mathcal{N}_{n-1}$ can be the above stated weights only. However, if the following conditions arise, then the similar actions are to be taken as in $\mathcal{R}_i$:

1. If $W_k(n-2, 4) = W_k(n-2, 0), k \in \{0, 1\}$, then set $\mathcal{R}_{n-2}[4] \leftarrow 0$. Otherwise, set $\mathcal{R}_{n-2}[4] \leftarrow 1$.
2. If $W_k(n-2, 3) = W_k(n-2, 7), k \in \{2, 3\}$, then set $\mathcal{R}_{n-2}[3] \leftarrow 1$. Otherwise, set $\mathcal{R}_{n-2}[3] \leftarrow 0$.
3. If $W_k(n-2, 2) < W_k(n-2, 6), k \in \{0, 1\}$, then set $\mathcal{R}_{n-2}[2] \leftarrow 0$ and $\mathcal{R}_{n-2}[6] \leftarrow 1$.
4. If $W_k(n-2, 5) > W_k(n-2, 1), k \in \{2, 3\}$, then set $\mathcal{R}_{n-2}[5] \leftarrow 1$ and $\mathcal{R}_{n-2}[1] \leftarrow 0$.

If all the conditions, related to weight are true, then the next states of all the RMTs of $\mathcal{R}_{n-2}$ can be set. However, if one or more conditions are not met, then some RMTs remain unfilled and a special care is to be taken. Following are the conditions, if arise, the next states of RMTs are set.

1. If $W_0(n-2,2) = W_0(n-2,6) = -1$, then set $\mathcal{R}_{n-2}[2] \leftarrow 0$ and $\mathcal{R}_{n-2}[6] \leftarrow 0$. Because both the RMTs contribute RMT 4 at level $(n-1)$ with weight $W_0(n-1,4) = 0$. If $\mathcal{R}_{n-2}[2]$ and $\mathcal{R}_{n-2}[6]$ are 1, then they contribute RMT 4 at level $(n-1)$ with weight $W_0(n-1,4) = -1$ that cannot contribute RMTs 0, 1 at leaves with weight 0.
2. If $W_1(n-2,2) = W_1(n-2,6) = 1$, then set $\mathcal{R}_{n-2}[2] \leftarrow 1$ and $\mathcal{R}_{n-2}[6] \leftarrow 1$. Here, both the RMTs contribute RMT 5 at level $(n-1)$ with weight $W_1(n-1,5) = 1$. If $\mathcal{R}_{n-2}[2]$ and $\mathcal{R}_{n-2}[6]$ are 0, then they contribute RMT 5 at level $(n-1)$ with weight $W_1(n-1,5) = 2$ that cannot contribute RMTs 2, 3 at leaves with weight 0.
3. If $W_2(n-2,1) = W_2(n-2,5) = -1$, then set $\mathcal{R}_{n-2}[1] \leftarrow 0$ and $\mathcal{R}_{n-2}[5] \leftarrow 0$. Because both the RMTs contribute RMT 2 at level $(n-1)$ with weight $W_2(n-1,2) = -1$. If $\mathcal{R}_{n-2}[1]$ and $\mathcal{R}_{n-2}[5]$ are 1, then they contribute RMT 2 at level $(n-1)$ with weight $W_2(n-1,2) = -2$ that cannot contribute RMTs 4, 5 at leaves with weight 0.
4. If $W_3(n-2,1) = W_3(n-2,5) = 1$, then set $\mathcal{R}_{n-2}[1] \leftarrow 1$ and $\mathcal{R}_{n-2}[5] \leftarrow 1$. Because they contribute RMT 3 at level $(n-1)$ with weight $W_3(n-1,3) = 0$. If $\mathcal{R}_{n-2}[1]$ and $\mathcal{R}_{n-2}[5]$ are 0, then they contribute RMT 3 at level $(n-1)$ with weight $W_3(n-1,3) = 1$, which cannot contribute RMTs 6, 7 at leaves with weight 0.
5. If $W_k(n-2,2) = W_k(n-2,6) = 0$, where $k \in \{0,1\}$, then arbitrarily set either $\mathcal{R}_{n-2}[2] \leftarrow \mathcal{R}_{n-2}[6] \leftarrow 0$ or $\mathcal{R}_{n-2}[2] \leftarrow \mathcal{R}_{n-2}[6] \leftarrow 1$.
6. If $W_k(n-2,1) = W_k(n-2,5) = 0$, where $k \in \{2,3\}$, then arbitrarily set either as $\mathcal{R}_{n-2}[1] \leftarrow \mathcal{R}_{n-2}[5] \leftarrow 0$ or as $\mathcal{R}_{n-2}[1] \leftarrow \mathcal{R}_{n-2}[5] \leftarrow 1$.

It can be verified that the conditions 1 and 2 (similarly, conditions 3 and 4) cannot hold simultaneously. So, the next states of all the RMTs of $\mathcal{R}_{n-2}$ can be set.

5.4.4 The Synthesis Algorithm

The synthesis algorithm (Algorithm 5) develops the *super node* of a level i, and finds the weights of RMTs of the node as in Algorithm 4. Individual rules of the CA, to be synthesized, are selected primarily based on these weights. As input, the algorithm takes $n \geq 5$ (the number of cells of the CA), and outputs an n-cell NCCA. The Γ_k is used to store the RMTs present in Γ_k^i of level i, and W_k stores the weight of RMTs in Γ_k for each $k \in \{0,1,2,3\}$.

The Step 1 forms the root node $\mathcal{N}_0$ of the reachability tree, and initializes the weights of RMTs. In Step 2, $\mathcal{R}_i[7] \leftarrow 1$ and $\mathcal{R}_i[0] \leftarrow 0$ are set for each rule $\mathcal{R}_i$ of rule vector $\mathcal{R}$ of the NCCA. A rule from $\mathcal{S}_{NCCA}$ is then selected randomly as $\mathcal{R}_0$ and the weights of the RMTs are found out that are present in $\mathcal{N}_1$ (Step 3). By Step 4, the $\mathcal{R}_1$ is found out. The weights of RMTs of $\mathcal{N}_2$ are found out in Step 5. To find the $\mathcal{R}_2$ to $\mathcal{R}_{n-3}$, the Steps 6, 7 and 8 are repeated. Step 9 and Step 10 are for selecting $\mathcal{R}_{n-2}$ and $\mathcal{R}_{n-1}$.

Algorithm 5: Synthesize an NCCA in periodic boundary

Input : $n(\geq 5)$

Output: An n-cell NCCA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in periodic boundary.

Step 1 : Set $i \leftarrow 2$, $\Gamma_0 \leftarrow \{0, 1\}$, $\Gamma_1 \leftarrow \{2, 3\}$, $\Gamma_2 \leftarrow \{4, 5\}$, $\Gamma_3 \leftarrow \{6, 7\}$, $W_k[r] \leftarrow 0$ if $r \in \Gamma_k$, for each k, $0 \leq k \leq 3$;

Step 2 : Set $\mathcal{R}_j[0] \leftarrow 0$, $\mathcal{R}_j[7] \leftarrow 1$, $0 \leq j \leq n-1$;

Step 3 : Randomly select a rule $\mathcal{R}_0$ from rule set {136, 170, 184, 192, 204, 226, 238, 240, 252};
Call *FindNextWeight*($\mathcal{R}_0$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;

Step 4 : Select the next states of RMTs of $\mathcal{R}_1$ according to the following:

```
if W_0[2] = −1 then
    R_1[2] ← 0;
    if W_1[4] = W_1[6] then
        R_1[3] ← R_1[4] ← 0, R_1[6] ← 1 ;
        R_1[1] ← R_1[5] ← α where α ∈ {0, 1} ;
    else
        R_1[5] ← 1, R_1[4] ← R_1[6] ← α where α ∈ {0, 1};
        if R_1[4] = 0 then
            R_1[1] ← R_1[3] ← 1;
        else
            R_1[1] ← R_1[3] ← 0;
        end
    end
else
    R_1[2] ← α where α ∈ {0, 1};
    if R_1[2] = 0 then
        if W_1[4] = W_1[6] then
            if W_1[4] = 0 or W_2[3] = 0 then
                R_1[3] ← 1, R_1[4] ← R_1[6] ← 0;
            else
                R_1[4] ← R_1[6] ← α where α ∈ {0, 1} ;
                if R_1[6] = 0 then
                    R_1[3] ← 1;
                else
                    R_1[3] ← 0;
                end
            end
            if R_1[4] = 0 then
                R_1[1] ← R_1[5] ← α where α ∈ {0, 1} ;
            else
                R_1[1] ← 0, R_1[5] ← 1;
            end
        else
            R_1[5] ← 1;
            R_1[4] ← 1, R_1[6] ← 0;
            R_1[1] ← 0, R_1[3] ← 1;
        end
    else
        R_1[3] ← 1, R_1[6] ← 1;
        if W_1[4] = W_1[6] then
            R_1[4] ← 0;
            R_1[1] ← R_1[5] ← α where α ∈ {0, 1} ;
        else
            R_1[5] ← 1;
            R_1[4] ← 1, R_1[1] ← 0;
        end
    end
end
```

Step 5 : Call *FindNextWeight*($\mathcal{R}_1$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;
Step 6 : Select the next states of RMTs 1, 4 and 5 of $\mathcal{R}_i$ according to the following:
For any $k \in \{0, 1, 2, 3\}$
if $W_k[4] = W_k[0]$ **then**
  $\mathcal{R}_i[4] \leftarrow 0$
else
  $\mathcal{R}_i[4] \leftarrow 1$
end
if $\mathcal{R}_i[4] \neq \mathcal{R}_i[0]$ **then**
  $\mathcal{R}_i[1] \leftarrow 0, \mathcal{R}_i[5] \leftarrow 1;$
else
  if *(*$W_0[5] = 1$*) or (*$W_1[5] = 2$*) or (*$W_2[5] = 1$*) or (*$W_3[5] = 1$*) or (*$W_3[1] = 1$*)* **then**
    $\mathcal{R}_i[1] \leftarrow 1, \mathcal{R}_i[5] \leftarrow 1$;
  else if $W_3[1] = -1$ **then**
    $\mathcal{R}_i[1] \leftarrow 0, \mathcal{R}_i[5] \leftarrow 0;$
  else
    $\mathcal{R}_i[1] \leftarrow \mathcal{R}_i[5] \leftarrow \alpha$ where $\alpha \in \{0, 1\}$;
  end
end
Step 7 : Select the next states of RMTs 2, 3 and 6 of $\mathcal{R}_i$ according to the following:
For any $k \in \{0, 1, 2, 3\}$
if $W_k[3] = W_k[7]$ **then**
  $\mathcal{R}_i[3] \leftarrow 1$
else
  $\mathcal{R}_i[3] \leftarrow 0$
end
if $\mathcal{R}_i[3] \neq \mathcal{R}_i[7]$ **then**
  $\mathcal{R}_i[2] \leftarrow 0, \mathcal{R}_i[6] \leftarrow 1;$
else
  if *(*$W_0[2] = -1$*) or (*$W_1[2] = -1$*) or (*$W_2[2] = -2$*) or (*$W_3[1] = -1$*) or (*$W_0[6] = -1$*)* **then**
    $\mathcal{R}_i[2] \leftarrow 0, \mathcal{R}_i[6] \leftarrow 0$;
  else if $W_0[6] = 1$ **then**
    $\mathcal{R}_i[2] \leftarrow 1, \mathcal{R}_i[6] \leftarrow 1;$
  else
    $\mathcal{R}_i[2] \leftarrow \mathcal{R}_i[6] \leftarrow \alpha$ where $\alpha \in \{0, 1\}$;
  end
end
Step 8 : Call *FindNextWeight*($\mathcal{R}_i$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;
Step 9 : $i \leftarrow i + 1$
if $i < n - 2$ **then** goto Step 6 Set $\Gamma_0 \leftarrow \Gamma_0 \cap \{0, 2, 4, 6\}$, $\Gamma_1 \leftarrow \Gamma_1 \cap \{0, 2, 4, 6\}$, $\Gamma_2 \leftarrow \Gamma_2 \cap \{1, 3, 5, 7\}$, $\Gamma_3 \leftarrow \Gamma_3 \cap \{1, 3, 5, 7\}$;
Set $\mathcal{R}_{n-2}[0] \leftarrow 0, \mathcal{R}_{n-2}[7] \leftarrow 1;$
Check the bellow conditions and set the next states of RMTs of $\mathcal{R}_{n-2}$
if $W_k[4] = W_k[0]$, $k \in \{0, 1\}$ **then**
  $\mathcal{R}_{n-2}[4] \leftarrow 0$;
else
  $\mathcal{R}_{n-2}[4] \leftarrow 1$;
end

if $\mathcal{R}_{n-2}[4] \neq \mathcal{R}_{n-2}[0]$ **then**
  $\mathcal{R}_{n-2}[1] \leftarrow 0, \mathcal{R}_{n-2}[5] \leftarrow 1;$
else
  if $W_2[1] = W_2[5] = -1$ **then**
    $\mathcal{R}_{n-2}[1] \leftarrow 0, \mathcal{R}_{n-2}[5] \leftarrow 0;$
  else if $W_3[1] = W_3[5] = 1$ **then**
    $\mathcal{R}_{n-2}[1] \leftarrow 1, \mathcal{R}_{n-2}[5] \leftarrow 1;$
  else
    $\mathcal{R}_{n-2}[1] \leftarrow \mathcal{R}_{n-2}[5] \leftarrow \alpha$ where $\alpha \in \{0, 1\}$;
  end
end
if $W_k[3] = W_k[7]$, $k \in \{2, 3\}$ **then**
  $\mathcal{R}_{n-2}[3] \leftarrow 1$;
else
  $\mathcal{R}_{n-2}[3] \leftarrow 0$;
end
if $\mathcal{R}_{n-2}[3] \neq \mathcal{R}_{n-2}[7]$ **then**
  $\mathcal{R}_{n-2}[2] \leftarrow 0, \mathcal{R}_{n-2}[6] \leftarrow 1;$
else
  if $W_0[2] = W_0[6] = -1$ **then**
    $\mathcal{R}_{n-2}[2] \leftarrow 0, \mathcal{R}_{n-2}[6] \leftarrow 0;$
  else if $W_1[2] = W_1[6] = 1$ **then**
    $\mathcal{R}_{n-2}[2] \leftarrow 1, \mathcal{R}_{n-2}[6] \leftarrow 1;$
  else
    $\mathcal{R}_{n-2}[2] \leftarrow \mathcal{R}_{n-2}[6] \leftarrow \alpha$ where $\alpha \in \{0, 1\}$;
  end
end
Call *FindNextWeight*($\mathcal{R}_{n-2}$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;
Step 10 : Set $\Gamma_0 \leftarrow \Gamma_0 \cap \{0, 4\}$, $\Gamma_1 \leftarrow \Gamma_1 \cap \{1, 5\}$, $\Gamma_2 \leftarrow \Gamma_2 \cap \{2, 6\}$, $\Gamma_3 \leftarrow \Gamma_3 \cap \{3, 7\}$;
Set $\mathcal{R}_{n-1}[0] \leftarrow 0, \mathcal{R}_{n-1}[7] \leftarrow 1;$
if $W_0[4] = 0$ **then**
  $\mathcal{R}_{n-1}[4] \leftarrow 0;$
else
  $\mathcal{R}_{n-1}[4] \leftarrow 1;$
end
if $W_3[3] = 0$ **then**
  $\mathcal{R}_{n-1}[3] \leftarrow 1;$
else
  $\mathcal{R}_{n-1}[3] \leftarrow 0;$
end
For each $r \in \{1, 5\}$
if $W_1[r] = 0$ **then**
  $\mathcal{R}_{n-1}[r] \leftarrow 0;$
else
  $\mathcal{R}_{n-1}[r] \leftarrow 1;$
end
For each $r \in \{2, 6\}$
if $W_2[r] = 0$ **then**
  $\mathcal{R}_{n-1}[r] \leftarrow 1;$
else
  $\mathcal{R}_{n-1}[r] \leftarrow 0;$
end
Call *FindNextWeight*($\mathcal{R}_{n-1}$, Γ_k, W_k) for each $k \in \{0, 1, 2, 3\}$;
Step 11 : Output the NCCA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$;

Complexity: The time requirement of Algorithm 5 depends only on n, the number of cells of the CA. Therefore, time complexity of Algorithm 5 is $O(n)$.

Example 5.3 Let consider the synthesis of an NCCA of 6 cells. The root node $\mathcal{N}_0$ of the reachability tree is formed and the weight of each RMTs is assigned to 0 (Step 1 of Algorithm 5). In Step 2, the next state of RMT 7 and RMT 0 are assigned as 1 and 0 respectively for each $\mathcal{R}_i$ ($i \in \{0, \cdots, n-1\}$). After that a rule is randomly selected from the rule set {136, 170, 184, 192, 204, 226, 238, 240, 252} as $\mathcal{R}_0$. Let $\mathcal{R}_0 = 192$. Now, we need to find the weights of RMTs of $\mathcal{N}_1$.

To get $\mathcal{R}_1$, the conditions of Step 4 are to be checked. As $W_0[2] = 0$, $W_3[5] = 0$, $W_2[3] = 0$ and $W_1[4] = 1$, the RMT 3 is 1. Now, if $\mathcal{R}_1[2] = 0$ is considered, then following the other conditions, we can set $\mathcal{R}_1[4] \leftarrow \mathcal{R}_1[6] \leftarrow 0$ and $\mathcal{R}_1[1] \leftarrow \mathcal{R}_1[5] \leftarrow 0$, which result in $\mathcal{R}_1 = 136$.

To select $\mathcal{R}_2$ and $\mathcal{R}_3$, Steps 6, 7 and 8 are executed. Following the conditions stated in these steps, we can choose $\mathcal{R}_2 = 184$ and $\mathcal{R}_3 = 252$. However, to find $\mathcal{R}_4$, Step 9 is to be followed, as level 4 is the $(n-2)$th level. From node $\mathcal{N}_4$, it can be noted that $W_k[4] = W_k[0]$ and $W_k[3] = W_k[7]$, for any k. So, the next states of RMT 4 and RMT 3 are set as 0 and 1 respectively. As per the other conditions, we can set $\mathcal{R}_4[2] \leftarrow \mathcal{R}_4[6] \leftarrow 1$ and $\mathcal{R}_4[1] \leftarrow \mathcal{R}_4[5] \leftarrow 0$ and, therefore, $\mathcal{R}_4 = 204$. Similarly, to find $\mathcal{R}_5$, Step 10 is executed. From node $\mathcal{N}_5$, it can be found that $W_0[4] = 0$ and $W_3[3] = 0$. So, the RMTs 4 and 3 are set as 0 and 1 respectively. Further, from node $\mathcal{R}_5$, it can be found that $W_1[1] = W_1[5] \neq 0$ and $W_2[2] = W_2[6] = 0$. So, $\mathcal{R}_5[2] \leftarrow \mathcal{R}_5[6] \leftarrow 1$ and $\mathcal{R}_5[1] \leftarrow \mathcal{R}_5[5] \leftarrow 1$. It results in $\mathcal{R}_5 = 238$. Finally, Step 11 returns the NCCA $\mathcal{R} = (192, 136, 184, 252, 204, 238)$.

5.5 Non-increasing and Non-decreasing Rules and Number Conservation

The framework elaborated in the earlier sections can be extended to characterize the number non-increasing and non-decreasing CAs. In case of a number non-increasing (resp. non-decreasing) CA, the weights of RMTs at leaves of the reachability tree are greater or equal (resp. less or equal) to 0. Recall that an RMT sequence, formed with the RMTs on edges of an edge sequence from the root to a leaf, represents a predecessor of the configuration represented by the edge sequence. For example, a 4-cell CA (136, 190, 64, 192), with periodic boundary, is a number non-increasing CA (Fig. 5.2). Reachability tree, along with weights, of the CA is shown in Fig. 5.8. The weights of all the RMTs at leaves are non-negative that proves the CA is number non-increasing.

Unlike NCCAs, the RMTs of number non-increasing and non-decreasing CAs may assume multiple weights. In Fig. 5.8, the RMTs of three nodes have multiple weights. The RMT 6 of a node at level 3 has weight 0 and 1 (marked in red). Two leaves have the RMTs (4 and 5) with multiple weights. The theories for analysis and

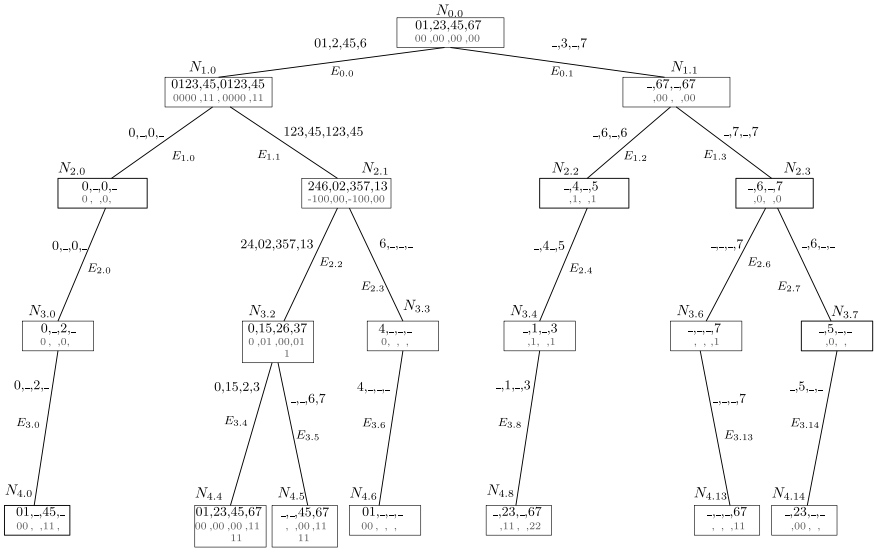


Fig. 5.8 Reachability tree of number non-increasing CA with weights of RMTs

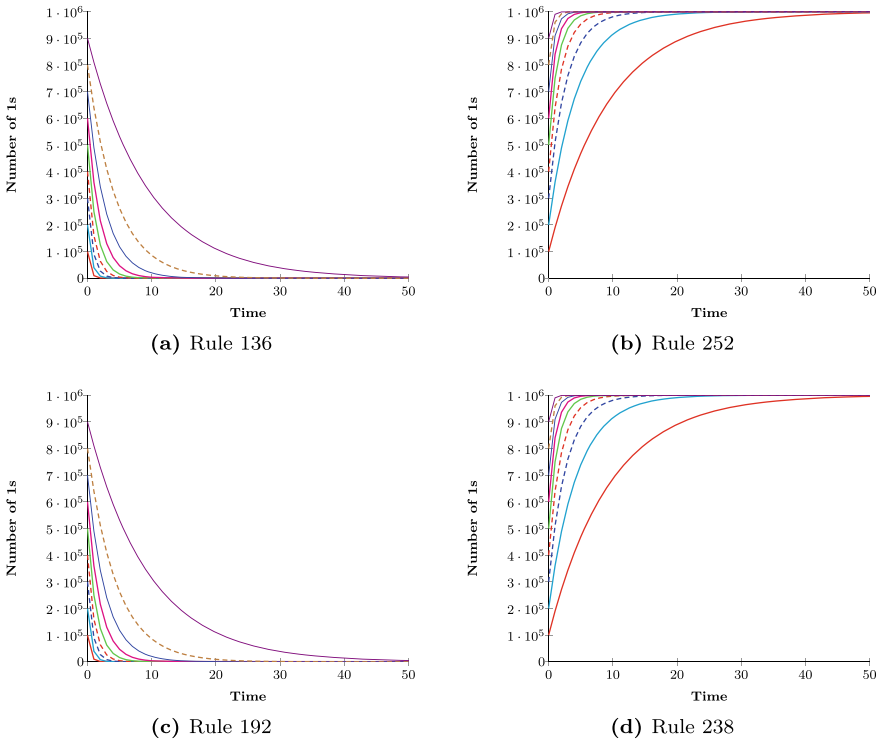
synthesis of number non-increasing and non-decreasing CAs are yet to be developed and left for future research.

It is already mentioned that the nine ECA rules—136, 170, 184, 192, 204, 226, 238, 240 and 252 can participate to form NCCA. Among these, the rule 136 and 192 are the number non-increasing, and 238 and 252 are the number non-decreasing rules. During evolution of the (uniform) CA of any number of cells with such a rule does not increase and decrease the number of 1s of initial configurations respectively. In fact, the CAs formed with these rules always decrease or increase the number of 1s until the CA becomes saturated. The fluctuations of 1s in the configurations, during the evolution, of such CA are plotted in Fig. 5.9a–d. Here, the lattice size is taken as 10^6, and the initial configurations are chosen randomly.

However, when the number non-decreasing and number non-increasing rules are used to form the non-uniform CA, the non-uniform CA may be an NCCA. In a non-uniform CA, if altrnate cells are configured with the rule 136 and 252 or rules 192 and 238, then the CA is an NCCA. But this is not the fact for the rule 2 (number non-increasing) and 247 (number non-decreasing).

Theorem 5.3 *A non-uniform CA* $(\mathcal{R}_1, \mathcal{R}_2)^k$ *in periodic boundary is an NCCA for any* $k \geq 1$ *iff*

1. $\mathcal{R}_1, \mathcal{R}_2 \in \{136, 170, 184, 192, 204, 226, 238, 240, 252\}$, *and*
2. $\mathcal{R}_1[r] \neq \mathcal{R}_2[s]$, *where* $r \in Equi_i = \{i, i+4\}$, $s \in Sibl_i = \{2i, 2i+1\}$, $i \in \{1, 2\}$



(a) Rule 136 (b) Rule 252

(c) Rule 192 (d) Rule 238

Fig. 5.9 Fluctuations of 1s in configurations, during evolutions, of ECAs

Proof By contradiction, let assume that the non-uniform CA is NCCA while at least one of the conditions 1 and 2 is violated. Here, $Equi_1 = \{1, 5\}$, $Equi_2 = \{2, 6\}$, $Sibl_1 = \{2, 3\}$ and $Sibl_2 = \{4, 5\}$. However, if the first condition is violated, then the CA cannot be an NCCA as per Theorem 5.2. When the second condition is not satisfied, then an RMT of $\Gamma_1^{N_{3,j}}$ can have multiple weights. This violates the condition of Theorem 5.1. Hence, if both the conditions are satisfied, then the CA is an NCCA. □

Applying the conditions of Theorem 5.3, the rule pairs obtained are the (136, 252), (238, 192) and (170, 240). Hence, there are three classes of CAs $(136, 252)^k$, $(238, 192)^k$ and $(170, 240)^k$ which are NCCAs. The members of the first two classes are the number non-decreasing and number non-increasing rules whereas, the rules of $(170, 240)^k$ are number conserving. Theorem 5.3 further confirms that only 4 non-increasing and non-decreasing rules can form NCCAs. Figure 5.10 describes the construction of NCCA. Any closed path of the graph represents the rule vector of a CA. This family of CAs can also be expressed by regular expression $((136, 252)^+, (238, 192)^+)^+$.

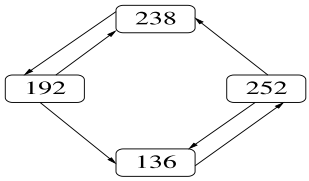


Fig. 5.10 NCCAs family $((136, 252)^+, (238, 192)^+)^+$

Corollary 5.7 *In periodic boundary, each CA of the family* $((136, 252)^*, (238, 192)^*)^+$ *is an NCCA.*

The proof directly follows from the above discussion.

5.6 Change of Boundary Condition

The theories presented in the earlier sections have considered that the CAs are in periodic boundary. The similar theory can be developed for other boundary conditions. The change of boundary condition can affect only the structure of reachability tree. In this section, the number conservation behavior of CAs with null boundary is examined.

In Sect. 5.2.1, it is noted that the weights are assigned to the RMTs in a node of a reachability tree to show the surplus or deficiency of 1s in the RMT sequence with respect to its next configuration. The same definition of weight with a minor modification can be used to decide number conservation of CAs in null boundary.

Since a node is a set of RMTs for null boundary (it is a collection of sets of RMTs for periodic boundary), a weight of an RMT r in a node at level i is denoted by $W(i, r)$. As per definition of reachability tree of an n-cell CA in null boundary (Definition 18), the nodes of level $n-1$ can contain only the even RMTs. So, while assigning weights to the RMTs of nodes at different levels, only even RMTs of nodes at level $n-1$ are considered. Further, leaves of the tree is empty. So, i in $W(i, r)$ varies from 0 to $n-1$. Following is the definition of weight for null boundary:

1. $W(0, r) = 0 \qquad : r \in \{0, 1, 2, 3\}$
2. $W(i, 2r \pmod 8) = W(i, 2r + 1 \pmod 8)$

$$= \begin{cases} W(i-1, r) + 1 : r \in \{2, 3, 6, 7\} \text{ and } \mathcal{R}_{i-1}[r] = 0 \\ W(i-1, r) - 1 : r \in \{0, 1, 4, 5\} \text{ and } \mathcal{R}_{i-1}[r] = 1 \\ W(i-1, r) \qquad\qquad\qquad\qquad\qquad\quad : \text{otherwise} \end{cases}$$

3. $W(n-1, 2r \pmod 8) =$

$$\begin{cases} W(n-2, r) + 1 : r \in \{2, 3, 6, 7\} \text{ and } \mathcal{R}_{n-2}[r] = 0 \\ W(n-2, r) - 1 : r \in \{0, 1, 4, 5\} \text{ and } \mathcal{R}_{n-2}[r] = 1 \\ W(n-2, r) \qquad\qquad\qquad\qquad\quad : \text{otherwise} \end{cases}$$

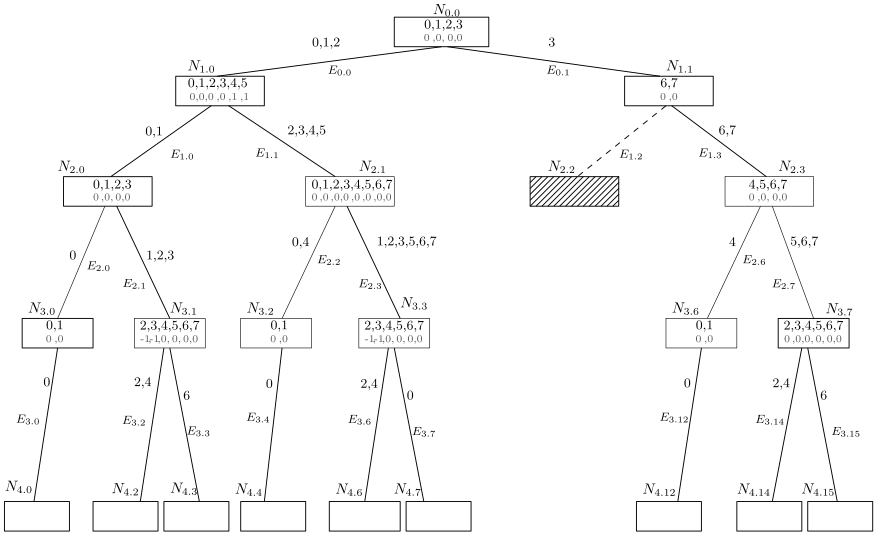


Fig. 5.11 Reachability tree for CA (136, 252, 238, 192) with null boundary

Let consider a 4-cell CA (136, 252, 238, 192) with null boundary, and the weights to the RMTs of nodes of the reachability tree are assigned as per the above definition. Figure 5.11 shows the weights which are noted at the bottom of the RMTs of different nodes in the tree. There is no RMT in the leaves, hence there is no weight. In case of periodic boundary, if weights of all the RMTs in the leaves of a tree are 0, the CA can be decided as the number conserving (Example 5.1). This is not possible for null boundary. An additional check is required. By applying the last rule ($\mathcal{R}_{n-1}$), we need to verify if each pair of consecutive configurations of the CA contains no surplus or deficiency of 1s. For the RMTs of each node at level $n-1$, the following condition is to be verified:

$$\begin{aligned} W(n-1,r)+1 &= 0 \quad \text{if } r \in \{2,6\} \text{ and } \mathcal{R}_{n-1}[r]=0 \\ W(n-1,r)-1 &= 0 \quad \text{if } r \in \{0,4\} \text{ and } \mathcal{R}_{n-1}[r]=1 \\ W(n-1,r) &= 0 \quad \text{otherwise} \end{aligned} \tag{5.1}$$

If the above condition is applied to Fig. 5.11, it can be noticed that the RMTs of each node at level 3 satisfies the condition. Hence the CA is an NCCA.

The result of Theorem 5.1 with adequate modification for null boundary is reproduced next. The proof is omitted as it is similar to that of Theorem 5.1.

Theorem 5.4 *The weight of an RMT in any node at a level of the reachability tree of an NCCA in null boundary is unique.*

At a level of a reachability tree, as in periodic boundary, some nodes may be identified as *sub-node* of other nodes (Sect. 5.3). A node is a sub-node of another node at the

same level if the first node is the subset of the second. In Fig. 5.11, $N_{2.0}$ is a sub-node of $N_{2.1}$ ($N_{2.0} \subseteq N_{2.1}$). For the same reason, as noted in Sect. 5.3, sub-nodes have no additional role in deciding the number conservation. So, they are removed during construction of the tree. Hence, a straightforward decision algorithm can be: Construct the reachability tree with weights, omit the sub-nodes at any level, and apply Condition 5.1 to the RMTs of nodes of level $n-1$ to decide on the number conservation.

The idea of *super node* can also be introduced, as it has been done in Sect. 5.3. Each node of a level is a sub-node of the super node. So, the decision algorithm can proceed only with the super node of each level. Following are the steps of such decision algorithm. Input to the algorithm is an n-cell CA with null boundary.

1. Form a node ($\mathcal{N}_0$) with RMTs 0, 1, 2 and 3. Assign weight 0 to each RMT.
2. Get the RMTs and weights of those RMTs from the node $\mathcal{N}_{i-1}$. If any RMT receives two different weights, the CA is not an NCCA. Otherwise, the RMTs form the super node $\mathcal{N}_i$ of level i.
3. Repeat Step 2 upto $i = n-2$. From $\mathcal{N}_{n-2}$, form $\mathcal{N}_{n-1}$ with even RMTs only and check if an RMT receives two different weights. If so, the CA is not NCCA.
4. Test if all the RMTs of $\mathcal{N}_{n-1}$ satisfy Condition 5.1. If yes, the CA is an NCCA.

The above algorithm runs in $O(n)$ time. An efficient implementation of it demands only $O(1)$ space. However, if possible weights of the RMTs are found out (like Sect. 5.3.1), further improvement is possible. If the weight of an RMT of $\mathcal{N}_i$ does not match with the possible weight, early conclusion of non-number conservation can be possible. This has been done in Algorithm 4.

It is also possible to find out the set of rules that can form a rule vector of an NCCA. For null boundary, the first and last rules are different than others. The RMTs 4, 5, 6 and 7 of the first rule ($\mathcal{R}_0$) and the RMTs 1, 3, 5 and 7 of the last rule ($\mathcal{R}_{n-1}$) are *don't care*, and have no role in the behavior of the CA. However, it can be shown that the remaining rules ($\mathcal{R}_1, \mathcal{R}_2, \cdots, \mathcal{R}_{n-2}$) are to be the member of $\mathcal{S}_{NCCA} = \{136, 170, 184, 192, 204, 226, 238, 240, 252\}$. This set of rules is obtained from Theorem 5.2. Similar theorem can be found for null boundary. The possible rules for $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$ can also be found out exploiting the weight relationship in the tree of a CA. Consequently, the rule sequence in tabular format (like Table 5.3) can be identified to decide the number conservation.

It can be noted that the set of rules $\mathcal{S}_{NCCA}$ remains invariant to the change of boundary condition. Other than some special cells like fist and last cell in null boundary, the rule of any cell of an NCCA is to be a member of $\mathcal{S}_{NCCA}$. This fact can even be proved without considering reachability tree. For example, in the all 0s configuration, if one cell is flipped to 1, then to conserve the number of 1s (here single 1), in the next configuration, the RMTs of the rules have to follow certain condition. From such conditions, the set of possible rules for NCCAs can be found as $\mathcal{S}_{NCCA}$.

A synthesis scheme for NCCAs with null boundary can be developed by exploiting the weight relationship among the RMTs. Such synthesis scheme is more efficient than that can be done by tabular approach. This has been adopted in Algorithm 5 for periodic boundary.

5.7 Scalability of Number Conserving CAs

The rule vectors of two NCCAs with number of cells n and m can be merged to get an $(n+m)$-cell NCCA. The converse can also be true. That is, the $(n+m)$-cell NCCA can be split to an n-cell and m-cell NCCAs. Presence of *left independent* and *right independent* rules enables the scalability.

Definition 34 A rule R is *left independent* if $R[0xy] = R[1xy]$ for each $x, y \in \{0, 1\}$. The R is *right independent* if $R[xy0] = R[xy1]$ for any $x, y \in \{0, 1\}$.

In other word, a rule R is left (resp. right) independent rule if each pair of equivalent (resp. sibling) RMTs r, s of R have the same next state—that is, $R[r] = R[s]$. For example, the rule 136 is left independent whereas, rule 192 is right independent. This can be verified from Table 5.1. Similarly, the rule 102 is left independent and 60 is right independent.

5.7.1 Merging of Number Conserving CAs

Let $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ and $(\mathcal{R}'_0, \mathcal{R}'_1, \cdots, \mathcal{R}'_{m-1})$ be the two NCCAs. Further assume that $\mathcal{R}_{n-1}$ is right independent and $\mathcal{R}'_0$ is left independent. Now, if an $(n+m)$-cell NCCA is formed by concatenating the two rule vectors, then the states of cells $n-1$ and n of the CA $(\mathcal{R}_0, \cdots, \mathcal{R}_{n-1}, \mathcal{R}'_0, \cdots, \mathcal{R}'_{m-1})$ remain unaffected.

Let now consider that the boundary condition for each of the n-cell and m-cell NCCAs is periodic and, $\mathcal{R}_0$ is left independent and $\mathcal{R}'_{m-1}$ is right independent. Then, the first and last cells of the merged CA $(\mathcal{R}_0, \cdots, \mathcal{R}_{n-1}, \mathcal{R}'_0, \cdots, \mathcal{R}'_{m-1})$ are independent of each other. In that case, the dynamical behavior of the n-cell and m-cell NCCAs are preserved in the paired up $(n+m)$-cell CA. Hence, the $(n+m)$-cell CA is number conserving.

Example 5.4 The 4-cell CA (136, 252, 238, 192) in periodic boundary is number conserving (Fig. 5.1). The rule 136 is left independent and rule 192 is right independent. The 6-cell CA (238, 192, 136, 184, 252, 204) in periodic boundary is also number conserving. The rule 238 is left independent and 204 is right independent. If these two NCCAs are merged to construct the 10-cell CA (136, 252, 238, 192, 238, 192, 136, 184, 252, 204), in periodic boundary, then it is also an NCCA.

Out of the nine rules of $\mathcal{S}_{NCCA} = \{136, 170, 184, 192, 204, 226, 238, 240, 252\}$, four rules are the left independent and four are the right independent. The left independent rules are 136, 170, 204 and 238, and the rule 192, 204, 240 and 252 are right independent.

In case of null boundary, while merging two NCCAs with the number of cells n and m to construct an $(n+m)$-cell NCCA, unlike periodic boundary, any two NCCAs can be merged If $\mathcal{R}_{n-1}$ of the NCCA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ is right independent and $\mathcal{R}'_0$ of the NCCA $(\mathcal{R}'_0, \mathcal{R}_1, \cdots, \mathcal{R}'_{m-1})$ is left independent, then the merged

CA $(\mathcal{R}_0, \cdots, \mathcal{R}_{n-1}, \mathcal{R}'_0, \cdots, \mathcal{R}'_{m-1})$ is number conserving. If $\mathcal{R}_{n-1}$ is not right independent, then an equivalent rule $\mathcal{R}^e_{n-1}$ of $\mathcal{R}_{n-1}$, where $\mathcal{R}^e_{n-1}$ is right independent, is to be set for the cell n of the $(n+m)$-cell CA. In fact, it is always possible to synthesize such $\mathcal{R}^e_{n-1}$ for a given $\mathcal{R}_{n-1}$. Similarly, if $\mathcal{R}'_0$ is not left independent, then the rule for the cell $(n+1)$ of the $(n+m)$-cell CA is to be replaced by the rule $\mathcal{R}^e_0$ which is equivalent to $\mathcal{R}'_0$ and also left independent.

Example 5.5 The CAs $(8, 252, 238, 64)$ and $(14, 226, 64)$ are NCCAs in null boundary. The rule 64 and 192 are equivalent if selected for the rightmost cell. For the RMTs 0, 2, 4 and 6, both the rules have the same next states. Further, the rule 192 is right independent. On the other hand, the rule 14 and 238 are equivalent if selected for the first cell. Because, for RMTs 0, 1, 2 and 3, these two rules have the same next states. The rule 238 is a left independent rule. Therefore, the 7-cell CA $(8, 252, 238, 192, 238, 226, 64)$ is an NCCA in null boundary.

5.7.2 Infinite Lattice

In the earlier sections, the number conservation of finite CAs has been explored. This section considers infinite lattice ($\mathcal{L} = \mathbb{Z}$) for the study of number conservation. For ease of presentation, let consider the following notations

$$\mathcal{L}^n_{-\infty} = \{\cdots, n-2, n-1, n\}, \text{ for } n \in \mathbb{Z}$$
$$\mathcal{L}^\infty_m = \{m, m+1, m+2, \cdots\}, \text{ for } m \in \mathbb{Z}$$
$$\mathcal{L}^n_m = \{m, m+1, \cdots, n-1, n\}, \text{ for } m, n \in \mathbb{Z} \text{ and } m \leq n$$

That is, $\mathcal{L} = \mathcal{L}^{n-1}_{-\infty} \cup \mathcal{L}^n_m \cup \mathcal{L}^\infty_{m+1}$ for some integers n and m, where $m \leq n$. Now a CA can be defined on $\mathcal{L}^{n-1}_{-\infty}$; similarly on $\mathcal{L}^n_m$ and $\mathcal{L}^\infty_{m+1}$. Further, consider that the three CAs defined on $\mathcal{L}^{n-1}_{-\infty}$, $\mathcal{L}^n_m$ and $\mathcal{L}^\infty_{m+1}$ are NCCAs and (i) the last rule of the first CA is right independent, (ii) the first and last rule of the second CA are left and right independent respectively, and (iii) the first rule of the third CA is left independent. If these three CAs are concatenated, then the final CA, defined over $\mathbb{Z}$, is also an NCCA.

Example 5.6 Let consider the CA $(\cdots, 184, 184, 252)$ defined over $\mathcal{L}^{n-1}_{-\infty}$ for some integer n. Here, all but the last rule are 184, and the last rule 252 is right independent. Now, consider the CA $(238, 226, 226, \cdots)$ defined over $\mathcal{L}^\infty_{m+1}$. The first rule (238) is left independent and rest of the rules are 226. It can be verified that both the CAs are number conserving. The 4-cell CA $(136, 252, 238, 192)$ is also number conserving (Fig. 5.1) and the rules 136 and 192 are left and right independent respectively. Now, the CA $(\cdots, 184, 184, 252, 136, 252, 238, 192, 238, 226, 226, \cdots)$ is also number conserving.

Table 5.4 Number conserving CAs on $\mathcal{L}^n_{-\infty}$ and $\mathcal{L}^\infty_m$

$\mathcal{L}^n_{-\infty}$	$\mathcal{L}^\infty_m$
$(\cdots, 184, 184, 252)$	$(136, 184, 184, \cdots)$
$(\cdots, 226, 226, 192)$	$(238, 226, 226, \cdots)$
$(\cdots, 170, 170, 0)$	$(0, 240, 240, \cdots)$
$(\cdots, 170, 170, 136, 12)$	$(68, 192, 240, 240, \cdots)$
$(\cdots, 170, 170, 34, 226, \cdots, 226, 192)$	$(136, 184, \cdots, 184, 48, 240, 240, \cdots)$

It is to be noted that there are a number of NCCAs defined over $\mathcal{L}^n_{-\infty}$ and $\mathcal{L}^\infty_m$ that can be found out analytically. Table 5.4 shows some of those CAs. These, along with other finite NCCAs, can be concatenated following the above approach to get new NCCAs over infinite grid. In this construction, a *wall* is formed between two blocks of NCCAs which stops the blocks to affect each other. Such a wall is implemented through the pair of right and left independent rules.

5.8 Universal Logic Design

Number conserving cellular automata are classically used to model traffic systems, such as road traffic on the highway and in the city, network traffic etc. The NCCAs can be used to emulate the traffic behavior in junctions. In fact, a phenomenon where the number of 1s in initial condition is to be preserved throughout its lifetime can be modelled by an NCCA.

An NCCA can also be explored to design logic circuits. Some NCCAs can realize the basic logic gates, such as AND, OR, NOT, NAND, XOR, etc. For example, let consider an n-cell uniform CA with rule 184 in periodic boundary. The CA can be considered as a computational circuit block which maps the n inputs $I_1, I_2, \cdots, I_n$ to n outputs $O_1, O_2, \cdots, O_n$ (Fig. 5.12). The desired output can be found after predefined evolution steps.

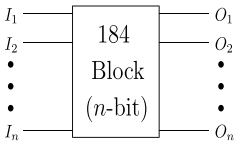


Fig. 5.12 The n-input and n-output 184 block

5.8.1 Rule 184—A Universal Gate Generator

Let us consider a 3-input and 3-output 184 block (Fig. 5.13). The rule 184 can be expressed as $184(x, y, z) = y \cdot z + x \cdot y'$. If X, Y and Z are three inputs of the 184 block, then after first time step of its evolution, the outputs can be

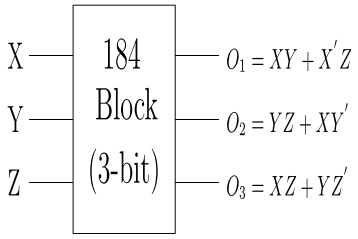
$$\begin{aligned} O_1 &= XY + X'Z \\ O_2 &= YZ + XY' \\ O_3 &= XZ + YZ' \end{aligned} \tag{5.2}$$

The 3-bit 184 block of Fig. 5.13a works as the multiplexers. The inputs Y and Z can be considered as the data inputs of a 2 to 1 multiplexer and X as the select line. The O_1 is the desired output. Similarly, O_2 (O_3) is the output of the 2 to 1 multiplexer where the inputs are X and Z (X and Y) and Y (Z) is the select line.

Realization of Elementary Gates: By proper selection of input, it is possible to implement AND, OR, NOT, FAN-OUT by a 3-input and 3-output 184 block. Let Z be set as logic 0. Then output O_1 of Eq. 5.2 results in XY—hence, a two input AND (X AND Y) is realized. Similarly, if Z is set to logic 1, then O_2 of Eq. 5.2 is $Y + XY'$ $= (Y + X)(Y + Y') = X + Y$.

Further, if Y and Z are set to 0, 1 respectively, then O_1 of Eq. 5.2 results in X' (NOT X) and, O_2 and O_3 are X (FAN-OUT). Figure 5.14 shows all such realization.

Realization of Universal Gates: To realize the universal logic gates, such as NAND and NOR, the 5-input and 5-output 184 blocks of Fig. 5.15a and b respectively are evolved for two time steps. Let the inputs be the V, W, X, Y and Z. Then after two time steps, the outputs can be:



(a) Block diagram

X	Y	Z	O_1	O_2	O_3
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

(b) Truth table

Fig. 5.13 The 3-input and 3-output 184 block with truth table

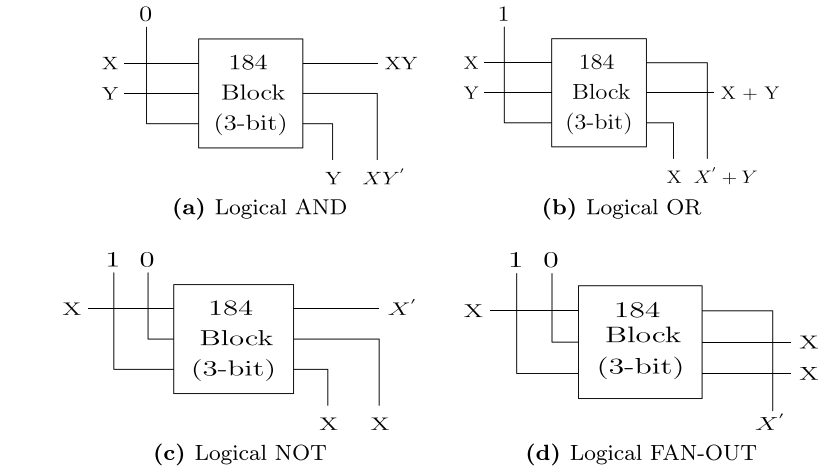


Fig. 5.14 Implementation of basic gates using 3-input and 3-output 184 block

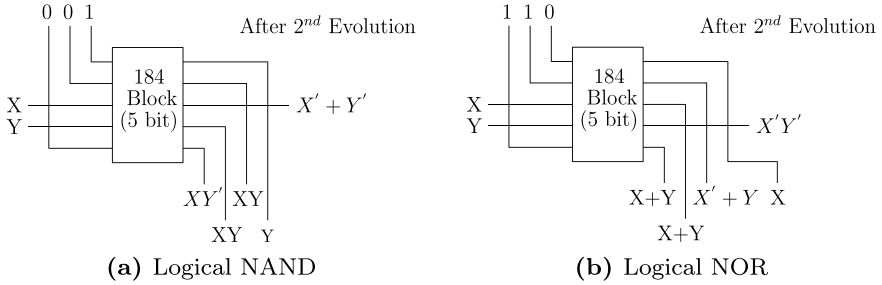


Fig. 5.15 Realization of universal gates with 5-input and 5-output 184 block

$$
\begin{aligned}
O_1 &= WXZ + VWX + VW'Z + V'YZ' + W'YZ' \\
O_2 &= WXY + VXY + V'W'Z + VWX' + V'X'Z \\
O_3 &= XYZ + WYZ + VW'X' + WXY' + VW'Y' \\
O_4 &= VYZ + VXZ + WX'Y' + XYZ' + WX'Z' \\
O_5 &= VWZ + VWY + V'YZ + V'XY' + XY'Z'
\end{aligned}
\tag{5.3}
$$

Now, if V, W and Z are set to 1, 0 and 0 respectively, then O_3 of Eq. 5.3 results in $X' + Y' = (X.Y)'$. That is, O_3 realizes 2-input NAND (Fig. 5.15a). Further, if V, W and Z are set to 0, 1 and 1 respectively, then O_4 is $X'Y' = (X + Y)'$—that is, O_4 realizes 2-input NOR (Fig. 5.15b).

Realization of XOR and XNOR: The 2-input XOR and XNOR gates are realized by evolving the 5-input and 5-input 184 block for four time steps. Let the inputs to the 5-bit 184 block be V, W, X, Y and Z. After four time steps, the outputs are

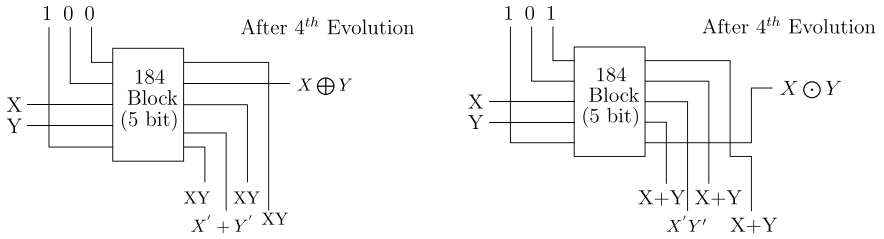


Fig. 5.16 Realization of XOR and XNOR gates using 184 block

$$
\begin{aligned}
O_1 &= V'W'XY + VWY'Z' + VYZ + VXZ + V'WZ + V'WX' \\
O_2 &= W'X'YZ + V'WXZ' + VWZ + VWY + VW'X + W'XY' \\
O_3 &= V'W'XY + VX'Y'Z + WXZ + VWX + WX'Y + X'YZ' \\
O_4 &= W'X'YZ + VWY'Z' + WXY + VXY + XY'Z + V'Y'Z \\
O_5 &= VX'Y'Z + V'WXZ' + XYZ + WYZ + VYZ' + VW'Z'
\end{aligned}
\tag{5.4}
$$

Figure 5.16a and b describe the realization of XOR and XNOR respectively. If V, W and Z are set to 0, 0 and 1 respectively, then O_2 (Eq. 5.4) results in $X'Y + XY'$–that is, XOR of X and Y. Further, if V, W and Z are set to 1, 0 and 1 respectively, then O_5 is $X'Y' + XY$ –that is, O_5 realizes XNOR of X and Y.

Though realization of 2-input functions are shown, it is possible to realize k-input gates with the n-cell ECA 184, where $n > k$.

Realization of Arbitrary Function: Let us consider a 4-input function $f(A, B, C, D) = A'B'C + A'BD' + AB'D + ABC'$. The logic diagram for f can be obtained by utilizing the basic gates, realized with 184 blocks. For f, the 2-input AND gates, 2-input OR gates, and NOT gates are needed. The design is shown in Fig. 5.17. Five time steps are needed to realize the f.

5.8.2 *Design of Adders*

The adders (half adder and full adder) can directly be implemented by evolving the 184 block.

Half Adder: To design a half adder 5-bit (5-input and 5-output) 184 block can be used (Fig. 5.18). If the CA is evolved for 4 time steps, then the desired carry and sum are obtained. After fourth time step, the outputs of the 184 block are as in Eq. 5.4.

Let the inputs V, W, Z be set to 0, 0, 1 respectively. Then the output O_2 of Eq. 5.4 is $X \oplus Y$—that is, the sum of the half adder. Similarly, the output O_1 of Eq. 5.4 is XY—that is, the carry.

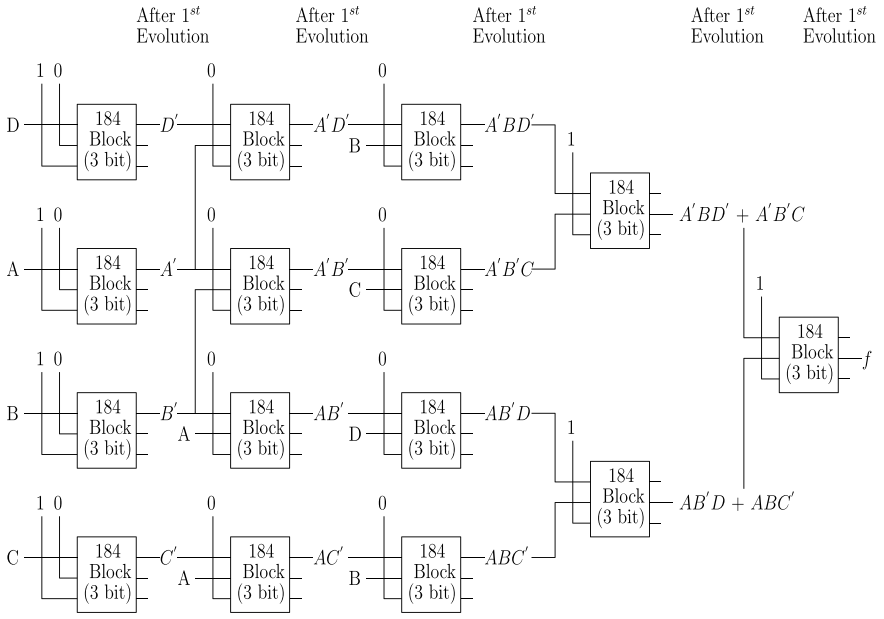


Fig. 5.17 Realization of logic circuit for $f(A, B, C, D) = A^{'}B^{'}C + A^{'}BD^{'} + AB^{'}D + ABC^{'}$

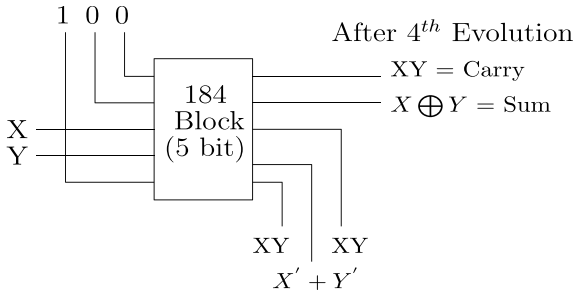


Fig. 5.18 Realization of half adder using 184 block

Full Adder: A 7-bit (7-input and 7-output) 184 block is used to realize the full adder. If the CA is evolved for 7 time steps, then the carry is generated. Similarly, after 8th time step, the sum is obtained. Let input to the 7-bit 184 block be t, u, v, w, x, y and z. After evolution of 7th time step, the outputs are

$$
\begin{array}{|l|l|}
\hline
O_1 = & t'uvx'y'z + t'u'vwxz' + t'uv'wxz' + tvw'x'y'z' + tuvwx' + t'uvyz' \\
\hline
 & + t'uvw'z' + tyz + txz + txy + twz + twy + tu'v' + tu'w' \\
\hline
O_2 = & t'u'v'wxy + t'u'vw'xy + tu'vwy'z' + t'uwx'y'z' + t'u'vwz + t'u'vwx' \\
\hline
 & + uvwxy' + uyz + uxz + tuz + tuy + tux + uv'w' + uv'x' \\
\hline
O_3 = & u'v'w'xyz + u'v'wx'yz + t'uv'wxz' + t'u'vxy'z' + tu'v'wx + u'v'wxy' \\
\hline
 & + vwxyz' + uvz + uvy + tvz + tvy + vw'x' + vw'y' \\
\hline
O_4 = & t'u'vw'xy + v'w'x'yz + tv'w'xy'z + t'u'v'wyz' + t'wxyz + uv'w'xy \\
\hline
 & + vw'xyz' + vwz + uwz + uvw + tvw + tuw + wx'y' + wx'z' \\
\hline
O_5 = & t'u'v'w'xz + u'v'wx'yz + tuw'x'y'z + tuw'x'yz' + tu'xyz + vw'x'yz \\
\hline
 & + t'w'x'yz + vwx + uwx + uvx + twx + tvx + t'xy' + xy'z' \\
\hline
O_6 = & tu'v'w'x'y + tv'w'xy'z + t'uvx'y'z + tuvx'y'z' + tuv'yz + twx'y'z \\
\hline
 & + tu'x'y'z + wxy + vxy + vwy + uxy + uwy + t'yz' + u'yz' \\
\hline
O_7 = & tuv'w'x'y' + tuw'x'yz' + t'uvwy'z' + tu'vwy'z' + tuvw'z + tuxy'z' \\
\hline
 & + tuxy'z' + tuv'y'z' + xyz + wyz + wxz + vyz + vxz + t'u'z + t'v'z \\
\hline
\end{array}
$$

Similarly, after 8th time step, these are

$$
\begin{array}{|l|l|}
\hline
O_1 = & t'uvwy'z' + tu'vwy'z' + tuz + tuy + tux + t'yz + t'xz + t'u'z + \\
\hline
 & tuv' + t'v'z \\
\hline
O_2 = & t'u'vwxz' + t'uv'wxz' + uvz + uvy + tuv + tu'z + tu'y + tu'v' + \\
\hline
 & uvw' + tu'w' \\
\hline
O_3 = & t'u'v'wxy + t'u'vw'xy + vwz + uvw + tuw + uv'z + tuv' + uv'w' + \\
\hline
 & vwx' + uv'x' \\
\hline
O_4 = & u'v'w'xyz + u'v'wx'yz + vwx + uwx + twx + uvw' + tuw' + vw'x' + \\
\hline
 & wxy' + vw'y' \\
\hline
O_5 = & tv'w'x'yz + tv'w'xy'z + wxy + vxy + uxy + vwx' + uwx' + wx'y' + \\
\hline
 & xyz' + wx'z' \\
\hline
O_6 = & tuw'x'y'z + tuw'x'yz' + xyz + wyz + vyz + t'yz + wxy' + vxy' + \\
\hline
 & t'xy' + xy'z' \\
\hline
O_7 = & t'uvx'y'z + tuvx'y'z' + tyz + txz + twz + tu'z + xyz' + wyz' + \\
\hline
 & t'yz' + u'yz' \\
\hline
\end{array}
$$

Now, if the inputs v, w, x and y are set 1, 1, 0 and 0 respectively, then after the 7th time step, O_3 is $tu + tz + uz$, which is the carry of full adder. Similarly, after the 8th time step, O_1 is $t \oplus u \oplus z$, which is the sum of full adder. Figure 5.19 shows the implementation of full adder.

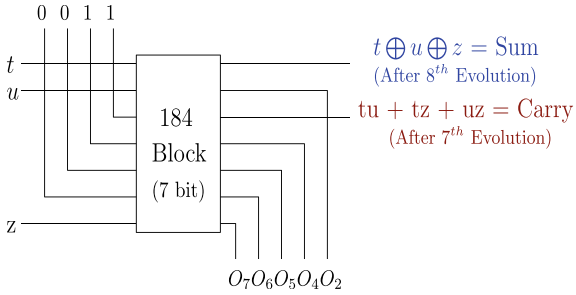


Fig. 5.19 Realization of full adder using 184 block

5.9 Chapter Notes

The study of number conservation in non-uniform cellular automata has been initiated in [1] and that has been continued in [2, 3]. In the Ph.D thesis of Hazari [4], the efficient schemes of analysis and synthesis of finite number conserving cellular automata in periodic boundary have been reported. The issue has been re-looked again in [5], and the schemes are extended for null boundary. The number conservation for infinite lattice has been addressed in [6]. Though, the issue of number conservation of non-uniform CAs on infinite lattice has been first introduced in [7]. Design of logic circuits using number conserving CAs has been treated in [4, 8].

Number conservation in non-uniform CAs is a flourishing topic of research. Following are the few issues that can be explored in future.

1. Analysis and synthesis of number non-increasing and number non-decreasing non-uniform ECAs.
2. Efficient algorithms for analysis and synthesis of non-uniform NCCAs with d states per cell.
3. Study on the computational ability of NCCAs through evolution.

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Chapter 6
Maximal Length Cellular Automata

Non-uniform finite cellular automata (CAs) with open boundary share a special property called *maximality* in cycle length. It means that all but one configuration of the cellular automaton (CA) form a single cycle. In case of an n-cell binary CA, the maximal cycle length is $2^n - 1$. Such a CA is traditionally referred to as the *maximal length* CA. Almost all the works on maximal length CAs revolve around the binary CAs with ECA rules. This chapter also primarily considers non-uniform ECAs. However, it reports some properties for non-binary cases (Sect. 6.7). The maximal length CAs are categorised as linear, complemented and non-linear. This chapter covers all such categories. Sections 6.2 and 6.3 provide the analysis and synthesis of linear maximal length CAs. Whereas, Sects. 6.5 and 6.6 respectively cover complementary and non-linear maximal length CAs. There is a relation between the primitive polynomials in Galois Field (GF) and linear maximal length CAs—one can be obtained from the other. The linear maximal length CAs can be synthesized from primitive polynomials (Sect. 6.3). On the other hand, the linear maximal length CAs can be used as a tool to generate primitive polynomials (Sect. 6.4).

6.1 Fundamental Properties

The maximal length CAs are the special class of reversible CAs, where all but one configuration form a single cycle.

Definition 35 A finite CA is a *maximal length CA* if all but one configuration of it are in a single cycle.

For an n-cell 2-state maximal length CA, the cycle length is $2^n - 1$. The remaining configuration forms a single length cycle. For example, the CA (150, 150, 90, 150) is a 4-cell maximal length CA in null boundary. It has a cycle of length $2^4 - 1 = 15$

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Complexity and Computation 49, https://doi.org/10.1007/978-3-032-00257-0_6

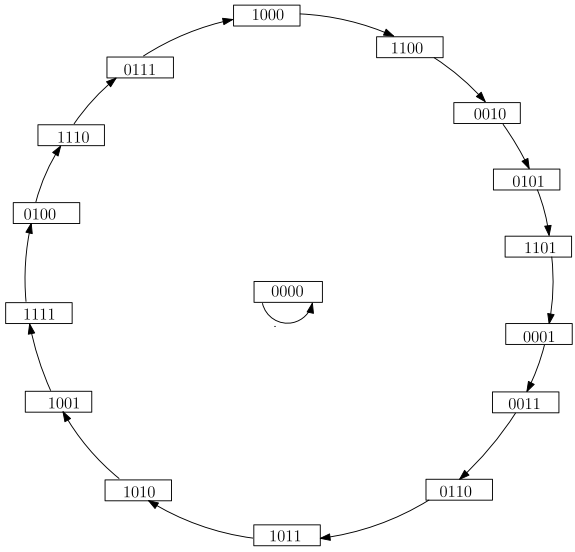


Fig. 6.1 Transition diagram of CA (150, 150, 90, 150) in null boundary

(Fig. 6.1). This is a linear CA, in which all 0s configuration forms a single length cycle. In case of complemented CAs and some of the non-linear maximal length CAs, a non-zero configuration forms a single length cycle. The initial focus of this section is on linear maximal length CAs that employ ECA rules.

6.1.1 Matrix Algebraic Properties

As discussed in Chap. 3 (see Sect. 3.2), an n-cell linear CA can be characterized by a matrix called characteristic matrix $[T]_{n\times n}$, where

$$T_{i,j} = \begin{cases} 1 \text{ if the next state of cell } i \text{ depends on the present state of cell } j, \\ 0 \text{ otherwise.} \end{cases}$$

Since the ECA rules follow 3-neighborhood dependency, the $T_{i,j}$ can have non-zero values only when $j = (i-1), i, (i+1)$. That is, the T-matrix is a tridiagonal matrix, where the main, upper and lower diagonal elements can only be non zero in null boundary.

$$T = \begin{bmatrix} d_0 & b_0 & 0 & \cdots & 0 & 0 \\ a_1 & d_1 & b_1 & \ddots & & 0 \\ 0 & a_2 & d_2 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & & & \ddots & d_{n-2} & b_{n-2} \\ 0 & 0 & \cdots & \cdots & a_{n-1} & d_{n-1} \end{bmatrix}_{n \times n}$$

For ease of reference, the block diagrams of null boundary, periodic boundary and intermediate boundary conditions are given in Fig. 6.2. The characteristic matrix of the 5-cell CA (90, 150, 90, 150, 150), in null boundary (T_{NB}), periodic boundary (T_{PB}) and intermediate boundary (T_{IB}), are shown below.

$$T_{NB} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad T_{PB} = \begin{bmatrix} 0 & 1 & 0 & 0 & \mathbf{1} \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ \mathbf{1} & 0 & 0 & 1 & 1 \end{bmatrix} \quad T_{IB} = \begin{bmatrix} 0 & 1 & \mathbf{1} & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & \mathbf{1} & 1 & 1 \end{bmatrix}$$

The characteristic polynomial of a T-matrix is $P(x) = \det(xI + T)$. All the operations defined here are the GF(2) operations. The $P(x)$ can be obtained following the recurrence relation given in Lemma 9. The notation $\mathcal{R}_{i,j}$ $(i \leq j)$ is adopted to refer to a sub-rule vector of $\mathcal{R}$: $\mathcal{R}_{i,j} = (\mathcal{R}_i, \mathcal{R}_{i+1}, \ldots, \mathcal{R}_j)$. That is, $\mathcal{R}_{i,j}$ consists of rules from cell i to cell j. The $\Delta_{i,j}$ denotes the characteristic polynomial of $\mathcal{R}_{i,j}$, but $\Delta_{0,j}$ is denoted as Δ_j. The $\Delta_{i,j}$ is a subpolynomial. The rule vector $\mathcal{R}$ is, therefore, denoted as $\mathcal{R}_{0,n-1}$ that corresponds to the characteristic polynomial Δ. Let T_k be a $k \times k$ submatrix of T. Then,

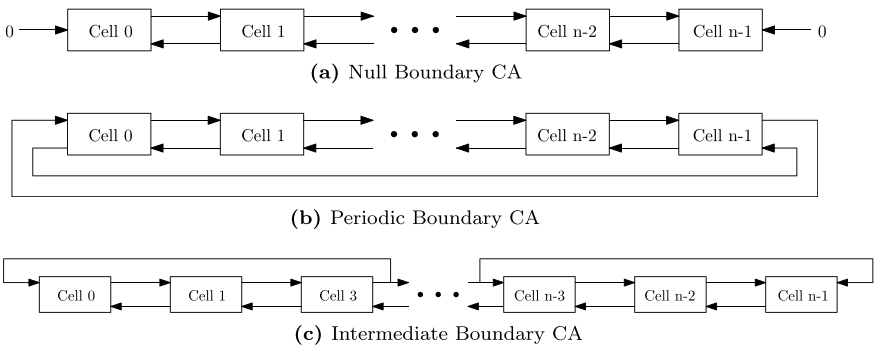


Fig. 6.2 Block diagram of CAs in null, periodic and intermediate boundary

$$T_k = \begin{bmatrix} d_0 & b_0 & 0 & \cdots & 0 & 0 \\ a_1 & d_1 & b_1 & \ddots & & 0 \\ 0 & a_2 & d_2 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & & & \ddots & d_{k-2} & b_{k-2} \\ 0 & 0 & \cdots & \cdots & a_{k-1} & d_{k-1} \end{bmatrix}_{k\times k}$$

Here, the corresponding CA is assumed to be in null boundary.

Lemma 9 *If T_k denotes the $k \times k$ submatrix of T and $\Delta_{k-1} = \det(xI + T_k)$, then Δ_{k-1} satisfies the following*

$$\begin{aligned} \Delta_{-1} &= 1, \\ \Delta_0 &= (x + d_0) \\ \Delta_{k-1} &= (x + d_{k-1})\Delta_{k-2} + b_{k-2}a_{k-1}\Delta_{k-3} \end{aligned}$$

Proof By expanding $(xI + T_k)$ along the last row, we can have $\Delta_{k-1} = (x + d_{k-1})\Delta_{k-2} + a_{k-1}\det(B)$, where

$$B = \begin{bmatrix} x + d_0 & b_0 & 0 & \cdots & 0 & 0 \\ a_1 & x + d_1 & b_1 & \ddots & & 0 \\ 0 & a_2 & x + d_2 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{k-3} & x + d_{k-3} & b_{k-3} \\ 0 & 0 & \cdots & \cdots & a_{k-2} & b_{k-2} \end{bmatrix}$$

Now, $\det(B) = b_{k-2}(xI + T_{k-2}) + a_{k-2} \cdot 0$.

Therefore,

$$\Delta_{k-1} = (x + d_{k-1})\Delta_{k-2} + b_{k-2}a_{k-1}\Delta_{k-3}$$

□

The characteristic polynomial of a CA can be found following the above recurrence relation.

Example 6.1 Consider a 4-cell CA (90, 150, 90, 150) in null boundary, which is represented by the following characteristic matrix:

$$T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}_{4\times 4}$$

Here,

$$\begin{aligned}\Delta_{-1} &= 1,\\ \Delta_0 &= (x+0) = x,\\ \Delta_1 &= (x+1)x+1 = x^2+x+1,\\ \Delta_2 &= (x+0)(x^2+x+1)+x = x^3+x^2,\\ \Delta_3 &= (x+1)(x^3+x^2)+(x^2+x+1) = x^4+x+1\end{aligned}$$

Hence, the characteristic polynomial of the CA is x^4+x+1.

6.1.2 Maximality and Primitivity

The characteristic polynomial of T-matrix can be reducible, irreducible or primitive. It depends on the two factors—the rules in the rule vector and the sequence of rules.

Definition 36 A nonzero polynomial $P(x)$ over GF(q) is said to be *irreducible* if it cannot be factored into two non-constant polynomials $G(x)$ and $H(x)$ over the same field—that is, $P(x) \neq G(x) \times H(x)$; otherwise, $P(x)$ is *reducible*.

Definition 37 The polynomial $P(x)$ over GF(q) with degree k is *primitive* if it is irreducible and $\min_{i\in\mathbb{N}} \{i | P(x) \text{ divides } x^i - 1\} = q^k - 1$. In this case, $P(x)$ has a root α in GF(q^k) such that, it generates all the elements of the extension field GF(q^k) from the base field GF(q) as successive power of α—that is, 0, 1, α, α^2, …, α^{q^k-2} are the elements of the field GF(q^k).

Example 6.2 $x^4+x^3+x^2+1$ is reducible in GF(2), as $x^4+x^3+x^2+1 = (x+1)(x^3+x+1)$. On the other hand, x^4+x^3+1 is irreducible as well as primitive polynomial in GF(2).

Theorem 6.1 *In the characteristic matrix T of an n-cell CA, if $a_i = 0$ for any $i \in \{1, 2, \ldots, n-1\}$ or $b_j = 0$ for any $j \in \{0, 1, \ldots, n-2\}$, then the characteristic polynomial Δ is reducible.*

Proof Suppose, $a_k = 0$ ($0 < k \leq n-1$). Then, according to Lemma 9, $\Delta_k = (x + d_k)\Delta_{k-1}$. Hence, Δ_{k-1} is a factor of Δ_k. Therefore, $\Delta_{k+1} = (x+d_{k+1})\Delta_k + \Delta_{k-1}$ is divisible by Δ_{k-1}. By induction, Δ is also divisible by Δ_{k-1}. Since Δ_{k-1} is a polynomial of degree $k \geq 1$, it is the case that Δ is reducible. Similarly, if $b_k = 0$ is considered, for any $k \in \{0, 1, \ldots, n-2\}$, then the Δ can be shown as reducible. Hence the proof. □

Out of 256 ECA rules, 8 rules are linear (see Table 2.2). All the linear rules except 90 and 150 are either independent of left neighbor ($a = 0$) or independent of right

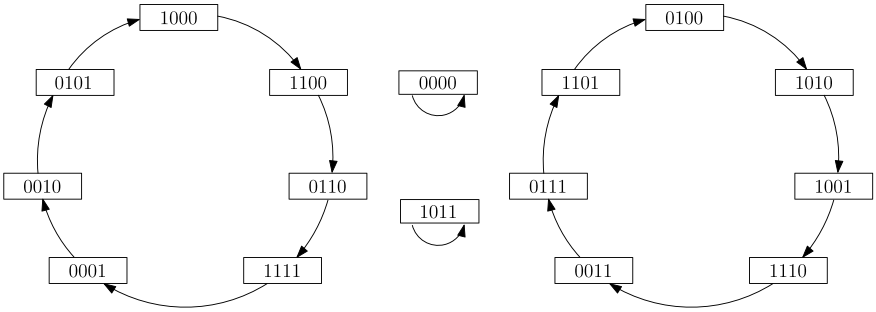


Fig. 6.3 Transition diagram of CA (150, 90, 90, 90)

neighbor ($b = 0$). Therefore, as per Theorem 6.1, these rules cannot be employed to design a maximal length CA. Hence, only the two rules, 90 and 150, have the potential to generate irreducible and primitive characteristic polynomials.

Theorem 6.2 *Only a specific sequence of the rules 90 and 150 can form a maximal length CA.*

Proof By contradiction, consider that any sequence of 90 and 150 of length $n \geq 2$ forms a maximal length CA. The sequence of length 4 (150, 90, 90, 90), however, contradicts the proposition, because the corresponding CA is not a maximal length CA. Transition diagram of the CA is shown in Fig. 6.3. Hence, a specific sequence of 90 and 150 can form a maximal length CA. □

The CA of Fig. 6.3 consists of 2 cycles of length 1 and 2 cycles of length 7. Hence, the *cycle structure* of the CA is identified as [2(1), 2(7)] (see Chap. 7 for details of cycle structure). The characteristic polynomial of the CA is $x^4 + x^3 + x^2 + 1$ which is reducible. The cycle structure of a CA with reducible characteristic polynomial is obtained from the irreducible factors of the characteristic polynomial (detail discussion is in Chap. 7). Cycle structure of a CA with irreducible characteristic polynomial, on the other hand, is of the form $[1(1), \mu(k)]$, where μ is the number of k-length cycles. If the characteristic polynomial Δ is irreducible, then μ and k maintain the following relation

$$\mu = \frac{2^n - 1}{k}$$

where n is the degree of Δ as well as the number of cells of the CA and $k = \min_{i \in \mathbb{N}} \{i | \Delta \text{ divides } x^i - 1\}$. If Δ is primitive, then $k = 2^n - 1$, which implies that the CA is having cycle structure $[1(1),1(2^n - 1)]$—that is, it is a maximal length CA. This discussion leads us the following theorem.

Theorem 6.3 *A linear CA is a maximal length CA iff its characteristic polynomial is primitive over GF(2).*

The algorithm to synthesize maximal length CA, in null boundary, from a primitive polynomial, is given in Sect. 6.3.

The CAs that involve only rules 90 and 150 are also called 90–150 CAs. In T-matrix of a 90–150 CA, upper and lower diagonals are all 1s. Whereas, the elements of main diagonal (d_i) can be 0 or 1—that is,

$$d_i = \begin{cases} 0 \text{ if cell } i \text{ uses rule } 90 \\ 1 \text{ if cell } i \text{ uses rule } 150 \end{cases}$$

The characteristic polynomial of such a 90–150 CA can be derived from the following recurrence relation, which is a simplification of the recurrence relation of Lemma 9.

$$\begin{aligned} &\Delta_{-1} = 1, \\ &\Delta_0 = (x + d_0) \\ &\Delta_1 = (x + d_1)\Delta_0 + \Delta_{-1}, \\ &\vdots \\ &\Delta_k = (x + d_k)\Delta_{k-1} + \Delta_{k-2}, \\ &\vdots \\ &\Delta = \Delta_{n-1} = (x + d_{n-1})\Delta_{n-2} + \Delta_{n-3}. \end{aligned} \tag{6.1}$$

6.2 Analysis of Maximality

There are two approaches available to decide on the maximality of a linear CA.

1. [Exhaustive Search] Test reversibility of a given 90–150 CA. If the CA is reversible, then check whether the cycle length is maximal.
2. [Primitivity Based] Find characteristic polynomial of the given 90–150 CA. If the polynomial is primitive, the CA is of maximal length.

Both the above approaches are of exponential time complexity. However, if primitive polynomials are known a priori, the second approach is efficient. In few cases, without even following any of the above approaches, decision on non-maximality can be possible. This section examines such cases, as well as few other cases which are related to the maximality of a CA.

6.2.1 Boundary Condition Dependence

The fact is, maximality of CA cannot be observed in periodic boundary. For an n-cell periodic boundary CA, with rules 90 and 150, either x or $(x+1)$ is a factor of the characteristic polynomial, or the characteristic polynomial is a perfect square. That is, the characteristic polynomial of a 90–150 CA in periodic boundary is always reducible.

A characteristic polynomial of a 90–150 CA with periodic boundary can be expressed by the characteristic polynomial of a 90–150 CA with null boundary, having the same rule vector. The only difference is in T matrices. The $T_{0,n-1}$ and $T_{n-1,0}$ for periodic boundary are 1. Let the characteristic polynomial of an n-cell CA with periodic boundary be $\Phi_{0,n-1}$. The following lemma relates $\Phi_{0,n-1}$ with $\Delta_{0,n-1}$ and $\Delta_{1,n-2}$.

Lemma 10 *The characteristic polynomial of a 90–150 CA with periodic boundary ($\Phi_{0,n-1}$) is related to the characteristic polynomial of the 90–150 CA in null boundary having the same rule vector. The relation is*

$$\Phi_{0,n-1} = \Delta_{0,n-1} + \Delta_{1,n-2}.$$

Proof Let A be the characteristic matrix of an n-cell 90–150 CA with periodic boundary, where

$$A = \begin{bmatrix} a_0 & 1 & 0 & \cdots & \cdots & 0 & 1 \\ 1 & a_1 & 1 & \ddots & & & 0 \\ 0 & 1 & a_2 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & a_{n-3} & 1 & 0 \\ 0 & & & \ddots & 1 & a_{n-2} & 1 \\ 1 & 0 & \cdots & \cdots & 0 & 1 & a_{n-1} \end{bmatrix}_{n\times n}$$

and a_i denotes $x + d_i$. By expanding the determinant along the first row, we can get.

$$\Phi_{0,n-1} = a_0\Delta_{1,n-1} + |B| + |D|,$$

where

$$B = \begin{bmatrix} 1 & 1 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & a_2 & 1 & \ddots & & & 0 \\ 0 & 1 & a_3 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & a_{n-3} & 1 & 0 \\ 0 & & & \ddots & 1 & a_{n-2} & 1 \\ 1 & 0 & \cdots & \cdots & 0 & 1 & a_{n-1} \end{bmatrix}, \quad D = \begin{bmatrix} 1 & a_1 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 1 & a_2 & 1 & \ddots & & \vdots \\ \vdots & \ddots & 1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & 1 & 0 \\ \vdots & & & \ddots & \ddots & a_{n-3} & 1 \\ 0 & & & & \ddots & 1 & a_{n-2} \\ 1 & 0 & \cdots & \cdots & \cdots & 0 & 1 \end{bmatrix}$$

Now, by expanding B along the first column and D along the last row, the following can be derived.

$$|B| = \Delta_{2,n-1} + |E|, \text{ and}$$
$$|D| = \Delta_{1,n-2} + |F|$$

where

$$E = \begin{bmatrix} 1 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ a_2 & 1 & \ddots & & & & 0 \\ 1 & a_3 & \ddots & \ddots & & & \vdots \\ 0 & 1 & \ddots & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & a_{n-4} & 1 & \ddots & \vdots \\ \vdots & & \ddots & 1 & a_{n-3} & 1 & 0 \\ 0 & \cdots & \cdots & 0 & 1 & a_{n-2} & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 1 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 1 & a_2 & 1 & \ddots & & \vdots \\ \vdots & \ddots & 1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & 1 & 0 \\ \vdots & & & \ddots & \ddots & a_{n-4} & 1 \\ \vdots & & & & \ddots & 1 & a_{n-3} \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & 1 \end{bmatrix}$$

Since E and F are respectively the lower and upper triangular matrices, with all 1s on the diagonal, $|E| = |F| = 1$. Therefore,

$$|B| = \Delta_{2,n-1} + 1$$
$$|D| = \Delta_{1,n-2} + 1$$

Hence,

$$\begin{aligned} \Phi_{0,n-1} &= a_0 \Delta_{1,n-1} + |B| + |D| \\ &= (x + d_0)\Delta_{1,n-1} + \Delta_{2,n-1} + 1 + \Delta_{1,n-2} + 1 \\ &= \Delta_{0,n-1} + \Delta_{1,n-2}, \end{aligned}$$

where $\Delta_{0,n-1} = (x + d_0)\Delta_{1,n-1} + \Delta_{2,n-1}$ □

A CA with periodic boundary has *rotational symmetry*. It implies that if the rule vector of the CA is rotated by any number of positions, the CA remains the same. However, for null boundary, such rotations result in different CAs. A crucial consequence of this is that the n (typically different) null boundary CAs, which are derived from an n-cell CA under periodic boundary, have the same $\Delta_{0,n-1} + \Delta_{1,n-2}$. Let w denote the weight of the rule vector depending on even and odd number of rules 90 and 150. If the rule vector contains an even number of rule 150, then $w = 0$ and 1 otherwise.

Theorem 6.4 *If 90–150 CA in periodic boundary, with characteristic polynomial $\Phi_{0,n-1}$, has weight w, then*

$$\Phi_{0,n-1} = \begin{cases} q^2 & \text{if } n \text{ is even and } w = 0 \\ (x+1)q^2 & \text{if } n \text{ is odd and } w = 1 \\ (x)q^2 & \text{if } n \text{ is odd and } w = 0 \\ (x)(x+1)q^2 & \text{if } n \text{ is even and } w = 1 \end{cases} \tag{6.2}$$

for some polynomial q.

Proof The proof is by induction on n. The base cases can be checked by finding out the characteristic polynomials of all the CAs in periodic boundary, for $n \leq 4$. The inductive step consists of ten cases.

The first eight cases arise when the CA under periodic boundary can be rotated in number of position to get $d_0 = d_{n-1}$ (i.e., the CA contains adjacent cells that have the same rule). The eight cases are due to (1) the n is even or odd, (2) the w is zero or one, and (3) the $d_0 (= d_{n-1})$ is zero or one. It is straightforward to show that

$$\Phi_{0,n-1} = (x + d_0)\Phi_{1,n-1} + \Phi_{1,n-2} \tag{6.3}$$

Let n be even, $w = 0$, and $d_0 = 0$. Then $[d_1, d_2, \ldots, d_{n-1}]$ has odd length and as weight is zero, by induction $\Phi_{1,n-1} = xq_y^2$ for some $q_y \in GF(2)[x]$. Similarly, $[d_1, d_2, \ldots, d_{n-2}]$ has even length and weight zero (as $d_0 = d_{n-1}$). So, $\Phi_{1,n-2} = q_z^2$ for some $q_z \in GF(2)[x]$. Applying it in Eq. 6.3, the following can be derived.

$$\begin{aligned} \Phi_{0,n-1} &= (x + d_0)\Phi_{1,n-1} + \Phi_{1,n-2} \\ &= x(xq_y^2) + q_z^2 \\ &= (xq_y + q_z)^2 \\ &= q^2 \end{aligned}$$

where $q = xq_y + q_z$. Thus the inductive hypothesis is satisfied. The other seven cases can be proven along the same line.

The last two cases arise when no two adjacent cells have the same rule. For these cases, n is necessarily even, but w could be zero or one. However, such a CA is of the form (90, 150, 90, 150, ..., 90, 150). This special structure gives

$$\Phi_{0,n-1} = x(x+1)\Phi_{2,n-1} + \Phi_{4,n-1}$$

By an argument similar to that for the eight cases, the inductive hypothesis can be shown to hold. This completes the proof of the theorem. □

Therefore, the characteristic polynomial of any CA in periodic boundary can be factored. The following result is a consequence of Theorem 6.4.

Corollary 6.1 *There is no 90–150 CA with periodic boundary that has the primitive characteristic polynomial.*

Implication of the above result is that the maximal length CA cannot be found in periodic boundary. However, there exist maximal length CAs against each primitive polynomial in intermediate boundary. Following theorem establishes the fact that there exists at least one CA in intermediate boundary, corresponding to a CA with null boundary.

Theorem 6.5 *For every 90–150 CA in null boundary, there exists at least one CA in intermediate boundary having the same characteristic polynomial.*

Proof The characteristic matrix of a 90–150 CA with null boundary is represented as

$$T_{NB} = \begin{bmatrix} d_0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & d_1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & d_2 & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & d_{n-3} & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & d_{n-2} & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & d_{n-1} \end{bmatrix}_{n \times n}$$

where $d_i = 0$ (resp. 1), if the rule of cell i is 90 (resp. 150). The characteristic polynomial of the CA is $\det(T_{NB} + Ix)$, where

$$T_{NB} + Ix = \begin{bmatrix} d_0 + x & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & d_1 + x & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & d_2 + x & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 + x & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & d_{n-3} + x & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & d_{n-2} + x & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & d_{n-1} + x \end{bmatrix}_{n \times n}$$

Let consider that the rows and columns of the above matrix are $Row(1)$, $Row(2)$, ..., $Row(n)$, and $Col(1)$, $Col(2)$, ..., $Col(n)$ respectively. The following are the matrices resulted out of the row and column operations noted against them.

$$\Rightarrow \begin{bmatrix} 1+d_0+x & 1+d_1+x & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & d_1+x & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & d_2+x & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3+x & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & d_{n-3}+x & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & d_{n-2}+x & 1 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1+d_{n-2}+x & 1+d_{n-1}+x \end{bmatrix}_{n\times n}$$

$$Row(1) \leftarrow Row(1) + Row(2)$$
$$Row(n) \leftarrow Row(n) + Row(n-1)$$

$$\Rightarrow \begin{bmatrix} 1+d_0+x & d_0+d_1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 1+d_1+x & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & d_2+x & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3+x & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & d_{n-3}+x & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1+d_{n-2}+x & 1 \\ 0 & 0 & 0 & 0 & \cdots & 1 & d_{n-2}+d_{n-1} & 1+d_{n-1}+x \end{bmatrix}_{n\times n}$$

$$Col.(2) \leftarrow Col.(1) + Col.(2)$$
$$Col.(n-1) \leftarrow Col.(n) + Col.(n-1)$$

$$\Rightarrow \begin{bmatrix} 1+d_0 & d_0+d_1 & \boxed{1} & 0 & \cdots & 0 & 0 & 0 \\ 1 & 1+d_1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & d_2 & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & d_{n-3} & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1+d_{n-2} & 1 \\ 0 & 0 & 0 & 0 & \cdots & \boxed{1} & d_{n-2}+d_{n-1} & 1+d_{n-1} \end{bmatrix}_{n\times n}$$

$$\Rightarrow T_{IB} + Ix$$

where T_{IB} is the matrix representation of CA with intermediate boundary. Hence the proof. □

6.2.2 Palindromic and Uniform CA

This section explores some special patterns of rule vectors, for which the CA in null boundary is always non-maximal. First of these is the palindromic, where the rule vector is identical to its own reversal—that is, $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1}) = (\mathcal{R}_{n-1}, \mathcal{R}_{n-2}, \ldots, \mathcal{R}_0)$. As an example, the CA (150, 90, 150, 90, 150) is a palindromic CA. The characteristic polynomial of an even length palindromic CA is a perfect square, and it is the product of a monomial and a perfect square for odd length. Hence, the characteristic polynomial of a palindromic CA is always reducible.

To facilitate the proof, let an n-cell CA be partitioned into two components—the first one consists of the cells $0, 1, \ldots, k-1$ and the second component consists of the cells $k, k+1, \ldots, n-1$ (Fig. 6.4).

Theorem 6.6 *If a CA in null boundary is partitioned into two components, one with cells* $0, 1, \ldots, k-1$ *and the other with the cells* $k, k+1, \ldots, n-1$*, then for any* k *with* $0 \leq k \leq n-1$*,*

$$\Delta_{0,n-1} = \Delta_{0,k-1}\Delta_{k,n-1} + \Delta_{0,k-2}\Delta_{k+1,n-1}.$$

Proof This is proved by induction. The base case of the inductive proof is for $k = 0$. By definition of $\Delta_{i,j}$ and from Lemma 9, $\Delta_{0,-1}$=Δ_{-1}=1, and $\Delta_{0,-2}$=Δ_{-2}=0. Then, as per the theorem

$$\begin{aligned}\Delta_{0,n-1} &= \Delta_{0,-1}\Delta_{0,n-1} + \Delta_{0,-2}\Delta_{1,n-1} \\ &= 1 \cdot \Delta_{0,n-1} + 0 \cdot \Delta_{1,n-1} \\ &= \Delta_{0,n-1}\end{aligned}$$

and hence is satisfied trivially.

Assuming that the theorem holds for k, it is to show that it holds for $k+1$. By the inductive hypothesis,

$$\Delta_{0,n-1} = \Delta_{0,k-1}\Delta_{k,n-1} + \Delta_{0,k-2}\Delta_{k+1,n-1}$$

A CA has the left-right symmetry with respect to its characteristics polynomial. It implies that the characteristic polynomial remains unchanged if the cell labelings are reversed. This is formally proved in Sect. 6.3.1 (see Corollary 6.4). Hence,

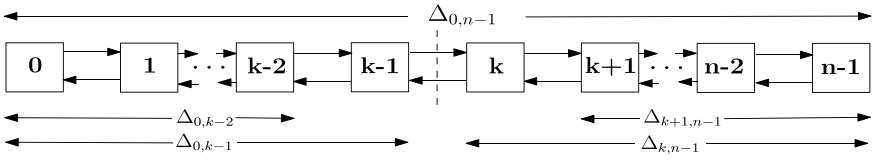


Fig. 6.4 Component CAs and their concatenation

Relation 6.1 can be rewritten considering the right-most rule as the first rule. Then we get

$$\Delta_{k,n-1} = (x + d_k)\Delta_{k+1,n-1} + \Delta_{k+2,n-1}$$

Hence,

$$\begin{aligned}\Delta_{0,n-1} &= \Delta_{0,k-1}\Delta_{k,n-1} + \Delta_{0,k-2}\Delta_{k+1,n-1} \\ &= \Delta_{0,k-1}((x + d_k)\Delta_{k+1,n-1} + \Delta_{k+2,n-1}) + \Delta_{0,k-2}\Delta_{k+1,n-1} \\ &= \Delta_{0,k-1}\Delta_{k+2,n-1} + (x + d_k)\Delta_{0,k-1}\Delta_{k+1,n-1} + \Delta_{0,k-2}\Delta_{k+1,n-1} \\ &= \Delta_{0,k-1}\Delta_{k+2,n-1} + ((x + d_k)\Delta_{0,k-1} + \Delta_{0,k-2})\Delta_{k+1,n-1}\end{aligned}$$

Applying Relation 6.1, where $\Delta_{0,k} = \Delta_k = (x + d_k)\Delta_{0,k-1} + \Delta_{0,k-2}$, we get

$$\Delta_{0,n-1} = \Delta_{0,k}\Delta_{k+1,n-1} + \Delta_{0,k-1}\Delta_{k+2,n-1}$$

Hence, the relation of the theorem holds for $k + 1$, given that it holds for k. Hence the proof. □

Example 6.3 Consider the 4-cell CA $(90, 150, 90, 150)$ in null boundary. With $k = 2$,

$$\begin{aligned}\Delta_{0,3} &= \Delta_{0,1}\Delta_{2,3} + \Delta_{0,0}\Delta_{3,3} \\ &= (x^2 + x + 1)(x^2 + x + 1) + x(x + 1) \\ &= x^4 + x + 1\end{aligned}$$

Hence, the characteristic polynomial of the CA is $x^4 + x + 1$.

Theorem 6.7 *The characteristic polynomial Δ ($\Delta_{0,n-1}$) of a palindromic CA is given by*

$$\Delta_{0,n-1} = \begin{cases} (\Delta_{0,\frac{n}{2}-1} + \Delta_{0,\frac{n}{2}-2})^2, & \text{if } n \text{ is even} \\ (x + d_{\frac{n-1}{2}})(\Delta_{0,\frac{n-1}{2}-1})^2, & \text{if } n \text{ is odd} \end{cases}$$

Proof Suppose that n is even, and $k = \frac{n}{2} - 1$. The symmetry of the rule vector $\mathcal{R}$ implies that—$(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1}) = (\mathcal{R}_0, \ldots, \mathcal{R}_k, \mathcal{R}_k, \ldots, \mathcal{R}_0)$.

Due to left-right symmetry of the CA with respect to its characteristic polynomial, the polynomial remains unchanged if the cell labelings are reversed. Applying it into the palindromic CA, we can get $\Delta_{0,k} = \Delta_{k+1,n-1}$ and $\Delta_{0,k-1} = \Delta_{k+2,n-1}$. While substituting these into Theorem 6.6, the relation can be

$$\begin{aligned}\Delta_{0,n-1} &= \Delta_{0,k}\Delta_{k+1,n-1} + \Delta_{0,k-1}\Delta_{k+2,n-1} \\ &= \Delta_{0,k}\Delta_{0,k} + \Delta_{0,k-1}\Delta_{0,k-1} \\ &= (\Delta_{0,k})^2 + (\Delta_{0,k-1})^2 \\ &= (\Delta_{0,k} + \Delta_{0,k-1})^2 \\ &= (\Delta_{0,\frac{n}{2}-1} + \Delta_{0,\frac{n}{2}-2})^2\end{aligned}$$

Now, if n is odd and $k = \frac{n-1}{2} - 1$, then the $\mathcal{R}$ is of the form $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ $= (\mathcal{R}_0, \ldots, \mathcal{R}_{k-1}, \mathcal{R}_k, \mathcal{R}_{k+1}, \mathcal{R}_k, \mathcal{R}_{k-1}, \ldots, \mathcal{R}_0)$. Hence, $\Delta_{0,k} = \Delta_{k+2,n-1}$ and $\Delta_{0,k-1}$ $= \Delta_{k+3,n-1}$. Now,

$$\begin{aligned}\Delta_{0,n-1} &= \Delta_{0,k}\Delta_{k+1,n-1} + \Delta_{0,k-1}\Delta_{k+2,n-1} \\ &= \Delta_{0,k}((x + d_{k+1})\Delta_{k+2,n-1} + \Delta_{k+3,n-1}) + \Delta_{0,k-1}\Delta_{0,k} \\ &= \Delta_{0,k}((x + d_{k+1})\Delta_{0,k} + \Delta_{0,k-1}) + \Delta_{0,k-1}\Delta_{0,k} \\ &= (x + d_{k+1})\Delta_{0,k}\Delta_{0,k} + \Delta_{0,k}\Delta_{0,k-1} + \Delta_{0,k-1}\Delta_{0,k} \\ &= (x + d_{k+1})(\Delta_{0,k})^2 \\ &= (x + d_{\frac{n-1}{2}})(\Delta_{0,\frac{n-1}{2}-1})^2\end{aligned}$$

□

The above theorem also states that the characteristic polynomial of palindromic CA is reducible. Therefore, a palindromic CA cannot be maximal length CA. A *uniform* CA is trivially palindromic and has a reducible characteristic polynomial. So, a uniform CA cannot be the maximal length CA. Following corollary is followed from the above theorem.

Corollary 6.2 *The characteristic polynomial of an n-cell uniform CA is always reducible.*

Example 6.4 Let us consider a 5-cell CA (150, 90, 150, 90, 150) in null boundary. This is a palindromic CA. The characteristic polynomial of this CA is $x^5 + x^4 + x^3 + x^2 + x + 1 = (x + 1)(x^2 + x + 1)^2$. It is a reducible polynomial. Let consider the 8-cell uniform CA with rule 90 (in null boundary). Its characteristic polynomial is $x^8 + x^6 + x^4 + 1 = (x + 1)^2(x^3 + x + 1)^2$ and, therefore, reducible.

6.2.3 Conjugate CA

It has been shown in Sect. 6.1.2 that with only the rules 90 and 150 a linear maximal length CA may be designed. There is no efficient algorithm to decide on the maximality of a 90–150 CA or primitivity of a GF(2) polynomial. Therefore, search for a pattern in the rule vectors of maximal length CAs may help to find the maximal

length CA. For an n, there are 2^n rule vectors of 90–150 CAs. Now the palindromic rule vectors can be excluded from the 2^n to reduce the search space.

Apart from the palindromic pattern in rule vectors, however, there is no specific pattern found which can help us to conclude the maximality or non-maximality of a 90–150 CA. For example, ratio of rules or existence of specific sequence in rule vectors do not guide us to decide the maximality of a 90–150 CA. However, a property of rule vectors of 90–150 CAs that is related to irreducible polynomials, helps to reduce the search space of maximal length CAs.

Definition 38 Two CAs $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ and $\mathcal{R}' = (\mathcal{R}'_0, \mathcal{R}'_1, \ldots, \mathcal{R}'_{n-1})$ are said to be *conjugate* to each other if the following condition is satisfied: $\mathcal{R}_i$ is 90 (resp. 150) iff $\mathcal{R}'_i$ is 150 (resp. 90), for each $i \in \{0, 1, \ldots, n-1\}$.

As an example, let consider the CA $\mathcal{R} = (90, 90, 90, 150, 150)$. Its conjugate CA is $\mathcal{R}' = (150, 150, 150, 90, 90)$.

Theorem 6.8 *The characteristic polynomial of a CA is irreducible iff the characteristic polynomial of its conjugate is irreducible.*

Proof If $p(x) = det(T + xI)$ is the characteristic polynomial of CA $\mathcal{R}$ then the characteristic polynomial of its conjugate $\mathcal{R}'$ is $det(T + I + xI) = det(T + (x + 1)I) = p(x + 1)$. If $p(x)$ is not irreducible and $p(x) = a(x)b(x)$, then also $p(x + 1)$ is not irreducible as $p(x + 1) = a(x + 1)b(x + 1)$. Symmetrically, if $p(x + 1)$ is not irreducible, then its conjugate $p(x)$ is not irreducible. □

Example 6.5 Let consider the pair of conjugate CAs: $\mathcal{R} = (90, 90, 90, 150, 150)$ and $\mathcal{R}' = (150, 150, 150, 90, 90)$. The characteristic polynomial of $\mathcal{R}$ is $x^5 + x^3 + x^2 + x + 1$, which is irreducible. For $\mathcal{R}'$, it is $x^5 + x^4 + x^3 + x + 1$, which is also irreducible.

Now, it indicates that there may exist some pairs of conjugate CAs where each member of a pair is a maximal length CA. A good share of maximal length CAs can be found from these pairs. Table 6.1 explains how much of maximal length CAs are conjugate maximal length CAs. Some of the rows ($n = 5, 7, 13, 17, 19$) of Table 6.1 show that all the maximal length CAs are conjugate maximal length CAs. The exponent of these numbers are, however, Mersenne prime.[1] It is known that, in these cases, all the irreducible polynomials are primitive.

6.2.4 Phase Shift Operation

Table 6.2 notes the consecutive configurations of a maximal length CA (150, 150, 90, 150) (see Fig. 6.1). Column 2 shows the bit sequence generated by cell 0. It can

[1] A Mersenne prime is a prime number which is one less than a power of two. It is a prime number of the form $M_p = 2^n - 1$ for some prime p.

Table 6.1 Distribution of conjugate maximal length CAs

# cells	# Maximal	# Conjugate maximal	%
4	4	2	50
5	12	12	100
6	12	6	50
7	36	36	100
8	32	18	56.25
9	96	84	87.50
10	120	68	56.66
11	352	332	94.31
12	288	110	38.19
13	1260	1260	100
14	1512	972	64.28
15	3600	2972	82.55
16	4096	2022	49.36
17	15420	15420	100
18	15552	8268	53.16
19	55188	55188	100
20	48000	22062	45.95

Table 6.2 Phase shift with respect to cell 0 of the CA (150, 150, 90, 150) in null boundary

Step	Cell			
	0	1	2	3
1	**1**	0	0	0
2	1	1	0	0
3	0	0	1	0
4	0	**1**	0	1
5	1	1	0	1
6	0	0	0	1
7	0	0	1	1
8	0	1	1	0
9	1	0	1	1
10	1	0	1	0
11	1	0	0	**1**
12	1	1	1	1
13	0	1	0	0
14	1	1	**1**	0
15	0	1	1	1

be observed that the same sequence is repeated in the third column from the 4th time step (start of a sequence is marked by bold face). The same sequence is also repeated in the remaining columns. Let consider that $(S_i^0)_{1\le i\le 15}$ is the bit sequence generated by cell 0, and σ is a left shift operator. Then,

$$(S_i^0)_{1\le i\le 15} = \sigma^3((S_i^1)_{1\le i\le 15}) = \sigma^{13}((S_i^2)_{1\le i\le 15}) = \sigma^{10}((S_i^3)_{1\le i\le 15})$$

These shifts in sequences with respect to cell 0 are called *phase shifts*. Hence, the phase shift of first, second and third cell with respect to cell 0 are 3, 13 and 10 respectively.

Definition 39 For an n-cell maximal length CA, there exists an integer k, $1 \le k \le 2^n - 1$, such that $(S_i^a)_{1\le i\le 2^n-1} = \sigma^k((S_i^b)_{1\le i\le 2^n-1})$, where a and b are the two cells of the CA ($0 \le a, b \le n-1$). Such a shift of k positions in the bit sequence is called *phase shift*.

The amount of phase shift can be found out from the T-matrix of a maximal length CA.

Theorem 6.9 *If T^k has a 1 at the jth column of the ith row (where $i \neq j$), with all other elements of the row as 0, then there exists a phase shift of k bits for the bit sequence generated by the ith cell with respect to that of the jth cell of the CA having T as its characteristic matrix.*

Proof Let the configuration of a CA with characteristic matrix T at a particular instant of time be S and after k time step, it is S^k. Assume that the ith row of T^k has a single 1 at the jth column. That is,

$$S^k = T^k \times S$$

That is,

$$\begin{bmatrix} s_0^k \\ s_1^k \\ \vdots \\ s_i^k \\ \vdots \\ s_{n-1}^k \end{bmatrix} = \begin{bmatrix} \vdots & \vdots\ \vdots\ \vdots & \vdots \\ 0\cdots 0 & 1 & 0\cdots 0 \\ \vdots & \vdots\ \vdots\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} s_0 \\ s_1 \\ \vdots \\ s_j \\ \vdots \\ s_{n-1} \end{bmatrix}$$

Hence, $s_i^k = s_j$.

This proves that the sequence appearing at the jth cell appears at the ith cell after k time step. Hence the proof. □

Let take the example CA of Table 6.2. As per Theorem 6.9, the positions of T^k have been marked (as shown below) to identify the shifting characteristic.

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad T^2 = \begin{bmatrix} 0 & 0 & \boxed{1} & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad T^3 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ \boxed{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \quad T^5 = \begin{bmatrix} 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

The first row of T^2 contains a single 1 in the third column, which means that the sequence generated by Cell 0 has a 2-bit ($k = 2$) phase shift with respect to the sequence generated by Cell 2. Similarly, $k = 3$ for the shift of Cell 1 with respect to Cell 0, and $k = 5$ for the shift of Cell 0 with respect to Cell 3. As this is a 4-cell maximal length CA, the sequence generated by any cell is also of length $2^4 - 1 = 15$. Considering circularity of the sequence, the phase shifts of all the cells with respect to the 0th cell is found out. For the above example, these shifts are 3, 13 and 10 for the cells 1, 2 and 3 respectively.

Based on the above characterization, a method can be developed to find the phase shift of any cell with respect to any other cell of a given n-cell maximal length CA. Firstly, a set of k (shifts) is to be found out such that T^k contains a row with a single 1. Then, by appropriately manipulating these shifts, the phase shift of the desired cell can be identified.

6.3 Synthesis of Maximal Length CA from Primitive Polynomial

The pair of polynomials Δ_{n-1} and Δ_{n-2} (defined in Sect. 6.1.1) uniquely determine the CA (using Euclid's GCD algorithm). Conversely, a CA uniquely determines Δ_{n-1} and Δ_{n-2}. If Δ_{n-1} is divided by Δ_{n-2}, then it results in the quotient $(x + d_{n-1})$ and the remainder Δ_{n-3} (as per Relation 6.1). A series of quotients $x + d_{n-1}$, $x + d_{n-2}$, $\ldots$, $x + d_0$ is, therefore, resulted in while computing the GCD of Δ_{n-1} and Δ_{n-2} by Euclid's algorithm. The sequence $(d_0,d_1,\ldots,d_{n-1})$ corresponds to the rule vector of a 90–150 CA (Lemma 11).

Definition 40 For an n-cell CA, $\Delta_{0,n-2}$ is called a CA *subpolynomial*, and the pair of polynomials $\Delta_{0,n-1}$ and $\Delta_{0,n-2}$ (hence Δ_{n-1} and Δ_{n-2}) is called a CA *polynomial-subpolynomial* pair. A CA subpolynomial is itself a CA polynomial.

Lemma 11 *Let p and q be two polynomials, with degree n and $n-1$ respectively. Then, there exists a CA with characteristic polynomial p and subpolynomial q ($\Delta_{n-1} = p$ and $\Delta_{n-2} = q$) iff, while applying Euclid's GCD algorithm to p and q, n quotients with degree 1 are generated in sequence.*

Proof Let there exists a CA with $\Delta_{n-1} = p$ and $\Delta_{n-2} = q$. Lemma 9 (Relation 6.1) ensures that the sequence of remainder polynomials obtained from Euclid's GCD algorithm is Δ_{n-3}, Δ_{n-4}, $\ldots$, Δ_1, Δ_0. As these polynomials have degrees differing by one, Euclid's algorithm produces n quotients, each of which is of degree 1. For the converse, suppose that Euclid's algorithm is applied to a degree-n polynomial p and

degree-$(n-1)$ polynomial q, and that n quotients $x + d_{n-1}$, $x + d_{n-2}$, …, $x + d_0$ are obtained in sequence. The last remainder is always zero. The n quotients imply that the last divisor (second last remainder) is one. Now, if the recurrence Relation 6.1 is applied considering $d_0, d_1, \ldots, d_{n-1}$, we can get the polynomials $\Delta_{n-1} = p$ and $\Delta_{n-2} = q$. Hence, there exists a CA with the desired properties. □

Example 6.6 Let consider the CA (90, 150, 150, 90, 90), where $\Delta_{n-1} = x^5 + x^3 + 1$ and $\Delta_{n-2} = x^4 + 1$. When Euclid's GCD algorithm is applied on Δ_{n-1} and Δ_{n-2}, following can be found during successive divisions.

Dividend	Divisor	Quotient	Remainder	d_i (Rule)
$x^5 + x^3 + 1$	$x^4 + 1$	x	$x^3 + x + 1$	0 (90)
$x^4 + 1$	$x^3 + x + 1$	x	$x^2 + x + 1$	0 (90)
$x^3 + x + 1$	$x^2 + x + 1$	$x + 1$	x	1 (150)
$x^2 + x + 1$	x	$x + 1$	1	1 (150)
x	1	x	0	0 (90)

The rules are extracted from the series of 5 quotients to get the rule vector of the CA. However, synthesis of a CA following such steps is the worst-case scenario for the GCD algorithm, because it requires maximum possible application of the division algorithm.

6.3.1 The Quadratic Congruence

Identification of a q, satisfying Lemma 11, can solve the CA synthesis problem. In order to find q $(= \Delta_{n-1})$, a congruence relation, referred to as the CA *quadratic congruence*, is developed. The relation is reported in Theorem 6.11, which follows Theorem 6.10.

Let represent a 90–150 CA by $(d_0, d_1, \ldots, d_{n-1})$. If the first (resp. last) rule—that is, d_0 (resp. d_{n-1}) is complemented, then the resulted CA is represented as $(d_0 + 1, d_1, \ldots, d_{n-1})$ (resp. $(d_0, d_1, \ldots, d_{n-1} + 1)$). Characteristic polynomials of these CAs (resulted out of complementation) can be expressed in terms of characteristic polynomials and subpolynomials of the original CA $(d_0, d_1, \ldots, d_{n-1})$. That is,

Case	Characteristic polynomial
original $(d_0, d_1, \ldots, d_{n-1})$	$\Delta_{0,n-1}$
complement d_{n-1} $(d_0, d_1, \ldots, d_{n-1} + 1)$	$\Delta_{0,n-1} + \Delta_{0,n-2}$
complement d_0 $(d_0 + 1, d_1, \ldots, d_{n-1})$	$\Delta_{0,n-1} + \Delta_{1,n-1}$
complement both $(d_0 + 1, d_1, \ldots, d_{n-1} + 1)$	$\Delta_{0,n-1} + \Delta_{0,n-2} + \Delta_{1,n-1} + \Delta_{1,n-2}$

Here, $d_i = 0$ when the rule for cell i is 90 and $d_i = 1$ when the rule is 150.

Theorem 6.10 *For any CA,*

$$\Delta_{0,n-2} + \Delta_{1,n-1} = \begin{cases} (x^2+x)\Delta'_{0,n-1} & \text{if } n \text{ is even, } w = 0 \\ \Delta_{0,n-1} + (x^2+x)\Delta'_{0,n-1} & \text{if } n \text{ is even, } w = 1 \\ (x+1)\Delta_{0,n-1} + (x^2+x)\Delta'_{0,n-1} & \text{if } n \text{ is odd, } w = 0 \\ x\Delta_{0,n-1} + (x^2+x)\Delta'_{0,n-1} & \text{if } n \text{ is odd, } w = 1 \end{cases}$$

Proof The proof requires inspection of four cases. Let n be even and w be zero.

As per Eq. 6.2, there exists a polynomial q_1 such that

$$\Phi_{0,n-1} = \Delta_{0,n-1} + \Delta_{1,n-2} = q_1^2$$

If d_{n-1} is complemented, $[d_1, \ldots, d_{n-2}]$ remains unchanged and the length is still even, but the weight becomes one. Hence, there exists a q_2 such that

$$(\Delta_{0,n-1} + \Delta_{0,n-2}) + \Delta_{1,n-2} = (x^2+x)q_2^2 \tag{6.4}$$

Similarly, if d_0 is complemented, there is a q_3 satisfying

$$(\Delta_{0,n-1} + \Delta_{1,n-1}) + \Delta_{1,n-2} = (x^2+x)q_3^2 \tag{6.5}$$

If both d_0 and d_{n-1} are complemented, the resulting CA is of even length and the weight is zero. Therefore, there can be a q_4 such that

$$\begin{aligned} (\Delta_{0,n-1} + \Delta_{0,n-2} + \Delta_{1,n-1} + \Delta_{1,n-2}) + \Delta_{1,n-2} &= q_4^2 \\ \Rightarrow \Delta_{0,n-1} + \Delta_{0,n-2} + \Delta_{1,n-1} &= q_4^2 \end{aligned} \tag{6.6}$$

By adding the Eqs. 6.4 and 6.5 and differentiating both sides:

$$\begin{aligned} \Delta_{0,n-2} + \Delta_{1,n-1} &= (x^2+x)(q_2+q_3)^2 \\ \Rightarrow (\Delta_{0,n-2} + \Delta_{1,n-1})' &= (q_2+q_3)^2 \end{aligned}$$

As a result, the following differential equation is obtained.

$$(\Delta_{0,n-2} + \Delta_{1,n-1}) = (x^2+x)(\Delta_{0,n-2} + \Delta_{1,n-1})' \tag{6.7}$$

Equation 6.6 denotes that $\Delta_{0,n-1} + \Delta_{0,n-2} + \Delta_{1,n-1}$ is a perfect square, and so

$$\Delta'_{0,n-1} = (\Delta_{0,n-2} + \Delta_{1,n-1})' \tag{6.8}$$

Combining the Eqs. 6.7 and 6.8, the following can be obtained.

$$\Delta_{0,n-2} + \Delta_{1,n-1} = (x^2+x)\Delta'_{0,n-1}$$

This completes the proof for Case 1. For the other cases, the similar argument holds. □

Theorem 6.11 *If a CA is having the characteristic polynomial $\Delta_{0,n-1}$ and characteristic subpolynomials $\Delta_{0,n-2}$ and $\Delta_{1,n-1}$, then both $y = \Delta_{0,n-2}$ and $y = \Delta_{1,n-1}$ satisfy the congruence*

$$y^2 + (x^2 + x)\Delta'_{0,n-1}y + 1 \equiv 0 \pmod{\Delta_{0,n-1}}$$

where $\Delta'_{0,n-1}$ is the formal derivative of $\Delta_{0,n-1}$.

Proof The proof applies the relation of Theorem 6.10. It gives the following congruence relation.

$$\Delta_{0,n-2} + \Delta_{1,n-1} \equiv (x^2 + x)\Delta'_{0,n-1} \pmod{\Delta_{0,n-1}} \tag{6.9}$$

The product of subpolynomials relation comes from Euclid's extended GCD algorithm. It states that

$$\Delta_{0,n-2}\Delta_{1,n-1} \equiv 1 \pmod{\Delta_{0,n-1}} \tag{6.10}$$

Solving Eq. 6.9 for $\Delta_{1,n-1}$ and substituting it into Eq. 6.10, we can get

$$(\Delta_{0,n-2})^2 + (x^2 + x)\Delta'_{0,n-1}\Delta_{0,n-2} + 1 \equiv 0 \pmod{\Delta_{0,n-1}}$$

Equivalently,

$$(\Delta_{1,n-1})^2 + (x^2 + x)\Delta'_{0,n-1}\Delta_{1,n-1} + 1 \equiv 0 \pmod{\Delta_{0,n-1}}$$

Hence the proof. □

Example 6.7 Let consider the CA (90, 90, 150, 150, 90) in null boundary. Its characteristic polynomial is $x^5 + x^3 + 1$. For this CA, $\Delta_{0,3} = x^4 + x$ and $\Delta_{1,4} = x^4 + 1$. Substitution of $y = x^4 + 1$ into the left-hand side of Theorem 6.8 gives

$$\begin{aligned} y^2 + (x^2 + x)\Delta'_{0,n-1}y + 1 &= (x^4 + 1)^2 + (x^2 + x)(x^4 + x^2)(x^4 + 1) + 1 \\ &= (x^8 + 1) + (x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^3) + 1 \\ &= 0 \pmod{x^5 + x^3 + 1} \end{aligned}$$

Similarly, substitution of $y = x^4 + x$ into the left-hand side results in zero.

Corollary 6.3 *If p is a polynomial of degree n, then p is a CA polynomial iff there exists some solution q for y of the congruence*

$$y^2 + (x^2 + x)p'y + 1 \equiv 0 \pmod{p} \tag{6.11}$$

where p' is the formal derivative of p.

The following algorithmic steps can synthesize a CA.

for each solution $q = y$ *to*
$y^2 + (x^2 + x)p'y + 1 \equiv 0 \pmod{p}$
if $GCD(p, q)$ *yields n quotients, each with degree 1*
then output the CA considering the n quotients

If no CA is found, then p cannot be the characteristic polynomial of a CA.

Corollary 6.4 *If p is an irreducible polynomial of degree n, then p is the characteristic polynomial of exactly two CAs.*

Proof As per Theorem 6.11, two solutions for a characteristic polynomial of a CA can be possible. This implies that two CAs can be found with the same characteristic polynomial. It can also be shown from the behavior of Euclid's GCD algorithm on GF(2) polynomials that the CA quadratic congruence (Theorem 6.11) has exactly two solutions for an irreducible polynomial. Hence, each irreducible polynomial is the characteristic polynomial of exactly two CAs. □

Example 6.8 Consider the irreducible polynomial $p(x) = x^5 + x^3 + 1$. For $q(x) = x^4 + x$, Eq. 6.11 is satisfied (see Example 6.7). So, it can be used to find a CA. If $GCD(p, q)$ is computed, the 5 quotients $x, x + 1, x + 1, x, x$ are generated in sequence. This corresponds to the CA (90, 90, 150, 150, 90). Now, let $q(x) = x^4 + 1$. This also satisfies Eq. 6.11 and $GCD(p, q)$ generates quotient sequence: $x, x, x + 1, x + 1, x$, which corresponds to the CA (90, 150, 150, 90, 90). For any other q, Eq. 6.11 is not satisfied. So, there can be only two CAs for the characteristic polynomial $p(x)$.

The following section provides an efficient procedure to find q for synthesizing a CA from the irreducible polynomial.

6.3.2 Synthesis of CA from Irreducible Polynomial

Let p be a polynomial of degree n. As per Theorem 6.11,

$$y^2 + (x^2 + x)p'y + 1 \equiv 0 \pmod{p}$$

Let introduce two polynomials for efficient deduction of q: $f(x) = (x^2 + x)p'$ and $g(x) = \left(\frac{1}{f(x)}\right)^2$.

To describe the synthesis algorithm, the *trace* (Tr(a)) of a polynomial $a(x)$ is defined with respect to the irreducib polynomial p, where

$$\begin{aligned} Tr(a) &= (a + a^2 + a^4 + \cdots + a^{2^{n-1}}) \pmod{p} \\ &= \left(\sum_{i=0}^{n-1} a^{2^i}\right) \pmod{p} \end{aligned}$$

Now, a polynomial $\theta(x)$ with trace one is required. If n is odd, then $\theta(x) = 1$ may be used. But if n is even, then such a polynomial is to be found out. Interestingly, any such $\theta(x)$ suffices. Now, a polynomial $\beta(x)$ is defined as

$$\begin{aligned}\beta(x) &= g\theta^2 + (g + g^2)\theta^2 + \cdots + (g + g^2 + \cdots + g^{2^{n-2}})\theta^{2^{n-1}} \\ &= \sum_{i=1}^{n-1} \left(\sum_{j=0}^{i-1} g^{2^j} \right) \theta^{2^i}\end{aligned}$$

If n is odd, the use of $\theta(x) = 1$ simplifies the expression.

$$\begin{aligned}\beta(x) &= g^2 + g^8 + \cdots + g^{2^{n-2}} \\ &= \sum_{i=1}^{(n-1)/2} g^{2^{2i-1}}\end{aligned}$$

The βf now can be considered as q. As p is irreducible, the quadratic has two distinct solutions (Theorem 6.11)—$q_1(x)$ and $q_2(x)$. The CA, obtained while considering q_2, is the reversal of the CA obtained for q_1. However, only one solution is needed. Euclid's GCD algorithm is then applied to get the solution. Algorithm 6 summarizes the process of computing a CA corresponding to a given irreducible polynomial p.

Algorithm 6: Synthesize CA from irreducible polynomial

Input : $p(x)$ – an n-degree irreducible polynomial in GF(2).
Output: An n-cell CA in null boundary.

Step 1 : Find $f = (x^2 + x)p'(x)$, where $p'(x)$ is formal derivative of $p(x)$.
Step 2 : Find $g = (\frac{1}{f})^2$
Step 3 : If n is even, find θ with trace one, and then calculate β.
Otherwise, calculate β.
Step 4 : Find $q = \beta f$.
Step 5 : Compute $\gcd(p, q)$.
Step 6 : Collect the constant terms of the n quotients, generated in Step 5, as $[d_{n-1}, \cdots, d_1, d_0]$.
Step 7 : Report the CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$, where $\mathcal{R}_i = 90$ if $d_i = 0$ and $\mathcal{R}_i = 150$ if $d_i = 1$ for $i \in \{0, 1, \cdots, n-1\}$.

Let consider the following irreducible (in this case, it is primitive) polynomial of degree $n = 5$.

$$p = x^5 + x^2 + 1$$

The derivative of p (reducing the coefficients modulo 2) is

$$p' = 5x^4 + 2x = x^4$$

Now f (all polynomials are reduced in modulo p) is

$$f = (x^2 + x)p' = x^6 + x^5 = x^3 + x^2 + x + 1$$

If the extended Euclidean GCD algorithm is applied, then the inverse of f is

$$1/f = x^3 + x^2 + 1$$

That is,

$$g = (1/f)^2 = x^6 + x^4 + 1 = x^4 + x^3 + x + 1$$

The trace of g is zero in this case, as expected. Since n is odd, the β, where $\theta = 1$, is computed as

$$\begin{aligned}\beta &= \sum_{i=1}^{2} g^{2^{2i-1}} \\ &= g^2 + g^8\end{aligned}$$

As $g = x^4 + x^3 + x + 1$, therefore,

$$\begin{aligned}g^2 = (x^4 + x^3 + x + 1)^2 = x^8 + x^6 + x^2 + 1 = x \\ g^4 = (g^2)^2 = x^2 \\ g^8 = (g^4)^2 = x^4\end{aligned}$$

That is,

$$\begin{aligned}\beta &= g^2 + g^8 \\ &= x + x^4\end{aligned}$$

Finally, a solution q to the quadratic is computed:

$$\begin{aligned}q &= (x^2 + x)p'\beta \\ &= (x^2 + x)(x^4)(x + x^4) \\ &= x^4 + x^2 + 1\end{aligned}$$

Euclid's algorithm is now applied to determine the CA, and

$$\begin{aligned} x^5 + x^2 + 1 &= (x)(x^4 + x^2 + 1) + x^3 + x^2 + x + 1 \\ x^4 + x^2 + 1 &= (x+1)(x^3 + x^2 + x + 1) + x^2 \\ x^3 + x^2 + x + 1 &= (x+1)(x^2) + x + 1 \\ x^2 &= (x+1)(x+1) + 1 \\ x + 1 &= (x+1)(1) + 0 \end{aligned}$$

It generates the five degree-one quotients $[x, x+1, x+1, x+1, x+1]$, and so the CA is (150, 150, 150, 150, 90).

6.3.3 Minimal Cost Maximal Length CA

There is no standardized procedure to estimate the cost of a CA. It involves assigning relative costs to addition, multiplication by constants, and the storage of elements of the field. However, the $cost_{CA} = (\sum_{i=1}^{n-1} a_i + \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-2} b_i)$ can be a realistic measure, where (a_i, d_i, b_i) is the triplet representing the rule $\mathcal{R}_i$. When $cost_{CA}$ is minimal, then the CA is called minimal cost CA. For example, the costs of two CAs (150, 90, 150, 150, 90) and (150, 90, 90, 90, 90) in null boundary are 11 and 9 respectively. The triplets for rule 150 and 90 are (1, 1, 1) and (1, 0, 1) respectively.

In case of binary CAs, the minimal cost maximal length CA is a maximal length CA with minimum number of rule 150 in its rule vector. An n-cell maximal length CA with a single 150 is the best choice while considering the cost only. A procedure can be developed to exhaustively search for the minimal cost maximal length CAs. The search can start with a single 150 rule at different cells. If the search remains unsuccessful, then the maximal length CAs having two cells with rule 150 are searched. The procedure stops when the first maximal length CA is found. To decide the maximality, primitivity of the corresponding characteristic polynomial can be checked.

Table 6.3 shows a list of minimal cost maximal length CAs upto $n = 500$ found from exhaustive search. The entries in the table indicate the cells that are with rule 150. For example, the entries 1 and 6, for $n = 10$, indicate that the cell 1 and cell 6 are with rule 150 and the rest are with rule 90. That is, the maximal length CA is (90, 150, 90, 90, 90, 90, 150, 90, 90, 90). From the experimental results, the following has been conjectured.

Conjecture 2 *For every n, there exists at least one maximal length CA having at most two cells with rule 150.*

Table 6.3 Minimal cost maximal length CA

n	CA	n	CA	n	CA	n	CA	n	CA	n	CA
(1)	0	(2)	0	(3)	0	(4)	0, 2	(5)	0	(6)	0
(7)	2	(8)	1, 2	(9)	0	(10)	1, 6	(11)	0	(12)	2, 6
(13)	4	(14)	0	(15)	2	(16)	0, 14	(17)	4	(18)	0, 16
(19)	2	(20)	1, 2	(21)	0, 9	(22)	4	(23)	0	(24)	7, 11
(25)	8	(26)	0	(27)	0, 19	(28)	2	(29)	0	(30)	0
(31)	10	(32)	0, 14	(33)	0	(34)	0, 18	(35)	0	(36)	5
(37)	8	(38)	6	(39)	0	(40)	7	(41)	0	(42)	18
(43)	2	(44)	3, 25	(45)	8	(46)	1, 9	(47)	12	(48)	14
(49)	0, 9	(50)	10	(51)	0	(52)	1, 28	(53)	0	(54)	8
(55)	16	(56)	3, 13	(57)	8	(58)	16	(59)	3, 14	(60)	1, 37
(61)	0, 9	(62)	4	(63)	30	(64)	2, 4	(65)	0	(66)	0, 18
(67)	14	(68)	7	(69)	0	(70)	0, 36	(71)	16	(72)	5, 54
(73)	8	(74)	0	(75)	6	(76)	1, 21	(77)	2, 43	(78)	0, 40
(79)	8	(80)	0, 70	(81)	0	(82)	0, 68	(83)	0	(84)	35
(85)	0, 45	(86)	0	(87)	12	(88)	4	(89)	0	(90)	0
(91)	14	(92)	2, 70	(93)	32	(94)	41	(95)	0	(96)	5
(97)	0, 81	(98)	7	(99)	12	(100)	0, 66	(101)	0, 19	(102)	32
(103)	14	(104)	1,39	(105)	0	(106)	29	(107)	18	(108)	0, 34
(109)	0, 3	(110)	12	(111)	26	(112)	1, 4	(113)	0	(114)	21
(115)	40	(116)	15	(117)	32	(118)	29	(119)	0	(120)	2, 72
(121)	44	(122)	13	(123)	50	(124)	20	(125)	12	(126)	39
(127)	14	(128)	0, 28	(129)	48	(130)	0, 26	(131)	0	(132)	17
(133)	0, 3	(134)	25	(135)	0	(136)	0, 96	(137)	0, 131	(138)	27
(139)	10	(140)	7	(141)	24	(142)	4	(143)	34	(144)	12
(145)	0, 45	(146)	0	(147)	0, 135	(148)	7	(149)	0, 107	(150)	1, 101
(151)	26	(152)	39	(153)	12	(154)	65	(155)	0	(156)	0, 23
(157)	32	(158)	0	(159)	60	(160)	0, 150	(161)	0, 115	(162)	72
(163)	0, 23	(164)	25	(165)	2, 129	(166)	71	(167)	0, 59	(168)	0, 112
(169)	80	(170)	21	(171)	36	(172)	2	(173)	0	(174)	15
(175)	1, 70	(176)	0, 44	(177)	0, 21	(178)	0, 78	(179)	0	(180)	18
(181)	80	(182)	33	(183)	0	(184)	1, 162	(185)	60	(186)	0, 124
(187)	44	(188)	1, 9	(189)	0	(190)	1, 7	(191)	0	(192)	2, 110
(193)	8	(194)	13	(195)	18	(196)	31	(197)	64	(198)	0, 144
(199)	40	(200)	10	(201)	0, 25	(202)	23	(203)	6	(204)	0, 102
(205)	56	(206)	0, 70	(207)	0, 129	(208)	73	(209)	0	(210)	0
(211)	16	(212)	1, 168	(213)	0, 55	(214)	58	(215)	28	(216)	3, 25
(217)	52	(218)	49	(219)	48	(220)	46	(221)	0	(222)	1, 11
(223)	8	(224)	63	(225)	32	(226)	0, 32	(227)	108	(228)	5
(229)	20	(230)	0	(231)	0	(232)	92	(233)	0	(234)	0, 144
(235)	68	(236)	60	(237)	32	(238)	34	(239)	0	(240)	2, 6
(241)	0, 161	(242)	3	(243)	0	(244)	1, 48	(245)	0	(246)	39
(247)	0, 101	(248)	0, 184	(249)	80	(250)	0, 216	(251)	0	(252)	3, 30

(continued)

Table 6.3 (continued)

n	CA	n	CA	n	CA	n	CA	n	CA	n	CA
(253)	92	(254)	0	(255)	54	(256)	0, 126	(257)	48	(258)	0, 180
(259)	20	(260)	3, 88	(261)	0	(262)	2, 188	(263)	94	(264)	1, 172
(265)	68	(266)	45	(267)	102	(268)	0, 72	(269)	0, 209	(270)	80
(271)	4	(272)	2, 108	(273)	0	(274)	1, 28	(275)	102	(276)	80
(277)	0, 71	(278)	43	(279)	68	(280)	1, 46	(281)	0	(282)	2, 18
(283)	34	(284)	48	(285)	0, 207	(286)	1, 166	(287)	108	(288)	0, 60
(289)	100	(290)	111	(291)	96	(292)	4	(293)	0	(294)	0, 78
(295)	56	(296)	1	(297)	2, 83	(298)	110	(299)	0	(300)	0, 250
(301)	80	(302)	1, 104	(303)	0	(304)	3, 129	(305)	72	(306)	0
(307)	122	(308)	13	(309)	0	(310)	0, 42	(311)	6	(312)	0, 294
(313)	56	(314)	2, 40	(315)	122	(316)	0, 24	(317)	1, 126	(318)	15
(319)	20	(320)	2, 78	(321)	96	(322)	1, 73	(323)	0	(324)	18
(325)	32	(326)	0	(327)	24	(328)	79	(329)	0	(330)	0
(331)	20	(332)	0, 64	(333)	0, 51	(334)	103	(335)	0, 11	(336)	3, 34
(337)	0, 99	(338)	0, 40	(339)	1, 8	(340)	1, 36	(341)	144	(342)	68
(343)	98	(344)	12	(345)	48	(346)	31	(347)	12	(348)	0, 24
(349)	116	(350)	1, 62	(351)	0, 63	(352)	0, 144	(353)	96	(354)	23
(355)	68	(356)	39	(357)	72	(358)	8	(359)	0	(360)	0, 96
(361)	0, 281	(362)	1, 153	(363)	50	(364)	26	(365)	84	(366)	0, 64
(367)	92	(368)	138	(369)	0, 73	(370)	74	(371)	0	(372)	0, 256
(373)	4	(374)	1, 312	(375)	0	(376)	61	(377)	148	(378)	1, 11
(379)	32	(380)	3, 101	(381)	0, 43	(382)	137	(383)	76	(384)	0, 214
(385)	160	(386)	0	(387)	0, 175	(388)	0, 338	(389)	88	(390)	1, 55
(391)	0, 251	(392)	172	(393)	32	(394)	58	(395)	0, 85	(396)	0, 112
(397)	112	(398)	0	(399)	60	(400)	197	(401)	184	(402)	153
(403)	178	(404)	67	(405)	116	(406)	115	(407)	94	(408)	1, 238
(409)	3, 96	(410)	2, 180	(411)	0	(412)	184	(413)	0	(414)	3, 136
(415)	0, 275	(416)	0, 218	(417)	128	(418)	35	(419)	0	(420)	0, 322
(421)	0, 107	(422)	0, 120	(423)	1, 220	(424)	1, 79	(425)	36	(426)	0
(427)	1, 90	(428)	183	(429)	0	(430)	2, 92	(431)	0	(432)	0, 48
(433)	44	(434)	85	(435)	1, 280	(436)	16	(437)	4	(438)	186
(439)	170	(440)	3, 134	(441)	0	(442)	137	(443)	0	(444)	183
(445)	196	(446)	0, 442	(447)	222	(448)	0, 152	(449)	108	(450)	1, 136
(451)	10	(452)	28	(453)	0	(454)	17	(455)	30	(456)	1, 208
(457)	0, 219	(458)	90	(459)	18	(460)	31	(461)	0, 89	(462)	137
(463)	32	(464)	81	(465)	0, 201	(466)	89	(467)	94	(468)	1, 377
(469)	112	(470)	0	(471)	1, 184	(472)	1, 211	(473)	0	(474)	60
(475)	44	(476)	24	(477)	0, 265	(478)	85	(479)	0, 139	(480)	7, 238
(481)	224	(482)	0, 284	(483)	0	(484)	2, 236	(485)	0, 271	(486)	0, 180
(487)	158	(488)	187	(489)	108	(490)	1, 286	(491)	0	(492)	14
(493)	0, 183	(494)	1, 2	(495)	0	(496)	2, 68	(497)	0, 199	(498)	62
(499)	0, 173	(500)	1, 77								

6.4 Synthesis of Primitive Polynomial from CA

Deciding a polynomial over GF(2) as primitive is one of the difficult problems. Till date, there is no efficient algorithm available to decide on the primitivity of a polynomial in polynomial time. This section introduces the 90–150 CAs as the tool to derive primitive polynomials over GF(2).

According to Theorem 6.3, if a (linear) CA is maximal length, then its characteristics polynomial is primitive. So, if a maximal length CA is found (obviously, without using primitive polynomial), then the primitive polynomial over GF(2) can be derived from the characteristic matrix of the CA.

6.4.1 Potential Candidates of Maximality

It has already been observed that there is no specific pattern in 90–150 CAs to be a maximal length CA. However, two types of 90–150 CAs have the higher potential to show maximality for some n (number of cells). Such CAs can be used as the generator of primitive polynomials. Since there are few false positive cases, these generators are referred to as the *imperfect* generators.

Definition 41 An n-cell CA with rule vector $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ is called

- $CA(90')$ if $\mathcal{R}_0 = 150$ and $\mathcal{R}_i = 90$, for each $i \in \{1, \ldots, n-1\}$.
- $CA(150')$ if $\mathcal{R}_0 = 90$ and $\mathcal{R}_i = 150$, for each $i \in \{1, \ldots, n-1\}$.

The $CA(90')$ and $CA(150')$ are *conjugate* (see Sect. 6.2.3) to each other. The $CA(90')$ shows maximality for $n = 3, 5, 6, 9, 11$, etc., and $CA(150')$ shows maximality for $n = 3, 5, 9, 11, 14$, etc. So, if such a set of n is identified for each type of CAs, it can be used as the generator of primitive polynomials. Characterization of these CAs is dependent on a special type of configurations consisting of all but one zeros. As the decimal equivalent of such a configuration is in power-of-2, it is referred to as *p-configurations*.

Definition 42 A configuration of a CA is a *p-configuration* if only one cell of the CA is in state 1 and the rest are in state 0.

The configuration 0001, for example, is a p-configuration. In an n-cell CA, there are n p-configurations—$p^0, p^1, p^2, \ldots, p^{n-1}$. Here, p^i indicates that cell $n-i-1$ is at state 1 and the rest are in state 0. For example, the configuration 0010 is p^1 and the configuration 1000 is p^3. It is, however, obvious that if a CA with n cells is a maximal length CA, then all the p-configurations are covered in the same cycle.

Definition 43 In a CA, $length(x, y) = k$ for two configurations x and y if $y = T^k.x$, where T is the characteristic matrix of the CA.[2]

[2] When multiplied with T, the configurations x and y are considered as column vector.

Lemma 12 *A p-configuration p^j of an n-cell CA is reachable from another p-configuration p^i with length$(p^i, p^j) = k \geq 1$ iff column c of T^k consists of a single 1 at position (r, c), where $c = n - 1 - i$, $r = n - 1 - j$ and $0 \leq i, j \leq n - 1$.*

Proof If p^i and p^j are represented as column vectors, then this is resulted from the property of matrix multiplication. □

6.4.2 Characterization of $CA(90')$

The following example shows some intrinsic behavior of the characteristic matrix of $CA(90')$.

Example 6.9 Let consider the 5-cell $CA(90')$. The rule vector of the CA is (150, 90, 90, 90, 90) and its characteristic matrix is

$$T = \begin{bmatrix} \mathbf{1}\,\mathbf{1}\,0\,0\,0 \\ \mathbf{1}\,0\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,0 \\ 0\,0\,\mathbf{1}\,0\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,0 \end{bmatrix}$$

Therefore,

$$T^2 = \begin{bmatrix} 0\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ \mathbf{1}\,0\,0\,\mathbf{1}\,0 \\ \mathbf{1}\,0\,0\,0\,\mathbf{1} \\ 0\,\mathbf{1}\,0\,0\,0 \\ 0\,0\,\mathbf{1}\,0\,\mathbf{1} \end{bmatrix}; \quad T^4 = \begin{bmatrix} 0\,0\,0\,\mathbf{1}\,\mathbf{1} \\ 0\,0\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,0\,0\,\mathbf{1} \\ \mathbf{1}\,0\,0\,\mathbf{1}\,0 \\ \mathbf{1}\,0\,\mathbf{1}\,0\,0 \end{bmatrix}; \quad T^8 = \begin{bmatrix} 0\,0\,\mathbf{1}\,\mathbf{1}\,0 \\ 0\,\mathbf{1}\,0\,0\,\mathbf{1} \\ \mathbf{1}\,0\,0\,0\,0 \\ \mathbf{1}\,0\,0\,0\,\mathbf{1} \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,0 \end{bmatrix}$$

$$T^{16} = \begin{bmatrix} 0\,0\,0\,0\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,\mathbf{1} \\ 0\,0\,\mathbf{1}\,\mathbf{1}\,0 \\ 0\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ \mathbf{1}\,\mathbf{1}\,0\,0\,0 \end{bmatrix}; \quad T^{32} = \begin{bmatrix} \mathbf{1}\,\mathbf{1}\,0\,0\,0 \\ \mathbf{1}\,0\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,0 \\ 0\,0\,\mathbf{1}\,0\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,0 \end{bmatrix} = T$$

If the sequence of p-configurations—00001, 00010, 01000, 00100, 10000 and 00001 are followed, then it can be observed that

$$T \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}; T^2 \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}; T^4 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}; T^8 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix};$$

$$T^{16} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

That is, $length(00001, 00010) = 2^0$, $length(00010, 01000) = 2^1$, $length(01000, 00100) = 2^2$, $length(00100, 10000) = 2^3$ and $length(10000, 00001) = 2^4$. This example validates Lemma 12. Therefore, to understand the dynamics of $CA(90')$ of size n, we need to study the matrices $T^2, T^4, T^8, \ldots$. Here, T, T^2, T^4, and T^8 are the symmetric matrices. All the matrices form a diamond like shape by the 1s (1s are in bold face), and only one column (and a row) of them is having a single 1. The position of this diamond shape depends on the power of T.

Theorem 6.12 *In $CA(90')$ with $n \geq 2$ cells, the following relation holds:*

$$\begin{aligned} &length(p^{m_0}, p^{m_1}) = 2^0, \\ &length(p^{m_1}, p^{m_2}) = 2^1, \\ &\vdots \\ &length(p^{m_i}, p^{m_{i+1}}) = 2^i, \\ &\vdots \\ &length(p^{m_k}, p^{m_0}) = 2^k \end{aligned}$$

where $m_0 = 0$ and the sequence $(m_i)_{0 \leq i \leq k}$, for some $k \leq n-1$, maintains the following relation:

$$m_{i+1} = \begin{cases} 2m_i + 1 & \text{if } 2m_i + 1 < n \\ 2(n - 1 - m_i) & \text{otherwise} \end{cases} \tag{6.12}$$

Proof Let consider that T is the characteristic matrix of an n-cell $CA(90')$. Since T is symmetric, the matrices $T^2, T^4, \ldots, T^{2^i}, \ldots$ are also symmetric. Due to the specific structure of T, no row (and no column) of these matrices consists of 0 only or more than two 1s. In fact, the 1s in these matrices form a *pattern*, which assumes a diamond like shape. The shape has four sides, called *left-down*, *right-down*, *right-up* and *left-up*. Following are the general properties of the pattern.

1. Suppose that $(0, i)$ is the position of first 1 from left side in row 0, where $0 \leq i \leq n-1$. Then, only the following positions of a matrix are filled up with 1s: $(0, i), (1, i-1), (2, i-2), \ldots, (i, 0)$ [the *left-up* side]

$(i+1, 0), (i+2, 1), \ldots, (n-1, n-2-i)$ [the *left-down* side]
$(n-1, n-i), (n-2, n-i+1), \ldots, (n-i, n-1)$ [the *right-down* side]
$(n-i-2, n-1), (n-i-3, n-2), \ldots, (0, i+1)$ [the *right-up* side].
The $(0, i)$ is the *first position*. Depending upon the value of i, all the four sides may not exist simultaneously.

2. Consider a position (a, b), which is 1 in the matrix. If the position is on the left-up side, then $a + b = i$. If it is on the left-down side, then $a - b = i + 1$. If on the right-down side, then $a + b = 2n - i - 1$, and if on the right-up side, then $b - a = i + 1$.
3. Only one column (hence only one row) of these matrices is having a single 1 and the rest rows and columns consist of two 1s. Assuming the first position of the matrix is $(0, i)$, the column $n - 1 - i$ is the column of single 1. The position of single 1 in column $n - 1 - i$ is the *key position*.

When $T^{2^{j+1}}$ from T^{2^j} is derived, the positions of the four sides are changed. In other word, the first position of a matrix (T^{2^j}) shifts to another position of row 0 in $T^{2^{j+1}}$. Consider that the first position in T^{2^j} is $(0, x)$ whereas, in $T^{2^{j+1}}$ the first position is $(0, y)$. If $x < n - 1$, then there are two consecutive 1s at $(0, x)$ and $(0, x + 1)$ in row 0.

Assume $x < n - 1$. As $T^{2^{j+1}}_{0,y} = T^{2^j}_{0,x}.T^{2^j}_{x,y} + T^{2^j}_{0,x+1}.T^{2^j}_{x+1,y}$, either (x, y) or $(x + 1, y)$ position of T^{2^j} is 1. If $n - x - 2 \geq x$, then one of these positions is on the right-up side. Let us now find out the position (x, y) considering the above pattern.

If (x, y) is on the right-up side, then $y - x = x + 1$, and if $(x + 1, y)$ is on the right-up side, then $y - x - 1 = x + 1$ (see point 2). Now if x is odd, y is to be odd, and if x is even, then also y is to be odd. Therefore, when $n - x - 2 \geq x$, then $y - x = x + 1$, which implies $y = 2x + 1$.

If $n - x - 2 < x$, then the right-down side is to be considered. Since $(0, y)$ is the first position of the second matrix, $T^{2^{j+1}}_{0,y} = T^{2^j}_{0,x+1}.T^{2^j}_{x+1,y}$. Otherwise, $(0, y)$ cannot be the first position. That is, $(x + 1, y)$ is on the right-down side, which gives $(x + 1) + y = 2n - x - 1$ by point 2. Hence, $y = 2(n - x - 1)$.

If $x = n - 1$, then the antidiagonal of T^{2^j} is all-1. In this case, $y = 0$. Now combining these all, we get the following.

$$y = 2x + 1 \text{ if } 2x + 1 < n; \quad \text{otherwise, } y = 2(n - 1 - x) \tag{6.13}$$

Let now consider that the key position of the matrix T^{2^j} is (r, c). Assume that $(0, x)$ is the first position of the matrix. Then $c = n - 1 - x$. We can now find out r keeping an eye on the above pattern. When $n - 1 - x \leq x$ then the key position is on the left side of the column x and the position is on the left-up side as stated in point 1 of the above. In this case, we get from point 2 that $r + n - 1 - x = x$, which implies $r = 2x - n + 1$.

Similarly, when $n-1-x > x$—that is, the key position is on the right side of column x, then the position is on the right-up side. Hence, by point 2, we find $n-1-x-r = x+1$, which gives $r = n-2x-2$. If these two cases are combined, then

$$r = \begin{cases} n-2x-2 & \text{if } 2x+1 < n \\ 2x-n+1 & \text{otherwise} \end{cases} \tag{6.14}$$

Hence, the following relation can be obtained.

$$\begin{aligned} n-1-c &= x \\ n-1-r &= \begin{cases} 2x+1 & \text{if } 2x+1 < n \\ 2(n-1-x) & \text{otherwise} \end{cases} \end{aligned} \tag{6.15}$$

That is, $n-1-c$ assumes a sequence of values by Relation 6.13. Further, $n-1-r$ is equal to the value that $n-1-c$ assumes in the next step. Hence, by Lemma 12 we get that $length(p^{m_i}, p^{m^{i+1}}) = 2^j$, where

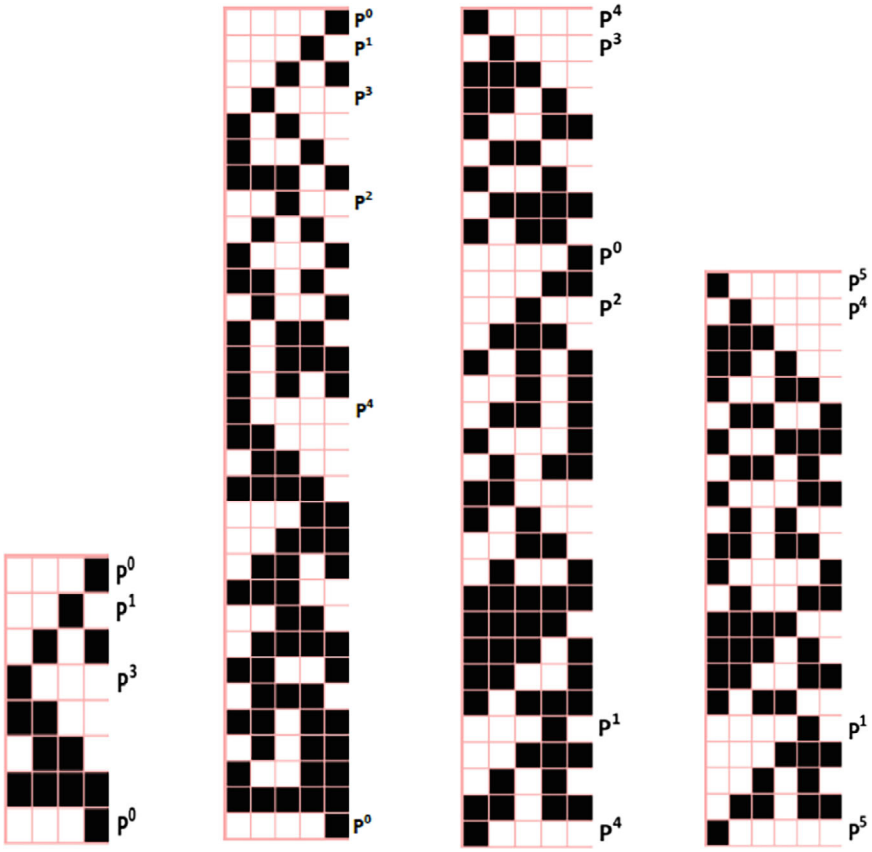
$$m_{i+1} = 2m_i + 1 \text{ if } 2m_i + 1 < n; \quad \text{otherwise, } m_{i+1} = 2(n-1-m_i)$$

As initially $x = 0$, $length(p^0, p^1) = 2^0$, and then $length(p^{m_i}, p^{m^{i+1}}) = 2^i$ for $i \geq 0$. Here, we can get a sequence of p-configuration $(m_0, m_1, \ldots)$. Since the CA is reversible for any n, the p^0 can be found after k, for $k < n$, steps. Hence the proof. □

Example 6.10 Let consider the $n = 5$-cell CA (see, Fig. 6.5b). Here, sequence of p-configurations is $(p^0, p^1, p^3, p^2, p^4)$, and $length(p^0, p^1) = 2^0$, $length(p^1, p^3) = 2^1$, $length(p^3, p^2) = 2^2$, $length(p^2, p^4) = 2^3$ and $length(p^4, p^0) = 2^4$. All p-configurations are covered in one cycle, and this CA is a maximal length CA. For the $n = 4$-cell CA of Fig. 6.5a, the p-configurations covered are the p^0, p^1 and p^3, and $length(p^0, p^1) = 2^0$, $length(p^1, p^3) = 2^1$ and $length(p^3, p^0) = 2^2$. However, this CA is not a maximal length CA.

Corollary 6.5 *For an n-cell $CA(90')$, if all the p-configurations are covered by a single cycle and two consecutive p-configurations p^x and p^y maintain the relation: $y = 2x+1$ if $(2x+1) < n$; otherwise, $y = 2(n-1-x)$, then the CA is a maximal length CA.*

Proof Let assume that the n p-configurations of an n-cell $CA(90')$ are in a single cycle and two consecutive p-configurations p^x and p^y of the cycle maintain the following relation: $y = 2x+1$ if $(2x+1) < n$; otherwise, $y = 2(n-1-x)$. Hence, we get from Theorem 6.12 that lengths between two consecutive p-configurations in the cycle, starting from p^0, are—$2^0, 2^1, \ldots, 2^{n-1}$. Therefore, length of the cycle is $length(p^0, p^0) = 2^0 + 2^1 + \cdots + 2^{n-1} = 2^n - 1$. Hence, the CA is a maximal length CA. □



(a) n=4 (b) n=5 (c) n=5 (d) n=6

Fig. 6.5 Space-time diagram of $CA(90')$ and $CA(150')$. **a** and **b** are for $CA(90')$, where the evolutions start from configuration $p^0(0^{n-1}1)$ and **c** and **d** are for $CA(150')$, where evolutions start from $p^{n-1}(10^{n-1})$. Black box represents state 1 and white is for state 0

In many occasions, it can be observed that the sequence of p-configurations $(p^0, p^{m_0}, p^{m_1}, \ldots, p^{m_{k-2}}, p^{n-1}, p^0)$, maintaining the condition of two consecutive p-configurations of Theorem 6.12, does not include all the p-configurations. The CA with such p-configurations is not a maximal length CA. On the other hand, in most of the cases when all the p-configurations are covered, maintaining the relation between two consecutive p-configurations as per Theorem 6.12, then it is a maximal length CA.

6.4.3 Characterization of $CA(150')$

In the case of $CA(150')$, the similar behavior as in $CA(90')$ is observed. For $CA(150')$, the p^{n-1} is considered as the initial configuration. The relations between the p-configurations are stated in Theorem 6.13. The following example highlights intrinsic properties of the characteristics matrix of $CA(150')$.

Example 6.11 Let consider the 5-cell $CA(150')$. The rule vector of the CA is (90, 150, 150, 150, 150) and its characteristic matrix is

$$T = \begin{bmatrix} 0\,\mathbf{1}\,0\,0\,0 \\ \mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0 \\ 0\,0\,\mathbf{1}\,\mathbf{1}\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,\mathbf{1} \end{bmatrix}$$

Now,

$$T^2 = \begin{bmatrix} \mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ \mathbf{1}\,\mathbf{1}\,0\,\mathbf{1}\,0 \\ \mathbf{1}\,0\,\mathbf{1}\,0\,\mathbf{1} \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,0 \\ 0\,0\,\mathbf{1}\,0\,0 \end{bmatrix}; \quad T^4 = \begin{bmatrix} \mathbf{1}\,0\,0\,\mathbf{1}\,\mathbf{1} \\ 0\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,\mathbf{1}\,0\,\mathbf{1} \\ \mathbf{1}\,0\,0\,0\,0 \\ \mathbf{1}\,0\,\mathbf{1}\,0\,\mathbf{1} \end{bmatrix}; \quad T^8 = \begin{bmatrix} \mathbf{1}\,0\,\mathbf{1}\,\mathbf{1}\,0 \\ 0\,0\,0\,0\,\mathbf{1} \\ \mathbf{1}\,0\,\mathbf{1}\,0\,0 \\ \mathbf{1}\,0\,0\,\mathbf{1}\,\mathbf{1} \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,\mathbf{1} \end{bmatrix}$$

$$T^{16} = \begin{bmatrix} \mathbf{1}\,0\,0\,0\,\mathbf{1} \\ 0\,\mathbf{1}\,0\,\mathbf{1}\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,0 \\ 0\,\mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0 \\ \mathbf{1}\,\mathbf{1}\,0\,0\,\mathbf{1} \end{bmatrix}; \quad T^{32} = \begin{bmatrix} 0\,\mathbf{1}\,0\,0\,0 \\ \mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0\,0 \\ 0\,\mathbf{1}\,\mathbf{1}\,\mathbf{1}\,0 \\ 0\,0\,\mathbf{1}\,\mathbf{1}\,\mathbf{1} \\ 0\,0\,0\,\mathbf{1}\,\mathbf{1} \end{bmatrix} = T$$

It can be observed that all the matrices are symmetric, and all but one element of the main diagonal of these matrices are 1. As in the case of $CA(90')$, a diamond like shape is observed in each of the matrices and only one column of them consists of a single 1. Further, the position of a matrix, through which the main diagonal and any sides of the diamond passes is 0.

Theorem 6.13 *In $CA(150')$ with $n \geq 2$ cells, the following relation holds:*

$$\begin{aligned} length(p^{m_0}, p^{o_0}) &= 2^0, \\ length(p^{m_1}, p^{o_1}) &= 2^1 \\ &\vdots \\ length(p^{m_i}, p^{o_i}) &= 2^i, \\ &\vdots \\ length(p^{m_k}, p^{o_k}) &= 2^k \end{aligned}$$

where $m_0 = n-1$, $o_0 = n-2$ *and the sequences* $(m_i)_{0\le i\le k}$ *and* $(o_i)_{0\le i\le k}$, *for some* $k \le n-1$ *obey the following relation:*

$$m_{i+1} = \begin{cases} 2m_i + 1 & \text{if } 2m_i + 1 < n \\ 2(n-1-m_i) & \text{otherwise} \end{cases}$$
$$o_{i+1} = \begin{cases} 2o_i + 1 & \text{if } 2o_i + 1 < n \\ 2(n-1-o_i) & \text{otherwise} \end{cases} \tag{6.16}$$

Proof Let consider the T represents an n-cell $CA(150')$. Since T is symmetric, the matrices $T^2, T^4, \ldots, T^{2^j}, \ldots$ are also symmetric. As all but one position in main diagonal of T are 1, there is only a single 0 in the main diagonal of T^{2^j} for any $j \ge 1$. Apart from this main diagonal, the rest 1s in these matrices form a *pattern* similar to $CA(90')$. Let us first note down the general properties of such pattern.

1. Suppose that $(0, i)$ is the position of first 1 from the main diagonal in row 0, where $1 \le i \le n-1$. Then, the following positions of T^{2^j} for any $j \ge 1$ are filled up by 1s:
   $(0, i), (1, i-1), (2, i-2), \ldots, (i, 0)$ [the *left-up* side]
   $(i+1, 0), (i+2, 1), \ldots, (n-1, n-2-i)$ [the *left-down* side]
   $(n-1, n-i), (n-2, n-i+1), \ldots, (n-i, n-1)$ [the *right-down* side]
   $(n-i-2, n-1), (n-i-3, n-2), \ldots, (0, i+1)$ [the *right-up* side].
   To maintain uniformity with the case of $CA(90')$, we call this $(0, i)$ as the *first position*. Generally, $T^{2^j}_{0,0} = 1$, but in T, the $(0, 0)$ position is 0. If $(0, 0)$ position in a matrix is 0, we consider the first position as $(0, 0)$. Depending on the first position, obviously, all the four sides may not exist simultaneously.
2. Consider a position (a, b), which is 1 in the matrix but $a \ne b$. If the position is on the left-up side, then $a + b = i$, if on the left-down side, then $a - b = i + 1$, if on the right-down side, then $a + b = 2n - i - 1$, and if on the right-up side, then $b - a = i + 1$.
3. Generally the rows and columns consist of three 1s—two 1s for the two sides of above shape and the other for the main diagonal. Column $n - 1 - i$ intersects only one side of the shape, so this column consists of two 1s.
4. Only one position in the main diagonal of T^{2^j} for any $j \ge 0$ is 0. Consider the position as (m, m). One side of the above pattern passes through (m, m). So the column m consists of a single 1. Like previous, we call the position of single 1 in column m as *key position*.
5. Occasionally there can exist a column with all 0s. This happens when $m = n - 1 - i$. In this case, the matrix has no key position.

Let us now find out the position (m, m) in a matrix (T^{2^j}). If we observe the pattern as stated in point 1, then we find that the left-up and right-down sides can only pass through (m, m). Consider that $(0, x)$ is the first position of the matrix. If x is even, then $(m, m) = (\frac{x}{2}, \frac{x}{2})$ is on the left-up side. If n is odd, then the (m, m) is on the right down side, and $m = n - \frac{x+1}{2}$.

As the structure of diamond shapes are same for $CA(90')$ and $CA(150')$, the shift of the first position when we find $T^{2^{j+1}}$ from T^{2^j} for $j \ge 0$ is also same. Consider

that the first positions in T^{2^j} and $T^{2^{j+1}}$ are $(0, x)$ and $(0, y)$ respectively. Then, we get following from the proof of Theorem 6.12:

$$y = 2x + 1 \text{ if } 2x + 1 < n; \quad \text{otherwise, } y = 2(n - 1 - x) \tag{6.17}$$

Let us now consider that the key position of a matrix is (r', c'). Here $c' = m$. So, if $(0, x)$ is the first position of the matrix, then

$$c' = \begin{cases} \frac{x}{2} & \text{if } x \text{ is even} \\ n - \frac{x+1}{2} & \text{otherwise} \end{cases} \tag{6.18}$$

We can now find out r' keeping an eye on the above pattern. If x is even, the position (m, m) is on the left-up side. Now if $n - 1 - x > \frac{x}{2}$, then the key position is on the left-down side; otherwise, the key position is the right-down side (see point 4 of the above). Hence, if $n - 1 - x > \frac{x}{2}$, then $r' - c' = i + 1$ (by point 2) which implies $r' = \frac{3x+2}{2}$. If $n - 1 - x < \frac{x}{2}$, then $r' + c' = 2n - x - 1$ (by point 2) which gives $r' = 2n - \frac{3x+2}{2}$. However, when $n - 1 - x = \frac{x}{2}$, then the matrix does not have the key position by point 4. That is, when $3x + 2 = 2n$ then the matrix is without a key position. Since the first position starts from $(0, 0)$ and it is updated by Relation 6.17, there is no first position in a matrix such that $3x + 2 = 2n$. Hence, in the matrix T^{2^j}, for $j > 0$, we can always find a key position.

Now consider that x is odd. In this case, if column c' is on the right side of column x, then the key position is on the right-up side; otherwise, it is on left-up side. Hence, if $n - \frac{x+1}{2} > x$, then $c' - r' = x + 1$ which gives $r' = n - \frac{3x+3}{2}$. Otherwise, $r' + c' = x$, that is, $r' = \frac{3x+1}{2} - n$. Combining all these cases, we get the following.

$$r' = \begin{cases} \frac{3x+2}{2} & \text{if } x \text{ is even and } 3x < 2(n-1) \\ 2n - \frac{3x+2}{2} & \text{if } x \text{ is even and } 3x > 2(n-1) \\ n - \frac{3x+3}{2} & \text{if } x \text{ is odd and } 3x < 2n - 1 \\ \frac{3x+1}{2} - n & \text{if } x \text{ is odd and } 3x \geq 2n - 1 \end{cases} \tag{6.19}$$

Now by Lemma 12, we get that $T^{2^j} \cdot p^{n-1-c'} = p^{n-1-r'}$ and $length(p^{n-1-c'}, p^{n-1-r'}) = 2^j$. Let us consider that the key position of $T^{2^{j+1}}$ is (r, c). Hence, $T^{2^{j+1}} \cdot p^{n-1-c} = p^{n-1-r}$ and $length(p^{n-1-c}, p^{n-1-r}) = 2^{j+1}$. We now find out relation between c and c', and between r and r'.

If $(0, y)$ is the first position of $T^{2^{j+1}}$, then we get c and r using Relations 6.18 and 6.19 respectively.

$$c = \begin{cases} \frac{y}{2} & \text{if } y \text{ is even} \\ n - \frac{y+1}{2} & \text{otherwise} \end{cases} \tag{6.20}$$

$$r = \begin{cases} \frac{3y+2}{2} & \text{if } y \text{ is even and } 3y < 2(n-1) \\ 2n - \frac{3y+2}{2} & \text{if } y \text{ is even and } 3y > 2(n-1) \\ n - \frac{3y+3}{2} & \text{if } y \text{ is odd and } 3y < 2n-1 \\ \frac{3y+1}{2} - n & \text{if } y \text{ is odd and } 3y \geq 2n-1 \end{cases} \tag{6.21}$$

Since x and y are related by Relation 6.17, we can get r and c as function of x. We get y as even iff $y = 2(n-1-x)$; otherwise y is odd. Putting this constraint in Relation (6.20), we get that $c = n-1-x$. Hence, considering the sequence of values that x can assume,

$$n-1-c = \begin{cases} 2(n-1-c')+1 & \text{if } 2(n-1-c')+1 < n \\ 2(n-1-(n-1-c')) & \text{otherwise} \end{cases} \tag{6.22}$$

Similarly, replacing y by x in Relation 6.21 we get the following:

$$r = \begin{cases} 3n-3x-2 & \text{and then } 3x > 2(n-1) \\ 3x+2-n & \text{and then } 3x < 2(n-1) \\ n-3x-3 & \text{and then } 3x < n-2 \\ 3x+2-n & \text{and then } 3x \geq n-2 \end{cases} \tag{6.23}$$

Now, when x and y both are odd and $3x < n-2$, we get the following considering the Relations 6.19, 6.21 and 6.23:

$$n-1-r = n-1-(n-3x-3) = 3x+2$$

$$n-1-r' = n-1-\left(n-\frac{3x+3}{2}\right) = \frac{3x+1}{2}$$

Hence,

$$n-1-r = 2(n-1-r')+1$$

When x is odd but y is even and $3x < 2n-2$, then

$$n-1-r = n-1-(3x+2-n) = 2n-3x-3$$

$$n-1-r' = n-1-\left(n-\frac{3x+3}{2}\right) = \frac{3x+1}{2}$$

This gives

$$n-1-r = 2(n-1-(n-1-r'))$$

Using similar rationale, we can establish relation between r and r' for other conditions. When x is odd but y is even and $3x > 2n-1$ then

$$n-1-r = 2(n-1-(n-1-r'))$$

If x is even but y is odd and $3x < n-2$, then

$$n-1-r = 2(n-1-r')$$

If x and y both are even and $3x < 2(n-1)$, then

$$n-1-r = 2(n-1-(n-1-r'))$$

Finally, if x and y both are even and $3x > 2(n-1)$, then

$$n-1-r = 2(n-1-r')+1$$

Combining all the above cases and considering the sequence values that x can assume by Relation 6.17, we can write the following:

$$n-1-r = \begin{cases} 2(n-1-r')+1 & \text{if } 2(n-1-r')+1 < n \\ 2(n-1-(n-1-r')) & \text{otherwise} \end{cases} \tag{6.24}$$

Hence from Relations 6.22 and 6.24, and using Lemma 12, we get:

$$length(p^{m_i}, p^{o_i}) = 2^j$$

$$length(p^{m_{i+1}}, p^{o_{i+1}}) = 2^{j+1}$$

where

$$m_{i+1} = 2m_i + 1 \text{ if } 2m_i + 1 < n; \quad \text{otherwise } m_{i+1} = 2(n-1-m_i)$$

$$o_{i+1} = 2o_i + 1 \text{ if } 2o_i + 1 < n; \quad \text{otherwise } o_{i+1} = 2(n-1-o_i)$$

Now when $x = 0$, then $c' = 0$ and $r' = 1$. Consider that $m_0 = 0$ and $o_0 = 1$. Then, $length(p^{n-1}, p^{n-2}) = 2^0$, and $length(p^{m_i}, p^{o_i}) = 2^i$ for $i \geq 0$. Further, we can always find a $k \leq n-1$ such that $m_{k+1} = m_0$ and $o_{k+1} = o_0$. Hence the proof. □

Example 6.12 Consider the $n = 5$-cell CA of Fig. 6.5c. For two sequences of p-configurations $(p^4, p^0, p^1, p^3, p^2, p^4)$ and $(p^3, p^2, p^4, p^0, p^1, p^3)$, it can be observed that the $length(p^4, p^3) = 2^0$, $length(p^0, p^2) = 2^1$, $length(p^1, p^4) = 2^2$, $length(p^3, p^0) = 2^3$ and $length(p^2, p^1) = 2^4$. From these two sequences, a sequence of p-configurations p^4, p^3, p^0, p^2, p^1, p^4 can be extracted. This sequence is in a single cycle with length between two consecutive p-configurations are 2^0, 2^3, 2^1, 2^4 and 2^2 respectively. Hence, it is a maximal length CA (see Fig. 6.5c). Now, for the $n = 6$-cell CA of Fig. 6.5d, the two sequences of p-configurations are $(p^5, p^0, p^1, p^3, p^4, p^2)$ and $(p^4, p^2, p^5, p^0, p^1, p^3)$. From these, a sequence of p-configurations—p^5, p^4, p^1, p^5 can be derived, where the $length(p^5, p^4) = 2^0$,

$length(p^4, p^1) = 2^4$, $length(p^1, p^5) = 2^2$. Since all the p-configurations are not in a single cycle, the CA is not maximal length (see Fig. 6.5d).

6.4.4 Greedy Strategies

In a maximal length CA, all the p-configurations are covered by a single cycle. The p-configurations of $CA(90')$ and $CA(150')$, on the other hand, follow the relationship mentioned in Theorems 6.12 and 6.13 respectively. If all the p-configurations are covered following Relation 6.12 for $CA(90')$ and Relation 6.16 for $CA(150')$, then the CA may be maximal length CA. However, these conditions are the necessary condition but not the sufficient to show maximality. It is illustrated in Example 6.13. The necessary condition, using Relations 6.12 and 6.16, is framed in the following two procedures.

```
1: Procedure CA90'(n)
2: i ← 0, count ← 1
3: while i ≠ n − 1 do
4:   if 2i + 1 < n then
5:     i ← 2i + 1
6:   else
7:     i ← 2(n − 1 − i)
8:   end if
9:   count ← count + 1
10: end while
11: if count ≠ n then
12:   return False
13: else
14:   return True
15: end if
```

```
1: Procedure CA150'(n)
2: A[n − 1] ← n − 2, m ← 0, o ← 2
3: while m ≠ n − 1 and o ≠ n − 2 do
4:   if 2m + 1 < n then
5:     m ← 2m + 1
6:   else
7:     m ← 2(n − 1 − m)
8:   end if
9:   if 2o + 1 < n then
10:    o ← 2o + 1
11:  else
12:    o ← 2(n − 1 − o)
13:  end if
14:  A[m] ← o
15: end while
16: j ← A[0], count ← 1
17: while j ≠ 0 do
18:   j ← A[j]
19:   count ← count + 1
20: end while
21: if count ≠ n then
22:   return False
23: else
24:   return True
25: end if
```

The above procedures return "True" if all the p-configurations can be covered by Relations 6.12 and 6.16. This implies from Theorems 6.12 and 6.13 that all the

p-configurations are in a single cycle. Obviously, if the above procedures return "False", the CA is non-maximal.

Example 6.13 Let consider an $n = 18$-cell $CA(90')$. For this n, Procedure $CA90'(n)$ returns "True", but the CA is not a maximal length CA. The sequence of p-configurations derived is 0, 1, 3, 7, 15, 4, 9, 16, 2, 5, 11, 12, 10, 14, 6, 13, 8, 17, 0. These configurations maintain the distance as stated in Theorem 6.12. However, the configurations do not follow this sequence in reality. In between p^{13} and p^{8}, p^{17} appears. So, the last four p-configurations of the CA are appeared as 13, 17, 8, 0. Therefore, the distance between p^{13} and p^{17}/between p^{17} and p^{8}/between p^{0} and p^{0} cannot be identified. If the CA could follow the sequence of p-configurations as per the procedure, the CA would be a maximal length CA. It can be found that the cycle is traversed thrice if the computed sequence is targeted to identify on the cycle.

It is found experimentally that, in most of the cases, the CA is maximal length when the corresponding procedure (Procedure $CA90'(n)$/ Procedure $CA150'(n)$) returns "True". Tables 6.4 and 6.5 report n (number of cells) for which the procedures return "True". Only a few are ('*' marked) *false positive*, which means that the procedure returns "True" but the CA is not the maximal length. It can be noted that the occurrence of false positive cases is higher for even n than that of the odd n. So, for higher accuracy, the even n can be excluded from the table.

Table 6.4 List of n when Procedure $CA90'(n)$ returns "True"

2	3	5	6	9	11	14	18[a]	23	26	29
30	33	35	39	41	50[a]	51	53	65	69	74
81	83	86	89	90	95	98[a]	99[a]	105	113	119
131	134[a]	135	146	155	158	173	174[a]	179	183	186[a]
189	191	194[a]	209	210	221	230	231	233	239	243
245	251	254	261	270[a]	273	278[a]	281	293	299	303
306	309	323	326	329	330	338[a]	350[a]	354[a]	359	371
375	378[a]	386	393[a]	398	410[a]	411	413	414[a]	419	426
429	431	438[a]	441	443	453	470	473	483	491	495
509	515	519	530	531	543	545	554	558	561	575
585	593	606[a]	611	614[a]	615	618	629	638	639	641
645[a]	650[a]	651	653	659	683	686[a]	690[a]	713	719	723
725	726	741[a]	743	746	749	755	761	765	771	774
779	783[a]	785	791	803[a]	809	810	818	831	833	834
846	866	870	873	879	891	893	911	923	930[a]	933
935	938	939	950[a]	953	965	974[a]	975	986[a]	989	993
998	1013	1014[a]	1019	1026[a]	1031	1034[a]	1041	1043	1049	1055
1065	1070	1103	1106[a]	1110[a]	1118	1119	1121	1133	1134[a]	1146
1154	1155	1166	1169	1178[a]	1185	1194				

[a] Marked are false positive

Table 6.5 List of n when Procedure $CA150'(n)$ returns "True"

2	3	5	9	11	14	23	26	29	35	39
41	50[a]	53	65	69	74	81	83	86	89	95
98[a]	99[a]	105	113	119	131	134[a]	146	155	158	173
179	189	191	194[a]	209	221	230	231	233	239	243
251	254	278[a]	281	293	299	303	323	326	329	338[a]
350[a]	359	371	375	386	398	410[a]	411	413	419	429
431	443	453	470	473	491	509	515	519	530	531
543	554	561	575	593	614[a]	615	629	638	639	641
645[a]	650[a]	653	659	683	686[a]	713	719	723	725	741[a]
743	746	749	761	779	783[a]	785	803[a]	809	818	831
833	866	873	893	911	923	950[a]	953	965	974[a]	975
986[a]	989	993	998	1013	1019	1031	1034[a]	1041	1043	1049
1070	1103	1106[a]	1118	1119	1121	1133	1154	1155	1166	1169
1178[a]	1185									

[a]Marked are false positive

6.5 Complemented Maximal Length CA

The advantage of studying linear CA is that such a CA can be characterized by the matrix algebra. This can, however, be extended to characterize the complemented CA by introducing an inversion vector (F). The F is defined as

$$F_i = \begin{cases} 1 & \text{if complemented rule is applied for the } i\text{th cell,} \\ 0 & \text{otherwise (linear rule)} \end{cases}$$

If c_t represents the configuration of the CA at the tth instant of time, then the next configuration—that is, the configuration at $(t+1)$th time instant, is given by

$$c_{t+1} = T \cdot c_t + F \Rightarrow c_{t+p} = T^p \cdot c_t + (I + T + T^2 + \cdots + T^{p-1})F \qquad (6.25)$$

where c_{t+p} is the configuration of CA at $(t+p)$th instant of time.

The complemented CAs exhibit more varieties in cyclic behavior than that of the linear CAs. A complemented CA is reversible iff its linear counterpart is reversible. However, the *cycle structure* of a complemented CA may differ from that of its linear counterpart. It is proved in Chap. 7 that the cycle structures of a complemented CA and its linear CA counterpart are identical if the characteristic polynomial of the T-matrix does not have a factor $(x+1)$.

Example 6.14 Let consider a 5-cell complemented CA with the following characteristic matrix T and an inversion vector $F \neq 0$.

$$T = \begin{bmatrix} 1\,1\,0\,0\,0 \\ 1\,0\,1\,0\,0 \\ 0\,0\,1\,1\,0 \\ 0\,0\,1\,1\,1 \\ 0\,0\,0\,1\,0 \end{bmatrix}$$

The characteristic polynomial of the T is $(x^3 + x + 1)(x^2 + x + 1)$. The cycle structure of the corresponding linear CA is [1(1), 1(3), 1(7), 1(21)]. The complemented CA has the identical cycle structure as that of the linear CA, irrespective of F.

The characteristic polynomial of a maximal length CA is primitive and, therefore, irreducible. An irreducible polynomial does not have any factor. Therefore, maximal length complemented CAs can be derived from the linear maximal length CAs. That is, for every linear maximal length CA, there are complemented maximal length CAs.

To generate complemented maximal length CA from a linear maximal length CA, at least one complemented rule 165 (105) in place of rule 90 (150) is to be applied. That is, if any 90 or 150 in the rule vector of a linear maximal length CA is replaced by its complement (that is, rule 165 or 105), then the complemented CA is maximal length. For example, the CA (90, 150, 90, 150) is a linear maximal length CA. Therefore, the complemented CA (165, 105, 90, 150) (Fig. 6.6), (90, 150, 165, 150), etc. are also maximal length. Any number of rules in a rule vector of a linear

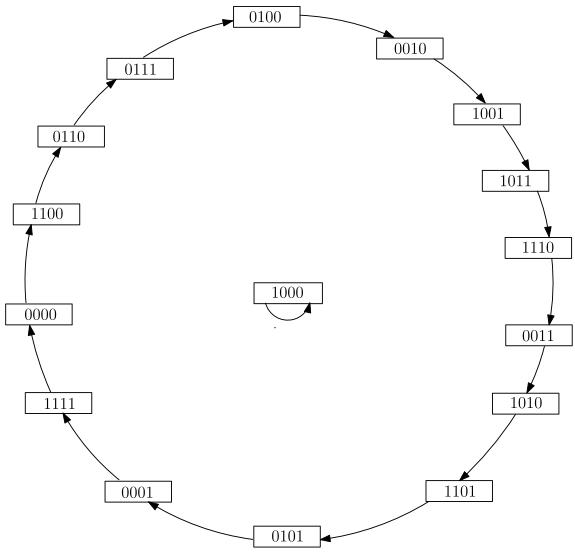


Fig. 6.6 Transition diagram of CA (165, 105, 90, 150) in null boundary

maximal length CA can be replaced by their complements to get a complemented maximal length CA. Therefore, for an n-cell linear maximal length CA, there are $2^n - 1$ number of maximal length complemented CAs.

6.6 Non-linearity in Maximal Length CAs

The maximal length CAs, explored so far in the earlier sections, are linear and complemented CAs. Out of 256 rules, only eight rules are linear and another eight rules are complemented. The rest 240 rules are non-linear. If any one cell of a CA uses a non-linear rule, then the CA is non-linear. This section explores the non-linear maximal length CAs. The algebraic framework reported in the earlier sections are not applicable to the non-linear CA. So, a different theoretical framework is developed for the study of non-linear maximal length CAs.

6.6.1 Information Flow

In a maximal length CA, excluding one configuration, all others are reachable from each other. Let consider the case, where all the cells of a CA are in stable state—that is, the RMTs of rules acting on the cells are self-replicating. Therefore, the corresponding configuration is in a single length cycle of the CA. Now, if only one cell is updated by inverting its state, then the new configuration is part of the other cycle. To be a maximal length CA, the rest configurations are to be reachable from the new configuration. That is, the update of one cell has to affect all the cells of the CA in the subsequent time steps. This flow of updates can be termed as the *Information Flow*. If this information flow is *blocked* at any position, then some configurations remain unreachable from the initial configuration, which confirms that the CA is non-maximal. Therefore, the task is to find whether any rule or rule sequence blocks the information flow. A similar concept is reported in Chap. 10 for the study of chaos in cellular automata. However, to introduce the blocking formally, *subconfiguration* of a configuration is defined first.

Definition 44 For an n-cell CA, a *subconfiguration* of length k ($1 \leq k \leq n-1$) of a configuration $c = (s_i)_{0 \leq i < n}$ is a string $(s_i)_{i \in [j, j+k-1]}$, where $j + k \leq n$.

For example, 10101 is a subconfiguration of the configuration 01011**10101**10 of a 12-cell CA. A subconfiguration may *block* the flow of information from one side of the CA to the other side. That is, a subconfiguration can act as a *blocking word*.

Definition 45 A subconfiguration $s = (s_i)_{i \in [j, j+k-1]}$ of a configuration $c = (s_i)_{0 \leq i < n}$ is a *blocking word* if for any $t \geq 1$, $G^t(c)|_i = s_i$, for each $i \in [j, j + k - 1]$, but $G^t(c)|_l \neq s_l$ for some $l \notin [j, j + k - 1]$. Here, G is the global transition function of the n-cell CA and $0 \leq l < n$.

To illustrate the blocking, let consider the CA (10, 90, 172, 150, 204, 20) in null boundary. For this CA, the subconfiguration 111 is a blocking word when it is in the second to fourth cells. The chain of configurations 00**111**0 → 01**111**1 → 11**111**0 → 10**111**1 → 00**111**0 forms a cycle of length 4. Obviously, there is no information flow from the first two cells (and from the last cell) to other cells. Maximum possible cycle length that can be formed with these configurations is bounded by 2^{n-k} where k is the length of blocking word (here $n = 6$ and $k = 3$). Further, a configuration having a blocking word is always non-reachable from any other configuration which does not have this particular blocking word. For example, the configuration 001010 is non-reachable from 001110 of the above CA. Therefore, the existence of such a blocking word ensures that the CA is not maximal length.

Theorem 6.14 *A CA is not a maximal length CA if it contains a blocking word.*

Proof Let consider that $s = (s_i)_{i\in[j,j+k-1]}$ is a blocking word of an n-cell CA. If s is a subconfiguration of $c = (s_i)_{0\le i<n}$, then, by definition, $G^t(c)|_i = s_i$, $\forall i \in [j, j + k - 1]$, for any $t \ge 1$. Therefore, the CA can never reach a configuration $c' = (s'_i)_{0\le i<n}$, where $s'_i \ne s_i$ for any $i \in [j, j + k - 1]$. Hence, the CA is not a maximal length CA. □

The blocking word in a given CA, if exists, can efficiently be identified. The cells corresponding to the blocking word do not update their states. Therefore, these cells act on self-replicating RMTs. Additionally, the left (right) cell of the first (last) cell against the blocking word does not affect the word. The following condition is to be satisfied for the existence of a blocking word.

Condition 6.1 *In an n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$, a blocking word $(s_i)_{i\in[j,j+k-1]}$ exists iff the following criteria are met:*

1. *Each RMT of the equivalent RMT set $\{0s_js_{j+1}, 1s_js_{j+1}\}$ of $\mathcal{R}_j$ is self-replicating. If $j = 0$, then consider the RMT $0s_js_{j+1}$ only (because of the null boundary condition).*
2. *For each l ($j < l < j + k - 1$), the RMT $s_{l-1}s_ls_{l+1}$ of $\mathcal{R}_l$ is self-replicating.*
3. *Each RMT of the sibling RMT set $\{s_{j+k-2}s_{j+k-1}0, s_{j+k-2}s_{j+k-1}1\}$ of $\mathcal{R}_{j+k-1}$ is self-replicating. If $j + k = n$, then consider the RMT $s_{j+k-2}s_{j+k-1}0$ only (because of the null boundary condition).*

Therefore, whether a given CA is having a blocking word that can be decided in polynomial time by checking Condition 6.1. However, there are some rules whose presence in a rule vector ensures that the CA is having a blocking word. For example, if a rule is having more than four self-replicating RMTs, then a blocking word is always formed. A blocking word is also formed if any rule of the rule vector is a *right* or *left independent* rule (Definition 34). A rule $\mathcal{R}_i : \{0, 1\}^3 \to \{0, 1\}$ is right (resp. left) independent if $\mathcal{R}_i(x, y, 0) = \mathcal{R}_i(x, y, 1)$ (resp. $\mathcal{R}_i(0, x, y) = \mathcal{R}_i(1, x, y)$) for any $x, y \in \{0, 1\}$.

Corollary 6.6 *A CA is not a maximal length CA if any of the following conditions is satisfied:*

1. *A rule of the rule vector is a right or left independent rule.*
2. *Any rule of the rule vector, except the first and last rule, has more than four self replicating RMTs.*

Proof **Case 1**: Let consider that the cell i of an n-cell CA ($0 \leq i \leq n-1$) uses a right independent rule (the proof is similar if the cell uses a left independent rule). If the CA does not have a single length cycle, then, by definition, it is not a maximal length CA. If the CA has a single length cycle and say, the configuration $x = (x_0 x_1 \cdots x_i \cdots x_{n-1})$ forms the cycle, then by Condition 6.1, the subconfiguration $(x_0 x_1 \cdots x_i)$ is a blocking word of length $i+1$. Hence, the CA is not a maximal length CA as per Theorem 6.14.

Case 2: Maximal length CAs are reversible. As per the discussion of Sect. 4.5, there are only 62 reversible rules (Table 4.3) that can form a rule vector of a reversible CA in null boundary. Out of these 62 rules, only sixteen rules (rule 92, 172, 197, 202, 108, 156, 198, 201, 77, 142, 212, 232, 78, 141, 216 and 228) have the six self replicating RMTs and only one rule (204) has eight self-replicating RMTs. The rest of the 62 rules consist of four or less than four self-replicating RMTs. Rule 204, the Identity rule, is left and right independent rule. So by Case 1, existence of rule 204 in a rule vector makes the CA non-maximal.

Let consider that the cell i of an n-cell CA uses any one of the sixteen rules having six self-replicating RMTs, where $1 \leq i \leq n-2$. Further, assume that the CA forms a single length cycle with configuration $x = (x_0 x_1 \cdots x_i \cdots x_{n-1})$. Now, it can be checked that the subconfiguration $(x_0 x_1 \cdots x_i)$ or the subconfiguration $(x_i \cdots x_{n-1})$ of the CA forms a blocking word satisfying Condition 6.1. □

A tabular approach has been reported in Sect. 4.5.3 to synthesize reversible CAs in null boundary. To synthesize reversible CAs excluding the rules that surely contribute in forming blocking words, Table 4.6 is modified. There are ten rules which are right or left independent (rule 51, 60, 195, 204, 15, 240, 85, 102, 153 and 170) and the sixteen rules that have six self-replicating RMTs are in Table 4.6. Table 6.6 is the revised table after excluding these rules.

However, special care is to be taken to identify left and right independent rules for the first and last cells. In null boundary, only four RMTs are *valid* (RMTs 0, 1, 2 and 3) for $\mathcal{R}_0$. Then, $\mathcal{R}_0$ is right independent if $\mathcal{R}_0(0, x, 0) = \mathcal{R}_0(0, x, 1)$ for any $x \in \{0, 1\}$. Under this constraint, only rules 3 and 12 of Table 4.7 are right independent. Likewise, the last rule $\mathcal{R}_{n-1}$, in null boundary, is left independent if $\mathcal{R}_{n-1}(0, x, 0) = \mathcal{R}_{n-1}(1, x, 0)$ for any $x \in \{0, 1\}$. This implies that the rules 17 and 68 of Table 4.8 are left independent. However, there is no left (resp. right) independent rules in Table 4.7 (resp. Table 4.8). Tables 6.7 and 6.8 are the revised tables for first and last rules respectively for reversible CAs.

However, exclusion of all these rules does not ensure the non-existence of blocking word in a given CA. Some sequences of the remaining rules may introduce blocking

Table 6.6 Revised rules table for class relationship of $\mathcal{R}_i$ and $\mathcal{R}_{i+1}$

Class of $\mathcal{R}_i$	$\mathcal{R}_i$	Class of $\mathcal{R}_{i+1}$
I	90, 165	II
	105, 150	III
	53, 58, 83, 163	IV
	54, 57, 99, 147	V
	86, 89, 101, 106, 149, 154, 166, 169	VI
II	30, 45, 75, 90, 105, 120,	I
	135, 150, 165, 180, 210, 225	
III	105, 150	II
	90, 165	III
	23, 43, 113, 178	IV
	27, 39, 114, 177	V
	86, 89, 101, 106, 149, 154, 166, 169	VI
IV	90, 165	IV
	105, 150	V
V	86, 89, 90, 101, 105, 106,	VI
	149, 150, 154, 165, 166, 169	
VI	105, 150	IV
	90, 165	V

Table 6.7 Revised first rule table

Rules for $\mathcal{R}_0$	Class of $\mathcal{R}_1$
5, 10	II
6, 9	III

Table 6.8 Revised last rule table

Rule class for $\mathcal{R}_{n-1}$	Rule set for $\mathcal{R}_{n-1}$
I	20, 65
II	5, 20, 65, 80
III	5, 80
IV	20, 65
V	–
VI	5, 80

word in the rule vectors. For example, in the 5-cell CA (6, 150, 30, 90, 80), there is a 3-bit blocking word *101*, applied on first three cells.

Further observation reveals that, out of thirty six rules of Table 6.6, sixteen rules (86, 89, 101, 106, 149, 154, 166, 169, 30, 45, 75, 120, 135, 180, 210 and 225) may trigger a blocking word in a CA. Because, in first eight rules of these sixteen, one pair of equivalent RMTs are self-replicating, and in the last eight rules, one pair of sibling RMTs are self-replicating. This hints that point 1 or point 3 of Condition 6.1 may get satisfied and a blocking word may be formed if these rules are used.

6.6.2 Degree of Non-linearity

A careful examination of the synthesis process following Tables 6.6, 6.7 and 6.8 reveals the fact that a CA has to use the rules 90, 150, 105 and 165 (which are linear and complemented rules). Further, how many non-linear rules can take part in such a rule vector can also be identified from these tables.

Lemma 13 *An n-cell reversible CA, synthesized following the Tables 6.6, 6.7 and 6.8, possesses following properties:*

1. *The rule vector can have at most three consecutive non-linear rules.*
2. *The rule vector can have at most* $\lfloor \frac{n-5}{2} \rfloor + 3$ *non-linear rules, and the rest rules are chosen from the set of four rules {90, 150, 105, 165}, where* $n \geq 5$.

The formal proof of the lemma is omitted as it can be verified from the tables. As an example, let choose rule 5 for $\mathcal{R}_0$ (rule 5 is equivalent to 165 when applied for the left most cell in null boundary), which indicates that the class of $\mathcal{R}_1$ is II. Now a non-linear rule from Class II (of Table 6.6) is chosen as $\mathcal{R}_1$. Then the class of $\mathcal{R}_2$ is to be I. If a non-linear rule is chosen as $\mathcal{R}_2$, then the class of $\mathcal{R}_3$ is IV, V or VI. Out of these three classes, only class V contains non-linear rules. So, a non-linear rule $\mathcal{R}_2$ has to be chosen in such a way that $\mathcal{R}_3$ is to be from Class V. However, after this a non-linear rule cannot be chosen for $\mathcal{R}_4$. This illustrates that the maximum possible number of consecutive non-linear rules is 3. If the process is continued carefully, then it is possible to synthesize a CA having $\lfloor \frac{n-5}{2} \rfloor + 3$ non-linear rules. For example, consider the 10-cell CA—(10, 30, 54, 86, 90, 86, 90, 89, 150, 20) which has five non-linear rules—that is, the number of non-linear rules is $\lfloor \frac{10-5}{2} \rfloor + 3 = 5$. Here, the first and last rules are considered as linear in null boundary.

As the maximal length CAs also follow the Tables 6.6, 6.7 and 6.8, Lemma 13 is also applicable for the maximal length CAs. Table 6.9 shows a sample list of non-linear maximal length CAs, which also validates the result of Lemma 13.

6.6.3 Linear Maximal Length CA to Non-linear Maximal Length CA

Transition diagrams of the n-cell maximal length CAs are isomorphic to each other. It can also be observed that applying a very few, say one or two non-linear rules, non-linear maximal length CAs can be obtained (Table 6.9). This section investigates whether a non-linear maximal length CA can be synthesized by replacing one or more linear rules of a linear maximal length CA with non-linear rules.

The RMTs are the building blocks of a rule, and hence of a rule vector. If the next states of some RMTs of rule 90 or 150 are appropriately changed, then the rule can be non-linear. For example, if rule 150 of a cell (say, cell i) is replaced by the non-linear rule 86 in a rule vector, then actually the next states of RMTs 6 and 7

Table 6.9 A list of non-linear maximal length CAs

n (# cells)	$\mathcal{R}$ (CA)
4	(6, 178, 90, 20)
5	(5, 150, 99, 165, 5)
6	(5, 90, 106, 90, 166, 5)
7	(6, 101, 90, 154, 105, 165, 65)
8	(9, 90, 105, 30, 54, 150, 105, 65)
9	(5, 180, 150, 105, 165, 149, 150, 90, 65)
10	(10, 105, 54, 154, 90, 166, 90, 86, 105, 65)
11	(5, 150, 165, 30, 58, 90, 150, 86, 105, 90, 65)
12	(6, 105, 165, 90, 180, 147, 165, 165, 105, 165, 150, 8)
13	(6, 86, 90, 169, 105, 150, 89, 90, 165, 150, 90, 90, 65)
14	(9, 177, 89, 90, 89, 90, 101, 105, 165, 90, 150, 90, 150, 65)
15	(10, 75, 90, 90, 166, 90, 86, 105, 90, 150, 166, 105, 90, 90, 20)
16	(6, 90, 178, 150, 154, 150, 105, 105, 90, 150, 150, 90, 165, 105, 90, 80)
17	(6, 165, 150, 165, 150, 150, 90, 101, 150, 165, 150, 105, 165, 169, 150, 165, 20)
18	(9, 165, 86, 150, 90, 90, 165, 150, 105, 150, 165, 150, 105, 105, 150, 149, 150, 20)
19	(10, 45, 58, 90, 165, 105, 165, 150, 165, 150, 149, 165, 90, 165, 90, 105, 105, 105, 5)
20	(10, 150, 165, 105, 149, 165, 165, 150, 150, 89, 90, 105, 105, 165, 105, 150, 150, 165, 165, 20)

are only changed. So, successor (y) of a configuration (x) remains same in both the CAs, unless RMT r_i is 6 or 7, where $\tilde{x} = (r_i)_{0 \le i \le n-1}$. In other words, replacement of linear rule (150) by non-linear one (86) *affects* the successor y of x if $r_i = 6$ or 7. If such affected configuration does not have impact on length of the cycle of the linear maximal length CA, the new non-linear CA remains maximal length.

Definition 46 Let G_1 and G_2 be two n-cell CAs. The G_2 is called *twin* of G_1 iff

1. Transition diagram of G_2 is isomorphic[3] to that of G_1.
2. There exist a non-empty set of configuration C such that $\forall x \in C, G_1(x) = G_2(x)$.

If G_1 and G_2 are the maximal length CAs, then their transition diagrams are isomorphic to each other. However, it does not mean that the CAs are twins. For example, the 4-cell CA (6, 90, 105, 80) is a maximal length CA, but not twin of the (linear) maximal length CA (6, 90, 150, 80) (see Fig. 6.7a). Because, point 2 of Definition 46 is not followed. However, the maximal length CA (6, 90, 86, 80) (see Fig. 6.7b) is a twin of the CA (6, 90, 150, 80) (see Fig. 6.7a). There are twelve

[3] Two graphs which contain the same number of vertices connected in the same way are said to be isomorphic.

configurations for which the next configurations in both the CAs are same (hence, point 2 of Definition 46 is verified).

It is, however, obvious that if G_2 is the twin of G_1 and G_1 is a maximal length CA, then G_2 is also a maximal length CA. Let consider that G_1 is a linear maximal length CA with rule vector $\mathcal{R} = (\mathcal{R}_i)_{0 \le i \le n-1}$ and G_2 is a non-linear reversible CA with rule vector $\widehat{\mathcal{R}} = (\widehat{\mathcal{R}}_i)_{0 \le i \le n-1}$ having the following properties:

- For a set of $k \ge 1$ positions, $p_1, p_2, \ldots, p_k$, $\mathcal{R}_i \ne \widehat{\mathcal{R}}_i$ if $i \in \{p_1, p_2, \ldots, p_k\}$. Here, $\widehat{\mathcal{R}}_i$ is a non-linear rule, and $p_1, p_2, \ldots, p_k \in \{1, 2, \ldots, n-2\}$.
- For any $i \in \{0, 1, \ldots, n-1\} \backslash \{p_1, p_2, \ldots, p_k\}$, $\mathcal{R}_i = \widehat{\mathcal{R}}_i$.

That is, G_2 is a non-linear reversible CA which is designed after replacing some linear rules of G_1 by the non-linear rules. Following example illustrates the design of $\widehat{\mathcal{R}}$ from $\mathcal{R}$.

Example 6.15 Consider the $\mathcal{R} = (6, 90, 150, 80)$ of Fig. 6.7a as G_1. Let design G_2 by replacing one linear rule. From Table 6.7, it is found that the class of second rule is III. Considering rule 90 as the second rule, the class of third rule is found as III (see Table 6.6). Let choose rule 86 in place of rule 150 of $\mathcal{R}$. Then the class of last rule is VI, and the rule 80 is considered as the last rule. Therefore, G_2 is $\widehat{\mathcal{R}} = (6, 90, 86, 80)$, which is a non-linear reversible CA.

The next task is to decide if the G_2 is isomorphic to G_1. Due to the replacement of the rule 150, however, there exist a non-empty set of configurations C' such that $\forall x \in C'$, $G_1(x) \ne G_2(x)$. Let call the member of C' as the *affected configurations*.

Definition 47 Let G_1 and G_2 be two CAs that are isomorphic to each other. Now, a configuration x is called an *affected configuration* if $G_1(x) \ne G_2(x)$.

For example, the configurations in bold face of Fig. 6.7b are the affected configurations when the CA of Fig. 6.7b is isomorphic to the CA of Fig. 6.7a. There are four affected configurations in Fig. 6.7b with $C' = \{0111, 0110, 1110, 1111\}$. However, whether a configuration is affected due to the replacement, can easily be identified. Let consider that $\mathcal{R}_i$ is replaced by a non-linear rule $\widehat{\mathcal{R}}_i$. Further consider that $\mathbf{r}_i$ denotes the set of RMTs that differ in the next states in $\mathcal{R}_i$ and $\widehat{\mathcal{R}}_i$, and $\mathcal{A}_i$ is the set of affected configurations due to this replacement. Following are two results relating $\mathcal{A}_i$ and $\mathbf{r}_i$.

Proposition 6.1 *At least two RMTs of $\mathcal{R}_i$ remain unaffected when the $\mathcal{R}_i$ is replaced by a non-linear rule $\widehat{\mathcal{R}}_i$. That is, $0 < |\mathbf{r_i}| < 8$, where $\mathbf{r_i}$ denotes the set of RMTs that differ in next states in $\mathcal{R}_i$ and $\widehat{\mathcal{R}}_i$.*

Proof Maximal length CAs are reversible and the reversible CAs use only balanced rules in the rule vectors. Hence, both the rules $\mathcal{R}_i$ and $\hat{\mathcal{R}}_i$ are balanced. Due to the balancedness, any change in the next state of an RMT enforces a change in another RMT to keep $\widehat{\mathcal{R}}_i$ balanced. Let $r_0, r_1, \ldots, r_7$ be the RMTs of $\mathcal{R}_i$ such that $\mathcal{R}_i[r_0] = \mathcal{R}_i[r_1] = \mathcal{R}_i[r_2] = \mathcal{R}_i[r_3] = 0$ and $\mathcal{R}_i[r_4] = \mathcal{R}_i[r_5] = \mathcal{R}_i[r_6] = \mathcal{R}_i[r_7] = 1$.

Without loss of generality, suppose $\widehat{\mathcal{R}}_i[r_0] = 1$. Then one of $\widehat{\mathcal{R}}_i[r_4]$, $\widehat{\mathcal{R}}_i[r_5]$, $\widehat{\mathcal{R}}_i[r_6]$ and $\widehat{\mathcal{R}}_i[r_7]$ is to be 0 to remain balanced. The number of affected RMTs, hence, is always even. Now, $|\mathbf{r_i}|$ cannot be eight, as this implies that the $\widehat{\mathcal{R}}_i$ is a complemented rule. Also $|\mathbf{r_i}|$ can not be zero. This proves the proposition. □

Proposition 6.2 $|\mathcal{A}_i| = |\mathbf{r_i}| \times 2^{n-3}$.

Proof Let express an $r \in \mathbf{r_i}$ as $xyz \in \{0, 1\}^3$. In the set of all configurations of the n-cell CA, the binary string xyz appears 2^{n-3} times. So, for every $r \in \mathbf{r_i}$, there exists 2^{n-3} affected configurations. □

For example, in Fig. 6.7b, $|\mathbf{r_i}| = 2$ and $|\mathcal{A}_i| = 2 * 2^{4-3}$. However, if $k \geq 1$ rules are replaced in G_1 to get G_2, then the set of affected configurations is

$$C' = \bigcup_{i \in \{p_1, p_2, \ldots, p_k\}} \mathcal{A}_i$$

The C' may cover the whole configuration space of G_2.

Proposition 6.3 *All the configurations of G_2 are affected if following conditions are satisfied:*

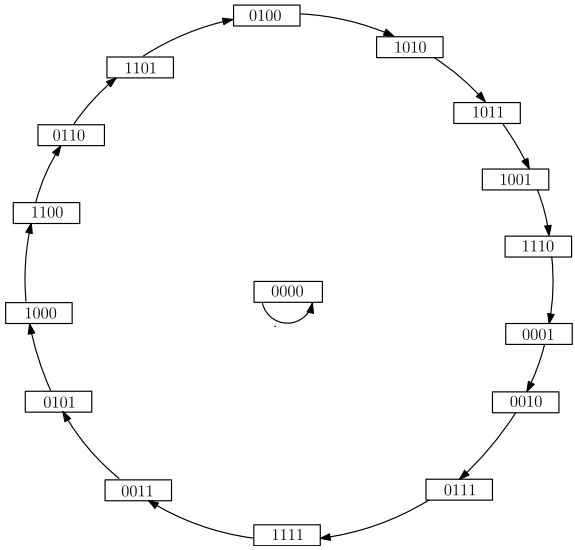
1. *Two consecutive rules $\widehat{\mathcal{R}}_i$ and $\widehat{\mathcal{R}}_{i+1}$ are non-linear.*
2. $\{2r, 2r + 1 \pmod 8 | r \in \{0, \ldots, 7\} \setminus \mathbf{r}_i\} \subseteq \mathbf{r}_{i+1}$

Proof Consider the two arbitrary configurations x and y. The RMT sequences of which are $\tilde{x} = (r_j)_{0 \leq j \leq n-1}$ and $\tilde{y} = (s_j)_{0 \leq j \leq n-1}$ respectively and $r_i \in \mathbf{r}_i$ and $s_i \notin \mathbf{r}_i$. By definition, x is an affected configuration. However, $s_i \notin \mathbf{r}_i$ implies $s_i \in \{0, \ldots, 7\} \setminus \mathbf{r}_i$ that means $s_{i+1} \in \mathbf{r}_{i+1}$. This implies that y is also an affected configuration. That is, there is no configuration that is not affected. □

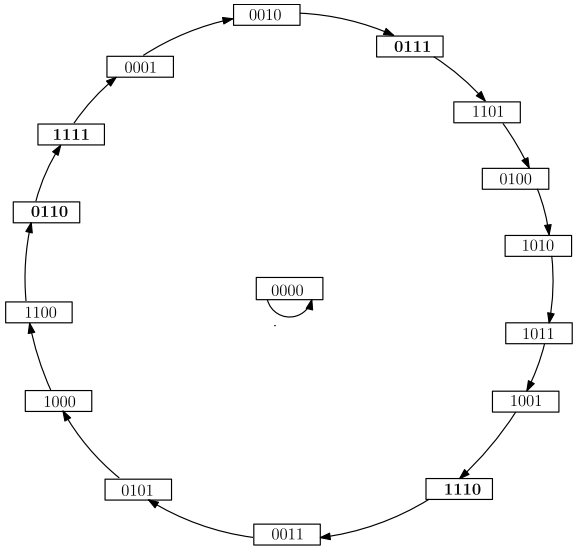
As an example, consider the G_2 is (6, 39, 89, 80) that can be designed from the CA (6, 90, 150, 80) (Fig. 6.7a). Here, all the configurations of G_2 are affected. The non-linear rules are applied on the second and third cells, and $\mathbf{r}_1 = \{0, 2, 3, 4, 5, 6\}$, $\mathbf{r}_2 = \{0, 1, 2, 3, 6, 7\}$. That is, the condition of Proposition 6.3 is satisfied. However, this CA is not a maximal length CA.

Now, if G_2 is a maximal length CA, then for any $x, y \in C'$, when $G_2(x) \neq x$ and $G_2(y) \neq y$, $G_2^t(x) = y$ for some $t \geq 1$. That is, the configurations x, y of C' form a *chain* of length $l \geq |C'|$. For example, in Fig. 6.7b, the members of C' form a chain of length ten, starting and ending with 1110 and 0111 respectively. The length of another chain is eleven if it starts with 01110. However, the formation of a single such chain is necessary and sufficient condition for G_2 to be a maximal length CA.

Theorem 6.15 *Let G_1 be a linear maximal length CA and G_2 be a reversible non-linear CA which is constructed by replacing some linear rules of G_1 with non-linear ones. Now, G_2 is a maximal length CA iff the configurations of $C' \setminus \{x\}$ form a single chain by G_2, where C' is the set of all affected configurations and $G_2(x) = x$.*



(a) Transition diagram of CA $(6, 90, 150, 80)$



(b) Transition diagram of CA $(6, 90, 86, 80)$

Fig. 6.7 Linear to non-linear: the affected configurations are in bold face

Proof *"If" part*: Let assume that the members of C' form a chain in G_2, which starts with a and ends with b; $a, b \in C'$. The configurations of G_2 that are not in the chain form another chain. This chain can also be found in G_1 as the predecessor-successor relation is preserved for these configurations in both G_1 and G_2. Suppose, this chain starts with w and ends with z; $w, z \notin C'$. Now, $G_2(z) \neq w$, otherwise, z is to be an affected configuration by definition. Since G_2 is reversible, the only option is $G_2(z) = a$ and $G_2(b) = w$. Hence, G_2 is a maximal length CA.

"Only if" part: Let G_2 be a maximal length CA. Then obviously, all the configurations, except x when $G_2(z) = x$, are in a single cycle. That is, all the configurations of $C'\backslash\{x\}$ are in the same cycle. □

Hence, to decide whether the G_2 is a non-linear maximal length CA, it is to be tested if the affected configurations form a single chain.

Example 6.16 Consider the CAs of Example 6.15 where G_1 is $\mathcal{R} = (6, 90, 150, 80)$, a linear maximal length CA, and G_2 is $\widehat{\mathcal{R}} = (6, 90, 86, 80)$. $\widehat{\mathcal{R}}$ does not satisfy Condition 6.1, hence it has no blocking word. Further, it has exactly one single length cycle (which is 0000), and all the configurations are not affected. Here $k = 1$, $p_1 = 2$ and $\mathbf{r_1} = \{6, 7\}$. Hence, the total number of affected configurations is $2^{4-3} \times 2 = 4$. Let now choose 1110 as the starting affected configuration. It can be observed that a chain of length ten cover all the affected configurations. Hence, G_2 is a twin of G_1 and it is a non-linear maximal length CA.

Table 6.10 lists a few non-linear maximal length CAs, that are obtained from the linear maximal length CAs. The second column of Table 6.10 shows the linear maximal length CAs, whereas, the third and fourth columns report the cell numbers of the linear CAs that are to be configured with the non-linear rules, and the non-linear rules respectively. The linear rule which is replaced are marked in bold face. For example, for $n = 9$, cell number 1 and 5 are to be configured with non-linear rules 30 and 58.

Conjecture 3 *For every $n \in \mathbb{N}$, there exists a non-linear maximal length CA.*

6.7 Maximality Over GF(q)

This section deals with the CAs defined over the finite filed GF(q), $q \geq 2$ is a prime. The cells can be in any of the states of $\mathcal{S} = \{0, 1, \ldots, q-1\}$. A configuration of the CA is a mapping c: $\mathcal{L} \rightarrow \mathcal{S}$. Let $\mathcal{C}$ be the set of all possible configurations of an n-cell CA. Then, $|\mathcal{C}| = q^n$. The following defines the linear rules of q-state CA.

Table 6.10 Linear maximal length CAs to non-linear maximal length CAs. Rule with bold face is replaced by non-linear one

n	G_1	G_2	
		Positions	Non-linear **rules**
4	(6, 90, **150**, 80)	2	89
5	(6, 150, **150**, 150, 80)	2	75
6	(6, 90, **90**, 150, 90, 20)	2	86
7	(10, 90, 150, **90**, 150, 90, 20)	3	169
8	(6, 150, 150, 90, 150, **150**, 150, 20)	5	154
9	(10, **150**, 150, 150, 90, **90**, 90, 90, 20)	1, 5	30, 58
10	(10, 150, 150, **90**, 90, 90, 90, 150, 150, 20)	3	101
11	(6, 90, 150, 150,**150**, 90, 150, 90, 90, 150, 20)	4	86
12	(10, 90, **150**, 150, 150, 150, 90, **90**, 90, 150, 150, 20)	2, 7	86, 149
13	(6, 90, 150, **90**, **90**, 90, 90, 150, 90, 150, 90, 150, 20)	3, 4	210, 101
14	(10, 150, 90, **150**, **90**, 150, 90, 90, 90, 90, 90, 90, 90, 20)	3, 4	45, 169
15	(10, 150, **90**, 90, 90, 150, 150, 90, 150, 90, 150, 90, 150, 150, 20)	2	53
16	(6, 150, 90, 90, 90, 90, 90, 90, 90, **150**, 90, 90, 150, 90, 90, 20)	9	154
17	(6, 90, 90, 90, **90**, 150, 90, 150, 90, 150, 90, 150, 90, 150, 90, 90, 20)	4	101
18	(6, 150, 150, 90, **150**, 90, 150, 90, 150, 90, 90, 150, 90, 90, 150, 90, 150, 80)	4	225
19	(10, 90, 150, 150, **90**, **90**, 150, 90, 90, 90, 150, 150, 150, 150, 90, 150, 90, 90, 20)	4, 5	30, 154
20	(10, 90, 150, 150, 150, 150, 150, 90, 90, 90, 90, 150, 150, **150**, 90, 90, 90, 150, 150, 20)	13	86
	(10, 90, 150, 150, 150, 150, 150, 90, 90, **90**, 90, 150, 150, 150, 90, 90, 90, **150**, 150, 20)	9, 17	45, 86
21	(10, 90, 90, 150, 150, 150, 150, 150, **90**, 150, 150, 90, 150, 90, 90, 90, 90, 90, 150, 90, 80)	8	86
	(10, 90, 90, 150, 150, 150, 150, 150, 90, 150, 150, 90, **150**, 90, 90, 90, 90, 90, 150, **90**, 80)	12, 19	30, 106
22	(10, 90, **150**, 90, 90, 90, 150, 150, 90, 90, 90, 90, 150, 150, 90, 90, 90, 90, 150, 90, 90, 20)	2	154
	(10, 90, 150, 90, 90, 90, 150, **150**, 90, 90, 90, 90, 150, 150, 90, **90**, 90, 90, 150, 90, 90, 20)	7, 15	30, 154
23	(10, 150, 150, 90, 150, 150, 150, 150, 90, 150, 150, 150, 90, 150, 150, 150, 90, **150**, 90, 90, 90, 150, 80)	17	169
	(10, 150, 150, 90, 90, 150, 150, 150, 90, 90, 90, 90, 90, 150, **90**, 90, 90, 90, 90, **150**, 150, 90, 20)	14, 19	30, 86
24	(6, 90, 150, 150, 90, 90, 150, 90, 150, 150, 150, 150, 90, 90, 150, 150, 150, 150, **150**, 150, 150, 150, 150, 20)	18	169
	(6, 90, 150, 150, 90, **90**, 150, 90, 150, 150, 150, 150, 90, 90, 150, 150, 150, 150, 150, 150, 150, **150**, 150, 20)	5, 21	30, 86

Definition 48 A rule $R : \mathcal{S}^3 \rightarrow \mathcal{S}$ is called *linear* if R can be expressed as $R(x, y, z) = ax + dy + bz \pmod q$, for some $a, b, d \in \mathcal{S}$.

The triplet (a, d, b) is used to state a rule. For a CA in GF(q), there are in total q^3 linear rules. For example, number of linear rules for GF(2), GF(3) and GF(5) are 8, 27 and 125 respectively. In this section, by a "CA", only the 1-dimensional q-state 3-neighborhood non-uniform CAs with null boundary is referred. The rule vector of an n-cell CA is $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_i, \ldots, \mathcal{R}_{n-1})$, where cell i is set with $\mathcal{R}_i \equiv (a_i, d_i, b_i)$.

As per discussion of Sect. 6.1.1, an n-cell 3-neighborhood linear CA can also be expressed by a characteristic matrix $T_{n\times n}$. In null boundary condition,

$$T = \begin{bmatrix} d_0 & b_0 & 0 & \cdots & 0 & 0 \\ a_1 & d_1 & b_1 & \ddots & & 0 \\ 0 & a_2 & d_2 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & & & \ddots & d_{n-2} & b_{n-2} \\ 0 & 0 & \cdots & \cdots & a_{n-1} & d_{n-1} \end{bmatrix}$$

The characteristic polynomial of T is $P(x) = det(xI + T) \pmod q$. The $P(x)$ in GF(q) can be derived following the similar recurrence relation $\Delta = det(xI + T) \pmod q$ as in Sect. 6.1.1. That is,

$$\Delta = \Delta_n = (x + d_{n-1})\Delta_{n-1} + b_{n-2}a_{n-1}\Delta_{n-2} \tag{6.26}$$

Example 6.17 Let consider a 3-state 3-cell CA with rule vector ((1, 2, 1), (2, 0, 1), (2, 0, 2)). The characteristic polynomial of this CA is $det(xI + T) \pmod 3 = x^3 + x^2 + 2x + 1$, where

$$T = \begin{bmatrix} 2 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

Figure 6.8 is the transition diagram of the 3-state 3-cell CA. Here, the next configuration c' of a configuration c can be found by multiplying c with T, where the configuration c is represented by a column vector. The CA of Fig. 6.8 is reversible and also a maximal length CA.

For an n-cell linear maximal length CA over GF(q), the length of the cycle is $q^n - 1$. Theorem 6.16 relates the maximal length CA and primitive polynomial over GF(q). The characteristic polynomial $x^3 + x^2 + 2x + 1 \pmod 3$ is primitive over GF(3).

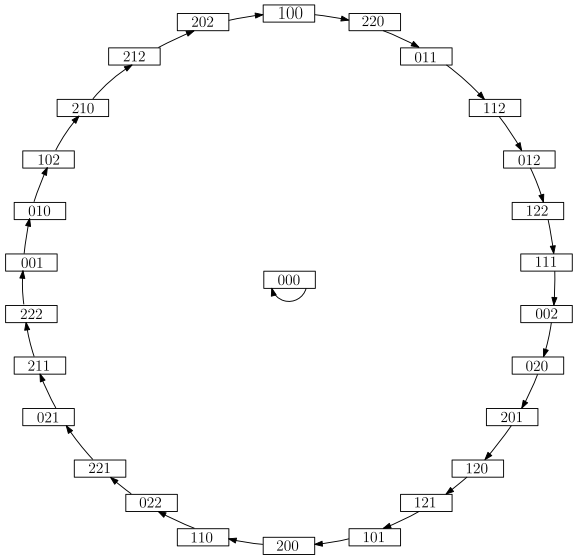


Fig. 6.8 Transition diagram of 3-state CA ((1, 2, 1), (2, 0, 1), (2, 0, 2))

Theorem 6.16 *A q-state linear CA is maximal length iff the characteristic polynomial of this CA is primitive over GF(q).*

For some conditions, the characteristic polynomials of CAs over GF(q) are always reducible. Following theorem is the generalization of Theorems 6.1 and 6.7.

Theorem 6.17 *The characteristic polynomial of an n-cell CA is always reducible over GF(q), if one of the following conditions is satisfied:*

1. *If $a_i = 0$ for any $0 < i \leq n-1$ or $b_j = 0$ for any $0 \leq j < n-1$ of a rule.*
2. *If the CA is palindromic.*

Now, the issue is whether for an irreducible polynomial $p(x)$, there exist a CA with characteristic polynomial $p(x)$. For CAs over GF(2), it is always possible to find a CA from an irreducible polynomial. However, there is no proof that a CA can be synthesized with characteristic polynomial $p(x)$ in GF(q). The empirical evidences suggest the following conjecture for the CAs over GF(q). The evidences are obtained by exhaustive enumeration of a large number of CAs for various n.

Conjecture 4 *If $p(x)$ is an irreducible polynomial over GF(q), then there exists a CA with characteristic polynomial $p(x)$.*

The CA is synthesized from a given irreducible polynomial over GF(2) in Sect. 6.3.2. However, there is no efficient method for synthesizing CAs over GF(q). In GF(q), the actual difficulty lies in the reverse process, that means, design of CA from a given primitive polynomial. Therefore, to design maximal length CA in GF(q), the only way available is the following.

1. For a given n, randomly generate a candidate CA.
2. Find characteristic polynomial of the CA following recurrence relation.
3. Test whether the polynomial is primitive, by prime factorization.

There are open problems related to maximal length CA over GF(q), such as the synthesis of a CA from a given polynomial in GF(q), finding number of CAs against each irreducible polynomial, etc. However, these problems are solved in GF(2).

6.8 Chapter Notes

The property of maximality has been first observed by Pries et al. [1]. In early phase of development, Hortensius et al. [2–4] have used (linear) maximal length CAs to generate pseudo-random numbers. In the same line, Serra et al. [5] have reported the utility of (linear) maximal length CAs as high quality pseudo-random pattern generators. A synthesis scheme from an irreducible polynomial, with 90/150 rules, has been reported in [6]. In [7, 8], Cattell and Muzio have developed an adequate algebraic framework for the study of maximal length CAs. In [9], Zhang et al. have presented an algorithm for finding minimal cost maximal length CA. The results have been extended in [10] and have identified such CAs upto degree 500. Based on the results of extensive simulations, Bardell [11, 12] conjectured that there is no maximal length CA in periodic boundary, which has been proved by Nandi et al. [13]. Nandi et al. [13] have also shown that there exist maximal length CAs against each primitive polynomial in intermediate boundary. The phase shifts of a maximal length CA has been investigated by Bardell [11]. Later, Nandi and Chaudhuri [14] have proposed a method to study the phase shifts based on matrix algebra. The fact that a complemented CA whose linear counter part is maximal length is also maximal length, has been proved by exploring their cycle structure in [15].

Whether there is a pattern in the rule vectors of (linear) maximal length CAs has been investigated in [16]. Although no specific pattern has been found, some special types of CAs have been considered to design maximal length CAs in [17, 18]. These CAs have been used to generate primitive polynomials in GF(2). The study on non-linear maximal length CAs has been reported in [19], which has been detailed in the PhD thesis of Sumit Adak [20]. From a different perspective, the synthesis of non-linear maximal length CA from a linear maximal length CA has been reported in [21, 22].

Exploring maximal length CAs is an ever-challenging research area. Following are few topics which can be explored:

1. Search for an efficient algorithm to decide a given CA as maximal length (without using primitive polynomial).
2. It can be observed in Tables 6.4 and 6.5 that, for a prime p, if $2p + 1$ is also prime then $CA(90')$ as well as $CA(150')$ of size p show maximality. That is, for Sophie Germain prime (a prime number p is a Sophie Germain prime if $2p + 1$ is also prime), $CA(90')$ and $CA(150')$ are maximal length CAs. A research can

be conducted to prove (or disprove) whether the n-cell $CA(90')$ and $CA(150')$ are maximal length with n and $2n + 1$ are both primes.
3. Identification of more effective sequence of 90–150 for generating maximal length CAs.
4. Synthesis of a CA from a given polynomial over GF(q).
5. Finding of number of CAs for an irreducible polynomial over GF(q).
6. Generation of non-linear maximal length CAs from a linear maximal length CA in polynomial time.
7. Study of existence of non-linear maximal length CA over GF(q).
8. Prove (or disprove) that there exist a CA for every irreducible polynomial in GF(q).

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Chapter 7
Cycle Structure of Linear and Complemented Cellular Automata

A finite cellular automaton (CA) forms cycles in its configuration space. The configurations, which are part of a cycle, are called *cyclic* configurations and the rest are *acyclic* configurations. In a reversible (finite) CA, all the configurations are cyclic. Whereas, an irreversible (finite) CA contains both the cyclic and acyclic configurations. This chapter deals with linear and complemented cellular automata (CAs) and characterizes the cycles formed by them. The non-uniform ECAs—that is, the CAs with only ECA rules are explored here. The vector-space theoretic analysis is adopted to study the *cycle structure* of these CAs. The chapter first proceeds with linear CAs and develops a theoretical framework for the analysis of linear CAs. The framework enables the development of an algorithm to find the cycle structure of a linear CA (Sect. 7.2). The same framework is utilized to synthesize a linear CA from a cycle structure (Sect. 7.3). The theoretical framework of the linear CAs is then extended to analyze (Sect. 7.4) and synthesize (Sect. 7.5) complemented CAs.

7.1 The Cycle Structure

The cycle structure of a CA refers to the information regarding the number and length of distinct cycles formed by the configurations of the CA. It is represented as $CS = [\mu_{k_1}(k_1), \mu_{k_2}(k_2), \cdots, \mu_{k_m}(k_m)]$, where μ_{k_i} represents the number of cycles of length k_i $(1 \leq i \leq m)$. The μ_{k_i} is also known as *cyclic component* corresponding to the cycle length k_i and also represented as $[\mu_{k_1}(k_1) + \mu_{k_2}(k_2) + \cdots + \mu_{k_m}(k_m)]$.

Let consider a 7-cell CA $(0, 240, 204, 102, 204, 170, 60)$ with null boundary. Figure 7.1 shows the transition diagram of the CA. It contains 4 cycles of length 1, 2 cycles of length 2, 4 cycles of length 3 and 2 cycles of length 6. Hence, its cycle structure is [4(1), 2(2), 4(3), 2(6)]. In case of an irreversible CA, there are some non-reachable configurations. The maximum number of transitions needed to reach

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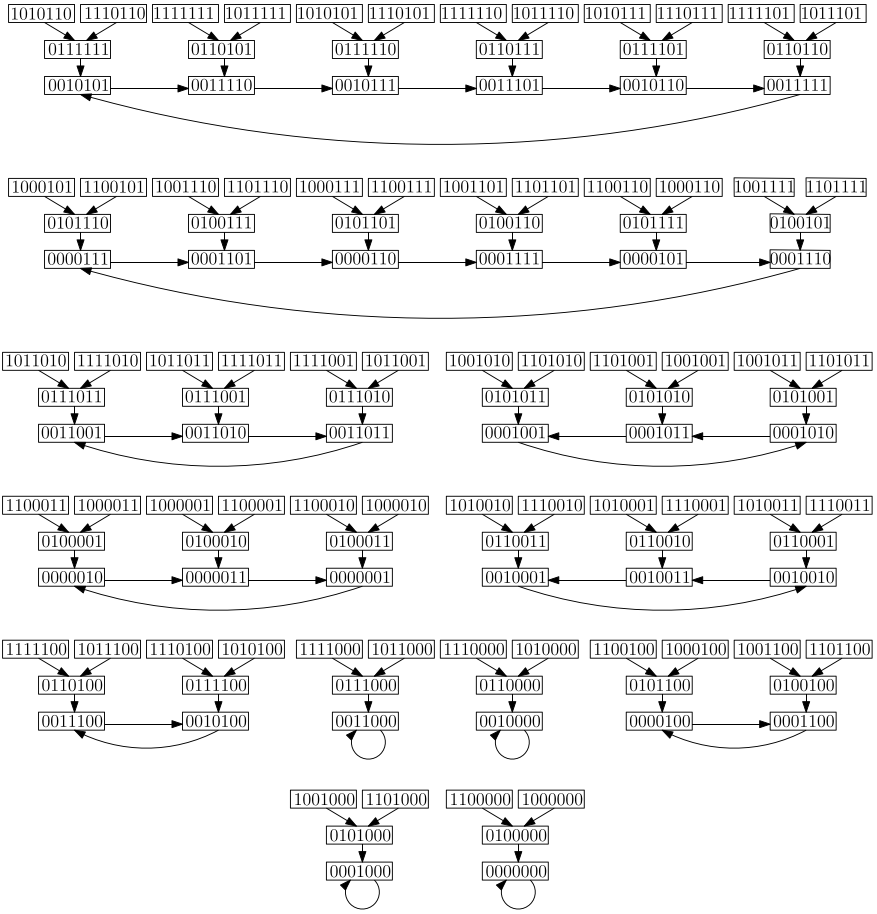


Fig. 7.1 Transition diagram of the 7-cell CA (0, 240, 204, 102, 204, 170, 60)

a cyclic configuration from a non-reachable configuration is referred to as the *depth* (d) of the CA. The depth of the CA of Fig. 7.1 is 2.

If an n-cell CA and an m-cell CA are merged to get an $(n + m)$-cell CA (Sect. 4.6) and the transition of configurations of both the n-cell and m-cell CAs can be traced in the $(n + m)$-cell CA, then the cycle structure of the $(n + m)$-cell CA can be obtained by *multiplying* the cycle structures of the n-cell and m-cell CAs. Let the cycle structures of the n-cell and m-cell CAs be

$$CS_1 = [\sum_{i=1}^{l} \mu_i(k_i)] \text{ and } CS_2 = [\sum_{j=1}^{l'} \mu'_j(k'_j)]$$

respectively, where $l, l' \geq 1$ are the number of cyclic components in the cycle structure. The cross product of CS_1 and CS_2 ($CS_1 \times CS_2$) is the product of each ith term of CS_1 and each jth term of CS_2. The product of $\mu_i(k_i)$ and $\mu'_j(k'_j)$ results in $\mu_k(k)$, where

$$\mu_k = \mu_i.\mu'_j.\ \gcd(k_i, k'_j) \text{ and } k = \text{lcm}(k_i, k'_j) \tag{7.1}$$

A detail discussion on the issue is provided in next chapter (see Sect. 8.1).

Example 7.1 Consider $CS_1 = [1(1), 1(3)]$ and $CS_2 = [1(1), 3(5)]$. Then, $CS_1 \times CS_2 = [1(1), 1(3), 3(5), 3(15)$.

In a linear CA, all 0s configuration always form a self loop. That is, a linear CA contains at least one cycle of length 1. Whereas, for the complemented CAs, there is no such restriction.

Definition 49 A cycle structure is called *linear* if there exists a linear CA with such a cycle structure. It is *complemented* if the cycle structure is not linear but there exists a complemented CA against the cycle structure. The cycle structure is *non-linear* if it is neither linear nor complemented.

It can be noted that a set of CAs can be found with the same cycle structure. This set may contain linear, complemented and non-linear CAs. Here, the set of CAs with a linear cycle structure contains at least one linear CA, whereas, the set of CAs against a complemented cycle structure contains no linear CA but at least one complemented CA. That is, a non-linear CA may have linear cycle structure, but a linear CA cannot have non-linear cycle structure. For example, maximal length CAs can be linear, complemented and non-linear. But their cycle structure is linear.

7.2 Cycle Structure of Linear Cellular Automata

This chapter deals with the CAs that are finite and use only ECA rules in their rule vectors. The configurations of such an n-cell linear CA form an n-dimensional vector space over GF(2) and the characteristic matrix T acts as the operator in this space (see Sect. 3.2). This space can be decomposed into a number of invariant subspaces. The decomposition boils down the study of behavior of the operator T in the whole space to the study of its behavior in various component subspaces. So, to get the cycle structure of an n-cell linear CA, the vector space of the CA is first decomposed into indecomposable invariant subspaces, and then the cycle structures of individual subspaces are computed.

A *minimal polynomial* of a vector space annihilates every vector of the space. For a linear CA, the minimal polynomial can be uniquely determined by $T + Ix$. Minimal polynomial is invariant to the operator T. A minimal polynomial $m(x)$ can further be decomposed into irreducible factors.

$$m(x) = [\phi_1(x)]^{n_1} \cdots [\phi_2(x)]^{n_2} \cdots [\phi_k(x)]^{n_k}$$

where $\phi_i(x)$ is an irreducible polynomial. Minimal polynomial is a factor of the *characteristic polynomial* $f(x)$ of the T matrix (that is, $f(x) = \det(T + Ix)$). In fact, there are a number of invariant factors of the characteristic polynomial $f(x)$ and each invariant factor can be decomposed into irreducible factors over GF(2). The characteristic polynomial is a product of such invariant factors. Hence, the characteristic polynomial of T can be expressed as

$$f(x) = [\phi_1(x)]^{n_1} \cdot [\phi_2(x)]^{n_2} \cdots [\phi_{\mathcal{N}}(x)]^{n_{\mathcal{N}}} \tag{7.2}$$

where $\phi_i(x)$ is an irreducible polynomial and $\mathcal{N}$ is the number of such polynomials. Here, each $[\phi_i(x)]^{n_i}$ is called *elementary divisor* of $T + Ix$. For an irreversible CA, however, the characteristic polynomial is expressed as

$$f(x) = x^{n_1} \cdot x^{n_2} \cdots x^{n_l} \cdot [\phi_{l+1}(x)]^{n_{l+1}} \cdots [\phi_{\mathcal{N}}(x)]^{n_{\mathcal{N}}} \tag{7.3}$$

where $\phi_i(x)$ is an irreducible polynomial ($i = l + 1, \cdots, \mathcal{N}$) other than x. As per the decomposition theorem of vector space, the whole space can be split into cyclic subspaces whose minimal polynomials are the elementary divisors of the operator T. The cycles of a CA are formed by a cyclic subspace with $[\phi_i(x)]^{n_i}$ as minimal polynomial. Whereas, for acyclic configurations, the depth of the CA can be obtained. To get the cycle structure of the CA, we need to consider only the subspaces against the elementary divisors $[\phi_{l+1}(x)]^{n_{l+1}}, [\phi_{l+2}(x)]^{n_{l+2}}, \cdots, [\phi_{\mathcal{N}}(x)]^{n_{\mathcal{N}}}$ of Eq. 7.3. The elementary divisors correspond to the component CAs that can be merged in sequence to get a CA having the same cycle structure as the given CA.

Two linear CAs with the same number of cells, however, can have the same characteristic polynomial as product of same elementary divisors. Then, cycle structure of these two CAs are same. In fact, there may exist more than one CA against an elementary divisor.

7.2.1 Finding Elementary Divisors

Elementary divisors are irreducible factors of invariant polynomials of a T. The invariant polynomials can be found by converting the matrix $T + Ix$ into Smith Normal Form.

Definition 50 *Smith Normal Form* is a diagonal form of the matrix $(T + Ix)$, with nonzero elements $a_1(x), a_2(x), \cdots, a_j(x), \cdots, a_n(x)$ of GF(2)[x] at the diagonal. Each element $a_j(x)$ is so arranged that $a_j(x)$ is a factor of $a_{j+1}(x)$. The elements $a_1(x), \cdots, a_n(x)$ are the invariant factors of T and $a_n(x)$ is the minimal polynomial of T.

Since each $a_j(x)$ in Smith Normal Form is an invariant polynomial, the irreducible factors of $a_j(x)$ can be denoted as $[\phi_l(x)]^{n_l}, [\phi_{l+1}(x)]^{n_{l+1}}, \cdots, [\phi_{l+m}(x)]^{n_{l+m}}$, where

$\phi_{l+i}(x)$ is an irreducible polynomial. Each $\phi_{l+i}(x)^{n_{l+i}}$ is an elementary divisor of T. So, with Smith Normal Form, the characteristic polynomial of T, in *Elementary Divisor Form (EDF)*, can be obtained as

$$f(x) = [\phi_1(x)]^{n_1}[\phi_2(x)]^{n_2} \cdots [\phi_{\mathcal{N}}(x)]^{n_{\mathcal{N}}}$$

where each $\phi_i(x)$ is an irreducible polynomial and factor of some $a_j(x)$. The matrix $T + Ix$ can be converted to its Smith Normal Form through elementary row and column operations. The procedure of such conversion can be found in standard text book on matrix algebra. The following example illustrates the conversion to Smith Normal Form to get the characteristic polynomial in the form of Eq. 7.3.

Example 7.2 Consider a 7-cell linear CA with rule vector (0, 240, 204, 102, 204, 170, 60) with null boundary (Fig. 7.1). The characteristic matrix of the CA is

$$T = \begin{pmatrix} \lceil 0\ 0 \rceil & 0 & 0 & 0 & 0 & 0 \\ \lfloor 1\ 0 \rfloor & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & [1] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lceil 1\ 1 \rceil & 0 & 0 \\ 0 & 0 & 0 & \lfloor 0\ 1 \rfloor & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lceil 0\ 1 \rceil \\ 0 & 0 & 0 & 0 & 0 & \lfloor 1\ 1 \rfloor \end{pmatrix}$$

The Smith Normal Form of $T + Ix$ is

$$\begin{pmatrix} 1\,0\,0\,0\,0\,0 & 0 & & 0 \\ 0\,1\,0\,0\,0\,0 & 0 & & 0 \\ 0\,0\,1\,0\,0\,0 & 0 & & 0 \\ 0\,0\,0\,1\,0\,0 & 0 & & 0 \\ 0\,0\,0\,0\,1\,0 & 0 & & 0 \\ 0\,0\,0\,0\,0\,1 & 0 & & 0 \\ 0\,0\,0\,0\,0\,0 & 1+x & & 0 \\ 0\,0\,0\,0\,0\,0 & 0 & x^2 \cdot (1+x)^2 \cdot (x^2+x+1) & \end{pmatrix}$$

The characteristic polynomial $f(x) = x^2 \cdot (1 + x) \cdot (1 + x)^2 \cdot (x^2 + x + 1)$ [in elementary divisors form].
The minimal polynomial $f_{min}(x) = x^2 \cdot (1 + x)^2 \cdot (x^2 + x + 1)$.
[Note : $a_1(x) = a_2(x) = a_3(x) = a_4(x) = a_5(x) = 1$, $a_6(x) = (1 + x)$ and $a_7(x) = x^2 \cdot (1 + x)^2 \cdot (x^2 + x + 1)$].

Once the characteristic polynomial is derived as the product of elementary divisors, then the next task is to find the cycle structure of each $[\phi_i(x)]^{n_i}$, where $\phi_i(x)$ is an irreducible polynomial.

7.2.2 Cyclic Subspace of Elementary Divisors

An elementary divisor $[\phi_i(x)]^{n_i}$ corresponds to a CA (referred to as $T(\phi_i(x)^{n_i})$) with $n_i \times r_i$ cells, where r_i is the degree of $\phi_i(x)$. The irreducible polynomial $\phi_i(x)$ corresponds to another CA (referred to as $T(\phi_i(x))$) of r_i cells. By augmenting $T(\phi_i(x))$

n_i times in a specific way, the $T(\phi_i(x)^{n_i})$ can be generated such that the configuration space of $T(\phi_i(x)^{n_i})$ is the direct sum of the configuration spaces of n_i number of $T(\phi_i(x))$. Then, the cycle structure of $T(\phi_i(x)^{n_i})$ can be computed from the cycle structure of $T(\phi_i(x))$ using Eq. 7.1. Hence, the cycle lengths of $T(\phi_i(x)^{n_i})$ is the multiple of that of $T(\phi_i(x))$. If k is the length of a cycle of $T(\phi_i(x))$, then there are cycles in $T(\phi_i(x)^{n_i})$ of length k, $2k$, 2^2k, etc. The cycles of length k in $T(\phi_i(x)^{n_i})$ are called *primary cycles* and the rest are *secondary cycles*.

Primary Cycle Structure: It consists of μ_k primary cycles of length k and denoted as $[\mu_k(k)]$, where

$$\mu_k = \frac{2^{r_i} - 1}{k} \tag{7.4}$$

r_i being the degree of $\phi_i(x)$, and k is the smallest integer such that $\phi_i(x)$ divides $x^k + 1$ (see Theorem 7.6). Both the μ_k and k are odd. So, primary cycle lengths are always odd.

Secondary Cycle Structure: The secondary cycle length is of the form $2^j \cdot k$, where k is the length of a primary cycle and j is an integer. If $n_i = 2$, then the length of secondary cycles is $2k$. If n_i is 3 or 4, in addition to the cycles of length $2k$, the cycles of length 2^2k are also included. It can be considered that the different factors of the elementary divisors contribute secondary cycles of different lengths. Then, we get that all the factors of an elementary divisor $\phi_i(x)^{n_i}$, starting from $\phi_i(x)^{2^{j-1}+1}$ to $\phi_i(x)^{2^j}$, contribute cycles of length $2^j \cdot k$. Let call these factors as the jth factor set. Therefore, the length of a secondary cycle is even.

Example 7.3 Let consider an elementary divisor $(x^4 + x^3 + x^2 + x + 1)^9$. The length and number of primary cycles are $k = 5$ and $\mu_k = 3$ respectively. Then, the factor $(x^4 + x^3 + x^2 + x + 1)^2$ contributes the cycles of length $2 \cdot k = 10$. The factors $(x^4 + x^3 + x^2 + x + 1)^3$ and $(x^4 + x^3 + x^2 + x + 1)^4$ contribute the cycle of length $2^2 \cdot k = 20$. Whereas, all the other factors $(x^4 + x^3 + x^2 + x + 1)^5$, $(x^4 + x^3 + x^2 + x + 1)^6$, $(x^4 + x^3 + x^2 + x + 1)^7$ and $(x^4 + x^3 + x^2 + x + 1)^8$ form cycles of length $2^3 \cdot 5 = 40$.

Definition 51 The set of factors of $[\phi_i(x)]^{n_i}$, $n_i \geq 1$, powers of which lie between $(2^{j-1} + 1)$ to 2^j, for some $j \geq 1$, is referred to as the *jth factor set* (F_j). Whereas, the factor $\phi_i(x)$ is the only member of 0th factor set. Each member of the jth factor set corresponds to a cycle of length $2^j \cdot k$, where k is the primary cycle length and the maximum value of j is $\lceil \log_2(n_i) \rceil$. The cardinality of the jth factor set in an elementary divisor $\phi_i(x)^{n_i}$ is denoted as $N(F_j)$ and can be determined by

$$N(F_j) = \begin{cases} 1 & \text{if } j = 0 \\ 0 & \text{if } j > 0 \ \& \ 2^{j-1} \geq n_i \\ 2^{j-1} & \text{if } j > 0 \ \& \ 2^j < n_i \\ n_i - 2^{j-1} & \text{if } j > 0 \ \& \ 2^{j-1} < n_i \leq 2^j \end{cases} \tag{7.5}$$

The following theorem guides the enumeration of cyclic components of cycle length $2^j \cdot k$. Its proof follows from the above definition.

Theorem 7.1 *The number of cyclic components corresponding to the cycle length* $2^j \cdot k$ *is given by the following relation*

$$\mu_{2^j \cdot k} = \frac{2^r \sum_{\mathcal{J}=0}^{j} N(F_{\mathcal{J}}) - 2^r \sum_{\mathcal{J}=0}^{j-1} N(F_{\mathcal{J}})}{2^j \cdot k} \tag{7.6}$$

where r *is the degree of the underlying irreducible polynomial* $\phi_i(x)$.

The union of the cycle structures of all the factors of the characteristic polynomial yields the complete cycle structure of the elementary divisor. It is represented as

$$CS_{(\phi_i(x)^{n_i})} = [1(1) + \sum_{j=0}^{m_k} \mu_{2^j \cdot k}(2^j \cdot k)] \tag{7.7}$$

where $m_k = \lceil \log_2(n_i) \rceil$ and 1(1) represents the all 0s vector, forming a self-loop.

Definition 52 The cycle structure corresponds to an elementary divisor $\phi_i(x)^{n_i}$ is called the *Elementary Cycle Structure (ECS)*. It is represented by the triplet $(\mu_k, k)^{n_i}$.

The jth factor set F_j is associated with an elementary divisor $\phi_i(x)^{n_i}$. Since the elementary cycle structure $CS_{\phi_i(x)^{n_i}}$ represents the elementary divisor $\phi_i(x)^{n_i}$, in rest of the discussion, the F_j will be associated with the elementary cycle structure as well.

Definition 53 All the cycles of the form $(2^j \cdot k)$, for an odd k, are the members of the family of cycles referred to as *k-cycle family*. If k is the length of a primary cycle, then it is called *primary cycle family*. The set of all cycle lengths of k-cycle family is represented by $PCF(k)$.

Example 7.4 Let the characteristic polynomial $f(x)$ of an example linear CA be $f(x) = (x^4 + x^3 + x^2 + x + 1)^9$. Therefore, degree $r_i = 4$.

Primary cycle: Since $x^4 + x^3 + x^2 + x + 1$ divides $x^5 + 1$, $k = 5$. So, $\mu_k = \frac{2^4-1}{5} = 3$. Therefore, $x^4 + x^3 + x^2 + x + 1$ constitutes the cycle structure 3(5). As $n_i = 9$, $\lceil \log_2(n_i) \rceil = 4$. So, j varies from 0 to 4.

Secondary cycle: As per the table noted below, the cardinality of $N(F_0) = 1$, $N(F_1) = 1$, $N(F_2) = 2$, $N(F_3) = 4$, and $N(F_4) = 1$.

j	$2^{j-1}+1$	2^j	$N(F_j)$
0		1	1 since $n_i = 9 > 2^0$
1	2	2	$2^{1-1} = 1$ since $9 > 2^1$
2	3	4	$2^{2-1} = 2$ since $9 > 2^2$
3	5	8	$2^{3-1} = 4$ since $9 > 2^3$
4	9	16	$9\text{-}2^{4-1} = 1$ since $2^{4-1} < 9 < 2^4$

Hence, by Relation 7.5, the cyclic components corresponding to $\mu_{2^1 \cdot k}$ is $\frac{2^{4(1+1)}-2^{4(1)}}{2^1 \cdot 5}$ = *24*, and corresponding to $\mu_{2^2 \cdot k}$ is $\frac{2^{4(2+2)}-2^{4(2)}}{2^2 \cdot 5}$ = *3264*.
Similarly, the cyclic components corresponding to $\mu_{2^3 \cdot k}$ is $\frac{2^{4(4+4)}-2^{4(4)}}{2^3 \cdot 5}$ = *107372544*, and corresponding to $\mu_{2^4 \cdot k}$ is $\frac{2^{4(1+8)}-2^{4(8)}}{2^4 \cdot 5}$ = *805306368*.
Hence the complete cycle structure $CS_{(x^4+x^3+x^2+x+1)^9}$ = [1(1), 3(5), 24(10), 3264(20), 107372544(40), 805306368(80)].
The elementary cycle structure (ECS) can also be represented by the triplet $(3, 5)^9$ as per Definition 52, where $\mu_k = 3$, $k = 5$, $n_i = 9$, and j varies from 0 to 4. So, 2^j varies from 1 to 16. As the primary cycle length (k) is 5, then by Definition 53, $PCF(5)$ = {5, 10, 20, 40, 80}.

7.2.3 The Complete Cycle Structure

The entire vector space of a linear CA is the direct sum of the sub-spaces generated by its elementary divisors. As the CA can be considered as a merger of component CAs, which correspond to the elementary divisors, the cycle structure of the CA can be obtained from the cross product ($\times$) of all the cycle structures of elementary divisors. If $\mu_1(k_1)$ of the first component CA is multiplied with $\mu_2(k_2)$ of the second component CA, then by Relation 7.1, the following cyclic component (μ_k) of length k can be found. That is,

$$\mu_k = \mu_{k_1} . \mu_{k_2} . \gcd(k_1, k_2) \text{ and } k = \text{lcm}(k_1, k_2)$$

Hence, the complete cycle structure of a linear CA, derived from the cross product of cycle structures of individual elementary divisors, is expressed as

$$CS = CS_{\phi_1(x)^{n_1}} \times CS_{\phi_2(x)^{n_2}} \times \cdots \times CS_{\phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}} \tag{7.8}$$

where $\mathcal{N}$ is the number of elementary divisors forming the linear CA.

Example 7.5 Let consider the CA of Example 7.2. The elementary divisors are x^2, $(x+1)$, $(x+1)^2$ and (x^2+x+1). Following are the cycle structures of all the elementary divisors except x^2.

$$CS_{(x+1)} = [1(1), 1(1)], \quad CS_{(x+1)^2} = [1(1), 1(1), 1(2)], \quad CS_{(x^2+x+1)} = [1(1), 1(3)]$$

Now, the complete cycle structure of the CA is

$$CS = [1(1), 1(1)] \times [1(1), 1(1), 1(2)] \times [1(1), 1(3)] = [4(1), 2(2), 4(3), 2(6)]$$

The transition diagram of the CA is shown in Fig. 7.1.

The following theorem, however, guides us for better understanding of cycle formation.

Theorem 7.2 *The cycle structure ($CS(\mathcal{N})$) of a linear CA with $\mathcal{N}$ elementary divisors can be represented as*

$$CS(\mathcal{N}) = [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)] \tag{7.9}$$

where k_i is odd and $m_{k_i} = \lceil \log_2 n_i \rceil$ with n_i as the power of the ith elementary divisor $\phi_i(x)^{n_i}$.

Proof Let $CS_{(\phi_i(x)^{n_i})} = [1(1), \alpha_i]$, where $\alpha_i = \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)$. Then $CS(\mathcal{N}) = [1(1), \alpha_1] \times [1(1), \alpha_2] \times \cdots \times [1(1), \alpha_{\mathcal{N}}]$. On performing successive cross products, the following can be resulted in.

$$CS(\mathcal{N}) = [1(1), (\alpha_1, \alpha_2, \cdots, \alpha_{\mathcal{N}}), (\alpha_1 \times \alpha_2, \cdots, \alpha_{\mathcal{N}-1} \times \alpha_{\mathcal{N}}), \cdots, (\alpha_1 \times \alpha_2 \times \cdots \times \alpha_{\mathcal{N}})]$$

That is, each member in the ith group (marked within '()') is formed by the cross product of i number of α's that form a component of CS($\mathcal{N}$). Each component, as illustrated below for $i = 2$, represents a primary cycle family (Definition 53). Generalization, for $i > 2$, follows directly.

Let $\tilde{CS} = (\alpha_1 \times \alpha_2)$ and [CL] be the set of cycle lengths formed from this cross product. The members of the [CL] can be found out by the relation:

$$\text{lcm}((2^{j_1} \cdot k_1), (2^{j_2} \cdot k_2)) = 2^{\max(j_1, j_2)} \text{lcm}(k_1, k_2) \tag{7.10}$$

Let $\text{lcm}(k_1, k_2) = k_3$. Consequently, all cycle lengths of the [CL] is a member of k_3 cycle family and the cycle structure can be expressed as

$$\tilde{CS} = \sum_{j=0}^{\max(m_{k_1}, m_{k_2})} \mu_{2^j \cdot k_3}(2^j \cdot k_3), \text{ where } m_{k_i} = \lceil \log_2 n_{k_i} \rceil, i \in \{1, 2\}$$

Both the k_1 and k_2 are lengths of the primary cycles and odd. So, k_3 is also odd. Hence the proof. □

It is to be noted that all the odd-length cycles are not primary cycles. But they can form the cycle families as like the primary cycles. Theorem 7.2 is used to replace the costly operation of successive cross products and to get the complete cycle structure of a linear CA.

7.2.4 Replacement of Cross Products

In a linear CA, two ECSs may contain the primary cycles of same length. To replace the cross products, following two steps are performed in sequence.

- Step A: Form groups of ECSs, where each group comprises of ECSs having the same length of their primary cycles. Evaluate the cycle structure of each group separately.
- Step B: Derive the final cycle structure considering a pair of groups at a time, each group having the different primary cycle structure.

Step A: Derivation of Cycle structure from the ECSs with same primary cycle length
Let consider a set of $\mathcal{N}$ ECSs that are having the same length of their primary cycles. The cycle structure of this group of ECSs is $ECS_1 \times ECS_2 \times \cdots \times ECS_{\mathcal{N}}$. This cycle structure is the Elementary Family Cycle Structure (EFCS).

Definition 54 *Elementary Family Cycle Structure (EFCS)* is a family of cycle structures that is resulted in from the cross products of elementary cycle structures, each having the same primary cycle length. That is, the primary cycle length (say k) is same for all the underlying elementary divisors generating the $EFCS$.

Let an EFCS be formed with $\mathcal{N}$ ECSs. Then,

$$EFCS = (\mu, k)^{n_1} \times (\mu, k)^{n_2} \times (\mu, k)^{n_{\mathcal{N}}}$$

The $ECS\ (\mu, k)^{n_i}$ corresponds to the elementary divisor $\phi_i(x)^{n_i}$. From Theorem 7.2, it is followed that an $EFCS$ consists of one primary cycle length (k) having the following form

$$EFCS = [1(1) + \sum_{j=0}^{m_k} \mu_{2^j \cdot k}(2^j \cdot k)] \tag{7.11}$$

where $m_k = \lceil \log_2(\max_{1 \le i \le \mathcal{N}}\{n_i\}) \rceil$. The following example illustrates the concept of elementary family cycle structure (EFCS).

Example 7.6 Let consider the elementary divisor form of the characteristic polynomial $f(x) = (x^2 + x + 1)^1 \cdot (x^2 + x + 1)^4 \cdot (x^2 + x + 1)^7 \cdot (x^2 + x + 1)^8$. The ECSs in triplet form (Definition 52) are $(1, 3)^1$, $(1, 3)^4$, $(1, 3)^7$ and $(1, 3)^8$ respectively.

The cycle structure $CS = (1, 3)^1 \times (1, 3)^4 \times (1, 3)^7 \times (1, 3)^8$.
In expanded form, CS = [1(1), 1(3)] × [1(1), 1(3), 2(6), 20(12)] × [1(1), 1(3), 2(6), 20(12), 672(24)] × [1(1), 1(3), 2(6), 20(12), 2720(24)]
Consequently, CS = [1(1), 85(3), 3688(6), 5591040(12), 4581088288(24)]. It shows CS has a primary cycle of length 3. The CS is the Elementary Family Cycle Structure (EFCS) derived out of the cross products of the four ECSs of k-cycle family (Definition 53) with $k = 3$.

It can be shown that performing cross product is not a necessity in deriving the EFCS out of the constituent ECS. Rather, it can be derived out of the jth factor set F_j, present in each constituent elementary divisor. The $EFCS$ and the Elementary Cycle Structure (ECS) play the key role in this derivation.

Derivation of $EFCS$ from F_j: $N(F_j)$ is the cardinality of F_j present in a single ECS. To compute the total number of jth factor set present in different ECSs, constituting the $EFCS$, a new term $\tilde{N}(F_j)$ is introduced. By Definition 51, F_j represents the factors having power between $2^{j-1} + 1$ and 2^j, present in any ECS $(\mu, k)^{n_i}$, and $N_i(F_j)$ denotes the cardinality of F_j present in the ith ECS.

Definition 55 $\tilde{N}(F_j)$ is the sum of the cardinalities of F_j, the jth factor sets, of all the constituent ECSs comprising the $EFCS$–that is,

$$\tilde{N}(F_j) = \sum_{i=1}^{\mathcal{N}} N_i(F_j) \tag{7.12}$$

where $\mathcal{N}$ is the number of ECS and $\mathcal{N}_i(F_j)$ is the cardinality of F_j of the ith ECS which is computed by Relation 7.5.

The maximum possible value of j is $\lceil \log_2(\max_{1 \le i \le \mathcal{N}} \{n_i\}) \rceil$. The following example illustrates the computation of $\tilde{N}(F_j)$.

Example 7.7 Let an $EFCS$ be formed from the following ECSs.

$$EFCS = (1, 3)^1 \times (1, 3)^4 \times (1, 3)^7 \times (1, 3)^8$$

The underlying elementary divisors, as noted in Example 7.6, are $(x^2 + x + 1)^1$, $(x^2 + x + 1)^4$, $(x^2 + x + 1)^7$, and $(x^2 + x + 1)^8$. The highest value of j is $\lceil \log_2(\max\{1, 4, 7, 8\}) \rceil = 3$.
While computing $\tilde{N}(F_j)$, for each $j \in \{0, 1, 2, 3\}$, it results in
$\tilde{N}(F_0) = 4$ (0th component)–that is, the irreducible polynomial $(x^2 + x + 1)$ is present in all the four ECSs.

$\tilde{N}(F_1)$ 1+1+1 = 3 (the 1^{st} component)–that is, the factors starting from $(2^{1-1}+1)$ to 2^1 are present in the last three $ECSs$.
$\tilde{N}(F_2)$ = 2+2+2 = 6 (the 2^{nd} component)–that is, the factors starting from $(2^{2-1}+1)$ to 2^2 are present only in the last three $ECSs$.
$\tilde{N}(F_3)$ = 3+4 = 7 (the 3^{rd} component)–that is, the factors starting from $(2^{3-1}+1)$ to 2^3 are present only in the last two $ECSs$.

The following theorem shows that the cycle length $(2^j \cdot k)$ can be directly computed from $\tilde{N}(F_{\mathcal{J}})$, $\mathcal{J} \in \{0, \cdots, j\}$.

Theorem 7.3 *The cycle structure of EFCS is given by* $[1(1), \mu_k(k), \cdots, \mu_{2^j \cdot k}(2^j \cdot k), \cdots]$, *where the number of cyclic components* $\mu_{2^j \cdot k}$ *follows the relation*

$$\mu_{2^j \cdot k} = \frac{2^{r\sum_{\mathcal{J}=0}^{j} \tilde{N}(F_{\mathcal{J}})} - 2^{r\sum_{\mathcal{J}=0}^{j-1} \tilde{N}(F_{\mathcal{J}})}}{2^j \cdot k} \tag{7.13}$$

The $\tilde{N}(F_{\mathcal{J}})$ *represents the sum of the cardinalities of* $\mathcal{J}$*th factor set of all the constituent ECSs and r is the degree of irreducible polynomial forming the ECS.*

Proof A cycle of length $2^j \cdot k$ can be found only by the cross product of the cycles of lengths $2^j \cdot k$ and $2^{\mathcal{J}} \cdot k$, where $\mathcal{J} \leq j$ (see Expression 7.10). Therefore, the number of cyclic components $\mu_{2^j \cdot k}$ depends only on the number of $\mathcal{J}$th factor set. So, the number of additional configurations required to form the cycles of length $2^j \cdot k$, after forming cycles of length $2^{j-1} \cdot k$, is

$$S = 2^{r\sum_{\mathcal{J}=0}^{j} \tilde{N}(F_{\mathcal{J}})} - 2^{r\sum_{\mathcal{J}=0}^{j-1} \tilde{N}(F_{\mathcal{J}})}$$

Hence, $\mu_{2^j \cdot k} = S/(2^j \cdot k)$, where $2^j \cdot k$ is the cycle length corresponding to the component $\mu_{2^j \cdot k}$. □

Example 7.8 Consider the

$$EFCS = (1,3)^1 \times (1,3)^4 \times (1,3)^7 \times (1,3)^8$$

of Example 7.7. Here, 8 is the highest power of the constituent ECSs. It is the fourth ECS and the value of r is 2. The values of $\tilde{N}(F_0)$, $\tilde{N}(F_1)$, $\tilde{N}(F_2)$ and $\tilde{N}(F_3)$ are 4, 3, 6 and 7 respectively. The cyclic components are

$$\mu_{2^0 \cdot 3} = \frac{2^{2(4)} - 2^{2 \cdot 0}}{2^0 \cdot 3} = 85, \quad \mu_{2^1 \cdot 3} = \frac{2^{2(4+3)} - 2^{2 \cdot 4}}{2^1 \cdot 3} = 3688$$

$$\mu_{2^2 \cdot 3} = \frac{2^{2(7+6)} - 2^{2 \cdot 7}}{2^2 \cdot 3} = 5591040 \quad \mu_{2^3 \cdot 3} = \frac{2^{2(13+7)} - 2^{2 \cdot 13}}{2^2 \cdot 3} = 4581088288$$

The complete cycle structure CS = [1(1), 85(3), 3688(6), 5591040(12), 4581088288(24)]. It can be observed that the cycle structure is same as that of Example 7.6.

Step B: Derivation of cycle structure from the set of EFCSs each having primary cycles of different length

The final cycle structure (CS) of a CA can be obtained by the successive cross products of the constituent EFCSs. That is,

$$CS = EFCS_1 \times EFCS_2 \times \cdots EFCS_i \times EFCS_{i+1} \cdots \times EFCS_{\tilde{\mathcal{N}}} \quad (7.14)$$

where $\tilde{\mathcal{N}}$ is the number of EFCSs. Like Step A, Step B targets to avoid the cross products. The EFCSs are arranged in ascending order of their primary cycle lengths. Then, the following procedure is applied to derive the cycle structure.

Let $CS = CS_1 \times CS_2$, where $CS_1 = EFCS_l$ and $CS_2 = EFCS_{l+1} \times \cdots \times EFCS_{\tilde{\mathcal{N}}}$. The CS_1 is expressed as per Expression 7.11, and its primary cycle (say k_1) is smaller than the primary cycles in CS_2. Since CS_2 is formed from the cross product of a number of EFCSs, its cycle structure can be represented as per Expression 7.9. So, CS_1 and CS_2 can be expressed as:

$$CS_1 = [1(1), \alpha] \qquad CS_2 = [1(1), \sum_{k_i} \beta_i],$$

where

$$\alpha = \sum_{j=0}^{m_{k_1}} \mu_{2^j \cdot k_1}(2^j \cdot k_1), \qquad \beta_i = \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i), \quad i > 1$$

Here, β_i represents the cycle structure contributed by k_i-cycle family. Therefore,

$$CS = CS_1 \times CS_2 = [1(1), \alpha] \times [1(1), \sum_i \beta_i] \quad (7.15)$$

$$\therefore CS = [1(1), \alpha, \sum_i (\beta_i, \alpha \times \beta_i)] \quad (7.16)$$

Theorem 7.4 *The cyclic component of CS, corresponding to the k_1 cycle family is same as the CS_1, where $CS = CS_1 \times CS_2$.*

Proof The α of CS_1 (Relation 7.16) contains the primary cycles of length k_1 whereas, β_i of CS_2, for any i, does not have the primary cycle of length k_1. If k_i is the length of primary cycle of β_i, then $\alpha \times \beta_i$ results in the cycles of length $\text{lcm}(k_1, k_i)$. Since k_i is greater than k_1, $\text{lcm}(k_1, k_i)$ is greater than k_1. So, α directly comes in CS from CS_1, and there is no cyclic component against the cycle length k_1 contributed by $\alpha \times \beta_i$, for any i. Hence the proof. □

Example 7.9 Let CS_1= [1(1), [3(1), 2(2)]] and CS_2 = [1(1), 1(3), 3(5), 3(15)]. The CS, formed from the cross product of CS_1 and CS_2, is [1(1), [3(1), 2(2)], [4(3), 2(6)], [12(5), 6(10)], [12(15), 6(30)]]. The cyclic components corresponding to the primary cycle 1 ([3(1), 2(2)]) is same in CS_1 and CS_2.

Let assume that the CS_1 and CS_2 consist of k_1 and k_2 cycle families respectively, and $k_3 = \text{lcm}(k_1, k_i)$. The following theorem provides the alternate formula for obtaining each member of the cyclic components $\mu_{2^j \cdot k_3}$. This result can be employed to replace the cross product $\alpha \times \beta_i$.

Theorem 7.5 *The cyclic component ($\mu_{2^j \cdot k_3}$) formed from the cross product of the cyclic component of k_1 and k_2 cycle family in CS_1 and CS_2 respectively is determined as*

$$\mu_{2^j \cdot k_3} = [A + B] \tag{7.17}$$

where

$$A = \frac{\mu_{2^j \cdot k_2} \cdot S_{2^j \cdot k_1} \cdot \gcd(k_1, k_2)}{k_1} \quad \textit{and} \quad B = \frac{\mu_{2^j \cdot k_1} \cdot S_{2^{j-1} \cdot k_2} \cdot \gcd(k_1, k_2)}{k_2}$$

The $S_{2^j \cdot k_i}$ ($i \in \{1, 2\}$) is the number of configurations, excluding the all 0s configuration, covered by the cycles of length $2^{\mathcal{J}} \cdot k_i$, for all $\mathcal{J} \in \{0, 1, \cdots, j\}$ –that is,

$$S_{2^j \cdot k_i} = \sum_{\mathcal{J}=0}^{j} \mu_{2^{\mathcal{J}} \cdot k_i} \times (2^{\mathcal{J}} \cdot k_i) \tag{7.18}$$

Proof The result of the cross product of the cycles of length $2^{j_1} \cdot k_1$ and $2^{j_2} \cdot k_2$ is the cycles of length $2^{\max(j_1, j_2)}(k_3)$, where $k_3 = \text{lcm}(k_1, k_2)$ (see Relation 7.10). Therefore, to get $2^j \cdot k_3$, either j_1 or j_2 is to be equal to j.

Enumeration of A : If $j_2 = j$ and $j_1 \leq j$, then by Relation 7.1, the Part A of the cyclic components of $\mu_{2^j \cdot k_3}$ can be estimated as

$$\mu_{2^j \cdot k_3} = \mu_{2^j \cdot k_2} \cdot [\mu_{k_1} + \mu_{2 \cdot k_1} \times 2 + \cdots + \mu_{2^j \cdot k_1} \times 2^j] \cdot \gcd(k_1, k_2)$$

That is,

$$\begin{aligned} \mu_{2^j \cdot k_3} &= \mu_{2^j \cdot k_2} \cdot [\frac{\sum_{\mathcal{J}=0}^{j} \mu_{2^{\mathcal{J}} \cdot k_1} \times (2^{\mathcal{J}} \cdot k_1)}{k_1}] \cdot \gcd(k_1, k_2) \\ &= \mu_{2^j \cdot k_2} \cdot S_{2^j \cdot k_1} \cdot \gcd(k_1, k_2) \end{aligned}$$

Enumeration of B: It is analogous to the enumeration of A. If $j_1 = j$ and $j_2 \leq j$, the cycles of length $2^j \cdot k_3$ can be evaluated. To avoid multiplication of lengths $(2^j \cdot k_1)$ and $(2^j \cdot k_2)$ twice, j_2 ends at $(j - 1)$ in the second case. Hence the proof. □

The following corollary, derived from the above theorem, characterizes the cycle structure generated by $(\alpha \times \beta_2)$. This structure, as noted earlier, is denoted as k_3-cycle family, where $k_3 = \text{lcm}(k_1, k_2)$.

Corollary 7.1 *The cycle structure of the k_3-cycle family has the form*

$$CS(k_3) = \sum_{j=0}^{m_{k_3}} \mu_{2^j \cdot k_3}(2^j \cdot k_3), \text{ where } m_{k_3} = \max(m_{k_1}, m_{k_2}) \tag{7.19}$$

Now, the methodology to be adopted is briefed based on the fact that whether the primary cycle of length k_2 (of β_2) and k_3 (of $\alpha \times \beta_2$) are same. It is to be noted that the value of k_3 can be either greater than or equal to k_2.

1. If $k_3 = k_2$, then both the β_2 and $\alpha \times \beta_2$ belong to the same cycle family. Consequently, the cyclic component for length k_2 ($\tilde{\mu}_{2^j \cdot k_2}$) in CS is

$$\tilde{\mu}_{2^j \cdot k_3} = \mu_{2^j \cdot k_2} + \mu_{2^j \cdot k_3} \tag{7.20}$$

   Since $\text{lcm}(k_1, k_2) = k_3 = k_2$, $gcd(k_1, k_2) = k_1$. Now, from Relation 7.17,

$$\tilde{\mu}_{2^j \cdot k_3} = \mu_{2^j \cdot k_2} \cdot (S_{2^j \cdot k_1} + 1) + \mu_{2^j \cdot k_1} \cdot S_{2^{j-1} \cdot k_2} \frac{k_1}{k_2} \tag{7.21}$$

2. If $k_3 > k_2$, then besides the cyclic component $\mu_{2^j \cdot k_2}$ of the k_2-cycle family from CS_2, the k_3-cycle family is also added to CS. Hence,

$$\tilde{\mu}_{2^j \cdot k_3}(CS) = \mu_{2^j \cdot k_3} = \frac{\mu_{2^j \cdot k_2} \cdot S_{2^j \cdot k_1}}{k_1} \cdot \text{gcd}(k_1, k_2) \tag{7.22}$$

The following example illustrates the computation of the two cases–$\text{lcm}(k_1, k_2) > k_2$ and $\text{lcm}(k_1, k_2) = k_2$.

Example 7.10 This example is to show the cases when $\text{lcm}(k_1, k_2) > k_2$ and $\text{lcm}(k_1, k_2) = k_2$.

Case 1: $\text{lcm}(k_1, k_2) > k_2$.
Let CS_1 = [1(1), 1(3)] and CS_2 = [1(1), 3(5)]. Therefore, $k_1 = 3$, $k_2 = 5$ and $\text{lcm}(3, 5) = 15$—that is, greater than 5. Now, as per Relation 7.22,

$\tilde{\mu}_{2^j \cdot k_3}(\text{CS}) = \mu_{2^j \cdot k_3} =$
$\frac{\mu_{2^j \cdot k_2} \cdot S_{2^j \cdot k_1}}{k_1} \cdot \text{gcd}(k_1, k_2)$.
Since for j = 0, $S_{2^j \cdot k_1} = 3$, and $\mu_{2^j \cdot k_2} = 3$, then
$\mu_{15} = \frac{\mu_3 \cdot S_3}{3} \cdot \text{gcd}(3, 5) = \frac{3 \cdot 3}{3} = 3$.
Therefore, the final CS = [1(1), 1(3), 3(5), 3(15)].

Case 2: $\text{lcm}(k_1, k_2) = k_2$
Let CS_1 =[1(1), 3(1), 2(2)] and CS_2 = [1(1), 1(3), 3(5), 3(15)]
For $k_1 = 1$ and $k_2 = 3$, $\text{lcm}(k_1, k_2) = k_2$.
The number of cyclic components $\tilde{\mu}_k$ with cycle length k can be derived from Relation 7.21—that is,
$\tilde{\mu}_{2^j \cdot k_3}(CS) = [\mu_{2^j \cdot k_2} \cdot (S_{2^j \cdot k_1} + 1) + \mu_{2^j \cdot k_1} \cdot S_{2^{j-1} \cdot k_2} \frac{k_1}{k_2}]$
Therefore,
$\tilde{\mu}_3 = [\mu_3 \cdot (S_1 + 1)] = 1 \times 4 = 4$. That is, there are 4 cycles, each of length 3.
$\tilde{\mu}_6 = [\mu_6 \cdot (S_2 + 1) + \mu_2 \cdot S_3 \cdot \frac{1}{3}] = [0 + 2 \cdot 3 \cdot \frac{1}{3}] = 2$; since for j = 0, $2^j \cdot k_2 = 3$, $2^j \cdot k_1 = 1$, and $S_1 = 3$.
Similarly, $\tilde{\mu}_5 = 12$, $\tilde{\mu}_{10} = 6$ and $\tilde{\mu}_{15} = 12$, $\tilde{\mu}_{30} = 6$; since for j = 1, $2^j \cdot k_2 = 6$, $2^j \cdot k_1 = 2$, $2^{j-1} \cdot k_2 = 3$ and $S_{2^{j-1} \cdot k_2} = 3$,

Therefore, the final CS = [1(1), [3(1), 2(2)], [4(3), 2(6)], [12(5), 6(10)], [12(15), 16(30)]].

The scheme to find cycle structure of a linear CA is summarized in Algorithm 7. It takes the characteristic matrix T of a linear CA as input, and outputs its cycle structure. Step 8 of the algorithm either executes cross products or uses Step A and Step B of Sect. 7.2.4 to get the final cycle structure.

Algorithm 7: Find cycle structure of linear CA

Input : Characteristic matrix T of a linear CA.
Output: Cycle structure of the CA.

Step 1: Convert $(T + Ix)$ into Smith Normal Form.
Step 2: Represent the characteristic polynomial of T in the elementary divisor form

$$f(x) = \phi_1(x)^{n_1} \phi_2(x)^{n_2} \cdots \phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$$

where each $\phi_i(x)$ is an irreducible polynomial, and $\deg(\phi_i(x)) \geq \deg(\phi_{i-1}(x))$ for each $i \in \{2, 3, \cdots, \mathcal{N}\}$.
Step 3: Let's say that, for first $l \geq 0$ polynomials of $f(x)$, $\phi_i(x) = x$. Then, report $d = \max_{1 \leq i \leq l} \{n_i\}$ as the depth of the CA.
Step 4: For each $i \in \{l + 1, l + 2, \cdots, \mathcal{N}\}$, repeat Step 5 to Step 7.
Step 5: Find the primary cycle structure $\mu_k(k)$ for $\phi_i(x)$, where k is the least possible value such that $\phi_i(x)$ divides $(x^k + 1)$ and $\mu_k = \frac{2^{r_i} - 1}{k}$ with $r_i = \deg(\phi_i(x))$.
Step 6: Find $\mu_{2^j \cdot k}(2^j \cdot k)$ for each $j \in \{1, 2, \cdots, \lceil \log_2(n_i) \rceil\}$, where $\mu_{2^j \cdot k}$ is calculated using Relation 7.6.
Step 7: Accumulate the results of Step 5 and Step 6 to get elementary cycle structure of $\phi_i(x)^{n_i}$. Name it as $CS_{\phi_i(x)^{n_i}}$.
Step 8: Report the CS, obtained either by cross product as $CS_{\phi_{l+1}(x)^{n_{l+1}}} \times CS_{\phi_{l+2}(x)^{n_{l+2}}} \times \cdots \times CS_{\phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}}$, or by executing Step A and Step B of Sect. 7.2.4.

7.3 Synthesis of Linear Cellular Automata

The synthesis problem is to find a linear CA (T matrix) for a specified cycle structure. The theories presented in Sect. 7.2 are the basis of this synthesis scheme.

A given linear cycle structure (Definition 49) can be decomposed into a number of Elementary Family Cycle Structures (EFCSs). Each EFCS can further be decomposed into Elementary Cycle Structures (ECSs) as a reverse process of getting EFCS from ECSs. Then the elementary divisor can be derived corresponding to each ECS that can result in a *component* linear CA. The final CA can be obtained by combining the component CAs. The following example illustrates this synthesis scheme.

Example 7.11 Let consider the synthesis of a linear CA having the cycle structure CS = [4(1), 2(2), 4(3), 2(6)] and depth = 2. The given CS can be expressed as a cross product of two EFCSs. That is,

$$CS = [1(1), 3(1), 2(2)] \times [1(1), 1(3)]$$

From these two EFCSs, three ECSs can be derived. While representing the ECSs in triplet form, the

$$CS = (1, 1)^1 \times (1, 1)^2 \times (1, 3)^1$$

From these ECSs, the corresponding elementary divisors $(x + 1)$, $(x + 1)^2$, $(x^2 + x + 1)$ are obtained. If the depth 2 is considered, then the elementary divisor x^2 is to be added. Hence, the characteristic polynomial of the desired linear CA in elementary divisor form is

$$x^2 \cdot (x + 1) \cdot (x + 1)^2 \cdot (x^2 + x + 1)$$

Now, the component CA for each of the elementary divisors are to be identified. The component CAs are represented by the submatrices of the final T matrix. The T matrix of the linear CA and the submatrices in block diagonal form are given below. The individual submatrix is marked within braces.

$$T = \begin{pmatrix} \lceil 0 & 0 \rceil & 0 & 0 & 0 & 0 & 0 \\ \lfloor 1 & 0 \rfloor & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & [1] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lceil 1 & 1 \rceil & 0 & 0 \\ 0 & 0 & 0 & \lfloor 0 & 1 \rfloor & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lceil 1 & 1 \rceil \\ 0 & 0 & 0 & 0 & 0 & \lfloor 1 & 0 \rfloor \end{pmatrix}$$

7.3.1 Decomposing a Cycle Structure into EFCSs

The objective of this decomposition is to express a given cycle structure (CS) into the following form

$$CS = [EFCS_1] \times [EFCS_2] \times \cdots \times [EFCS_{\tilde{\mathcal{N}}}]$$

where $\tilde{\mathcal{N}}$ is the number of elementary family cycle structures (EFCSs) present in the CS. The above expression can be obtained recursively–first, extract the first EFCS as CS_1 so that $CS = CS_1 \times CS_2$ for an appropriate CS_2, and then repeat the first step considering CS_2 as CS until $CS_2 = [1(1)]$.

Generation of CS_1: Since the input cycle structure is linear, the smallest primary cycle is same in CS and CS_1. Hence, the cycle family with the smallest primary cycle of the CS is extracted and declared as CS_1. The following example illustrates the generation of CS_1.

Example 7.12 Let CS = [4(1), 2(2), 4(3), 12(5), 2(6), 6(10)], 12(15),6(30)]. It can also be represented as CS = [1(1), [3(1), 2(2)], [4(3), 2(6)], [12(5), 6(10)], [12(15), 6(30)]]. Then CS_1 consists of 1-cycle family, as 1 is the smallest cycle length in CS. Hence, CS_1 = [1(1), 3(1), 2(2)].

Generation of CS_2: To get CS_1 and CS_2 from the CS, the PCFs constituting the CS are arranged in ascending order of k_i. The k_1-cycle family has already been treated as CS_1. So, k_2-cycle family is to be included in CS_2. From the remaining cyclic components of the CS, the components which can be obtained as the cross product of CS_1 and k_2-cycle family are to be removed. Suppose, k_1 and k_2 are the lengths of the two cycles from CS_1 and k_2-cycle family, and $k_3 = \text{lcm}(k_1, k_2)$. Then, cycle(s) of length k_3 is to be removed from the CS. Here, two cases may arise.

- **Case 1 :** $k_3 = \text{lcm}(k_1, k_2) > k_2$
- **Case 2 :** $k_3 = \text{lcm}(k_1, k_2) = k_2$

Case 1 : The methodology to address this case, when $\text{lcm}(k_1, k_2) > k_2$, is illustrated through the following example.

Example 7.13 Consider the CS = [1(1), [1(3)], [3(5)], [3(15)]]. Then, k_1 = 3 and CS_1 = [1(1), 1(3)]. The remaining cycles in CS have the length k_i = 5, 15. For the first cycle of length k_2 = 5, k_1 is not a factor of k_2. So, k_2-cycle family is to be part of CS_2. So, at this stage, CS_2 = [1(1), 3(5)]. Now, $k_3 = \text{lcm}(k_1, k_a)$ = 15, and $\mu_{15} = \mu_3 \cdot \mu_5 \cdot \gcd(3, 5) = 3$. Hence, 3(15) is to be dropped from CS, where CS_2 = [1(1), 3(5)]. Since there is no remaining element in CS, CS_2 = [1(1), 3(5)].

The following actions, as illustrated in Example 7.13, are taken to handle *Case 1*.

- Since the k_2-cycle family is not arisen as a result of cross product, the entire cycle family (of k_2) is the member of CS_2.

- The components of k_3-cycle family are obtained from Relation 7.17.
- This k_3-cycle family, produced as a result of the cross product of k_2- and k_1-cycle families, are removed from CS.

Case 2: In this case, the cross product of k_1-cycle family and k_2-cycle family yields a cycle family with primary cycle length k_2. So, instead of adding the entire cycle family to CS_2 from CS, as in Case 1, the components of k_2 cycle family are divided into two parts (say γ and δ) – (i) the first part, denoted as $\gamma(CS_2)$, is added to CS_2 and (ii) the remaining part $\delta(CS_2)$ is resulted from the cross product–that is, $\delta(CS_2) = CS_1 \times \gamma(CS_2)$. The component $\gamma(CS_2)$ is computed by reframing Relation 7.21 as

$$\gamma(CS_2) = \frac{\mu_{2^j \cdot k_2}(CS) - [\mu_{2^j \cdot k_1}(CS_1)] \cdot [S_{2^{j-1} \cdot k_2}(CS_2)] \cdot \frac{k_1}{k_2}}{[S_{2^j \cdot k_1}(CS_1)] + 1} \tag{7.23}$$

where

- $\mu_{2^j \cdot k_2}(CS)$ is the cyclic component corresponding to k_2-cycle family in CS. It is marked as $\tilde{\mu}_{2^j \cdot k_3}$ in Relation 7.21.
- $\mu_{2^j \cdot k_1}(CS_1)$ is the cyclic component corresponding to k_1-cycle family in CS_1.
- $S_{2^j \cdot k_1}(CS_1)$ and $S_{2^{j-1} \cdot k_2}(CS_2)$ are the sum of configurations, as per Relation 7.18, covered by CS_1 and CS_2 respectively.

The following example illustrates the use of Relation 7.23 to extract the cyclic components.

Example 7.14 Let the cycle structure CS = [1(1), [3(1), 2(2)], [4(3), 2(6)], [12(5), 6(10)], [12(15), 6(30)]] (Example 7.10). Here $k_1 = 1$ (the smallest cycle length). Therefore, CS_1 = [1(1), 3(1), 2(2)], and so, $\mu_1(CS_1) = 3 + 1 = 4$, $S_{2^j \cdot k_1}(CS_1) = 3 \times 1 = 3$ (for $j = 0, k_1 = 1$) and for $j = 1, k_1 = 1$, $S_{2^j \cdot k_1}(CS_1) = 3 \times 1 + 2 \times 2 = 7$. After the subtraction of CS_1 from CS, CS = [[4(3), 2(6)], [12(5), 6(10)], [12(15), 6(30)]]. The remaining primary cycles in CS are $k_i = 3, 5, 15$, where for each k_i, lcm$(k_i, 1) = k_i$. For $k_2 = 3$, $\mu_3(CS) = 4$ whereby, Relation 7.23 leads to

$\frac{\tilde{\mu}_3(CS) - \mu_1(CS_1) \cdot S_0(CS_2) \cdot \frac{1}{3}}{3+1} = \frac{4 - 4 \times 0 \times \frac{1}{3}}{3+1} = 1$

For $\mu_6(CS) = 2$, $\frac{\tilde{\mu}_6(CS) - \mu_2 \cdot S_3 \cdot \frac{1}{3}}{(7+1} = \frac{2 - 2 \times 3 \times \frac{1}{3}}{7+1} = 0$.

That is, there is 1 cycle of length 3 and no cycle of length 6. Similarly, for $k_2 = 5$, there are 3 cycles of length 5 and no cycle of length 10. For $k_2 = 15$, there are 3 cycles of length 15 and no cycle of length 30.
Therefore, CS_2 = [1(1), 1(3), 3(5), 3(15)].

Once the set of elementary family cycle structures (EFCSs) is obtained, the next task is to generate the elementary cycle structures (ECSs) set from each EFCS. Each ECS corresponds to an elementary divisor (ED). In the subsequent discussions, the ECS and ED are used interchangeably.

7.3.2 Generation of ECS from EFCS

This step accepts an $EFCS$ of k-cycle family as input and generates the corresponding $ECSs$ in triplet form. That is,

$$EFCS = (\mu_k, k)^{n_1} \times (\mu_k, k)^{n_2} \times \cdots \times (\mu_k, k)^{n_{\mathcal{N}}}$$

In order to implement this step, two tasks are to be executed–(i) Finding of μ_k, and (ii) Computation of $\mathcal{N}$ and $n_1, n_2, \cdots, n_{\mathcal{N}}$.

Let consider the EFCS = [1(1), 85(3), 3688(6), 5591040(12), 4581088288(24)]. The corresponding ECS = [$(1, 3)^1$, $(1, 3)^4$, $(1, 3)^7$, $(1, 3)^8$]. That is, in this step, the value of μ_k ('1' in this example), the total number of ECSs $\mathcal{N}$ (= 4) and the value of each n_i (here, they are 1, 4, 7 and 8) are to be determined.

Finding μ_k: The μ_k is the number of cyclic components with cycle length k generated by an irreducible polynomial $\phi(x)$ and $\mu_k = (2^l - 1)/k$, where l is the smallest integer for which k divides $2^l - 1$. Here, l becomes the degree of $\phi(x)$.

Theorem 7.6 *The polynomial $\phi(x)$ generating the cycle structure [1(1), $\mu_k(k)$] is irreducible iff $\mu_k \times k = 2^l - 1$, where l is the smallest integer for which k divides $2^l - 1$. Here, l is the degree of $\phi(x)$.*

Proof Assume that $\mu_k \times k = 2^l - 1$, where l is the smallest integer for which k divides $2^l - 1$. To prove $\phi(x)$ as irreducible by contradiction, consider that $\phi(x)$ is reducible and $\phi(x) = \phi_1(x).\phi_2(x)$. Since the resultant cycle structure for $\phi(x)$ is a primary cycle structure, each of $\phi_1(x)$ and $\phi_2(x)$ corresponds to a primary cycle of length k. So, cycle structures for $\phi_1(x)$ and $\phi_2(x)$ are to be [1(1), $\mu_1(k)$] and [1(1), $\mu_2(k)$] respectively. Hence, $\mu_1 \times k = 2^{l_1} - 1$, where $l_1 < l$. But this is not possible as l is the least value. Hence, $\phi(x)$ is irreducible. Then, by definition, l becomes the degree of $\phi(x)$. On the other hand, if $\phi(x)$ is irreducible, then also by definition, $\mu_k \times k = 2^l - 1$ with l as the smallest integer which is the degree of $\phi(x)$. Hence the proof. □

Example 7.15 For $k = 3$, $\mu_k = \frac{2^2-1}{3} = 1$, as 2 is the least value of l for which k divides $(2^l - 1)$ and for $k = 9$, $\mu_k = \frac{2^6-1}{9} = 7$, where 6 is the least value of l for which k divides $2^l - 1$.

Computation of $\mathcal{N}$ and $n_1, n_2, \cdots, n_{\mathcal{N}}$: This involves (1) the computation of the total number of jth factor set $\tilde{N}(F_j)$ (Relation 7.12), and (2) identification of the ECS covering the cycle structure corresponding to $\tilde{N}(F_j)$ factors and also the power of each ECS.

The relationship between the primary cycle family and the underlying elementary divisors is in Sect. 7.2.3. Relation 7.13 is reproduced below.

$$\mu_{2^j.k} = \frac{2^{r\sum_{\mathcal{J}=0}^{j} \tilde{N}(F_{\mathcal{J}})} - 2^{r\sum_{\mathcal{J}=0}^{j-1} \tilde{N}(F_{\mathcal{J}})}}{2^j \cdot k}.$$

Therefore, from the relation,

$$\tilde{N}(F_j) = \frac{1}{r}\log_2(\mu_{2^j.k} \times k \times 2^{(j-r\cdot\sum_{\mathcal{J}=0}^{j-1}\tilde{N}(F_{\mathcal{J}}))} + 1) \tag{7.24}$$

where $\mu_{2^j.k}$ is the given cyclic component from which the cardinality of F_j (the jth factor set) has to be derived, and r is the degree of the polynomial $\phi(x)$ obtained from Relation 7.4.

Example 7.16 Consider CS = [1(1), 85(3), 3688(6), 5591040(12), 4581088288(24)]. For this case, $r = 2$. Therefore, from Relation 7.24
$\tilde{N}(F_0) = \frac{1}{2}\log_2(85 \times 3 \times 2^{0-0} + 1) = \frac{\log_2(256)}{2} = 4$,
$\tilde{N}(F_1) = \frac{1}{2}\log_2(3688 \times 3 \times 2^{1-4\times 2} + 1) = \frac{\log_2(64)}{2} = 3$,
$\tilde{N}(F_2) = \frac{1}{2}\log_2(5591040 \times 3 \times 2^{2-(4+3)\times 2} + 1) = \frac{\log_2(4096)}{2} = 6$,
$\tilde{N}(F_3) = \frac{1}{2}\log_2(4581088288 \times 3 \times 2^{3-(4+3+6)\times 2} + 1) = \frac{\log_2(16384)}{2} = 7$.

Subsequent discussions report the distribution of the members of F_j in an ECS. As defined (Definition 51), at most 2^{j-1} factors against an ECS can be the members of F_j. So, a parameter $N(ECS(j))$ is introduced to denote the least number of ECSs required to accommodate $\tilde{N}(F_j)$. This can be calculated by assuming that each ECS contributes the maximum number of members. However, if F_j is the last factor set corresponding to an ECS, then it can have less than 2^{j-1} members.

Definition 56 $N(ECS(j))$ is the number of elementary divisors/ECSs with non-empty jth factor set.

Example 7.17 Let $f(x) = \phi(x)^1 \cdot \phi(x)^4 \cdot \phi(x)^7 \cdot \phi(x)^8$. Then for $j = 3$, the 3^{rd} factor set (Definition 51), that is, the factors with power 5 ($2^{3-1}+1$) to 8 (2^3) are present in the two elementary divisors $\phi(x)^7$ and $\phi(x)^8$. The last elementary divisor has all the possible factors, namely the factors with power 5, 6, 7 and 8, in its 3^{rd} factor set, whereas, the second last has only the first three factors. Hence, $N(ECS(j)) = 2$.

The following lemma formalizes the above discussions, where $\tilde{N}(F_j)$ is the cardinality of F_j present in all the constituent ECSs comprising the EFCS.

Lemma 14 *The relation between the number of elementary cycle structures* $N(ECS(j))$ *and* $\tilde{N}(F_j)$ *is*

$$N(ECS(j)) = \begin{cases} \lceil \frac{\tilde{N}(F_j)}{2^{j-1}} \rceil & j \geq 1 \\ \tilde{N}(F_j) & j = 0 \end{cases} \tag{7.25}$$

where the j*th factor set of the last ECS contains* L_j *factors with*

$$L_j = \begin{cases} \tilde{N}(F_j) \mod 2^{j-1} & \tilde{N}(F_j) \mod 2^{j-1} > 0 \\ 2^{j-1} & \tilde{N}(F_j) \mod 2^{j-1} = 0 \end{cases} \tag{7.26}$$

Proof The maximum number of the jth factor that can be present in an elementary divisor is 2^{j-1} (Definition 51). Therefore, the number of elementary divisors required to accommodate $\tilde{N}(F_j)$ is given by Relation 7.25, where $\tilde{N}(F_0)$ represents the number of elementary divisors.

The relation is obtained from simply dividing $\tilde{N}(F_j)$ by 2^{j-1}. The integer remainder provides the cardinality of the jth factor set in the last ECS. If the remainder is zero, the last ECS has the full 2^{j-1} factors. Therefore, the remainder L_j can be expressed by Relation 7.26. □

Example 7.18 Let the given parameters $\tilde{N}(F_0) = 4$, $\tilde{N}(F_1) = 3$, $\tilde{N}(F_2) = 6$ and $\tilde{N}(F_3) = 7$ (Example 7.16). The following are the $N(ECS(j))$s from Relation 7.25.
$N(ECS(0)) = 4$—that is, there are 4 irreducible polynomials. Hence, the number of $ECS = 4$.
$N(ECS(1)) = 3$—that is, 3 ECSs, out of 4, have the factors ranging from $(2^{1-1} + 1)$ to 2^1.
$N(ECS(2)) = 3$—that is, 3 ECSs have the factors ranging from $(2^{2-1} + 1)$ to 2^2.
$N(ECS(3)) = 2$—that is, 2 ECSs have the factors ranging from $(2^{3-1} + 1)$ to 2^3.
It can be observed that $N(ECS(a)) \geq N(ECS(b))$, when $a < b$. This is because if the lower factor is not present in an ECS, then the higher factors cannot be present.

On computation of N(ECS(j)), the constituent factors of each ECS get specified. Therefore, the next task is to find the powers $n_1, n_2, \cdots n_{\mathcal{N}}$ of the elementary divisors/ECSs. The following lemma sets the power of each ECS by identifying the highest $N(ECS(j))$ of the ECS.

Lemma 15 *The number of ECSs (say $\mathcal{N}_j$) with the range of powers from $2^{j-1} + 1$ to 2^j is*

$$\mathcal{N}_j = N(ECS(j)) - N(ECS(j+1)) \tag{7.27}$$

If $\mathcal{N}_j > 0$ and $P_{j,i}$ is the power of ith such ECS, $i \in \{0, 1, \cdots, \mathcal{N}_j\}$, then

$$P_{j,i} = \begin{cases} 1 & \text{if } \mathcal{N}_j > 0 \text{ and } j = 0 \\ 2^j & \text{if } \mathcal{N}_j > 0 \text{ and } j > 0 \text{ and } i < \mathcal{N}_j \\ 2^{j-1} + L_j & \text{if } \mathcal{N}_j > 0 \text{ and } j > 0 \text{ and } i = \mathcal{N}_j \end{cases} \tag{7.28}$$

where the value of L_j is derived from Relation 7.26.

Proof The number of EDs/ECSs having jth factor set is given by N(ECS(j)). Whereas, a fewer number of ECSs, given by N(ECS($j+1$)), contain $(j+1)$th factor set. Therefore, $(N(ECS(j))$–$N(ECS(j+1)))$ denotes the ECS having the highest factor set j—that is, Relation 7.27 determines $\mathcal{N}_j$.

As per Lemma 14, all but one of the $\mathcal{N}_j$ elementary divisors contribute 2^{j-1} factors in the jth factor set. So, the power of each of these $(\mathcal{N}_j - 1)$ elementary divisors is $P_{j,i} = 2^j$, $i \in \{0, 1, \cdots, \mathcal{N}_{j-1}\}$ (derived through Relation 7.26). The rest ED contributes L_j number of factors in the jth factor set (Relation 7.26). Therefore, the rest ED has the power $2^{j-1} + L_j$. For $j = 0$, each of the $\mathcal{N}_0$ elementary divisors/ECSs has power 1. However, if $\mathcal{N}_j = 0$, there is no question of power. Hence the proof. □

The following example summarizes the above discussions on generation of ECS in triplet form. To impart completeness, the input is the CS of Example 7.16 and the intermediate step producing $N(ECS(j))$, illustrated in Example 7.18, are reproduced in this example.

Example 7.19 Let the cycle structure $EFCS = [1(1), 85(3), 3688(6), 5591040(12), 4581088288(24)]$ for which ECS is to be identified (Example 7.8). The following table shows step by step calculation of each of the parameters to arrive at the final result.

j	$\tilde{N}(F_j)$	$N(ECS(j))$	$\mathcal{N}_j$	$P_{j,i}$
0	4	4	1	1
1	3	3	0	–
2	6	3	1	4
3	7	2	2	7, 8

Since N(ECS(3)) is 2 and N(ECS(4)) = 0 (as there is no N(ECS(4))), there are 2 [N(ECS(3))–N(ECS(4))] ECSs with power in between $2^{3-1}+1$ and 2^3. Out of which, one has power $P_{3,0} = 8$ (2^3) (Relation 7.28) and the other has $P_{3,1} = 2^{3-1} + L_3 = 2^2 + N(F_3) \mod 4 = 4 + 7 \mod 4 = 7$. Therefore, the EFCS can be expressed as $(1,3)^1 \times (1,3)^4 \times (1,3)^7 \times (1,3)^8$.

7.3.3 *Finding Elementary Divisors from an ECS*

An elementary divisor $\phi_i(x)^{n_i}$, as noted in Definition 52, has the ECS in triplet form $(\mu_k, k)^{n_i}$, where the irreducible polynomial $\phi_i(x)$ corresponds to the cycle structure $[1(1), \mu_k(k)]$. This subsection reports the process of generation of the polynomials from the given cycle structure $[1(1), \mu_k(k)]$, where $\mu_k \times k = 2^r - 1$ and r is the degree of $\phi_i(x)$. Now, if a database of r-degree irreducible polynomials with their possible cycle structure is available, then the corresponding polynomial can be found by matching the given cycle structure. A database of primitive polynomials may also be used. Such a database is much smaller than that of the irreducible polynomials.

The cycle structure of a CA with characteristic polynomial as a primitive polynomial of degree r is $[1(1), 1(2^r - 1)]$. Whereas, it is $[1(1), \mu_k(k)]$ if the characteristic polynomial is the irreducible polynomial of degree r and $\mu_k \times k = 2^r - 1$. The following theorem illustrates the method of finding an irreducible polynomial of degree r from an r-degree primitive polynomial. The primitive polynomial is represented by a linear operator T in Rational Canonical Form.

Theorem 7.7 *If T is the characteristic matrix of a primitive polynomial that corresponds to the cycle structure $CS = [1(1), 1(\tilde{k})]$, then the characteristic polynomial of T^{μ_k} corresponds to the cycle structure $[1(1), \mu_k(k)]$, where $\mu_k \times k = \tilde{k}$.*

Proof If the cycle structure of T is $[1(1), 1(\tilde{k})]$, then $T^{\tilde{k}} = I$, that is, $T^{\mu_k \times k} = I$. Therefore, the matrix T^{μ_k} has the cycle of length k or sub-multiple of k. Let k_1

be a factor of k and for state x, $(T^{\mu_k})^{k_1} \cdot x = x$. Therefore, $T^{(\mu_k \times k_1)} \cdot x = x$. But this is not possible. Hence, T^{μ_k} has the cycle structure $[1(1), \mu_k(k)]$. Consequently, characteristic polynomial $f(x) = det(T^{\mu_k} + Ix)$ corresponds to the cycle structure $[1(1), \mu_k(k)]$. □

Following example illustrates the synthesis of an irreducible polynomial as elementary divisor corresponds to the cycle structure $[(1), \mu_k(k)]$ from a primitive polynomial of the same degree.

Example 7.20 To generate the elementary divisors for ECS = $(3, 5)^2$ (Example 7.3), the following steps are executed.

Step 1: Find $r = (\log_2(\mu_k \times k + 1))$ –that is, $r = 4$.
Step 2: Find a primitive polynomial of degree 4 in $GF(2)$. Here it is assumed as $x^4 + x + 1$.
Step 3: Construct the T matrix from the polynomial using Rational Canonical Form.

So, $$T = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Step 4: Compute $$T^3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Step 5: Find the characteristic polynomial of T^{μ_k} to obtain $\phi(x) = det[T^3 + Ix] = x^4 + x^3 + x^2 + x + 1$ which is an irreducible polynomial. Hence the corresponding elementary divisor is $x^4 + x^3 + x^2 + x + 1$.

Once the elementary divisors are obtained, the linear operator T of the linear CA can be generated.

7.3.4 CA Synthesis from Elementary Divisors

The synthesis of a linear cellular automaton from the elementary divisors is accomplished by the following tasks.

1. Task 1: Convert each elementary divisor $\phi_i(x)^{n_i}$ into a characteristic matrix $T(\phi_i(x)^{n_i})$ for the ith component CA.
2. Task 2: Merge $T(\phi_i(x)^{n_i})$s to get T –that is, the characteristic matrix of the linear CA.

Task 1: Under the assumption of 3-neighborhood dependency of the cells of a linear CA, a tridiagonal matrix is formed corresponding to an elementary divisor $\phi_i(x)^{n_i}$. Then the following two cases may arise.

Case 1—*The elementary divisor representing a cycle structure:* For this case, the underlying irreducible polynomial $\phi(x)$ is converted into a 3-neighborhood T matrix. The scheme discussed in Sect. 6.3 is used to get the T. The formation of $T(\phi_i(x)^{n_i})$ with the characteristic and minimal polynomial as $\phi_i(x)^{n_i}$ is next undertaken employing the result of the following theorem. This result can be found in any standard text book on Matrix Algebra (see, for example, [1]).

Theorem 7.8 *If $d_{n-1}(x)$ is the greatest common divisor of all the minors of order $(n-1)$ in the matrix $(T+Ix)$ of order n and $\phi(x)$ is the characteristic polynomial of T, then the minimal polynomial of T is $\frac{\phi(x)}{d_{n-1}(x)}$.*

The next theorem provides the methodology on how to derive $T(\phi_i(x)^{n_i})$ corresponding to $\phi_i(x)^{n_i}$.

Theorem 7.9 *If $T(\phi_i(x))$ is the tridiagonal matrix corresponding to the characteristic polynomial $\phi_i(x)$, then $T(\phi_i(x)^{n_i})$ has the characteristic and minimal polynomial both as $\phi_i(x)^{n_i}$, where*

$$T(\phi_i(x)^{n_i}) = \begin{bmatrix} T(\phi_i(x)) & 1 & 0 & \cdots \\ 0 & T(\phi_i(x)) & 1 & 0 \cdots \\ \cdot & & & \\ \cdot & & & \\ 0 & \cdots & 0 & T(\phi_i(x)) \end{bmatrix}$$

Proof The proof is presented for $n_i = 2$. It can be easily generalized for $n_i > 2$. The characteristic polynomial of

$$T(\phi_i(x))^2 = \begin{bmatrix} T(\phi_i(x)) & 1 \\ 0 & T(\phi_i(x)) \end{bmatrix}$$

is

$$det \begin{bmatrix} T(\phi_i(x)) + Ix & 1 \\ 0 & T(\phi_i(x)) + Ix \end{bmatrix} = \phi(x) \cdot \phi(x) - 1 \cdot 0 = \phi(x)^2.$$

The minimal polynomial of $T(\phi_i(x)^2)$ can be found from Theorem 7.8. The elements of the super-diagonal and sub-diagonal of $T(\phi_i(x))$ are non-zero, when $\phi_i(x)$ is irreducible (Theorem 6.1). Placement of 1 in the junction provides a non-zero minor (d_i), whose value can be shown as 1. Therefore, gcd of all the minors $d(x) = \gcd(d_1(x), \cdots, d_n(x)) = 1$, since $d_i = 1$. Hence the minimal polynomial is $\frac{\phi(x)^2}{1} = \phi(x)^2$. □

Example 7.21 Consider $(x^2+x+1)^2$ as an elementary divisor. The T matrix for x^2+x+1 is $T(x^2+x+1) = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. Then $T(x^2+x+1)^2 = \begin{bmatrix} \lceil 1 & 1 \rceil & 0 & 0 \\ \lfloor 1 & 0 \rfloor & 1 & 0 \\ 0 & 0 & \lceil 1 & 1 \rceil \\ 0 & 0 & \lfloor 1 & 0 \rfloor \end{bmatrix}$.
The $T(x^2+x+1)^2$ has the characteristic and minimal polynomial as $(x^2+x+1)^2$.

Case 2—*The elementary divisor representing the depth d*: The elementary divisor which represents the depth d of a CA is given by x^d (follows from Relation 7.3). If lower diagonal of a matrix (say D) of dimension d is assigned 1 and the rest elements of D are 0, then the characteristic polynomial of D is x^d.

Example 7.22 Let $d = 3$. Then the $[D]_{3\times 3}$ is $D = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. The characteristic polynomial of D is $\det(D + Ix) = x^d$.

Once all the $T_i s$ for elementary divisors are obtained, the next task is to merge them to form the final T matrix of the linear CA.

Task 2: This operation involves the construction of the final T with characteristic polynomial $f(x)$, where $f(x) = \phi_1(x)^{n_1} \cdots \phi_i(x)^{n_i} \cdots \phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$. The T for the characteristic polynomial $f(x)$ can be formed following the next theorem. This result can be found in a standard text book on Matrix Algebra (see, for example, [1]).

Theorem 7.10 *If the characteristic matrices* $T(\phi_1(x)^{n_1})$, $T(\phi_2(x)^{n_2})$, $\cdots$, $T(\phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}})$, *with characteristic polynomials* $\phi_1(x)^{n_1}$, $\phi_2(x)^{n_2}$, $\cdots$, $\phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$ *respectively are arranged in block diagonal form, then the resultant matrix*

$$T = \begin{bmatrix} T(\phi_1(x)^{n_1}) & 0 & \cdots \\ 0 & T(\phi_2(x)^{n_2}) & 0 \cdots \\ \cdot & & \\ \cdot & & \\ \cdot & & \\ 0 & \cdots & T(\phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}) \end{bmatrix}$$

has the characteristic polynomial $f(x) = \phi_1(x)^{n_1} \cdots \phi_i(x)^{n_i} \cdots \phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$

Example 7.23 Let us consider the characteristic polynomial $f(x) = x^2 \cdot (x+1) \cdot (x+1)^2 \cdot (x^2 + x + 1)$ (Example 7.2). The matrices corresponding to x^2, $(x+1)$, $(x+1)^2$ and $(x^2 + x + 1)$ are $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ respectively.

Therefore, the final T, by merging these matrices, is

$$T = \begin{bmatrix} \lceil 0\ 0 \rceil & 0 & 0 & 0 & 0 & 0 \\ \lfloor 1\ 0 \rfloor & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & [1] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lceil 1\ 1 \rceil & 0 & 0 \\ 0 & 0 & 0 & \lfloor 0\ 1 \rfloor & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lceil 1\ 1 \rceil \\ 0 & 0 & 0 & 0 & 0 & \lfloor 1\ 0 \rfloor \end{bmatrix}$$

Algorithm 8 describes the complete process of synthesizing the linear CA (T) from a given linear cycle structure.

Algorithm 8: Synthesize linear CA

Input : Linear cycle structure (CS) and depth d.
Output: Characteristic matrix T of a linear CA.

Step 1: Arrange the given cycle structure as $CS = [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i} (2^j \cdot k_i)]$ such that the cycle families appear in ascending order.

Step 2: Identify $\sum_{j=0}^{m_{k_1}} \mu_{2^j \cdot k_1} (2^j \cdot k_1)$ in CS, and consider $CS_1 = [1(1) + \sum_{j=0}^{m_{k_1}} \mu_{2^j \cdot k_1} (2^j \cdot k_1)]$. Here, k_1 is the smallest odd-length cycle in CS.

Step 3: Compute CS_2 by the following lines of this step such that $CS = CS_1 \times CS_2$.
Consider $CS_2 = [1(1)]$ and $CS = CS - CS_1$.

foreach *cycle family of CS in ascending order* **do**
  Let k_2 be the smallest primary cycle in CS, and the family is $\sum_{j=0}^{m_{k_2}} \mu_{2^j \cdot k_2} (2^j \cdot k_2)$.
  if $\text{lcm}(k_1, k_2) = k_2$ **then**
    foreach $j \in \{0, 1, \cdots, m_{k_2}\}$ **do**
      Obtain $\gamma(CS_2)$ from Relation 7.23
      $CS_2 \leftarrow CS_2 + \gamma(CS_2)$
      $CS \leftarrow CS - (\gamma(CS_2) + \delta(CS_2))$, where $\delta(CS_2) = CS_1 \times \gamma(CS_2)$
    end
  end
  else
    foreach $j \in \{0, 1, \cdots, m_{k_2}\}$ **do**
      $CS_2 \leftarrow CS_2 + \mu_{2^j \cdot k_2} (2^j \cdot k_2)$
      Obtain $\tilde{\mu}_{2^j \cdot k_3}$ from Relation 7.17 ($k_3 = \text{lcm}(k_1, k_2)$)
      $CS \leftarrow CS - \tilde{\mu}_{2^j \cdot k_3} (2^j \cdot k_3)$
      $CS \leftarrow CS - \mu_{2^j \cdot k_2} (2^j \cdot k_2)$
    end
  end
end

Step 4: CS_1 is an EFCS. Repeat Step 2 and Step 3, assuming $CS = CS_2$ until $CS = CS_1$ to identify all EFCSs. Suppose $\tilde{\mathcal{N}}$ EFCSs are identified, which are $EFCS_1, EFCS_2, \cdots, EFCS_{\tilde{\mathcal{N}}}$.

Step 5: For each of the $\tilde{\mathcal{N}}$ EFCSs, find ECSs, using the following lines, in triplet format:
$EFCS = (\mu_k, k)^{n1} \times (\mu_k, k)^{n2} \times \cdots \times (\mu_k, k)^{n\mathcal{N}}$, where k is the primary cycle's length.
Find $\mu_k = (2^l - 1)/k$ such that l is the smallest integer for which k divides $2^l - 1$.

foreach $j \in \{0, 1, \cdots, \lceil \log_2(\max_{1 \le i \le \mathcal{N}} \{n_i\}) \rceil\}$ **do**
  Find $\mathcal{N}_j$ by Relation 7.27
  if $\mathcal{N}_j > 0$ **then**
    foreach $i \in \{0, 1, \cdots, \mathcal{N}_j\}$ **do**
      Find $P_{j,i}$ by Relation 7.28
      Consider $(\mu_k, k)^{P_{j,i}}$ as an ECS.
    end
  end
end

Step 6: For each ECS $(\mu_k, k)^{n_i}$, identified above, find elementary divisor $\phi_i(x)^{n_i}$ by the following substeps.
Choose a primitive polynomial of degree $r = \log_2(\mu_k \times k + 1)$.
Construct a matrix (T) from the polynomial in Rational Canonical Form.
Find T^{μ_k}, for which $\phi_i(x) = \det(T^{\mu_k} + Ix)$

Step 7: For each elementary divisor $\phi_i(x)^{n_i}$,
Synthesize the matrix $T(\phi_i(x))$ corresponding to $\phi_i(x)$ using Algorithm 6.
Place $T(\phi_i(x))$ n_i times following Theorem 7.9 to form $T(\phi_i(x)^{n_i})$.

Step 8: If $d > 0$, form a D matrix of dimension d, with only lower diagonal elements as 1 and rest as 0.

Step 9: Place all the matrices, synthesized in Step 7 and Step 8, in Block Diagonal Form to get the final T matrix.

7.4 Cycle Structure of Complemented Cellular Automata

A complemented CA consists of linear and complemented rules, and is characterized by the characteristic matrix (T) and an inversion vector (F). The complemented CAs can show more variations in cycle structure than the linear CAs. Figure 7.2 shows an example of cycle structure which cannot be of any linear CA. A complemented CA has its linear counterpart, which can be obtained by replacing complemented rules with the corresponding linear rules (see Sect. 3.2). This section focuses on the variations of cycle structure of a complemented CA over its linear linear counterpart. Let represent a complemented CA by C' and its linear counterpart by C. So, C' and C have the same T matrix with $F \neq 0$ and $F = 0$ respectively.

To analyze the cycle structure of complemented CAs, the following issues are to be focused.

1. To check if a cycle of length k is present or not.
2. To identify the special class of complemented CAs having the same cycle structure as that of their linear counterpart.
3. To identify the class of complemented CAs having the cycle structure that is different from that of their linear counterpart.
4. To find the cycle structure of C'.

7.4.1 A Cycle of Length k

Cycle length of a linear CA is related to the *null space* of T. The null space of a matrix consists of all such vectors that are transformed to the all 0s vector when premultiplied by the matrix.

Theorem 7.11 *If a linear CA with a characteristic matrix T has a cycle of length k, then the cardinality of the null space of $(T^k + I)$ denotes the number of configurations forming the cycles of length k or submultiple of k.*

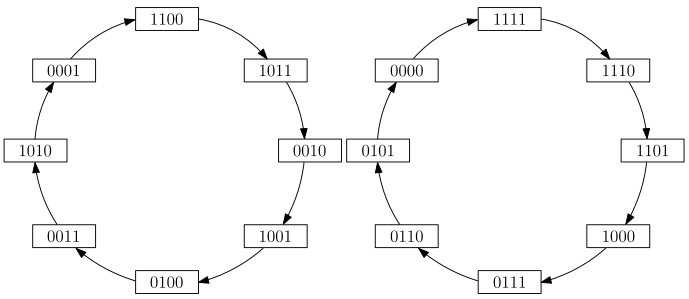


Fig. 7.2 Cycle structure of 4-cell complemented CA (153, 153, 153, 153)

Proof Suppose x is a configuration on a cycle of length k of the CA. Then, $T^k \cdot x = x$ (assuming x is presented as a column vector). Hence, $(T^k + I) \cdot x = 0$. This is also true if the cycle length is submultiple of k. So, the cardinality of the null space of $(T^k + I)$ is the total number of configurations that form the cycles of length k or submultiple of k. □

The following theorem is to determine whether a cycle of length k exists in a complemented CA.

Theorem 7.12 *A complemented CA (C') with characteristic matrix T and the inversion vector F contains a cycle of length k if*

$$rank([T^k + I]) = rank([T^k + I, \mathcal{F}]), \text{ where } \mathcal{F} = [I + T + T^2 + \cdots + T^{k-1}]F$$

Proof Let x be a configuration in a cycle of length k of C'. Hence,

$$x = T^k x + [I + T + T^2 + \cdots + T^{k-1}]F$$

That is,

$$[T^k + I]x = \mathcal{F}, \text{ where } \mathcal{F} = [I + T + T^2 + \cdots + T^{k-1}]F \tag{7.29}$$

If a cycle of length k exists in the C', then Relation 7.29 should be consistent. The condition for consistency is

$$rank([T^k + I]) = rank([T^k + I, \mathcal{F}]) \tag{7.30}$$

(here, $[T^k + I, \mathcal{F}]$ represents the augmented matrix). Hence the proof. □

7.4.2 Cycle Structure Identical to Linear Counterpart

Theorem 7.12 is utilized to explore a special class of C' that has the cycle structure identical to that of C, irrespective of the inversion vector F of C'. The following theorem defines such a class of C'.

Theorem 7.13 *The cycle structures of the CAs C' and C are identical if the characteristic polynomial of the T matrix of C' and C does not have the factor $(x + 1)$.*

Proof Let k be the length of a cycle of C with characteristic matrix T and characteristic polynomial $f(x)$. So, $f(x)$ has the factor that divides $x^k + 1$. Assume that $f(x)$ does not have a factor $(x + 1)$. As per Theorem 7.11, the number of vectors forming cycle of length k or submultiple of k is related to the cardinality of the null space of $\alpha_1 = (T^k + I)$.

Let $(x^k + 1) = g(x) \cdot \phi_c(x)$, where $\phi_c(x)$ is the highest factor of $f(x)$ that divides $(x^k + 1)$. Therefore, $g(x)$ and $f(x)$ do not have a common factor. Hence, for each

configuration c with $(T^k + I) \cdot c = 0$, there is a unique configuration c' such that $\phi_c(T) \cdot c' = 0$. Hence, the cardinality of null space of $\alpha_2 = \phi_c(T)$ is same as that of α_1.

Since $f(x)$ does not have a factor $(x + 1)$ and $x^k + 1 = (x + 1)(1 + x + x^2 + \cdots + x^{k-1})$, the null space of $\alpha_3 = [I + T + T^2 \cdots T^{k-1}]$ is same as of α_1 and α_2. Hence,

$$rank(T^k + I) = rank(T^k + I, I + T + T^2 + \cdots + T^{k-1}).$$

That is,

$$rank(T^k + I) = rank(T^k + I, \mathcal{F})$$

directly follows for any F.

Therefore, for any cycle of length l in C, there also exists a cycle of length l in C'. Since the number of vectors forming each cycle length is same for both (directly derived from cardinality of null space), the cycle structures of both the CAs are identical. Hence the proof. □

Example 7.24 Let consider a complemented CA C' with characteristic matrix

$$T = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

and an inversion vector $F \neq 0$. The characteristic polynomial of T is $(x^3 + x + 1)(x^2 + x + 1)$ and the cycle structure of the linear counterpart C is [1(1), 1(3), 1(7), 1(21)].

The complemented CA, as per Theorem 7.13, has the same cycle structure as that of the corresponding linear CA, irrespective of the value of F. Let consider a particular cycle of length 7. As per the theorem, $\alpha_1 = T^7 + I$, $\alpha_2 = T^3 + T + 1$ and $\alpha_3 = T^6 + T^5 + T^4 + T^3 + T^2 + T + I$, where

$$T^7 + I = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad T^3 + T + 1 = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ and}$$

$$T^6 + T^5 + T^4 + T^3 + T^2 + T + I = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

All the three matrices, having the third, fourth and fifth row as all 0s, have the same null space with cardinality 8. Therefore, rank(α_1) = rank(α_2,α_3) and rank(α_1) = rank(α_2, $\mathcal{F}$), where $\mathcal{F} = (I + T + T^2 + T^3 + T^4 + T^5 + T^6) \cdot F$. The F is a 5-dimensional inversion vector. Therefore, both the C' and C have the cycles of length 7. The cyclic components of cycle length 7 = $\frac{(8-1)}{7} = 1$.

The complete cycle structure of $C^{'}$ can be found as [1(1), 1(3), 1(7), 1(21)], which is same as that of C.

7.4.3 Cycle Structure Different from Linear Counterpart

From Theorem 7.13, it is obvious that the cycle structure of a complemented CA (C') can differ from its linear counterpart (C) only if the characteristic polynomial of T has the factor $(x + 1)$. The following terminologies and the theorems are important while identifying the role of the factor $(x + 1)$ in forming the cycle structure of C' and C.

- $C\phi(x)$ and $C'\phi(x)$ represent the linear and complemented CA respectively with characteristic polynomial $\phi(x)$.
- $C(x + 1)^n$: The linear CA, having characteristic matrix T and the characteristic and minimal polynomial $(x + 1)^n$, has the cycle structure

$$CS = [1(1) + \sum_{j=0}^{m} \mu_{2^j}(2^j)], \text{ where } m = \lceil \log_2(n) \rceil \tag{7.31}$$

Therefore,

$$rank\ (T + I) = n - 1, \qquad rank\ (T + I)^i = n - i. \tag{7.32}$$

- $[F^k]$: The set of inversion vectors which annihilates only $(x + 1)^k$ –that is,

$$(T + I)^k \cdot x = 0, \text{ where } x \in [F^k] \text{ and for any } k' < k, \ (T + I)^{k'} x \neq 0 \tag{7.33}$$

Based on the results of Theorem 7.12, the exact condition can be specified that is responsible for the difference in the cycle structure of $C(x + 1)^n$ and $C'(x + 1)^n$.

Theorem 7.14 *The complemented CA $C'(x + 1)^n$ with characteristic matrix T and an inversion vector F is having the cycle structure identical to its linear counterpart if*

$$rank(T + I) = rank(T + I, F) \tag{7.34}$$

Proof In order to test whether the cycle structure of $C(x+1)^n$ and $C'(x+1)^n$ are identical, the consistency of Relation 7.29 is checked for the presence of a cycle of length k in C'. The linear CA C has cycle(s) of length $k = 2^j$, $j \in \{0, 1, \cdots, m\}$, where $m = \lceil \log_2(n) \rceil$ (Relation 7.31). Hence, Relation 7.29 can be rewritten as

$$(T^{2^j} + I)x = (I + T + T^2 + \cdots + T^{2^j-1})F \tag{7.35}$$

The relation has to be consistent for all values of j to establish the equivalence of cycle structures of C and C'. Since $(T^{2^j} + I) = (T + I)^{2^j}$ and $(I + T + T^2 + \cdots + T^{2^j-1}) = (T + I)^{2^j-1}$, then

$$(T + I)^{2^j} x = (T + I)^{2^j-1} F \tag{7.36}$$

$$\Rightarrow (T + I)^{2^j-1}[(T + I)x] = (T + I)^{2^j-1} F \tag{7.37}$$

which implies that the consistency of

$$(T + I)x = F \tag{7.38}$$

has to be checked to determine the existence of cycles of length 2^j ($j \in \{0, 1, 2, \cdots, m\}$). Therefore, the consistency of the relation is determined by checking whether $rank(T + I) = rank(T + I, F)$. Hence the proof. □

The next theorem characterizes the vector F which causes the difference in the cycle structure between C and C'.

Theorem 7.15 *The cycle structure of $C'(x+1)^n$ differs from its linear counterpart $C(x+1)^n$ iff the inversion vector $F \in [F^n]$.*

Proof In order to prove this, the consistency of Relation 7.38 is checked for the inversion vector (1) $F \in [F^{n'}]$, where $n' < n$, and (2) $F \in [F^n]$. The proof establishes the fact that in the first case both C and C' have the same cycle structure while in the second case the cycle structure of C' differs from that of C. The checking of consistency for every cycle length $k = 2^j$, $j \in \{0, 1, 2, \cdots, m\}$ is to be done.
Case 1: Inversion vector $F \in [F^{n'}]$, where $n' < n$: It is to be shown that Relation 7.38 is consistent iff $x \in [F^{n'+1}]$.

If $x \in [F^{n'+1}]$, then from Relation 7.33,

$$(T + I)^{n'+1} \cdot x = 0 \Rightarrow (T + I)^{n'} \cdot [(T + I)x] = 0.$$

As per the definition of $[F^{n'+1}]$, the constituent vectors become zero only when premultiplied by $(T + I)^{n'+1}$ and its higher factors. Therefore, $(T + I) \cdot x \neq 0$ and there exists a vector y, where $(T + I)x = y$. Moreover, Relation 7.37 can be rewritten as

$$(T + I)^{n'} \cdot y = 0 \quad \Rightarrow \quad y \in [F^{n'}] \quad i.e. \quad (T + I)^{n''} \cdot y \neq 0 \quad n'' < n$$

Since rank of $(T + I)$ is $(n - 1)$, there exist a pair $x_1, x_2 \in [F^{n'+1}])$ such that

$$(T + I) \cdot x_1 = (T + I) \cdot x_2 = y$$

Hence, exploiting all such possible pairs, a set $Y = \{y = (T + I)x \mid x \in [F^{n'+1}]\}$ can be obtained with the cardinality of $\frac{|[F^{n'+1}]|}{2}$, where $|[F^{n'+1}]|$ is the cardinality of $[F^{n'+1}]$.

Since rank of $(T + I)^{n'+1}$ is one less than that of $(T + I)^{n'}$, therefore, from Relation 7.32, we get

$$|[F^{n'+1}]| = 2 \times |[F^{n'}]|.$$

That is, $|Y| = |[F^{n'}]|$ –that is, the full set of $[F^{n'}]$ is represented by Y. It implies that

$$(T + I)x = F$$

is consistent for all $F \in [F^{n'}]$ and consequently, Relation 7.37 is consistent for all $F \in [F^{n'}], n' < n$.

Case 2: *Checking consistency of cycle length for complemented CA $C'(x + 1)^n$ with the inversion vector $F \in [F^n]$*: It shows that Relation 7.38 is inconsistent for all the cycle lengths $2^j \leq n$. Multiplying both sides of Relation 7.38 with $(T + I)^{n-1}$, the following can be obtained.

$$(T + I)^n x = (T + I)^{n-1} F. \tag{7.39}$$

It is inconsistent since the left hand side (LHS) = 0 while the right hand side (RHS) $\neq 0$. Since the characteristic and minimal polynomial of C is $(x + 1)^n$, Relation 7.37 is consistent if $2^j - 1 \geq n$, when both the LHS and RHS are 0. □

The earlier theorem leads to the following corollary which specifies the cycle structure of $C'(x + 1)^n$. It shows that the cycle structure of a CA is guided by a factor referred to as the *minimum additive factor* $\mathcal{M}$ defined next.

Definition 57 *Minimum Additive Factor* ($\mathcal{M}$): The least cycle length of any primary cycle family $PCF(k)$ in CS' is given by $(2^{\mathcal{M}} \cdot k)$, where $\mathcal{M}$ is the *Minimum Additive Factor*.

Corollary 7.2 *The cycle structure of $C'(x + 1)^n$ with $F \in F^n$ is*

$$CS' = \mu'(2^{\mathcal{M}}), \quad \mu' = 2^{n-\mathcal{M}} \quad \textit{and} \quad \mathcal{M} = \lfloor \log_2(n) \rfloor + 1 \tag{7.40}$$

where the cycle structure of $C(x + 1)$ (Relation 7.31) is

$$CS = [1(1) + \sum_{j=0}^{m} \mu_{2^j}(2^j)], \quad \textit{where } m = \lceil \log_2(n) \rceil$$

Proof As per Theorem 7.15, Relation 7.37 becomes consistent if $2^j - 1 \geq n$. That is, a cycle of length 2^j exists in C' if $2^j - 1 \geq n$. The minimum value of j, for which the equation becomes consistent, is represented by $\mathcal{M}$, where $2^{\mathcal{M}-1} \leq n < 2^{\mathcal{M}}$.
(i) If $n = 2^{\lceil \log_2 n \rceil}$, then n is of the form 2^j, where j is an integer. Therefore, $\lceil \log_2 n \rceil = \lfloor \log_2 n \rfloor = \log_2 n$. In that case, $\mathcal{M} = \lfloor \log_2 n \rfloor + 1$.
(ii) If $n < 2^{\lceil \log_2 n \rceil}$, then $\mathcal{M} = \lceil \log_2 n \rceil = \lfloor \log_2 n \rfloor + 1$. Combining both the cases (i) and (ii), $\mathcal{M} = \lfloor \log_2 n \rfloor + 1$.
Since all the configurations form the null space of $(T + I)^{2^{\mathcal{M}}}$ (Theorem 7.11), the complemented CA can have only the cycles of length $2^{\mathcal{M}}$ and the number of cyclic components (μ') is $2^n/2^{\mathcal{M}} = 2^{n-\mathcal{M}}$. □

The following example is to illustrate the results of Theorem 7.15 and Corollary 7.2.

Example 7.25 Let the T matrix and the two inversion vectors F_1 and F_2 of the 4-cell complemented CAs are

$$T = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad F_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad F_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

The characteristic and minimal polynomial of the T is $(x + 1)^4$.
The cycle structure(CS) of the corresponding linear CA is [2(1), 1(2), 3(4)].
The inversion vector F_1 annihilates $(T + I)^4$. The complemented CA with T and F_1 changes the cycle structure to [2(8)] (Fig. 7.2).
The inversion vector F_2 does not annihilate $(T + I)^4$. The complemented CA with T and F_2 has the cycle structure CS' identical to CS of the linear CA.

Theorem 7.15 and Corollary 7.2 can be extended for the generalized class of linear and complemented CAs. The cycle structure of C', as per Theorem 7.13, can differ from C only if the characteristic polynomial of T-matrix contains a factor of $(x + 1)$. The characteristic polynomial of the CA can be represented in Elementary Divisor Form (EDF) as

$$f(x) = (x + 1)^{n_1} \cdots (x + 1)^{n_l} \phi_{l+1}(x)^{n_{l+1}} \cdots \phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$$

where (i) each $\phi_i(x)^{n_i}$, $i \in \{l, l + 1, \cdots, \mathcal{N}\}$ is an elementary divisor, and (ii) the irreducible factor of the first l elementary divisors is $(x + 1)$.

The following theorem states the nature of cycle structure of such linear and complemented CAs.

Theorem 7.16 *The cycle structure of a linear CA with characteristic matrix T and having $(x + 1)$ as a factor in its characteristic polynomial is*

$$CS = [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)]; \quad k_1 = 1, \tag{7.41}$$

whereas, the cycle structure of complemented CA, with characteristic matrix T and the inversion vector F, and the largest $(x+1)$-elementary divisor annihilated by F is $(x+1)^{n_i}$, is of the form

$$CS' = \sum_{k_i} \sum_{j=\mathcal{M}}^{m_{k_i}} \mu'_{2^j \cdot k_i}(2^j \cdot k_i), \quad k_1 = 1 \tag{7.42}$$

where

$$\mu'_{2^{\mathcal{M}} \cdot k_i} = \begin{cases} \sum_{j=0}^{\mathcal{M}} \dfrac{(\mu_{2^j \cdot k_i} \times 2^j \cdot k_i) + 1}{2^{\mathcal{M}} \cdot k_i} & \textit{for } k_i = 1 \\ \sum_{j=0}^{\mathcal{M}} \dfrac{\mu_{2^j \cdot k_i} \times 2^j \cdot k_i}{2^{\mathcal{M}} \cdot k_i} & \textit{otherwise} \end{cases}$$

$$\textit{and} \quad \mu'_{2^j \cdot k_i} = \mu_{2^j \cdot k_i} \quad j > \mathcal{M}, \qquad \mathcal{M} = \lfloor \log_2 n_i \rfloor + 1$$

Proof The proof is developed with the assumption that $(x+1)^{n_i}$ is the *only* factor annihilated by F. It can be generalized to prove the case where $(x+1)^{n_i}$ is the *largest* factor annihilated by F.
The characteristic polynomial $f(x)$ of the CA is

$$f(x) = (x+1)^{n_i} \times \tilde{\phi}(x); \quad \tilde{\phi}(x) = \phi_1(x)^{n_1} \cdots \phi_{i-1}(x)^{n_{i-1}} \cdot \phi_{i+1}(x)^{n_{i+1}} \cdots \phi_{\mathcal{N}}(x)^{n_{\mathcal{N}}}$$

The cycle structure $CS(\tilde{\phi}(x))$, corresponding to $\tilde{\phi}(x)$, is same for both the linear CA and complemented CAs. Let F be the inversion vector which annihilates the factor $(x+1)^{n_i}$.

As per Theorem 7.15, the cycle structures generated due to the factor $(x+1)^{n_i}$ differ in C and C'. If the cycle structures are respectively $CS'((x+1)^{n_i})$ and $CS((x+1)^{n_i})$, then by Corollary 7.2,

$$CS'(x+1)^{n_i} = \mu'(2^{\mathcal{M}}), \quad \mu' = 2^{n_i - \mathcal{M}} \quad \& \quad \mathcal{M} = \lfloor \log_2(n_i) \rfloor + 1$$

$$CS(x+1)^{n_i} = [1(1) + \sum_{j=0}^{m} \mu_{2^j}(2^j)], \qquad m = \lceil \log_2(n_i) \rceil$$

The cycle structure of $C(\tilde{\phi}(x))$, follows from Relation 7.9 as it is a linear cycle structure, is

$$CS(\tilde{\phi}(x)) = [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)]$$

Therefore, the cycle structure of CS and CS' are represented as

$$CS = CS(\tilde{\phi}(x)) \times CS((x+1)^{n_i}) = [1(1) + \sum_{j=0}^{m} \mu_{2^j}(2^j)] \times [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)]$$

$$CS' = CS(\tilde{\phi}(x)) \times CS'((x+1)^{n_i}) = [\mu'(2^{\mathcal{M}})] \times [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)]$$

The above two cross products result in CS and CS', specified in Relations 7.41 and 7.42 respectively. □

Example 7.26 Let T matrix of a 5-cell CA be

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

Further, consider the two complemented CAs having the above T of the 5-cell CA and the inversion vectors

$$F_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and } F_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

- The characteristic polynomial $f(x)$ of the CA in Elementary Divisor Form is

$$f(x) = (x+1)(x+1)^2(x^2+x+1).$$

- The inversion vector F_1= *[01100]* annihilates only $(x+1)^2$. Therefore, $\mathcal{M} = \lfloor \log_2(2) \rfloor + 1 = 2$.
- $f(x)$ can be expressed as $\tilde{\phi}(x) \times (x+1)^2$, where $\tilde{\phi}(x) = (1+x) \cdot (x^2+x+1)$.
- The cycle structure $CS(\tilde{\phi}(x)) = CS(1+x) \times CS(x^2+x+1) = [1(1), 1(1)] \times [1(1), 1(3)] = [2(1), 2(3)]$.
- The cycle structure $CS((1+x)^2) = [1(1), 1(1), 1(2)]$ whereas $CS'((1+x)^2) = [1(4)]$.
- The resultant structure $CS = CS(\tilde{\phi}(x)) \times CS((1+x)^2) = [4(1), 2(2)], [4(3), 2(6)]$ while $CS' = CS(\tilde{\phi}(x)) \times CS'((1+x)^2) = [2(4), 2(12)]$.
- Similarly, $F_2 = [111100]$ annihilates both the elementary divisors $(x+1)$ and $(x+1)^2$. This is the case where $(x+1)^2$ is the largest factor annihilated by F_2 and not the only factor annihilated. Therefore, $CS' = CS'(x+1) \times CS'((x+1)^2) \times CS(x^2+x+1) = [1(4)] \times [1(2)] \times [1(1), 1(3)] = [2(4), 2(12)]$, is same as the cycle structure generated by F_1.
- The changes in cycle structure of each cycle family in CS and CS' respectively are as per Relation 7.42 of Theorem 7.16. For example, for 2(4) where $k_1 = 1$ and

$2^{\mathcal{M}} = 4$, the corresponding cycle structure in C is [4(1),2(2)]. The cyclic component of $\mu_{2^2 \cdot 1}$ has been formed as per Relation 7.42. Here, $\mu_{2^2 \cdot 1} = \frac{4 \times 1 + 2 \times 2}{4} = 2$.

7.4.4 Complete Cycle Structure of Complemented CA

Algorithm 9 is developed based on the formulation reported in the earlier subsections to find the cycle structure of the complemented CA C'. The next example illustrates the execution of the steps of the algorithm.

Algorithm 9: Find cycle structure of complemented CA

Input : Characteristic matrix T and an inversion vector F of a Complemented CA.
Output: Cycle Structure and depth (if any) of the CA.

Step 1: Use Algorithm 7 to get cycle structure (CS) and depth (d) against T. Output d as the depth of the CA. Here,

$$CS = [1(1) + \sum_{k_i} \sum_{j=0}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)];\ k_i\ is\ odd$$

Suppose, $[CL]$ is the set of all such odd lengths k_i.
Step 2: If characteristic polynomial of T doesn't have a factor of $(x + 1)$, then Output CS as the cycle structure of the CA and exit.
Step 3: Find $\mathcal{M}$: Initially assume $\mathcal{M} = 0$, then repeatedly increment $\mathcal{M}$ by 1 until $rank[T^{2^{\mathcal{M}-1}} + 1] = rank[T^{2^{\mathcal{M}-1}} + I, \mathcal{F}]$, where $\mathcal{F} = [I + T + T^2 + ... + T^{2^{\mathcal{M}-1}}]F$
Step 4: For each $k_i \in [CL]$, evaluate $\mu'_{2^{\mathcal{M}} \cdot k_i}$ by Relation 7.42
Step 5: Output $CS' = \sum_{k_i} \sum_{j=\mathcal{M}}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)$ with $k_1 = 1$ as the cycle structure of the CA.

Example 7.27 Let the characteristic matrix of a 7-cell complemented CA C' (as in Example 7.2) be

$$T = \begin{pmatrix} \lceil 0\ 0 \rceil & 0 & 0 & 0 & 0 & 0 \\ \lfloor 1\ 0 \rfloor & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & [1] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lceil 1\ 1 \rceil & 0 & 0 \\ 0 & 0 & 0 & \lfloor 0\ 1 \rfloor & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lceil 1\ 1 \rceil \\ 0 & 0 & 0 & 0 & 0 & \lfloor 1\ 0 \rfloor \end{pmatrix} \quad \text{and} \quad F = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

The matrix $[T^k + I, \mathcal{F}]$ is the *augmented matrix*.
Step 1 and 2 : Find the Characteristic polynomial and cycle structure.
Characteristic polynomial = $x^2 \cdot (x + 1)^3 \cdot (x^2 + x + 1)$.
The cycle structure of the linear CA C is [4(1), 2(2), 4(3), 2(6)].
Depth of C is 2. It implies that the depth of C' is 2.

The transition diagram of C is shown in Fig. 7.1.
Step 3: Find the value of $\mathcal{M}$.

- Rank of $(T^1 + I)$ is 3, while the rank of the augmented matrix is 4. Hence, the cycle of length 1 does not exist in the C'
- Rank of $(T^2 + I)$ is 2, while the rank of the augmented matrix is 3. Hence, the cycle of length 2 also does not exist in the C'
- Rank of $(T^4 + I)$ is 2, while the rank of the augmented matrix is 2. Hence, the cycle of length 4 exists in C'.
  So, the $\mathcal{M}$ (as per Step 3) is 2.

Step 4 : Find the cyclic components.
1. $k_i = 1$, $\mu_{2^2 \cdot 1} = \frac{4\times1+2\times2+0\times4}{2^2 \cdot 1} = 2$.
2. $k_i = 3$, $\mu_{2^2 \cdot 3} = \frac{4\times3+2\times6+0\times12}{2^2 \cdot 3} = 2$.
Hence, the cycle structure of the C' is [2(4), 2(12)]. That is, it has 32 cyclic configurations.

7.5 Synthesis of Complemented Cellular Automata

The cycle structure of a complemented CA follows a well-defined relationship with that of its linear counterpart. This section exploits that relationship to synthesize a complemented CA from a *complemented cycle structure* (Definition 49). For the sake of clarity, the terms CS, $EFCS$ etc., used for linear CA synthesis, are noted as CS', $EFCS'$, etc. for the complemented CA. Following is the basic overview of the synthesis scheme.

1. To synthesize the linear operator T for a complemented CA C', the ECS is derived from a given CS'. The $EFCS'$ acts as an intermediate step in this derivation process.
2. Once ECS is formed, the synthesis of T follows the same methodology as in Sect. 7.3.
3. The inversion vector F of C' is next synthesized to meet the input specification.

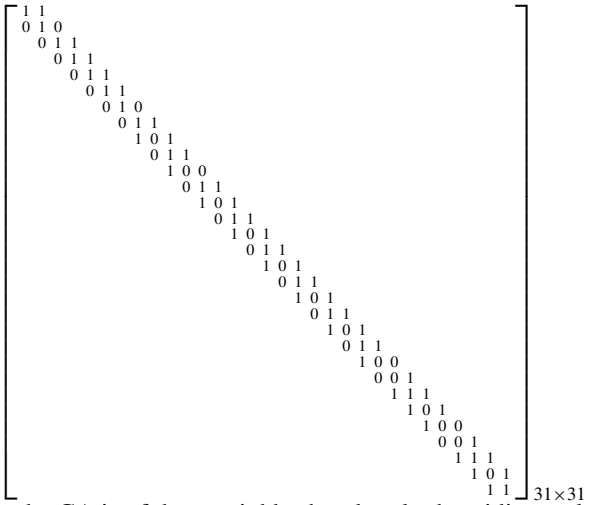
The above steps of the synthesis scheme is now illustrated through the following example. The input cycle structure (CS') taken in the example satisfies the constraints stated in Theorem 7.16. Hence, the characteristic polynomial of the synthesized T matrix contains a factor $(x + 1)$ which is responsible for the difference in cycle structure between C' and C. The steps of synthesis of C', other than the design of F-vector, are similar to that of linear CA synthesis scheme.

Example 7.28 Let CS' = [16(4), 24(8), 1397760(12), 3072(20), 44039680(24), 4608(40), 71302144(60), 2246048256(120)] and the $\mathcal{M} = 2$ (see Theorem 7.16). That is, the smallest cycle of any primary cycle family $PCF(k)$ (Definition 53) is $2^{\mathcal{M}}$, while synthesizing the input cycle structure.

Step 1: Separating each primary cycle family, $CS' = [64[4], 24(8)] \times [1(1), 4095[12], 2560(24)] \times [1(1), 15[20]] \times [1(1), 15[60]]$ can be obtained, where each group represents an $EFCS'$.

Step 2: From each $EFCS'$, the ECS in triplet form is obtained. That is, $CS = (1, 1)^2 \times (1, 1)^5 \times (1, 3)^2 \times (1, 3)^6 \times (3, 5)^1 \times (1, 15)^1$. Now, the linear operator T can be constructed from the ECSs following the procedure as followed for the linear CA:

$T =$



(Since the CA is of three neighborhood, only the tridiagonal elements of the matrix are shown, the rest of the elements are zero.)

Step 3: Corresponding to T, the following inversion vector F is synthesized.

$F = [1\ 1\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1]$.

The above example points to the fact that similar to the divide and conquer policy adopted for linear CA synthesis, synthesis of the complemented CA corresponding to a given CS' proceeds through the identification of each participating elementary cycle structure. This identification process (Step 1 of the above outline) goes through several intermediate steps. To facilitate further discussions, the term Cycle Information (CI) is introduced.

Definition 58 *Cycle information (CI)* is denoted as $S_{2^{\mathcal{M}}\cdot k}[2^{\mathcal{M}} \cdot k]$, where $S_{2^{\mathcal{M}}\cdot k}$ represents the number of configurations covered by the cycles of length $2^j \cdot k$ ($j = 0$ to $\mathcal{M}$).

The necessity of treating two different cases for $k_i = 1$ and $k_i > 1$ has been explained through Relation 7.42 in Theorem 7.16. The terms $EFCS'$ and CS' for these two cases are redefined with reference to the cycle information.

Let $\beta'_i = [\ S_{2^{\mathcal{M}}\cdot k_i}[2^{\mathcal{M}} \cdot k_i], \sum_{j=\mathcal{M}+1}^{m_{k_i}} \mu_{2^j \cdot k_i}(2^j \cdot k_i)]$. Then,

$$EFCS' = \begin{cases} \beta'_i, & k_i = 1 \\ 1(1) + \beta'_i, & k_i > 1 \end{cases} \quad \text{and} \quad CS' = \begin{cases} \sum_i \beta'_i, & if\ k_i = 1 \\ 1(1) + \sum_i \beta'_i, & if\ k_i > 1 \end{cases}.$$

Example 7.29 The CS' of Example 7.28 is [16(4), 24(8), 1397760(12), 44039680(24), 3072(20), 4608(40), 71302144(60), 2246048256(120)]. In line with the new definition of CS' mentioned above, the $CS' =$ [64[4], 24(8), 16773120[12], 44039680(24), 61440[20], 4608(40), 4278128640[60], 2246048256(120)]. That is, instead of representing in the form of $\mu(k)$–that is, 16(4), it is represented in the form of $S[k]$, where S is the total number of configurations–that is, 64[4].

7.5.1 Generation of $EFCS'$ from Cycle Structure

Generation of $EFCS'$ from the cycle structure CS' is almost same as that of the derivation of $EFCS$ from the CS for linear CA (Sect. 7.3.1). The only difference is the process of generating cycle information–that is, the information regarding the number of configurations covered by the cycles $\leq 2^{\mathcal{M}} \cdot k_i$, where $\mathcal{M}$ is the minimum additive factor and k_i is the length of the odd cycle.

The identical cyclic components for linear CA and complemented CA: From Theorem 7.16, it is found that the cyclic components of cycle length greater than $2^{\mathcal{M}} \cdot k_i$, are the same in CS and CS'. Therefore, identification of the CI is needed, as it only involves cycles of length less than or equal to $2^{\mathcal{M}} \cdot k_i$. Deriving the CI from CS' demands the partition of CS' corresponding to its $EFCS'$s–that is,

$$CS' = [EFCS'_1] \times [EFCS'_2] \times \cdots \times [EFCS'_{\tilde{\mathcal{N}}}]$$

where $\tilde{\mathcal{N}}$ denotes the number of $EFCS'$ comprising the CS'.

To separate out each $EFCS'$ the recursive technique, noted in Sect. 7.3.1 for the linear CA, is employed. It divides $CS' = CS'_1 \times CS'_2$. Here, CS'_1 is the $EFCS'$ with the smallest primary cycle. It is worth noting the fact that $CS'_1 = \beta'_1$, if $k_1 = 1$ and $[1(1) + \beta'_1)$, if $k_1 > 1$, where $\beta'_1 = [\, S_{2^{\mathcal{M}} \cdot k_1}[2^{\mathcal{M}} \cdot k_1], \sum_{j=\mathcal{M}+1}^{m_{k_1}} \mu_{2^j \cdot k_1}(2^j \cdot k_1)]$ and k_1 is the length of the primary cycle. While $\tilde{S}$ denotes CI in CS', S denotes CI in CS'_1 and CS'_2. Instead of using $S_{2^{\mathcal{M}} \cdot k}[2^{\mathcal{M}} \cdot k]$, the $S_{2^{\mathcal{M}} \cdot k}$ can be used to represent the CI in the subsequent discussions. Further, for the cycles of length $2^j \cdot k_i$, $j > \mathcal{M}$, the steps of evaluation are exactly the same as found for the linear CA synthesis.

Characterization of CS' in terms of CS'_1 and CS'_2: $CS'_1 = [1(1) + \beta'_1]$ if $k_1 > 1$; and $[\beta'_1]$ if $k_1 = 1$, $CS'_2 = [1 + \sum_i \beta'_i]$, where $i > 1$ and k_i is the length of primary cycle corresponding to β_i. So,

$$CS' = CS'_1 \times CS'_2 = [1 + \beta'_1 + \sum_{i>1}(\beta'_i + \beta'_1 \times \beta'_i)];\ k_1 > 1 \tag{7.43}$$

$$CS' = CS'_1 \times CS'_2 = [\beta'_1 + \sum_{i>1}(\beta'_1 \times \beta'_i)]; \qquad k_1 = 1 \tag{7.44}$$

Therefore, when the cross product of each cycle family k_2 of CS'_2 is done with k_1-cycle family of CS_1 to derive a cycle family k_3, besides β'_1, (i) the components $\beta'_3 = (\beta'_1 \times \beta'_2)$ and β'_2 are computed, in case of $k_1 > 1$, and (ii) the components $\beta'_3 = \sum(\beta'_1 \times \beta'_2)$ are computed if $k_1 = 1$.

The following theorem guides the derivation of CI of cycle family k_3 that results in from the cross product of k_1-cycle family of CS'_1 and k_2-cycle family of CS'_2. The theorem is in line with Theorem 7.5.

Theorem 7.17 *The cycle information (CI) of k_3-cycle family, derived from the cross product of the cyclic component of k_1- and k_2-cycle families in CS'_1 and CS'_2 respectively, is*

$$S_{2^{\mathcal{M}}.k_3} = S_{2^{\mathcal{M}}.k_1} \times S_{2^{\mathcal{M}}.k_2} \tag{7.45}$$

Proof As per Relation 7.10, the cross product of the cycles of lengths $2^{j_1} \cdot k_1$ and $2^{j_2} \cdot k_2$ results in cycle(s) of length $2^{\max(j_1,j_2)} \cdot k_3$, where $k_3 = \text{lcm}(k_1, k_3)$. Therefore, the only cycles those are of length less than or equal to $2^{\mathcal{M}} \cdot k_1$ and $2^{\mathcal{M}} \cdot k_2$ participate to form CI of k_3. It can be observed that the CI of k_1 and k_3 result in the CI of k_3, when cross product is done. Therefore, the entire configuration space produced by k_1- and k_2-cycle families becomes $S_{2^{\mathcal{M}}.k_3}$ –that is, $S_{2^{\mathcal{M}}.k_3} = S_{2^{\mathcal{M}}.k_1} \times S_{2^{\mathcal{M}}.k_2}$. □

The CI added to the CS' due to the cross product of CS'_1 and CS'_2 can be as follows.

- For $k_1 = 1$, the only component added to CS' is $S_{2^{\mathcal{M}}.k_3}$. It directly follows from Relation 7.44.
- For $k_1 > 1$, both the β'_2 and $(\beta'_1 \times \beta'_2)$ are added to CS' (depending on the value of k_3). As in the linear CAs, $k_3 = \text{lcm}(k_1, k_2)$ can be (i) greater than k_2 or (ii) equal to k_2.
  For $k_3 > k_2$, the components added to CS' are $S_{2^{\mathcal{M}}.k_2}$ and $\tilde{S}_{2^{\mathcal{M}}.k_3}$, where $\tilde{S}_{2^{\mathcal{M}}.k_3} = S_{2^{\mathcal{M}}.k_3}$ (as derived in Relation 7.45).
  For $k_3 = k_2$, the components added to CS' are $\tilde{S}_{2^{\mathcal{M}}.k_2}$, where

$$\tilde{S}_{2^{\mathcal{M}}.k_2}(CS') = S_{2^{\mathcal{M}}.k_1} \times S_{2^{\mathcal{M}}.k_2} + S_{2^{\mathcal{M}}.k_2} = [S_{2^{\mathcal{M}}.k_1}(CS'_1) + 1] \times S_{2^{\mathcal{M}}.k_2}(CS'_2) \tag{7.46}$$

Generation of CS'_2 from CS': The following action is taken while deriving CS'_2 from CS'.

- If $k_1 = 1$, then $S_{2^{\mathcal{M}}.k_2} = (\tilde{S}_{2^{\mathcal{M}}.k_3} / S_{2^{\mathcal{M}}.k_1})$ added to CS'_2 and $\tilde{S}_{2^{\mathcal{M}}.k_3}$ is subtracted from CS'.
- If $k_1 > 1$, then

Case 1 If $k_3 = \text{lcm}(k_1, k_2) > k_2$, then $\tilde{S}_{2^{\mathcal{M}}.k_2}$ is added to CS'_2 and removed from CS'. The resultant CI $\tilde{S}_{2^{\mathcal{M}}.k_3}$ $(= S_{2^{\mathcal{M}}.k_3})$ of k_3-cycle family is also removed from CS'.

Case 2 If $k_3 = \text{lcm}(k_1, k_2) = k_2$, then $S_{2^{\mathcal{M}}.k_2}$, added to CS'_2, is computed as (by reorienting Relation 7.46).

$$S_{2^{\mathcal{M}}.k_2}(CS'_2) = \frac{\tilde{S}_{2^{\mathcal{M}}.k_2}(CS')}{[S_{2^{\mathcal{M}}.k_1} + 1](CS'_1)} \tag{7.47}$$

Here, $S_{2^{\mathcal{M}}.k_2}(CS'_2)$ signifies the CI for CS'_2.

Example 7.30 This example illustrates both the cases for evaluation of CS'_2, where $k_1 > 1$. Let assume that the given cycle structure CS' = [1(1), 4095[12], 2560(24), 15[20], 1044465[60], 130560(120)] and the minimum additive factor $\mathcal{M} = 2$. The CS'_1 = [1(1), 4095[12], 2560(24)] and $k_1 = 3$. The cycle structure CS' after removal of CS'_1 becomes [15[20], 1044465[60], 130560(120)]. The lengths of the remaining primary cycles are $k_i = 5$ and 15.

- Case 1: Consider $k_2 = 5$. For this case, $\text{lcm}(k_1, k_2) > k_2$.
  Therefore, CS'_2 = [1(1), 15[20]] –that is, the entire cycle structure, arising from the primary cycle 5, becomes the member of CS'_2. The cycle family arising out of a cross product of k_1 and k_2 is $k_3 = \text{lcm}(k_1, k_2) = 15$. Therefore, from Relation 7.45, the cross product of cyclic components, having the cycle lengths [12] and [20], produces $S_{60} = S_{12} \times S_{20} = 4095 \times 15 = 61425$–that is, $CI = 61425[60]$ and the cross product of 24 and [20] yields $\mu_{120} = \frac{\mu_{8.3} \times S}{5} = \frac{2560 \times 15}{5} = 7680$ (Relation 7.17). Both the cycles are removed from the CS'. Therefore, the remaining cycle structure is [1(1), 983040[60], 122880(120)].
- Case 2: Now consider $k_2 = 15$, so $\text{lcm}(k_1, k_2) = k_2$.
  Therefore, following Relation 7.47, the cross product of the cyclic components, having the cycle lengths [12] and [60], produces $S_{60} = \frac{\tilde{S}_{60}}{S_{12}+1} = \frac{983040}{4095+1} = 240$–that is, $CI = 240[60]$, and the cross product of the cyclic components having the cycle lengths 24 and [60] produces $\mu_{120} = \frac{122880 - 2560 \times 240 \times \frac{3}{15}}{65536} = 0$ (Relation 7.23). Hence CS'_2 = [1(1), 15[20], 240[60]].

7.5.2 Generation of ECS from EFCS′

The methodology to generate ECS from $EFCS'$ is similar to the approach described in Sect. 7.3.2 for linear CAs. The only difference is the enumeration of $\tilde{N}(F_j)$ from the CI of the $EFCS'$, where $\tilde{N}(F_j)$ is the sum of the number of F_j factor set present in all the constituent ECS comprising the $EFCS$ [Note: F_j (Definition 51) are the set of factors having the power between $2^{j-1} + 1$ and 2^j present in any ECS $(\mu, k)^{n_i}$]. The conversion of $EFCS'$ to a set of ECSs demands computation of $\tilde{N}(F_j)$ from CI. The CI $S_{2^{\mathcal{M}}.k}[2^{\mathcal{M}} \cdot k]$ represents the aggregate number of configurations $(S_{2^{\mathcal{M}}.k})$ generated by a set of elementary divisors, each of which has the primary cycle of length k. These elementary divisors give $f(x) = \phi(x)^{n_1} \cdot \phi(x)^{n_2} \cdots \phi(x)^{n_{\mathcal{N}}}$, where

$\phi(x)$ generating cycles of length k is responsible for generating the CI. Therefore, the first task is to derive the elementary divisors' set which is responsible for the formation of CI. Once the elementary divisors are known, then the computation of $\tilde{N}(F_j)$ can follow. The following inferences guide this computation.

- Inference 1: One of the elementary divisors consists of an irreducible polynomial $(x+1)$. Because, the presence of $(x+1)$ factor is a prerequisite (Theorem 7.13) for the generation of cycle structure for a complemented CA that is different from that of its linear counterpart.
- Inference 2: As the cycle structure of a complemented CA differs from that of its linear counterpart due to the presence of an elementary divisor $(x+1)^{n_i}$ with $2^{\mathcal{M}-1} \leq n < 2^{\mathcal{M}}$ (Theorem 7.15), the power (n_i) of at least one member of the elementary divisors is to be in between $2^{\mathcal{M}-1}$ and $2^{\mathcal{M}}-1$. The least one, $2^{\mathcal{M}-1}$, plays a critical role in the synthesis process.
- Inference 3: For any other irreducible polynomial $\phi(x)$, the power of each member $\phi(x)$ of the elementary divisors $[\phi(x)]$ is $\leq 2^{\mathcal{M}}$. Because, the factors higher than $2^{\mathcal{M}}$ in $\phi(x)$ produces cycles of length $> 2^{\mathcal{M}} \cdot k$ (Theorem 7.3).

As per the above formulation, the elementary divisors generating the CI (that is, $S_{2^{\mathcal{M}}.k}$) can be directly evaluated. For the case of $(x+1)^{n_i}$, since there is an extra precondition (*Inference 2*), its modelling is marginally different as noted below:

1. The underlying elementary divisors giving $f(x)$ for $\phi(x) \neq (x+1)$, which generates the configurations of size $S_{2^{\mathcal{M}}.k}$, is given by the relation

$$f(x) = \phi(x)^{2^{\mathcal{M}}} \cdot \phi(x)^{2^{\mathcal{M}}} \cdot \phi(x)^{2^{\mathcal{M}}} \cdot \phi(x)^{L}, \text{ where } L \leq 2^{\mathcal{M}} \tag{7.48}$$

2. In case of $\phi(x) = (x+1)$, the total number of configurations $S_{2^{\mathcal{M}}.k}$ are generated by the $f(x)$, where

$$f(x) = \phi(x)^{2^{\mathcal{M}-1}} \cdot \phi(x)^{2^{\mathcal{M}}} \cdot \phi(x)^{2^{\mathcal{M}}} \cdot \phi(x)^{L} \tag{7.49}$$

   As one elementary divisor $(x+1)$ must have the power between $2^{\mathcal{M}-1}$ and $2^{\mathcal{M}}-1$ and the least one is considered.

Now, the next task is to compute the value of $\mathcal{N}$ –that is, the number of associated elementary divisors and the L. The following lemma formalizes this evaluation for a specific $S_{2^{\mathcal{M}}.k}$.

Lemma 16 *For a given $S_{2^{\mathcal{M}}.k}$, the number of elementary divisors $\mathcal{N} = \hat{N} + B$, where*

$$\hat{N} = \begin{cases} \lfloor \frac{\log_{2^r}(S_{2^{\mathcal{M}}.k}) + 2^{\mathcal{M}-1}}{2^{\mathcal{M}}} \rfloor & k = 1; \\ \lfloor \frac{\log_{2^r}(S_{2^{\mathcal{M}}.k}+1)}{2^{\mathcal{M}}} \rfloor & k > 1; \quad r \text{ is the degree of underlying } \phi(x) \end{cases}$$

and the power of the last elementary divisor is

$$L = \begin{cases} (\log_{2^r}((S_{2^{\mathcal{M}}.k}) - 2^{\mathcal{M}-1}) \mod 2^{\mathcal{M}} & k = 1 \\ \log_{2^r}(S_{2^{\mathcal{M}}.k} + 1) \mod 2^{\mathcal{M}} & k > 1 \end{cases} \text{ and } B = \begin{cases} 0 & L = 0 \\ 1 & L > 0 \end{cases}$$

Proof The proof is shown only for $\phi(x) \neq (x+1)$. The proof for $\phi(x) = (x+1)$ directly follows.

Since the number of configurations generated by the $\phi(x)$ is 2^r, the total number of configurations generated by $f(x)$ as per Relation 7.48, is given by $(2^{r \cdot 2^{\mathcal{M}}} \cdots 2^{r \cdot 2^{\mathcal{M}}} \cdot 2^{rL})$ –that is, $2^{r \cdot 2^{\mathcal{M}} \cdot \hat{N} + L}$ (the r is obtained from Relation 7.4).

Therefore, the $\hat{N}$ and L can be obtained by equating the number of configurations covered by $f(x)$ with $S_{2^{\mathcal{M}}.k}$. That is,

$$S_{2^{\mathcal{M}}.k} = 2^{r \cdot 2^{\mathcal{M}} \cdot \hat{N} + L} - 1;$$

where subtraction of '1' implies the exclusion of all 0s configuration. Hence,

$$\log_{2^r}(S_{2^{\mathcal{M}}.k}) + 2^{\mathcal{M}-1} = 2^{\mathcal{M}} \cdot \hat{N} + L$$

Therefore, performing integer division and taking the remainder, the

$$\hat{N} = \lfloor \frac{\log_{2^r}(S_{2^{\mathcal{M}}.k} + 1)}{2^{\mathcal{M}}} \rfloor \text{ and } L = \log_{2^r}(S_{2^{\mathcal{M}}.k} + 1) \mod 2^{\mathcal{M}}$$

The $\mathcal{N}$ is naturally one less when L is 0. Hence the proof. □

Example 7.31 This example illustrates the computation of $\mathcal{N}$ and L (see Example 7.19).There are two cases – $k > 1$ and $k = 1$.

For $k > 1$, it is given that $EFCS' = [1(1), 4095[12], 2560(24)]$.
Then $CI = [4095[12]]$, $\mathcal{M} = 2$, $r = 2$, $S = 4095$, $k = 3$;
$\hat{N} = \lfloor \frac{\log_{2^2}(4095+1)}{2^2} \rfloor = 1$; $L = \log_{2^2}(4095+1) \pmod{2^2} = 2$. Therefore, $B = 1$. That is, $\mathcal{N} = 2$ and $L = 2$.

For $k = 1$, the underlying irreducible polynomial is $(x+1)$.
Given the cycle structure $CS = [16(4), 24(8)]$
$\mathcal{M} = 2$, r=1; $CI = [64[4]]$
The value of $\hat{N} = \lceil \frac{\log_{2^1}(16) + 2^1}{2^2} \rceil = 2$; $L = (\log_{2^1}(16) - 2^1) \pmod{2^2} = 0$. Hence, $B = 0$. Therefore, $\mathcal{N} = 2$ and $L = 0$.

From the earlier derivation, it is known that the elementary divisors are responsible for generating CI. Since the μ_k and k are also known, the CS' can be expressed in triplet format.
For $k = 1$,

$$CS' = (1,1)^{2^{\mathcal{M}-1}} \cdot (1,1)^{2^{\mathcal{M}}} \cdot (1,1)^{2^{\mathcal{M}}} \cdot (1,1)^{L}$$

For $k > 1$,

$$CS' = (\mu_k, k)^{2^{\mathcal{M}}} \cdot (\mu_k, k)^{2^{\mathcal{M}}} \cdot (\mu_k, k)^{2^{\mathcal{M}}} \cdot^{L} \quad L \leq 2^{\mathcal{M}}$$

The derivation of $\tilde{N}(F_j)$ now can be straightforward. The same process, as it is for the linear CAs, can be followed.

Example 7.32 This example illustrates the derivation of $\tilde{N}(F_j)$, for both $k = 1$ and $k > 1$.
For $k > 1$:
The given CI = [4095[12]], $\mathcal{M} = 2$, $\mu_k = 1$, $k = 3$, $\hat{N} = 1$ and $L = 2$;
Therefore, the ECS set comprises of $[(1, 3)^4, (1, 3)^2]$.
Hence, $\tilde{N}(F_0) = 2$; $\tilde{N}(F_1) = 2$; $\tilde{N}(F_2) = 2$.
For $k = 1$:
The given $\mathcal{M} = 2$, r=1; CI = [64[4]], $\mathcal{N} = 2$ and $L = 0$. Therefore, the ECS set is $[(1, 1)^2, (1, 1)^4]$.
The $\tilde{N}(F_j)$ are derived as $\tilde{N}(F_0) = 2$, $\tilde{N}(F_1) = 2$ and $\tilde{N}(F_2) = 2$.

The method of synthesizing the inversion vector F to complete the synthesis of complemented CA, specified in terms of its cycle structure and depth, follows next.

7.5.3 Synthesis of Inversion Vector F

The synthesis of inversion vector F for the complemented CA is guided by the principle laid down in Theorems 7.15 and 7.16. These state that the inversion vector F should lie only in the null space of $(x + 1)^{n_i}$, where $\lfloor \log_2(n_i) \rfloor + 1 = \mathcal{M}$. To synthesize F, satisfying this property, first F is synthesized for the characteristic polynomial $f(x)$ having $(x + 1)^{n_i}$ as the single elementary divisor–that is, $f(x) = (x + 1)^{n_i}$. The T matrix representing the $(x + 1)^{n_i}$ is

$$T(x+1)^{n_i} = \begin{bmatrix} 1 & 1 & 0 & \cdots & \\ 0 & 1 & 1 & 0 & \cdots \\ \cdots & & & & \\ \cdots & & & & \\ 0 & \cdots & 0 & 1 & \end{bmatrix}_{n_i \times n_i}$$

Given this T, it is known that the F with all 1s lies only in the null space of $(T + I)^{n_i}$. Because, none of the expressions $(T + I)^j \cdot F$ is 0 unless $j = n_i$. That is, $F = [11 \cdots 11]^T_{n_i \times 1}$ lies only in the null space of $(T + I)^{n_i}$.

If $T(x + 1)^{n_i}$ occupies the Block (L,n_i) in T, then the following lemma defines the method for synthesizing the desired F.

During synthesis of T matrix, the characteristics matrices corresponding to the elementary divisors are placed in block diagonal form (Theorem 7.10). Let consider that the submatrix $T((x+1)^{n_i})$ corresponding to the elementary divisor $(x+1)^{n_i}$ is placed at (l, l) of the T. Then, the following result is easy to verify.

Lemma 17 *The inversion vector F having all 1s in its lth to $(l+n_i-1)$th row belongs to the null space of $(T+I)^{n_i}$.*

Proof Since the matrices are arranged in block diagonal form, the expression $((T+I)^j \cdot F)$ only depends on the multiplication of the $n_i \times n_i$ submatrix, placed at (l, l) of T, with the lth to $(l+n_i-1)rmth)$ rows of the F-vector. Therefore, whatever be the elements in the other rows, $(T+1)^j \cdot F = 0$ only when $j = n_i$. Hence the proof. □

Example 7.33 Consider the characteristic polynomial $f(x) = x^2(x+1)(x+1)^2(x^2+x+2)$. Its corresponding characteristic matrix in block diagonal form is

$$T = \begin{pmatrix} \lceil 0\ 0 \rceil & 0 & 0 & 0 & 0 & 0 \\ \lfloor 1\ 0 \rfloor & 0 & 0 & 0 & 0 & 0 \\ 0\ \ 0 & [1] & 0 & 0 & 0 & 0 \\ 0\ \ 0 & 0 & \lceil 1\ 1 \rceil & 0 & 0 \\ 0\ \ 0 & 0 & \lfloor 0\ 1 \rfloor & 0 & 0 \\ 0\ \ 0 & :0 & 0 & 0 & \lceil 0\ 1 \rceil \\ 0\ \ 0 & 0 & 0 & 0 & \lfloor 1\ 1 \rfloor \end{pmatrix}$$

The submatrix $T((x+1)^2)$ corresponding to the elementary divisor $(x+1)^2$ occupies a block at (4, 4). The F vector, lies only in the null space of $T(x^2(x+1)(x+1)^2(x^2+x+1))$, is given by $F = (* * * 1\ 1 * *)^T$, where * implies that the elements can be either 0 or 1.

The summary of synthesizing the F vector for a complemented CA is, therefore, follows.
Step 1: Identify the block in T matrix representing $(x+1)^{n_i}$, where $\mathcal{M} = \lceil \log_2(n_i) \rceil + 1$.
Step 2: Set 1 in the corresponding rows of F vector.
Step 3: Set either 1 or 0 in the other rows of F vector.

For ease of reference, the steps of the complete algorithm for synthesizing the complemented CA are given in Algorithm 10.

Algorithm 10: Synthesize complemented CA

Input : Complemented Cycle Structure (CS') and depth d.
Output: Characteristic matrix T and F vector of a complemented CA.

Step 1: Arrange CS' such that the cycle families appear in ascending order of odd cycles.
Step 2: Identify β'_1 in CS' with k_1 as the smallest odd-length cycle, and consider $CS'_1 = [1(1) + \beta'_1]$ if $k_1 > 1$; and $CS'_1 = [\beta'_1]$ if $k_1 = 1$.
Step 3: Compute CS'_2 by the following lines of this step such that $CS' = CS'_1 \times CS'_2$.
Consider $CS'_2 = [1(1)]$ and $CS' = CS' - CS'_1$.
foreach *cycle family of CS in ascending order* **do**
Let k_2 be the smallest primary cycle in CS', and the family is $\sum_{j=0}^{m_{k_2}} \mu_{2^j \cdot k_2}(2^j \cdot k_2)$.
if $\text{lcm}(k_1, k_2) = k_2$ **then**
Obtain $S_{2\mathcal{M} \cdot k_2}(CS'_2)$ from Relation 7.47
$CS'_2 \leftarrow CS'_2 + S_{2\mathcal{M} \cdot k_2}(CS'_2)$; $CS' \leftarrow CS' - \tilde{S}_{2\mathcal{M} \cdot k_2}(CS'_2)$
foreach $j \in \{\mathcal{M}, \mathcal{M}+1, \cdots, m_{k_2}\}$ **do**
Obtain $\gamma(CS'_2)$ from Relation 7.23
$CS'_2 \leftarrow CS'_2 + \gamma(CS'_2)$
$CS' \leftarrow CS' - (\gamma(CS'_2) + \delta(CS'_2))$, where $\delta(CS'_2) = CS'_1 \times \gamma(CS'_2)$
end
end
else
$CS'_2 \leftarrow CS'_2 + \tilde{S}_{2\mathcal{M} \cdot k_2}$
Obtain $\tilde{S}_{2\mathcal{M} \cdot k_3} (= S_{2\mathcal{M} \cdot k_3}$, as derived in Relation 7.45) ($k_3 = \text{lcm}(k_1, k_2)$)
$CS' \leftarrow CS' - \tilde{S}_{2\mathcal{M} \cdot k_3}$; $CS' \leftarrow CS' - \tilde{S}_{2\mathcal{M} \cdot k_2}$
foreach $j \in \{\mathcal{M}, \mathcal{M}+1, \cdots, m_{k_2}\}$ **do**
$CS_2 \leftarrow CS_2 + \mu_{2^j \cdot k_2}(2^j \cdot k_2)$
Obtain $\tilde{\mu}_{2^j \cdot k_3}$ from Relation 7.17 ($k_3 = \text{lcm}(k_1, k_2)$)
$CS \leftarrow CS - \tilde{\mu}_{2^j \cdot k_3}(2^j \cdot k_3)$; $CS \leftarrow CS - \mu_{2^j \cdot k_2}(2^j \cdot k_2)$
end
end
end
Step 4: CS'_1 is an EFCS. Repeat Step 2 and Step 3, assuming $CS' = CS'_2$ until $CS' = CS'_1$ to identify all EFCSs. Suppose $\tilde{\mathcal{N}}$ EFCSs are identified, which are $EFCS'_1, EFCS'_2, \cdots, EFCS'_{\tilde{\mathcal{N}}}$.
Step 5: For each of the $\tilde{\mathcal{N}}$ EFCSs, find ECSs, using the following lines, in triplet format:
$EFCS' = (\mu_k, k)^{n_1} \times (\mu_k, k)^{n_2} \times \cdots \times (\mu_k, k)^{n_{\mathcal{N}}}$, where k is the primary cycle's length.
Find $\mu_k = (2^l - 1)/k$ such that l is the smallest integer for which k divides $2^l - 1$.
foreach $j \in \{0, 1, \cdots, \lceil \log_2(\max_{1 \le i \le \mathcal{N}} \{n_i\}) \rceil\}$ **do**
if $j \le \mathcal{M}$ **then**
Find $\tilde{N}(F_j)$ from CI (Sect. 7.5.2)
end
else
Find $\tilde{N}(F_j)$ for $j > \mathcal{M}$ by Relation 7.24
end
Find $\mathcal{N}_j$ by Relation 7.27
if $\mathcal{N}_j > 0$ **then**
foreach $i \in \{0, 1, \cdots, \mathcal{N}_j\}$ **do**
Find $P_{j,i}$ by Relation 7.28
Consider $(\mu_k, k)^{P_{j,i}}$ as an ECS.
end
end
end

Step 6: For each ECS $(\mu_k, k)^{n_i}$, identified above, find elementary divisor $\phi_i(x)^{n_i}$ by the following substeps.
Choose a primitive polynomial of degree $r = \log_2(\mu_k \times k + 1)$.
Construct a matrix (T) from the polynomial in Rational Canonical Form.
Find T^{μ_k}, for which $\phi_i(x) = \det(T^{\mu_k} + Ix)$
Step 7: For each elementary divisor $\phi_i(x)^{n_i}$,
Synthesize the matrix $T(\phi_i(x))$ corresponding to $\phi_i(x)$ using Algorithm MAXLEN (page 151).
Place $T(\phi_i(x))$ n_i times following Theorem 7.9 to form $T(\phi_i(x)^{n_i})$.
Step 8: If $d > 0$, form a D matrix of dimension d, with only lower diagonal elements as 1 and rest as 0.
Step 9: Place all the matrices, synthesized in Step 7 and Step 8, in Block Diagonal Form to get the final T matrix.
Step 10: Synthesize F-vector obeying Lemma 17.

7.6 Chapter Notes

Study of linear machines based on the characteristic and minimal polynomials of the linear operator has been initiated in [2]. Further characterization can be found in [3]. Vector-space theoretic analysis of linear cellular automata, a special class of linear machines, has been initiated by Paul [4, 5]. The cycle structure of complemented cellular automata has been discussed using the same framework in [6]. The problem has been dealt in detail by Ganguly [7].

Following are the few research topics, related to this chapter, which may be carried out in future.

1. Efficient identification of CAs, transition diagrams of which are isomorphic to each other.
2. Extension of the framework, presented in this chapter, for the CAs beyond linear and complemented systems.

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Chapter 8
Cycles in Non-linear Cellular Automata

Vector-space theoretic analysis that has been used to deal with the cycle structure of (finite) linear and complemented cellular automata in Chap. 7 is not applicable to non-linear cellular automata (CAs). A different method is needed to explore the configuration space of a non-linear cellular automaton (CA) and to find cycles in it. Reachability tree, a tool to characterize non-linear CA (see Sect. 3.3), is explored in this chapter for the purpose. The chapter concentrates only on the non-uniform finite CAs with ECA rules.

The reachability tree represents reachable configurations of a finite CA through its sequences of edges from the root to leaves. An edge sequence in a reachability tree represents more than one configuration–a reachable configuration and its predecessor(s). However, the tree cannot directly identify the predecessors of a predecessor of a configuration. Therefore, the *links* between edge sequences in a reachability tree are established to trace such predecessors and effectively trace the cycles in the CA.

This chapter first introduces the basic principle of cycle formation (Sect. 8.1), and then describes how to establish the *links* among the edges of a reachability tree to trace the predecessors of a configuration (Sect. 8.2). Characterization of links enables us to address the Configuration REachability Problem (CREP) in CAs. The problem is dealt in Sect. 8.3. The *cross link*, a special type of link, is the key ingredient of cycle formation in CAs. It is characterized in Sect. 8.4 to find the cycle structure of a CA (Sect. 8.5). A stochastic approach is then reported to synthesize reversible CAs with large cycle (Sect. 8.6). As the schemes are equally applicable to linear and complemented CAs, therefore, in this chapter, by "CA" we refer to the linear, complemented and non-linear CA.

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8.1 Formation of Cycles

A sequence of configurations of a CA forms a cycle when every bit of the sequence flips zero or even number of times. Let consider the 3-length cycle of the 4-cell CA $(9, 150, 45, 65)$ of Fig. 8.1 with null boundary. Here, each of the bits of the configurations is flipped twice.

Theorem 8.1 *A sequence of l configurations of an n-cell CA forms a cycle $\langle c_1, c_2, \cdots, c_l, c_1\rangle$ of length $l \geq 1$ iff the number of flips of each bit of the configurations is either 0 or even, and $c_i \neq c_j$ when $i \neq j$ for any $i, j \in \{1, 2, \cdots, l\}$.*

Proof Let assume that a sequence of l configurations of an n-cell CA forms a cycle of length $l \geq 1$ and the ith bit of the configuration c_1, in the cycle, is x. Now, the x may be the same in all the configurations of the cycle. But if the bit is flipped to x' in some c_j, then it has to be flipped again in some c_{j+k} to get back x at the ith bit position, where $1 < j < j + k \leq l + 1$. So, two transitions are there. If another such j exists, then corresponding k also exists. Hence, even number of flips are required. Further, the configurations are unique as the l configurations form a cycle of length l. Conversely, suppose that the number of flips of each bit of the l unique configurations is either 0 or even. Following the above argument, the configurations form a cycle of length l. Hence the proof. □

8.1.1 The Signature of Cycles

It has been observed in Sect. 4.6 that an $(n + m)$-cell CA can be obtained by merging rule vectors of an n-cell CA and an m-cell CA. Similarly, the rule vector of an N-cell CA can contain rule vector of an n-cell CA where $N \geq n$. If a trace of transitions

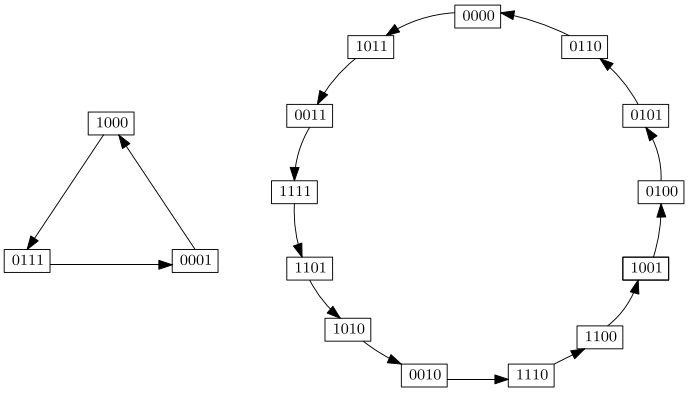

Fig. 8.1 Transition diagram of the 4-cell CA $(9, 150, 45, 65)$ in null boundary

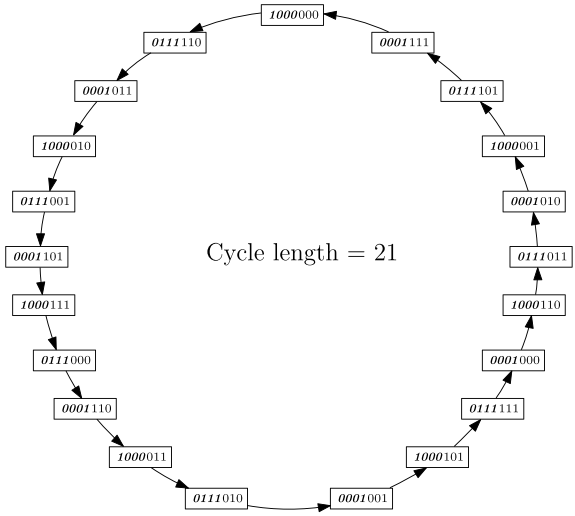

Fig. 8.2 Transition diagram (partial) of CA (9, 150, 45, 195, 85, 105, 20) in null boundary

of configurations of the n-cell CA in that of the N-cell CA can be identified, then a relationship between the cycle lengths of n-cell and N-cell CAs can be drawn. The following is defined to extract such a relationship.

Definition 59 The configuration space of an N-cell CA contains the *signature* of the configuration space of an n-cell CA with $N \geq n$, if each chain of configurations in the n-cell CA is identifiable at least in a chain of configurations of the N-cell CA.

Figure 8.2 is a cycle of the 7-cell CA (9, 150, 45, 195, 85, 105, 20) with null boundary. In its rule vector, a 4-cell CA (9, 150, 45, 65(195)) and a 3-cell CA (5(85), 105, 20) with null boundary can be identified. Here, the rules 65 and 195 as the right most rule and the rules 5 and 85 as the left most rule in a CA under null boundary condition are equivalent. The signature of the configuration space (Fig. 8.1) of the 4-cell CA is contained in the configuration space of the 7-cell CA. The first four bits (*emphasized* in Fig. 8.2) of each configuration of the 7-cell CA represent a configuration of the 4-cell CA.

Theorem 8.2 *The cycle lengths of N-cell CA are the multiple of the lengths of cycles of an n-cell CA ($n \leq N$) if the configuration space of N-cell CA contains the signature of n-cell CA.*

Proof Let consider that the signature of configuration space of the n-cell CA is contained, in between i and $(i + n - 1)$ bit positions, in the configuration space of the N-cell CA. If the configurations of an l-length cycle of the n-cell CA are identifiable on the configurations of an L-length cycle of the N-cell CA, the number of flips at each of the n bit positions (from i to $i + n - 1$) is even at a distance of l configurations (Theorem 8.1). However, as the cycle length of N-cell CA is L, the number of flips at each of the N-bit positions including i to $i + n - 1$ positions is also even. That is, L is to be divisible by l. □

For example, the cycle of Fig. 8.2 is of length 21, which is multiple of 3. The cycle of the 4-cell CA (Fig. 8.1), which is contained in the cycle of Fig. 8.2, is of length 3.

Consider that the cycle structure of an n-cell CA is $CS_1 = [\mu_1(k_1), \mu_2(k_2), \cdots, \mu_i(k_i), \cdots]$, where $\mu_i(k_i)$ represents that the CA has μ_i cycles of length k_i. Let the signature of configuration space of the n-cell CA is contained in the configuration space of N-cell CA. Therefore, length of each cycle of the N-cell CA is the multiple of some k_i (Theorem 8.2).

It can be observed that to get the signature of the n-cell CA, the $(n-1)$th bit of each of the configurations of the N-cell CA should not be affected by the nth cell. That is, the $(n-1)$th cell is to be independent of its right neighbor (Definition 34). Similarly, if $N = n + m$ and if the signature of the configuration space of the m-cell CA is to be contained, the nth cell is to be independent of its left neighbor. Following example illustrates the fact.

Example 8.1 The cycle structures of the 4-cell CA (9, 150, 45, 65) and the 3-cell CA (5, 105, 20) are [1(3), 1(13)] and [1(1), 1(7)] respectively. The 5th cell (rule 85) of the 7-cell CA (9, 150, 45, 105, 85, 105, 20) is independent of its left neighbor, as rule 85 remains invariant to the states of its left neighbor. As a result, the sequence of configurations in the 3-cell CA remains unaffected in the configuration space of the 7-cell CA. So, the signature of the configuration space of the 3-cell CA is contained in the configuration space of the 7-cell CA. The rules 65 and 105 are equivalent while they are treated as the right most cell rule in null boundary. The cycle structure of the 7-cell CA is [1(1), 3(7), 1(15), 1(21), 1(28), 1(42)]. The length of its each cycle is the multiple of a cycle length of the 3-cell CA.

On the other hand, the 7-cell CA (9, 150, 45, 195, 165, 105, 20), in which the 4th cell (rule 195) is independent of its right neighbor, contains the signature of configuration space of the 4-cell CA (9, 150, 45, 65). The cycle structure of this 7-cell CA is [1(3), 1(13), 1(21), 1(91)]. The length of each of the cycles of the 7-cell CA is the multiple of the length of a cycle (3 or 13) of the 4-cell CA.

8.1.2 Product of Cycles

Let consider that the N-cell CA is obtained by merging an n-cell CA and an m-cell CA, and the signatures of the configuration spaces of both the CAs are contained in the configuration space of the N-cell CA. The cycle structure of the $N = (n + m)$-cell CA is the product of cycle structures of the n-cell and m-cell CAs. That is,

$$\begin{aligned} CS &= CS_1 \times CS_2 \\ &= [\mu_1(k_1), \mu_2(k_2), \cdots, \mu_i(k_i), \cdots] \times [\mu'_1(k'_1), \mu'_2(k'_2), \cdots, \mu'_j(k'_j), \cdots] \end{aligned} \tag{8.1}$$

where $CS_1 = [\mu_1(k_1), \mu_2(k_2), \cdots, \mu_l(k_l)]$ and $CS_2 = [\mu'_1(k'_1), \mu'_2(k'_2), \cdots, \mu'_m(k'_m)]$ are the cycle structures of n-cell and the m-cell CAs respectively.

Proposition 8.1 $[\mu_i(k_i)] \times [\mu'_j(k'_j)]$ *is* $[\mu_i \mu'_j \gcd(k_i, k'_j)(\text{lcm}(k_i, k'_j))]$.

Proof If the configurations of two cycles, one of length k_i from n-cell CA and the other of length k'_j from the m-cell CA, are concatenated to get a cycle of $(n+m)$-cell CA, then the length (k) of such cycle is to be the multiple of both the k_i and k'_j. Because, the number of flips in each bit position of each of the three cycles is 0 or even (Theorem 8.1). So, k is the least common multiple (lcm) of k_i and k'_j. Now consider, the number of k-length cycles is μ. Therefore, $\mu k = \mu_i k_i \mu'_j k'_j$

$\Rightarrow \mu = \mu_i \mu'_j \frac{k_i k'_j}{k}$

$\Rightarrow \mu = \mu_i \mu'_j \gcd(k_i, k'_j)$

Here, $\gcd(k_i, k'_j)$ denotes the greatest common divisor of k_i and k'_j. Hence, $[\mu_i(k_i)] \times [\mu'_j(k'_j)] = [\mu_i \mu'_j \gcd(k_i, k'_j)(\text{lcm}(k_i, k'_j))]$. □

To ensure the signatures of both the configuration spaces of n-cell and m-cell CAs contained in the configuration space of $N = (n+m)$-cell CA, its nth and $(n+1)$th cells are to be independent to each other.

By applying Proposition 8.1, CS of Relation 8.1 can be evaluated. Let express CS_1 and CS_2 as

$$CS_1 = \sum_{i=1}^{l} \mu_i(k_i), \qquad CS_2 = \sum_{j=1}^{m} \mu'_j(k'_j)$$

Then, the product of these two cycle structures is

$$CS = \sum_{i=1}^{l} \sum_{j=1}^{m} \mu_i \mu'_j \gcd(k_i, k'_j)(\text{lcm}(k_i, k'_j))$$

Example 8.2 The 4th and 5th cells (rule 195 and 85) of the 7-cell CA (9, 150, 45, 195, 85, 105, 20) are independent of each other. The cycle structures of the CAs (9, 150, 45, 65) and (5, 105, 20) are [1(3),1(13)] and [1(1), 1(7)] respectively. The cycle structure of the 7-cell CA is [1(3), 1(13)] × [1(1), 1(7)] = [1(3), 1(13), 1(21), 1(91)]. The signatures of both the configuration spaces of the 4-cell and 3-cell CAs are contained in the 7-cell CA (Fig. 8.2).

In the configuration space of an $(n+m)$-cell CA, if neither the signature of configuration space of the n-cell nor that of the m-cell CA is contained, then the cycle structure of $(n+m)$-cell CA cannot be obtained following the above method.

Example 8.3 The 7-cell CA (9, 150, 45, 105, 165, 105, 20) can be obtained from the 4-cell CA ((9, 150, 45, 65)) and 3-cell CA ((5, 105, 20)). Its fourth cell (rule 105) and fifth cell (rule 165) are dependent on each other. That is, the signature of configuration spaces of the 4-cell CA and/or 3-cell CA cannot be contained in the configuration space of the 7-cell CA. The cycle structure of the 7-cell CA is [1(1), 1(7), 1(22), 1(98)]. Direct mapping with the cycle structures [1(3), 1(13)]/[1(1), 1(7)] of its 4-cell/3-cell CA segments cannot be possible.

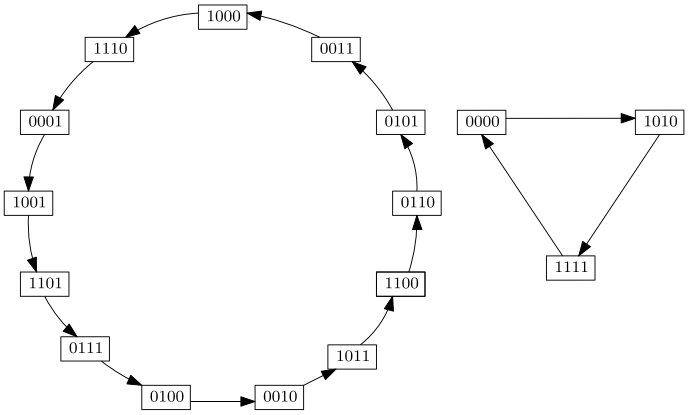

Fig. 8.3 Transition diagram of 4-cell CA (5, 120, 53, 20) in null boundary

Definition 60 Two n-cell CAs are said to be isomorphic iff transition diagrams of the CAs are isomorphic.

For example, the 4-cell CA (5, 120, 53, 20) is isomorphic to the CA (9, 150, 45, 65) in null boundary, as their transition diagrams (Figs. 8.3 and 8.1) are isomorphic. The cycle structures of the isomorphic CAs are the same. In other word, the isomorphic CAs form an equivalence class with respect to the *isomorphism* relation over the set of all n-cell CAs for any $n \in \mathbb{N}$, and the cycle structures of all the CAs in an equivalence class are the same.

Definition 61 An N-cell CA is called *reducible* if there exists an n-cell CA ($n < N$) such that the configuration space of the N-cell CA or any of its isomorphic CAs contains the signature of the configuration space of the n-cell CA. Otherwise, the N-cell CA is called as an *irreducible* CA.

Both of the 7-cell CAs (9, 150, 45, 105, 85, 105, 20) and (9, 150, 45, 195, 165, 105, 20) of Example 8.1 are reducible. Whereas, any maximal length CA (Chap. 6) is irreducible. Reducibility (or irreducibility) of a CA is, therefore, dependent on the cycle structure of the CA. Reducible CAs can further be classified as strictly reducible and weakly reducible.

Definition 62 An N-cell reducible CA is strictly reducible if there exist two CAs of n and m cells such that $N = n + m$ and the configuration space of the N-cell CA or any of its isomorphic CAs contains the signatures of both the n-cell and m-cell CAs side by side at 0 to $n - 1$ and n to $(n + m)$ cells respectively. Otherwise, the N-cell CA is weakly reducible.

Definition 63 An n-cell CA is called a component CA of an N-cell CA ($n < N$) if the signature of the configuration space on the n-cell CA is contained in the configuration space of the N-cell CA.

Reducibility of a CA has a role to find the cycle structure of the CA from the cycle structure of its component CAs. To get the cycle structure of an irreducible CA, the reachability tree is explored.

8.2 Links in Reachability Tree

Let consider G as the global transition function of an n-cell CA. For any configuration x of the CA, if $y = G(x)$, then x and y are paired up in the reachability tree of the CA, and they are represented by an edge sequence. Here, $x = (x_0 x_1 \cdots x_i \cdots x_{n-1})$ and $y = (y_0 y_1 \cdots y_i \cdots y_{n-1})$, where $x_i, y_i \in \{0, 1\}$. In the edge sequence that represents (x, y) pair in the reachability tree, x is found as RMT sequence, $\tilde{x} = (r_0, r_1, \cdots, r_i, \cdots, r_{n-1})$ (see Definition 12). Another edge sequence can be found in the tree that represents x as $(x_0 x_1 \cdots x_i \cdots x_{n-1})$. Hence, there is a correspondence between the edge on which r_i exists and the edge that represents x_i. This correspondence between the two edges is referred to as the *link*. If $x = G(z)$ and $y = G(x)$ –that is, if $y = G^2(z)$, then using the links, the z which is the predecessor of predecessor of x can be identified.

In a reachability tree, an edge can participate in a number of edge sequences. For example, the edges $E_{0.0}$ and $E_{0.1}$ of a reachability tree participate in all the edge sequences, present in the tree. This implies that an edge may link to more than one edge.

Let consider a set of ordered pairs $(E_{i.j}, r)$, where $E_{i.j}$ is the edge and r is an RMT in the label of the edge; $r \in l_{i.j}$ for null boundary condition, and $r \in \Gamma_k^{E_{i.j}}$ ($k \in \{0, 1, 2, 3\}$) for periodic boundary condition. That is, in the tree for null boundary, the set is $\{E_{i.j}\} \times l_{i.j}$; and it is $\{E_{i.j}\} \times (\cup_{k=0}^{3} \Gamma_k^{E_{i.j}})$ for periodic boundary. Then, the link $link_{i.j}$, for an edge $E_{i.j}$, is a binary relation:

$$link_{i.j} : \{(E_{i.j}, r)\} \ \ to \ \ E_i$$

where $\{(E_{i.j}, r)\}$ is the set of ordered pairs as stated above, and $E_i = \{E_{i.j} \mid j \in \{0, 1, \cdots, 2^{i+1} - 1\}\}$ is the set of all edges at level i. Let consider that the edge $E_{i.j}$ links to edge $E_{i.k}$ for RMT r. That is, $(E_{i.j}, r) \ \ link_{i.j} \ \ E_{i.k}$ holds. This is denoted as $E_{i.j}(r) \rightarrow E_{i.k}$.

Further, in a reachability tree, the edges and edge labels of level i are dependent on the edges and edge labels of level $i - 1$ when $i > 0$, and on the rule $\mathcal{R}_i$. So, to get the links at level i, the links of level $i - 1$, when $i > 0$, and $\mathcal{R}_i$ are needed. When $i = 0$, the links are formed by $\mathcal{R}_0$ only. As per Definition 21, an edge $(E_{i.j})$ is called the *edge of level i* if it connects a node of level i to its child. In the same line, level of a link is defined below.

Definition 64 A link between two edges of level i—that is, $E_{i.j}(r) \rightarrow E_{i.k}$, is called a link of level i.

The configuration x is found in the tree as RMT sequence $(r_0, r_1, \cdots, r_{n-1})$ and as bit sequence $(x_0, x_1, \cdots, x_{n-1})$. If r_0 is on $E_{0.0}$ (resp. $E_{0.1}$) and x_0 is represented by $E_{0.1}$ (resp. $E_{0.0}$), then the link $E_{0.0}(r_0) \rightarrow E_{0.1}$ (resp. $E_{0.1}(r_0) \rightarrow E_{0.0}$) is formed. If r_0 and x_0 are on the same edge, the link is $E_{0.j}(r_0) \rightarrow E_{0.j}$ $(j \in \{0, 1\})$. Let consider a link $E_{i-1.j}(r_{i-1}) \rightarrow E_{i-1.k}$ at level $i-1$, where $E_{i-1.k}$ represents x_{i-1}. By construction of the tree, x_i is represented by either $E_{i.2k}$ (if $x_i = 0$) or $E_{i.2k+1}$ (if $x_i = 1$), and r_i is either on $E_{i.2j}$ (if $r_i = 2r_{i-1} \pmod 8$) or on $E_{i.2j+1}$ (if $r_i = 2r_{i-1} + 1 \pmod 8$). That is, there can be four possible links: $E_{i.2j}(r_i) \rightarrow E_{i.2k}$, $E_{i.2j+1}(r_i) \rightarrow E_{i.2k}$, $E_{i.2j}(r_i) \rightarrow E_{i.2k+1}$ and $E_{i.2j+1}(r_i) \rightarrow E_{i.2k+1}$.

Let consider the 4-cell CA $(5, 73, 200, 80)$ with null boundary, as shown in Fig. 8.4, and the configuration 1100. The corresponding RMT sequence is 3640. Figure 8.5 is the reachability tree of the CA that shows the 4 links. The RMT sequence 3640 is identified by the edge sequence $(E_{0.0}, E_{1.1}, E_{2.2}, E_{3.4})$, and $(E_{0.1}, E_{1.3}, E_{2.6}, E_{3.12})$ identifies the bit sequence of the configuration. The dotted arrows in Fig. 8.5 represent the links, and the dotted edges are non-reachable edges.

Definition 65 A link $E_{i.j}(r) \rightarrow E_{i.k}$ is called a (i) *forward link* if $j < k$, (ii) *backward link* if $j > k$ and (iii) *self link* if $j = k$.

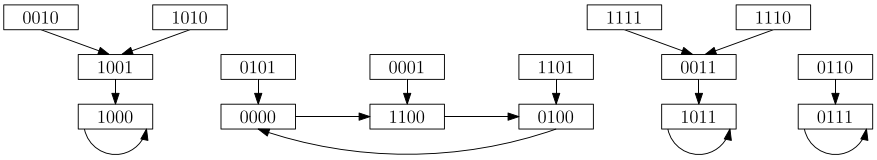

Fig. 8.4 Transition diagram of CA $(5, 73, 200, 80)$ in null boundary

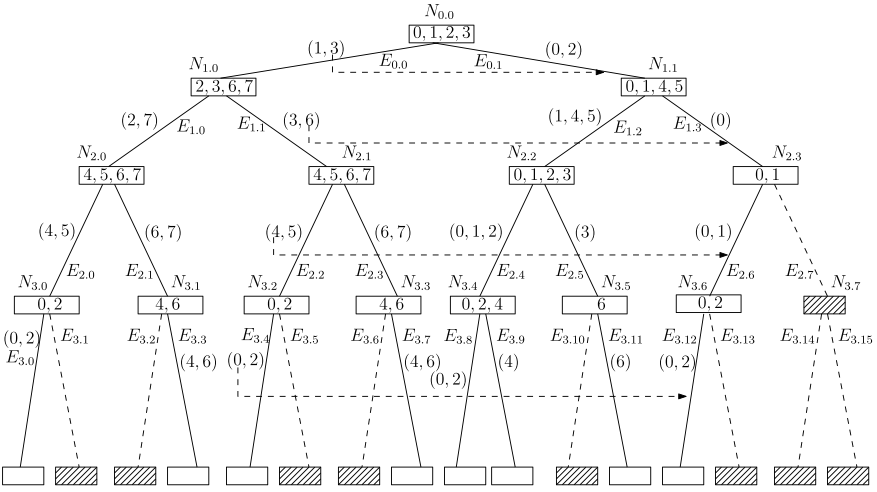

Fig. 8.5 Reachability tree of CA $(5, 73, 200, 80)$ in null boundary

8.2.1 Rules of Links

The summary of earlier discussions are noted below for null boundary. For the edges of level 0 (root),

$$\begin{aligned} &E_{0.j}(r) \rightarrow E_{0.0} \quad \text{if } r \in \{0, 1\}, \\ &E_{0.j}(r) \rightarrow E_{0.1} \quad \text{if } r \in \{2, 3\}, \\ &\text{where } j \in \{0, 1\},\ r \in l_{0.j}. \end{aligned} \tag{8.2}$$

For the edges of other levels,

$$\begin{aligned} &\text{Let } E_{i-1.j}(r) \rightarrow E_{i-1.k}. \text{ Then,} \\ &\quad E_{i.m}(s) \rightarrow E_{i.2k} \quad \text{if } s \in \{0, 1, 4, 5\}, \\ &\quad E_{i.m}(s) \rightarrow E_{i.2k+1} \quad \text{if } s \in \{2, 3, 6, 7\}, \\ &\text{where } m \in \{2j, 2j+1\},\ s \in l_{i.m} \cap \{2r \pmod 8, 2r+1 \pmod 8\}. \end{aligned} \tag{8.3}$$

Definition 66 An RMT r of a rule $\mathcal{R}_i$ is self-replicating if $\mathcal{R}_i[xyz] = y$, where xyz is a 3-bit binary representation of r. That is, r is self-replicating if $\mathcal{R}_i[r] = 0$ when $r \in \{0, 1, 4, 5\}$, and $\mathcal{R}_i[r] = 1$ when $r \in \{2, 3, 6, 7\}$.

The rules of links are sensitive to the fact that whether an RMT is self-replicating or not. The rules can be elaborated based on this fact. Following are the elaborated rules of links for null boundary. For periodic boundary, $l_{i.j}$ is to be replaced by $\Gamma_k^{E_{i.j}}$, $k \in \{0, 1, 2, 3\}$.

(R1) If RMT $r \in l_{0.j}$ is self-replicating ($j \in \{0, 1\}$),
then $E_{0.j}$ is self linked for RMT r.
Otherwise, if $j = 0$,
then there is a forward link from $E_{0.0}$ to $E_{0.1}$ for RMT r;
else there is a backward link from $E_{0.1}$ to $E_{0.0}$ for RMT r.

(R2) If $E_{i-1.j}$ is self linked for RMT $r \in l_{i-1.j}$,
and if s is self-replicating,
where $s \in l_{i.2j}$ (resp. $s \in l_{i.2j+1}$) is $2r \pmod 8$ or $2r+1 \pmod 8$,
then $E_{i.2j}$ (resp. $E_{i.2j+1}$) is self linked.
But if s is not self-replicating,
then there is a forward link from $E_{i.2j}$ to $E_{i.2j+1}$ (resp. backward link from $E_{i.2j+1}$ to $E_{i.2j}$).

(R3) If there is a link from $E_{i-1.j_1}$ to $E_{i-1.j_2}$ ($j_1 \neq j_2$), for RMT $r \in l_{i-1.j_1}$,
and $s \in l_{i.2j_1}$ (resp. $s \in l_{i.2j_1+1}$) is $2r \pmod 8$ or $2r+1 \pmod 8$,
then there is a link from $E_{i.2j_1}$ (resp. $E_{i.2j_1+1}$) to $E_{i.2j_2}$ while $s \in \{0, 1, 4, 5\}$ or to $E_{i.2j_2+1}$ while $s \in \{2, 3, 6, 7\}$.
[It is forward link if $j_1 < j_2$ and backward link if $j_1 > j_2$]

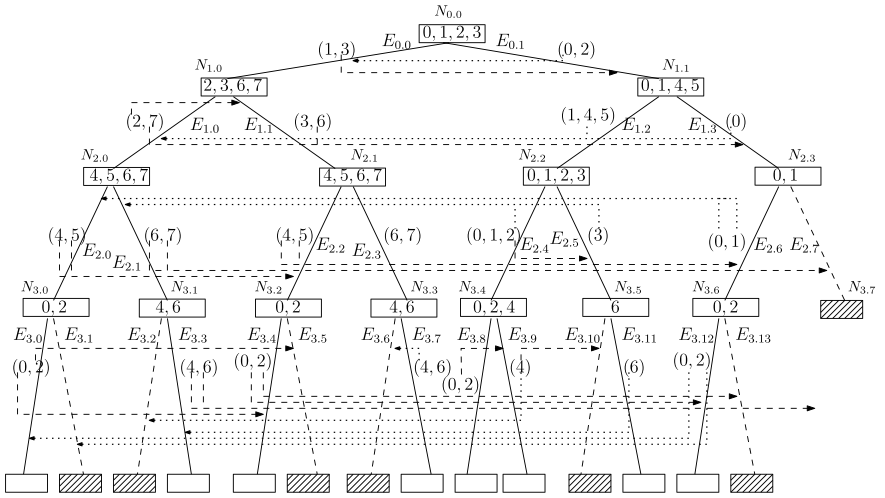

Fig. 8.6 The links of reachability tree of Fig. 8.5

Figure 8.6 shows the links of edges for the CA (5, 73, 200, 80) in null boundary. There is
(i) a forward link from $E_{0.0}$ to $E_{0.1}$ for RMT 3 (following **R1** and a dotted line is drawn from RMT 3 of $E_{0.0}$ to $E_{0.1}$),
(ii) a forward link from $E_{1.1}$ to $E_{1.3}$ for RMT 6 (following **R3**).
(iii) The links from $E_{2.2}$ to $E_{2.6}$ for RMT 4, and from $E_{3.4}$ to $E_{3.12}$ for RMT 0 (following **R3**).

However, no line is shown in Fig. 8.6 for the RMTs involved in self links. Therefore, for the RMT sequence 3640, following the sequence of links, a sequence of edges $\langle E_{0.1}E_{1.3}E_{2.6}E_{3.12}\rangle$ can be extracted. It points to the configuration 1100. The RMT sequence 3640 also corresponds to the configuration 1100. The edge sequence $\langle E_{0.0}E_{1.1}E_{2.2}E_{3.4}\rangle$ associates the configuration 0100, as well as the RMT sequence 3640. The RMT sequence 3640, hence the configuration 1100 is, therefore, the predecessor of 0100.

There are some configurations, predecessors of which are non-reachable configurations. As no edge sequence exists in the tree for a non-reachable configuration, some edges can be found that are supposed to link to some non-existent edges (non-reachable edges). For example, there should be a link from $E_{2.1}$ to $E_{2.7}$ (see Fig. 8.6) for the RMTs 6 and 7. Since $E_{2.7}$ does not exist, no link for the RMTs 6 and 7 exists. The predecessor (of a configuration) that involves non-reachable edge is the non-reachable configuration.

The above rules can also be applied to the reachability trees for periodic boundary condition. Let us consider a 4-cell CA (204, 240, 48, 0) with periodic boundary. The links in the tree of the CA, following the same rules as applied for the null boundary condition is shown in Fig. 8.7. Both the edges of level 0, $E_{0.0}$ and $E_{0.1}$,

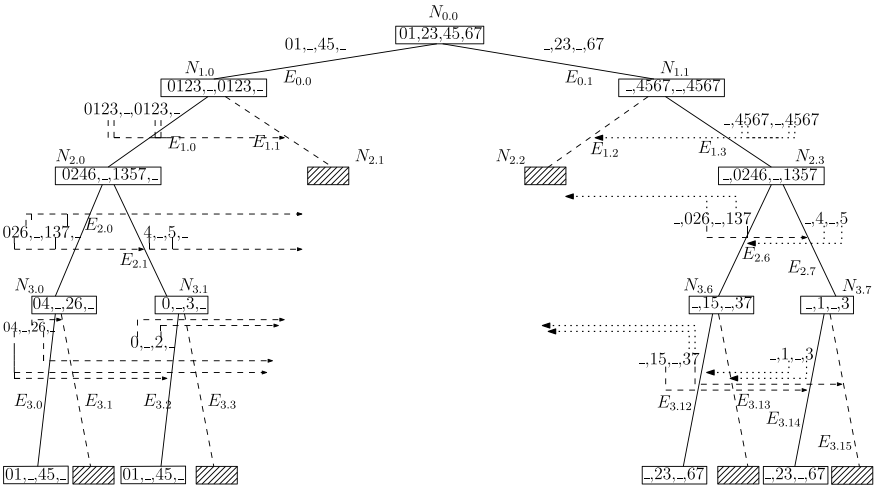

Fig. 8.7 The reachability tree of CA (204, 240, 48, 0) with links in periodic boundary

are self linked for each RMT present in their labels. In level 1, $\Gamma_0^{E_{1.0}} = \{0, 1, 2, 3\}$ and $\Gamma_2^{E_{1.0}} = \{0, 1, 2, 3\}$. For the RMTs 2 and 3 of both the $\Gamma_0^{E_{1.0}}$ and $\Gamma_2^{E_{1.0}}$, there are forward links from $E_{1.0}$ to $E_{1.1}$. Similarly, other links in the tree can be found. The links that do not point to an edge link to some non-reachable edges, all of which are not shown in the figure.

8.2.2 Weight of an RMT

Let consider a reachability tree for null boundary condition. An RMT r on an edge of the tree may be involved in more than one link. For example, consider an edge with label $l_{i.j} = \{0, 4\}$, where RMT 0 is involved in the self link and RMT 4 is involved in a forward link to $E_{i.k}$. Then $l_{i+1.2j}$ or $l_{i+1.2j+1}$ contains RMT 0 (or RMT 1). Here, RMT 0 or 1 is contributed by both the RMTs 0 and 4 of $l_{i.j}$. Therefore, as per rules of link, RMT 0 is involved in two links. In Fig. 8.6, the RMT 2 of $E_{2.4}$ and RMT 3 of $E_{2.5}$ are having two links: $E_{2.4}(2) \rightarrow E_{2.5}$, $E_{2.4}(2) \rightarrow E_{2.1}$, $E_{2.5}(3) \rightarrow E_{2.1}$, $E_{2.5}(3) \rightarrow E_{2.5}$.

To represent the fact, a weight $w_{i.j}^r$ of RMT $r \in l_{i.j}$ is introduced.[1] The $w_{i.j}^r$ is the total number of links from $E_{i.j}$ to itself or to some other edges for RMT r. If for each r $\in l_{i.j}$, $w_{i.j}^r = 0$, then $l_{i.j} = \emptyset$. If $l_{i.j} = \emptyset$ (for some j), the edge $E_{i.j}$ is non-reachable. Now, for each $s \in l_{i.k}$, the $w_{i.k}^s$ is decremented by 1 if $E_{i.k}(s) \rightarrow E_{i.j}$ for any k. After such operation, if the weights of all the RMTs of $l_{i.k}$ become 0, then the edge can be considered as non-reachable. The $w_{i.k}^s$ can never be 0 if RMT s is

[1] The weight of an RMT is also used in Sect. 5.2.1, but for different purpose.

involved in a self or *cross link* (see Sect. 8.4). Here, the RMTs ($s \in l_{i.j}$) for which the weights are 0 ($w^s_{i.j} = 0$) are not considered. So, the weight of an RMT may be more than one in some cases. In Fig. 8.6, the weights of all the RMTs in labels of edges at levels 0 and 1 are 1. But, in $l_{2.4}$ and $l_{2.5}$, the RMTs 2 and 3 have the weights 2 ($w^2_{2.4} = w^3_{2.5} = 2$). So, there are two links for these two RMTs.

To establish links among the edges at a level of a reachability tree, an edge can be considered as a structure of three parameters—its label, a list of weights of RMTs and a set of links of each RMT. To form links at level i, the rule $\mathcal{R}_i$ and the links of edges at level $i-1$ are needed. The weight of an RMT on the label of an edge at level 0 is 1. However, the weight of an RMT for the other levels can increase as per the above discussion.

For periodic boundary, the label of an edge is to be changed to the list of four sets of RMTs ($l_{i.j} = (\Gamma_k^{E_{i.j}})_{0 \le k \le 3}$) and the weight of an RMT is to be redefined to accommodate the set index (k in $\Gamma_k^{E_{i.j}}$) in the label. The weight of an RMT $r \in \Gamma_k^{E_{i.j}}$ is denoted as $w^r_{i.j.k}$ and is updated as per the above rule. For example, at edge $E_{3.0}$ of Fig. 8.7, the RMT 4 has the weight 2, because it comes from RMT 2 and RMT 6 of the previous level. Similarly, weight of RMT 7 on $E_{2.12}$ has weight 2. That is, $w^4_{3.0.0} = w^7_{3.12.3} = 2$. However, the basic logic of link formation remains same and the rule of link formation in the reachability tree of CAs in periodic boundary is to be followed.

8.3 The Reachability Problem

The links are to trace the configuration transitions in reachability tree by identifying the predecessor(s) of each configuration. Through the links, the predecessor of predecessor of a configuration, if exists, can be identified. Hence, the links follow *transitivity property*.

Proposition 8.2 *If* $E_{i.j}(r_1) \to E_{i.k}$ *and* $E_{i.k}(r_2) \to E_{i.p}$, *then* $E_{i.j}(r_1) \to E_{i.k}(r_2) \to E_{i.p}$.

Therefore, whether a configuration is reachable from other configuration that can be decided through the links of reachability tree. That is, the reachability of one configuration from another can effectively be analyzed by the links of reachability tree. This problem is known as the Configuration REachability Problem (CREP) in CA. This section develops an effective scheme to deal with the CREP.

8.3.1 Properties of Links

Let first identify the fundamental properties of links that can be derived from the rules of link formation. In some cases, the boundary condition affects the behavior of links.

It is already discussed that the links of level 0 are formed due to the first rule ($\mathcal{R}_0$) only. Whereas, the links of level $i+1$ ($i \geq 0$) are dependent on the rule $\mathcal{R}_{i+1}$ and the links of level i. Indeed, a link of level i *triggers* link(s) of level $i+1$.

Proposition 8.3 *A link of level i generally triggers two links at level $i+1$. In the reachability tree of an n-cell CA in null boundary, a link of level $n-2$ triggers one link at level $n-1$ whereas, in periodic boundary, a link of levels $n-3$ and $n-2$ trigger only one link to the next level.*

For example, the link $E_{1.3}(0) \rightarrow E_{1.0}$ of Fig. 8.6 triggers two links $E_{2.6}(0) \rightarrow E_{2.0}$ and $E_{2.6}(1) \rightarrow E_{2.0}$. However, an RMT may contribute only one RMT to the next level depending on the boundary condition, which is the reason of the exception.

Proposition 8.4 *For an n-cell CA, the total number of links in its reachability tree (forward, backward and self) in level i is:*

- For null boundary: 2^{i+2} *if* $0 \leq i \leq n-2$, *and it is* 2^{i+1} *if* $i = n-1$.
- For periodic boundary: 2^{i+3} *when* $0 \leq i \leq n-3$, 2^{i+2} *when* $i = n-2$, *and* 2^{i+1} *when* $i = n-1$.

In the reachability tree of a CA in null boundary, there are four RMTs in total in the labels of $E_{0.0}$ and $E_{0.1}$. So, for level 0, the total number of links is $4 = 2^{0+2}$. Since each link triggers two links to the next level, the total number of links in level 1 is $8 = 2^{1+2}$. Consequently, the number of links in level i, when $i \leq n-2$, is 2^{i+2}. For level $n-1$, however, the total number of links is 2^n, as one link of level $n-2$ triggers only one link at level $n-1$ (see Fig. 8.6 for verification). The similar argument is applicable to periodic boundary condition too. Since there are eight RMTs at root, the number of links formed among the edges at level 0 is $8 = 2^{0+3}$. Hence, any level i contains 2^{i+3} links when $i \leq 3$. The number is 2^{i+2} when $i = n-2$, and 2^{i+1} when $i = n-1$. That is, the number of links for both the levels are 2^n (see Fig. 8.7 for verification). The following property follows as the consequence of Propositions 8.3 and 8.4.

Proposition 8.5 *In a reachability tree of an n-cell CA,*

- For null boundary: *there are two links to $E_{i.j}$ from $E_{i.2p}$ or $E_{i.2p+1}$ or both for a pair of sibling RMTs $r, s \in l_{i.2p} \cup l_{i.2p+1}$, when $0 \leq i \leq n-2$, and there is only one link to $E_{i.j}$ from $E_{i.m}$, when $i = n-1$ ($0 \leq p \leq 2^i$ and $0 \leq j, m \leq 2^{i+1}-1$),*
- For periodic boundary: *(1) there are four links to $E_{i.j}$ – two from $E_{i.2p}$ or $E_{i.2p+1}$ or both for a pair of sibling RMTs $r, s \in \Gamma_k^{E_{i.2p}} \cup \Gamma_k^{E_{i.2p+1}}$ and other two from $E_{i.2q}$ or $E_{i.2q+1}$ or both, for a pair of sibling RMTs $r', s' \in \Gamma_{k+2}^{E_{i.2q}} \cup \Gamma_{k+2}^{E_{i.2q+1}}$ (k is either 0 or 1), when $0 \leq i \leq n-3$ ($0 \leq p, q \leq 2^i - 1$ and $0 \leq j \leq 2^{i+1}-1$),*
  (2) there are two links to $E_{n-2.j}$ from $E_{n-2.m}$, for RMT $r \in \Gamma_k^{E_{n-2.m}}$ and from $E_{n-2.m'}$, for RMT $r' \in \Gamma_{k+2}^{E_{n-2.m'}}$ ($0 \leq j, m, m' \leq 2^{n-1}-1$), and
  (3) there is only one link to $E_{n-1.j}$ from $E_{n-1.m}$ ($0 \leq j, m \leq 2^n - 1$).

If $E_{i-1.j}(r) \rightarrow E_{i-1.k}$, then for the pair of sibling RMTs $2r \pmod 8$ and $2r+1 \pmod 8$, $E_{i.2j}$ or $E_{i.2j+1}$ or both link to a single edge, either $E_{i.2k}$ or $E_{i.2k+1}$. Hence, there are two links to an edge from a given link. Since there are four pairs of sibling RMTs at the root, two pairs of them link to an edge of the root. As it continues to the next levels, the number of links to an edge remains as four.

Proposition 8.6 *Let consider a self link $E_{i.j}(r) \rightarrow E_{i.j}$ in the reachability tree of an n-cell CA, where $0 \leq i \leq n-1$, $0 \leq j \leq 2^{i+1}-1$. Then,*

i $r \in \{0,1\}$, *iff* $j \equiv 0 \pmod 4$.
ii $r \in \{2,3\}$, *iff* $j \equiv 1 \pmod 4$.
iii $r \in \{4,5\}$, *iff* $j \equiv 2 \pmod 4$.
iv $r \in \{6,7\}$, *iff* $j \equiv 3 \pmod 4$.

This result is independent of the boundary condition of the CA, and can be verified for $i = 0$ and 1. To formally prove the result by induction, let consider that it is true for $i = k$. That is, all the four cases hold in $E_{k.j}(r) \rightarrow E_{k.j}$. We now show that the result holds for the edges at level $k+1$.

Suppose $r = 0$, which implies $j \equiv 0 \pmod 4$. So, it produces RMTs 0 and 1 for the next level, which can participate in self link if they are on 0-edge –that is, on $E_{k+1.2j}$. If $E_{k+1.2j}$ is in a self link for RMT 0 or 1, then $2j \equiv 0 \pmod 4$, as $j \equiv 0 \pmod 4$. Hence, when $E_{k+1.2j}(s) \rightarrow E_{k+1.2j}$, then $s \in \{0,1\}$ iff $2j \equiv 0 \pmod 4$. Now consider $r = 1$, which also implies $j \equiv 0 \pmod 4$. The RMT 1 contributes RMTs 2 and 3 to the next level, which can participate in self link if they are on 1-edge ($E_{k+1.2j+1}$). In this case, when $E_{k+1.2j+1}(s) \rightarrow E_{k+1.2j+1}$, then $s \in \{2,3\}$ iff $2j+1 \equiv 1 \pmod 4$.

Hence, the above result holds for $j = k+1$ for case i. Therefore, it is true for any i $(0 \leq i \leq n-1)$. In the same line of argument, other cases can also be proved.

8.3.2 The CREP

The CREP is to check whether a configuration D of an n-cell CA is reachable from a configuration S. The configurations are denoted as $S = (s_i)_{0 \leq i \leq n-1}$ and $D = (d_i)_{0 \leq i \leq n-1}$. The S and D can also be identified in the reachability tree as sequences of edges. For ease of understanding, let define the sequences of edges as $(s_i)_{0 \leq i \leq n-1}$ representing S, and as $(d_i)_{0 \leq i \leq n-1}$ representing the D. Now, a path (link) from d_i to s_i is searched in the reachability tree. If no path exists, then D is not reachable from S.

Theorem 8.3 *For an n-cell CA, D is reachable from S, iff there exists a path from d_{n-1} to s_{n-1}, where $S = (s_i)_{0 \leq i \leq n-1}$ and $D = (d_i)_{0 \leq i \leq n-1}$.*

Proof Let consider that there is a path from d_{n-1} to s_{n-1} of length m –that is, $E_{n-1.j_1}(r_1) \rightarrow \cdots \rightarrow E_{n-1.j_q}(r_q) \rightarrow \cdots \rightarrow E_{n-1.j_m}$, where $d_{n-1} = E_{n-1.j_1}$ and $s_{n-1} = E_{n-1.j_m}$. Now, there can be a sequence of edges from the root to $E_{n-1.k}$ for each $k \in \{j_1, j_2, \cdots, j_m\}$ which represents a reachable configuration. Here, two reachable configurations which are represented by the edge sequences that end with $E_{n-1.j_p}$ and $E_{n-1.j_{p+1}}$ respectively are two consecutive configurations. Hence, a sequence of consecutive configurations can be obtained from S to D. Hence, D is reachable from S.

Let D be reachable from S. Obviously, there is a path from d_i to s_i, $0 \leq i \leq n$. Hence the proof. □

Example 8.4 Let consider $S = 0001$ and $D = 0100$ for the CA $(5, 73, 200, 80)$ in null boundary. Now, from Fig. 8.6, it can be noted that $d_3 = E_{3.4}$ and $s_3 = E_{3.1}$ (which is non-reachable here). From the tree with links, the path $E_{3.4}(2) \rightarrow E_{3.12}(2) \rightarrow E_{3.1}$ $(length(d_3, s_3) = 2)$ can be traced. Therefore, D is reachable from S, and $D = G^2(S)$ (see Fig. 8.4 for verification).

For the same CA, if $S = 0001$ and $D = 1010$, then $d_3 = E_{3.10}$ and $s_3 = E_{3.1}$. It can be observed from Fig. 8.6 that there is no path from d_3 to s_3. So, 1010 is not reachable from 0001.

To decide on the reachability in CA, form the root of the reachability tree (using $\mathcal{R}_0$), get edges from the root and identify links between the edges. If there exists any path from d_0 to s_0, then the process continues; otherwise, it is concluded that D is not reachable from S. If a path exists, then form the next level (using $\mathcal{R}_1$) and the links are to be identified as well as a path from d_1 to s_1. If no path exists, then D is not reachable from S. Otherwise, repeat the process for d_2 and s_2 and so on. Finally, if there exists a path from d_{n-1} to s_{n-1}, then the D is reachable from S.

However, the non-reachability of D from S, without tracing the path of d_i and s_i can be decided by checking some conditions related to d_i and s_i.

Condition 8.1 *In the reachability tree of an n-cell CA, if the edge* d_i $(0 \leq i \leq n-1)$ *is non-reachable, then D is not reachable from S, where* $D = (d_i)_{0 \leq i \leq n-1}$ *and* $S = (s_i)_{0 \leq i \leq n-1}$.

Reason: As per Theorem 8.3, the $D = (d_i)_{0 \leq i \leq n-1}$ is reachable from the $S = (s_i)_{0 \leq i \leq n-1}$ if there exists a path from d_{n-1} to s_{n-1}. From Proposition 8.3, it can be noted that the path at leaf level is triggered from the root. That is, if there is a path from d_{n-1} to s_{n-1}, then there are paths from d_0 to s_0, d_1 to s_1, $\cdots$, and d_{n-2} to s_{n-2}. If at any level, d_i is non-reachable, then there is no link from this edge. Hence, there is no path from d_{i+1} to s_{i+1}, $\cdots$, and d_{n-1} to s_{n-1}. Therefore, there is no path at leaf level and it can be concluded that D is non-reachable from S. □

Example 8.5 Consider the configurations $S = 0000$ and $D = 1011$ for the CA $(9, 170, 195, 80)$ in null boundary. From Fig. 8.8, it can be observed that $d_0 = E_{0.1}$ and $s_0 = E_{0.0}$, and there is a path $E_{0.1}(0) \rightarrow E_{0.0}$. At level 1, $d_1 = E_{1.2}$ and $s_1 = E_{1.0}$ and there is also a path $E_{1.2}(0) \rightarrow E_{1.0}$. Now, at level 3, $d_2 = E_{2.5}$ and $s_2 = E_{2.0}$

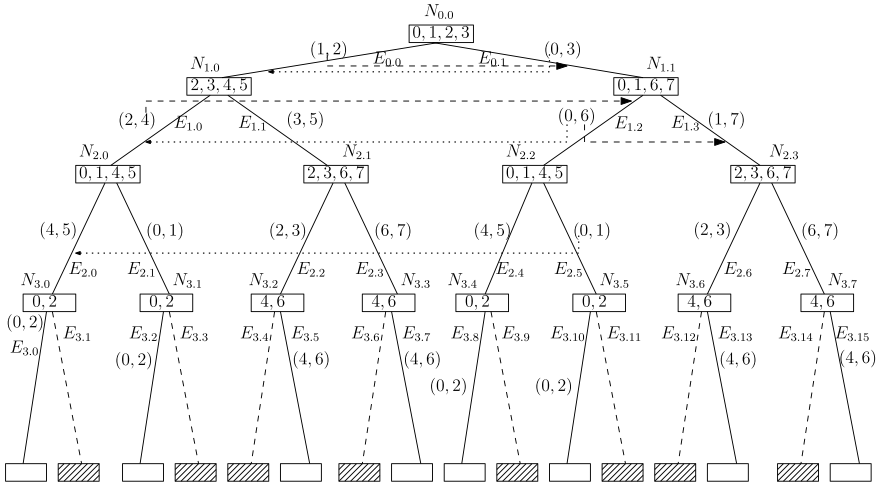

Fig. 8.8 Reachability tree of CA (9, 170, 195, 80) in null boundary

and there is a path $E_{2.5}(0) \rightarrow E_{2.0}$. At the last level, $d_3 = E_{3.11}$ and $s_3 = E_{3.0}$, but the edge d_3 is non-reachable. So, there is no path from d_3. Therefore, D is not reachable from S.

Condition 8.2 *In the reachability tree of an n-cell CA, if the edge s_i is self linked for two sibling RMTs and $d_i \neq s_i$, then $D = (d_i)_{0 \leq i \leq n-1}$ is not reachable from $S = (s_i)_{0 \leq i \leq n-1}$.*

Reason: As per Theorem 8.3, $D = (d_i)_{0 \leq i \leq n-1}$ is reachable from $S = (s_i)_{0 \leq i \leq n-1}$ if there exists a path from d_{n-1} to s_{n-1}, which implies that the paths from d_0 to s_0, d_1 to $s_1, \cdots$, and d_{n-2} to s_{n-2} exist. According to Proposition 8.5, there exist two links to $E_{i.j}$ from any edge (except leaf level). If the edge s_i is self linked for two RMTs, then no other edge can link to s_i. So, s_i can be reached only from the edge s_i and in that case, if $d_i \neq s_i$, there is no path from d_i to s_i. □

Example 8.6 Consider the configurations $S = 1111$ and $D = 0000$ of the CA (9, 170, 195, 80) in null boundary. From Fig. 8.8, it can be observed that $d_0 = E_{0.0}$ and $s_0 = E_{0.1}$, and there is path $E_{0.0}(2) \rightarrow E_{1.0}$. Now, at the next level, $d_1 = E_{1.0}$ and $s_1 = E_{1.3}$ and there is also a path $E_{1.0}(4) \rightarrow E_{1.2}(6) \rightarrow E_{1.3}$. At the next level, $d_2 = E_{2.0}$ and $s_2 = E_{2.7}$, and there is no path from d_2 to s_2 ($d_i \neq s_i$). The edge s_2 is self linked for the RMTs 6 and 7 of rule 195. Hence, the edge is not reachable from any other edge.

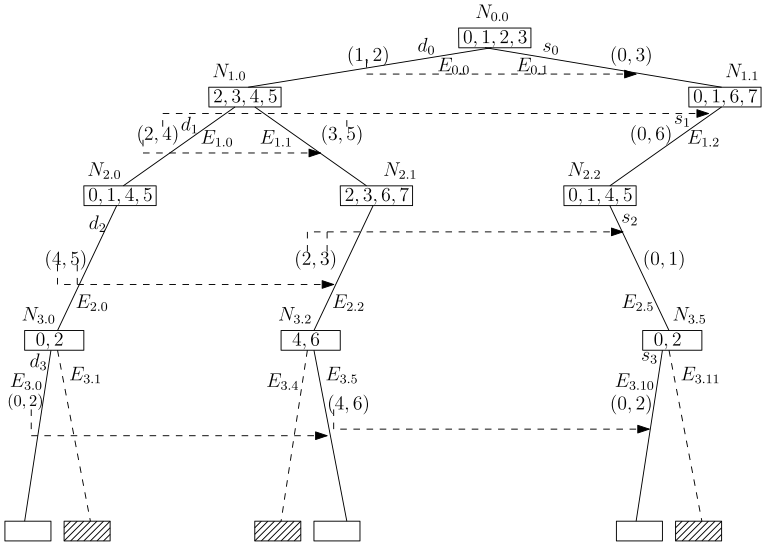

Fig. 8.9 Reachability tree of CA (9, 170, 195, 80) in null boundary. Irrelevant edges for $S = 1010$ and $D = 0000$ are removed

By definition, the reachability tree grows exponentially. In this particular problem, however, all the edges are not considered during processing. The edges, not in the path of d_i and s_i, are irrelevant also. To reduce the number of edges/nodes in the proposed decision procedure, such irrelevant edges are removed.

Example 8.7 Let consider the CA (9, 170, 195, 80) in null boundary and its two configurations $S = 1010$ and $D = 0000$. Figure 8.9 shows that D is reachable from S. The paths of d_i and s_i are shown in the figure. The edge $E_{1.3}$ is not in the path of d_1 and s_1, and so it is irrelevant in this particular case. Hence, $E_{1.3}$ is removed, and the corresponding subtree is not explored. Similarly, $E_{2.1}$, $E_{2.3}$, $E_{2.4}$, etc. are irrelevant, and hence removed.

Algorithm 11 is to decide on whether a configuration D of a CA in null boundary is reachable from a configuration S of the CA. The algorithm deals only with the labels of edges, and does not form the whole tree at a time. The two sets of labels $\{l_{i.0}, l_{i.1}, \cdots, l_{i.2^i-1}\}$ and $\{l_{i+1.0}, l_{i+1.1}, \cdots, l_{i+1.2^{i+1}-1}\}$ are noted in the algorithm as $l_0,\ l_1, \cdots$ and $l'_0,\ l'_1, \cdots$ respectively. Here, l_j corresponds to the label of $E_{i.j}$ and l'_k corresponds to the label of $E_{i+1.k}s$ $(0 \leq i \leq n-1)$. The input of the algorithm is the CA (*rule vector*), S (*Source*) and D (*Destination*). The output is 'Yes' if D is reachable from S; 'No' otherwise.

Algorithm 11: Decide reachability of D from S

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary, $S = (s_i)_{0 \le i \le n-1}$ and $D = (d_i)_{0 \le i \le n-1}$
Output: "Yes" if D is reachable from S; "No" otherwise.

Step 1: (a) Put RMT $r \in \{0, 1, 2, 3\}$ of $\mathcal{R}_0$ in l'_0 (resp. l'_1) if $\mathcal{R}_0[r] = 0$ (resp. 1), and get links for each RMT.
(b) Set $s \leftarrow s_0$ and $d \leftarrow d_0$
Step 2: Set $Count \leftarrow 1$, $i \leftarrow 0$ and goto Step 6
Step 3: If $i \ge n$, report "Yes", and exit.
Step 4: For label l_k, $0 \le k \le Count - 1$
Find l'_{2k} and l'_{2k+1} so that, if $r \in l_k$ and $s = 2r \pmod 8$ or $2r + 1 \pmod 8$, then $s \in l'_{2k}$ (resp. l'_{2k+1}) when $\mathcal{R}_i[s] = 0$ (resp. $\mathcal{R}_i[s] = 1$) and get the links for each RMT.
Step 5: Set $s \leftarrow 2 \cdot s + s_i$ and $d \leftarrow 2 \cdot d + d_i$
Step 6: Verify the following.
(a) If $l'_d = \emptyset$, report "No" and exit (Condition 8.1).
(b) If l'_s is self linked for two RMTs, report "No" and exit. (Condition 8.2).
Step 7: Search for paths from l'_d to l'_s. If no path exists, report "No" and exit.
Step 8: (a) Mark the labels (l'_js) which are not in any path, computed in Step 7, as *irrelevant*.
(b) $Count \leftarrow 2 \cdot Count - \#\, irrelevant\ labels$.
Step 9: (a) Assign the elements of l' to l without irrelevant labels, and accordingly update s and d.
(b) Set $i \leftarrow i + 1$, and goto Step 3.

Example 8.8 Let consider the CA $(9, 170, 195, 80)$ in null boundary, $S = 1010$ and $D = 0000$ (Fig. 8.9) as input to Algorithm 11. Here $l'_0 = \{1, 2\}$, $l'_1 = \{0, 3\}$, $s = 1$ and $d = 0$. A path from l'_0 to l'_1 exists (Step 7). Since there is no irrelevant label, so $Count = 2$. Next, 4 labels (Fig. 8.8) $l'_0 = \{2, 4\}, l'_1 = \{3, 5\}, l'_2 = \{0, 6\}$ and $l'_3 = \{1, 7\}$ are obtained (Step 4). Now, $s = 2$ and $d = 0$. The conditions of Step 6 are not satisfied, so the algorithm searches for a path from l'_0 to l'_2. There exists a path involving l'_0, l'_1 and l'_2 (see Fig. 8.9). That is, l'_3 is irrelevant. Hence, $Count = 3$ (Step 8(b)). Now, assign the following: $l_0 \leftarrow l'_0, l_1 \leftarrow l'_1, l_2 \leftarrow l'_2$, and update s and d as $s = 2$ and $d = 0$ (Step 9(a)). As the next step, the algorithm finds $l'_0, l'_1, \cdots, l'_5$ (Step 4) and sets $s = 5$ and $d = 0$. There exists a path involving l'_0, l'_2 and l'_5. So, l'_1, l'_3 and l'_4 are irrelevant. Hence, $Count = 3$ (Step 8(b)). Next, assign $l_0 \leftarrow l'_0, l_1 \leftarrow l'_2$, $l_2 \leftarrow l'_5$, and update $s = 2$ and $d = 0$ (Step 9(a)). Following execution of such steps, the algorithm reports "Yes"—that is, $D = 0000$ is reachable from $S = 1010$.

Complexity: In Algorithm 11, the worst case occurs if D is reachable from S and no labels (hence, edges) can be removed. That is, the reachability tree contains all the possible leaves. Therefore, the space requirement to store the two arrays l_i and l'_i is exponential. The time requirement is also exponential. However, Algorithm 11 performs well on an average. Because, in many cases, a large number of edges are removed, as well as before reaching the leaf of the tree, non-reachability can be decided.

8.4 The Cross Link

Applying the transitivity property of links (Proposition 8.2), a collection of forward and backward links can be identified which collectively form a loop. That is, a cycle of links like $E_{i.j_1}(r_1) \rightarrow \cdots \rightarrow E_{i.j_q}(r_q) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_1}$ can be observed. This cycle of links is referred to as the *cross link*. The length of a cross link is the number of edges involved in the link. Here, it is m. However, an edge may be visited more than once in a cross link.

Example 8.9 Let consider the reachability tree of Fig. 8.6. The edges at level 0 form a cross link of length 2: $E_{0.0}(2) \rightarrow E_{0.1}(1) \rightarrow E_{0.0}$. In level 1, the cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}(7) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ of length 5 is formed. Here, $E_{1.0}$ and $E_{1.3}$ are visited twice in the cross link. A cross link $E_{3.0}(0) \rightarrow E_{3.4}(0) \rightarrow E_{3.12}(0) \rightarrow E_{3.0}$ among the edges at level 3 can also be observed. The length of this cross link is 3, and the corresponding CA $\langle 5, 73, 200, 80\rangle$ contains a cycle of length 3 (Fig. 8.4).

Theorem 8.4 *An n-cell CA contains a cycle of length $m > 1$ iff a cross link of length m among the edges at level $n-1$ exists.*

Proof Let consider a cross link of length m ($m > 1$) among the edges at level $n-1$: $E_{n-1.j_1}(r_1) \rightarrow \cdots \rightarrow E_{n-1.j_q}(r_q) \rightarrow \cdots \rightarrow E_{n-1.j_m}(r_m) \rightarrow E_{n-1.j_1}$. So, there exist m sequences of edges from root to $E_{n-1.j_1}, E_{n-1.j_2}, \cdots$, and $E_{n-1.j_m}$, which represent m reachable configurations, say, $c_1, c_2, \cdots$, and c_m respectively. Then, by construction of link, c_2 is the predecessor of c_1, c_3 is the predecessor of c_2, and so on. Hence these m configurations form a cycle of length m.

Conversely consider that the CA contains a cycle of length m, which is formed with the configurations $c_1, c_2, \cdots$, and c_m. So, there exist m edges at level $n-1$ in the reachability tree against these m configurations. Since each of these configurations is a predecessor of other, the m edges at level $n-1$ form a cross link. Obviously, the length of the cross link is m. Hence the proof. □

8.4.1 Classification of Cross Links

A sequence of edges $\langle E_{i.j_1}, E_{i.j_2}, \cdots, E_{i.j_m}\rangle$ that forms a cross link among the edges at level i can be of two types–(1) no edge repeats in the sequence, and (2) at least one edge repeats. The first one is called *elementary cross link* and the other is the *compound cross link*.

Definition 67 A cross link is said to be *elementary cross link* if each of the edges is unique in the sequence of edges of the cross link. Otherwise, it is a *compound cross link*.

For example, in Fig. 8.6 at level 1, the cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ is elementary. On the other hand, $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}(7) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ is a compound cross link. Because, the edge $E_{1.3}$ appears twice in the cross link. Two or more elementary cross links can collectively form a compound cross link. For example, the compound cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}(7) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ consists of two elementary cross links $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ and $E_{1.3}(0) \rightarrow E_{1.0}(7) \rightarrow E_{1.3}$. Further, a compound cross link can also involve self links. For example, in Fig. 8.6, $E_{0.0}(3) \rightarrow E_{0.1}(2) \rightarrow E_{0.1}(0) \rightarrow E_{0.0}$ is a compound cross link involving the self link $E_{0.1}(2) \rightarrow E_{0.1}$.

Proposition 8.7 *A compound cross link is either a collection of elementary cross links, or a collection of elementary cross links and self links.*

An edge appears only once in an elementary cross link. Whereas, in compound cross link, repetition of edges is necessary. This repetition occurs through a pair of sibling RMTs.

Theorem 8.5 *Let $E_{i.j_1}(r_1) \rightarrow E_{i.j_2}(r_2) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_1}$ be a compound cross link in a reachability tree of an n-cell CA. Then, there exists at least one pair of sibling RMTs r and s in the labels of $E_{i.2p}$ and $E_{i.2p+1}$ such that $r, s \in \{r_1, r_2, \cdots, r_m\}$ and $2p, 2p+1 \in \{j_1, j_2, \cdots, j_m\}$. Here, $r, s \in l_{i.2p} \cup l_{i.2p+1}$ and $i < n-1$ for null boundary, and $r, s \in \Gamma_k^{E_{i.2p}} \cup \Gamma_k^{E_{i.2p+1}}$ and $i < n-2$ for periodic boundary ($k \in \{0, 1, 2, 3\}$).*

Proof Let consider a compound cross link $E_{i.j_1}(r_1) \rightarrow E_{i.j_2}(r_2) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_1}$ in a reachability tree of an n-cell CA. As at least one edge repeats in a compound cross link, there exist two links in the cross link such that $E_{i.u}(r) \rightarrow E_{i.t}$ and $E_{i.v}(s) \rightarrow E_{i.t}$. According to Proposition 8.5, such links form when the RMTs r and s are sibling to each other and $r, s \in l_{i.2k} \cup l_{i.2k+1}$ for null boundary ($r, s \in \Gamma_k^{E_{i.2p}} \cup \Gamma_k^{E_{i.2p+1}}$ for periodic boundary ($k \in \{0, 1, 2, 3\}$)). Hence proved. □

Example 8.10 In Fig. 8.6, among the edges at level 0, a compound cross link $E_{0.0}(3) \rightarrow E_{0.1}(2) \rightarrow E_{0.1}(0) \rightarrow E_{0.0}$ exists. Here, the RMTs $2 \in l_{0.1}$ and $3 \in l_{0.0}$ are sibling to each other. For the cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}(7) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$ at level 1, the RMTs $6 \in l_{1.1}$ and $7 \in l_{1.0}$ are also the sibling to each other.

Corollary 8.1 *In an elementary cross link $E_{i.j_1}(r_1) \rightarrow E_{i.j_2}(r_2) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_1}$, no pair of sibling RMTs r and s can exist such that $r, s \in l_{i.2p} \cup l_{i.2p+1}$ (in null boundary) or $r, s \in \Gamma_k^{E_{i.2p}} \cup \Gamma_k^{E_{i.2p+1}}$ (periodic boundary), and $r, s \in \{r_1, r_2, \cdots, r_m\}$, where $2p, 2p+1 \in \{j_1, j_2, \cdots, j_m\}$.*

Proof Let consider the cross link $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}(r_1) \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_0}$. As per Theorem 8.5, the existence of one pair of sibling RMTs $r, s \in \{r_0, r_1, \cdots r_m\}$ on edges $E_{i.2p}$ and $E_{i.2p+1}$ implies that the cross link is compound. Hence, in elementary cross link such conditions are not satisfied. □

Theorem 8.6 *All cross links among edges at level $n-1$ are the elementary cross links.*

Proof A cross link among the edges at level $n-1$ represents a cycle of length greater than 1 of the n-cell CA (Theorem 8.4). On the other hand, a sequence of edges, ending with an edge $E_{n-1.j}$, represents a reachable configuration. Now an edge $E_{n-1.j}$, included in a cross link, is unique as the configurations are unique in a cycle. Obviously, the edge $E_{n-1.j}$ cannot be part of two cross links. Hence, the cross links are elementary cross links. □

Example 8.11 Figure 8.6 shows a cross link $E_{3.0}(0) \rightarrow E_{3.4}(0) \rightarrow E_{3.12}(0) \rightarrow E_{3.0}$ at level 3. The edges of this cross link are not involved in any other cross links and it is an elementary cross link.

Since elementary cross links form compound cross links (Proposition 8.7), any change in the elementary cross links is reflected in the compound cross links. Due to this fact, only elementary cross links are considered for the characterization of cross links.

As cross links are derived from the forward links and backward links, the properties of links are also followed in cross links. Say, a cross link among edges at level i is $E_{i.j_1}(r_1) \rightarrow E_{i.j_2}(r_2) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_1}$. Now, each of the links (forward, backward and self) in the cross link triggers link(s) to the next level (Proposition 8.3). Suppose, $E_{i+1.p}(t) \rightarrow E_{i+1.q}$ is such a link, where $p, q \in \{2j_1, 2j_1+1, 2j_2, 2j_2+1, \cdots, 2j_m, 2j_m+1\}$ and $t \in \{2r_1 \pmod 8, 2r_1+1 \pmod 8, \cdots, 2r_m \pmod 8, 2r_m+1 \pmod 8\}$. If a cross link is formed at level $i+1$ that involves a set of the triggered links, then it is referred to as the cross link at level $i+1$ *triggered* by the cross link of level i. A cross link of level i may trigger more than one cross link at the level $i+1$. On the other hard, the cross link may also be discontinued at level $i+1$. The discontinuation of a cross link is known as *dissolving of cross link*.

Theorem 8.7 *In an n-cell CA, an elementary cross link of length m at level i either gets dissolved or it triggers a cross link, at least of length m, at level $i+1$, where $0 \leq i < n-1$.*

Proof Let consider an elementary cross link of length m at level i: $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}(r_1) \rightarrow \cdots \rightarrow E_{i.j_{m-1}}(r_{m-1}) \rightarrow E_{i.j_0}$. Each forward and backward link of the cross link, $E_{i.j_p}(r_t) \rightarrow E_{i.j_q}$ $(p, q, t \in \{0, 1, \cdots, m-1\})$, triggers two links for two RMTs $2r_t \pmod 8, 2r_t+1 \pmod 8$ from $E_{i+1.2j_p}$ or $E_{i+1.2j_p+1}$ to $E_{i+1.2j_q}$ $(E_{i+1.2j_q+1})$. So, there are only m edges like $E_{i+1.2j_q}$ $(E_{i+1.2j_q+1})$ which are linked from $E_{i+1.2j_p}(s_t)$ or $E_{i+1.2j_p+1}$. If any of these m edges is non-reachable, then no cross link forms and the cross link at level i dissolves. On the other hand, to form a cross link at level $i+1$, all of these m edges are to be involved for which m forward and backward links are needed. Hence, the length of the cross link, triggered by the cross link at level i, is at least m. □

Example 8.12 Let consider the CA (5, 73, 200, 80) with null boundary. In the reachability tree of the CA (Fig. 8.6), the cross link $E_{0.0}(3) \rightarrow E_{0.1}(0) \rightarrow E_{0.0}$ exists. It triggers two elementary cross links at level 1 –that is,

1. $E_{1.0}(7) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$, and
2. $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$

Here, the first cross link maintains the same length (2) whereas, the length of the second cross link is 3. It can be observed that at level 2, the first cross link gets dissolved, but the second cross link contributes two cross links of the same length—that is, $E_{2.0}(4) \rightarrow E_{2.4}(4) \rightarrow E_{2.6}(0) \rightarrow E_{2.0}$ and $E_{2.0}(5) \rightarrow E_{2.4}(5) \rightarrow E_{2.6}(1) \rightarrow E_{2.0}$. Finally, the second cross link triggers a cross link of length 3 at level 3.

8.4.2 Expansion of Elementary Cross Link

Expansion of cross link refers to the increment in length of a cross link at level $i + 1$ that is triggered by a cross link at level i. The lengths of elementary cross links and the number of cross links appearing at level i are inter-related. Further, the initial length of an elementary cross link is 2.

Proposition 8.8 *The length of elementary cross link is* 2 *at level* i*, where it appears first. If length of elementary cross link is greater than* 2 *at level* i*, then the cross link is triggered by a cross link at level* k*, where* $k < i$ *and* $0 \leq i \leq n - 2$.

Only self links at a level can trigger a forward link and a backward link, which can jointly form a cross link. There is no other way of forming cross link. If this cross link occurs, then its length is 2. Now, this cross link can trigger other cross links, length of which can increase.

According to Theorem 8.4, cross links at level $n - 1$ are responsible of forming cycles of length $m > 1$. Theorem 8.6 states that all the cross links at the last level are the elementary cross links. Hence, the length of an elementary cross link can increase. In case of an n-cell maximal length CA (Chap. 6), only one cross link of length $2^n - 1$ is formed among the edges at level $n - 1$.

Proposition 8.9 *The maximum length of any elementary cross link at level* i *(*$0 \leq i \leq n - 2$*) is the maximum number of possible edges of that level—that is,* 2^{i+1}.

In an elementary cross link, the edges involved are unique. The number of edges at level i is 2^{i+1}, $(0 \leq i \leq n - 1)$. Thus, the maximum number of edges of the cross link is 2^{i+1}.

In Fig. 8.10, at level 2, there are more than one elementary cross link—$E_{2.3}(1) \rightarrow E_{2.4}(7) \rightarrow E_{2.3}$, $E_{2.2}(0) \rightarrow E_{2.4}(5) \rightarrow E_{2.6}(5) \rightarrow E_{2.2}$, etc. But at the last level, there is only one cross link $E_{3.1}(4) \rightarrow E_{3.11}(0) \rightarrow \cdots \rightarrow E_{3.1}$ and it is elementary. Thus, elementary cross links can be combined in the next level, and can form an elementary cross link of larger length. A notable fact is that whenever length of cross

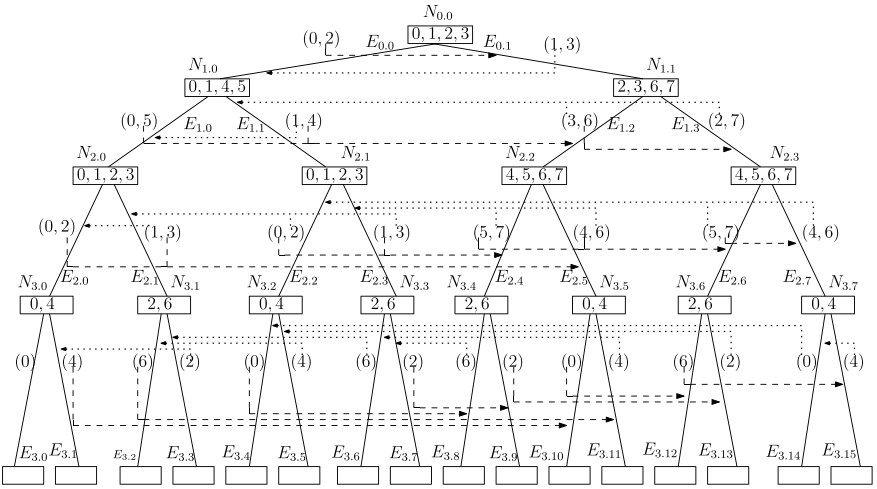

Fig. 8.10 Reachability tree with links of CA (90, 150, 90, 150) in null boundary

links of level i is increased in level $i+1$, at least one adjacent self link of level i forms a forward or backward link at level $i+1$, or two cross links are combined.

Lemma 18 *Length of an elementary cross link increases at level $i+1$, if there is a compound cross link at level i.*

Proof Let assume that there is an elementary cross link $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}(r_1) \rightarrow \cdots \rightarrow E_{i.j_m}(r_m) \rightarrow E_{i.j_0}$ at level i. Each link of the cross link $E_{i.j_p}(r_t) \rightarrow E_{i.j_q}$ $(p, q, t \in \{0, 1, \cdots, m\})$ triggers two links for the two RMTs $2r_t \pmod 8$, $2r_t + 1 \pmod 8$: from $E_{i+1.2j_p}$ or $E_{i+1.2j_p+1}$ to $E_{i+1.2j_q}$ $(E_{i+1.2j_q+1})$. To form an elementary cross link, any one of the links are to be considered. Involvement of both the links indicates that an edge appears twice in the cross link, which is not allowed in the elementary cross link. Hence, another cross link or self link at level i is necessary to form the cross link of greater length at level $i+1$. If there exist another self link or cross link among the edges $E_{i.j_0}, E_{i.j_1}, \cdots$, and $E_{i.j_m}$, then obviously there is a compound cross link at ith level. □

Hence, a compound cross link at level i can cause the increase of the length of cross link at level $i+1$. Let an elementary cross link be formed among the edges at level i, some of which are also self linked. Hence, these edges participate in a compound cross link. This is a situation when the length of the cross link at level $i+1$ can increase. If an edge from the same set of edges is also in self link, then the self link can trigger forward or backward link, which can cause the increase in length of the cross link at level $i+1$ by 1.

Example 8.13 Let consider the compound cross link $E_{0.0}(3) \rightarrow E_{0.1}(0) \rightarrow E_{0.0}(1) \rightarrow E_{0.0}$ at level 0 of Fig. 8.6. At level 1, the cross link triggers an elementary cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{0.0}$ of length 3. The self link

$E_{0.0}(1) \rightarrow E_{0.0}$ at level 0 results in a forward link $E_{1.0}(2) \rightarrow E_{1.1}$ that takes part in the cross link.

Theorem 8.8 *Two elementary cross links of length m at level i trigger an elementary cross link of length 2m at level* $i+1$ *(*$0 \leq i < n-1$*) if*
1. both the cross links involve same set of edges,
2. two RMTs r and s on an edge $E_{i.j}$*, participate in the cross links at level i, are sibling to each other, and*
3. $\mathcal{R}_{i+1}[r] \neq \mathcal{R}_{i+1}[s]$ *for each pair of sibling RMTs r and s of* $\mathcal{R}_{i+1}$*.*

Proof Let $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}(r_1) \rightarrow \cdots \rightarrow E_{i.j_{m-1}}(r_{m-1}) \rightarrow E_{i.j_0}$ and $E_{i.j_0}(s_0) \rightarrow E_{i.j_1}(s_1) \rightarrow \cdots \rightarrow E_{i.j_{m-1}}(s_{m-1}) \rightarrow E_{i.j_0}$ be the two cross links of length m. Since $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}$ and $E_{i.j_0}(s_0) \rightarrow E_{i.j_1}$ are the two links (forward or backward), r_0 and s_0 are sibling to each other (Proposition 8.5). Similarly, r_1 and s_1, r_2 and s_2, etc. are pairs of sibling RMTs. Now, the link $E_{i.j_0}(r_0) \rightarrow E_{i.j_1}$ triggers two links and the link $E_{i.j_0}(s_0) \rightarrow E_{i.j_1}$ triggers other two links at level $i+1$. Since r_0 and s_0 are sibling to each other, if the first two triggered links point to $E_{i+1.2j_1}$ (resp. $E_{i+1.2j_1+1}$) then the other two triggered links point to $E_{i+1.2j_1+1}$ (resp. $E_{i+1.2j_1}$). Similarly, $E_{i.j_1}(r_1) \rightarrow E_{i.j_2}$ and $E_{i.j_1}(s_1) \rightarrow E_{i.j_2}$ trigger links which point to both $E_{i+1.2j_2}$ and $E_{i+1.2j_2+1}$. In this way, all the $2m$ edges $E_{i+1.2j_1}$, $E_{i+1.2j_1+1}$, $E_{i+1.2j_2}$, $E_{i+1.2j_2+1}, \cdots$ are inter-linked and form a cross link. Hence the length of the cross link is $2m$. □

8.4.3 Dissolving of Cross Links

Cross links are not elementary links. They are derived from the forward links and backward links. Hence, the properties of these links are also applicable to cross links. By definition, the links at level i depend on the links at level $i-1$ ($1 \leq i \leq n-1, n$ is the number of cells of CA). A cross link at level i is also triggered by a cross link (or self link) at level $i-1$. However, a cross link at level i may not trigger a cross link at level $i+1$. When a cross link at level i does not trigger any cross link at level $i+1$, then the cross link of level i is dissolved.

Proposition 8.10 *An elementary cross link at level i gets dissolved at level* $i+1$*, if there exists a link* $E_{i+1.j}(r) \rightarrow E_{i+1.k}$ *which is triggered by a forward or backward link of the cross link at level i and* $E_{i+1.k}$ *is a non-reachable edge.*

A cross link at level i can also be dissolved without the presence of a non-reachable edge. Let $E_{i.j}(r) \rightarrow E_{i.k}(s) \rightarrow \cdots$ be an elementary cross link at level i. Now, the cross link can be dissolved if the RMTs $2r \pmod 8$, $2r+1 \pmod 8$ and RMT $2s \pmod 8$, $2s+1 \pmod 8$ of $\mathcal{R}_{i+1}$ fulfil some criteria. Let partition the eight RMTs as $S_0 = \{0, 1, 4, 5\}$ and $S_1 = \{2, 3, 6, 7\}$, and consider $a = 2r \pmod 8$ and $b = 2s \pmod 8$. Now, in any of the following cases, the cross link gets dissolved.

- Case 1: If the RMTs $a, a+1, b$ and $b+1$ are either from S_0 or from S_1, then a pair of sibling RMTs $a, a+1$ or $b, b+1$ of $\mathcal{R}_{i+1}$ are to be non-self-replicating.
- Case 2: If RMTs $a, a+1 \in S_1(S_0)$ and $b, b+1 \in S_0(S_1)$, then the RMTs $a, a+1$ (resp. $b, b+1$) of $\mathcal{R}_{i+1}$ are to be self-replicating and the RMTs $b, b+1$ (resp. $a, a+1$) are to be non-self-replicating.

In both of the above cases, either $E_{i+1.2j}$ or $E_{i+1.2j}$ links to either $E_{i+1.2k}$ or $E_{i+1.2k+1}$ for a pair of sibling RMTs, but no link between $E_{i+1.2k}$ and $E_{i+1.2k+1}$ is formed. Hence, the cross link gets dissolved. So, the existence of one or more pairs of sibling RMTs r' and s' of $\mathcal{R}_{i+1}$ with $\mathcal{R}_{i+1}[r'] = \mathcal{R}_{i+1}[s']$ can dissolve the cross link at level i.

In a right independent rule R, every pair of sibling RMTs r and s has the same next state—that is, $R[r] = R[s]$ (see Definition 34). Therefore, the right independent rules are the best candidate as $\mathcal{R}_{i+1}$ to dissolve the cross links at level i.

Proposition 8.11 *If there is a cross link at level i and $\mathcal{R}_{i+1}[r] \neq \mathcal{R}_{i+1}[s]$ (r, s are the sibling RMTs) but the cross link gets dissolved at level $i+1$ for $\mathcal{R}_{i+1}$, then there exists a set of rules that makes level $i+1$ cross link free.*

8.5 Cycle Structure

During its evolution, a finite CA always settles in a cycle. Number of configurations in the cycle defines the length of the cycle. If there are μ_1 cycles of length l_1, μ_2 cycles of length l_2, etc. in the configuration space of a CA, then this information of cycles can be written as $[\mu_1(l_1), \mu_2(l_2), \cdots]$, which is known as *cycle structure* of the CA (see also Sect. 7.1). For example, the CA (5, 73, 200, 80) in null boundary consists of four cycles, out of which 3 cycles have length 1 and the remaining cycle has length 3 (see Fig. 8.4). Hence, the cycle structure of the CA is [3(1), 1(3)].

However, all the configurations of a CA may not participate in cycles. Only *cyclic configurations* form cycles. A configuration of a CA is cyclic if it is in a cycle. So, to get the cycle structure of a CA, its cyclic configurations are to be considered.

8.5.1 Cyclic Configurations

Whether a configuration is cyclic or not that can be decided by testing its reachability from itself (see Sect. 8.3.2). The theories described till now can be used to extract cyclic configurations of a CA. Next two theorems further characterize the reachability tree to identify cyclic configurations.

Theorem 8.9 *An RMT in the label of an edge cannot be a part of the RMT sequence of a cyclic configuration if the RMT is not involved in a self link or cross link.*

Proof Let consider that an RMT r, in the label of an edge $E_{i.j}$ in the reachability tree of an n-cell CA, is not involved in self or cross link. This implies that $E_{i.j}(r) \rightarrow E_{i.k}$ for some k ($j \neq k$) and the link is not involved in cross link. If $i = n-1$, then $E_{i.j}$ cannot be a part of a cross link for RMT r. Now, if $i < n-1$, the link can trigger (forward or backward) link(s) at level $i+1$ that cannot participate in a cross link. Finally, a (forward or backward) link $E_{n-1.u}(s) \rightarrow E_{n-1.v}$, triggered by the link, can be observed for some u and v, where RMTs r and s are in an RMT sequence. If $E_{n-1.v}$ does not exist, then RMT s is not involved in a cross link. On the other hand, if both the $E_{n-1.u}$ and $E_{n-1.v}$ exist, then also no link can be found from $E_{n-1.v}$ to $E_{n-1.u}$ after applying the rules of link formation, due to the fact that there is no path from $E_{i.k}$ to $E_{i.j}$. Therefore, $E_{n-1.u}$ does not participate in a cross link for RMT s. However, there can be another case when the RMT r is not involved in a link (that is, $w^r_{i.j} = 0$). In that case, there is no question of getting the link $E_{n-1.u}(s) \rightarrow E_{n-1.v}$. Therefore, it can be concluded by following the argument of the proof of Theorem 8.4 that the RMT sequence involving the RMTs r and s is not in a cycle, because in no case $E_{n-1.k}$ is a part of the cross link for RMT s. Hence the proof. □

Example 8.14 Let consider the tree of Fig. 8.6 for the CA (5, 73, 200, 80) in null boundary. For the RMTs 6,7 $\in l_{2.1}$ at level 2, the edge $E_{2.1}$ is linked to $E_{2.7}$, which is a non-reachable edge. So, the RMT sequences (configurations) that involve the RMTs 6,7 $\in l_{2.1}$ – 3764 (1110) and 3776 (1111) are non-reachable, hence acyclic. Now, consider the link $E_{2.4}(2) \rightarrow E_{2.1}$ and $E_{2.5}(3) \rightarrow E_{2.1}$. As $E_{2.1}$ is linked to a non-reachable edge, it cannot participate in the cross link. Thus, the RMT sequences 0124 (0010) and 0136 (0011), involving these two RMTs, are acyclic. Figure 8.4 confirms that the configurations 0010, 0011, 1110 and 1111 are acyclic.

Corollary 8.2 *An n-cell CA contains m point attractors iff there exist m self-linked edges at level $n-1$ in the reachability tree of the CA.*

Proof The proof follows from the fact that a self link in an edge at level $n-1$ corresponds to a configuration whose predecessor is the configuration itself. □

Example 8.15 In Fig. 8.6, there are 3 self linked edges at level 3—$E_{3.7}(6)$, $E_{3.8}(0)$ and $E_{3.11}(6)$. So, the CA contains 3 point attractors. From the transition diagram of the CA (Fig. 8.4), it can be confirmed.

The following theorem is analogous to Condition 8.2.

Theorem 8.10 *If an edge $E_{i.j}$ in a reachability tree of an n-cell CA in null boundary is self linked for the two sibling RMTs $r, s \in l_{i.j}$, then the RMT sequences that involve the RMT $t \in l_{i.j} \setminus \{r, s\}$ represent acyclic configurations ($0 \leq i \leq n-2$, $0 \leq j \leq 2^{i+1}-1$).*

Proof Since the two links point to $E_{i.j}$ (self link) for the RMTs r and s, no more link can point to $E_{i.j}$ (Proposition 8.5). So, an RMT $t \in l_{i.j} \setminus \{r, s\}$ cannot participate in the self link. Further, $E_{i.j}$ cannot take part in any cross link. Hence, as per Theorem 8.9, the RMT t cannot be a part of an RMT sequence of a cyclic configuration. □

Theorem 8.9 points to the fact that an RMT r on a label of an edge cannot be a part of a cycle if it is not involved in a self or cross link. So, the RMTs in the label are excluded that are not involved in self or cross links from further processing. The RMT sequences that involve these excluded RMTs represent acyclic configurations. Finally, the RMTs that remain in the labels of the edges at level $n - 1$ participate in the self and cross links. The RMT sequences that involve only the RMTs in these labels represent cyclic configurations.

Example 8.16 Let check whether the configuration 0100 of the CA (5, 73, 200, 80) in null boundary is cyclic. The RMT sequence of the configuration is 1240. In level 0, self link is formed by RMT $1 \in l_{0.0}$ at edge $E_{0.0}$. Thus, verification goes to next level, where RMT $2 \in l_{1.0}$ forms a forward link at edge $E_{1.1}$, and becomes a part of cross link $E_{1.0}(2) \rightarrow E_{1.1}(6) \rightarrow E_{1.3}(0) \rightarrow E_{1.0}$. Hence, for the level 1, it also satisfies the condition, and is forwarded to level 2. In level 2, RMT $4 \in l_{2.0}$ participates in the cross link $E_{2.0}(4) \rightarrow E_{2.2}(4) \rightarrow E_{2.6}(0) \rightarrow E_{2.0}$. Finally, in level 3, RMT $0 \in l_{3.0}$ becomes the part of the cross link of length 3: $E_{3.0}(0) \rightarrow E_{3.4}(0) \rightarrow E_{3.12}(0) \rightarrow E_{3.0}$. Therefore, the rightmost RMT of the RMT sequence participates in a cross link at level 3, which is the level $n - 1$ as $n = 4$. Thus, the configuration 0100 is cyclic. Figure 8.4 verifies the result.

To count the cyclic configurations of a CA in null boundary, the weight $w^r_{i.j}$ of an RMT $r \in l_{i.j}$ is to be considered. The $w^r_{i.j}$ notes the total number of links from $E_{i.j}$ to itself or to some other edges for the RMT r (see Sect. 8.2.2). Algorithm 12 that counts cyclic configurations of a CA considers the weight of an RMT in a label. It deals with the labels of edges, as the formation of links depends on the RMTs of labels. If $E_{i.j}(r) \rightarrow E_{i.k}$ where $E_{i.k}$ is a non-reachable edge, then the weight of the RMT is decreased by 1. An RMT $r \in l_{i.j}$ is discarded from $l_{i.j}$, if $w^r_{i.j} = 0$. As discussed before, the weight of an RMT that participates in the self or cross link cannot be 0. The algorithm proceeds with the RMTs of a label having non-zero weights, as these RMTs can only take part in the RMT sequences of cyclic configurations (Theorem 8.9). If all the RMTs are discarded from a label, then the corresponding edge is considered as a non-reachable edge. Step 5 and Step 6 of Algorithm 12 recursively identify the new non-reachable edges at a level. The algorithm proceeds only with the non-empty labels (Step 7).

Algorithm 12: Counting of cyclic configurations

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: Number of cyclic configurations.

Step 1: Find the labels of the edges at level 0 using $\mathcal{R}_0$.
Set weight of each RMT of l_0 and l_1 as 1.
Step 2: Identify the links at level 0 of the reachability tree.
Step 3: For $i = 0$ to $n - 1$, repeat Step 4 to Step 8
Step 4: Mark the non-reachable edge of level i.
Step 5: If an edge at level i links a non-reachable edge for an RMT r, then
Decrease the weight of the RMT r by 1.
Step 6: If the weights of all the RMTs of a label becomes 0, then
Mark the corresponding edge as non-reachable.
Step 7: Proceed with unmarked edges to find the labels of edges at level $i + 1$,
using $\mathcal{R}_{i+1}$, if $i < n - 1$.
Step 8: Identify links among edges at level $i + 1$, if $i < n - 1$.
Step 9: Report the number of marked edges.

Example 8.17 This example illustrates the counting of cyclic configurations of a CA in null boundary. Let consider the CA $(5, 73, 200, 80)$ in null boundary. The reachability tree for the CA after establishing the links is shown in Fig. 8.6 (self links are not shown). The edges at a level are numbered from left to right. Here, $w^1_{0.0} = w^2_{0.1} = 1$ (RMTs 1 and 2 are involved in self links), and $w^3_{0.0} = w^0_{0.1} = 1$ (RMTs 3 and 0 are involved in a cross link). Further, $w^2_{1.0} = w^7_{1.0} = w^3_{1.1} = w^6_{1.1} = w^1_{1.2} = w^4_{1.2} = w^5_{1.2} = w^0_{1.3} = 1$ (RMTs 3, 4 and 5 are involved in self links, RMT 1 is involved in backward link and the rest RMTs are involved in the cross links—see Fig. 8.6 for verification). In the next level, a non-reachable edge is first found out. The RMTs 6 and 7 of $l_{2.1}$ are supposed to link the non-reachable edge. So, $w^6_{2.1} = w^7_{2.1} = 0$. Hence, the edge $E_{2.1}$ can be considered as a new non-reachable edge. However, $w^2_{2.4} = w^3_{2.5} = 2$, because $E_{2.4}(2) \rightarrow E_{2.1}$ and $E_{2.4}(2) \rightarrow E_{2.5}$, and $E_{2.5}(3) \rightarrow E_{2.1}$ and $E_{2.5}(3) \rightarrow E_{2.5}$. Since $E_{2.1}$ is now non-reachable, the links from $E_{2.4}$ and from $E_{2.5}$ to $E_{2.1}$ are to be removed, and $w^2_{2.4}$ and $w^3_{2.5}$ are to be decreased by 1. So, $w^4_{2.0} = w^5_{2.0} = w^4_{2.2} = w^5_{2.2} = w^6_{2.3} = w^7_{2.3} = w^0_{2.4} = w^1_{2.4} = w^2_{2.4} = w^3_{2.5} = w^0_{2.6} = w^1_{2.6} = 1$. Here, two edges (out of eight edges) are declared as non-reachable. Figure 8.11a shows these two non-reachable edges. Hence, at least $\frac{2}{}8$ configurations of the CA are acyclic. However, from the six remaining edges, there are seven leaves. Here, the weights of RMTs of labels of edges at level 3 are either 1 or 0. That is, $E_{3.9}$ is declared as non-reachable (Fig. 8.11b) because the weight of only RMT of $l_{3.9}$ (RMT 4) is 0. Finally, there are 6 leaves and, therefore, the number of cyclic configurations is 6 (see Fig. 8.4 for verification).

Complexity: The complexity of Algorithm 12 depends on n and on the number of edges of each level. Say m is the maximum number of edges of a level that are dealt with during the processing. So, the time requirement for handling the m edges is $O(m)$. Hence, the complexity of the algorithm is $O(mn)$.

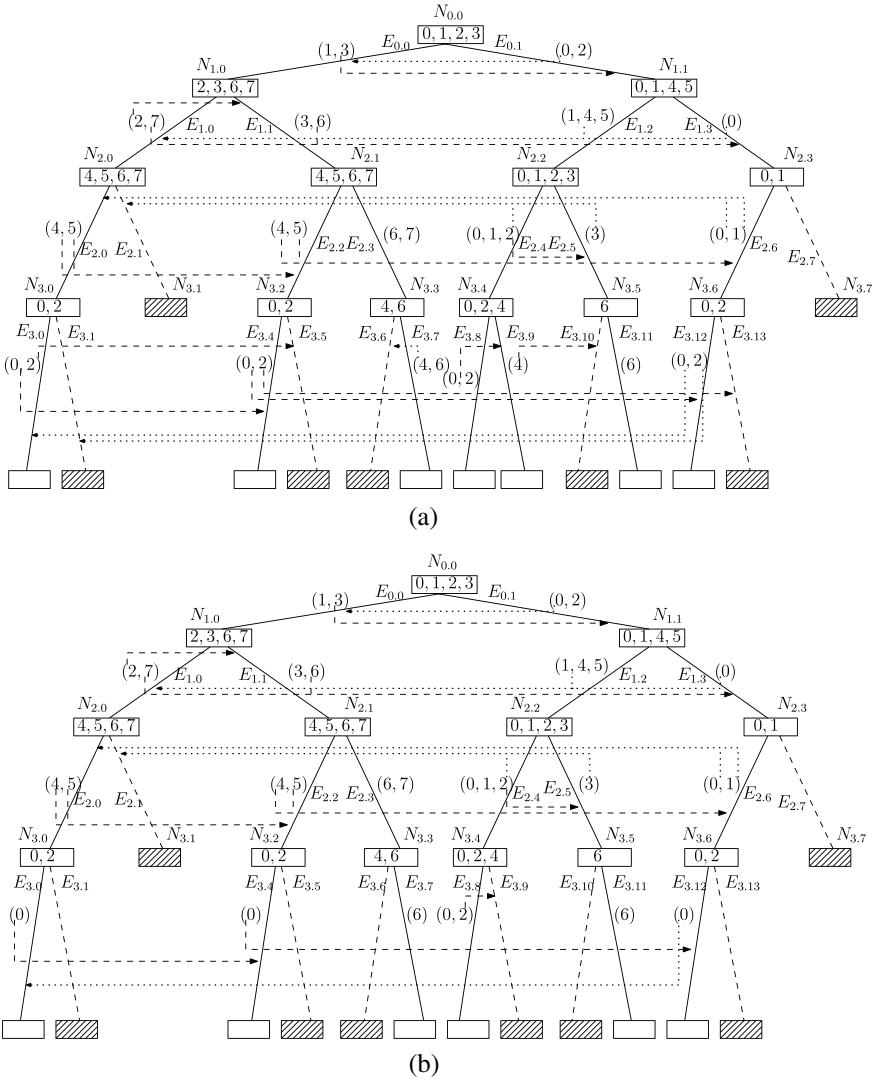

Fig. 8.11 Finding cyclic configurations of the (5, 73, 200, 80) in null boundary

8.5.2 Finding Cycle Structure

Algorithm 12 terminates after identification of self and cross links of level $n-1$. So, it can be modified to get the cycle structure of an n-cell CA. The length of each cross link is to be measured to output the cycle structure of the CA. However, complexity of such scheme is dependent on the number of cyclic configurations (m). If m is exponential for a CA (which is the case for reversible CAs), the complexity of the

algorithm becomes exponential. This can be improved if the CA is a *strictly reducible* CA (see Definition 62).

An n-cell strictly reducible CA (null boundary) can be considered as a composition of n_1-cell and n_2-cell CAs, where $n = n_1 + n_2$. Let introduce an operator '$\odot$' to denote this composition. Suppose, $\mathcal{R}$ is the rule vector of the n-cell CA, and $\mathcal{R}^{n_1}$ and $\mathcal{R}^{n_2}$ are the rule vectors of other two CAs. Then, $\mathcal{R} = \mathcal{R}^{n_1} \odot \mathcal{R}^{n_2}$. If $\mathtt{CS}_{\mathcal{R}}, \mathtt{CS}_{\mathcal{R}^{n_1}}, \mathtt{CS}_{\mathcal{R}^{n_2}}$ are the cycle structures of these three CAs respectively, then following relation holds as per discussion in Sect. 8.1.2:

$$\mathtt{CS}_{\mathcal{R}} = \mathtt{CS}_{\mathcal{R}^{n_1}} \times \mathtt{CS}_{\mathcal{R}^{n_2}}$$

Let $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n_1-1}, \mathcal{R}_{n_1}, \cdots, \mathcal{R}_{n-1})$ where $n_2 = n - n_1$. The simplest way to claim that the CA is a strictly reducible if $\mathcal{R}_{n_1-1}$ and $\mathcal{R}_{n_1}$ are right and left independent rules respectively. If it is found that there are k such component CAs with number of cells $n_1, n_2, \cdots, n_k$ ($n = n_1 + n_2 + \cdots + n_k$), the rule vectors of which are the $\mathcal{R}^{n_1}, \mathcal{R}^{n_2}, \cdots, \mathcal{R}^{n_k}$ respectively, then the $\mathcal{R}$ can be

$$\mathcal{R} = \mathcal{R}^{n_1} \odot \mathcal{R}^{n_2} \odot \cdots \odot \mathcal{R}^{n_k}$$

Therefore, to get the cycle structure of $\mathcal{R}$, the cycle structure of individual component CAs are to be computed.

If the CA is irreducible, then the above process cannot work. For irreducible CAs, as well as the component CAs, cycle structure can be found out by exploring the entire reachability tree. So, Algorithm 12 is to be utilized to get self and cross links among edges at level $n - 1$. From the self and cross links, the cycle structure can be identified. Algorithm 13 consolidates these steps to get the cycle structure of a CA in null boundary.

Algorithm 13: Find cycle structure of a CA in null boundary

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_i, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: The cycle structure of the CA.

Step 1: Find the labels of the edges at level 0 using $\mathcal{R}_0$.
Set weight of each RMT of l_0 and l_1 as 1.
Step 2: Identify the links at level 0 of the reachability tree.
Step 3: For $i = 0$ to $n - 1$, repeat Step 4 to Step 8
Step 4: Mark the non-reachable edge of level i.
Step 5: If an edge at level i links a non-reachable edge for an RMT r, then
Decrease the weight of the RMT r by 1.
Step 6: If the weights of all the RMTs of a label becomes 0, then
Mark the corresponding edge as non-reachable.
Step 7: Proceed with unmarked edges to find the labels of edges at level $i + 1$,
using $\mathcal{R}_{i+1}$, if $i < n - 1$.
Step 8: Identify links among edges at level $i + 1$, if $i < n - 1$.
Step 9: Find the number of self links and length of each cross link.
Report the cycle structure.

Table 8.1 Experimental results for evaluating cycle structures using Algorithm 13

CA size n	The rule vector $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$	Cycle structure $\mathtt{CS}_{\mathcal{R}}$
10	$(10, 150, (90, 150)^3, 90, 20)$	[1(1), 1(1023)]
12	$(14, 30^3, 45^2, 30^4, 45, 5)$	[1(1), 2(3), 1(4), 1(8)]
18	$(10, 150, 90, 150, 240^2, (90, 150)^2, 240^2, (90, 150)^2, 240, 80)$	[1(1), 1(3), 2(6), 20(12), 160(24)]
25	$((1, 32, 76, 44, 89)^4, 1, 32, 76, 44, 81)$	[297(2), 360(4), 1341(6), 216(12)]
28	$(6, 195, 8, 10, (102, 195, 8, 10)^5, 102, 195, 8, 0)$	[1(1)]
30	$((8^2, 4^2, 90^2)^4, 8^2, 4^2, 90, 80)$	[1024(1), 15878(2)]
35	$(4, 186, 15, 240, 80)^7$	[1280(4)]
42	$((8, 45, 30, 120, 64, 48)^6, 8, 45, 30, 120, 64, 16)$	[1(1), 5461(3)]
55	$((1, 21, 35, 95, 48)^{10}, 1, 21, 35, 95, 16)$	[1024(2)
100	$(8^{50}, 48^{49}, 16)$	[1(1)
500	$(10, 90, 240^2, (90^2, 240^2)^{123}, 90^2, 240, 80)$	[1(1)]

Example 8.18 Let consider the CA $(5, 73, 200, 80)$ in null boundary (Example 8.17). After Step 8, only the edges with self and cross links at level 3 remain (Fig. 8.6). There are 3 self links and one cross link among the remaining edges at length 3. So, the cycle structure of the CA is [3(1), 1(3)].

A sample output of Algorithm 13 is given in Table 8.1. The first column represents the number of cells (n) of a CA. The second and third columns refer to the rule vector of a CA and the cycle structure respectively. Although in many cases, the scheme can output the cycle structure, the worst case complexity of the algorithm is exponential.

Remark: If a CA is not strictly reducible, it can be weakly reducible. An n-cell weakly reducible CA contains signature of an n_1-cell CA, where $n > n_1$. If cycle structure of the n_1-cell CA is known, the length of the cycles of the n-cell CA can be predicted. The cycle lengths of the n-cell CA are the multiple of cycle lengths the n_1-cell CA.

8.6 Reversible CA with Large Length Cycle

A reversible CA contains cycles of different lengths. A cycle is referred to as the *large length cycle* if it covers more than half of its configurations.

Definition 68 An n-cell CA is said to have a large length cycle if the length of the longest cycle of the CA is greater than 2^{n-1}.

For example, a maximal length CA has the large length cycle. If a CA is reversible, performance of Algorithm 13 becomes the worst. And, if the CA is irreducible, then it becomes difficult to get the cycle structure of the CA. On the other hand, it is equally difficult to synthesize a reversible CA with large length cycle. In this section, a stochastic scheme of designing reversible CA which is expected to have large length cycle is described.

In an n-cell CA, the next state of a cell depends on the present states of the cell and its neighbors. Nevertheless, for some rules, this dependency may be limited. If some of the rules are independent of their neighbors, the lengths of cycles of the CA may not be very long. For example, if all the cells of a CA are self dependent –that is, the rules are either 204 or 51, then the cycle lengths of the CA can be either 1 or 2. So, dependency of a cell to its neighbors has a role on the cycle lengths of a CA. If each cell is *influenced* by its neighbors, the CA has higher chance to have larger cycle. To measure the dependence of cells of a CA, a parameter is next introduced.

8.6.1 Parameter for Measuring Dependence

If the RMTs of an equivalent RMT pair (that is, RMTs 0 and 4, 1 and 5, 2 and 6, and 3 and 7) of a rule $\mathcal{R}_i$ have the different next states, then the cell i is dependent on its left neighbor. If it is true for all such pairs, then the cell is *completely left dependent*. If some of the pairs hold this property, then the cell is *partially left dependent*. To measure the *degree of left dependence*, the parameter $\mathtt{P}_1(\mathcal{R}_i)$ is defined, where

$$\mathtt{P}_1(\mathcal{R}_i) = \frac{\text{No. of equivalent RMT pairs of } \mathcal{R}_i \text{ with different next states}}{4}$$

The $\mathtt{P}_1(90)$ is 1, because for rule 90 all the equivalent RMT pairs have different next states. But $\mathtt{P}_1(1) = 0.25$ and $\mathtt{P}_1(204) = 0$. However, if only the reversible CAs are considered, rules of which are always balanced, then $\mathtt{P}_1(\mathcal{R}_i)$ can only be 0, 0.5 and 1. Table 8.2 notes the $\mathtt{P}_1$ parameter of all the reversible rules in null boundary

Table 8.2 Categories of reversible rules on parameter $\mathtt{P}_1$

Category ($\mathtt{P}_1$)	$\mathcal{R}_i$
Completely left dependent (1)	60, 195, 15, 240, 90, 165, 150, 105, 30, 45, 75, 120, 135, 180, 210, 225
Partially left dependent (0.5)	53, 58, 83, 92, 163, 172, 197, 202, 54, 57, 99, 108, 147, 156, 198, 201, 23, 43, 77, 113, 142, 178, 212, 232, 27, 39, 78, 114, 141, 177, 216, 228, 86, 89, 101, 106, 149, 154, 166, 169
Left independent (0)	51, 204, 85, 170, 102, 153

(Sect. 4.5). Based on the value of $\mathtt{P}_l$, the rules are categorized as completely left dependent, partially left dependent and left independent.

In similar fashion, parameter $\mathtt{P}_r(\mathcal{R}_i)$ is defined to measure the *degree of right dependence* of a cell. In this case, the pairs of sibling RMTs (that is, RMTs 0 and 1, 2 and 3, 4 and 5, and 6 and 7) of a rule are to be considered.

$$\mathtt{P}_r(\mathcal{R}_i) = \frac{\text{No. of sibling RMT pairs of } \mathcal{R}_i \text{ with different next states}}{4}$$

The $\mathtt{P}_r$ can also assume only three values – 0, 0.5 and 1 for the reversible rules. Table 8.3 shows the $\mathtt{P}_r$ of the reversible rules in null boundary.

Based on the $\mathtt{P}_l$ and $\mathtt{P}_r$ parameters, the parameter $\mathtt{P}(\mathcal{R}_i)$ is defined, where

$$\mathtt{P}(\mathcal{R}_i) = \mathtt{P}_r(\mathcal{R}_i) * \mathtt{P}_l(\mathcal{R}_i).$$

Therefore, $\mathtt{P}(\mathcal{R}_i)$ of reversible rules can be 0, 0.25, 0.5 or 1 and considered as the measure of neighborhood dependence of a cell with rule $\mathcal{R}_i$. The reversible rules are classified into four categories based on $\mathtt{P}(\mathcal{R}_i)$—that is, *completely dependent*, *partially dependent*, *weakly dependent* and *independent* corresponding to the $\mathtt{P}$ values of 1, 0.5, 0.25 and 0 respectively (see Table 8.4).

Table 8.3 Categories of reversible rules on parameter $\mathtt{P}_r$

Category ($\mathtt{P}_r$)	$\mathcal{R}_i$
Completely right dependent (1)	102, 153, 85, 170, 90, 165, 150, 105, 86, 89, 101, 106, 149, 154, 166, 169
Partially right dependent (0.5)	53, 58, 83, 92, 163, 172, 197, 202, 54, 57, 99, 108, 147, 156, 198, 201, 23, 43, 77, 113, 142, 178, 212, 232, 27, 39, 78, 114, 141, 177, 216, 228, 30, 45, 75, 120, 135, 180, 210, 225
Right independent (0)	51, 204, 60, 195, 15, 240

Table 8.4 Four categories of reversible rules on the parameter $\mathtt{P}$

Category ($\mathtt{P}$)	$\mathcal{R}_i$
Completely dependent (1)	90, 165, 150, 105
Partially dependent (0.5)	30, 45, 75, 120, 135, 180, 210, 225, 86, 89, 101, 106, 149, 154, 166, 169
Weakly dependent (0.25)	53, 58, 83, 92, 163, 172, 197, 202, 54, 57, 99, 108, 147, 156, 198, 201, 23, 43, 77, 113, 142, 178, 212, 232, 27, 39, 78, 114, 141, 177, 216, 228
Independent (0)	51, 204, 85, 170, 102, 153, 60, 195, 15, 240

Table 8.5 Categories of $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$ based on $\mathtt{P}$

Category	$\mathcal{R}_0$	$\mathcal{R}_{n-1}$
Completely Dependent	5, 6, 9, 10	5, 20, 65, 80
Independent	3, 12	17, 68

In null boundary, only four RMTs are valid for the first and last rules—RMTs 0, 1, 2 and 3 for $\mathcal{R}_0$ and RMTs 0, 2, 4 and 6 for $\mathcal{R}_{n-1}$. So, only $\mathtt{P}_r$ is effective for the first cell whereas, only $\mathtt{P}_l$ is effective for the last cell. So, the final parameter for the first and last rules are defined as $\mathtt{P}(\mathcal{R}_0) = \mathtt{P}_r(\mathcal{R}_0)$ and $\mathtt{P}(\mathcal{R}_{n-1}) = \mathtt{P}_l(\mathcal{R}_{n-1})$ and can be either 0 (independent) or 1 (completely dependent) for a reversible CA. As discussed in Sect. 4.5.1, only sixteen rules for the first cell and another sixteen rules for the last cell are applicable. Out of these sixteen rules, only six rules can participate in reversible CAs (see Tables 4.7 and 4.8 of Chap. 4). Table 8.5 notes the categories of $\mathcal{R}_0$ and $\mathcal{R}_{n-1}$.

8.6.2 Stochastic Synthesis Scheme

The target of this scheme is to synthesize CAs with large length cycle. The hypothesis is that the more dependence of cells on their neighbors, there are more chances of getting CAs with large length cycle. If rules are chosen only from completely dependent category (see Table 8.4), then overall dependency of cells gets maximized. However, by this approach, only linear and complemented CAs are synthesized. No non-linear CA can be synthesized as the rules of this category are either linear or complemented. However, non-linear CAs can have large cycles, which can even be maximal length CAs (see Sect. 6.6). Non-linear maximal length CAs contain a small percentage of partially dependent rules, sometime even weakly dependent rules (see Table 6.9 for reference). Hence, in the stochastic synthesis scheme, although most rules are taken from the completely dependent category, a provision is kept to probabilistically choose rules from the partially and weakly dependent categories.

As discussed in Sect. 4.5.3, Tables 4.6, 4.7 and 4.8 are to be followed to synthesize reversible CAs in null boundary. The stochastic synthesis scheme follows these tables and selects rules considering the $\mathtt{P}$. The first and last rules are chosen from the completely dependent category. The rest $n-2$ rules of an n-cell reversible CA are chosen following a Gaussian distribution, such that maximum rules are chosen from completely dependent category, some are from partially dependent category and a few are from the weakly dependent category.

An issue with this approach is that the category of the rules takes discrete values whereas, the Gaussian distribution works only for continuous variables. Thus let us introduce X, a continuous random variable, which follows normal distribution, in particular with mean $\frac{1}{2}$ and standard deviation $\frac{1}{6}$. The mean and standard deviation

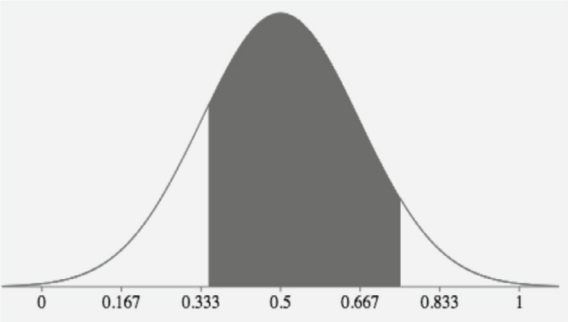

Fig. 8.12 Normal distribution of a random variable with mean 0.5 and standard deviation = 0.167

are so selected that most of the values of X (99.74%) lies between 0 and 1. The probability distribution function of X is given in Fig. 8.12.

The process of selecting each of the $(n-2)$ rules is–

1. Pick up a random number X following this normal distribution.
2. If X is less than 0.35, choose a rule from the partially dependent category.
3. Otherwise, if X is in between 0.35 and 0.75, select a rule from the completely dependent category; else, pick a rule from the weakly dependent category.

The probability that the selected rule belongs to the partially dependent category is, therefore,

$$Prob[X \leq 0.35] = Prob[Z \leq \frac{0.35 - 0.5}{0.167}] = \Phi[-0.898] = 0.1841$$

where Z and Φ are the standardized normal variable and the cumulative distribution function of the standard normal distribution. Similarly, probability that the selected rule belongs to the completely dependent category (shown by the shaded region in Fig. 8.12) is

$$\begin{aligned} Prob[0.35 \leq X \leq 0.75] &= Prob[\frac{0.35 - 0.5}{0.167} \leq Z \leq \frac{0.75 - 0.5}{0.167}] \\ &= \Phi[1.497] - \Phi[-0.898] = 0.7491 \end{aligned}$$

and finally, the probability that the selected rule belongs to the weakly dependent category is

$$Prob[X \geq 0.75] = Prob[Z \geq \frac{0.75 - 0.5}{0.167}] = 1 - \Phi[1.497] = 0.0668.$$

Since reversible CAs follow a specific sequence of rules, it may so happen that no rule of the chosen category is allowed at a particular position of the rule sequence (as per Table 4.6). In such cases, rules from the completely dependent category are

chosen, because each class contains at least two rules from completely dependent category. Following example illustrates the complete synthesis scheme.

Example 8.19 Let consider a 4-cell CA which conforms to the above mentioned stochastic synthesis process. Let $\mathcal{R}_0$ be 9. Therefore, $\mathcal{R}_1$ is to be selected from Class III (following Table 4.6). To choose $\mathcal{R}_1$, generate X randomly. Let $X = 0.25$. Therefore, $\mathcal{R}_1$ should be selected uniformly at random from the set of rules {23, 43, 77, 113, 142, 178, 212, 232, 27, 39, 78, 114, 141, 177, 216, 228}, which are of Class III and partially dependent rules. Let $\mathcal{R}_1 = 113$. Next, the class of $\mathcal{R}_2$ is found to be IV (Table 4.6). Now, let the randomly generated $X = 0.95$. Therefore, $\mathcal{R}_2$ should be selected uniformly at random from Class IV rules that are also from weakly dependent category. But, Class IV does not contain such a rule. So, $\mathcal{R}_2$ is chosen uniformly at random from the set {90, 105, 150, 165}. Let $\mathcal{R}_2 = 90$. So, $\mathcal{R}_3$ is to be the member of Class IV. Let $\mathcal{R}_3 = 20$ be randomly selected from Table 4.8. Therefore, the stochastically synthesized CA is (9, 113, 90, 20).

In experimental runs of the above scheme, it is observed that around 50% of the synthesized CAs have a cycle of length more than 2^{n-1}. For example, out of 2077 stochastically synthesized CAs generated in a sample experiment, for $n = 10$, 1000 CAs have the large length cycles. Table 8.6 shows some sample CAs with large

Table 8.6 The n-cell CAs having cycle of length more than 2^{n-1}

n	Cycle length	CA
10	1023	(10, 90, 105, 166, 105, 165, 105, 150, 150, 65)
	827	(9, 43, 105, 154, 105, 165, 150, 90, 150, 20)
	1017	(9, 90, 89, 150, 165, 150, 106, 90, 89, 80)
	761	(5, 90, 165, 180, 154, 165, 106, 165, 90, 80)
	1003	(10, 105, 90, 150, 57, 150, 90, 105, 150, 65)
11	1771	(6, 150, 135, 169, 105, 90, 105, 90, 150, 165, 65)
	1959	(5, 90, 149, 165, 105, 150, 165, 90, 150, 105, 80)
	2003	(5, 90, 166, 105, 150, 86, 105, 165, 150, 90, 80)
	2044	(10, 165, 54, 149, 105, 105, 165, 165, 165, 150, 65)
12	3890	(5, 150, 57, 89, 165, 154, 105, 105, 105, 150, 90, 20)
	3961	(5, 90, 58, 165, 90, 90, 165, 90, 165, 105, 165, 80)
	3999	(10, 105, 58, 105, 105, 165, 165, 150, 105, 101, 150, 65)
	3939	(10, 45, 58, 90, 165, 105, 166, 90, 165, 165, 90, 5)
13	8021	(6, 90, 177, 105, 150, 105, 150, 165, 150, 105, 150, 154, 5)
	8143	(10, 30, 149, 150, 150, 165, 90, 89, 105, 90, 90, 165, 20)
	8161	(9, 150, 30, 105, 90, 154, 165, 101, 150, 105, 105, 105, 65)
	8179	(5, 180, 101, 90, 150, 150, 105, 105, 165, 101, 165, 90, 5)
	8191	(9, 113, 165, 90, 150, 105, 105, 105, 150, 150, 90, 90, 65)
14	16381	(10, 135, 166, 165, 165, 165, 150, 105, 165, 105, 105, 165, 165, 5)
	16293	(6, 165, 101, 90, 150, 165, 165, 90, 90, 90, 154, 90, 90, 80)
	16118	(6, 113, 90, 105, 86, 165, 165, 150, 105, 154, 165, 105, 105, 65)

length cycle, explored in this process, for $n = 10$ to 14. It is observed that the largest cycle length of these CAs varies from 2^{n-1} to $2^n - 1$.

The stochastic synthesis scheme can give reversible CAs with large length cycle. However, to get the exact cycle structure, Algorithm 13 is to be employed. It is highly probable that the synthesized CAs are irreducible and the algorithm may encounter its worst case. Therefore, if the n is not small, it is difficult to find cycle structure of these CAs.

8.6.3 Concatenation of Cellular Automata

Let us now extend the $\odot$ operator which has been introduced in Sect. 8.5.2 to concatenate multiple CAs for getting CAs with longer cycles. Let $\mathcal{R}^{n_1}, \mathcal{R}^{n_2}, \cdots, \mathcal{R}^{n_k}$ be the rule vectors of the CAs with number of cells $n_1, n_2, \cdots, n_k$ respectively. Now, the rule vector $\mathcal{R}$ of an n-cell CA, where $n = n_1 + n_2 + \cdots + n_k$, can be synthesized by applying the $\odot$ operator $(k-1)$ times. That is,

$$\mathcal{R} = \mathcal{R}^{n_1} \odot \mathcal{R}^{n_2} \odot \mathcal{R}^{n_3} \odot \cdots \odot \mathcal{R}^{n_k}$$

Suppose l_{m_i} is the length of the longest cycle of CA $\mathcal{R}^{n_i}$ $(i \in 1, 2, \cdots, k)$. Then, the n-cell CA $(\mathcal{R})$ contains cycle(s) of length $lcm(l_{m_1}, l_{m_2}, \cdots, l_{m_k})$ (see Sect. 8.1.2). If $l_{m_1}, l_{m_2}, \cdots, l_{m_k}$ are mutually co-prime, then the longest cycle of $\mathcal{R}$ is $l_{m_1} \times l_{m_2} \times \cdots \times l_{m_k}$. Hence, by suitably choosing the component CAs $(\mathcal{R}^{n_i})$, an n-cell CA can be synthesized that has a cycle of length greater than 2^{n-1}. For ease of computation, a database of component CAs with large length cycles can be created. The stochastic synthesis scheme of the above may be used to prepare such a database. Then deterministically a CA with large length cycle can be synthesized by suitably concatenating the component CAs.

Further, for an $l > 0$, an n-cell reversible CA can be synthesized that has a cycle of length at least l, following the approach described above. The target, however, is to be to get such a CA with smallest possible n. The l and n maintain the relation $n \geq \lfloor \log_2 l \rfloor + 1$. Ideally, the desired CA should have $\lfloor \log_2 l \rfloor + 1$ cells.

If $\lfloor \log_2 l \rfloor + 1$ is not large, then the stochastic synthesis scheme can be used to address the problem. Otherwise, the synthesis can be done with the component CAs. The following steps describe the synthesis of CA with cycles of length l.

1. Select two component CAs with cycles of length l_{m_1} and l_{m_2} such that l_{m_1} and l_{m_2} are mutually co-prime.
2. Concatenate the component CAs to get a CA with a cycle of length $l_{m_1} \times l_{m_2}$.
3. If $l_{m_1} \times l_{m_2} \geq l$, report the CA as a solution. Otherwise, choose another component CA having cycle of length l_{m_3} such that $l_{m_1} \times l_{m_2}$ and l_{m_3} are mutually co-prime.
4. Concatenate them to get a new CA, and repeat the process until a solution is found.

8.7 Chapter Notes

The link in reachability tree has been first proposed in [1]. An elaboration can be found in the PhD dissertation of Nazma Naskar [2]. Reachability tree to solve the reachability problem has been explored in [3]. Synthesis of non-linear reversible CAs with large length cycles has been reported in [4–6] and later refined in [7]. Further investigation of the CAs with large length cycles, as well as classifications of CAs as reducible and irreducible can be found in the dissertation of Mukherjee [8].

There are several directions to extend the work in future. Some of them are indicated below.

1. Efficient identification of reducible and irreducible CAs. Component CAs of a reducible CA are to be identified.
2. (A variation of the above problem) Efficiently decide on the isomorphism of two n-cell CAs.
3. Estimation of cycle structure of a weakly reducible CA from the cycle structure of component CAs.
4. Defining efficient schemes to find cycle structure of a non-linear CA.

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Chapter 9
Convergence

During its evolution, an irreversible finite cellular automaton (CA) approaches to a set of configurations that form a cycle called *attractor*. Attractors can be of different lengths. If the length of an attractor is 1, then it is a *point attractor*, which is also called as *point state attractor*, *fixed point*, *single length cycle attractor*. When all the attractors of a CA are point attractors, the CA is known as Multiple Attractor Cellular Automaton (MACA). As a special case, if the number of attractors is one and it is a point attractor, then the CA is a Single Attractor Cellular Automaton (SACA). This chapter deals with the cellular automata (CAs) that form only point attractors in their configuration space. Such a CA is converging—that is, if initialized with a configuration, the CA eventually reaches a point attractor .

This chapter first explores the linear CAs following theories of Chap. 7. Then the characterization of the CA with point attractors are studied through reachability tree. The non-linear CAs with point attractors only are analyzed and synthesized exploiting the *links* of the reachability tree, reported in Chap. 8. Finally, the density classification problem and SACAs are studied in this chapter.

9.1 Multiple Attractor Cellular Automata

A multiple attractor cellular automaton (MACA) forms only point attractors in its configuration space. The rest configurations are acyclic. The configurations of an MACA form trees in its transition diagram. Such a tree, rooted at a point attractor, say α, is known as the α-basin. The leaves of the tree are non-reachable configurations. The height of a basin is the maximum number of hops a CA needs to reach the point attractor from a non-reachable configuration of the basin. The depth (d) of a CA is the maximum height of all basins of the CA.

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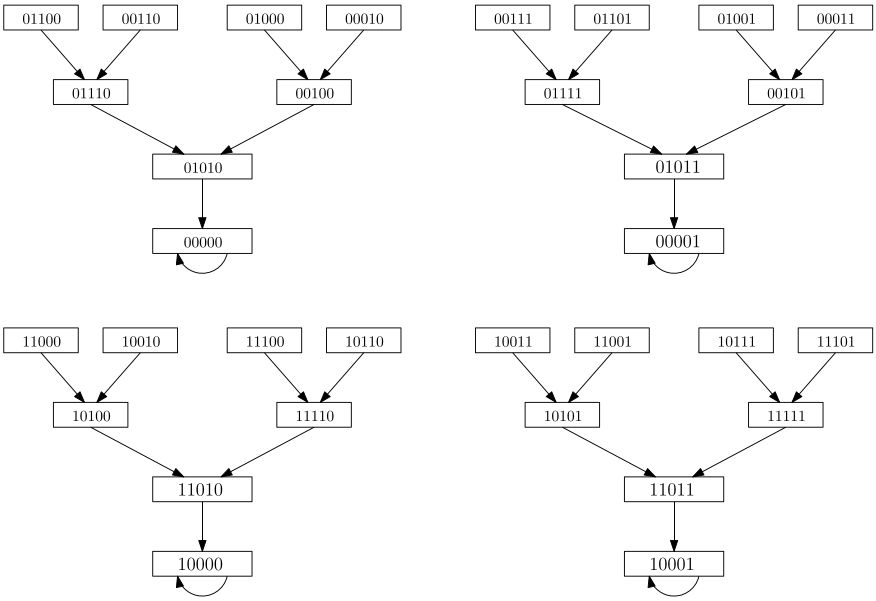

Fig. 9.1 Transition diagram of 5-cell MACA (204, 170, 90, 240, 204) with null boundary

Definition 69 The *depth* of a CA is the maximum number of hops that the CA needs to reach a cyclic configuration from a non-reachable configuration.

Let consider a 5-cell MACA (204, 170, 90, 240, 204) in null boundary. Figure 9.1 shows the transition diagram of the CA. It is a linear CA. The CA has four basins, which form four trees, rooted at four point attractors 00000, 00001, 10000 and 10001. Height of each basin is 4, so the depth of the MACA is 4. If the point attractors are at level 0 of the tree, then their predecessors are at level 1, and so on. For example, the configuration 01110 is at level 2 of the 0-basin of the MACA.

9.1.1 Linear MACA

As noted in Sect. 7.2, the characteristic polynomial of an irreversible CA is of the form

$$f(x) = x^{n_1} \cdot x^{n_2} \cdots x^{n_l} \cdot [\phi_{l+1}(x)]^{n_{l+1}} \cdots [\phi_{\mathcal{N}}(x)]^{n_{\mathcal{N}}} \tag{9.1}$$

where $\phi_i(x)$ is an irreducible polynomial ($i = l+1, \ldots, \mathcal{N}$) other than x, and each term of the $f(x)$ is an elementary divisor. The irreducible polynomials correspond to the cycles that are the attractors here.

If each of the irreducible polynomial is $(x+1)$—that is, $\phi_i(x) = x+1$ for each $i \in \{l+1, \ldots, \mathcal{N}\}$, then the CA contains only the point attractors, and the CA is

an MACA. In that case, the characteristic polynomial of an n-cell CA has the form $f(x) = x^{n-m}(x+1)^m$ and Eq. 9.1 can have the form

$$x^{d_1}.x^{d_2}\cdots x^{d_p}(x+1).(x+1)\cdots m \text{ times}$$

where $d_1 \geq d_2 \geq \cdots \geq d_p$ and $d_1 + d_2 + \cdots + d_p = n - m$. The minimal polynomial of such an MACA is, therefore, $x^d \cdot (x+1)$, where $d = \max\{d_1, d_2, \ldots, d_p\}$ is the depth of the MACA. Here, the factor $(x+1)$ corresponds to the point attractor in the configuration space of the CA.

The factor $(x+1)$ in characteristic polynomial can cause to form multi-length cycles if the CA is a complemented CA with a non-zero inversion vector (see Theorems 7.13 and 7.15). So, such complemented CAs cannot be MACAs. On the other hand, the complemented CA, having an irreducible polynomial, other than $(x+1)$, as factor cannot be an MACA. Such a CA contains attractor of length more than one. The complemented CAs with rules 0 and 255 contain no multi-length cycle attractors, but a single point attractor. So they are SACAs.

In an n-cell linear MACA, however, the point attractors are uniquely identifiable by m bits ($m \leq n$). These bits appear exhaustively in the set of point attractors of the CA and are known as *pseudo-exhaustive* bits.

Definition 70 An m-bit field of an n-bit pattern set is said to be pseudo-exhaustive (*PEF*) if all possible 2^m patterns appear in the set.

Theorem 9.1 *In an n-cell linear MACA with 2^m point attractors, there exist m-bit positions at which the attractors generate pseudo-exhaustive 2^m patterns.*

Proof Let T be the characteristic matrix of the n-cell linear MACA that has 2^m point attractors. Since it has 2^m point attractors only, the rank of $(T \oplus I)$ is $(n-m)$, where I is the $n \times n$ identity matrix. The null space of $(T \oplus I)$ gives the point attractors of the CA (see Theorem 7.11). Now, the dimension of the null space is m, which implies that there exist m linearly independent basis vectors. Hence, the all possible combinations of these m vectors are present in the null space. So, $(T \oplus I)$ produces pseudo-exhaustive bit patterns at m positions. □

Theorem 9.2 *The modulo-2 sum of two configurations at level p and q $(p > q)$ of the same basin of an MACA is a configuration of level p of 0-basin.*

Proof Let x and y be the two configurations at level p and q $(p > q)$ of α-basin of an MACA. If T is the characteristic matrix of the CA, then $T^p \cdot x = \alpha$ and $T^q \cdot y = \alpha$. Since $p > q$, $T^p \cdot y = T^{p-q}(T^q \cdot y) = T^{p-q}(\alpha) = \alpha$. Therefore, $T^p(x \oplus y) = \alpha \oplus \alpha = 0$. This implies that $x \oplus y$ is a configuration of 0-basin. As x is a configuration at level p, $T^{p-1} \cdot x \neq \alpha$. So, $T^{p-1}(x \oplus y) \neq 0$. Hence, $x \oplus y$ is at level p. □

Example 9.1 The linear MACA (204, 170, 90, 240, 204) (Fig. 9.1) with following characteristic matrix (T) illustrates the results of above theorems.

$$T = \begin{bmatrix} 1\,0\,0\,0\,0 \\ 0\,0\,1\,0\,0 \\ 0\,1\,0\,1\,0 \\ 0\,0\,1\,0\,0 \\ 0\,0\,0\,0\,1 \end{bmatrix}$$

- The characteristic polynomial of the MACA is $x^3 \cdot (1+x)^2$. Therefore, $m = 2$. The number of attractors $k = 2^m = 4$, depth $d_1 = 3$. The elementary divisor form of the characteristic polynomial is $x^3 \cdot (1+x) \cdot (1+x)$ and the minimal polynomial is $x^3 \cdot (1+x)$.
- The most significant and least significant bit positions of the MACA constitute the pseudo-exhaustive field (*PEF*).
- As per Theorem 9.2, modulo-2 sum of a pair of configurations 00101 and 00111 of the 00001-attractor basin is 00010 which is a configuration in the 0-basin. By contrast, if the configurations are 00011 and 11000 of two different attractor basins 00001 and 10000 respectively, then their modulo-2 sum is 11011 which is a configuration in a non-zero attractor (10001) basin.
- In the 0-basin, (i) the all 0s vector is always present, (ii) the modulo-2 sum of any two configurations in the 0-basin lies in the 0-basin only. These two properties make the 0-basin a vector subspace.

The above discussion guides synthesis of an n-cell MACA with 2^m point attractors, called $MACA(n, m)$.

Theorem 9.3 *A linear $MACA(n, m)$ can be synthesized through concatenation of m number of n_i-cell ($i \in \{1, 2, \ldots, m\}$) 2-attractor MACA such that $n_1 + n_2 + \cdots + n_m = n$.*

Proof The characteristic polynomial of an $MACA(n, m)$, in elementary divisor form, is $\langle x^{d_1} \cdot x^{d_2} \cdots x^{d_m} \cdot (1+x) \cdot (1+x) \cdots m \text{ times } \rangle$. The elementary divisors can be clubbed into m clusters each containing one $(1+x)$. That is, each cluster has the characteristic and minimal polynomial of the form $x^{d_i}(1+x)$ that forms an MACA with 2-attractor basins, where $x^{d_i}(1+x)$ corresponds to $MACA(n_i, 1)$, $i \in \{1, 2, \ldots, m\}$. Therefore, the $MACA(n, m)$ can be synthesized by synthesizing $MACA(n_i, 1)$ for all $i \in \{1, 2, \ldots, m\}$ and then merging them. □

As per Theorem 7.10, the T matrix corresponding to the $MACA(n, m)$ is formed from block diagonal placement of $T_1, T_2, \ldots, T_m$ where each T_i represents the $MACA(n_i, 1)$.

Example 9.2 Let the characteristic polynomial of an MACA be $x^5(1+x)^2$. The minimal polynomial is $x^3(1+x)$, and its elementary divisor form representation

is $([x^3 \cdot (1+x)] \cdot [x^2 \cdot (1+x)])$. The T matrix representing the MACA is formed from the block diagonal placement of T_1 and T_2, where

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \; T_2 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } T = \begin{bmatrix} \lceil 1 & \overline{0} & \overline{0} & 0 \rceil & 0 & 0 & 0 \\ \lfloor 0 & 0 & 1 & 0 \rfloor & 0 & 0 & 0 \\ \lfloor 0 & 1 & 0 & 1 \rfloor & 0 & 0 & 0 \\ \lfloor 0 & \underline{0} & \underline{1} & 0 \rfloor & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lceil 1 & \overline{1} & 0 \rceil \\ 0 & 0 & 0 & 0 & \lfloor 1 & 1 & 1 \rfloor \\ 0 & 0 & 0 & 0 & \lfloor 0 & \underline{0} & 1 \rceil \end{bmatrix}.$$

Definition 71 The set of all linear $MACA(n, m)$, for a pair of n and m, obtained by merging the m 2-attractor $MACA(n_i, 1)$, forms the MACA family.

The MACA family is important to realize hash function.

9.1.2 MACA as Hash Function

The pseudo exhaustive field (PEF) of linear MACA enables us to realize a hash function. For the purpose of hashing, a node in an MACA tree (Fig. 9.1) is viewed as the key to be hashed. The address for a key is given by the *PEF* of the attractor containing the key. That is,

$$H_{MACA}(key) = PEF(attractor) \tag{9.2}$$

where $H_{MACA}(key)$ is the hashed value of the given key. The hash table size is k, the number of attractors of the MACA.

Example 9.3 The MACA of Fig. 9.1 provides a hash table of size 4 with table entry (0–3). For example, all the patterns of basin 11111 are hashed to table entry 3(11).

The MACA family depicts a unique property while hashing. It forms a *Hamming Hash Family (HHF)*—a hash family where the probability of collision between two patterns varies inversely with their Hamming distance. In view of this property, each basin of an MACA contains patterns close to each other in terms of Hamming distance—that is, MACA basins form natural cluster. It enables an MACA to behave as the pattern recognizer.

Example 9.4 Let consider the 10001-basin of Fig. 9.1. If the all possible pair of patterns/configurations (since there are 8 patterns in the basin, the number of possible pairs is ${}^8C_2 = 28$) is considered, then the average of their Hamming distances is 1.7. The average Hamming distance for all other basins of Fig. 9.1 are also 1.7.

The Hamming distance between a pair of randomly chosen 5-bit patterns is 2.5 on an average. Therefore, the average Hamming distance 1.7 between a pair of patterns of a basin is less than that of arbitrarily chosen pair of patterns. Conversely, if two

random patterns are taken with minimum Hamming distance, then they are more likely to fall in the same attractor basin.

Since a pattern formed by modulo-2 addition of a pair of patterns of a basin falls in the 0-basin (Theorem 9.2), the 0-basin holds a unique characteristic. Indeed, the pattern distribution of the 0-basin determines the probability of collision between two random patterns, hashed with the MACA.

Definition 72 The *weight* of a binary pattern is the number of 1s in the pattern.

For example, the weight of the pattern 01110 is 3. The lesser Hamming distance between patterns of the same α-basin signifies that the patterns of the 0-basin of MACA should have lesser weight. Therefore, a member (MACA) of *HHF* has its 0-basin with definite bias to the patterns with lower weight. The bias gets reflected in the ratio of two probabilities defined below.

1. The probability of a pattern, of particular weight w, to fall in the 0-basin.
2. The probability that a pattern (chosen from an unbiased pattern pool of cardinality 2^{n-m} (the cardinality of the 0-basin)) have the weight w.

The bias can also be expressed in terms of ratio of (i) average number of patterns of weight w falling in the 0-basin of an $MACA(n, m)$ referred to as $|P(w, n, m)|$ and (ii) $|\hat{P}(w, n, m)|$—the number of patterns of weight w in the pattern pool of cardinality 2^{n-m} (provided each pattern in the pool is chosen unbiasedly from the entire 2^n pattern pool).

Therefore, the Expected Occurrence (EO) of patterns $P(w, n, m)$ with particular weight w, in the 0-basin of an $MACA(n, m)$, is

$$EO(P(w, n, m)) = \frac{|P(w, n, m)|}{|\hat{P}(w, n, m)|} \tag{9.3}$$

The computation of EO provides a quantitative analysis of the gradient of bias of 0-basin towards the patterns of lesser weights. This result, in conjunction with Theorem 9.2, establishes the clustering property with low Hamming distance between a pair of patterns in an MACA basin.

9.1.3 Non-linear MACA

It has already been pointed out that complemented MACAs are not generally observed. However, there are variants of non-linear MACAs. Figures 9.2 and 9.3 are the two examples of non-linear MACAs with rule vectors $(8, 112, 44, 68)$ and $(12, 102, 192, 85)$ in null boundary respectively. Non-linear MACAs do not always show very symmetrical structure in transition diagram as that of the linear MACAs.

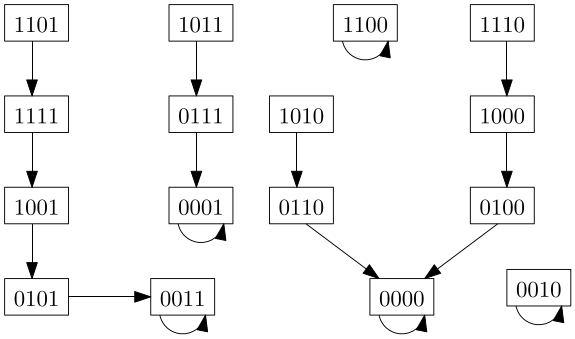

Fig. 9.2 Transition diagram of MACA (8, 112, 44, 68) in null boundary

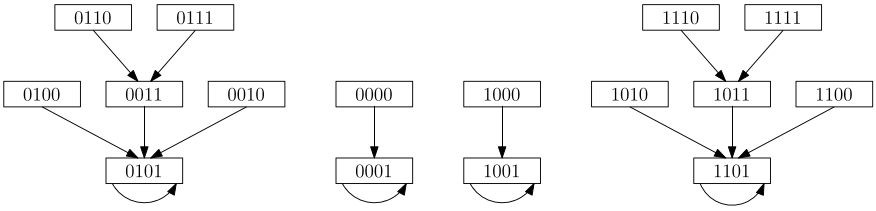

Fig. 9.3 Transition diagram of MACA (12, 102, 192, 85) in null boundary

Presence of multiple point attractors in a CA is the foremost condition of the CA to be an MACA. However, presence of point attractors is not sufficient to conclude that the CA is an MACA. To be an MACA, the CA should not have any multi-length cycle attractor.

9.2 The Point Attractors

The self-replicating RMTs of a rule play primary role to form the point attractors. An RMT $r = 4x + 2y + z$ ($x, y, z \in \{0, 1\}$) of a rule R is self-replicating if $R[r] = y$ (Definition 66). For example, RMT 6 of rule 112 is self-replicating, but this RMT of rule 44 is not self-replicating (see Table 9.1). Since the next configuration of a point attractor is the attractor itself, there has to be at least one self-replicating RMT in each rule $\mathcal{R}_i$ of an n-cell CA ($0 \leq i \leq n-1$) to form the point attractor.

Lemma 19 *A configuration (c) of an n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ is a point attractor if RMT r_i of rule $\mathcal{R}_i$ is self-replicating, where $\tilde{c} = (r_0 r_1 \cdots r_{n-1})$ is the RMT sequence of the configuration and $0 \leq i \leq n-1$.*

Proof The proof follows from the fact that the next configuration of a point attractor is the attractor itself and the self-replicating RMT allows a cell to stick to its previous state. □

Table 9.1 RMTs of the rules 8, 112, 44 and 68

Present state	111	110	101	100	011	010	001	000	Rule
RMT	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)	
Next state	0	0	0	0	1	0	0	0	8
Next state	0	1	1	1	0	0	0	0	112
Next state	0	0	1	0	1	1	0	0	44
Next state	0	1	0	0	0	1	0	0	68

9.2.1 Reachability Tree for Attractors

The reachability tree, introduced in Chap. 3, can efficiently identify the point attractors of a CA. To identify the point attractors, only self-replicating RMTs of the rules are to be considered.

Figure 9.4 is the reachability tree for the CA (8, 112, 44, 68) in null boundary. The RMTs of the rules are noted in Table 9.1. In the tree, out of 16, 12 configurations are reachable and the rest 4 are non-reachable. To identify the point attractors, all the non-self-replicating RMTs are removed from the nodes and labels of edges of the tree. Figure 9.5 is resulted out after removal of such RMTs from Fig. 9.4. Figure 9.5 is the *reachability tree for attractors* and shows only point attractors of the CA (8, 112, 44, 68). A sequence of edges from the root to a leaf in the tree represents an attractor. The 0011 is a point attractor and it can be identified in Fig. 9.5 as the edge sequence $\langle E_{0.0}, E_{1.0}, E_{2.1}, E_{3.3} \rangle$.

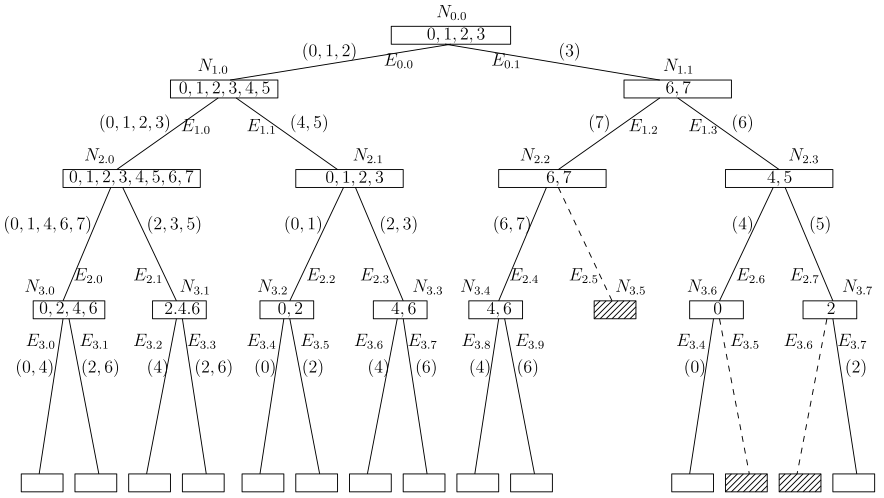

Fig. 9.4 Reachability tree for CA (8, 112, 44, 68) in null boundary

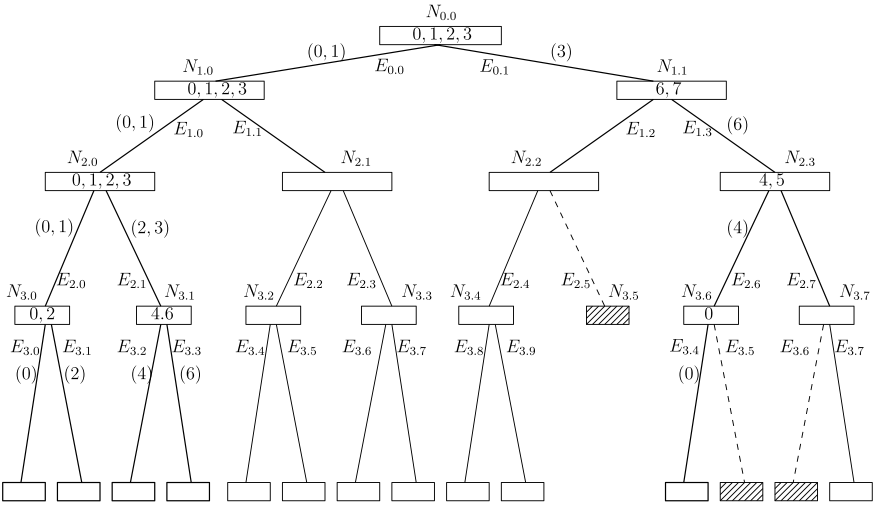

Fig. 9.5 Reachability tree for attractors of CA (8, 112, 44, 68) in null boundary

Theorem 9.4 *If* $B = b_0b_1 \cdots b_{i-1}b_ib_{i+1} \cdots b_{n-1}$ *and* $C = c_0c_1 \cdots c_{i-1}c_ic_{i+1} \cdots c_{n-1}$ *are two point attractors of an n-cell CA, where* $b_{i-1} = c_{i-1}$, $b_i = c_i$ *and* $b_{i+1} \oplus c_{i+1} = 1$, *then* $B' = b_0b_1 \cdots b_{i-1}b_ic_{i+1} \cdots c_{n-1}$ *and* $C' = c_0c_1 \cdots c_{i-1}c_ib_{i+1} \cdots b_{n-1}$ *are also the attractors of the CA.*

Proof If $b_{i-1} = c_{i-1}$, $b_i = c_i$ and $b_{i+1} \oplus c_{i+1} = 1$, then the RMTs $b_{i-1}b_ib_{i+1}$ and $c_{i-1}c_ic_{i+1}$ are sibling to each other. Since B and C are point attractors, each RMT of the RMT sequences of B and C is self-replicating. So, the RMT sequences are represented by the two edge sequences in the reachability tree for attractors. Now, as the RMTs $b_{i-1}b_ib_{i+1}$ and $c_{i-1}c_ic_{i+1}$ are sibling pair, they appear together in a node of level i. So, another two edge sequences can be identified which represent the RMT sequences of B' and C' in the tree for attractors. Hence the proof. □

Example 9.5 Let $B = 010101$ and $C = 000111$ be the two point attractors of a 6-cell CA. For the attractors B and C, the third and fourth bits are the same ($b_{i-1} = c_{i-1}$ and $b_i = c_i$). Whereas, the fifth bits (b_{i+1} and c_{i+1}) are different. Therefore, the derived point attractors are $B' = 010111$ and $C' = 000101$.

Corollary 9.1 *The derived point attractors (*B' *and* C'*) of a CA are same as the original (*B *and* C*) if* $b_k = c_k$, *for each* $k \in \{0, 1, \ldots, i-2\}$.

Proof If $b_k = c_k$ for each $k \in \{0, 1, \ldots, i-2\}$, then $B' = b_0b_1 \cdots b_{i-1}b_ic_{i+1} \cdots c_{n-1} = c_0c_1 \cdots c_{i-2}b_{i-1}b_ic_{i+1} \cdots c_{n-1} = c_0c_1 \cdots c_{i-2}c_{i-1}c_ic_{i+1} \cdots c_{n-1} = C$. Similarly, it can be shown that $C' = B$. Hence the proof. □

Corollary 9.2 *An n-cell CA that has at least 2 point attractors, can have maximum* 2^{m+1} *point attractors, where* $m = \lfloor \frac{n-1}{3} \rfloor$.

Proof Let consider a CA that has the point attractors B and C. For one set of cells $i-1, i$ and $i+1$, maintaining Theorem 9.4, the number of point attractors is doubled. That is, two pairs (2^2) of point attractors are there. If there is another such set of cells $j-1$, j, $j+1$, then another pair of point attractors can be derived from each pair of point attractors. Hence, for two such sets of cells, the number of attractors is 2^{2+1}. However, as per Theorem 9.4, if $i=1$, then $B=B'$ and $C=C'$. That is, no new attractor is derived. Therefore, excluding the left most bit, the maximum number of possible triplets $(i-1, i, i+1)$ of cells is $\lfloor \frac{n-1}{3} \rfloor$. Therefore, the maximum number of possible point attractors is 2^{m+1}, where $m = \lfloor \frac{n-1}{3} \rfloor$. □

The steps to find the number of point attractors of a CA, in null boundary, is given in Algorithm 14. The algorithm deals with the labels of edges of the tree. A weight is associated with each label denoting the possible number of point attractors that it carries. Initially, all the configurations are assumed to be the potential point attractors. When the tree for attractors grows, some edges are removed, and accordingly the weight is reduced to note that the number of point attractors is also reduced. At root level, weight of each of the two possible edges (hence RMT sets) is $\frac{2^n}{2}$. For level i, the weight of each label is $\frac{2^n}{2^{i+1}}$. However, if $N_{i.j} = N_{i.k}$ for any level $i < n-1$ in the tree, then two subtrees rooted at $N_{i.j}$ and $N_{i.k}$ exhibit the same number of point attractors. In Algorithm 14, if two labels are found same, then only one is considered for further processing with its weight as the sum of the two weights.

Complexity: Algorithm 14 runs a loop at Step 3, which depends on n (number of cells of the CA). There are two more loops at Step 4 and Step 5, which are inside the loop of the Step 3. The inner loops (of Step 4 and Step 5) are dependent on the *NoS*. Since the number of RMTs is 8, the *NoS* is limited to 8. Hence the complexity is $O(n)$.

Example 9.6 Let consider that the 4-cell CA (8, 112, 44, 68) in null boundary is the input to Algorithm 14. Each rule of the CA has some self-replicating RMTs, so the algorithm proceeds to Step 2. The rule 8 has three self-replicating RMTs—0, 1 and 3, which are distributed as $l_0 = \{0, 1\}$ and $l_1 = \{3\}$ (Step 2). Since both the l_0 and l_1 are non-empty sets, the sub-trees of the reachability tree at this level can generate at most $W_0 = W_1 = 2^{4-1} = 8$ point attractors (Fig. 9.5). In Step 4 of Algorithm 14, the nodes of level 1 of the reachability tree ($N_{1.0}$ and $N_{1.1}$ of Fig. 9.5) are found out. For the first node, $l'_0 = \{0, 1\}$ and $l'_1 = \emptyset$. Therefore, the right subsub-tree of $N_{1.0}$ cannot carry a point attractor. The left subsub-tree can carry attractors and its weight is $\frac{8}{2} = 4$. On the other hand, the node $N_{1.1}$ has only its right subsub-tree ($l'_0 = \emptyset$ and $l'_1 = \{6\}$). Hence, the weight of this right subsub-tree is $\frac{8}{2} = 4$. Here, $NoS = 2$. In the next level (level 2), two nodes are found—$N_{2.0} = \{0, 1, 2, 3\}$ and $N_{2.3} = \{4, 5\}$. The first node is having two children, but the node $N_{2.3}$ has one and the weights (2, 2 and 2) are calculated accordingly. After processing of the last cell rule (68), 5 sub-trees are constructed, each with single node. However, the labels of edges $E_{3.0}$ and $E_{3.12}$ of Fig. 9.5 are the same. Therefore, these two are replaced by a single one with weight = 2. Finally, the sum of weights is 5 $(2+1+1+1)$, which is the number of point attractors of the CA.

Algorithm 14: Count point attractors

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: Number of point attractors.

Step 1: If there is at least one rule $\mathcal{R}_i$ $(0 \leq i \leq n-1)$ that does not have any self-replicating RMT, then report the number of attractors is 0. Return.
Step 2: Distribute the self-replicating RMTs of $\mathcal{R}_0$ in the labels l_0 and l_1 of the edges at level 0 of the reachability for attractors.
(i) If $l_0 \neq \emptyset$, then set $W_0 \leftarrow 2^{n-1}$.
(ii) If $l_1 \neq \emptyset$, then set $W_1 \leftarrow 2^{n-1}$.
(iii) If $l_0 \neq \emptyset \neq l_1$, then set $NoS \leftarrow 2$. Otherwise, set $NoS \leftarrow 1$.
Step 3: For $i = 1$ to $n-1$, repeat Step 3 to Step 6
Step 4: For each l_j, $j \in \{0, 1, \cdots, NoS-1\}$
(i) Find RMTs of the corresponding node using Point 3 (or, Point 4 if $i = n-1$) of Definition 18.
(ii) Remove the RMTs (of $\mathcal{R}_i$) which are not self-replicating.
(iii) Distribute the rest RMTs in the labels l'_{2j} and l'_{2j+1} of edges at level i using $\mathcal{R}_i$ (Point 2 of Definition 18).
(iv) Set $W'_{2j} \leftarrow W'_{2j+1} \leftarrow \frac{W_j}{2}$.
(v) Replace RMTs 4, 5, 6, and 7 of l'_{2j} and l'_{2j+1}, if any, by RMTs 0, 1, 2 and 3 respectively.
Step 5: Set $k \leftarrow 0$. For $j = 0$ to $2NoS-1$, repeat Step 6.
Step 6: If $l'_j \neq \emptyset$, then do the following
(i) Set $l_k \leftarrow l'_j$, $W_k \leftarrow W'_j$.
(ii) If there is any set l'_m in the list $l'_{j+1}, l'_{j+2}, \cdots l'_{2NoUL-1}$ so that $l_k = l'_m$, then set $W_k \leftarrow W_k + W'_m$ and $l'_m \leftarrow \emptyset$.
(iii) Set $k \leftarrow k+1$.
Step 7: Set $NoS \leftarrow k$.
Step 8: Report $W_0 + W_1 + \cdots + W_{NoS-1}$ as the number of point attractors.

9.2.2 *Identification Bits and Pseudo-Exhaustive Bits*

To represent point attractors of an n-cell CA, n bits are needed for each point attractor. However, a point attractor can be identified from the set of point attractors by checking a lesser number of bits, say m bit positions ($m \leq n$). This implies that the number of point attractors is less than or equal to 2^m. These bits are the *identification bits* for point attractors. If the identification bits appear exhaustively, then these are the pseudo-exhaustive (PE) bits. In linear MACAs, the point attractors are always identified by PE bits. However, for non-linear MACAs, the PE bits as well as the non-PE identification bits can be found.

Example 9.7 The MACA $(12, 102, 192, 85)$ in null boundary has four point attractors—0101, 0001, 1001 and 1101 (see Fig. 9.3). The attractors can uniquely be identified by first two bits which appear exhaustively—01, 00, 10 and 11. These two are PE bits for the attractors. On the other hand, the MACA $(8, 112, 44, 68)$ in null boundary has five attractors—0000, 0001, 0010, 0011 and 1100 (see Fig. 9.2). Here, the first bit and last two bits can identify the attractors—that is, these are

identification bits. The 000, 001, 010, 011 and 100 identify the point attractors. The identification bits do not appear exhaustively.

Theorem 9.5 *All the K point attractors of an n-cell CA can be uniquely identified by the m bit positions, where $m \leq n$ and $K \leq 2^m$, and there exists r sets of PE bit positions that are the subset of m bit positions with cardinality $p_1, p_2, \ldots, p_r$, where $2^{p_1} + 2^{p_2} + \cdots + 2^{p_r} = K$.*

Proof Consider the reachability tree of a CA for K point attractors. The first node, starting from the root, having both the left and right children, splits the set of point attractors into two subsets. The bit corresponding to that node is the identification bit and can exhaustively identify the two sub-trees (subsets). Now, for each sub-tree, there exists another identification bit that splits the sub-tree into two and also can exhaustively identify these. This is true till the leaves are reached. That is, each attractor can be identified by a set of identification bits, and the maximum number of bits required to uniquely identify all the K attractors is m, where $m \leq n$.

Now, a number of attractors can be grouped in such a way that a subset of identification bits appear exhaustively to identify all the attractors of the group. Hence, the subset of identification bits is the PE bits for that particular subset of point attractors. Let consider that the number of such subsets of point attractors is r. Therefore, $2^{p_1} + 2^{p_2} + \cdots + 2^{p_r} = K$, where $p_1, p_2, \cdots, p_r$ are the cardinalities of these subsets. Hence the proof. □

Algorithm 15 finds the identification bits of a CA. It implicitly constructs the reachability tree for attractors. If a node with both the children is found at level i, then the ith bit is marked as an identification bit. As in Algorithms 14 and 15 also deals with the labels of edges while constructing the reachability tree.

Complexity; The complexity of Algorithm 15 depends on n (number of cells of the CA) and NoS. Since the number of RMTs is 8, the maximum possible value of NoS is limited to 8. Hence, the complexity of Algorithm 15 is $O(n)$.

Example 9.8 This example illustrates the execution of Algorithm 15 considering the CA $(8, 112, 44, 68)$ in null boundary as the input. Since each of the rules has some self-replicating RMTs, the algorithm proceeds to Step 1. Here, $l_0 = \{0, 1\}$ and $l_1 = \{3\}$ while considering $\mathcal{R}_1 = 8$ (Step 2). Hence, the first bits of point attractors is marked as an identification bit (Fig. 9.5). However, l_1' and l_2' are empty while considering the next level nodes for rule 112. Therefore, the second bits of the point attractors cannot be the identification bit (Step 3). Similarly, it can be found that the third (while considering rule 44) and fourth (while considering rule 68) bits are the identification bits (see Fig. 9.5).

Algorithm 15 can also be modified to find the PE bits of a CA. The reachability tree for attractors corresponding to a CA contains 2^m leaves while the CA has m PE bits to uniquely identify all such attractors. Therefore, if level i of the tree corresponds to a PE bit, all the nodes of the level i have to have exactly two children. The CA cannot have the PE bits for attractors if one or more nodes have a single child.

Algorithm 15: IdentificationBits

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: Identification bits.

Step 1: If there is at least one rule $\mathcal{R}_i$ $(0 \leq i \leq n-1)$ that does not have any self-replicating RMT, then return.
Step 2: Distribute the self-replicating RMTs of $\mathcal{R}_0$ in the labels l_0 and l_1 of the edges at level 0 of the reachability for attractors.
If $l_0 \neq \emptyset \neq l_1$, then mark the 0th bit and set $NoS \leftarrow 2$.
Otherwise, set $NoS \leftarrow 1$.
Step 3: For $i = 1$ to $n-1$, repeat Step 3 to Step 6
Step 4: For each l_j, $j \in \{0, 1, \cdots, NoS-1\}$
(i) Find RMTs of the corresponding node using Point 3 (or, Point 4 if $i = n-1$) of Definition 18.
(ii) Remove the RMTs (of $\mathcal{R}_i$) which are not self-replicating.
(iii) Distribute the rest RMTs in the labels l'_{2j} and l'_{2j+1} of edges at level i using $\mathcal{R}_i$ (Point 2 of Definition 18).
(iv) If $l'_{2j} \neq \emptyset \neq l'_{2j+1}$, then mark the ith bit
(v) Replace RMTs 4, 5, 6, and 7 of l'_{2j} and l'_{2j+1}, if any, by RMTs 0, 1, 2 and 3 respectively.
Step 5: Set $k \leftarrow 0$. For $j = 0$ to $2NoS-1$, repeat Step 6.
Step 6: If $l'_j \neq \emptyset$, then do the following
(i) Set $l_k \leftarrow l'_j$, $W_k \leftarrow W'_j$.
(ii) If there is any set l'_m in the list $l'_{j+1}, l'_{j+2}, \cdots l'_{2NoUL-1}$ so that $l_k = l'_m$, then set $W_k \leftarrow W_k + W'_m$ and $l'_m \leftarrow \emptyset$.
(iii) Set $k \leftarrow k+1$.
Step 7: Set $NoS \leftarrow k$.
Step 8: Report the marked bits as Identification-bits.

9.3 Multi-length Cycle Attractors

Presence of point attractors in a CA does not ensure that the CA is having only point attractors. Along with point attractors, a CA may have multi-length cycles (a cycle of length more than one) in its configuration space. Therefore, to decide whether a CA is MACA or SACA, it is to be tested for multi-length cycles in its configuration space. The reachability tree with links (Sect. 8) can be exploited to identify the presence of multi-length cycle. As per Theorem 8.4, the presence of cross link among the edges at level $n-1$ in the reachability tree ensures that the CA has a multi-length cycle attractor. It is also noted in Sect. 8.4 that a cross link of level i may propagate to the lower levels, as well as may get dissolved depending on some conditions. Therefore, if a cross link appears in level i $(i < n-1)$, it is to be observed whether the cross link propagates to level $n-1$. Since an elementary cross link (Sect. 8.4.1) at level i may get dissolved at level j $(j > i)$, the existence of cross link at level i is not enough to decide that the CA is having a multi-length cycle. However, in some cases, early decision can be possible.

Theorem 9.6 *An n-cell CA is having at least one multi-length cycle attractor, if a set of edges at level i ($< n-1$) participates only in forward and backward links among themselves but not in self link, and no other edge which is non-member of the set can link to an edge of the set.*

Proof During its evolution, a CA either settles to a point attractor or in a multi-length cycle attractor for any initial configuration. Let the edges $E_{i.j_1}, E_{i.j_2}, \ldots, E_{i.j_m}$ participate only in forward and backward links among themselves and none of them has the self link. Further, no edge outside of this set links to an edge of these m edges. Now, consider an initial configuration corresponding to an edge sequence that involves any of these edges. If the CA evolves from the initial configuration, it has to reach either a point attractor or a multi-length cycle attractor. Further, the CA cannot reach a configuration which does involve any of these m edges, because no edge which is not a member of this set of m edges links to an edge of this set. Since there is no self link in the edges $E_{i.j_1}, E_{i.j_2}, \ldots, E_{i.j_m}$, the CA cannot reach a point attractor. Hence, there is a multi-length cycle attractor in the CA. □

Example 9.9 Let consider an n-cell CA in null boundary having the first two rules 10 and 60. Here, $60[0] = 60[1] = 60[6] = 60[7] = 0$ and $60[2] = 60[3] = 60[4] = 60[5] = 1$. The edges at level 1, $E_{1.1}$, $E_{1.2}$ and $E_{1.3}$, form a cross link, but they do not participate in self link. It does not matter what are the other rules, the CA has to have at-least one multi-length cycle attractor. That is, the CA (10, 60, ...) contains multi-length cycle attractor(s).

Corollary 9.3 *An n-cell CA in null boundary is having at least one multi-length cycle attractor if there exists a cross link among the edges at level i ($< n-1$), and every edge of the cross link links to another edge for a pair of sibling RMTs.*

Proof Consider that $E_{i.j}(r) \rightarrow E_{i.k}$ and $E_{i.j}(s) \rightarrow E_{i.k}$, where $E_{i.j}$ and $E_{i.k}$ are the two edges of cross link ($i < n-1$). Further, consider that RMTs r and s of $\mathcal{R}_i$ are sibling to each other. Since $E_{i.k}$ is linked by two edges, no other link can point to $E_{i.k}$ (Proposition 8.5). Hence, no edge can participate in the self link. Therefore, as per Theorem 9.6, the CA contains at least one multi-length cycle attractor. □

Let consider Fig. 9.6. It shows links at levels $i-1$, i and $i+1$ of a reachability tree. Let rename the edges at level i as $E_{i.0}, E_{i.1}, \ldots$. The edges $E_{i.1}$, $E_{i.2}$ and $E_{i.3}$ participate in cross link: $E_{i.1}(4) \rightarrow E_{i.2}(6) \rightarrow E_{i.3}(2) \rightarrow E_{i.1}$ and $E_{i.1}(5) \rightarrow E_{i.2}(7) \rightarrow E_{i.3}(3) \rightarrow E_{i.1}$. Here, the RMT pairs 4 and 5, 6 and 7, and 2 and 3 are the sibling RMT pairs. If such a situation arises in the reachability tree of a CA, then as per Corollary 9.3, the CA contains a multi-length cycle attractor. For example, in the reachability tree of the 4-cell CA (10, 60, 154, 5) in null boundary, the above situation arises, and the CA contains a multi length cycle attractor.

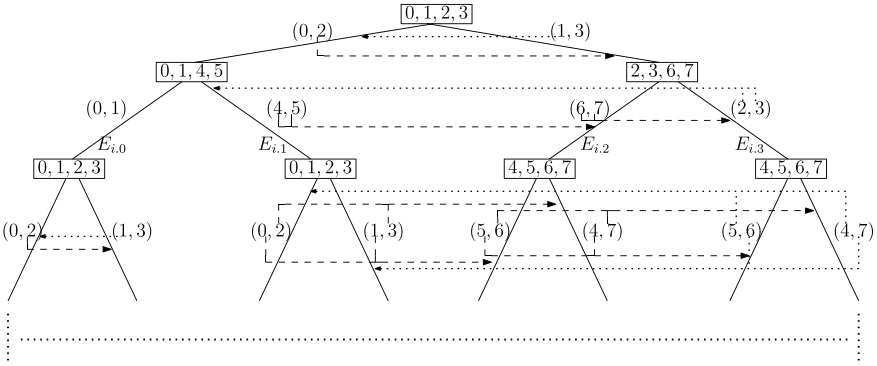

Fig. 9.6 A situation in reachability tree for the existence of multi-length cycle

9.3.1 Testing Convergence of a CA

Algorithm 14, in conjunction with Algorithm 12, can be employed to test whether a CA in null boundary converges to a point attractor from an arbitrary initial configuration. Algorithm 12 outputs total number of cyclic configurations (say n_1) of a CA whereas, Algorithm 14 counts the number of point attractors (say n_2) of the CA. Now, if $n_1 = n_2$, then the CA contains only the point attractors, which ensures that the CA converges to point attractor from any initial configuration. But, if $n_1 > n_2$, the CA is having atleast one multi-length cycle attractor.

However, a single scan of rule vector is sufficient to decide on the presence of a multi-length cycle attractor in a CA. The algorithmic steps are summarized below:

1. Repeat Steps 2–4 for each level i ($0 \leq i \leq n-2$).
2. Identify the links at level i ($0 \leq i \leq n-1$).
3. Identify the *weight* of each RMT $r \in l_{i.j}$, where $0 \leq i \leq n-1, 0 \leq j \leq 2^{i+1}-1$ (Sect. 8.2.2).
4. Discard the RMTs that participate either only in backward link or only in forward link, but not in a cross link.
5. If any cross link exists among $E_{n-1.j}$s, then the CA has multi-length cycle attractor.

The above steps are framed in Algorithm 16 for null boundary condition. The algorithm is very similar to Algorithm 12. It takes a rule vector $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ in null boundary as the input, and outputs "Yes" if the CA contains only point attractors, and "No" otherwise. The algorithm deals with the labels of edges, recursively identifies the edges (in Steps 5 and 6) that do not participate in self and cross link to remove them, and finally tests if a cross link among the edges at level $n-1$ is formed. If no, the algorithm outputs "Yes".

Complexity: The complexity of this algorithm depends on the number of cells (n) of the input CA and on the number of reachable edges at each level. Say, m is the maximum number of edges at a level that are dealt with during processing. The time

Algorithm 16: Decide if the CA is having only point attractors

Input : An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary.
Output: Yes/No

Step 1: Find the labels of the edges at level 0 using $\mathcal{R}_0$.
Set weight of each RMT of l_0 and l_1 as 1.
Step 2: Identify the links at level 0 of the reachability tree.
Step 3: For $i = 0$ to $n - 1$, repeat Step 4 to Step 8
Step 4: Mark the non-reachable edge at level i.
Step 5: If an edge at level i links a non-reachable edge for an RMT r, then
Decrease the weight of the RMT r by 1.
Step 6: If the weights of all the RMTs of a label becomes 0, then
Mark the corresponding edge as non-reachable.
Step 7: Proceed with unmarked edges to find the labels of edges at level $i + 1$,
using $\mathcal{R}_{i+1}$, if $i < n - 1$.
Step 8: Identify links among edges at level $i + 1$, if $i < n - 1$.
Step 9: If a cross link exists among the unmarked edges at level $n - 1$, output "No".
Otherwise, output "Yes".

to handle the m edges in Steps 4–6 of Algorithm 16 is $O(m)$. Hence complexity of Algorithm 16 is $O(mn)$.

Example 9.10 Let consider the CA (5, 73, 200, 80) in null boundary of Fig. 8.4. The reachability tree and links (Fig. 8.6) are formed for level 0. The weight of each RMT is assigned: $w^0_{0.1} = w^1_{0.0} = w^2_{0.1} = w^3_{0.1} = 1$. The RMTs 1 and 2 are involved in self links, while the RMTs 0 and 3 participate in cross link. In the next level, $E_{1.2}$ is self linked for the RMTs 4, $5 \in l_{1.2}$ and RMT $1 \in l_{1.2}$ is part of acyclic configurations, hence it is removed. The level 2 is also derived with $\mathcal{R}_2$. Here, a non-reachable edge ($E_{2.7}$) can be observed. All the RMTs have weight 1. However, the RMTs 6 and 7 of $l_{2.1}$ are supposed to link the non-reachable edge. So, $w^6_{2.1} = w^7_{2.1} = 0$. Here, one edge (out of eight edges) is declared as non-reachable. Figure 8.11a shows these two non-reachable edges. Hence, at least 4 configurations of the CA are acyclic (that is, one fourth of the configurations are discarded). This process is repeated until no new non-reachable edge is found. At level 2, $E_{2.0}$, $E_{2.2}$, $E_{2.3}$, $E_{2.4}$, $E_{2.5}$ and $E_{2.6}$ are left. At the last level, only six leaves are left. Among them three are self linked which form point attractors and the rest form cross link ($E_{3.0}(0) \rightarrow E_{3.4}(0) \rightarrow E_{3.12}(0) \rightarrow E_{3.0}$). So, the CA is having a multi-length cycle and the output of Algorithm 16 is *No*. However, the output is *Yes* for the CA (12, 102, 192, 85) which contains only point attractors (see Figs. 8.4 and 9.3 for verification).

Search for multi-length cycle attractor in periodic boundary: With a minor modification of Algorithm 16, the existence of multi-length cycle in the configuration space of a CA in periodic boundary can also be decided. Here, the label of an edge is to be considered as a collection of four RMT sets. Following example illustrates the case.

Example 9.11 Let consider the CA (204, 240, 48, 0) in periodic boundary. The reachability tree of the CA, after establishing the links, is shown in Fig. 8.7. Here, $w^{r_1}_{0.0,0} = w^{r_2}_{0.0,2} = w^{r_3}_{0.1,1} = w^{r_4}_{0.1,3} = 1$, where $r_1 \in \{0, 1\}, r_2 \in \{4, 5\}, r_3 \in \{2, 3\}$ and $r_4 \in \{6, 7\}$—that is, all the RMTs are involved in self links. Further, $w^0_{1.0,0} = w^1_{1.0,0} = w^6_{1.3,1} = w^7_{1.3,1} = 1$ and $w^2_{1.0,0} = w^3_{1.0,0} = w^4_{1.3,1} = w^5_{1.3,1} = 1$ (RMTs 0, 1, 6 and 7 are involved in self links, the RMTs 2 and 3 are involved in forward link and the RMTs 4 and 5 are involved in backward links, see Fig. 8.7 for verification). At this level, there are two non-reachable edges $E_{1.1}$ and $E_{1.2}$. The other two edges $E_{1.0}$ and $E_{1.2}$ link to the non-reachable edges for the RMTs 2 and 3 of $l_{1.0}$ and the RMTs 4 and 5 of $l_{1.3}$ respectively. So, $w^2_{1.0,0} = w^3_{1.0,0} = w^4_{1.3,1} = w^5_{1.3,1} = 0$. At level 2, a cross link is formed which triggers another cross link at level 3. Finally, one self link ($E_{3.0}(0) \rightarrow E_{3.0}$) and one cross link ($E_{3.12}(5) \rightarrow E_{3.14}(1) \rightarrow E_{3.12}$) are formed. So, the CA contains a multi-length cycle attractor. Now, consider the CA (157, 205, 213, 220) in periodic boundary. Figure 9.8 shows the reachability tree of the CA with links. The tree does not have a multi-length cycle among the edges at level 3. So, the CA contains only point attractors (see Fig. 9.7 for verification).

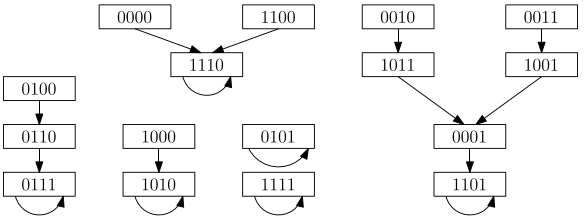

Fig. 9.7 Transition diagram of CA (157, 205, 213, 220) in periodic boundary

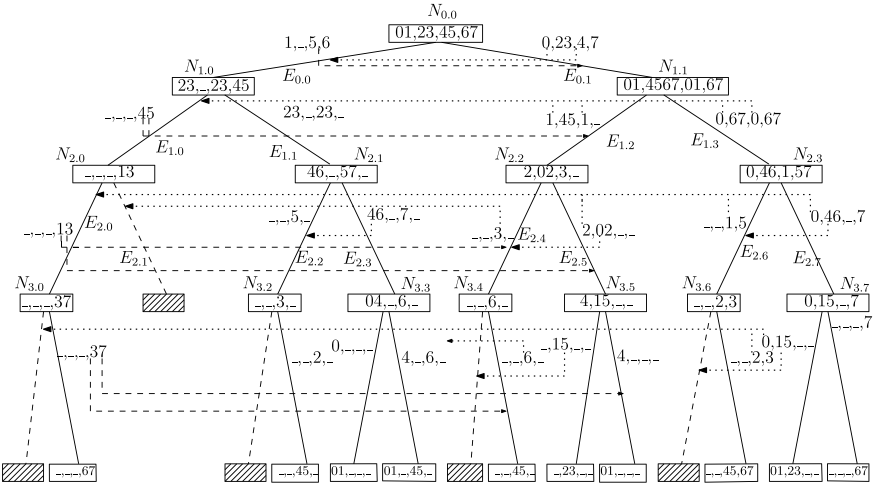

Fig. 9.8 Reachability tree of CA (157, 205, 213, 220) in periodic boundary

9.3.2 Scalability of Convergent CAs

It has been shown in Chaps. 4 and 5 that the CAs are scalable with respect to reversibility and number conservation. This section shows that the CAs are also scalable with respect to the convergence to point attractors. The rule vector of an n-cell CA which converges to point attractor from any initial configuration, can be extended to get the rule vector of an $(n+m)$-cell CA that also converges to point attractor from an arbitrary initial configuration. Similarly, from two given CAs with n and m cells that converge to point attractors only, the $(n+m)$-cell CA can be obtained which also converges to point attractor.

The $(n+1)$-cell CA from n-cell CA: Consider that an n-cell CA in null boundary converges to point attractor. Therefore, the corresponding reachability tree has no cross link at level $n-1$. It is easy to show that if level i is cross link free, then there exist at least one rule as $\mathcal{R}_{i+1}$ which ensures that the level $i+1$ is cross link free. However, this logic cannot directly be applied to the levels $n-1$ and n of the tree of $(n+1)$-cell CA. Because, the odd RMTs of $\mathcal{R}_{n-1}$ are ignored in the n-cell CA (due to null boundary condition) but they are used in the $(n+1)$-cell CA. Therefore, the odd RMTs of $\mathcal{R}_{i-1}$ may trigger a cross link at level $n-1$, which may not get dissolved at level n of the tree of $(n+1)$-cell CA.

Lemma 20 *If an n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ in null boundary converges to point attractor from an arbitrary configuration, then there exists a rule $\mathcal{R}_n$ such that the $(n+1)$-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}'_{n-1}, \mathcal{R}_n)$ in null boundary also converges to point attractor from an arbitrary configuration, where $\mathcal{R}'_{n-1}$ is a right independent rule and equivalent to $\mathcal{R}_{n-1}$ when act as the last rule of the n-cell CA.*

Proof Let consider an n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ that converges to point attractor from any configuration. As null boundary is considered, the odd RMTs of the last rule $\mathcal{R}_{n-1}$ are ignored. Now, consider a last rule $\mathcal{R}'_{n-1}$ such that $\mathcal{R}'_{n-1}[r] = \mathcal{R}'_{n-1}[r+1] = \mathcal{R}_{n-1}[r]$ for $r \in \{0, 2, 4, 6\}$. Clearly, the $\mathcal{R}'_{n-1}$ is a right independent rule. As the same edge is linked for a pair of sibling RMTs (Proposition 8.5), no cross link is formed. at level $n-1$. Then, a rule as $\mathcal{R}_n$ can always be found, for which the level n in the reachability tree of the $(n+1)$-cell CA is cross link free. □

Example 9.12 The CA (12, 102, 192, 85) in null boundary is having only point attractors (see Fig. 9.3). Let design a 5-cell CA from the 4-cell CA. Let arbitrarily choose rule 87, in place of rule 85, both of which have same next state for even RMTs. Hence, the 87 can replace the 85 in the 4-cell CA without affecting its behavior. However, the 5-cell CA (12, 102, 192, 87, 84) contains multi-length cycle attractor. Now, say the fourth rule $\mathcal{R}_3$ is chosen in such a way that $\mathcal{R}_3[r] = \mathcal{R}_3[s]$, where RMTs r and s are sibling to each other. That is, $\mathcal{R}_3 = 255$ can be a choice. The CA (12, 102, 192, 255, 84) converges to point attractor from any initial configuration.

The $(n+m)$-cell CA from n-cell and m-cell CA: An n-cell CA $\mathcal{R}^n = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ and an m-cell CA $\mathcal{R}^m = (\mathcal{R}'_0, \mathcal{R}'_1, \ldots, \mathcal{R}'_{m-1})$ can configure the $(n+$

m)-cell CA $\mathcal{R} = (\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}^{''}_{n-1}, \mathcal{R}^{''}_0, \mathcal{R}^{'}_1, \ldots, \mathcal{R}^{'}_{m-2}, \mathcal{R}'_{m-1})$. To append the rule vector of the m-cell CA with the rule vector of n-cell CA, $\mathcal{R}_{n-1}$ is replaced by a right independent rule $\mathcal{R}^{''}_{n-1}$, in which $\mathcal{R}^{''}_{n-1}[r+1] = \mathcal{R}^{''}_{n-1}[r] = \mathcal{R}_{n-1}[r]$ for each $i \in \{0, 2, 4, 6\}$. If $\mathcal{R}^{'}_0$ is now replaced by a left independent rule $\mathcal{R}^{''}_0$ such that $\mathcal{R}^{''}_0[r+4] = \mathcal{R}^{''}_0[r] = \mathcal{R}^{'}_0[r]$ for each $i \in \{0, 1, 2, 3\}$, then the dynamical behavior of the n-cell and m-cell CAs remain intact in the $(n+m)$-cell CA. As per Sect. 8.5.2, $\mathcal{R}$ can be represented as $\mathcal{R} = \mathcal{R}^n \odot \mathcal{R}^m$.

The $\mathcal{R}^{''}_0$ and $\mathcal{R}^{'}_0$ need not necessarily be the different rules. If the nodes at level n of the reachability tree of the $(n+m)$-cell CA are *root equivalent nodes* (Definition 73) and are subset of $\{0, 1, 2, 3\}$, then $\mathcal{R}^{''}_0$ and $\mathcal{R}^{'}_0$ can be the same rule. For example, the CAs $(10, 252, 140, 0)$ and $(12, 240, 68)$ in null boundary have only the point attractors. The CA $(10, 252, 140, 0, 12, 240, 68)$ is also having only the point attractors. Here, $\mathcal{R}^{'}_0 = \mathcal{R}^{''}_0$ and as rule 0 itself a right independent rule, no replacement is required for rule 0.

Lemma 21 *The $(n+m)$-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}^{''}_{n-1}, \mathcal{R}^{''}_0, \mathcal{R}^{'}_1, \ldots, \mathcal{R}^{'}_{m-1})$ converges to point attractor for any initial configuration iff the n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ and m-cell CA $(\mathcal{R}^{'}_0, \mathcal{R}^{'}_1, \ldots, \mathcal{R}^{'}_{m-2}, \mathcal{R}'_{m-1})$ converge to point attractor for any initial configuration, where $\mathcal{R}_0$ and $\mathcal{R}^{'}_{n-1}$ are replaced by $\mathcal{R}^{''}_0$ and $\mathcal{R}^{''}_{n-1}$ as per the above stated rule.*

9.4 The Synthesis Problem

The synthesis problem finds an n-cell CA, for a given n, that is having only the point attractors in its configuration space. The synthesis scheme for linear MACAs is introduced in Sect. 9.1.1. This section reports synthesis of both the linear and non-linear CAs.

The necessary properties of a reachability tree, representing an n-cell CA, having only the point attractors are the following.

1. There exists at least one self-linked edge at level $n-1$.
2. There is no cross-link among the edges at level $n-1$.

To get a CA having only the point attractors, only self links at the edges of the reachability tree can be allowed. If only self links are allowed and no cross link exists at any level, a very limited number of CAs can be synthesized and that is not desirable.

To get a larger number of CAs having only point attractors, a synthesis scheme should allow forward and backward links in the reachability tree. If at level i, the backward and forward links do not participate in cross link, then they can be ignored (Theorem 8.9). Concern arises when backward and forward links form cross link at level i, because this may be an indication of formation of a multi-length cycle. So, dissolving the cross link at level j ($j > i$, $0 \leq i \leq n-2$) is a requirement. In the synthesis scheme, where cross links can appear, two issues are to be considered.

1. *Dissolving of cross link:* As per discussion of Sect. 8.4.3, the right independent rules can play a pivotal role.
2. *Growth of cross link:* The growth should not be exponential.

If level i is cross link free, then at level $i+1$, a few nodes behave like the root. These are the *root equivalent nodes*. The corresponding sub-tree is the *connection-less sub-tree*.

Definition 73 A node $N_{i.j}$ is said to be the *root equivalent* if each RMT $r \in l_{i-1.j}$ does not participate in a cross link, where $1 \leq i < n-1$, $0 \leq j \leq 2^{i+1}-1$. The sub-tree, rooted at $N_{i.j}$, is called the *connection-less sub-tree*.

In Fig. 9.9, the nodes $N_{1.0}$ and $N_{1.1}$ are the root equivalent nodes and the corresponding sub-trees are connection-less sub-trees. In level 3, there are four root equivalent nodes—$N_{3.0}$, $N_{3.2}$, $N_{3.4}$ and $N_{3.6}$. If a root equivalent node $N_{i.j}$ is subset of another root equivalent node $N_{i.k}$, then the synthesis process can proceed only with the $N_{i.k}$. Because, potential of developing cross links in the sub-tree, rooted at $N_{i.k}$, is same or higher than that of the sub-tree rooted at $N_{i.j}$. The node $N_{i.j}$ is called a sub-node of $N_{i.k}$.

Definition 74 The root equivalent node $N_{i.j}$ is said to be the *sub-node* of a root equivalent node $N_{i.k}$ if $N_{i.j} \subseteq N_{i.k}$.

In Fig. 9.9, $N_{3.0} = N_{3.4}$ and $N_{3.2} = N_{3.6}$. Further, $N_{3.0} \subseteq N_{3.2}$. Therefore, $N_{3.0}$ and $N_{3.4}$ are the sub-node of $N_{3.2}$ (or $N_{3.6}$). Hence, for the next level, the synthesis process can proceed only with $N_{3.2}$ (or $N_{3.6}$), and drop the other nodes. So, the presence root equivalent nodes and their sub-nodes restrict the growth of the reachability tree, and hence the growth of cross links.

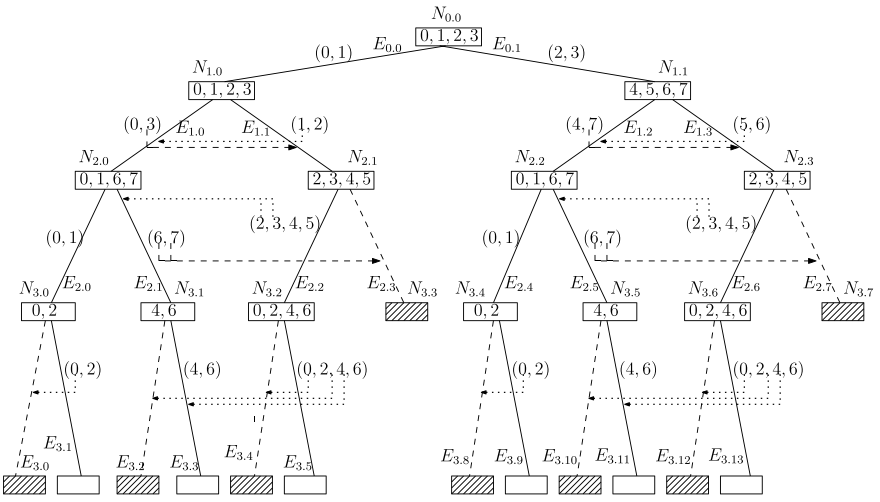

Fig. 9.9 Reachability tree of CA (12, 102, 192, 85) in null boundary

Lemma 22 *Maximum number of root equivalent nodes that are not the sub-nodes of any other node, at any level, is 4.*

Proof The nodes of any level are the subset of eight RMTs. Moreover, sibling RMTs are always paired in a node. Since there are four pairs of sibling RMTs, they can form at most four sets which are not the subset of others and can form root equivalent nodes. Hence the proof. □

9.4.1 Generic Synthesis Scheme

This synthesis scheme has two requirements—first, for each level i ($0 \leq i \leq n-1$), there is at least one self linked edge $E_{i.j}$ and the second, there exists no cross link among the edges at level $n-1$. It provisionally selects a rule for level i such that at least one self link is formed at the level i. If no cross link forms, the rule is considered as $\mathcal{R}_i$. Otherwise, the rule $\mathcal{R}_i$ is selected based on the fact that whether the cross links can be dissolved in the next levels. Following are the steps of the generic synthesis scheme.

1. Set $i \leftarrow 0$
2. Repeat Steps 3–9 until $i \leq n-2$.
3. Select $\mathcal{R}_{temp}$ randomly for level i so that at least one self link exists at level i.
4. Identify the cross links of level i.
5. If no cross link exists, then set $\mathcal{R}_i \leftarrow \mathcal{R}_{temp}$.
6. If cross link exists at level i, check if any rule $\mathcal{R}_j$ ($j > i$) can dissolve the cross links.
   If yes goto Step 7. Otherwise, goto Step 8.
7. Set $\mathcal{R}_i \leftarrow \mathcal{R}_{temp}$, goto Step 9.
8. Set $\mathcal{R}_i$ such that no cross link can exist but at least one self link is formed at level i.
9. Set $i \leftarrow i+1$.
10. Set $\mathcal{R}_{n-1}$ so that no cross link exists but one self link exists at level $n-1$.

Example 9.13 This example illustrates the synthesis of a 4-cell CA in null boundary with point attractors only, following the above scheme. Let first choose rule 12 as $\mathcal{R}_0$. Both the edges at level 0 of the corresponding reachability tree participate only in self link. Suppose rule 178 is selected as $\mathcal{R}_1$. Cross links at level 1 in the tree are then identified. There is no rule as $\mathcal{R}_2$ for which cross links of level 1 can be dissolved at level 2. So, choose another rule, say 207, as $\mathcal{R}_1$. Now, no cross link exists at level 1. So, the scheme proceeds to choose the next rule. Suppose, rule 5 is selected as the next rule. A cross link is observed at level 2, but the cross link can be dissolved at the next level. So, $\mathcal{R}_2 = 5$. Finally, a rule is selected for the right most cell so that no cross link among the edges at level 3 in the tree can exist. This criterion is satisfied if $\mathcal{R}_3 = 85$. Hence, the synthesized CA is $(12, 207, 5, 85)$.

Complexity: The synthesis scheme can synthesize a large variety of CAs having only the point attractors. But, its computational complexity is high. The complexity of this algorithm becomes exponential in worst case if a cross link grows exponentially. Step 6 is executed for all possible rules, until a rule is qualified to dissolve the cross links. In worst case, all the rules except rule 51 are to be tested. That is, Step 6 is executed for $2^8 - 1$ times. Hence, the upper bound of the worst case complexity is $1 + \sum_{i=1}^{n-1} 2^{i+1} \times (2^8 - 1) = O(2^{n+9})$.

Instead of using 255 rules, only right independent rules can be considered. There are 15 such eligible rules. So, the complexity for such synthesis scheme is $1 + \sum_{i=1}^{n-1} 2^{i+1} \times 2^4 - 1 = O(2^{n+5})$. That is also exponential.

9.4.2 A Simple Synthesis Algorithm

The complexity of the generic synthesis scheme, presented in the earlier subsection, can be reduced by adopting the following measures.

1. Whenever a cross link is identified at ith level, a rule $\mathcal{R}_{i+1}$ is selected so that it makes $(i + 1)$th level as cross link free level $(0 \leq i \leq n - 1)$.
2. The *overlapping situation* is not allowed.

The overlapping situation is a situation when two root equivalent nodes with different sets of RMTs are obtained. Figure 9.10 shows such a situation with {0, 1, 2, 3} and {4, 5, 6, 7} as two root equivalent nodes. The two subtrees, rooted at these nodes, may contain some cross links at a later level which may not be dissolved by any right independent rule at the next level. This is because of the presence of multiple cross links involving the same RMT of the rule, which requires different next state to be dissolved—an impossible requirement!

However, following some restrictions on the self link, it is possible to get sub-nodes with similar cross links and self links. To get sub-nodes, if more than one root equivalent node exist, then $\mathcal{R}_i$ with $\mathcal{R}_i[r] = \mathcal{R}_i[s]$ is selected, where r and s are the pair of equivalent RMTs.

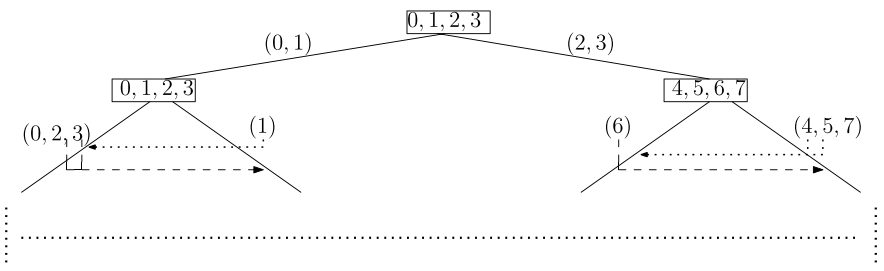

Fig. 9.10 Case of overlapping situation

This synthesis scheme is formally described in Algorithm 17. Input to the algorithm is the number of cells of the CA (n), and output is the desired rule vector. The algorithm selects rules arbitrarily so that at least one self link always exists, and either no cross link exists, or if exists, the cross link is dissolved immediately. The cross links are identified using weights of RMTs, as it has been done in Algorithm 16.

Algorithm 17: Simple synthesis scheme

Input : n
Output: An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary.

Step 1: Select $\mathcal{R}_0$ randomly, provided that at least one of the RMTs 0, 1, 2 and 3 of $\mathcal{R}_0$ is self-replicating.
(i) Find the labels of the edges at level 0 using $\mathcal{R}_0$.
(ii) Set weight of each RMT of l_0 and l_1 as 1.
Step 2: Identify the links at level 0 of the reachability tree.
Step 3: For $i = 0$ to $n - 2$, repeat Step 4 to Step 10
Step 4: Mark the non-reachable edge at level i.
Step 5: If an edge at level i links a non-reachable edge for an RMT r, then
Decrease the weight of the RMT r by 1.
Step 6: If the weights of all the RMTs of a label becomes 0, then
Mark the corresponding edge as non-reachable.
Step 7: Identify cross link, if exists, among the unmarked edges at level i.
Get the nodes of level $i + 1$ from the unmarked edges.
Step 8: If no cross link exists at level i, then
(i) Identify the root equivalent nodes at level $i + 1$ and remove sub-nodes.
(ii) Choose $\mathcal{R}_{i+1}$ randomly, so that (a) at least one self link is formed in an edge at level $i + 1$, and (b) $\mathcal{R}_{i+1}[r] = \mathcal{R}_{i+1}[s]$, if the RMTs r and s are equivalent to each other and they are in two different root equivalent nodes.
Step 9: If a cross link exists among the unmarked edges at level i, then
Choose $\mathcal{R}_{i+1}$ randomly, so that (a) at least one self link is formed in an edge at level $i + 1$, and (b) all the cross links of level i gets dissolved.
Step 10: Identify links among edges at level $i + 1$.
Step 11: Report $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ as the CA.

Complexity: The complexity of Algorithm 17 depends on n only (Step 3). For each level i, if cross links exist, the maximum length of cross link is 2. Therefore, complexity of the algorithm is $O(n)$.

Example 9.14 Let consider the synthesis of a 4-cell CA in null boundary, having point attractors only. Suppose, rule 12 as $\mathcal{R}_0$ is selected randomly. So, each RMT of the labels of $E_{0.0}$ and $E_{0.1}$ is involved in the self link. Next, the rule 102 is selected as $\mathcal{R}_1$. In rule 102, the equivalent RMTs have the same next state. Here, two cross links are observed at level 1: $E_{1.0}(3) \rightarrow E_{1.1}(1) \rightarrow E_{1.0}$ and $E_{1.2}(7) \rightarrow E_{1.3}(5) \rightarrow E_{1.2}$. Now, suppose rule 192 is selected as $\mathcal{R}_2$. As a result, there is no cross link among the edges at level 2, but there are self links at edges $E_{2.0}$ (for RMTs 0 and 1), $E_{2.2}$ (for RMTs 4 and 5), $E_{2.4}$ (for RMTs 0 and 1), $E_{2.6}$ (involving RMTs 4 and 5). Note that the labels of $E_{2.0}$ and $E_{2.4}$ are the same. Similarly, the labels of $E_{2.2}$ and $E_{2.6}$

are also same. Algorithm 17 proceeds with unique nodes only (Step 8(i)). Finally, the rule 85 is selected as $\mathcal{R}_3$. Therefore, the synthesized CA is (12, 102, 192, 85), which is having only point attractors. See Fig. 9.3 for verification.

9.4.3 An Optimized Synthesis Scheme

The generic synthesis scheme (Sect. 9.4.1) results in a huge number of CAs with point attractors only. But, the worst case complexity of this scheme is exponential. A simple synthesis scheme (Sect. 9.4.2), with linear complexity, on the other hand, greatly compromises with the number of synthesized CAs. As a trade off between these two schemes, the following measures are taken in an optimized synthesis scheme.

1. The cross links are allowed to grow. A cross link at level i may propagate to the level $n-2$ but it has to be dissolved at level $n-1$ ($0 \leq i \leq n-3$).
2. Identify the condition which restricts the overlapping situation. The situation indicates that more than one cross link exist, involving the same (or equivalent) RMT at level i, but to dissolve the cross links at level i, different next states are required for the same RMT at level $i+1$.
3. If cross links exist at level i, there is to be a rule $\mathcal{R}_{i+1}$ which ensures the level $i+1$ cross link free.

To have only the point attractors, the overlapping situation should not be allowed. The RMTs involved in both the self link and cross link are responsible for the overlapping situation. Figure 9.10 depicts such a scenario. In Algorithm 17, this has been ensured by choosing the rules, in which the pairs of equivalent RMTs have the same next state. This preventive measure works well as the cross link at level i gets dissolved at the level $i+1$, and a number of root equivalent nodes and their sub-nodes are formed. However, in this scheme, a cross link at level i may propagate to the level $n-2$. So, Condition 9.1 is imposed.

Condition 9.1 *If the RMTs $r_1, r_2, \ldots, r_k$ of $\mathcal{R}_i$ participate in the self links, then either the RMTs $2r_1, 2r_2, \ldots, 2r_k \pmod 8$ or RMTs $2r_1+1, 2r_2+1, \ldots, 2r_k+1 \pmod 8$ of $\mathcal{R}_{i+1}$ are self-replicating.*

Since the resulted CA does not have multi-length cycle, the condition of Corollary 9.3 is to be avoided. The following condition ensures existence of only one cross link at ith level. Moreover, the overlapping situation is also restricted.

Condition 9.2 *If the RMTs r_1 and r_2 of $\mathcal{R}_i$ participate in a cross link, then RMTs $2r_1$ and $2r_2+1 \pmod 8$ of $\mathcal{R}_{i+1}$ are the self-replicating (resp. non-self-replicating), and RMTs $2r_1+1$ and $2r_2 \pmod 8$ of $\mathcal{R}_{i+1}$ are non-self-replicating (resp. self-replicating).*

Corollary 9.3 states that if two cross links exist, which share sibling RMTs, then there exists a multi-length cycle attractor. Both the sibling RMTs have the same next

state to maintain Corollary 9.3. But, due to this condition, two RMTs of a sibling pair have different next states. Following are the algorithmic steps for synthesis of CA.

1. For each level, a rule $\mathcal{R}_i$ is chosen so that a self link exists at the ith level.
2. For the RMTs that participate in self links at ith level, follow Condition 9.1 for $(i+1)$th level.
3. For the RMTs that participate in cross links at ith level, follow Condition 9.2 for the $(i+1)$th level.
4. For choosing the rule for $(i+1)$th level, if any RMT is overwritten for the use of Conditions 9.1 or 9.2, then appropriately choose $\mathcal{R}_{i+1}$ so that no cross link is formed at $(i+1)$th level.
5. Remove the sub-nodes and proceed with unique nodes only.
6. Repeat Steps 1–5 upto leaf level.

The above steps are detailed out in Algorithm 18. It also deals with the labels of edges. The input to the algorithm is the number of cells of the CA to be synthesized and the output is the CA with point attractors only.

Example 9.15 Let consider the synthesis of 4-cell CA in null boundary, having only point attractors. Suppose, rule 10 is chosen as $\mathcal{R}_0$. Its RMTs 0 and 3 are self-replicating and, therefore, satisfies the requirement of at least one self-replicating RMT. One cross link is formed by $E_{0.0}$ and $E_{0.1}$ $[E_{0.0}(2) \rightarrow E_{0.1}(1) \rightarrow E_{0.0}]$ (see Fig. 9.11). Now, following Steps 8 and 9 of Algorithm 18, rule 86 is selected as $\mathcal{R}_1$. Following cross links $E_{1.1}(4) \rightarrow E_{1.2}(3) \rightarrow E_{1.1}$, $E_{1.1}(4) \rightarrow E_{1.2}(7) \rightarrow E_{1.3}(2) \rightarrow E_{1.1}$, $E_{1.0}(5) \rightarrow E_{1.2}(3) \rightarrow E_{1.1}(1) \rightarrow E_{1.0}$ and $E_{1.0}(5) \rightarrow E_{1.2}(7) \rightarrow E_{1.3}(2) \rightarrow E_{1.1}(1) \rightarrow E_{1.0}$, and the self links at $E_{1.0}(0)$ and $E_{1.3}(6)$ are formed. The cross

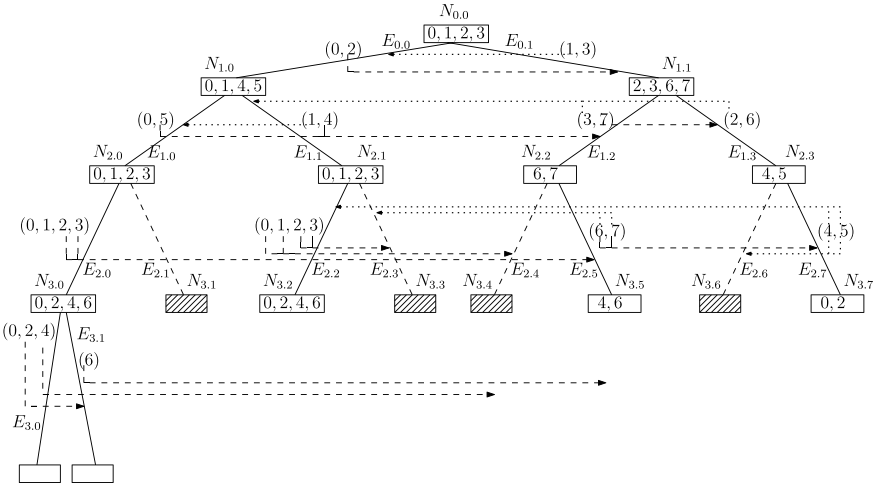

Fig. 9.11 Synthesis of CA (10, 86, 240, 64) following Algorithm 18

Algorithm 18: Optimised synthesis scheme

Input : n (CA size)
Output: $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ (null boundary)

Step 1: Select $\mathcal{R}_0$ randomly, provided that at least one of the RMTs 0, 1, 2 and 3 of $\mathcal{R}_0$ is self-replicating.
(i) Find the labels of the edges at level 0 using $\mathcal{R}_0$.
(ii) Set weight of each RMT of l_0 and l_1 as 1.
Step 2: Identify the links at level 0 of the reachability tree.
Step 3: For $i = 0$ to $n - 3$, repeat Step 4 to Step 13
Step 4: Mark the non-reachable edge at level i.
Step 5: If an edge at level i links a non-reachable edge for an RMT r, then
Decrease the weight of the RMT r by 1.
Step 6: If the weights of all the RMTs of a label becomes 0, then
Mark the corresponding edge as non-reachable.
Step 7: Identify self link and cross link, if exists, among the unmarked edges at level i.
Step 8: Use Condition 9.1 to set next states of some RMTs of $\mathcal{R}_{i+1}$.
Step 9: Use Condition 9.2 to set next states of some RMTs of $\mathcal{R}_{i+1}$.
Step 10: If conflict arises during the setting of next states of RMTs in Steps 8 and 9, then
Choose a rule as $\mathcal{R}_{i+1}$ so that no cross link but at least one self link can be formed.
Step 11: Otherwise, if any RMT is left from assigning next state in Steps 8 and 9, then arbitrarily assign the next state.
Step 12: Get the nodes of level $i + 1$ from the unmarked edges.
Identify the root equivalent nodes at level $i + 1$ and remove sub-nodes.
Step 13: Identify links among edges at level $i + 1$.
Step 14: Select $\mathcal{R}_{n-1}$ such that at least one RMT is self link but no cross link is formed.
Step 15: Report $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ as the CA.

links of level 1 should not propagate to level 2, as for RMTs are overwritten due to Condition 9.2. So, the $\mathcal{R}_2$ has to dissolve all the cross links among the edges at level 2. Hence, the rule 240 is selected as $\mathcal{R}_2$ which makes the level 2 cross link free. Further, four root equivalent nodes are generated, out of which only one ($\{0, 2, 4, 6\}$) is unique and the rest are the subset of it. So, the algorithm proceeds with one node only. Finally, the rule 64 is selected as $\mathcal{R}_3$. Therefore, the CA is (10, 86, 240, 64), which is having only point attractor 0000 (Fig. 9.11 has no cross link at leaf level).

Theorem 9.7 *Maximum length of cross link, as per Algorithm 18, is 8.*

Proof Without repeating an RMT, maximum possible length of a cross link at a level, say i can be 8. For further increment in length, other cross links and self links at the same level are required. In this scenario, to find the rule for cell $i + 1$, some RMTs of $\mathcal{R}_{i+1}$ are overwritten due to Conditions 9.1 and 9.2 (Steps 8 and 9). So, as per Step 10, $\mathcal{R}_{i+1}$ is chosen in such a way that all the cross links get dissolved. Hence, the maximum length of a cross link in the tree of a CA, synthesized by Algorithm 18, cannot be more than 8. □

Complexity: The complexity of Algorithm 18 depends on n only (Step 3). The rules are set in Steps 8–11. The algorithm identifies the self link and cross link, if any.

According to Theorem 9.7, the maximum length of a cross link is 8. If cross link does not occur, then according to Lemma 22, there are maximum 4 root equivalent nodes. Therefore, the maximum number of checking is constant and, therefore, the complexity of Algorithm 18 is $O(n)$.

9.4.4 Synthesis of CA with PE Bits

The synthesis algorithms, reported in the earlier subsections, do not consider the number of point attractors. This section reports a scheme that synthesizes a CA with specified PE bits—that is, with fixed number of point attractors.

Theorem 9.8 *In an n-cell CA with null boundary, having only point attractors, if the ith bit is PE bit, then the level $i-1$ of the reachability tree is free of cross link.*

Proof If level i of the reachability tree corresponds to a PE bit, then for each node $N_{i.j}$ of level i, there exist two edges $E_{i.2j}$ and $E_{i.2j+1}$ which participate in self links. At $E_{i.2j}$ ($E_{i.2j+1}$), a set of RMTs $S_0 \subseteq \{0, 1, 4, 5\}$ ($S_1 \subseteq \{2, 3, 6, 7\}$) participate in self links. The RMTs of S_0 (S_1) are obtained from the even (odd) RMTs of $l_{i-1.j}$. That is, at level $i-1$, the sibling RMTs r, $s \in l_{i-1.j}$ are in self link. According to Theorem 8.10, if the edge $E_{i-1.j}$ is self linked for two sibling RMTs r, $s \in l_{i-1.j}$, then the RMT sequence that involves RMT $t \in l_{i-1.j} \setminus \{r, s\}$ represents acyclic configurations. Thus, $E_{i-1.j}$ is not involved in any cross link. □

The algorithm to synthesize CAs with specified PE bits needs to address the four cases, as noted below.

(i) **The ith and $(i+1)$th bits are PE bits**: As ith bit is PE bit, each node of level i in the reachability tree for attractors has to have two children. Since $(i+1)$th bit is also an PE bit, each node of level $i+1$ also has to have two children. Hence, a label of an edge that connects the nodes of levels i and $i+1$ has to contain a pair of sibling RMTs. This implies that the RMTs 0, 1, 2 and 3, or RMTs 4, 5, 6 and 7 of $\mathcal{R}_i$ are self-replicating RMTs.

(ii) **The ith bit is PE bit but the $(i+1)$th is not a PE bit:** There are at least two RMTs $r \in S_0$ (S_1) and $s \in S_1$ (S_0). Theorem 9.6 and Corollary 9.3 are not to be verified for $\mathcal{R}_i$.

(iii) **The ith bit is not a PE bit but $(i+1)$th is the PE bit**: Theorem 9.8 is followed by $\mathcal{R}_i$ such that there exist self links at edge $E_{i.j}$ for the sibling RMT pair r and s. The cross links of $(i-1)$th level are dissolved by $\mathcal{R}_i$. Hence, at least the RMTs of one of the sets—{0, 1}, {4, 5}, {2, 3} and {6, 7} of $\mathcal{R}_i$ are self-replicating.

(iv) **The ith and $(i+1)$th bits are not the PE bits**: The set of RMTs {0, 4}, {1, 5}, {2, 6} and {3, 7} have to have the same next state in $\mathcal{R}_i$.

The synthesis schemes reported above (Algorithms 17 and 18) can be extended to adopt PE bits in the synthesis process. Following example illustrates the synthesis process with specified PE bits.

Example 9.16 Let consider the synthesis of a 4-cell CA in null boundary, where the bit 0 is the PE bit. Since the bit 0 is the PE bit but the bit 1 is not a PE bit, rule 4 is selected as $\mathcal{R}_0$. For rule 4, the RMTs 0 and 1 (belong to S_0) and RMT 2 (belongs to S_1) are self-replicating. So, the RMTs 0, 1, 4 and 5 from S_0 of $\mathcal{R}_1$ and the RMTs 2 and 3 from S_1 are the candidate RMTs of $\mathcal{R}_1$ that can generate point attractors. Suppose, rule 0 is chosen as $\mathcal{R}_2$. Then, RMTs 0, 1, 4 and 5 are self-replicating, and all these are in S_0. As the second and third bits are not the PE bits and the RMTs from S_0 form self link in level 1, the rule 31 is selected as $\mathcal{R}_2$. The RMTs 2 and 3 of rule 31 are self-replicating. Finally, the rule 85 is selected as $\mathcal{R}_3$. Hence, the synthesized CA is (4, 0, 31, 85).

9.5 Density Classification Problem

The density classification problem has been introduced in 1978 by G'acs, Kurdyumov and Levin. It is also known as the majority task. The task is to find a binary CA such that, if an initial configuration has more 1s than 0s (density of 1s is higher than density of 0s), the automaton should converge to all 1s point attractor. Otherwise, it converges to all 0s point attractor. This section reports the synthesis of CAs with only two point attractors—all 0s (0^n) and all 1s (1^n). These CAs are of interest while addressing the density classification problem.

Let first consider the synthesis of a CA in null boundary with 0^n and 1^n point attractors only. If RMT 0 of all the rules of the CA, and RMT 3 of the first rule, RMT 6 of the last rule and RMT 7 of rest rules of the CA are self-replicating, then the configurations 0^n and 1^n of the CA are point attractors. That is, the CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ has to satisfy the following condition: $\mathcal{R}_i[0] = 0$ for all $i \in \{0, 1, \ldots, n-1\}$, $\mathcal{R}_j[7] = 1$ for all $j \in \{1, 2, \ldots, n-2\}$, $\mathcal{R}_0[3] = 1$ and $\mathcal{R}_{n-1}[6] = 1$. Hence, only four rules—8, 10, 12 and 14 can act as $\mathcal{R}_0$ in null boundary, considering the next states of the RMTs 4, 5, 6 and 7 as 0. However, to ensure that there is no other point attractor or multi-length cycle, the reachability tree is to be explored. The tree is to be free from cross link among the edges at level $n-1$ and there should not be self linked edges other than the $E_{n-1.0}$ and $E_{n-1.2^{n-1}-1}$. The following are some of the schemes to synthesize such a CA.

Scheme 1: Design an n-cell CA in which the next configurations of all the configurations is either 0^n or 1^n. That is, all the configurations of the CA, other than the 0^n or 1^n, are non-reachable. The first rule of the CA can be any one of the four rules—8, 10, 12 and 14. While constructing the reachability tree of the CA, no cross link is formed at level 0. From the level 1, if the tree is partitioned into two sub-trees, then the left sub-tree can have only the forward link and the right sub-tree has only the backward link. Figure 9.12 shows the possible links in the reachability tree of such

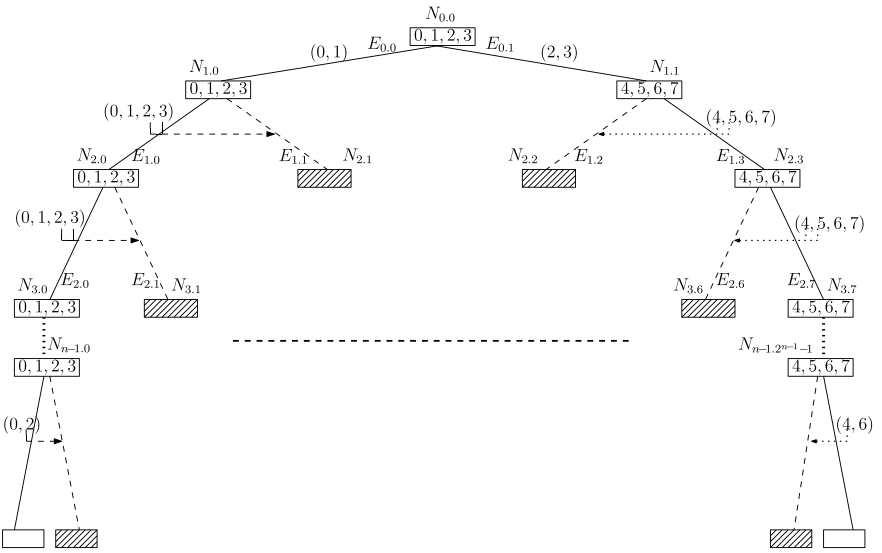

Fig. 9.12 Trivial case of CA having point attractors 0^n and 1^n

a CA. The RMTs that are not involved in self links are removed from the nodes. Figure 9.12 corresponds to the n-cell CA (12, 240, 240, ..., 240) for any $n > 1$.

Scheme 2: This is to synthesize other sets of CAs that are having some reachable configurations other than 0^n and 1^n. These configurations are acyclic. The scheme to synthesize such CAs utilizes the fact that the cross link of level $i-1$ in the reachability tree can be dissolved at level i. To restrict formation of point attractors other than 0^n and 1^n, the following condition is imposed on rules.

Condition 9.3 *The RMTs 3 and 4 of $\mathcal{R}_i$ do not participate in self link at level i, $1 \leq i \leq n-1$.*

The synthesis scheme is summarised in Table 9.2. For the rules of cell 3 to cell $n-2$, the RMTs 0, 1, 3, 4, 6 and 7 have fixed next states. Whereas, the RMTs 2 and 5 can have any next states. Hence, there are only four possible rules for the $\mathcal{R}_3, \mathcal{R}_4, \ldots, \mathcal{R}_{n-2}$. To find a rule vector for an n-cell CA, (i) $\mathcal{R}_0$ is chosen randomly from the set of four rules (8, 10, 12, 14). Based on the $\mathcal{R}_0$, $\mathcal{R}_1$ is selected from Table 9.2. (ii) $\mathcal{R}_2$ is always 240. (iii) $\mathcal{R}_i$ is selected depending on $\mathcal{R}_{i-1}$, where $3 \leq i \leq n-2$, and (iv) $\mathcal{R}_{n-1}$ is always 80. For example, the 7-cell CA (8, 208, 240, 212, 240, 244, 80) is a desired CA and it is synthesized following Table 9.2.

Table 9.2 Rules of Scheme 2

First three cells			Intermediate cells		Last cell
$\mathcal{R}_0$	$\mathcal{R}_1$	$\mathcal{R}_2$	$\mathcal{R}_{i-1}$	$\mathcal{R}_i$	$\mathcal{R}_{n-1}$
8	192, 208, 224, 240	240	240	240, 212, 208, 244	80
10	192, 196, 200, 204		212	240	
12	240		208	208, 240	
14	240, 244, 248, 252		244	244, 240	

Scheme 3: This scheme synthesizes the CAs obeying Condition 9.3 and the other properties of CA for having point attractors only. From cell 3 onward in the Scheme 2, the common structure of rule vectors is resulted irrespective of the first rule. But in this scheme, the CA depends on the first rule. The following are the desired CAs with 0^n and 1^n as the only point attractors.

1. $(12, (212/208/244)^{n-1})$,
2. $(14, (242/250/246/254)^{n-1}, 80)$,
3. $(8, (128/160/144/176)^{n-2}, 80)$,
4. $(10, 206, (242/246/250/254)^{n-3}, 80)$,
5. $(10, 130, (128/160/176/144)^{n-3}, 80)$.

Scheme 4: It clubs the schemes 2 and 3 enabling a wider variety of CAs with only two point attractors 0^n and 1^n. This construction of a CA is done in two parts. The first part contains the rules of Scheme 3 (excluding the last rule), then the rule 240 is added. The second part contains the rules of Scheme 2 (Table 9.2) that are considered for the intermediate cells in Scheme 2. Table 9.3 shows the rules and their relation to this scheme. An 8-cell CA can be synthesized as follows: (i) for the first part, consider (14, 254, 250, ...), and (ii) then consider the rule 240, and (iii) then recursively synthesize the rules for remaining cells (using Table 9.3) to finally get the CA (14, 254, 250, 240, 244, 244, 240, 80).

Table 9.3 Rules of Scheme 4

Upto cell k			Cell $k+1$	Cell $k+2$ to Cell $n-2$		Cell $n-1$
$\mathcal{R}_0$	$\mathcal{R}_1$	$\mathcal{R}_2$ to $\mathcal{R}_k$	$\mathcal{R}_{k+1}$	$\mathcal{R}_{i-1}$	$\mathcal{R}_i$	$\mathcal{R}_{n-1}$
10	206	242, 246, 250, 254	240	240	240, 212, 208, 244	80
	130	128, 160, 176, 144				
8	128, 160, 144, 176			212	240	
12	212, 208, 244			208	208, 240	
14	242, 250, 246, 254			244	244, 240	

None of the CAs, synthesized from the above schemes, can completely solve the density classification problem. So, the accuracy of the classification is to be found out for understanding the efficacy of the synthesized CAs.

A minor modification to the above schemes can synthesize CAs in periodic boundary. For periodic boundary condition, RMTs 0 and 7 of each rule of the rule vector of a CA are to be self-replicating. An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ in periodic boundary, where $\mathcal{R}_i$ is either 232 or 184 (226), can be such a CA.

9.6 Single Attractor Cellular Automata

A single attractor CA (SACA) contains only one point attractor in its configuration space to which the CA converges from any initial configuration. These CAs are also known as nilpotent CAs. The CAs with rules 0 and 255 are SACAs. There are other SACAs which can be characterized by the theories developed in earlier sections. The necessary properties of the reachability tree of an n-cell SACA are the following.

1. At least one edge at each level of the reachability tree is self linked.
2. There exists exactly one self-linked edge at level $n-1$.
3. There is no cross-link among the edges at level $n-1$.

Theorem 9.9 *A CA is not an SACA if for each self link $E_{i.j}(r) \rightarrow E_{i.j}$, there exists another self link $E_{i.k}(s) \rightarrow E_{i.k}$, where $j \neq k$, and either $r = s$ or r and s are equivalent to each other $(0 \leq i \leq n-1)$.*

Proof When $i = n-1$, the theorem is trivially followed. Another trivial case is that there is no self link in the edges at level i. Obviously, then the CA is not an SACA.

For the proof of non-trivial cases, let $E_{i.j}(r) \rightarrow E_{i.j}$ and $E_{i.k}(s) \rightarrow E_{i.k}$ $(j \neq k$ and $i < n-1)$ be two self links, where either $r = s$, or r and s are equivalent to each other. From the definition of links, two links at the next level for each of the links can be observed in null boundary when $i < n-2$, and only one link when $i = n-2$ (see Sect.. 8.2.1). Now, if finally a self link is received at leaf level from any one of these two links, then another self link can also be received due to the other one. That is, if a point attractor is resulted from one link at level i, then another point attractor can always be received.

Same conclusion can be drawn if there exist another such pair of self links. Hence, if all the self links can be paired up in the above fashion at any level i, the CA can have either no point attractor, or more than one point attractor. □

Example 9.17 Consider the 4-cell CA $(77, 252, 197, 10)$ in null boundary. The reachability tree (Fig. 9.13) of the CA contains two self links at level 1—$E_{1.1}(2) \rightarrow E_{1.1}$ and $E_{1.3}(6) \rightarrow E_{1.3}$. Both the self links result in the self links $E_{3.4}(0) \rightarrow E_{3.4}$ and $E_{3.12}(0) \rightarrow E_{3.12}$ respectively at level 3 as RMTs 2 and 6 are equivalent to each other. There exist another two self links at level 1—$E_{1.1}(3) \rightarrow E_{1.1}$ and $E_{1.3}(7) \rightarrow E_{1.3}$.

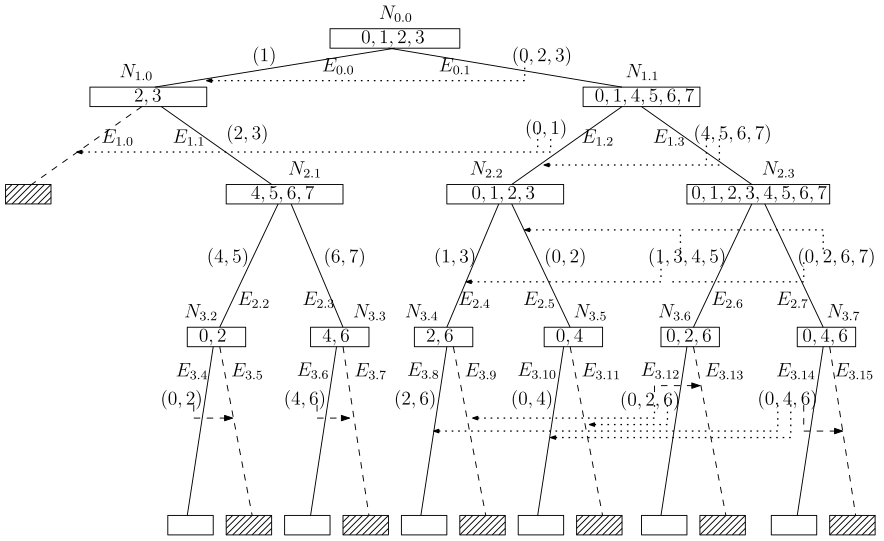

Fig. 9.13 Reachability tree of CA (77, 252, 197, 10) in null boundary

These self links also result in self links $E_{3.6}(4) \rightarrow E_{3.6}$ and $E_{3.14}(4) \rightarrow E_{3.14}$ respectively at level 3 as RMTs 3 and 7 are equivalent to each other. Hence, the CA contains more than one point attractor and thus, it is not an SACA.

Theorem 9.10 *An n-cell CA in null boundary is not an SACA if there exist at least two edges $E_{i.j}$ and $E_{i.k}$, $j \neq k$, each of which are self linked for a pair of sibling RMTs.*

Proof When $i = n - 1$, the theorem is trivially proved. For the proof of non-trivial cases, let us consider that in the reachability of a CA in null boundary, there are two self linked edges $E_{i.j}$ and $E_{i.k}$, $j \neq k$, for two pairs of sibling RMTs. That is, $E_{i.j}(r) \rightarrow E_{i.j}$ and $E_{i.j}(s) \rightarrow E_{i.j}$ for sibling RMTs r are s, and $E_{i.k}(r') \rightarrow E_{i.k}$ and $E_{i.k}(s') \rightarrow E_{i.k}$ for another pair of sibling RMTs r' and s'.

Now, a set of edge sequences, including $E_{i.j}$, can be obtained in the reachability tree, which represents a set of reachable configurations. Say, this set is S_1. Since $E_{i.j}$ is self linked twice for two RMTs, no other edge at level i can link $E_{i.j}$ (Proposition 8.5). This implies that the next configuration of any configuration in S_1 is also in S_1.

Similarly, another set S_2 against the edge $E_{i.k}$ can be obtained. Here, $S_1 \cap S_2 = \emptyset$, as $j \neq k$. That is, the CA with the configurations of S_1 approaches to an attractor (of length one or more), and the CA with configurations of S_2 approaches to another attractor (of length one or more). Hence, the CA is not an SACA. □

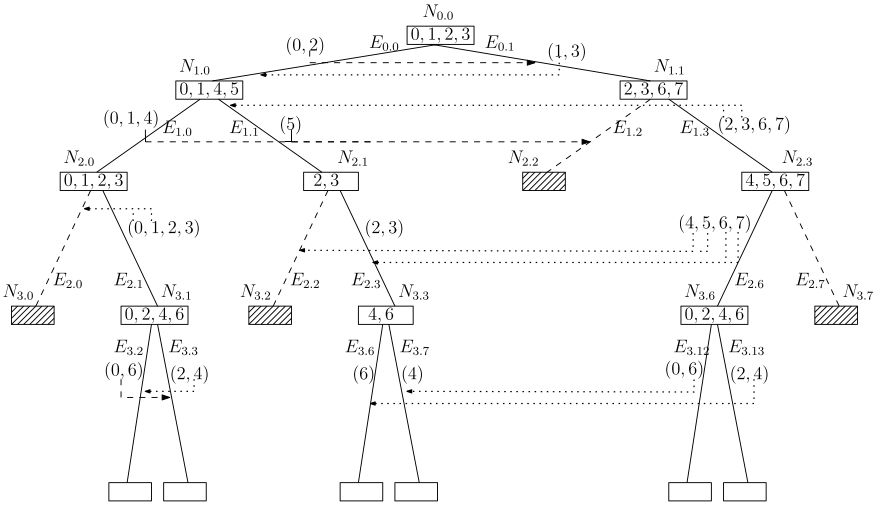

Fig. 9.14 Reachability tree of CA (138, 236, 15, 20) in null boundary

Example 9.18 The reachability tree of a 4-cell CA (138, 236, 15, 20) is shown in Fig. 9.14. The edge $E_{1.0}$ is self linked for the sibling RMTs 0 and 1. The self links on this edge result in a cross link $E_{3.2}(6) \rightarrow E_{3.3}(4) \rightarrow E_{3.2}$ at level 3. The edge $E_{1.3}$ is also self linked for the sibling RMTs 6 and 7 at the second level. The self links on this edge $E_{1.3}$ trigger two self links at level 3 that are $E_{3.12}(0) \rightarrow E_{3.12}$ and $E_{3.13}(2) \rightarrow E_{3.13}$. Thus, the CA is not an SACA.

Now, to decide whether a given CA is SACA, the above theories, as well as the theories reported in Sect. 9.3, are to be applied in the reachability tree of the CA. In some special cases, it is possible to decide an SACA by observing the RMTs of every rule.

Theorem 9.11 *An n-cell CA* $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ *in null boundary is an SACA if* $\mathcal{R}_i[0] = \mathcal{R}_i[2] = \mathcal{R}_i[4] = \mathcal{R}_i[6] = 0$, *for all* $i \in \{0, 1, \ldots, n-1\}$.

Proof Let consider that the next states of RMTs 0, 2, 4 and 6 of each rule are 0. That is, $\mathcal{R}_i[0] = \mathcal{R}_i[2] = \mathcal{R}_i[4] = \mathcal{R}_i[6] = 0$, for each $i \in \{0, 1, \ldots, n-1\}$. Hence, 0^n is a point attractor of the CA. No other point attractor can be formed in the CA. Now, to get a multi-length cycle, a cross link among the edges at level $n-1$ of the reachability tree is to be formed. However, in this CA, only 0-edges cannot form cross link at any level. Since 1-edges are not present at the last level (as $\mathcal{R}_{n-1}[0] = \mathcal{R}_{n-1}[2] = \mathcal{R}_{n-1}[4] = \mathcal{R}_{n-1}[6] = 0$), no cross link can be formed. Hence the proof. □

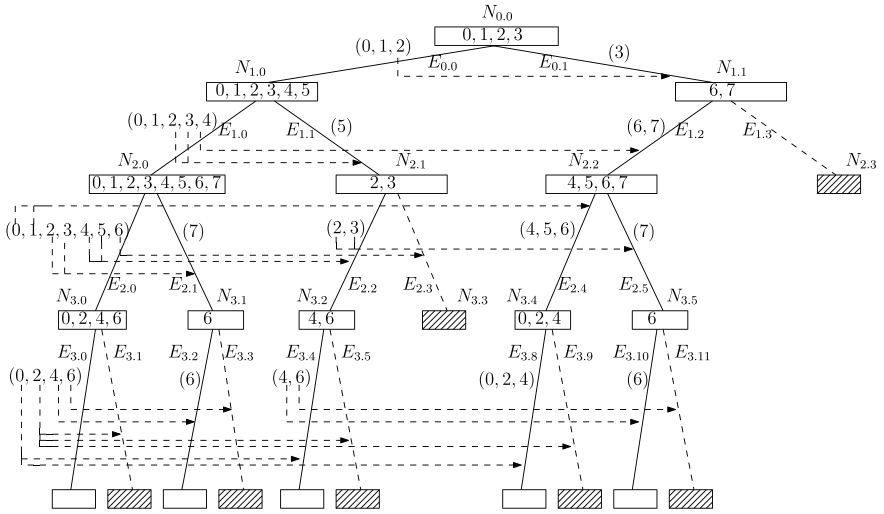

Fig. 9.15 Reachability tree of CA (8, 32, 128, 10) in null boundary

Example 9.19 Figure 9.15 shows a reachability tree of the CA (8, 32, 128, 10) in null boundary. The rules satisfy the condition of Theorem 9.11. The self links at edge $E_{0.0}$ for RMTs 0, 1 and edge $E_{0.1}$ for RMT 3 are present at level 0. At level 1, all but RMTs 0 and 1 participate only in forward links. A self linked edge $E_{2.0}$ for RMTs 0 and 1 exists. The level 3 contains only a self linked edge $E_{3.0}$ for the RMT 0, and there is no cross link among the edges at level 3. Hence, the CA is an SACA.

Now, to get SACAs, the rules are to be selected in such a way that only one self link remains in the reachability tree, but no cross link is formed. For this purpose, Conditions 9.1 and 9.2 (Sect. 9.4.3) are to be utilized. In addition to these two, another condition is required which enforces the CA to violate the condition of Theorem 9.9.

Condition 9.4 *If there exists a self linked edge for an RMT r of $\mathcal{R}_i$ at level i, then there is to be no other edge at the same level that participates in self link for r or for equivalent RMT of r.*

Algorithm 19 uses these conditions to synthesize an SACA in null boundary. The algorithm deals with the labels of edges. The output of the algorithm is a rule vector of SACA with n rules, where n is the number of cells of the CA.

Algorithm 19: Synthesis of SACA

Input : n
Output: An n-cell CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$ in null boundary

Step 1: Select $\mathcal{R}_0$ randomly so that at least one, but not all of the RMTs 0, 1, 2 and 3 of $\mathcal{R}_0$ are self-replicating.
Find $l_0 = \{r \in \{0, 1, 2, 3\} \mid \mathcal{R}_0[r] = 0\}$ and $l_1 = \{r \in \{0, 1, 2, 3\} \mid \mathcal{R}_0[r] = 1\}$.
Step 2: If l_0 (resp. l_1) consists of a sibling pair of RMTs (r_1, r_2) and l_1 (resp. l_0) consists of an RMT r_3 where r_1, r_2 and r_3 participate in self link, then RMTs $2r_3$ and $2r_3 + 1$ of $\mathcal{R}_1$ are not to be self-replicating.
Step 3: For $i = 1$ to $n - 2$, repeat Steps 4 to 11
Step 4: Find and store $2r \pmod 8$ and $2r + 1 \pmod 8$ for all RMTs r, that are self linked at $(i - 1)$th level and set RMTs at ith level using Condition 9.1.
Step 5: If RMTs r_1 and r_2 at ith level participate in self link and $2r_1 \equiv 2r_2 \pmod 8$, then follow Condition 9.4.
Step 6: Check whether a cross link exists for any RMTs of l_js.
If exists, goto Step 7.
Otherwise, goto Step 8.
Step 7: For each cross link at $(i - 1)$th level, set RMTs at ith level using Condition 9.2.
Step 8: Set all unfilled RMTs of $\mathcal{R}_i$ arbitrarily, goto Step 10 and discard all forward and backward links.
Step 9: Check whether any overlapping situation occurs for any RMT of $\mathcal{R}_i$. If such a situation occurs then find a $\mathcal{R}_i$ that discard all cross link.
Step 10: For each non-empty labels l_k
Find l'_{2k} and l'_{2k+1} so that, if $r \in l_k$ and $s = 2r \pmod 8$ or $2r + 1 \pmod 8$, then $s \in l'_{2k}$ (resp. l'_{2k+1}) when RMT s of $\mathcal{R}_i$ is 0 (resp. 1).
Step 11: Assign non-empty and unique l'_j to l_k so that the links among l'_j for each RMT is maintained.
Step 12: Select $\mathcal{R}_{n-1}$ such that all the cross link are discarded and exactly one self link is kept.
Step 13: Report the CA $(\mathcal{R}_0, \mathcal{R}_1, \cdots, \mathcal{R}_{n-1})$.

Example 9.20 Let illustrate the synthesis of the 4-cell CA (232, 38, 90, 186) (see Fig. 9.16). At first, the rule 232 is selected randomly in Step 1. Here, $l_{0.0}$ consists of the RMTs 0, 1 and 2, and $l_{0.1}$ consists of the RMT 3. The sibling pair 0 and 1, and RMT 3 participate in self links. Hence, at the next level, the RMTs 6 and 7 become non-self-replicating (Step 2). As the RMT 2 participates only in forward link at the first level, this RMT with the corresponding link has not been considered at the further levels. The values of the rest of the RMTs of rule $\mathcal{R}_1$ are chosen using Steps 4, 5 and 8. A cross link $(E_{0.0}(3) \rightarrow E_{0.1}(1) \rightarrow E_{0.0})$ exists at level 1. Therefore, along with Steps 4 and 5, Step 7 is also used to choose the rule $\mathcal{R}_2$—that is, rule 90. At the last level, the rule 186 is selected arbitrarily following Step 12 such that the last level contains exactly one self link with RMT 0 $(E_{3.0}(0) \rightarrow E_{3.0})$ and no cross link.

Complexity: A level is left with only self links if it does not contain any cross link. This is because all the forward and backward links are discarded if they do not participate in cross link. Moreover, as the algorithm follows Condition 9.4, there

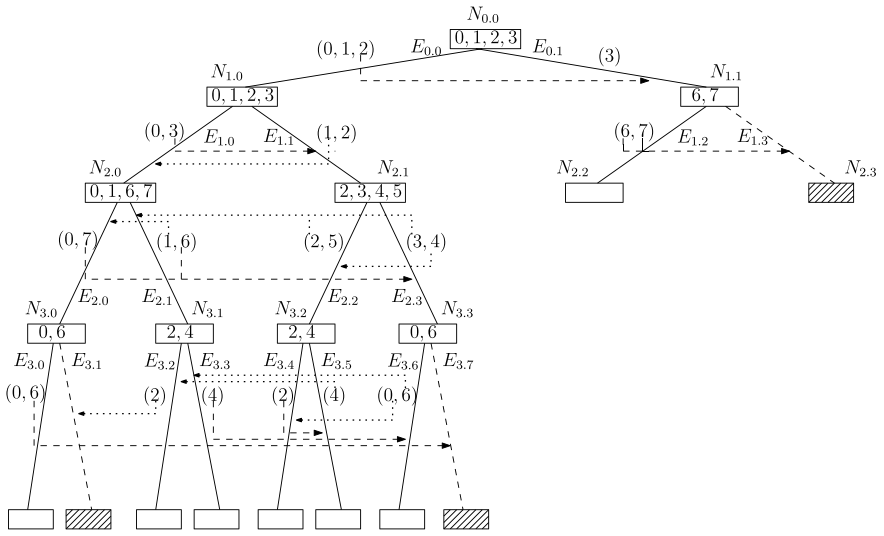

Fig. 9.16 Reachability tree of CA (232, 38, 90, 186) in null boundary

cannot be identical or equivalent RMTs at a level. There can be at most four RMTs at a level that participate in self link, provided that there is no sibling pair of RMTs to participate in the self link. If there exists any sibling pair of RMTs that participates in self link, then the number of self links at that level is 2 for the sibling pair as only that sibling pair converge to self link at the last level (as given in Step 2 of Algorithm 19). Therefore, in any case the maximum number of self links as well as the maximum number of edges becomes constant. Hence, the complexity depends only on the size of the CA (n)—that is, $O(n)$.

9.7 Chapter Notes

The MACAs have been first observed and studied in [1]. The work has later been extended in [2, 3]. Reachability tree has been explored to identify point attractors, PE-bits and identification bits in cellular automata [4]. The links in reachability tree have been conceptualized to study non-linear MACAs in [5, 6], which have been detailed out in [7]. The SACAs have been explored in [8, 9]. Nilpotency of non-uniform CAs on infinite lattice has been addressed in [10].

The following are few research directions, related to this chapter, for the future.

1. Develop a theoretical framework to understand the ability of non-linear MACA as hash function.
2. Analysis and synthesis of non-linear CAs with only attractors of length $k \geq 2$.

3. Analysis and synthesis of non-linear CAs that converge to attractors of length less than or equal to k, for a given k ($k \geq 2$).
4. Synthesis of non-linear MACAs with a given number of point attractors.
5. Analysis and synthesis of non-linear MACAs and SACAs with a specific height.

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Chapter 10
Chaos

Chaos can only be observed in those cellular automata (CAs) which are defined over infinite lattice. For the study of chaos in non-uniform CAs, one-dimensional 3-neighborhood CAs with $d \geq 2$ states per cell, defined over $\mathbb{Z}$ are considered in this chapter. The chaotic systems are unpredictable. To understand the potential of a cellular automaton (CA) to be unpredictable, a thought experiment is arranged (Sect. 10.2). The experiment records that for a small change in a cell, if the whole system can be affected, then the CA can be unpredictable as well as chaotic. That is, there is an *information flow* from the affected cell. To capture the notion of information flow, a binary relation over the set of cells is introduced (Sect. 10.3). The relation leads to a new definition of chaos in CAs. However, sometime the information flow is blocked due to the the presence of a *blocking word* in the configuration space of the CA. Such a blocking word restricts the CA to be chaotic. Presence of a blocking word in a CA is identified in Sect. 10.4. Finally, a parameter to predict whether the behavior of a CA is chaotic is reported in Sect. 10.5.

10.1 Definitions

This chapter deals with the one-dimensional d-state CAs which consist of infinitely many cells placed over $\mathbb{Z}$. A cell $i \in \mathbb{Z}$ assumes a state from $S = \{0, 1, \ldots, d-1\}$ and moves to its next state following a rule $\mathcal{R}_i : S^3 \to S$. Hence, the CA is $(\mathbb{Z}, S, 1, \mathcal{R}_i)$. The set of local rules $\{\mathcal{R}_i\}_{i\in\mathbb{Z}}$ induces the global transition function $G : S^{\mathbb{Z}} \to S^{\mathbb{Z}}$, where $S^{\mathbb{Z}}$ is the set of all possible configurations of the CA.

Although the cells of a CA, in general, can follow any rule, a finite sequence of rules may repeat in some CAs. Suppose a sequence of n rules $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})$ repeats over $\mathbb{Z}$. Then, the rule sequence of the non-uniform CA is expressed as $(\mathcal{R}_0, \mathcal{R}_1, \ldots, \mathcal{R}_{n-1})^{\mathbb{Z}}$. Further, if a rule f repeats in k consecutive places in a

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sequence, then the rules of k such places are abbreviated as f^k. Let consider a sequence of ECA rules (90, 90, 90, 90, 90, 90, 90, 90, 150, 150, 150, 150, 150, 150, 150). This sequence can also be represented as $(90^8, 150^7)$. If this sequence repeats over $\mathbb{Z}$, then it is denoted as $(90^8, 150^7)^{\mathbb{Z}}$.

The configuration space $S^{\mathbb{Z}}$ forms a *compact metric space* for a suitable distance metric *dis*. The $dis : S^{\mathbb{Z}} \times S^{\mathbb{Z}} \to \mathbb{R}$ finds distance between two configurations of the CA. Further, the global transition function G is continuous. It means that if two configurations are close to each other in terms of *dis*, then their images are also close to each other. These two properties (compactness and continuity) make the CAs as topological dynamical systems. Hence, the concepts and results of topological dynamics can be used in the study of chaos in CAs.

Many definitions of chaos consider the notion of *sensitivity to initial condition*. In a sensitive CA, a (small) change in the initial condition would greatly affect the CA in future. That is, for a sensitive CA, if x and y are two configurations, one of which is marginally different than the other (mathematically, $dis(x, y) < \epsilon$ for any $\epsilon > 0$), then there exists a $\delta > 0$ such that $dis(G^n(x), G^n(y)) > \delta$ for some $n \in \mathbb{N}$. Sensitivity to initial condition makes a system *unpredictable*. On the other hand, a CA is called *transitive* iff, for any two non-empty open subsets U and V of $S^{\mathbb{Z}}$, there exists an $n \in \mathbb{N}$ so that $G^n(U) \cap V \neq \emptyset$. For CAs, however, transitivity implies sensitivity to initial condition. A CA has *dense periodic points* iff the set $\{x \in \mathcal{S}^{\mathbb{Z}} \mid \exists k \in \mathbb{N} : G^k(x) = x\}$ of all periodic points is a dense subset of $\mathcal{S}^{\mathbb{Z}}$.

Definition 75 A CA is chaotic if (1) it is transitive, (2) its temporally periodic points are dense, and (3) it is sensitive to initial condition.

This definition of chaos is due to Devaney [1]. Apart from the Devaney's definition, there are restrictive definitions of chaos, such as Knudsen chaos, positively expansive chaos, etc. A CA is *chaotic* according to the definition of Knudsen iff it has a *dense orbit* and it is sensitive to initial condition. On the other hand, a CA is *positively expansive chaotic* iff it is transitive, has dense periodic points, and positively expansive. Positive expansivity is a stronger form of sensitivity. However, all such definitions of chaos are related.

For the study of chaos in CAs, *sibling* and *equivalent* RMTs (see Sect. 3.1) have an important role. Recall that, for a rule $\mathcal{R}_i : S^3 \to S$, each element of S^3 is called an RMT, which is generally represented by its decimal equivalent (Definition 11). A set of d RMTs $r_1, r_2, \ldots, r_d$ of rule $\mathcal{R}_i$ are equivalent to each other if $r_1 d \equiv r_2 d \equiv \cdots \equiv r_d d \pmod{d^3}$. If k $(0 \leq k \leq d^2 - 1)$ is an RMT, then $d^2 + k, 2d^2 + k, \ldots, (d-1)d^2 + k$ are its equivalent RMTs. This set of d equivalent RMTs is denoted as $Equi_{i.k} = \{k, d^2 + k, 2d^2 + k, \ldots, (d-1)d^2 + k\}$. For example, $Equi_{i.0} = \{0, 4\}$ when $\mathcal{R}_i$ is an ECA rule $(d = 2)$, $Equi_{i.0} = \{0, 9, 18\}$ when $d = 3$.

Similarly, a set of d RMTs $s_1, s_2, \ldots, s_d$ of $\mathcal{R}_i$ are called sibling to each other if $\lfloor \frac{s_1}{d} \rfloor = \lfloor \frac{s_2}{d} \rfloor = \cdots = \lfloor \frac{s_d}{d} \rfloor$. For RMT j $(0 \leq j \leq d^2 - 1)$, the set of sibling RMTs is $Sibl_{i.j} = \{d.j, d.j + 1, \ldots, d.j + (d-1)\}$. So, $Sibl_{i.0} = \{0, 1\}$, for $d = 2$, and $Sibl_{i.0} = \{0, 1, 2\}$ for $d = 3$.

In addition to sibling RMTs and equivalent RMTs, two more sets of RMTs, with respect to an RMT r, are to be considered for the study of chaos. These sets are left independent RMT set (L-set_i) and right independent RMT set (R-set_i).

Definition 76 Let r be the decimal equivalent of the RMT $xyz \in S^3$ of rule $\mathcal{R}_i$. Then, L-$set_i(r)$ = {RMT $xy'z' \in S^3 \mid y' \neq y$ and $z' \neq z$}, and R-$set_i(r)$ = {RMT $x'y'z \in S^3 \mid x' \neq x$ and $y' \neq y$}.

For example, in case of a 3-state CA ($d = 3$), for RMT 0 of a rule L-$set_i(0)$ = {4, 5, 7, 8} and R-$set_i(0)$ = {12, 15, 21, 24}.

There is a link between chaotic behavior and *permutivity* of a CA. Permutivity refers to the dependence of a cell of a CA to its neighbors. The rules are classified as *left* and *right* permutive.

Definition 77 A rule $\mathcal{R}_i$ is left (resp. right) permutive iff for any two equivalent (resp. sibling) RMTs r and s of rule $\mathcal{R}_i$, $\mathcal{R}_i[r] \neq \mathcal{R}_i[s]$.

That is, all the RMTs of $Equi_{i.k}$ (resp. $Sibl_{i.j}$) of a left (resp. right) permutive rule have different next states. For example, ECA rule 90 has such sibling and equivalent RMT sets. So, rule 90 is the left as well as the right permutive.

Many of the definitions of chaos, including Devaney's definition, declare some CAs as chaotic which is against the intuitive understanding of chaos. The *shift* CAs (for example, ECA 170) exhibit *predictable* behavior in some sense, but are claimed as chaotic by these definitions. But, a chaotic system is to be *unpredictable*, and the system destroys all the information contained in the initial configuration during evolution.

10.2 The Source of Unpredictability

Chaotic systems are unpredictable in long run. Knowing the current configuration of a chaotic system, it is not practically possible to understand how the system can behave after a long time. This section explores the properties of a CA that cause a CA unpredictable. As the cells of a CA interact among themselves during evolution, a change in a cell may or may not affect its neighbors. In other words, a cell may be *sensitive* or *insensitive* to its neighbors.

Consider a CA, where all of its cells are insensitive to their neighbors. That is, any change in the states of the neighbors of a cell does not affect the cell. This CA is stable, and so predictable. Moreover, it is not sensitive to initial condition, because a small change in initial configuration of the CA, which implies change of states of some cells, cannot remarkably affect the future configurations of the CA. On the other hand, if the cells of a CA are sensitive to their neighbors, change of states of some cells would affect their neighbors, and eventually all the cells can be affected. In this case, a small change in initial configuration matters in the course of

evolution of the CA. Therefore, whether a CA is unpredictable—that is, sensitive to the initial condition, is dependent on the fact that whether the cells are sensitive to their neighbors.

However, if a cell is sensitive to its neighbors, then we need to find the *degree* of sensitivity. This is measured as—out of total possible changes of states of the neighbors, in how many cases the cell gets affected. If the cells are *more* sensitive to their neighbors, the CA would be *more* sensitive to initial condition.

10.2.1 Information Propagation

Let carry out a thought experiment to understand the source of unpredictability in a CA. Assume that the state of a cell has been changed. Since this cell is also neighbor of other cells, the experiment examines how this update affects its neighbors at the next time steps.

Consider that the present state of cell $i \in \mathbb{Z}$ is s_i, which is updated to s_i'. If cell $i-1$ reacts to this update then it can be said that the information propagates to the left. Now if $\mathcal{R}_{i-1}(s_{i-2}, s_{i-1}, s_i)$ and $\mathcal{R}_{i-1}(s_{i-2}, s_{i-1}, s_i')$ are same, then the change of state of cell i from s_i to s_i' has no effect on cell $i-1$. But if the change of state of cell i is s_i to s_i'' and $\mathcal{R}_{i-1}(s_{i-2}, s_{i-1}, s_i) \neq \mathcal{R}_{i-1}(s_{i-2}, s_{i-1}, s_i'')$, then this change has an effect on its left neighbor. Observe that the RMTs $s_{i-2}s_{i-1}s_i$, $s_{i-2}s_{i-1}s_i'$ and $s_{i-2}s_{i-1}s_i''$ are sibling to each other. Therefore, to measure the effect of cell i on its left neighbor—that is, to measure the flow of information to the left, it is to be observed that in how many cases a set of sibling RMTs of $\mathcal{R}_{i-1}$ ($Sibl_{i-1\cdot j}$) have different next states. Let $s_{i-2}s_{i-1}s_i \in Sibl_{i-1\cdot j}$ and $\delta^{(-1)}_{i-1\cdot j}$ be a binary function over $Sibl_{i-1\cdot j}$ which notes this differences for the two sibling RMTs. The function $\delta^{(-1)}_{i-1\cdot j} : Sibl_{i-1\cdot j} \times Sibl_{i-1\cdot j} \to \{0, 1\}$ is defined below.

$$\delta^{(-1)}_{i-1\cdot j}(r, s) = \begin{cases} 1 & \text{if } \mathcal{R}_{i-1}[r] \neq \mathcal{R}_{i-1}[s], r, s \in Sibl_{i-1\cdot j} \\ 0 & \text{otherwise} \end{cases} \tag{10.1}$$

The function $\delta^{(-1)}_{i-1}$, therefore, enables us to understand the number of times cell $i-1$ changes its state when the state of cell i changes its state to any other $d-1$ possible states. The following parameter ($A^{(-1)}_{i-1\cdot j}$) is defined to capture this fact.

$$A^{(-1)}_{i-1\cdot j} = \frac{1}{d^2 - d} \sum_{\substack{r,s \in Sibl_{i-1\cdot j} \\ r \neq s}} \delta^{(-1)}_{i-1\cdot j}(r, s) \tag{10.2}$$

Hence, when the states of cell $i-2$ and $i-1$ are stable at s_{i-2} and s_{i-1}, but the state of cell i changes, then $A^{(-1)}_{i-1\cdot j}$ measures the possibility of getting affected of cell

Table 10.1 A left and right permutive rule for $d = 3$

Present state	222	221	220	212	211	210	202	201	200	122	121	120	112	111	110
(RMT)	(26)	(25)	(24)	(23)	(22)	(21)	(20)	(19)	(18)	(17)	(16)	(15)	(14)	(13)	(12)
Next state	1	0	2	1	0	2	1	0	2	0	2	1	0	2	1
Present state	102	101	100	022	021	020	012	011	010	002	001	000			
(RMT)	(11)	(10)	(9)	(8)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(0)			
Next state	0	2	1	2	1	0	2	1	0	2	1	0			

$i - 1$ by cell i. Now, to find out the total possibility of information propagation to the left of cell i, the following parameter is defined.

$$\Lambda_{i-1}^{p} = \frac{1}{d^2} \sum_{j=0}^{d^2-1} A_{i-1\cdot j}^{(-1)} \tag{10.3}$$

Here, Λ_{i-1}^{p} is a measure of information propagation to the left neighbor of cell i. If Λ_{i-1}^{p} attains its highest value (that is, 1), then $\mathcal{R}_{i-1}$ is right permutive rule (see Definition 77).

Example 10.1 Let consider a 3-state CA, where the $(i-1)$th rule is shown in Table 10.1. Here, $Sibl_{i-1.0} = \{0, 1, 2\}$ and $\mathcal{R}_{i-1}[0] \neq \mathcal{R}_{i-1}[1] \neq \mathcal{R}_{i-1}[2]$. RMT 0 and RMT 1 of the rule have the different next states. So, $\delta_{i-1.0}^{(-1)}(0, 1) = 1$. Similarly, $\delta_{i-1.0}^{(-1)}(1, 2) = 1$ and $\delta_{i-1.0}^{(-1)}(0, 2) = 1$. Hence, $A_{i-1.0}^{(-1)} = \frac{1}{3^2-3} \sum_{\substack{r,s\in\{0,1,2\} \\ r\neq s}} \delta_{i-1.0}^{(-1)}(r, s) = \frac{6}{6} = 1$. In this fashion, $A_{i-1\cdot j}^{(-1)}$ for $j = 1, 2, \ldots, 8$ can be found out. So, $\Lambda_{i-1}^{p} = \frac{1}{3^2} \sum_{j=0}^{8} A_{i-1\cdot j}^{(-1)} = 1$. Note that, all the sibling RMTs of a sibling RMT set have the different next states. So, the rule is a right permutive rule.

Using the same rationale, the possibility of information propagation to the right cell $i + 1$, say η_{i+1}^{p}, can be found out. As that of $\delta_{i-1}^{(-1)}$ (Eq. 10.1), let define $\delta_{i+1\cdot k}^{(+1)} : Equi_{i+1\cdot k} \times Equi_{i+1\cdot k} \to \{0, 1\}$, where

$$\delta_{i+1\cdot k}^{(+1)}(r, s) = \begin{cases} 1 & \text{if } \mathcal{R}_{i+1}(r) \neq \mathcal{R}_{i+1}(s), r, s \in Equi_{i+1\cdot k} \\ 0 & \text{otherwise} \end{cases} \tag{10.4}$$

Here, the equivalent RMT sets for the right neighbor of cell i is used. The number of occurrences that the cell $i + 1$ changes its state with the change of cell i is denoted by $A_{i+1\cdot k}^{(+1)}$ and it is defined as

$$A_{i+1\cdot k}^{(+1)} = \frac{1}{d^2 - d} \sum_{\substack{r,s \in Equi_{i+1\cdot k} \\ r \neq s}} \delta_{i+1\cdot k}^{(+1)}(r, s) \tag{10.5}$$

Finally, the effect of cell i on cell $i + 1$, when only the state of cell i is changed, can be calculated as

$$\eta_{i+1}^{p} = \frac{1}{d^2} \sum_{k=0}^{d^2-1} A_{i+1\cdot k}^{(+1)} \tag{10.6}$$

Example 10.2 Let consider the rule, given in Table 10.1, at cell $i + 1$. The information flow in the right direction of cell i (i.e. η_{i+1}^{p}) is computed following Eq. 10.6. For the given rule, the $\eta_{i+1}^{p} = 1$. So, any change in cell i affects the cell $i + 1$, and so $\mathcal{R}_{i+1}$, the rule of cell $i + 1$, is a left permutive rule.

10.2.2 Information Cooking

The update of cell i does not only affect cell $i - 1$ and cell $i + 1$, it can also affect the cell itself. When the left neighbor or the right neighbor is updated in a reaction to the update of cell i, the cell i can further be updated in the next time step in response to the updates of its neighbors. Such dependence on neighbors is required for a CA to be unpredictable. The updates of cell i in the subsequent time steps in reaction to the update of cell i is called *Information Cooking*.

To guess the impact of the update of i on itself, the RMTs of L-set_i and R-set_i of $\mathcal{R}_i$ are to be considered. Let consider that the states s_{i-1} and s_i are updated to s'_{i-1} and s'_i respectively. Now if $\mathcal{R}_i(s_{i-1}, s_i, s_{i+1}) \neq \mathcal{R}_i(s'_{i-1}, s'_i, s_{i+1})$, then it indicates that the updates of cell i and its left neighbor have an impact on itself. It may be viewed as *information flow* from left neighbor to the cell. Here, $s_{i-1}s_is_{i+1}$ and $s'_{i-1}s'_is_{i+1}$ are the two RMTs of R-set_i. To measure the impact, a function $\delta_{i.r}^{(+2)} : \text{R-}set_i(r) \rightarrow \{0, 1\}$ is defined, where

$$\delta_{i.r}^{(+2)}(s) = \begin{cases} 1 & \text{if } \mathcal{R}_i(r) \neq \mathcal{R}_i(s),\ s \in \text{R-}set_i(r) \\ 0 & \text{otherwise} \end{cases} \tag{10.7}$$

Using the parameter $\delta_{i.r}^{(+2)}$, the number of times cell i changes its state following the change of cell $i - 1$ and cell i can be found out. It is

$$B_{i.r}^{(+2)} = \frac{1}{(d-1)^2} \sum_{s \in \text{R-}set_i(r)} \delta_{i.r}^{(+2)}(s) \tag{10.8}$$

Hence, the possibility that cell i changes its state following the changes in cell $i-1$ and i can be defined as,

$$\eta_i^c = \frac{1}{d^3} \sum_{r=0}^{d^3-1} B_{i.r}^{(+2)}. \tag{10.9}$$

Example 10.3 Let consider a 3-state CA, where the ith rule is shown in Table 10.1. To know the $\delta_{i.r}^{(+2)}(s)$, we need to know the next states of the RMTs of R-$set_i(r)$. Here, R-$set_i(0) = \{12, 15, 21, 24\}$ and $\delta_{i.0}^{(+2)}(12) = \delta_{i.0}^{(+2)}(15) = \delta_{i.0}^{(+2)}(21) = \delta_{i.0}^{(+2)}(24) = 1$. Thus, $B_{i.0}^{(+2)} = 1$. Similarly, the other values of $\delta_{i.r}^{(+2)}(s)$ and $B_{i.r}^{(+2)}$ can be computed. Finally, it can be found that the $\eta_i^c = 0.98$. That is, the changes of cell $i-1$ and cell i affect the cell i with a probability of 0.98.

Using the same rationale, another measure, say Λ_i^c, for the information flow to the left when the cells $i+1$ and cell i are updated, can be found out. The function for information cooking $\delta_{i.r}^{(-2)}$: L-$set_i(r) \to \{0, 1\}$ can be used to measure in how many cases an L-set_i RMTs show different next states. Hence,

$$\delta_{i.r}^{(-2)}(s) = \begin{cases} 1 & \text{if } \mathcal{R}_i(r) \neq \mathcal{R}_i(s),\ s \in \text{L-}set_i(r) \\ 0 & \text{otherwise} \end{cases} \tag{10.10}$$

The number of times when cell i changes its state due to the change in cell $i+1$ and cell i is denoted by the parameter $B_{i.r}^{(-2)}$, where

$$B_{i.r}^{(-2)} = \frac{1}{(d-1)^2} \sum_{s \in \text{L-}set_i(r)} \delta_{i.r}^{(-2)}(s) \tag{10.11}$$

Hence, the possibility that the left neighbor of cell i gets affected for the changes in cell i and cell $i+1$ is

$$\Lambda_i^c = \frac{1}{d^3} \sum_{l=0}^{d^3-1} B_{i.r}^{(-2)} \tag{10.12}$$

In case of a CA, however, all the cells are updated in parallel. Even then these parameters give the measures of affecting the neighbors of cells. If all the cells affect their neighbors—that is, the Λ_i^p, η_i^c, Λ_i^c and η_i^p are non-zero for each $i \in \mathbb{Z}$, then any change in a cell makes the system unpredictable.

10.3 The Communication Class and Chaos

Considering the ability of cells to communicate among themselves, a *communication relation* (denoted as $\rightsquigarrow$) can be defined over the set of all cells of a CA. That is, for the two cells $i, j \in \mathbb{Z}$, "$i \rightsquigarrow j$" indicates that "cell i communicates with cell j". Intuitively, this relation reflects the fact that a state change in cell i has an influence on the state of cell j.

Definition 78 A communication relation ($\rightsquigarrow$) is a relation from $\mathbb{Z}$ to $\mathbb{Z}$ where $i \rightsquigarrow j$ holds iff

1. $\eta_k^p > 0$ or $\eta_k^c > 0$, for each $k \in \{i+1, i+2, \ldots, j\}$, when $j > i$,
2. $\Lambda_k^p > 0$ or $\Lambda_k^c > 0$, for each $k \in \{i-1, i-2, \ldots, j\}$, when $i > j$,
3. both (1) and (2) are true when $i = j$

For any two arbitrary cells i and j, where $j > i$, $i \rightsquigarrow j$ holds iff $i \rightsquigarrow i+1$, $i+1 \rightsquigarrow i+2, \ldots, j-1 \rightsquigarrow j$. Similarly, in the case $i > j$, the communication from i to j holds iff, $i \rightsquigarrow i-1$, $i-1 \rightsquigarrow i-2, \ldots, j+1 \rightsquigarrow j$. Therefore, to know whether a communication exists between any two arbitrary cells, one needs to check the communication for individual cells of the CA.

The relation $\rightsquigarrow$ is transitive. That is, if $i \rightsquigarrow j$ and $j \rightsquigarrow k$, then $i \rightsquigarrow k$. Whether $\rightsquigarrow$ is reflexive or symmetric, however, depends on the CA. If $i \rightsquigarrow j$ but $j \not\rightsquigarrow i$, then it signifies that any change in cell i can influence the state of cell j but any perturbation in cell j has no effect on cell i. To be a chaotic system, a tiny perturbation in initial condition should largely affect the future configurations of the system. Therefore, for each pair of $i, j \in \mathbb{Z}$, "if $i \rightsquigarrow j$, then $j \rightsquigarrow i$" has to hold in a chaotic CA. Hence, the relation '$\rightsquigarrow$' has to be an equivalence relation, and all the cells have to form a single equivalence class. An equivalence class is referred to as the *communication class*.

Let now consider $\rightsquigarrow$ as an asymmetric relation. That is, $\forall i, j$, if $i \rightsquigarrow j$ then $j \not\rightsquigarrow i$. In that case, each cell can communicate in only one direction and thus no communication class can be created.

Example 10.4 Let consider the CA $(170^{10}, 85^{10})^{\mathbb{Z}}$ (Fig. 10.1). The initial configuration (when $t = 0$) is set randomly. Each cell of the CA communicates only with its left neighbor. Thus the CA does not form any communication class.

Now, consider the relation $\rightsquigarrow$ is an anti-symmetric relation. That is, if $i \rightsquigarrow j$ and $j \rightsquigarrow i$, then $i = j$ (reflexive) or if $i \rightsquigarrow j$ holds then $j \rightsquigarrow i$ does not hold (asymmetric). The relation $\rightsquigarrow$ in a CA cannot be reflexive (if it is not symmetric) because the relation depicts the effect of a cell on its neighbors and not on itself. Thus, the relation in the CA of Fig. 10.1 is also anti-symmetric. Therefore, a CA to be chaotic, the relation $\rightsquigarrow$ is to be an equivalence relation.

Further, the configuration space of a CA is not to be split into two or more non-empty and disjoint subsets. It may sometime happen that, when the cells form a single communication class, then a set of configurations remain unreachable from other set

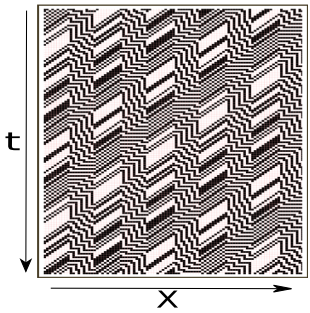


Fig. 10.1 A space-time diagram of $(170^{10}, 85^{10})^{\mathbb{Z}}$ where $\rightsquigarrow$ is asymmetric

of configurations. In this case, the system becomes predictable to some extent. That is, the CA is to be transitive to remain unpredictable. The notion of communication class, therefore, leads to a new definition of chaos in CA.

Definition 79 A CA is chaotic iff the CA is transitive and all the cells of the CA form a single communication class.

Let consider three ECAs with rules 30, 90 and 150. All these (uniform) CAs are chaotic as per the standard definition of chaos. They are also chaotic by the above definition.

In case of rule 30, $\Lambda_i^p = 0.5$, $\Lambda_i^c = 0.5$, $\eta_i^p = 1.0$, $\eta_i^c = 0.5$. For rule 90, $\Lambda_i^p = \Lambda_i^c = \eta_i^p = \eta_i^c = 1.0$. For rule 150, $\Lambda_i^p = \eta_i^p = 1$ and $\Lambda_i^c = \eta_i^c = 0$. As each such CA is uniform, the parameter values are same for all $i \in \mathbb{Z}$. Therefore, the relation $\rightsquigarrow$ is an equivalence relation for each of these cells, and a single communication class among the cells is formed. Further, the CAs are transitive as they are already shown as Devaney chaotic. Hence, these CAs are chaotic as per Definition 79.

Let consider the ECA with rule 51. For rule 51, $\Lambda_i^p = \eta_i^p = 0$, and $\Lambda_i^c = \eta_i^c = 1$. That is, the cells of the CA form a single communication class. But this CA is not transitive, because its configuration space is divided into infinitely many disjoint subsets. Hence, the CA is not chaotic.

All the Devaney chaotic CAs are not chaotic as per Definition 79. As an example, consider a CA with the rule 170. It simply shifts the state of a cell to its left neighbor. The CA is chaotic as per the Devaney's definition. However, the rule 170 CA is not chaotic as per Definition 79. For rule 170, $\Lambda_i^p = \Lambda_i^c = 1$ and $\eta_i^p = \eta_i^c = 0$, for any $i \in \mathbb{Z}$. This indicates that, for any two cells i and j $(i > j)$, $i \rightsquigarrow j$ and $j \not\rightsquigarrow i, \forall i, j \in \mathbb{Z}$. Hence, the communication relation is not symmetric and thus not an equivalence relation. That is, the cells of the CA does not form a single communication class. Therefore, as per Definition 79, the CA with rule 170 is not chaotic.

Proposition 10.1 *There exist CAs which are Devaney chaotic but not chaotic as per Definition 79.*

Therefore, to decide chaos in CAs, whether the cells of a CA form a single communication class is to be tested. In case of uniform CAs, it is a trivial task. The problem becomes difficult for the non-uniform CAs. A non-uniform CA can use arbitrary rule sequence $(\mathcal{R}_i)_{i\in\mathbb{Z}}$. So, a cell i is having its own η_i^p, η_i^c, Λ_i^p and Λ_i^c. To decide whether "$i \rightsquigarrow j$" holds for two arbitrary cells, these parameter values for all the cells, in between i and j, are to be identified. This leads to the fact that the problem of deciding single communication class formed by cells of an arbitrary CA is an undecidable problem.

However, for a class of non-uniform CAs, it is possible to decide their dynamical behavior. For example, if a finite sequence of rules is repeated over $\mathbb{Z}$, then one can decide if the cells form a single communication class. Let consider a CA with two rules where the first one is applied to the cells of the CA when cell $i < 0$ and the other is applied to the cells when $i \geq 0$. Therefore, if there is an effective way to decide the rule of an arbitrary cell, then we can decide if the cells form a single communication class.

10.4 The Blocking Words

In some cases, the cells of a CA form a single communication class, but the CA is not transitive. One reason of this is the presence of *blocking words* in the configuration space of the CA. A *blocking word* of a CA blocks the flow of information from one part of the system to other part. Such a case may arise if Λ_i^p, η_i^c, Λ_i^c and η_i^p for some $\mathcal{R}_i$ of the CA are <1. If it exists, then the left cells of the blocking word are separated from the right cells and makes the CA non-transitive. A blocking word is nothing but a *subconfiguration*.

Definition 80 A subconfiguration of length $k \geq 1$ of a configuration $\mathrm{c} = (s_i)_{i\in\mathbb{Z}}$ is a string $(s_i)_{i\in[j,j+k-1]}$, where $j \in \mathbb{Z}$.

For example, 0110 is a subconfiguration of length 4 of a configuration $\cdots$ 0011001110 $\cdots$ of a binary CA.

Definition 81 Let $\mathrm{X} = \{\cdots s_j s_{j+1} \cdots s_{j+k-1} \cdots\}$ be a set of configurations of a CA. The subconfiguration $\mathrm{s} = (s_i)_{i\in[j,j+k-1]}$ is called a blocking word if for any $c \in \mathrm{X}$ and any $t \geq 1$, $F^t(c)|_i = s_i, \forall i \in [j, j+k-1]$.

Therefore, during the evolution if a CA reaches a configuration of X, then the blocking word remains intact in the future configurations of the CA.

Example 10.5 Figure 10.3 shows a space-time diagram of a CA $(146^{20}, 126, 105, 30, 105, 45, 146, 150, 146, 90^{10}, 150^{50})^{\mathbb{Z}}$. The subconfiguration 10001 is a blocking word in the CA that remains intact during the evolution of the CA.

10.4.1 Search for Blocking Words

In a CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, a blocking word $(s_i)_{i\in[j,j+k-1]}$ exists iff the following conditions are satisfied:

1. Each RMT of the equivalent RMT set $\{0s_js_{j+1}, 1s_js_{j+1}, \ldots, (d-1)s_js_{j+1}\}$ of $\mathcal{R}_j$ is passive.
2. For each l, $j < l < j+k-1$, the RMT $s_{l-1}s_ls_{l+1}$ of $\mathcal{R}_l$ is passive.
3. Each RMT of the sibling RMT set $\{s_{j+k-2}\ s_{j+k-1}\ 0, s_{j+k-2}\ s_{j+k-1}\ 1, \ldots, s_{j+k-2}\ s_{j+k-1}\ (d-1)\}$ of $\mathcal{R}_{j+k-1}$ is passive.

To check the presence of blocking word, we need to search for a rule in the CA, where the rule has at least one equivalent RMT set, each RMT of which is passive. Let denote it as the *start rule*. Say, $\mathcal{R}_j$ is the start rule and the RMTs of $Equi_{j\cdot m} = \{0s_js_{j+1}, 1s_js_{j+1}, \ldots, (d-1)s_js_{j+1}\}$ are passive. It indicates that whatever be the rule in the left cell of cell j, it does not affect the cell j. Now, we need to find out whether the RMT $s_js_{j+1}s_{j+2}$ of rule $\mathcal{R}_{j+1}$ is passive. If it is passive, then the next rule is to be checked. Once a rule, say, $\mathcal{R}_{j+k-1}$ where all the RMTs of $Sibl_{j+k-1.m} = \{s_{j+k-2}s_{j+k-1}0, s_{j+k-2}s_{j+k-1}1, \ldots, s_{j+k-2}s_{j+k-1}(d-1)\}$ are passive, is found, it is referred to as the *end rule*. It indicates that cell $j+k-1$ cannot be affected by its right neighbor. Hence, it can be concluded that there exists a blocking word in the CA.

Therefore, an effective procedure can be developed to decide if there exists a blocking word in a given CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$. Following example illustrates the decision process.

Example 10.6 Let consider a CA, $\mathcal{R} = (146^{20}, 126, 105, 30, 105, 45, 146, 150, 146, 90^{10}, 150^{50})^{\mathbb{Z}}$, where the rules are ECA rules. Rule 126 is selected as a start rule because the RMTs of the equivalent RMT set {010, 110} are passive. Now, it is to be checked whether RMT 100 or 101 of rule 105 is passive or not. RMT 100 is passive for rule 105 whereas, RMT 101 is active. Let denote the RMT along with the rule as (100, 105). Hence, for the next rule (i.e. rule 30), the RMT 000 or 001 should be passive. The search for such passive RMTs of the rules is shown in Fig. 10.2. It shows the end rule as 45 for which all the RMTs of the sibling RMT set {010, 011} are passive. Therefore, it can be decided that a blocking word exists in the CA and the blocking word is 10001. The space-time diagram for the same CA is shown in Fig. 10.3a.

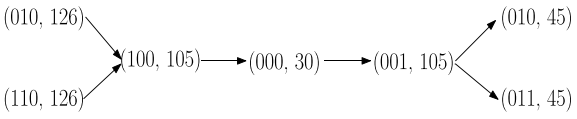


Fig. 10.2 Searching for a blocking word of the CA $(146^{20}, 126, 105, 30, 105, 45, 146, 150, 146, 90^{10}, 150^{50})^{\mathbb{Z}}$. The start rule is 126, end rule is 45 and blocking word is 10001

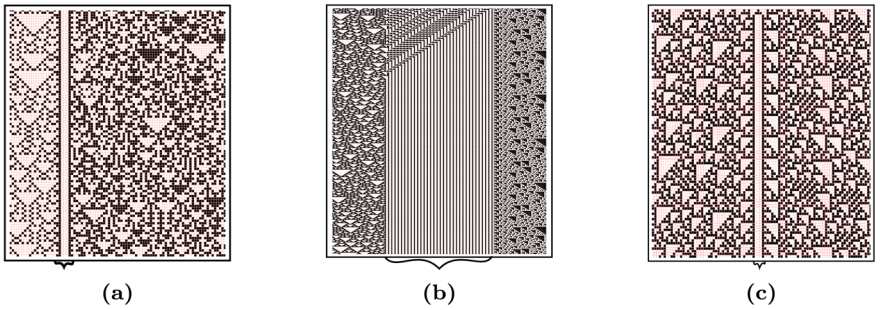


Fig. 10.3 Space-time diagram of different CAs having blocking words. The blocking words are marked by the braces. **a** CA $(146^{20}, 126, 105, 30, 105, 45, 146, 150, 146, 90^{10}, 150^{50})^{\mathbb{Z}}$, **b** CA of Example 10.7, **c** CA $(\cdots 102, 60 \cdots)^{\mathbb{Z}}$

However, it may happen that in the rule sequence $(\mathcal{R}_j, \ldots, \mathcal{R}_{j+k-1})$, some contiguous subsequence is repeated more than once. In that case, to decide the presence of blocking word, we need not always require to consider all the occurrences of the subsequence. Following examples explains this point.

Example 10.7 Consider the binary CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, where

$$\mathcal{R}_i = \begin{cases} 126 & i \leq 0 \\ 32 & i \in \{1, 4, 7, \ldots, 97\} \\ 57 & i \in \{2, 5, 8, \ldots, 98\} \\ 119 & i \in \{3, 6, 9, \ldots, 99\} \\ 195 & i \geq 100 \end{cases}$$

The contiguous subsequence of rules (32, 57, 119) is repeated for 99 times and ultimately the right neighbor of cell 99 (with rule 119) is associated with rule 195. Here, rule 126 can be considered as a start rule because the equivalent RMTs 010 and 110 are passive. In case of rule 32, RMT 100 is passive whereas, RMT 101 is active. Therefore, the RMT 100 of rule 32 is considered and it is denoted as (100, 32) as shown in Fig. 10.4. Similarly, the next rules are tested as shown in the figure. The RMT 001 is passive for rule 57 and RMT 010 is passive for rule 119. As the same subsequence repeats, the rule of the right neighbor of cell 3 (with rule 119) is 32 and there exists the same sequence $(100, 32) \rightarrow (001, 57) \rightarrow (010, 119)$. So, the

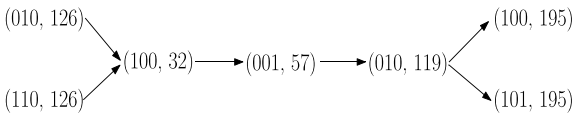


Fig. 10.4 Searching for a blocking word for the CA of Example 10.7

repeated subsequences of rules can be avoided by checking the subsequence only once. After cell 99, the rule 195 is tested. In rule 195, both the RMTs 100 and 101 are passive. Thus rule 195 can be considered as the end rule and it can be concluded that a blocking word exists in the CA. Figure 10.3b shows the space-time diagram of the CA, which confirms the existence of blocking word.

Example 10.8 Consider a CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$ such that,

$$\mathcal{R}_i = \begin{cases} 102 & \text{if } i < 0 \\ 60 & \text{otherwise} \end{cases}$$

In case of rule 102, the equivalent RMTs 000 and 100 are passive. Therefore, the rule can be considered as the start rule. Assume the start rule when $i = -1$. When $i = 0$, there are sibling RMTs 000 and 001 which are passive for rule 60. Hence, rule 60 can be considered as the end rule, and it can be concluded that there exists a blocking word in the CA. Figure 10.3c shows the blocking word of the CA.

Example 10.9 Consider a CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, where

$$\mathcal{R}_i = \begin{cases} 60 & \text{if } i < 0 \\ 102 & \text{otherwise} \end{cases}$$

In this case, the rule 60 cannot be a start rule because there is no such equivalent RMT set where all the RMTs of the set are passive. Similarly, rule 102 cannot be an end rule as there is no sibling RMT set where all the RMTs of the set are passive. Therefore, the CA cannot have a blocking word. Figure 10.5 shows the space-time diagram of the CA.

10.4.2 The Partial Block

The concept of partial blocking is an extended idea of blocking, where a subconfiguration can only block information flow based on some particular neighboring states of the subconfiguration. If any change of state in the left neighbor of a cell cannot affect the cell but change of state of right neighbor can affect the cell, the information flows to the left only and the flow in the right cell is blocked. Such blocking word is called *left blocking word*. On the other hand, if any change in right neighbor cannot affect a cell but a change in the left neighbor affects the cell, then it creates a *right blocking word*.

Definition 82 Let $\mathrm{X} = \{\cdots s_{j-1}s_j s_{j+1}\cdots s_{j+k-1}\, s_{j+k}\cdots\}$ be a set of configuration. The subconfiguration $s = (s_i)_{i\in[j,j+k-1]}$ is called a left blocking word if, for each state of cell $j-1$, there exists at least a state in cell $j+k$ for which $F^t(c)|_i = s_i$,

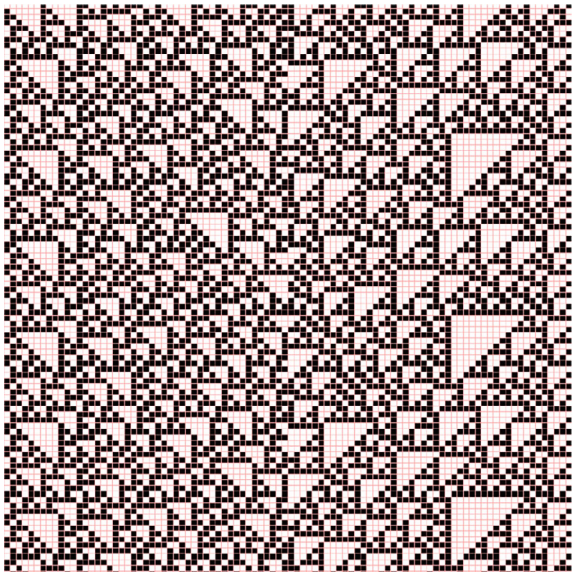


Fig. 10.5 Space-time diagram of a CA where blocking word does not exist

where $c \in X$ and $t \geq 1, \forall i \in [j, j+k-1]$. The blocking word is a right blocking word if for each state of cell $j+k$, there exists a state in cell $j-1$ for which $F^t(c)|_i = s_i$.

In a CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, a left blocking (resp. right blocking) word $(s_i)_{i\in[j,j+k-1]}$ exists iff the following conditions are satisfied:

1. All (resp. some) RMTs of the equivalent RMT set $\{0s_js_{j+1}, 1s_js_{j+1}, \ldots, (d-1)s_js_{j+1}\}$ of $\mathcal{R}_j$ are passive.
2. For each l, $j < l < j+k-1$, the RMT $s_{l-1}s_ls_{l+1}$ of $\mathcal{R}_l$ is passive.
3. Some (resp. all) RMTs of the sibling RMT set $\{s_{j+k-2}\ s_{j+k-1}\ 0, s_{j+k-2}\ s_{j+k-1}1, \ldots, s_{j+k-2}\ s_{j+k-1}\ (d-1)\}$ of $\mathcal{R}_{j+k-1}$ are passive.

The above discussion points to the fact that (1) if a CA is having a blocking word, then it is having the left and right blocking, and (2) if a CA is having left (resp. right) blocking word $(s_i)_{i\in[j,j+k-1]}$, then $(s_i)_{i\in[j,j+l-1]}$ (resp. $(s_i)_{i\in[m,j+k-1]}$) is also a left (resp. right) blocking word for any $l < k$ (resp. $m > j$).

Example 10.10 Let consider the CA $(186, 18, 45, \ldots, 18, 45)^{\mathbb{Z}}$ of Fig. 10.6. The figure indicates the existence of a blocking word. However, it is a partial blocking because rule 18 is considered as the start rule, where RMT 101 is passive whereas, the RMT 001 is active. To be a left blocking word, the RMTs 010 or 011 of rule 45 should be passive. In case of rule 45, both the RMTs are passive. Hence, this rule can be considered as the end rule. Thus a blocking word 01 can be considered. It indicates that, even if there is a change in the state of the right neighbor, this cannot affect the cell. Whereas, change of state in the left neighbor may affect the cell. Hence, the information may flow towards the right direction. In Fig. 10.7, it can be observed.

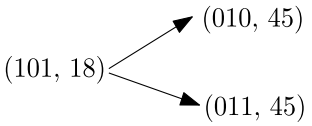
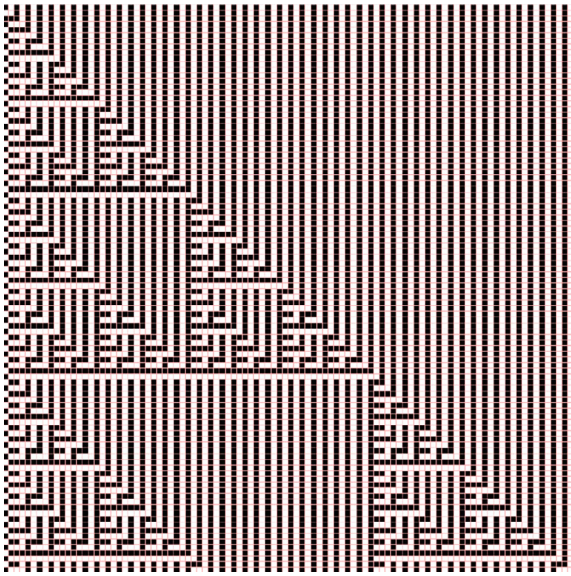


Fig. 10.6 Searching for a partial blocking word in CA $(186, 18, 45, \ldots, 18, 45)^{\mathbb{Z}}$ where the blocking word is 01

Fig. 10.7 Space-time diagram of the CA $(186, 18, 45, \ldots, 18, 45)^{\mathbb{Z}}$



The existence of partial blocking word (left or right blocking word) in a CA may point to the direction of information flow. But, it does not indicate the transitivity. Whereas, the existence of blocking word in a CA can denote the non-transitivity of a CA. Based on the communication relations among the cells and the blocking word, a parameter is defined in the next section to predict the chaotic behavior of a CA.

10.5 Parametrization

The parameter, defined in this section, is an ordered pair, first element of which is a *measure of transitivity*, and the other is the *degree* of communication among the cells to form a single communication class. This degree is in turn related to transitivity and unpredictability of the CAs.

There is a relation between permutivity and transitivity of a CA. A left or right permutive uniform CA (that is, the rule of the CA is left or right permutive by Definition 77) is Devaney chaotic, hence the CA is transitive. For left and right

permutive rule, $\Lambda_i^p = 1$ and $\eta_i^p = 1$ respectively (see Eqs. 10.3 and 10.6 of Sect. 10.2). If $\Lambda_i^p = 1$ (resp. $\eta_i^p = 1$), then the cell is substantially affected by the right (resp. left) neighbor. In case of non-uniform CAs, if $\Lambda_i^p > 0$ or $\eta_i^p > 0$, for any $i \in \mathbb{Z}$, then there is a chance that all the cells can be affected due to their right or left neighbors. Let express this *probability* with which all the cells get affected by their neighbors as

$$\bigwedge_{i\in\mathbb{Z}} (\Lambda_i^p \vee \eta_i^p)$$

In general, if the value of this expression is high, then it is highly probable that the CA is transitive. However, the CA may have a blocking word even it is true. As the presence of a blocking word makes a CA non-transitive, 0 is assigned to the first argument of the parameter when such a word exists. To express the first argument, the following function (p_1) is used, where

$$p_1 = \begin{cases} 0 & \text{if blocking word exists} \\ \bigwedge_{i\in\mathbb{Z}} (\Lambda_i^p \vee \eta_i^p) & \text{otherwise} \end{cases} \tag{10.13}$$

For the second argument of the parameter, it is to be checked if $i \rightsquigarrow i+1$ and $i+1 \rightsquigarrow i$ hold for each $i \in \mathbb{Z}$. If each pair $(i, i+1)$ of cells communicate between themselves, then the cells can form a single communication class. Let express this communication by

$$a_{i,i+1} = (\eta_{i+1}^p \vee \eta_{i+1}^c) \wedge (\Lambda_i^p \vee \Lambda_i^c)$$

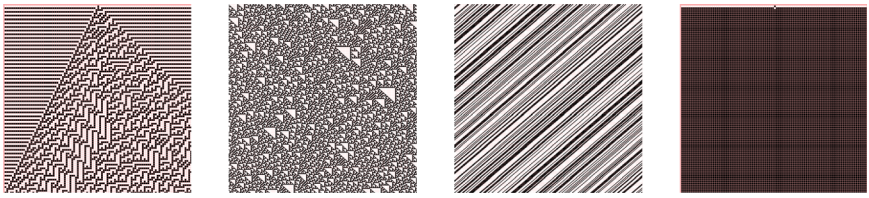
which can directly be obtained from Definition 78. Now, to get $a_{i,i+1}$, the conjunction ($\wedge$) and disjunction ($\vee$) operations have to be implemented. Following the method of Kleen-Dienes [2], the '$\wedge$' and '$\vee$' are considered as *min* and *max* operations respectively.

To form a single communication class, $a_{i,i+1}$ for each $i \in \mathbb{Z}$ is to be non-zero. When they are all non-zero, an average of $a_{i,i+1}$s can be considered as the degree of communication among the cells of a CA. So, a function p_2 can be used as the second argument of the parameter, where

$$p_2 = \begin{cases} 0 & \exists i \in \mathbb{Z},\ a_{i,i+1} = 0 \\ \text{avg}\{a_{i,i+1}\}_{i\in\mathbb{Z}} & \forall i \in \mathbb{Z},\ a_{i,i+1} \neq 0 \end{cases} \tag{10.14}$$

If $p_2 = 0$, then the cells do not form a single communication class. The p_2 is also related to transitivity of CAs. Further, if p_2 is high, then all the cells of the CA are substantially dependent on their neighbors, which makes the system unpredictable. However, in some cases, the p_1 is high but p_2 is low, and the system is not chaotic. Therefore, both the p_1 and p_2 are to be high for a CA to be chaotic. The parameter p is then defined, where

$$p = (p_1, p_2) \tag{10.15}$$



(a) ECA 45, $= (1.0, 0.5)$ (b) ECA 60, $p = (1.0, 1.0)$ (c) ECA 170, $p = (1.0, 0)$ (d) ECA 251, $p = (0, 0.25)$

Fig. 10.8 Space-time diagram and parameter p of some uniform CAs. The ECAs 45 and 60 are chaotic. Whereas, ECAs 170 and 251 other two are not chaotic

In case of uniform CAs, Λ_i^p (similarly $\Lambda_i^c, \eta_i^p, \eta_i^c$) is same for each $i \in \mathbb{Z}$. So, the subscript i can be removed, and new notations are Λ_p, Λ_c, η_p and η_c. For the first argument of p (of Eq. 10.15), if no blocking word exists, then $p_1 = (\Lambda_p \vee \eta_p)$. That is, $p_1 = \max\{\Lambda_p, \eta_p\}$. The second argument of p (of Eq. 10.15) becomes $p_2 = \min\{\max\{\Lambda_p, \Lambda_c\}, \max\{\eta_p, \eta_c\}\}$. If both the arguments of p are high then the (uniform) CA is chaotic in true sense. Figure 10.8 shows the behaviors of some sample CAs. In case of binary CAs, it is experimentally observed that if the first argument of p is ≥ 0.75 and the second argument is ≥ 0.5, then the (uniform) CA is, in general, chaotic.

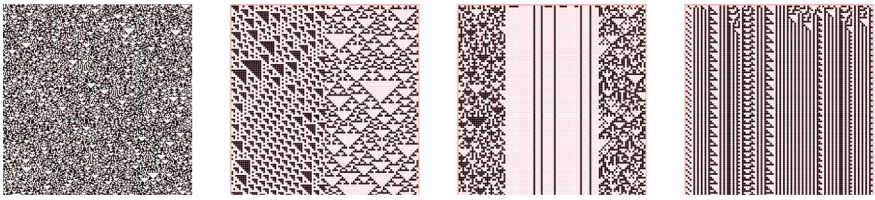
However, there are some CAs which are Devaney chaotic, but not chaotic in true sense. These CAs can also be successfully identified by the parameter p. For example, ECA 170 is Devaney chaotic, but the CA is not chaotic, because its behavior is predictable to some extent (Fig. 10.8c). As per the parameter, the CA is not chaotic as the second argument of the parameter is 0. Figure 10.8d shows the first argument of p is 0.

In the literature, a number of parameters, such as λ, Z, etc. have been considered to predict chaos in uniform CAs. However, the parameter p, defined in this section, provides more accurate prediction on chaotic behavior of a uniform CAs.

Example 10.11 Consider the binary CA, $\mathcal{R} = (\mathcal{R}_i)_{i \in \mathbb{Z}}$, where

$$\mathcal{R}_i = \begin{cases} 90 & \text{if } i \text{ is even} \\ 150 & \text{otherwise} \end{cases}$$

Now, when i is even then $\Lambda_i^p = \eta_i^p = \Lambda_i^c = \eta_i^c = 1$. For odd i, $\Lambda_i^p = \eta_i^p = 1$ and $\Lambda_i^c = \eta_i^c = 0$. Moreover, there is no blocking word in the CA. Hence, $p_1 = p_2 = 1$, and $p = (1, 1)$. So, the CA is chaotic as per the definition of p. Figure 10.9a substantiates it.



(a) p = (1, 1) (b) p = (0.75, 0.75) (c) p = (0, 1) (d) p = (0, 0.5)

Fig. 10.9 Space-time diagram and parameter p of some non-uniform CAs. The first two CAs are chaotic. The other two are not chaotic

Example 10.12 Consider a binary CA, $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, where

$$\mathcal{R}_i = \begin{cases} 193 & \text{if } i < 0 \\ 22 & \text{otherwise} \end{cases}$$

If $i < 0$, then $\Lambda_i^p = \eta_i^c = 0.25$ and $\Lambda_i^c = \eta_i^p = 0.75$, when $i \geq 0$, then $\Lambda_i^p = \eta_i^p = 0.75$ and $\Lambda_i^c = \eta_i^c = 0.25$. Hence, $\max(0.75, 0.25) = 0.75$ for all $i \in \mathbb{Z}$. The minimum of all these values is also 0.75. Further, no blocking word exists in the CA. Hence, $p_1 = 0.75$. For computing p_2, the $a_{i,i+1}$ is needed. Here, $a_{i,i+1} = 0.75$ for all $i \in \mathbb{Z}$. Therefore, the average is also 0.75. In this case, all the cells form a single communication class. So, $p_2 = 0.75$ and $p = (0.75, 0.75)$. So, it can be concluded that the CA is chaotic. Figure 10.9b substantiates the chaotic behavior of the CA.

Example 10.13 Consider the binary CA $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$, where

$$\mathcal{R}_i = \begin{cases} 150 & \text{if } i < 0 \\ 4 & \text{if } 0 \leq i \leq 100 \\ 90 & i > 100 \end{cases}$$

Now when $i < 0$, then $\Lambda_i^p = \eta_i^p = 1$ and $\Lambda_i^c = \eta_i^c = 0$, when $i \geq 0$ and $i < 100$, then $\Lambda_i^p = \eta_i^p = \Lambda_i^c = \eta_i^c = 0.25$, and when $i > 100$, then $\Lambda_i^p = \eta_i^p = \Lambda_i^c = \eta_i^c = 1$. The CA consists of blocking words. Hence, $p_1 = 0$, $p_2 = 1$, and $p = (0, 1)$. So, the CA can be concluded as non-chaotic. Figure 10.9c supports the fact.

Example 10.14 Consider the binary CA, $\mathcal{R} = (\mathcal{R}_i)_{i\in\mathbb{Z}}$ where

$$\mathcal{R}_i = \begin{cases} 80 & \text{if } |i| \text{ is prime} \\ 30 & \text{otherwise} \end{cases}$$

Now when i is prime, then $\Lambda_i^p = \eta_i^p = \Lambda_i^c = \eta_i^c = 0.5$; when i is not prime, then $\Lambda_i^p = \Lambda_i^c = \eta_i^c = 0.5$ and $\eta_i^p = 1$. However, the CA is having blocking words. Hence, $p_1 = 0$ and $p_2 = 0.5$. So the CA can be concluded as non-chaotic. Figure 10.9d justifies it.

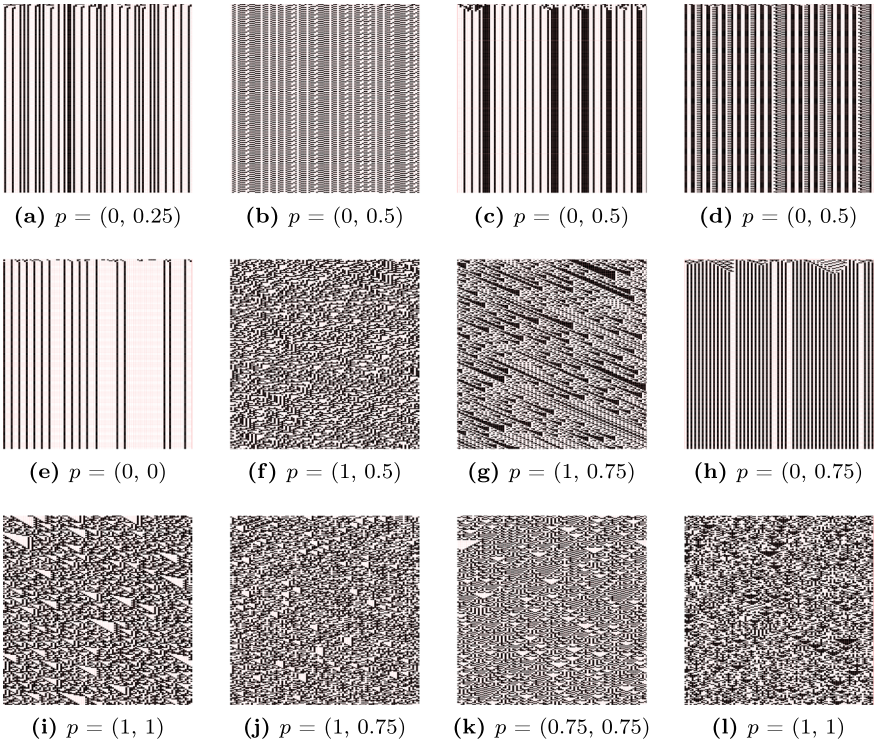
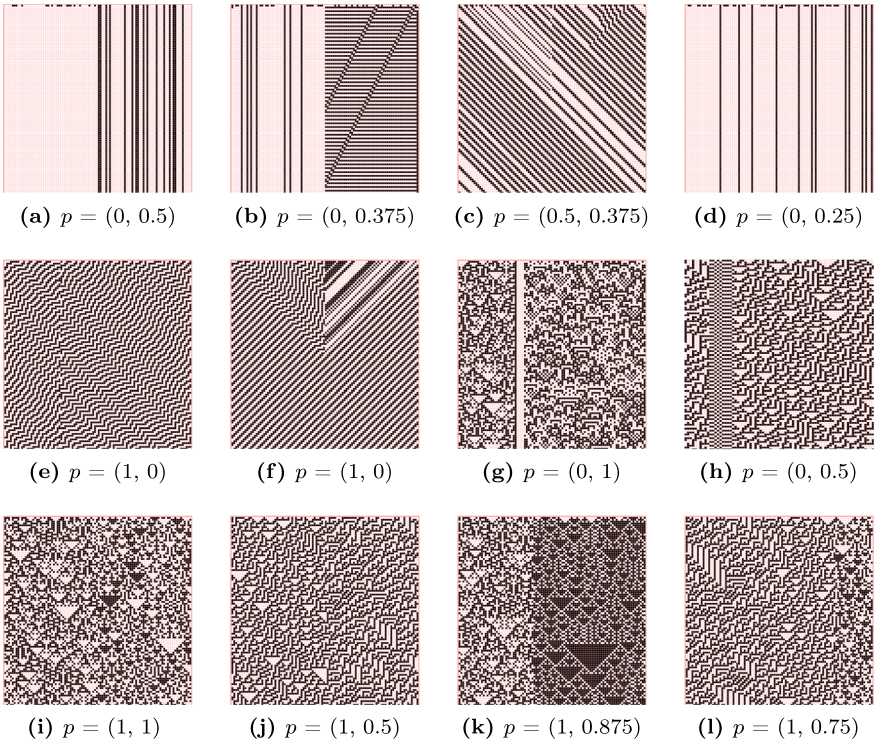


Fig. 10.10 Space-time diagrams and parameter p of CAs: **a** $(4, 30, 15, 100)^{\mathbb{Z}}$. **b** $(10, 18, 19, 41)^{\mathbb{Z}}$. **c** $(232, 156, 150, 65)^{\mathbb{Z}}$. **d** $(118, 130, 17, 75)^{\mathbb{Z}}$. **e** $(6, 0, 120, 52)^{\mathbb{Z}}$. **f** $(30, 45, 90)^{\mathbb{Z}}$. **g** $(210, 195)^{\mathbb{Z}}$. **h** $(193, 22)^{\mathbb{Z}}$. **i** $(90, 60, 150)^{\mathbb{Z}}$. **j** $(165, 30, 150)^{\mathbb{Z}}$. **k** $(82, 182)^{\mathbb{Z}}$. **l** $(150, 165)^{\mathbb{Z}}$

Some more examples of CAs along with their parameter values are given in Figs. 10.10 and 10.11. In the space-time diagrams, the initial configurations are taken arbitrarily, and the evolution of appropriately-chosen 100 cells of the CAs are shown. Although the p parameter is a measure for predicting chaotic behavior of CAs, but there are cases where it fails to predict correctly. For example, ECA 18 cannot be correctly identified by the p parameter.

10.6 Chapter Notes

In an early work, Hudlund showed that the configuration space of a CA forms a compact metric space with respect to Cantor topology and the global transition function of the CA is continuous in this topology [3]. Since then, the (uniform) CAs have been studied as topological dynamical systems and chaos therein [4].



(a) $p = (0, 0.5)$ **(b)** $p = (0, 0.375)$ **(c)** $p = (0.5, 0.375)$ **(d)** $p = (0, 0.25)$

(e) $p = (1, 0)$ **(f)** $p = (1, 0)$ **(g)** $p = (0, 1)$ **(h)** $p = (0, 0.5)$

(i) $p = (1, 1)$ **(j)** $p = (1, 0.5)$ **(k)** $p = (1, 0.875)$ **(l)** $p = (1, 0.75)$

Fig. 10.11 Space-time diagrams and parameter p of CAs: **a** $(\cdots 0, 204 \cdots)$. **b** $(\cdots 4, 119 \cdots)$. **c** $(\cdots 80, 84 \cdots)$. **d** $(\cdots 4, 150^3, 4 \cdots)$. **e** $(\cdots 15, 90, 15 \cdots)$. **f** $(\cdots 15, 170 \cdots)$. **g** $(\cdots 90, 204^5, 105 \cdots)$. **h** $(\cdots 45, 51^{10}, 30 \cdots)$. **i** $(\cdots 90, 150 \cdots)$. **j** $(\cdots 30, 45 \cdots)$. **k** $(\cdots 90, 182 \cdots)$. **l** $(\cdots 45, 30^{30}, 150 \cdots)$

Consequently, the study of non-uniform CA as topological dynamical system has been initiated [5].

There are many definitions of chaos, such as, Devaney's definition, Lyapunov's definition, Knudsen's definition, etc. In an interesting discussion, Cattaneo et al. [6] have pointed out that many of the definitions of chaos, including Devaney's definition, declare some CAs as chaotic which is the against of our intuitive understanding of chaos. An alternative definition of chaos is outlined in [7] to overcome this issue. In the same line of this work, chaos in non-uniform CAs has been studied in [8]. A consolidated discussion on chaos in CAs can be found in the PhD work of Supreeti Kamilya [9]. However, following are few research directions for future research on chaos in cellular automata.

1. Development of further improved parameter for predicting chaos in CAs (uniform and non-uniform).
2. Development of similar framework, as reported in this chapter, for two and higher dimensional CAs.

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